Oracle® AI Database

Oracle AI Vector Search User's Guide

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Preface

Oracle AI Database AI Vector Search User's Guide provides information about querying semantic and business data with Oracle AI Vector Search.

- Audience

Audience

This guide is intended for application developers, database administrators, data users, and others who perform the following tasks:

- Implement artificial intelligence (AI) solutions for websites and unstructured or structured data
- Build query applications by using natural language processing and machine learning techniques
- Perform similarity searches on content, such as words, documents, audio tracks, or images

To use this document, you must have a basic familiarity with vector embedding and machine learning concepts, SQL, SQL*Plus, and PL/SQL.

1 What's New for Oracle AI Vector Search

This chapter lists the following changes in *Oracle AI Database AI Vector Search User's Guide* for Oracle AI Database 26ai:

- Oracle AI Database 26ai Release Updates
  The following sections include new AI Vector Search features introduced in Oracle AI Database 26ai as part of the listed Release Update.
- Autonomous AI Database Updates
  The following sections include new AI Vector Search features introduced in Oracle Autonomous AI Database as part of the listed update.
- Deprecated Features
  The following features are deprecated, and may be desupported in a future release.

Oracle AI Database 26ai Release Updates

The following sections include new AI Vector Search features introduced in Oracle AI Database 26ai as part of the listed Release Update.

- July 2024, Release Update 23.5
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.5.
- October 2024, Release Update 23.6
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.6.
- January 2025, Release Update 23.7
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.7.
- April 2025, Release Update 23.8
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.8.
- July 2025, Release Update 23.9
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.9.
- January 2026, Release Update 23.26.1
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.1.
- April 2026, Release Update 23.26.2
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.2.
- July 2026, Release Update 23.26.3
  Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.3.

July 2024, Release Update 23.5

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.5.

Feature	Description
Binary Vectors	You can use the `BINARY` vector dimension format to declare vectors and vector columns. Create Tables Using the VECTOR Data Type
LangChain Integration	Oracle AI Vector Search's integration with LangChain allows you to use LangChain’s powerful open source orchestration framework seamlessly with vector search capabilities. Oracle AI Vector Search Integration with LangChain
Optimizer Plans for Vector Indexes	More information has been added about optimizer plans for HNSW and IVF indexes along with vector index hints. • Optimizer Plans for HNSW Vector Indexes • Optimizer Plans for IVF Vector Indexes • Vector Index Hints
Documentation Map to GenAI Prompts	Use the documentation map to find prompts that will help you get tailored answers from your preferred GenAI Chatbot to your Oracle AI Vector Search questions. Your Vector Documentation Map to GenAI Prompts

October 2024, Release Update 23.6

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.6.

> **Note**
>
> Certain features require the `COMPATIBLE` initialization parameter to be manually updated in order to be available for use. For information about the `COMPATIBLE` parameter and how to change it, see *Oracle AI Database Upgrade Guide*.
>
> If you are using RAC instances, certain features must be enabled using Oracle RAC two-stage rolling updates to ensure that any patches are enabled on all nodes. For more information, see *Oracle Real Application Clusters Administration and Deployment Guide*.

Feature	Description
Hybrid Search using Hybrid Vector Indexes	You can use a hybrid vector index to index and query data using a combination of full-text search and vector similarity search. Hybrid searches can enhance the relevance (quality) of your search results by integrating the keyword matching capabilities of Oracle Text indexes with the semantic precision of vector indexes. An end-to-end indexing pipeline facilitates the indexing of documents without requiring you to be an expert in various text processing, chunking, or embedding strategies. As part of this feature, new preference helper procedures have been added to the existing `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN` PL/SQL packages to manage vector index-related settings. A new PL/SQL package, `DBMS_HYBRID_VECTOR`, has also been added. The package contains a `SEARCH` API that allows you to query against hybrid vector indexes in multiple ways. A new dictionary view `<index name>$VECTORS` provides information on row identifiers, chunks, and embeddings for all indexed documents. • Manage Hybrid Vector Indexes • Perform Hybrid Search • DBMS_HYBRID_VECTOR • CREATE_PREFERENCE • DROP_PREFERENCE • <index name>$VECTORS
Updated AI Vector Search Workflow	The AI Vector Search Workflow included in this guide has been updated to include support for hybrid vector indexes and hybrid search. Oracle AI Vector Search Workflow
Updated Documentation Map to GenAI Prompts	The documentation map to GenAI prompts has been updated to include the new features available in Release Update 23.6. Your Vector Documentation Map to GenAI Prompts
Extended SQL Quick Start	The *SQL Quick Start Using a Vector Embedding Model Uploaded into the Database* has been extended to include steps for creating a hybrid vector index and running a hybrid search. SQL Quick Start Using a Vector Embedding Model Uploaded into the Database
Support for Ollama	You can use open embedding models (such as mxbai-embed-large, nomic-embed-text, or all-minilm) and large language models (such as Llama 3, Phi 3, Mistral, or Gemma 2) using the local host REST endpoint provider, Ollama. Ollama is a free and open-source command-line interface tool that allows you to run these models locally and privately on your Linux, Windows, and macOS systems. • Convert Text String to Embedding Using the Local REST Provider Ollama • Generate Summary Using the Local REST Provider Ollama • Generate Text Using the Local REST Provider Ollama
Support for `SPARSE` Vectors	In addition to `DENSE` vectors, you have the option to create `SPARSE` vectors. Sparse vectors are vectors that typically have a large number of dimensions but only a few of those dimensions have non-zero values. Because sparse vectors only store non-zero values, their use can improve efficiency and save storage space. The `COMPATIBLE` initialization parameter must be set to 23.6.0 to use this feature. Create Tables Using the VECTOR Data Type

Feature	Description
Integration with LlamaIndex	Oracle AI Vector Search's integration with LlamaIndex allows you to use LlamaIndex's open-source data framework with vector search capabilities. This provides a powerful foundation for building sophisticated AI applications that can leverage both structured and unstructured data within the Oracle ecosystem. Oracle AI Vector Search Integration with LlamaIndex
Jaccard Distance	`JACCARD` is available as a distance metric and `JACCARD_DISTANCE` as a shorthand for the `VECTOR_DISTANCE` function to calculate the Jaccard distance between two `BINARY` vectors. • Jaccard Similarity • JACCARD_DISTANCE
Hamming Distance	`HAMMING_DISTANCE` is available as a shorthand for the `VECTOR_DISTANCE` function to calculate the Hamming similarity between two vectors. HAMMING_DISTANCE
Transaction Support for HNSW Indexes	Transactions are maintained on tables with Hierarchical Navigable Small World (HNSW) index graphs by using data structures, such as private journal and shared journal. A private journal records vectors that are added or deleted by a transaction, whereas a shared journal records all the commit system change numbers (SCNs) and corresponding modified rows. If using RAC instances, you must enable Patch 36932885 using Oracle RAC two-stage rolling updates. HNSW Index Architecture: Transaction Support and Persistence
HNSW Index Duplication and Reload	By default, a duplication mechanism is used to create HNSW indexes in an Oracle RAC environment or when an HNSW index repopulation operation is triggered. When an Oracle AI Database instance starts again, a reload mechanism is triggered to recreate the HNSW graph in memory as quickly as possible. HNSW full checkpoints are used to reload the graph in memory. You can use the `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` initialization parameter to manage duplication and reload mechanisms for HNSW indexes. If using RAC instances, you must enable Patch 36932885 using Oracle RAC two-stage rolling updates. HNSW RAC Duplication
Update to the default for `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD`	The default value for the `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` initialization parameter has been changed from `OFF` to `RESTART`. With this update, HNSW duplication and reload mechanisms are enabled by default. HNSW RAC Duplication
Global and Local Partitioning for IVF Indexes	Inverted File Flat (IVF) vector indexes support both global and local indexes on partitioned tables. By default, IVF indexes are globally partitioned by centroid. You can choose to create a local IVF index, which provides a one-to-one relationship between the base table partitions or subpartitions and the index partitions. If using RAC instances, you must enable Patch 36932885 using Oracle RAC two-stage rolling updates. • Partition Maintenance Operations with GLOBAL Vector Indexes • Inverted File Flat CREATE INDEX

Feature	Description
GET_INDEX_STATUS, ENABLE_CHECKPOINT, DISABLE_CHECKPOINT, and INDEX_VECTOR_MEMORY_ADVISOR Procedures	New procedures are available with the DBMS_VECTOR PL/SQL package: • GET_INDEX_STATUS to query the status of a vector index creation • ENABLE_CHECKPOINT and DISABLE_CHECKPOINT to enable or disable the Checkpoint feature for HNSW indexes • INDEX_VECTOR_MEMORY_ADVISOR to determine the vector memory size needed for a vector index Vector Index Status, Checkpoint, and Advisor Procedures
VECSYS.VECTOR$INDEX$CHECKPOINTS View	A new dictionary view VECSYS.VECTOR$INDEX$CHECKPOINTS provides information about HNSW full checkpoints at the database level. VECSYS.VECTOR$INDEX$CHECKPOINTS
Relational Data Vectorization	You can use Oracle Machine Learning (OML) Feature Extraction algorithms with the VECTOR_EMBEDDING() SQL operator to vectorize sets of relational data, build similarity indexes, and perform similarity searches. These algorithms help in extracting the most informative features or columns from the data, making it easier to analyze correlations and redundancies in that data. Vectorize Relational Tables Using OML Feature Extraction Algorithms
Document Reranking	You can use the UTL_TO_RERANK function, added to the existing DBMS_VECTOR and DBMS_VECTOR_CHAIN PL/SQL packages, to enhance the relevancy of your query results in Retrieval Augmented Generation (RAG) scenarios. Third-party reranking models are used to reassess and reorder an initial list of retrieved documents. • Use Reranking for Better RAG Results • DBMS_VECTOR: UTL_TO_RERANK • DBMS_VECTOR_CHAIN: UTL_TO_RERANK
BLOB Support for UTL_TO_GENERATE_TEXT()	In addition to the existing textual input, the UTL_TO_GENERATE_TEXT() PL/SQL function accepts BLOB as input. You can provide binary data, such as an image file, to generate a textual analysis or meaningful description of the contents of an image. Describe Images Using Public REST Providers
BLOB Support for UTL_TO_EMBEDDING()	In addition to the existing textual input, the UTL_TO_EMBEDDING() PL/SQL function accepts BLOB as input. This lets you directly generate a vector embedding from image files using REST API calls to third-party image embedding models. Convert Image to Embedding Using Public REST Providers
List of REST Endpoints	You can refer to the list of REST endpoints and corresponding REST calls supported for all third-party REST providers. Supported Third-Party Provider Operations and Endpoints

January 2025, Release Update 23.7

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.7.

> **Note**
>
> If you are using RAC instances, certain features must be enabled using Oracle RAC two-stage rolling updates to ensure that any patches are enabled on all nodes. For more information, see *Oracle Real Application Clusters Administration and Deployment Guide*.

Feature	Description
External Table support	Columns of type VECTOR can be included in external tables, providing the option to store vector embeddings used for AI workloads outside of the database while still using the database as a semantic search engine. If using RAC instances, you must enable Patch 37244967 using Oracle RAC two-stage rolling updates. Vectors in External Tables
Arithmetic and aggregate function support	The arithmetic operators for addition, subtraction, and multiplication along with the aggregate functions SUM and AVG can be used with vectors and columns of type VECTOR. Arithmetic operators are supported for use with vectors in both SQL and PL/SQL. If using RAC instances, you must enable Patch 37289468 using Oracle RAC two-stage rolling updates for support in PL/SQL. Arithmetic Operators Aggregate Functions
PL/SQL BINARY vector support	In addition to FLOAT32 (the default format for comma-separated string representations of vectors), FLOAT64, and INT8 dimension formats, you can also use the BINARY dimension format. A BINARY vector represents each dimension as a single bit (0 or 1). BINARY vectors can now be created and declared in PL/SQL, supported in the same way as they are in SQL. If using RAC instances, you must enable Patch 37289468 using Oracle RAC two-stage rolling updates. BINARY Vectors
PL/SQL JACCARD distance metric support	The distance metric JACCARD can be used in PL/SQL for similarity searches with BINARY vectors. In PL/SQL, there is no standalone shorthand JACCARD_DISTANCE function. Use JACCARD as the distance metric parameter of the VECTOR_DISTANCE function instead. If using RAC instances, you must enable Patch 37289468 using Oracle RAC two-stage rolling updates. VECTOR_DISTANCE For additional information about VECTOR operations supported by PL/SQL, see *Oracle AI Database PL/SQL Language Reference*
Globally Distributed Database support	Columns of type VECTOR can be included in sharded tables and duplicated tables in a distributed database, and vector indexes are supported on sharded tables in a distributed database, with some restrictions. Vectors in Distributed Database Tables Vector Indexes in a Globally Distributed Database "Restrictions" in Create Tables Using the VECTOR Data Type

<table>
<tr><th>Feature</th><th>Description</th></tr>
<tr><td>In-Database Algorithms Support for VECTOR Data Type Predictors</td><td>The VECTOR data type is supported and can be used as an input to database machine learning algorithms such as classification, anomaly, regression, clustering and feature extraction.

Overview of Oracle AI Vector Search

More information on using VECTOR data type in machine learning can be found in Vector Data Type Support</td></tr>
<tr><td>Support for Image Transformer Models with AI Vector Search Using the In-Database ONNX Runtime</td><td>The Oracle AI Database ONNX runtime engine includes support of models for encoding images.

The ONNX Pipeline models provide methods for:
• Text Embedding - ONNX Pipeline Models: Text Embedding
• Image Embedding - ONNX Pipeline Models: Image Embedding
• Multi-modal Embedding - ONNX Pipeline Models: CLIP Multi-Modal Embedding
• Text Classification - ONNX Pipeline Models: Text Classification
• Reranking - ONNX Pipeline Models: Reranking Pipeline</td></tr>
<tr><td>Hybrid Vector Index for JSON</td><td>New PATHS field is added to the DBMS_VECTOR_CHAIN.VECTORIZER that allows specification of an array of path objects : CREATE_PREFERENCE

New syntax in the DBMS_HYBRID_VECTOR.SEARCH API to allow searches restricted to paths. The VECTOR field accepts an INPATH parameter which an array of valid JSON paths to restrict the search : SEARCH

Support for the creation of a LOCAL index on a HVI when the underlying vector index_type is IVF : CREATE HYBRID VECTOR INDEX

Support for running optimization only on the $VR table for an HVI : Hybrid Vector Index Maintenance Operations</td></tr>
<tr><td>Included Columns in Neighbor Partition Vector Indexes</td><td>Included columns permit additional table (non-vector) columns to be stored in a Neighbor Partition vector index. The benefit is that query execution is optimized by removing the need to access the underlying base table to retrieve these columns.

If using RAC instances, you must enable Patch 37306139 using Oracle RAC two-stage rolling updates.

Included Columns with IVF Indexes</td></tr>
</table>

April 2025, Release Update 23.8

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.8.

> **Note**
>
> If you are using RAC instances, certain features must be enabled using Oracle RAC two-stage rolling updates to ensure that any patches are enabled on all nodes. For more information, see *Oracle Real Application Clusters Administration and Deployment Guide*.

Feature	Description
Custom JavaScript Distance Function	In addition to the availability of built-in vector distance functions, you can now create your own custom distance function. The Multilingual Engine (MLE) is used to create a JavaScript function that defines the distance operation of your choice. The custom distance function can be used in similarity searches and in the creation of HNSW vector indexes. Custom Distance Function
Automatic Vector Pool sizing with on-premises deployments	The vector memory pool can be configured to dynamically grow to meet the sizing needs for creating HNSW indexes. This is now supported for both Oracle Autonomous AI Databases and non-Autonomous AI databases. Size the Vector Pool
`SPARSE` support in PL/SQL	PL/SQL now natively supports the creation and declaration of `SPARSE` vectors. If using RAC instances, you must enable Patch 37535524 using Oracle RAC two-stage rolling updates. SPARSE Vectors *Oracle AI Database PL/SQL Language Reference*
New table function: `DBMS_HYBRID_VECTOR.SEARCHPIPELINE`	In addition to the existing `DBMS_HYBRID_VECTOR.SEARCH` API, the new table function `DBMS_HYBRID_VECTOR.SEARCHPIPELINE` returns a pipeline of row records. SEARCHPIPELINE
New function: `DBMS_HYBRID_VECTOR.GET_SQL`	`DBMS_HYBRID_VECTOR.GET_SQL` function displays the internal SQL query that is executed for `DBMS_HYBRID_VECTOR.SEARCH`. GET_SQL
`FILTER_BY` support in `DBMS_HYBRID_VECTOR.SEARCH`	A `FILTER_BY` field is added in the `SEARCH` API of the `DBMS_HYBRID_VECTOR` package. The `FILTER_BY` field allows you to narrow the search results using standard relational logical constraints. FILTER_BY
Using JSON in Included Columns with IVF Indexes	Restrictions have been removed on JSON, `BLOB`, and `CLOB` data types, enabling broader and more flexible usage of these data types in included columns. Included Columns with IVF Indexes

July 2025, Release Update 23.9

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.9.

Feature	Description
IVF Index Reorganization	The quality of an IVF index may degrade over time if updates are made to the base table after the general vector distribution. IVF index reorganization permits the index to be reorganized while it remains available for DMLs and queries. Inverted File Flat Vector Indexes Online Rebuild
HNSW Index Snapshots	Incremental snapshots are introduced as an HNSW graph refresh mechanism optimized for minimal, frequent updates, balancing performance and memory use. HNSW Index Architecture: Transaction Support and Persistence

Feature	Description
Partition Maintenance Operation support for global IVF and HNSW Indexes	Partition Maintenance Operations can now be performed on partitioned tables that have global IVF and HNSW Indexes, including operations such as adding, dropping, merging, and splitting partitions. Partition Maintenance Operations and Vector indexes • Partition Maintenance Operations with GLOBAL Vector Indexes
SPARSE HNSW Indexes	Hierarchical Navigable Small World (HNSW) vector indexes support SPARSE vectors. SPARSE Vectors
Creating Hybrid Vector Index using DBMS_SEARCH API	Hybrid Vector Indexes (HVI) can now be created using the DBMS_SEARCH PL/SQL package. Creating Hybrid Vector Indexes using the DBMS_SEARCH API
Enhancements for altering Hybrid Vector Indexes	Existing Hybrid Vector Indexes can be altered using the SQL ALTER INDEX syntax to change settings such as replacing the vectorizer preference, the embedding model, or the type of vector index. ALTER INDEX
Enhancements for searching Hybrid Vector Indexes using the DBMS_HYBRID_VECTOR.SEARCH API	The SEARCH API for Hybrid Vector Indexes adds support for Weighted Reciprocal Rank Fusion (WRRF) as a method to search scores from keyword and semantic search results. It is also possible to define custom score calculations using a new parameter: score_calc. Additional enhancements include a new query parameter for keyword searches of Hybrid Vector Indexes as well as the ability to filter results using standard relational logic. SEARCH

January 2026, Release Update 23.26.1

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.1.

Feature	Description
Distributed HNSW Indexes with RAC	Distributed HNSW indexes on Real Application Cluster (RAC) databases can now seamlessly scale across the total memory available in all instances of a RAC database. With this enhancement, the vector memory pool can now span multiple RAC instances, meaning larger HNSW indexes are supported and the number of CPUs available for parallel operations has increased. HNSW Distribution on Oracle RAC
Online Build of Vector Indexes	It is now possible to create or rebuild a vector index online, with the underlying base table remaining open and updatable during the operation. Hierarchical Navigable Small World Index Syntax and Parameters
Scalar Quantized HNSW Indexes	Scalar quantization can now be used to compress HNSW indexes, reducing storage requirements and improving efficiency without significant loss of accuracy. Scalar Quantized HNSW Indexes

Feature	Description
Included Columns with HNSW Indexes	Included columns permit additional non-vector columns in a base table to be stored in an HNSW vector index. This means that the query execution no longer needs additional table access to retrieve the underlying columns from the base table, resulting in faster similarity search queries. Included Columns with HNSW Indexes
Automatic IVF Index Reorganization	IVF vector indexes can now be automatically reorganized after the database system has determined that a rebuild is necessary, without human intervention. This means that you do not have to manually decide the thresholds determining whether a rebuild is necessary as the system does it for you. Inverted File Flat Vector Indexes Online Rebuild

April 2026, Release Update 23.26.2

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.2.

Feature	Description
Function-Based Hierarchical Navigable Small World Indexes	HNSW vector indexes can now be created on an expression instead of a vector column. The index expression can be an arithmetic, conditional expression or any other expression that produces a vector. Function-Based Hierarchical Navigable Small World Index
Local Hierarchical Navigable Small World Indexes	HNSW vector indexes can now be created for each partition or sub partition of a partitioned table. Instead of building a single, global HNSW index across all vectors in a large, partitioned table, this enhancement enables the creation of individual HNSW indexes for each (sub)partition. Local Hierarchical Navigable Small World Index
Multi-Vector Search Using IVF Indexes	IVF indexes accelerate multi-vector search by partitioning the vector space into a configurable set of centroids. At query time, the search is limited to the most relevant partitions instead of scanning all vectors, delivering significantly faster response times for multi-vector workloads compared to a full scan. Multi-Vector Search Using IVF Indexes
IVF Vector Index on Iceberg External Tables	Oracle AI Vector Search now supports creating IVF vector indexes on Iceberg external tables stored in cloud object storage (OCI Object Storage, Amazon S3). IVF Indexing on External Iceberg Tables

July 2026, Release Update 23.26.3

Included are some notable Oracle AI Vector Search updates with Oracle AI Database 26ai, Release Update 23.26.3.

Feature	Description
Scalar Quantization Support for Distributed HNSW Indexes	Distributed HNSW indexes now support scalar quantization. Scalar quantization enables the compression of vectors into a more compact representation while preserving similarity characteristics. HNSW Distribution on Oracle RAC

Autonomous AI Database Updates

The following sections include new AI Vector Search features introduced in Oracle Autonomous AI Database as part of the listed update.

- June 2024
  Included are some notable Oracle AI Vector Search updates with the Autonomous AI Database, June 2024 release.
- July 2024
  Included are some notable Oracle AI Vector Search updates with the Autonomous AI Database, July 2024 release.
- October 2024
  New features are available for Oracle AI Database in October with Release Update 23.6.
- January 2025
  New features are available for Oracle AI Database in January with Release Update 23.7.
- April 2025
  New features are available for Oracle AI Database in April with Release Update 23.8.
- July 2025
  New features are available for Oracle AI Database in July with Release Update 23.9.
- January 2026
  New features are available for Oracle AI Database in January with Release Update 23.26.1.
- April 2026
  New features are available for Oracle AI Database in April with Release Update 23.26.2.

June 2024

Included are some notable Oracle AI Vector Search updates with the Autonomous AI Database, June 2024 release.

Feature	Description
HNSW Index Support in RAC Environments	It is possible to use Hierarchical Navigable Small World (HNSW) indexes with `SELECT` statements in RAC environments. For more information, see HNSW RAC Duplication.
Sizing the Vector Pool with ADB-S	With Autonomous AI Database Serverless (ADB-S) services, the vector pool dynamically grows and shrinks. It cannot be explicitly set. For more information, see Size the Vector Pool.

Feature	Description
SQL Quick Start with Vector Generator	A SQL scenario is provided to get you started using Oracle AI Vector Search. The quick start includes a PL/SQL program that creates a vector generator, offering a simple alternative to using a vector embedding model. For the full tutorial, see SQL Quick Start Using a FLOAT32 Vector Generator.
Similarity Search and Index Creation Syntax	• The `APPROX` and `APPROXIMATE` keywords are now optional. If omitted while connected to an ADB-S instance, an approximate search using a vector index is attempted if one exists. For more information about approximate search with indexes, see Approximate Search Using HNSW and Approximate Search Using IVF. • The `NEIGHBOR` keyword in the `ORGANIZATION` clause of `CREATE VECTOR INDEX` is now optional. For the full syntax to create HNSW and IVF indexes, see Hierarchical Navigable Small World Index Syntax and Parameters and Inverted File Flat CREATE INDEX.

July 2024

Included are some notable Oracle AI Vector Search updates with the Autonomous AI Database, July 2024 release.

Feature	Description
New Vector Index Views	The following vector index views are now available with Autonomous AI Database Serverless (ADB-S) services: • V$VECTOR_INDEX • V$VECTOR_GRAPH_INDEX • V$VECTOR_PARTITIONS_INDEX

October 2024

New features are available for Oracle AI Database in October with Release Update 23.6.

For information, see October 2024, Release Update 23.6.

January 2025

New features are available for Oracle AI Database in January with Release Update 23.7.

For information, see January 2025, Release Update 23.7.

April 2025

New features are available for Oracle AI Database in April with Release Update 23.8.

For information, see April 2025, Release Update 23.8.

July 2025

New features are available for Oracle AI Database in July with Release Update 23.9.

For information, see July 2025, Release Update 23.9.

January 2026

New features are available for Oracle AI Database in January with Release Update 23.26.1.

For information, see January 2026, Release Update 23.26.1.

April 2026

New features are available for Oracle AI Database in April with Release Update 23.26.2.

For information, see April 2026, Release Update 23.26.2.

Deprecated Features

The following features are deprecated, and may be desupported in a future release.

Starting with Oracle AI Database 26ai, Release Update 23.7, the python packages EmbeddingModel and EmbeddingModelConfig are deprecated. These packages are replaced with ONNXPipeline and ONNXPipelineConfig respectively.

- Details and utility of the deprecated package can be found here : Python Classes to Convert Pretrained Models to ONNX Models (Deprecated)
- Oracle recommends that you use the latest version. You can find the details of the updated python classes in Import Pretrained Models in ONNX Format

2 Overview

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.

Overview of Oracle AI Vector Search

Oracle AI Vector Search is designed for Artificial Intelligence (AI) workloads and allows you to query data based on semantics, rather than keywords.

VECTOR Data Type

The `VECTOR` data type is introduced with the release of Oracle AI Database 26ai, providing the foundation to store vector embeddings alongside business data in the database. Using embedding models, you can transform unstructured data into vector embeddings that can then be used for semantic queries on business data. In order to use the `VECTOR` data type and its related features, the `COMPATIBLE` initialization parameter must be set to 23.4.0 or higher. For more information about the parameter and how to change it, see *Oracle AI Database Upgrade Guide*.

See the following basic example of using the `VECTOR` data type in a table definition:

```
CREATE TABLE docs (doc_id INT, doc_text CLOB, doc_vector VECTOR);
```

For more information about the `VECTOR` data type and how to use vectors in tables, see Create Tables Using the VECTOR Data Type.

Due to the numerical nature of the `VECTOR` data type, you can use it as an input to the machine learning algorithms such as classification, anomaly, regression, clustering and feature extraction. More details on using `VECTOR` data type in machine learning could be found in Vector Data Type Support.

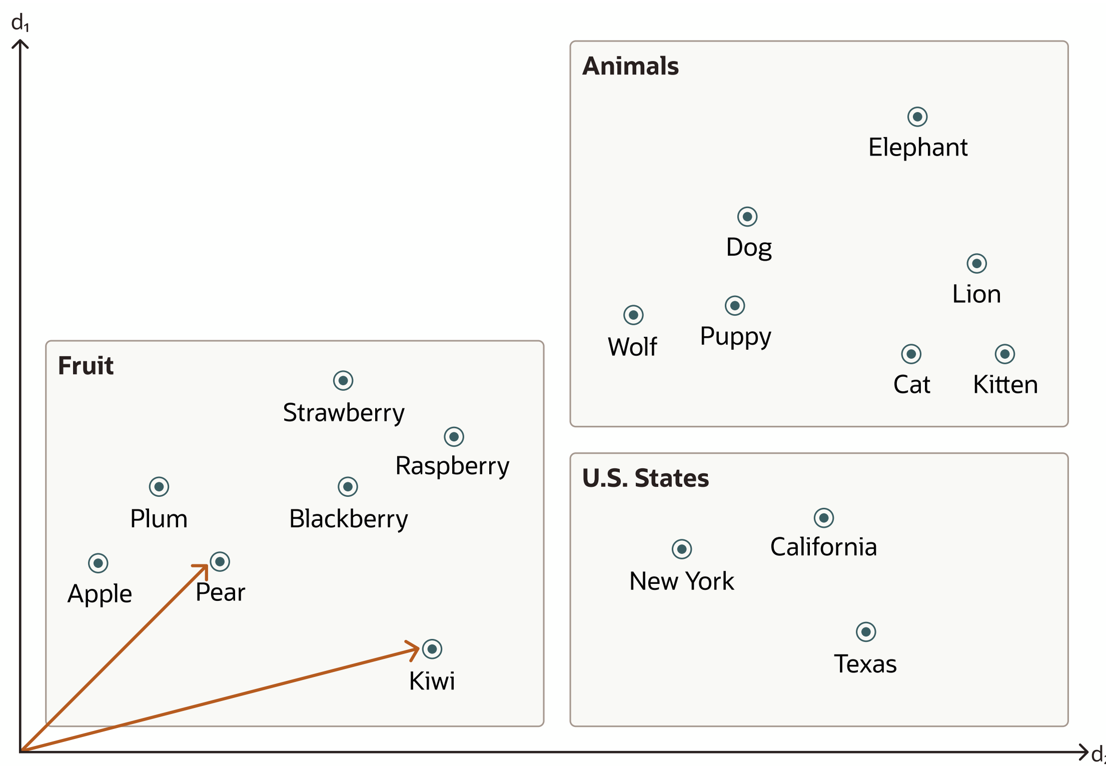
> **Note**
>
> Support for `VECTOR` data type machine learning is available in all versions starting with 23.7.

Vector Embeddings

If you've ever used applications such as voice assistants, chatbots, language translators, recommendation systems, anomaly detection, or video search and recognition, you've implicitly used vector embeddings features.

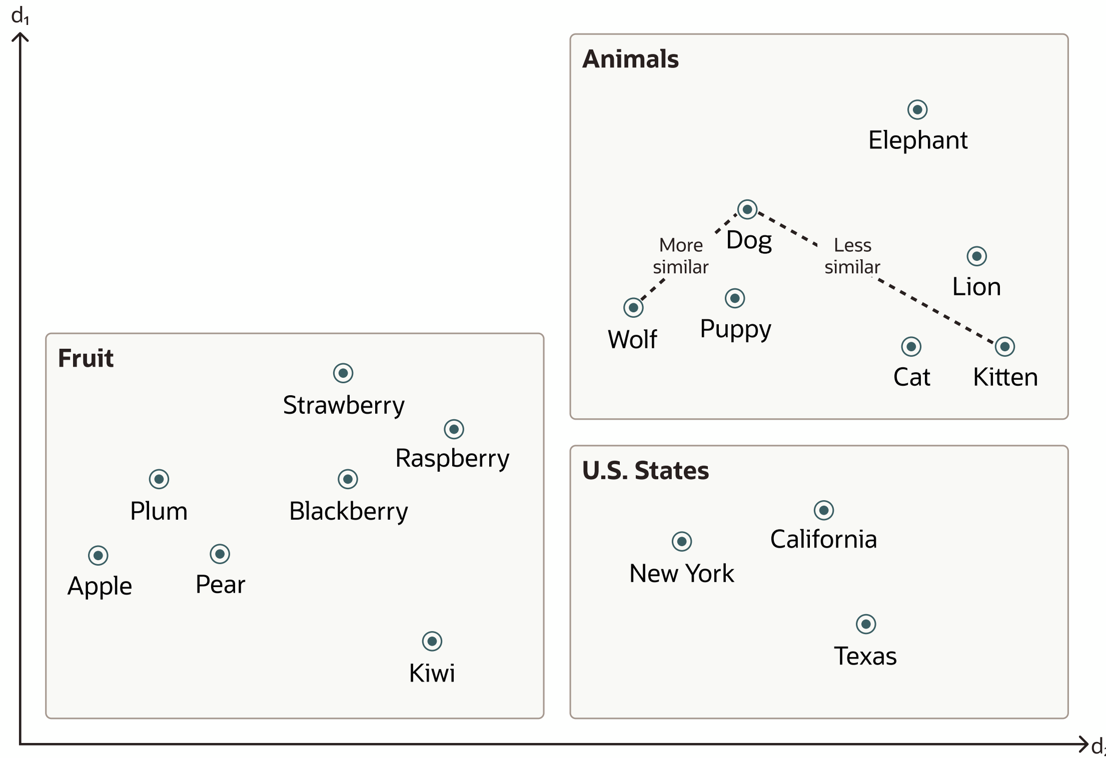
Oracle AI Vector Search stores vector embeddings, which are mathematical vector representations of data points. These vector embeddings describe the semantic meaning behind content such as words, documents, audio tracks, or images. As an example, while doing text based searches, vector search is often considered better than keyword search as vector search is based on the meaning and context behind the words and not the actual words themselves. This vector representation translates semantic similarity of objects, as perceived by humans, into proximity in a mathematical vector space. This vector space usually has many hundreds, if not thousands, of dimensions. Put differently, vector embeddings are a way of representing almost any kind of data, such as text, images, videos, users, or music as points in a multidimensional space where the locations of those points in space, and proximity to others, are semantically meaningful.

This simplified diagram illustrates a vector space where words are encoded as 2-dimensional vectors.



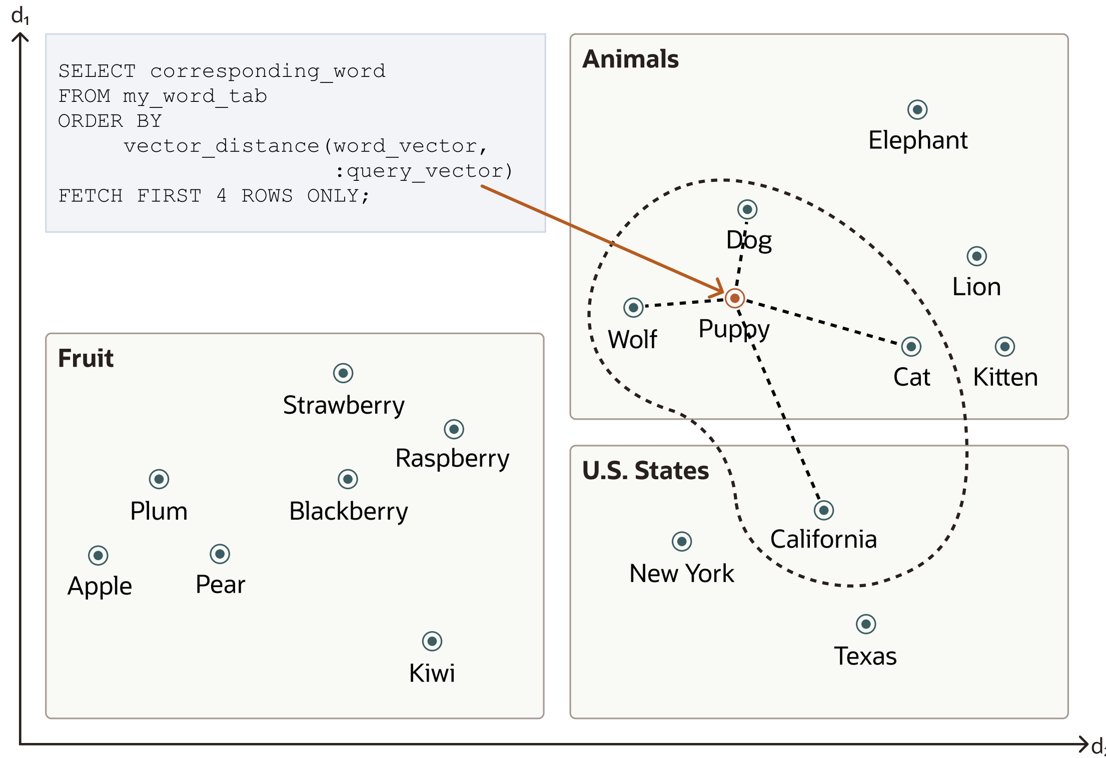
Similarity Search

Searching semantic similarity in a data set is now equivalent to searching nearest neighbors in a vector space instead of using traditional keyword searches using query predicates. As illustrated in the following diagram, the distance between *dog* and *wolf* in this vector space is shorter than the distance between *dog* and *kitten*. In this space, a dog is more similar to a wolf than it is to a kitten. See Perform Exact Similarity Search for more information.



Vector data tends to be unevenly distributed and clustered into groups that are semantically related. Doing a similarity search based on a given query vector is equivalent to retrieving the K-nearest vectors to your query vector in your vector space. Basically, you need to find an ordered list of vectors by ranking them, where the first row in the list is the closest or most similar vector to the query vector, the second row in the list is the second closest vector to the query vector, and so on. When doing a similarity search, the relative order of distances is what really matters rather than the actual distance.

Using the preceding vector space, here is an illustration of a semantic search where your query vector is the one corresponding to the word *Puppy* and you want to identify the four closest words:



Similarity searches tend to get data from one or more clusters depending on the value of the query vector and the fetch size.

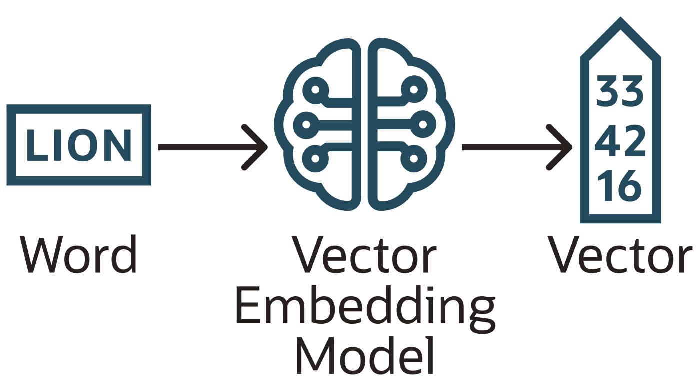
Approximate searches using *vector indexes* can limit the searches to specific clusters, whereas exact searches visit vectors across all clusters. See Use Vector Indexes for more information.

Vector Embedding Models

One way of creating such vector embeddings could be to use someone's domain expertise to quantify a predefined set of features or dimensions such as shape, texture, color, sentiment, and many others, depending on the object type with which you're dealing. However, the efficiency of this method depends on the use case and is not always cost effective.

Instead, vector embeddings are created via neural networks. Most modern vector embeddings use a transformer model, as illustrated by the following diagram, but convolutional neural networks can also be used.

Figure 2-1 Vector Embedding Model



Depending on the type of your data, you can use different pretrained, open-source models to create vector embeddings. For example:

- For textual data, sentence transformers transform words, sentences, or paragraphs into vector embeddings.
- For visual data, you can use Residual Network (ResNet) to generate vector embeddings.
- For audio data, you can use the visual spectrogram representation of the audio data to fall back into the visual data case.

Each model also determines the number of dimensions for your vectors. For example:

- Cohere's embedding model embed-english-v3.0 has 1024 dimensions.
- OpenAI's embedding model text-embedding-3-large has 3072 dimensions.
- Hugging Face's embedding model all-MiniLM-L6-v2 has 384 dimensions

Of course, you can always create your own model that is trained with your own data set.

Import Embedding Models into Oracle AI Database

Although you can generate vector embeddings outside the Oracle AI Database using pretrained open-source embeddings models or your own embeddings models, you also have the option to import those models directly into the Oracle AI Database if they are compatible with the Open Neural Network Exchange (ONNX) standard. Oracle AI Database implements an ONNX runtime directly within the database. This allows you to generate vector embeddings directly within the Oracle AI Database using SQL. See Generate Vector Embeddings for more information.

Why Use Oracle AI Vector Search?

One of the biggest benefits of Oracle AI Vector Search is that semantic search on unstructured data can be combined with relational search on business data in one single system.

This is not only powerful but also significantly more effective because you don't need to add a specialized vector database, eliminating the pain of data fragmentation between multiple systems.

For example, suppose you use an application that allows you to find a house that is similar to a picture you took of one you like that is located in your preferred area for a certain budget. Finding a good match in this case requires combining a semantic picture search with searches on business data.

With Oracle AI Vector Search, you can create the following table:

```
CREATE TABLE house_for_sale (house_id     NUMBER,
                             price        NUMBER,
                             city         VARCHAR2(400),
                             house_photo  BLOB,
                             house_vector VECTOR);
```

The following sections of this guide describe in detail the meaning of the `VECTOR` data type and how to load data in this column data type.

With that table, you can run the following query to answer your basic question:

```
SELECT house_photo, city, price
FROM   house_for_sale
```

```
WHERE  price <= :input_price AND
       city  = :input_city
ORDER BY VECTOR_DISTANCE(house_vector, :input_vector);
```

Later sections of this guide describe in detail the meaning of the `VECTOR_DISTANCE` function. This query is just to show you how simple it is to combine a vector embedding similarity search with relation predicates.

In conjunction with Oracle AI Database 26ai, Oracle Exadata System Software release 24.1.0 introduces AI Smart Scan, a collection of Exadata-specific optimizations capable of improving the performance of various AI vector query operations by orders of magnitude.

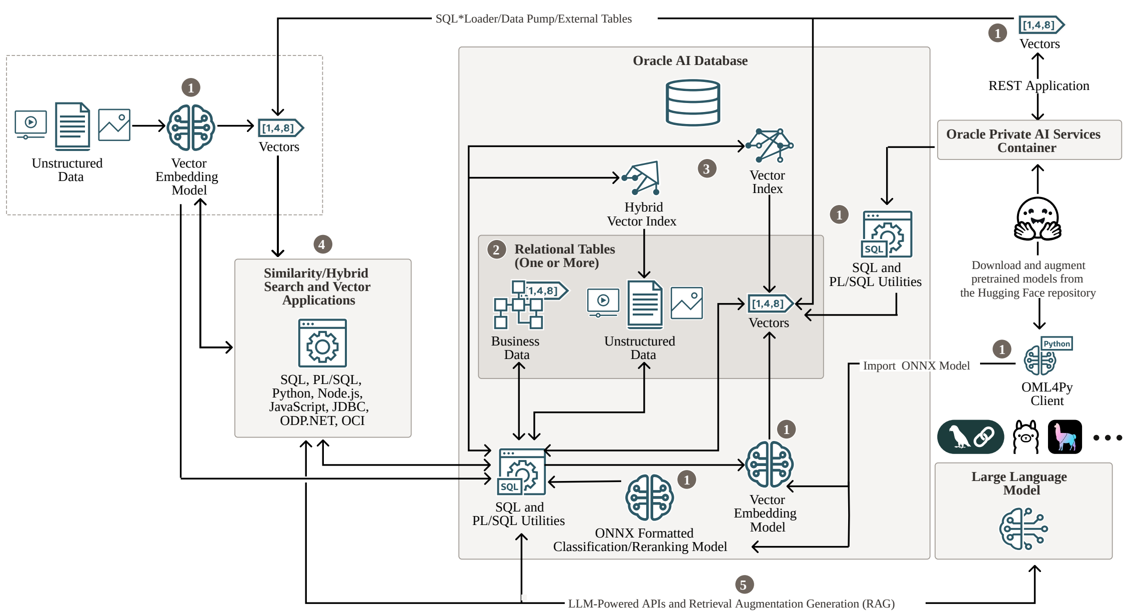
AI Smart Scan automatically accelerates Oracle AI Database 26ai AI Vector Search with optimizations that deliver low-latency parallelized scans across massive volumes of vector data. AI Smart Scan processes vector data at memory speed, leveraging ultra-fast Exadata RDMA Memory (XRMEM) and Exadata Smart Flash Cache in the Exadata storage servers, and performs vector distance computations and top-K filtering at the data source, avoiding unnecessary network data transfer and database server processing.

Oracle AI Vector Search Workflow

A typical Oracle AI Vector Search workflow follows the included primary steps.

This is illustrated in the following diagram:

Figure 2-2   Oracle AI Vector Search Use Case Flowchart



> **Note**
>
> Find all of our Interactive Architecture Diagrams on the Oracle Help Center.

Oracle AI Vector Search is designed for Artificial Intelligence (AI) workloads. It allows you to query data based on semantics and image similarity rather than simply keywords. The preceding diagram shows the possible steps you must take to manage vector embeddings with Oracle AI Vector Search.

Primary workflow steps:

1. **Generate Vector Embeddings from Your Unstructured Data**
   You can perform this step either outside or within Oracle AI Database. To perform this step inside Oracle AI Database, you must first import a vector embedding model using the ONNX standard. Your unstructured data can reside within or outside Oracle AI Database.

   For more information, see Generate Vector Embeddings and *Oracle Private AI Services Container User's Guide*.

2. **Store Vector Embeddings, Unstructured Data, and Relational Business Data in Oracle AI Database**
   After you have generated the vector embeddings, you can store them along with the corresponding unstructured and relational business data. If vector embeddings are stored outside Oracle AI Database, you can use SQL*Loader or Data Pump to load the vector embedding inside a relational table within Oracle AI Database. It is also possible to access vector embeddings stored outside the database through external tables.

   For more information, see Store Vector Embeddings.

3. **Create Vector Indexes and Hybrid Vector Indexes**
   Similar to how you create indexes on regular table columns, you can create vector indexes on vector embeddings, and you can create hybrid vector indexes (a combination of Oracle Text index and vector index) on your unstructured data. This is beneficial for running similarity searches over huge vector spaces.

   For more information, see Create Vector Indexes and Hybrid Vector Indexes.

4. **Query Data with Similarity and Hybrid Searches**
   You can then use Oracle AI Vector Search native SQL operations to combine similarity with traditional relational key searches. In addition, you can run hybrid searches, an advanced information retrieval technique that combines both the similarity and keyword searches to achieve highly relevant search results. SQL and PL/SQL provide powerful utilities to transform unstructured data, such as documents, into chunks before generating vector embeddings on each chunk.

   For more information, see Query Data With Similarity and Hybrid Searches and Supported Clients and Languages.

5. **Generate a Prompt and Send it to an LLM for a Full RAG Inference**
   You can use vector utility PL/SQL APIs for prompting large language models (LLMs) with textual prompts and images using LLM-powered interfaces. LLMs inherently lack the ability to access or incorporate new information after their training cutoff. By providing your LLM with up-to-date facts from your company, you can minimize the probability that an LLM will make up answers (hallucinate). Retrieval Augmented Generation (RAG) is an approach developed to address the limitations of LLMs. RAG combines the strengths of pretrained language models, including reranking ones, with the ability to retrieve information from a dataset or database in real time during the generation of responses. Oracle AI Vector Search enables RAG and LLM integration using popular frameworks like LangChain, Ollama, and LlamaIndex.

   For more information, see Work with LLM-Powered APIs and Retrieval Augmented Generation.

3

Get Started

To get started, review the steps for the different tasks that you can do with Oracle AI Vector Search.

- SQL Quick Start Using a Vector Embedding Model Uploaded into the Database
  A set of SQL commands is provided to run a particular scenario that will help you understand Oracle AI Vector Search capabilities.
- SQL Quick Start Using a FLOAT32 Vector Generator
  A PL/SQL program that creates a vector generator is included along with example queries and results, providing a simple way to get started with Oracle AI Vector Search without a vector embedding model.
- SQL Quick Start Using a BINARY Vector Generator
  A set of procedures generate `BINARY` vectors, providing a simple way to get started with Oracle AI Vector Search without a vector embedding model.
- Your Vector Documentation Map to GenAI Prompts
  Follow the included steps to increase your chance of getting better and more consistent answers about this document from your preferred Generative AI Chatbot with internet search capabilities.

SQL Quick Start Using a Vector Embedding Model Uploaded into the Database

A set of SQL commands is provided to run a particular scenario that will help you understand Oracle AI Vector Search capabilities.

This quick start scenario introduces you to the `VECTOR` data type, which represents the semantic meaning behind your unstructured data. You will also use Vector Search SQL operators, allowing you to perform a similarity search to find vectors (and thereby content) that are similar to each other. Vector indexes are also created to help you accelerate similarity searches in an approximate manner. In addition to pure similarity searches, you will create Hybrid Vector Indexes and use them to run hybrid searches.

See Overview of Oracle AI Vector Search for more introductory information if needed.

The script chunks two Oracle AI Database Documentation books, assigns them corresponding vector embeddings, and shows you some similarity searches using vector indexes.

To run this script you need three files similar to the following:

- `my_embedding_model.onnx`, which is an ONNX export of the corresponding embedding model. To create such a file, see Convert Pretrained Models to ONNX Model: End-to-End Instructions or download the provided Hugging Face all-MiniLM-L12-v2 model in ONNX format .
- `json-relational-duality-developers-guide.pdf`, which is the PDF file for *JSON-Relational Duality Developer's Guide*
- `oracle-database-26ai-new-features-guide.pdf`, which is the PDF file for *Oracle AI Database New Features* .

> **Note**
>
> You can use other PDF files instead of the ones listed here. If you prefer, you can use another model of your choice as long as you can generate it as an `.onnx` file.

Let's start.

1. Copy the files to your local server directory or Oracle Cloud Infrastructure (OCI) Object Storage.

   If you have downloaded the Hugging Face all-MiniLM-L12-v2 model in ONNX format, run the following before completing the rest of this step:

   ```
   unzip all-MiniLM-L12-v2_augmented.zip
   ```

   Result:

   ```
    Archive:   all-MiniLM-L12-v2_augmented.zip
    inflating: all_MiniLM_L12_v2.onnx
    inflating: README-ALL_MINILM_L12_V2-augmented.txt
   ```

   For more information about the all-MiniLM-L12-v2 model, see the blog post Now Available! Pre-built Embedding Generation model for Oracle Database 23ai.

   You have the option to use the `scp` command for the local case and Oracle Command Line Interface or cURL for the Object Storage case. In the Object Storage case, the relevant command using Oracle Command Line Interface is `oci os object put`.

   If using Object Storage, you can also mount the bucket on your database server using the following steps (executed as a `root` user). This will allow you to easily copy files to Object Storage.

   a. Install the `s3fs-fuse` package.

      ```
      yum install -y s3fs-fuse
      ```

   b. On OCI, create a Customer Secret key. Make sure to save the `ACCESS_KEY` and `SECRET_KEY` in your notes.

   c. Create a folder that will be the mount point for the object storage bucket.

      ```
      mkdir /mnt/bucket
      chown oracle:oinstall /mnt/bucket
      ```

   d. Put your Customer Secret key in a file that will be used to authenticate to OCI Object Storage.

      ```
      echo $ACCESS_KEY:$SECRET_KEY > .passwd=s3fs
      chmod 400 .passwd-s3fs
      ```

   e. Mount your object storage bucket in the mount point folder (note this is a one line command).

      ```
      s3fs ${BUCKET} /mnt/bucket -o passwd_file=.passwd-s3fs
      -o url=https://${NAMESPACE}.compat.objectstorage.$
      ```

```
{REGION}.oraclecloud.com
-onomultipart -o use_path_request_style -o endpoint=${REGION} -ouid=$
{ORAUID},
gid=${ORAGID},allow_other,mp_umask=022
```

f. If you want to make the mount permanent after reboot, you can create a crontab entry (note this is a one line command).

```
echo "@reboot s3fs ${BUCKET} /mnt/bucket -o passwd_file=.passwd-s3fs -o
url=https://${NAMESPACE}.compat.objectstorage.${REGION}.oraclecloud.com
-onomultipart
-o use_path_request_style -o endpoint=${REGION} -ouid=${ORAUID},gid=$
{ORAGID},
allow_other,mp_umask=022" > crontab-fragment.txt
```

g. Add the crontab entry to your server crontab.

```
crontab -l | cat - crontab-fragment.txt >crontab.txt && crontab
crontab.txt

rm -f crontab.txt crontab-fragment.txt
```

As an alternative to the preceding Object Storage instructions, you also have the option to use the `LOAD_ONNX_MODEL_CLOUD` procedure of the `DBMS_VECTOR` package to load an ONNX embedding model from cloud Object Storage. For more information about the procedure, see *Oracle AI Database PL/SQL Packages and Types Reference*.

2. Create storage, user, and privileges.

   Here you create a new tablespace and a new user. You grant that user the `DB_DEVELOPER_ROLE` and create an Oracle directory to point to the PDF files. You grant the new user the possibility to read and write from/to that directory.

```
sqlplus / as sysdba

CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;

drop user vector cascade;

create user vector identified by vector DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;

grant DB_DEVELOPER_ROLE to vector;

create or replace directory VEC_DUMP as '/my_local_dir/';

grant read, write on directory vec_dump to vector;
```

3. Load your embedding model into the Oracle AI Database using one of the following methods, depending where your ONNX file is stored.

   a. If your ONNX file is stored locally:

Using the `DBMS_VECTOR` package, load your embedding model into the Oracle AI Database. You must specify the directory where you stored your model in ONNX format as well as describe what type of model it is and how you want to use it.

For more information about downloading pretrained embedding models, converting them into ONNX format, and importing the ONNX file into Oracle AI Database, see Import Pretrained Models in ONNX Format.

i.
```
connect vector/<vector user password>@<pdb instance network name>

EXEC DBMS_VECTOR.DROP_ONNX_MODEL(model_name => 'doc_model', force
=> true);

EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL('VEC_DUMP',
                                    'my_embedding_model.onnx',
                                    'doc_model');
```

> **Note**
>
> If you have downloaded the Oracle pre-built ONNX file for the Hugging Face all-MiniLM-L12-v2 model, you can use the following simplified command to load the file in the Database:
>
> ```
> BEGIN
>    DBMS_VECTOR.LOAD_ONNX_MODEL(
>         directory => 'VEC_DUMP',
>         file_name => 'all_MiniLM_L12_v2.onnx',
>         model_name => 'ALL_MINILM_L12_V2');
> END;
> /
> ```
>
> Note that the `model_name` is chosen when you load the model into the database. The model name should be unique so that it can be easily distinguishable from other models.

b. If your ONNX file is already stored on object store:

i. Create the credentials:

```
SET DEFINE OFF;
BEGIN
    DBMS_CLOUD.create_credential(
    credential_name => 'MY_CLOUD_CREDENTIAL',
    username => '<username>',
    password => '<user password>';
END;
/
```

ii. List the ONNX model saved on Object Storage. The `location_uri` should correspond to the location of the ONNX file:

```
SELECT mod.object_name,
       round(mod.bytes/1024/1024,2) size_mb,
       to_date(substr(mod.last_modified,1,18), 'DD-MON-RR
```

```
HH24.MI.SS') modified
  FROM table(dbms_cloud.list_objects(credential_name =>
'CLOUD_CREDENTIAL',
          location_uri => 'https://
objectstorage.<region>.oraclecloud.com/n/<namespace>/b/
<bucketName>/o/')) mod
  WHERE mod.object_name = 'my_embedding_model.onnx';
```

iii. Import the model into the database:

```
DECLARE
  model_source BLOB := NULL;
BEGIN
  model_source := DBMS_CLOUD.get_object(credential_name =>
'MY_CLOUD_CREDENTIAL',
    object_uri => 'https://objectstorage.<region>.oraclecloud.com/n/
<namespace>/b/<bucketName>/o/my_embedding_model.onnx');
  DBMS_VECTOR.load_onnx_model('doc_model',
                              model_source);
END;
/
```

iv. List the model inside the database:

```
SELECT MODEL_NAME, ATTRIBUTE_NAME, ATTRIBUTE_TYPE, DATA_TYPE,
VECTOR_INFO
  FROM USER_MINING_MODEL_ATTRIBUTES WHERE MODEL_NAME = 'DOC_MODEL';

SELECT MODEL_NAME, MINING_FUNCTION, ALGORITHM, ALGORITHM_TYPE,
MODEL_SIZE
  FROM USER_MINING_MODELS WHERE MODEL_NAME = 'DOC_MODEL';
```

4. (Optional) Verify that the loaded embedding model is working.

   You can call the `VECTOR_EMBEDDING` SQL function to test your embedding model by passing a sample text string (`hello`) as the input.

```
SELECT VECTOR_EMBEDDING(doc_model USING 'hello' as data) AS embedding;
```

5. Create a relational table to store books in the PDF format.

   You now create a table containing all the books you want to chunk and vectorize. You associate each new book with an ID and a pointer to your local directory where the books are stored.

```
drop table documentation_tab purge;
create table documentation_tab (id number, data blob);
insert into documentation_tab values(1, to_blob(bfilename('VEC_DUMP',
'json-relational-duality-developers-guide.pdf')));
insert into documentation_tab values(2, to_blob(bfilename('VEC_DUMP',
'oracle-database-26ai-new-features-guide.pdf')));
commit;
select dbms_lob.getlength(data) from documentation_tab;
```

6. Create a relational table to store unstructured data chunks and associated vector embeddings using `my_embedding_model.onnx`.

You start by creating the table structure using the `VECTOR` data type. For more information about declaring a table's column as a `VECTOR` data type, see Create Tables Using the VECTOR Data Type .

The `INSERT` statement reads each PDF file from `DOCUMENTATION_TAB`, transforms each PDF file into text, chunks each resulting text, then finally generates corresponding vector embeddings on each chunk that is created. All that is done in one single `INSERT SELECT` statement.

Here you choose to use Vector Utility PL/SQL package `DBMS_VECTOR_CHAIN` to convert, chunk, and vectorize your unstructured data in one end-to-end pipeline. Vector Utility PL/SQL functions are intended to be a set of chainable stages (using table functions) through which you pass your input data to transform into a different representation. In this case, from PDF to text to chunks to vectors. For more information about using chainable utility functions in the `DBMS_VECTOR_CHAIN` package, see About Chainable Utility Functions and Common Use Cases.

```
drop table doc_chunks purge;
create table doc_chunks (doc_id number, chunk_id number, chunk_data
varchar2(4000), chunk_embedding vector);

insert into doc_chunks
select dt.id doc_id, et.embed_id chunk_id, et.embed_data chunk_data,
to_vector(et.embed_vector) chunk_embedding
from
    documentation_tab dt,
    dbms_vector_chain.utl_to_embeddings(

dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(dt.data),
json('{"normalize":"all"}')),
       json('{"provider":"database", "model":"doc_model"}')) t,
    JSON_TABLE(t.column_value, '$[*]' COLUMNS (embed_id NUMBER PATH
'$.embed_id', embed_data VARCHAR2(4000) PATH '$.embed_data', embed_vector
CLOB PATH '$.embed_vector')) et;

commit;
```

> **See Also**
>
> - *Oracle AI Database JSON Developer's Guide* for information about the `JSON_TABLE` function, which supports the `VECTOR` data type

7. Generate a query vector for use in a similarity search.

   For a similarity search you will need query vectors. Here you enter your query text and generate an associated vector embedding.

   For example, you can use the following text: 'different methods of backup and recovery'. You use the `VECTOR_EMBEDDING` SQL function to generate the vector embeddings from the input text. The function takes an embedding model name and a text string to generate the corresponding vector. Note that you can generate vector embeddings outside of the database using your favorite tools. For more information about using the `VECTOR_EMBEDDING` SQL function, see About SQL Functions to Generate Embeddings.

In SQL*Plus, use the following code:

```
ACCEPT text_input CHAR PROMPT 'Enter text: '
VARIABLE text_variable VARCHAR2(1000)
VARIABLE query_vector VECTOR
BEGIN
  :text_variable := '&text_input';
  SELECT vector_embedding(doc_model using :text_variable as data)
into :query_vector;
END;
/

PRINT query_vector
```

In SQLCL, use the following code:

```
DEFINE text_input = '&text'

SELECT '&text_input';

VARIABLE text_variable VARCHAR2(1000)
VARIABLE query_vector CLOB
BEGIN
  :text_variable := '&text_input';
  SELECT vector_embedding(doc_model using :text_variable as data)
into :query_vector;
END;
/

PRINT query_vector
```

8. Run a similarity search to find, within your books, the first four most relevant chunks that talk about backup and recovery.

   Using the generated query vector, you search similar chunks in the `DOC_CHUNKS` table. For this, you use the `VECTOR_DISTANCE` SQL function and the `FETCH` SQL clause to retrieve the most similar chunks.

   For more information about the `VECTOR_DISTANCE` SQL function, see Vector Distance Functions and Operators.

   For more information about exact similarity search, see Perform Exact Similarity Search.

```
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY;
```

   You can also add a `WHERE` clause to further filter your search, for instance if you only want to look at one particular book.

```
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
WHERE doc_id=1
```

```
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY;
```

9. Use the `EXPLAIN PLAN` command to determine how the optimizer resolves this query.

```
EXPLAIN PLAN FOR
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY;

select plan_table_output from
table(dbms_xplan.display('plan_table',null,'all'));


PLAN_TABLE_OUTPUT
----------------------------------------------------------------------------
---
Plan hash value: 1651750914
----------------------------------------------------------------------------
----------------------
| Id  | Operation              | Name          | Rows  | Bytes |TempSpc|
Cost (%CPU)| Time     |
----------------------------------------------------------------------------
----------------------
|   0 | SELECT STATEMENT       |               |     4 |   104 |
|   549   (3)| 00:00:01 |
|*  1 |  COUNT STOPKEY         |               |       |       |
|             |          |
|   2 |   VIEW                 |               |  5014 |   127K|
|   549   (3)| 00:00:01 |
|*  3 |    SORT ORDER BY STOPKEY|              |  5014 |   156K|
232K|   549   (3)| 00:00:01 |
|   4 |     TABLE ACCESS FULL  | DOC_CHUNKS    |  5014 |   156K|
|   480   (3)| 00:00:01 |
----------------------------------------------------------------------------
----------------------
```

10. Run a multi-vector similarity search to find, within your books, the first four most relevant chunks in the first two most relevant books.

    Here you keep using the same query vector as previously used.

    For more information about performing multi-vector similarity search, see Perform Multi-Vector Similarity Search.

```
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 2 PARTITIONS BY doc_id, 4 ROWS ONLY;
```

11. Create an In-Memory Neighbor Graph Vector Index on the vector embeddings that you created.

    When dealing with huge vector embedding spaces, you may want to create vector indexes to accelerate your similarity searches. Instead of scanning each and every vector embedding in your table, a vector index uses heuristics to reduce the search space to accelerate the similarity search. This is called approximate similarity search.

For more information about creating vector indexes, see Create Vector Indexes and Hybrid Vector Indexes.

```
CREATE VECTOR INDEX docs_hnsw_idx ON doc_chunks(chunk_embedding)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
DISTANCE COSINE
WITH TARGET ACCURACY 95;

SELECT INDEX_NAME, INDEX_TYPE, INDEX_SUBTYPE
FROM USER_INDEXES WHERE INDEX_NAME = 'DOCS_HNSW_IDX';
```

Result:

```
INDEX_NAME     INDEX_TYPE  INDEX_SUBTYPE
-------------- ----------- ----------------------------
DOCS_HNSW_IDX  VECTOR      INMEMORY_NEIGHBOR_GRAPH_HNSW


SELECT INDEX_ORGANIZATION, NUM_VECTORS, DISTANCE_TYPE,
    INDEX_DIMENSIONS, INDEX_DIM_TYPE, DEFAULT_ACCURACY
FROM V$VECTOR_INDEX
WHERE INDEX_NAME = 'DOCS_HNSW_IDX';
```

Result:

```
INDEX_ORGANIZATION        NUM_VECTORS DISTANCE_TYPE   INDEX_DIMENSIONS
INDEX_DIM_TYPE  DEFAULT_ACCURACY
------------------------- ----------- --------------- ----------------
--------------- ----------------
INMEMORY NEIGHBOR GRAPH   5014        COSINE          384
FLOAT32         95
```

12. Determine the memory allocation in the vector memory area.

    To get an idea about the size of your In-Memory Neighbor Graph Vector Index in memory, you can use the `V$VECTOR_MEMORY_POOL` view. See Size the Vector Pool for more information about sizing the vector pool to allow for vector index creation and maintenance.

    > **Note**
    >
    > You must have explicit `SELECT` privilege to select from the `V$VECTOR_MEMORY_POOL` view, which gives you detailed information about the vector pool.

```
select CON_ID, POOL, ALLOC_BYTES/1024/1024 as ALLOC_BYTES_MB,
USED_BYTES/1024/1024 as USED_BYTES_MB
from V$VECTOR_MEMORY_POOL order by 1,2;
```

13. Run an approximate similarity search to identify, within your books, the first four most relevant chunks.

Using the previously generated query vector, you search chunks in the `DOC_CHUNKS` table that are similar to your query vector. For this, you use the `VECTOR_DISTANCE` function to retrieve the most similar chunks using your vector index.

For more information about approximate similarity search, see Perform Approximate Similarity Search Using Vector Indexes.

```
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY WITH TARGET ACCURACY 80;
```

You can also add a `WHERE` clause to further filter your search, for instance if you only want to look at one particular book.

```
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
WHERE doc_id=1
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY WITH TARGET ACCURACY 80;
```

14. Use the `EXPLAIN PLAN` command to determine how the optimizer resolves this query.

See **Optimizer Plans for Vector Indexes** for more information about how the Oracle AI Database optimizer uses vector indexes to run your approximate similarity searches.

```
EXPLAIN PLAN FOR
SELECT doc_id, chunk_id, chunk_data
FROM doc_chunks
ORDER BY vector_distance(chunk_embedding , :query_vector, COSINE)
FETCH FIRST 4 ROWS ONLY WITH TARGET ACCURACY 80;

SELECT plan_table_output FROM
TABLE(dbms_xplan.display('plan_table',null,'all'));


PLAN_TABLE_OUTPUT
---------------------------------------------------------------------------
-----
Plan hash value: 2946813851
---------------------------------------------------------------------------
----------------------------
| Id  | Operation                        | Name           | Rows  | Bytes |
TempSpc| Cost (%CPU)| Time     |
---------------------------------------------------------------------------
----------------------------
|   0 | SELECT STATEMENT                 |                |     4 |   104
|       |   12083 (2)| 00:00:01 |
|*  1 |  COUNT STOPKEY                   |                |       |
|       |             |           |
|   2 |   VIEW                           |                |  5014 |
127K|       |   12083 (2)| 00:00:01 |
|*  3 |    SORT ORDER BY STOPKEY         |                |  5014 |
19M|    39M|   12083 (2)| 00:00:01 |
|   4 |     TABLE ACCESS BY INDEX ROWID| DOC_CHUNKS     |  5014 |
19M|       |    1    (0)| 00:00:01 |
```

```
|   5 |      VECTOR INDEX HNSW SCAN    | DOCS_HNSW_IDX |  5014 |
19M|       |    1    (0)| 00:00:01 |
---------------------------------------------------------------------------
-----------------------------
```

15. Determine your vector index performance for your approximate similarity searches.

    The index accuracy reporting feature allows you to determine the accuracy of your vector indexes. After a vector index is created, you may be interested to know how accurate your approximate vector searches are.

    The `DBMS_VECTOR.INDEX_ACCURACY_QUERY` PL/SQL procedure provides an accuracy report for a top-K index search for a specific query vector and a specific target accuracy. In this case you keep using the query vector generated previously. For more information about index accuracy reporting, see Index Accuracy Report.

```
SET SERVEROUTPUT ON
DECLARE
    report VARCHAR2(128);
BEGIN
    report := dbms_vector.index_accuracy_query(
        OWNER_NAME => 'VECTOR',
        INDEX_NAME => 'DOCS_HNSW_IDX',
        qv => :query_vector,
        top_K => 10,
        target_accuracy => 90 );
    dbms_output.put_line(report);
END;
/
```

    The report looks like the following: Accuracy achieved (100%) is 10% higher than the Target Accuracy requested (90%).

16. You are now going to create a new table that will be used to run hybrid searches.

```
DROP TABLE documentation_tab2 PURGE;

CREATE TABLE documentation_tab2 (id NUMBER, file_name VARCHAR2(200));

INSERT INTO documentation_tab2 VALUES(1, 'json-relational-duality-
developers-guide.pdf');

INSERT INTO documentation_tab2 VALUES(2, 'oracle-database-26ai-new-
features-guide.pdf');

COMMIT;
```

17. To create a hybrid vector index, you need to specify your data store where your files reside. Here we are using the `VEC_DUMP` directory where the PDF files are stored:

```
BEGIN
  ctx_ddl.create_preference('DS', 'DIRECTORY_DATASTORE');
  ctx_ddl.set_attribute('DS', 'DIRECTORY', 'VEC_DUMP');
END;
/
```

18. You can now create a hybrid vector index on the `documentation_tab2` table. Here, you indicate to the hybrid vector index where to find the files that need to be indexed and the types that will be filtered, as well as which embedding model to use and which type of vector index. For more information, see Understand Hybrid Vector Indexes.

```
CREATE HYBRID VECTOR INDEX my_hybrid_vector_idx ON
documentation_tab2(file_name)
  PARAMETERS ('
                DATASTORE      DS
                FILTER         CTXSYS.AUTO_FILTER
                MODEL          doc_model
                VECTOR_IDXTYPE ivf
');
```

19. Once the hybrid vector index has been created, check the following internal tables that were created:

```
SELECT COUNT(*) FROM DR$my_hybrid_vector_idx$VR;
```

Result:

```
   COUNT(*)
___________
       9581
```

or

```
SELECT COUNT(*) FROM my_hybrid_vector_idx$VECTORS;
```

Result:

```
   COUNT(*)
___________
       9581
```

```
DESC my_hybrid_vector_idx$VECTORS
```

Result:

```
Name                      Null?       Type
_______________________   _________   _________________
DOC_ROWID                 NOT NULL    ROWID
DOC_CHUNK_ID              NOT NULL    NUMBER
DOC_CHUNK_COUNT           NOT NULL    NUMBER
DOC_CHUNK_OFFSET          NOT NULL    NUMBER
DOC_CHUNK_LENGTH          NOT NULL    NUMBER
DOC_CHUNK_TEXT                        VARCHAR2(4000)
DOC_CHUNK_EMBEDDING                   VECTOR
```

```
SELECT COUNT(*) FROM DR$MY_HYBRID_VECTOR_IDX$I;
```

Result:

```
  COUNT(*)
----------
      4927
```

20. You can run your first hybrid search by specifying the following parameters:
    - The hybrid vector index name
    - The search scorer you want to use (this scoring function is used after merging results from keyword search and similarity search)
    - The fusion function to use to merge the results from both searches
    - The search text you want for your similarity search
    - The search mode to use to produce the vector search results
    - The aggregation function to use to calculate the vector score for each document identified by your similarity search
    - The score weight for your vector score
    - The `CONTAINS` string for your keyword search
    - The score weight for your keyword search
    - The returned max values you want to see
    - The maximum number of documents and chunks you want to see in the result

    For complete `DBMS_HYBRID_VECTOR.SEARCH` syntax information, see SEARCH.

```
SET LONG 10000

SELECT json_serialize(
    DBMS_HYBRID_VECTOR.SEARCH(
      JSON(
           '{
              "hybrid_index_name" : "my_hybrid_vector_idx",
              "search_scorer"     : "rsf",
              "search_fusion"     : "UNION",
              "vector":
                        {
                          "search_text"   : "How to insert data in json
format to a table?",
                          "search_mode"   : "DOCUMENT",
                          "aggregator"    : "MAX",
                          "score_weight"  : 1,
                        },
              "text":
                        {
                         "contains"      : "data AND json",
                         "score_weight"  : 1,
                        },
              "return":
                        {
                          "values"        : [ "rowid", "score",
"vector_score", "text_score" ],
                          "topN"          : 10
                        }
```

```
            }'
      )
   ) RETURNING CLOB pretty);
```

Result:

```
JSON_SERIALIZE(DBMS_HYBRID_VECTOR.SEARCH(JSON('{"HYBRID_INDEX_NAME":"MY_HYB
RID_VECTOR_IDX","SEARCH_SCORER":"RSF","SEARCH_FUSION":"UNION","VECTOR":
{"SEARCH_TEXT":"HOWTOSEARCHUSINGVECTORINDEXES?","SEARCH_MODE":"DOCUMENT","A
GGREGATOR":"MAX","SCORE_WEIGHT
___________________________________________________________________________
___________________________________________________________________________
___________________________________________________________________________
_____________________________
[
  {
    "rowid" : "AAASd4AALAAAFjYAAB",
    "score" : 88.77,
    "vector_score" : 77.53,
    "text_score" : 100
  }
]
```

> **Note**
>
> Before executing the following statement, replace the rowid value that you got from running the previous statement.

```
SELECT file_name FROM documentation_tab2 WHERE rowid='AAASd4AALAAAFjYAAB';
```

Result:

```
FILE_NAME
__________________________________________
json-relational-duality-developers-guide.pdf
```

> **Note**
>
> In this step, a lot of parameters were explicitly indicated. However, you can stick to the defaults and run the following statement instead.

```
SELECT json_serialize(
     DBMS_HYBRID_VECTOR.SEARCH(
        JSON(
             '{
               "hybrid_index_name" : "my_hybrid_vector_idx",
               "search_text"       : "How to insert data in json format to
a table?"
             }'
```

```
            )
    ) RETURNING CLOB pretty);
```

21. Hybrid vector index maintenance is done automatically in the background at two levels. Every 3 seconds, an index synchronization is run to maintain the index with ongoing DMLs. The index maintenance can take much more time depending on the number of DMLs and files added. So, you may not see your modifications right away as you will experience in this scenario. A scheduler job is also run every night at midnight to optimize the index. As you run DMLs on your index, the index gets fragmented. The optimize job defragments your index for better query performance.

    To experiment with hybrid vector index maintenance, insert a new row in your base table and run the same hybrid search query as you did in the previous step:

```
INSERT INTO documentation_tab2 VALUES(3, 'json-relational-duality-
developers-guide.pdf');

COMMIT;


SELECT json_serialize(
    DBMS_HYBRID_VECTOR.SEARCH(
      JSON(
           '{
              "hybrid_index_name" : "my_hybrid_vector_idx",
              "search_scorer"     : "rsf",
              "search_fusion"     : "UNION",
              "vector":
                        {
                          "search_text"   : "How to insert data in json
format to a table?",
                          "search_mode"   : "DOCUMENT",
                          "aggregator"    : "MAX",
                          "score_weight"  : 1,
                        },
              "text":
                        {
                         "contains"      : "data AND json",
                         "score_weight"  : 1,
                        },
              "return":
                        {
                          "values"        : [ "rowid", "score",
"vector_score", "text_score" ],
                          "topN"          : 10
                        }
            }'
      )
   ) RETURNING CLOB pretty);
```

You should observe no changes in the result:

```
JSON_SERIALIZE(DBMS_HYBRID_VECTOR.SEARCH(JSON('{"HYBRID_INDEX_NAME":"MY_HYB
RID_VECTOR_IDX","SEARCH_SCORER":"RSF","SEARCH_FUSION":"UNION","VECTOR":
{"SEARCH_TEXT":"HOWTOSEARCHUSINGVECTORINDEXES?","SEARCH_MODE":"DOCUMENT","A
GGREGATOR":"MAX","SCORE_WEIGHT
```

```
[
  {
    "rowid" : "AAASd4AALAAAFjYAAB",
    "score" : 88.77,
    "vector_score" : 77.53,
    "text_score" : 100
  }
]
```

22. Wait several minutes and rerun your hybrid search.

```
SELECT json_serialize(
    DBMS_HYBRID_VECTOR.SEARCH(
      JSON(
           '{
              "hybrid_index_name" : "my_hybrid_vector_idx",
              "search_scorer"     : "rsf",
              "search_fusion"     : "UNION",
              "vector":
                        {
                          "search_text"   : "How to insert data in json 
format to a table?",
                          "search_mode"   : "DOCUMENT",
                          "aggregator"    : "MAX",
                          "score_weight"  : 1,
                        },
              "text":
                        {
                         "contains"      : "data AND json",
                         "score_weight"  : 1,
                        },
              "return":
                        {
                          "values"        : [ "rowid", "score", 
"vector_score", "text_score" ],
                          "topN"          : 10
                        }
            }'
      )
   ) RETURNING CLOB pretty);
```

By this point, the result has been updated:

```
JSON_SERIALIZE(DBMS_HYBRID_VECTOR.SEARCH(JSON('{"HYBRID_INDEX_NAME":"MY_HYB
RID_VECTOR_IDX","SEARCH_SCORER":"RSF","SEARCH_FUSION":"UNION","VECTOR":
{"SEARCH_TEXT":"HOWTOSEARCHUSINGVECTORINDEXES?","SEARCH_MODE":"DOCUMENT","A
GGREGATOR":"MAX","SCORE_WEIGHT
```

```
[
  {
    "rowid" : "AAASd4AALAAAFjYAAB",
    "score" : 88.77,
    "vector_score" : 77.53,
    "text_score" : 100
  },
  {
    "rowid" : "AAASd4AALAAAFjYAAC",
    "score" : 88.77,
    "vector_score" : 77.53,
    "text_score" : 100
  }
]
```

SQL Quick Start Using a FLOAT32 Vector Generator

A PL/SQL program that creates a vector generator is included along with example queries and results, providing a simple way to get started with Oracle AI Vector Search without a vector embedding model.

The generator is a PL/SQL program that allows you to randomly generate vectors with a specified number of dimensions and clusters. Each dimension value is generated using specified minimum and maximum possible values. The output of the generation process is the population of a table called `genvec` that you can then use, for example, to experiment with similarity searches.

The following instructions assume you already have access to a database account with sufficient privileges (minimally the `DB_DEVELOPER_ROLE` role).

> **Note**
>
> Do not use the vector generator on production databases. The program is made available for testing and demo purposes.

1. Create the `genvec` table.

```
DROP TABLE genvec PURGE;

CREATE TABLE genvec (
  id number,            -- id of the generated vector
  v VECTOR,             -- generated vector
  name VARCHAR2(500),   -- name for the generated vector: C1 to Cn are
centroids, Cx_y is vector number y in cluster number x
  nv VECTOR,            -- normalized version of the generated vector
  ly number             -- random number you can use to filter out rows in
addition to similarity search on vectors
);
```

2. Create the package `vector_gen_pkg` and its associated package body.

Here you create the vector generator package:

```
CREATE OR REPLACE PACKAGE vector_gen_pkg AS

  TYPE t_vectors IS TABLE OF vector INDEX BY PLS_INTEGER;

  FUNCTION get_coordinate(
    input_string CLOB,
    i PLS_INTEGER
  ) RETURN NUMBER;

  PROCEDURE generate_vectors(
    num_vectors IN PLS_INTEGER,   -- Number of vectors to generate
    dimensions IN PLS_INTEGER,    -- Number of dimensions of each vector
    num_clusters IN PLS_INTEGER, -- Number of clusters to create
    cluster_spread IN NUMBER,     -- Relative closeness of each vector in
each cluster (using standard deviation)
    min_value IN NUMBER,          -- Minimum value for a vector coordinate
    max_value IN NUMBER           -- Maximum value for a vector coordinate
  );

END vector_gen_pkg;
/
```

And the package body:

```
CREATE OR REPLACE PACKAGE BODY vector_gen_pkg AS

  -------------------------------------------
  -------------------------------------------
  ---- V E C T O R   G E N E R A T O R -----
  -------------------------------------------
  -------------------------------------------
  -- Version 1.0                   ---------
  -------------------------------------------
  -------------------------------------------
  ---- DO NOT USE ON PRODUCTION DATABASES ---
  ---- ONLY FOR TESTING AND DEMO PURPOSES ---
  -------------------------------------------


  FUNCTION get_coordinate(
    input_string CLOB,
    i PLS_INTEGER
  ) RETURN NUMBER IS
    start_pos NUMBER;
    end_pos NUMBER;
    comma_pos NUMBER;
    coord VARCHAR2(100);
    comma_count NUMBER := 0;
    commas NUMBER;
    working_string CLOB;
  BEGIN
    -- Remove leading and trailing brackets
    working_string := input_string;
```

```
    working_string := TRIM(BOTH '[]' FROM working_string);
    commas := LENGTH(working_string) - LENGTH(REPLACE(working_string, ',', 
''));

    -- Initialize positions
    start_pos := 1;
    end_pos := INSTR(working_string, ',', start_pos);

    IF i<=0 OR i>commas+1 THEN RETURN NULL;
    END IF;

    -- Loop through the string to find the i-th coordinate
    LOOP
      IF comma_count + 1 = i THEN
        IF end_pos = 0 THEN
          -- If there's no more comma, the coordinate is the rest of the 
string
          coord := SUBSTR(working_string, start_pos);
        ELSE
          coord := SUBSTR(working_string, start_pos, end_pos - start_pos);
        END IF;
        RETURN coord;
      END IF;

      -- Move to the next coordinate
      comma_count := comma_count + 1;
      start_pos := end_pos + 1;
      end_pos := INSTR(working_string, ',', start_pos);

      -- Exit loop if no more coordinates
      EXIT WHEN start_pos > LENGTH(working_string);
    END LOOP;

    -- If the function hasn't returned yet, the index was out of bounds
    RETURN NULL;

  END;



  PROCEDURE generate_random_vector(
    dimensions IN PLS_INTEGER,
    min_value IN NUMBER,
    max_value IN NUMBER,
    vec OUT vector
    ) IS
    e CLOB;
  BEGIN
    e := '[';
    FOR i IN 1..dimensions-1 LOOP
      e := e || DBMS_RANDOM.VALUE(min_value, max_value) ||',';
    END LOOP;
    e := e || DBMS_RANDOM.VALUE(min_value, max_value) ||']';
    vec := VECTOR(e);
  END generate_random_vector;
```

```
  PROCEDURE generate_clustered_vector(
    centroid IN vector,
    cluster_spread IN NUMBER,
    vec OUT vector
  ) IS
    e CLOB;
    d number;
    BEGIN
      d := VECTOR_DIMENSION_COUNT(centroid);
      e := '[';
      FOR i IN 1 .. d-1 LOOP
        e := e || (get_coordinate(to_clob(centroid),i) + 
(DBMS_RANDOM.NORMAL * cluster_spread)) ||',';
      END LOOP;
      e := e || 
(get_coordinate(to_clob(centroid),VECTOR_DIMENSION_COUNT(centroid)) + 
(DBMS_RANDOM.NORMAL * cluster_spread)) || ']';
      vec := VECTOR(e);
  END generate_clustered_vector;



  FUNCTION normalize_vector(vec IN vector) RETURN vector IS
    e CLOB;
    v CLOB;
    n number;
    d number;
  BEGIN
    n := VECTOR_NORM(vec);
    v := to_clob(vec);
    d := VECTOR_DIMENSION_COUNT(vec);
    e := '[';
    FOR i IN 1 .. d-1 LOOP
      e := e || (get_coordinate(v,i)/n) ||',';
    END LOOP;
    e := e || (get_coordinate(v,d)/n) || ']';
    RETURN VECTOR(e);
  END normalize_vector;



  PROCEDURE generate_vectors(
    num_vectors IN PLS_INTEGER,   -- Must be 1 or above
    dimensions IN PLS_INTEGER,    -- Must be above 1 but less than 500
    num_clusters IN PLS_INTEGER, -- Must be 1 or above
    cluster_spread IN NUMBER,     -- Must be grather than 0
    min_value IN NUMBER,
    max_value IN NUMBER
  ) IS
    centroids t_vectors;
    vectors_per_cluster PLS_INTEGER;
    remaining_vectors PLS_INTEGER;
    vec vector;
    idx PLS_INTEGER := 1;
```

```
    max_id NUMBER;
    working_vector VECTOR;

  BEGIN
    IF (num_vectors) <=0 OR (num_clusters < 1) OR (num_vectors <
num_clusters) OR (dimensions <= 0) OR (dimensions > 500) OR
(cluster_spread <= 0) OR (min_value >= max_value) THEN RETURN;
    END IF;

    SELECT MAX(id) INTO max_id FROM genvec;
    IF max_id IS NULL THEN max_id := 0;
    END IF;

    -- Generate cluster centroids
    FOR i IN 1..num_clusters LOOP

      generate_random_vector(dimensions, min_value, max_value,
centroids(i));
      working_vector := normalize_vector(centroids(i));
      INSERT INTO genvec VALUES (max_id + idx, centroids(i), 'C'||i,
working_vector, DBMS_RANDOM.VALUE(3,600000000));
      idx := idx + 1;

    END LOOP;

    -- Calculate vectors per cluster
    vectors_per_cluster := TRUNC(num_vectors / num_clusters);
    remaining_vectors := num_vectors MOD num_clusters;

    -- Generate vectors for each cluster
    IF vectors_per_cluster > 1 THEN
      FOR i IN 1..num_clusters LOOP
        FOR j IN 1..(vectors_per_cluster - 1) LOOP
          generate_clustered_vector(centroids(i), cluster_spread, vec);
          working_vector := normalize_vector(vec);
          INSERT INTO genvec VALUES (max_id + idx, vec, 'C'||i||'-'||j,
working_vector, DBMS_RANDOM.VALUE(3,600000000));
          idx := idx + 1;
        END LOOP;
      END LOOP;
    END IF;

    -- Handle remaining vectors: all associated with cluster 1
    IF remaining_vectors > 0 THEN
      FOR j IN 1..remaining_vectors LOOP
        generate_clustered_vector(centroids(1), cluster_spread, vec);
        working_vector := normalize_vector(vec);
        INSERT INTO genvec VALUES (max_id + idx, vec, 'C1-'||idx,
working_vector, DBMS_RANDOM.VALUE(3,600000000));
        idx := idx + 1;
      END LOOP;
    END IF;
    COMMIT;
  END generate_vectors;
```

```
END vector_gen_pkg;
/
```

3. After you have your vector generator set up, you can run this and the following steps to understand how to use it.

   Run the `generate_vectors` procedure of the `vector_gen_pkg` package with sample values:

```
BEGIN
  vector_gen_pkg.generate_vectors(
    num_vectors => 100,    -- Number of vectors to generate. Must be 1 or
above
    dimensions => 3,       -- Number of dimensions of each vector. Must be
above 1 but less than 500
    num_clusters => 6,     -- Number of clusters to create. Must be 1 or
above
    cluster_spread => 1,  -- Relative closeness of each vector in each
cluster (using standard deviation). Must be grather than 0
    min_value => 0,        -- Minimum value for a vector coordinate
    max_value => 100       -- Maximum value for a vector coordinate. Min
value must be smaller than max value
  );
END;
/
```

4. Run a `SELECT` statement to view the newly generated vectors.

```
SELECT name, v FROM genvec;
```

   Example output:

```
NAME     V
_______ ____________________________________________________
C1       [6.35792809E+001,5.28954163E+001,5.16500435E+001]
C2       [5.67991257E+001,5.00640755E+001,2.3642437E+001]
C3       [2.42510891E+001,5.36970367E+001,6.88145638E+000]
C4       [8.13146515E+001,2.88190498E+001,4.09245186E+001]
C5       [6.70646744E+001,2.53395119E+001,6.14522667E+001]
C6       [5.60192604E+001,8.31662598E+001,2.93592377E+001]
C1-1     [6.4469986E+001,5.25044632E+001,5.22250557E+001]
C1-2     [6.31295433E+001,5.21443062E+001,5.02242126E+001]
C1-3     [6.25154915E+001,5.25730362E+001,5.2617794E+001]
C1-4     [6.3491375E+001,5.21440697E+001,5.06722069E+001]
C1-5     [6.21516266E+001,5.32161064E+001,5.23233032E+001]
C1-6     [6.2913269E+001,5.16970291E+001,5.16683655E+001]
C1-7     [6.22267456E+001,5.40408363E+001,5.0272541E+001]
C1-8     [6.1414093E+001,5.28870888E+001,5.27458E+001]

NAME      V
________ ____________________________________________________
C1-9      [6.29652252E+001,5.32767754E+001,5.27030106E+001]
C1-10     [6.35940704E+001,5.27265244E+001,5.23180656E+001]
C1-11     [6.34133224E+001,5.39401283E+001,5.29368248E+001]
C1-12     [6.18856697E+001,5.31113129E+001,5.18861504E+001]
C1-13     [6.32378883E+001,5.30308647E+001,5.04571724E+001]
```

```
C1-14    [6.18148689E+001,5.33705482E+001,5.29123802E+001]
C1-15    [6.43224258E+001,5.23124084E+001,5.21299057E+001]
C2-1     [5.59053535E+001,5.20054626E+001,2.28595486E+001]
C2-2     [5.71644516E+001,5.13243408E+001,2.31167526E+001]
C2-3     [5.66626244E+001,5.00615959E+001,2.27138176E+001]
C2-4     [5.73383865E+001,5.04509125E+001,2.36539135E+001]
C2-5     [5.6621357E+001,5.01576767E+001,2.38867531E+001]
C2-6     [5.59768562E+001,5.17590942E+001,2.49088764E+001]
C2-7     [5.64437904E+001,4.71531525E+001,2.23245487E+001]

NAME     V
________ ____________________________________________________
C2-8     [5.81449051E+001,5.09049644E+001,2.29072056E+001]
C2-9     [5.37190132E+001,4.87386665E+001,2.28188381E+001]
C2-10    [5.77416382E+001,4.93461685E+001,2.32014389E+001]
C2-11    [5.68353958E+001,5.11093979E+001,2.43693123E+001]
C2-12    [5.79631157E+001,5.0297657E+001,2.28039799E+001]
C2-13    [5.57930183E+001,5.11965866E+001,2.35887661E+001]
C2-14    [5.57345848E+001,5.03228951E+001,2.30780907E+001]
C2-15    [5.69435997E+001,4.8590435E+001,2.58747597E+001]
C3-1     [2.40239315E+001,5.2352993E+001,5.63517284E+000]
C3-2     [2.39717846E+001,5.30635986E+001,5.86633539E+000]
C3-3     [2.70314407E+001,5.48788643E+001,7.96345377E+000]
C3-4     [2.39875908E+001,5.39634552E+001,5.87654877E+000]
C3-5     [2.47772026E+001,5.2187336E+001,6.83652115E+000]
C3-6     [2.32920208E+001,5.41494293E+001,6.40737772E+000]

NAME     V
________ ____________________________________________________
C3-7     [2.46129742E+001,5.32308769E+001,6.29999685E+000]
C3-8     [2.51000671E+001,5.33271561E+001,8.86797047E+000]
C3-9     [2.4337059E+001,5.26281281E+001,6.9616766E+000]
C3-10    [2.39770508E+001,5.42386856E+001,5.63018417E+000]
C3-11    [2.59837551E+001,5.34013176E+001,6.97773361E+000]
C3-12    [2.40400314E+001,5.25649719E+001,7.2636981E+000]
C3-13    [2.13184013E+001,5.28633308E+001,8.3834734E+000]
C3-14    [2.50075855E+001,5.21548729E+001,6.88196087E+000]
C3-15    [2.53695087E+001,5.60495186E+001,6.76059389E+000]
C4-1     [8.28819885E+001,2.95163822E+001,4.03809738E+001]
C4-2     [8.18269348E+001,2.95735188E+001,3.99435768E+001]
C4-3     [8.2709259E+001,2.90755043E+001,4.07345886E+001]
C4-4     [8.18622665E+001,2.88013916E+001,4.1822567E+001]
C4-5     [7.99165421E+001,2.89941139E+001,4.09653854E+001]

NAME     V
________ ____________________________________________________
C4-6     [8.12936707E+001,2.98655643E+001,4.00380211E+001]
C4-7     [8.21705704E+001,2.90163479E+001,3.94858704E+001]
C4-8     [8.20081329E+001,2.89751148E+001,4.1045887E+001]
C4-9     [8.25486298E+001,2.84143009E+001,4.15654945E+001]
C4-10    [8.22034149E+001,2.92223415E+001,4.20033302E+001]
C4-11    [8.2048996E+001,2.98751106E+001,4.09612732E+001]
C4-12    [8.09316025E+001,2.7799057E+001,4.12611198E+001]
C4-13    [8.04624023E+001,2.88711109E+001,4.07331085E+001]
C4-14    [8.13773575E+001,2.97510109E+001,4.09169846E+001]
C4-15    [8.35310364E+001,2.971031E+001,4.16878052E+001]
```

```
C5-1      [6.87114258E+001,2.53504581E+001,6.11055298E+001]
C5-2      [6.73569031E+001,2.35163498E+001,6.01617432E+001]
C5-3      [6.78224869E+001,2.61236534E+001,6.0729248E+001]
C5-4      [6.76432266E+001,2.56426888E+001,6.35400085E+001]

NAME      V
________  ____________________________________________________
C5-5      [6.75377045E+001,2.60873699E+001,6.35584145E+001]
C5-6      [6.84944687E+001,2.51576214E+001,6.24934502E+001]
C5-7      [6.79246216E+001,2.53722992E+001,6.32098122E+001]
C5-8      [6.84075165E+001,2.63778133E+001,6.10950584E+001]
C5-9      [6.73214798E+001,2.70551453E+001,6.27835197E+001]
C5-10     [6.50006485E+001,2.67408028E+001,6.07828026E+001]
C5-11     [6.68869705E+001,2.3982399E+001,6.13440819E+001]
C5-12     [6.55524521E+001,2.42231808E+001,6.07235756E+001]
C5-13     [6.72140808E+001,2.42842178E+001,6.21546478E+001]
C5-14     [6.89587936E+001,2.67715569E+001,6.08621559E+001]
C5-15     [6.68405685E+001,2.44039059E+001,6.12652893E+001]
C6-1      [5.4925251E+001,8.28179474E+001,3.0869236E+001]
C6-2      [5.52922363E+001,8.23375549E+001,2.94804363E+001]
C6-3      [5.60466652E+001,8.18454132E+001,2.99774895E+001]

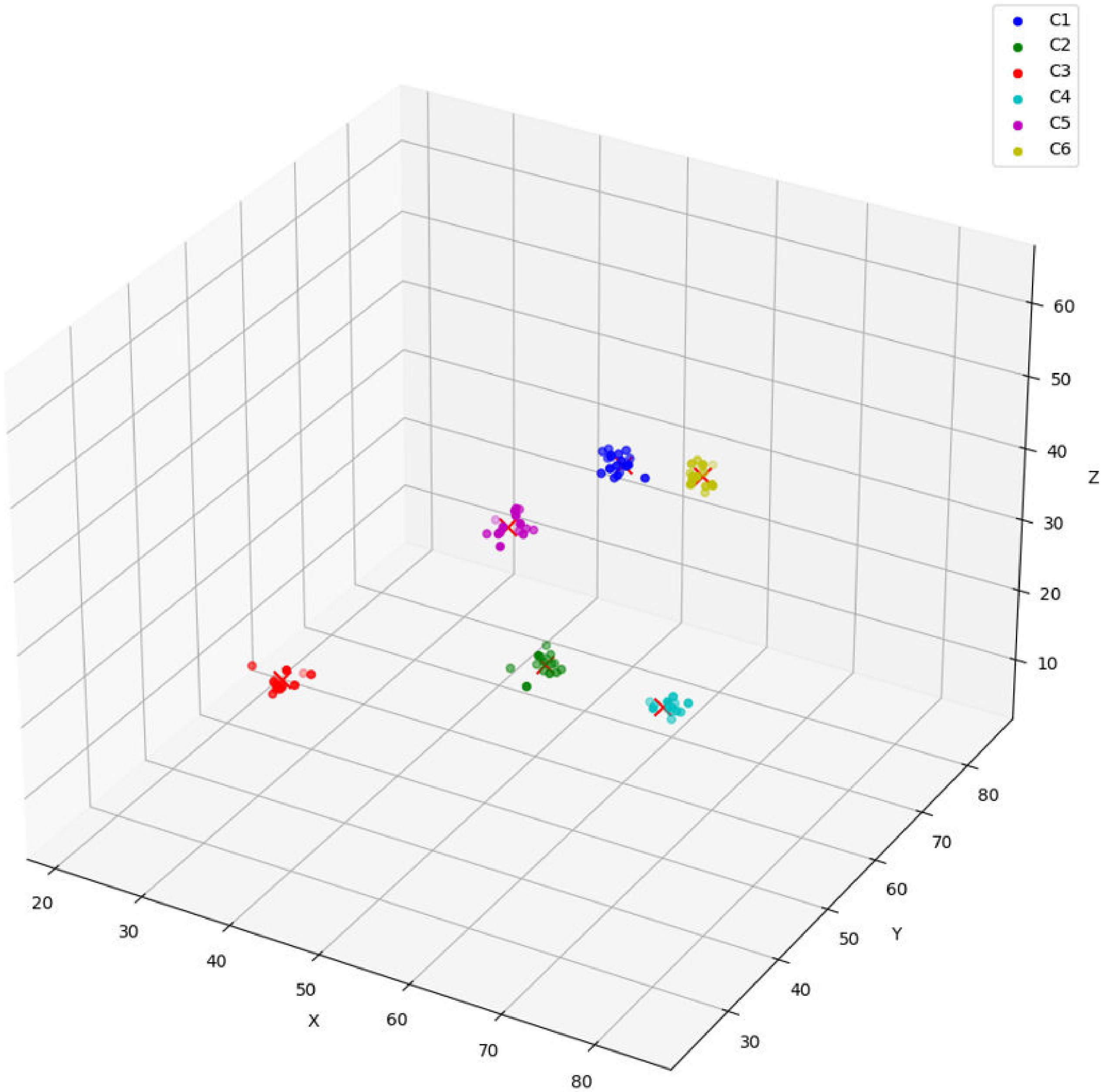
NAME      V
________  ____________________________________________________
C6-4      [5.74460373E+001,8.26830368E+001,2.86887722E+001]
C6-5      [5.57439041E+001,8.14622726E+001,2.94924259E+001]
C6-6      [5.4913372E+001,8.48766251E+001,2.92711105E+001]
C6-7      [5.66876144E+001,8.25907898E+001,2.84199276E+001]
C6-8      [5.6253479E+001,8.3280838E+001,2.69524212E+001]
C6-9      [5.50792351E+001,8.37676392E+001,3.08755417E+001]
C6-10     [5.57719955E+001,8.11036758E+001,2.92569256E+001]
C6-11     [5.60834808E+001,8.3103096E+001,3.09748001E+001]
C6-12     [5.58962059E+001,8.3612648E+001,2.95026093E+001]
C6-13     [5.73348083E+001,8.26950226E+001,2.88242455E+001]
C6-14     [5.45099411E+001,8.33315659E+001,2.90559101E+001]
C6-15     [5.57930641E+001,8.5720871E+001,2.92863998E+001]
C1-97     [6.3716671E+001,5.41518326E+001,5.18371048E+001]
C1-98     [6.58774261E+001,5.32223206E+001,5.05089798E+001]

NAME       V
_________  ____________________________________________________
C1-99      [6.31867676E+001,5.25712204E+001,5.16621819E+001]
C1-100     [6.20503845E+001,5.15550919E+001,5.08155479E+001]

100 rows selected.
```

To visualize the resulting vectors in space, consider the following graph:

Figure 3-1 Vector Clusters



5. Run a similarity search on the generated vectors in the `genvec` table.

   First define a variable called `cluster_number` to be used to form the name of the vector in the query.

```
DEFINE cluster_number = '&clusterid'
```

   You will be prompted to enter a value for `clusterid`. In this example, we use `5`:

```
Enter value for clusterid: 5
```

Run the following query to perform a similarity search on the generated vectors:

```
SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v,(SELECT v FROM genvec WHERE
name='C'||'&cluster_number'),EUCLIDEAN)
  FETCH EXACT FIRST 20 ROWS ONLY;
```

Example output:

```
NAME
________
C5
C5-15
C5-13
C5-3
C5-11
C5-1
C5-8
C5-6
C5-7
C5-12
C5-9
C5-4
C5-2
C5-5

NAME
________
C5-14
C5-10
C4-5
C4-12
C4-4
C4-13

20 rows selected.
```

6. Here is another example of running a similarity search on the generated vectors, this time using the Cosine distance metric.

```
SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v,(SELECT v FROM genvec WHERE
name='C'||'&cluster_number'),COSINE)
  FETCH EXACT FIRST 20 ROWS ONLY;
```

Example output:

```
NAME
________
C5
C5-6
C5-12
```

```
C5-15
C5-7
C5-4
C5-5
C5-13
C5-11
C5-3
C5-1
C5-8
C5-9
C5-2

NAME
________
C5-14
C5-10
C4-5
C4-4
C4-10
C4-12

20 rows selected.
```

7. Create a variable called `query_vector` and then use `SELECT INTO` to store a vector value in the variable.

```
VARIABLE query_vector CLOB

BEGIN
  SELECT v INTO :query_vector
  FROM genvec
  WHERE name='C'||'&cluster_number';
END;
/

PRINT query_vector;
```

Example output:

```
QUERY_VECTOR
--------------------------------------------------------------------
[6.70646744E+001,2.53395119E+001,6.14522667E+001]
```

8. Create an explain plan for a similarity search using the query vector created in the previous step.

```
EXPLAIN PLAN FOR
  SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v, :query_vector, EUCLIDEAN)
  FETCH EXACT FIRST 20 ROWS ONLY;

SELECT plan_table_output
  FROM table(dbms_xplan.display('plan_table',null,'all'));
```

Example output:

```
PLAN_TABLE_OUTPUT
______________________________________________________________________________
__________
Plan hash value: 1549136425

---------------------------------------------------------------------------
-------
| Id  | Operation                | Name   | Rows  | Bytes | Cost (%CPU)|
Time     |
---------------------------------------------------------------------------
-------
|   0 | SELECT STATEMENT         |        |    20 |  5040 |     4  (25)|
00:00:01 |
|*  1 |  COUNT STOPKEY           |        |       |       |
|           |
|   2 |   VIEW                   |        |   100 | 25200 |     4  (25)|
00:00:01 |
|*  3 |    SORT ORDER BY STOPKEY|        |   100 | 25800 |     4  (25)|
00:00:01 |
|   4 |     TABLE ACCESS FULL   | GENVEC |   100 | 25800 |     3   (0)|
00:00:01 |
---------------------------------------------------------------------------
-------

Query Block Name / Object Alias (identified by operation id):
-------------------------------------------------------------

PLAN_TABLE_OUTPUT
______________________________________________________________

   1 - SEL$2
   2 - SEL$1 / "from$_subquery$_002"@"SEL$2"
   3 - SEL$1
   4 - SEL$1 / "GENVEC"@"SEL$1"

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=20)
   3 - filter(ROWNUM<=20)

Column Projection Information (identified by operation id):
-----------------------------------------------------------

PLAN_TABLE_OUTPUT
__________________________________________________________________________

   1 - "from$_subquery$_002"."NAME"[VARCHAR2,500]
   2 - "from$_subquery$_002"."NAME"[VARCHAR2,500]
   3 - (#keys=1) VECTOR_DISTANCE("V" /*+ LOB_BY_VALUE */ ,
       VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ),
       EUCLIDEAN)[BINARY_DOUBLE,8], "NAME"[VARCHAR2,500]
   4 - "NAME"[VARCHAR2,500], VECTOR_DISTANCE("V" /*+ LOB_BY_VALUE */ ,
       VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ),
```

```
       EUCLIDEAN)[BINARY_DOUBLE,8]

Note
-----
   - dynamic statistics used: dynamic sampling (level=2)

41 rows selected.
```

9. Create an Hierarchical Navigable Small World (HNSW) index.

```
CREATE VECTOR INDEX genvec_hnsw_idx ON genvec(v)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  DISTANCE EUCLIDEAN
  WITH TARGET ACCURACY 95;

SELECT INDEX_NAME, INDEX_TYPE, INDEX_SUBTYPE FROM USER_INDEXES;
```

Example output:

```
INDEX_NAME                   INDEX_TYPE    INDEX_SUBTYPE
____________________________ _____________ ________________________________
DM$CEDOC_MODEL               NORMAL
SYS_IL0000073592C00002$$     LOB
GENVEC_HNSW_IDX              VECTOR        INMEMORY_NEIGHBOR_GRAPH_HNSW
SYS_C008694                  NORMAL

4 rows selected.
```

10. Query information about the HNSW index from the `V$VECTOR_INDEX` view.

```
SELECT JSON_SERIALIZE(IDX_PARAMS RETURNING VARCHAR2 PRETTY)
  FROM VECSYS.VECTOR$INDEX
  WHERE IDX_NAME = 'GENVEC_HNSW_IDX';

SELECT INDEX_ORGANIZATION, NUM_VECTORS, DISTANCE_TYPE,
       INDEX_DIMENSIONS, INDEX_DIM_TYPE, DEFAULT_ACCURACY
  FROM V$VECTOR_INDEX
  WHERE INDEX_NAME = 'GENVEC_HNSW_IDX';
```

Example output:

```
INDEX_ORGANIZATION        NUM_VECTORS DISTANCE_TYPE   INDEX_DIMENSIONS
INDEX_DIM_TYPE  DEFAULT_ACCURACY
------------------------ ----------- -------------- ----------------
--------------- ----------------
INMEMORY NEIGHBOR GRAPH   100         EUCLIDEAN       384
FLOAT32         95
```

For information about the columns of the `VECSYS.VECTOR$INDEX` view, see VECSYS.VECTOR$INDEX.

11. With the HNSW index created, create another explain plan for a similarity search on the genvec table.

```
EXPLAIN PLAN FOR
  SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v, :query_vector, EUCLIDEAN)
  FETCH FIRST 20 rows only;

SELECT plan_table_output
FROM table(dbms_xplan.display('plan_table',null,'all'));
```

Example output:

```
PLAN_TABLE_OUTPUT
___________________________________________________________________________
__________________________________
Plan hash value: 1202819565

---------------------------------------------------------------------------
-------------------------------
| Id  | Operation                        | Name             | Rows  | Bytes |
TempSpc| Cost (%CPU)| Time     |
---------------------------------------------------------------------------
-------------------------------
|   0 | SELECT STATEMENT                 |                  |    20 |  5040
|       |   165   (2)| 00:00:01 |
|*  1 |  COUNT STOPKEY                   |                  |       |
|       |             |          |
|   2 |   VIEW                           |                  |   100 | 25200
|       |   165   (2)| 00:00:01 |
|*  3 |    SORT ORDER BY STOPKEY         |                  |   100 |
425K|   808K|   165   (2)| 00:00:01 |
|   4 |     TABLE ACCESS BY INDEX ROWID| GENVEC           |   100 |
425K|       |     1   (0)| 00:00:01 |
|   5 |      VECTOR INDEX HNSW SCAN    | GENVEC_HNSW_IDX |   100 |
425K|       |     1   (0)| 00:00:01 |
---------------------------------------------------------------------------
-------------------------------

Query Block Name / Object Alias (identified by operation id):

PLAN_TABLE_OUTPUT
________________________________________________________________
-------------------------------------------------------------

   1 - SEL$2
   2 - SEL$1 / "from$_subquery$_002"@"SEL$2"
   3 - SEL$1
   4 - SEL$1 / "GENVEC"@"SEL$1"
   5 - SEL$1 / "GENVEC"@"SEL$1"

Predicate Information (identified by operation id):
---------------------------------------------------
```

```
   1 - filter(ROWNUM<=20)
   3 - filter(ROWNUM<=20)


PLAN_TABLE_OUTPUT
____________________________________________________________________________
__________________________
Column Projection Information (identified by operation id):
-----------------------------------------------------------

   1 - "from$_subquery$_002"."NAME"[VARCHAR2,500]
   2 - "from$_subquery$_002"."NAME"[VARCHAR2,500]
   3 - (#keys=1) VECTOR_DISTANCE("V" /*+ LOB_BY_VALUE */ ,
VECTOR(:QUERY_VECTOR, *, * /*+
       USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)[8], "NAME"[VARCHAR2,500]
   4 - "NAME"[VARCHAR2,500], VECTOR_DISTANCE("V" /*+ LOB_BY_VALUE */ ,
VECTOR(:QUERY_VECTOR, *, *
       /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)[8]
   5 - "GENVEC".ROWID[ROWID,10], VECTOR_DISTANCE("V" /*+ LOB_BY_VALUE */ ,
VECTOR(:QUERY_VECTOR,
       *, * /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)[8]

Note
-----

PLAN_TABLE_OUTPUT
____________________________________________________________
   - dynamic statistics used: dynamic sampling (level=2)

43 rows selected.
```

As you can see, the explain plan now includes information about the HNSW index.

12. Again perform a similarity search on the vectors in the `genvec` table. Note that it is possible for query results to vary based on the indexing technique used. The results included in this scenario are simply an example.

```
SELECT name
  FROM genvec
  ORDER BY vector_distance(v, :query_vector, EUCLIDEAN)
  FETCH FIRST 20 ROWS ONLY;
```

Example output:

```
NAME
________
C5
C5-15
C5-13
C5-3
C5-11
C5-1
C5-8
C5-6
C5-7
```

```
C5-12
C5-9
C5-4
C5-2
C5-5

NAME
________
C5-14
C5-10
C4-5
C4-12
C4-4
C4-13

20 rows selected.
```

13. Drop the HNSW index and create an Inverted File Flat (IVF) vector index.

```
DROP INDEX genvec_hnsw_idx;

CREATE VECTOR INDEX genvec_ivf_idx ON genvec(v)
  ORGANIZATION NEIGHBOR PARTITIONS
  DISTANCE EUCLIDEAN
  WITH TARGET ACCURACY 95;

SELECT INDEX_ORGANIZATION, NUM_VECTORS, DISTANCE_TYPE,
       INDEX_DIMENSIONS, INDEX_DIM_TYPE, DEFAULT_ACCURACY
  FROM V$VECTOR_INDEX where INDEX_NAME = 'GENVEC_IVF_IDX';
```

Example output:

```
INDEX_ORGANIZATION     NUM_VECTORS DISTANCE_TYPE   INDEX_DIMENSIONS
INDEX_DIM_TYPE  DEFAULT_ACCURACY
---------------------- ----------- --------------- ----------------
--------------- ----------------
NEIGHBOR PARTITIONS    100         EUCLIDEAN       384
FLOAT32         95
```

14. Again create an explain plan, which now includes information about the newly created IVF index.

```
EXPLAIN PLAN FOR
  SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v, :query_vector, EUCLIDEAN)
  FETCH FIRST 20 ROWS ONLY;

SELECT plan_table_output
  FROM table(dbms_xplan.display('plan_table',null,'all'));
```

Example output:

```
PLAN_TABLE_OUTPUT
____________________________________________________________________________
____________________________________________________________________________
____________________
Plan hash value: 2965029064

----------------------------------------------------------------------------
----------------------------------------------------------------------------
-----------------
| Id  | Operation                   |
Name                                                                   | Rows  |
Bytes | Cost (%CPU)| Time      | Pstart| Pstop |
----------------------------------------------------------------------------
----------------------------------------------------------------------------
-----------------
|   0 | SELECT STATEMENT
|                                                                     |     5
|  1260 |    29  (14)| 00:00:01 |       |       |
|   1 |  VIEW
|                                                                     |     5
|  1260 |    29  (14)| 00:00:01 |       |       |
|   2 |   SORT ORDER BY
|                                                                     |     5
| 21910 |    29  (14)| 00:00:01 |       |       |
|*  3 |    HASH JOIN
|                                                                     |     5
| 21910 |    28  (11)| 00:00:01 |       |       |
|   4 |     VIEW                    |
VW_IVPSR_11E7D7DE                                                    |    20
|   320 |    24   (9)| 00:00:01 |       |       |
|*  5 |      COUNT STOPKEY
|                                                                     |
|       |            |          |       |       |
|   6 |       VIEW                  |
VW_IVPSJ_578B79F1                                                    |    25
|   450 |    24   (9)| 00:00:01 |       |       |
|*  7 |        SORT ORDER BY STOPKEY
|                                                                     |    25
|   550 |    24   (9)| 00:00:01 |       |       |
|*  8 |         HASH JOIN
|                                                                     |    25
|   550 |    23   (5)| 00:00:01 |       |       |

PLAN_TABLE_OUTPUT
____________________________________________________________________________
____________________________________________________________________________
____________________
|   9 |          PART JOIN FILTER CREATE
| :BF0000                                                             |     6
|    18 |     4  (25)| 00:00:01 |       |       |
|  10 |           VIEW              |
VW_IVCR_B5B87E67                                                     |     6
|    18 |     4  (25)| 00:00:01 |       |       |
|* 11 |            COUNT STOPKEY
```

```
|                                                                   |
|       |             |          |       |       |
|  12 |             VIEW                  |
VW_IVCN_9A1D2119                                                  |    24
|   312 |     4  (25)| 00:00:01 |       |       |
|* 13 |              SORT ORDER BY STOPKEY
|                                                                   |    24
|   216 |     4  (25)| 00:00:01 |       |       |
|  14 |               TABLE ACCESS FULL     |
VECTOR$GENVEC_IVF_IDX$87355_87370_0$IVF_FLAT_CENTROIDS            |    24
|   216 |     3   (0)| 00:00:01 |       |       |
|  15 |          PARTITION LIST JOIN-
FILTER|
|   100 |  1900 |      3   (0)| 00:00:01 |:BF0000|:BF0000|
|  16 |           TABLE ACCESS FULL         |
VECTOR$GENVEC_IVF_IDX$87355_87370_0$IVF_FLAT_CENTROID_PARTITIONS |   100
|  1900 |     3   (0)| 00:00:01 |:BF0000|:BF0000|
|  17 |      TABLE ACCESS FULL              |
GENVEC                                                            |   100
|   426K|     3   (0)| 00:00:01 |       |       |
---------------------------------------------------------------------------
---------------------------------------------------------------------------
-----------------

Query Block Name / Object Alias (identified by operation id):
-------------------------------------------------------------


PLAN_TABLE_OUTPUT
___________________________________________________________
   1 - SEL$94C0F189 / "from$_subquery$_002"@"SEL$2"
   2 - SEL$94C0F189
   4 - SEL$E731354C / "VW_IVPSR_11E7D7DE"@"SEL$1"
   5 - SEL$E731354C
   6 - SEL$0C00A749 / "VW_IVPSJ_578B79F1"@"SEL$E731354C"
   7 - SEL$0C00A749
  10 - SEL$700CE8F1 / "VW_IVCR_B5B87E67"@"SEL$0C00A749"
  11 - SEL$700CE8F1
  12 - SEL$E5326247 / "VW_IVCN_9A1D2119"@"SEL$700CE8F1"
  13 - SEL$E5326247
  14 - SEL$E5326247 / "VTIX_CENTRD"@"SEL$E5326247"
  16 - SEL$0C00A749 / "VTIX_CNPART"@"SEL$0C00A749"
  17 - SEL$94C0F189 / "GENVEC"@"SEL$1"


PLAN_TABLE_OUTPUT
___________________________________________________________________________
___
Predicate Information (identified by operation id):
---------------------------------------------------

   3 - access("GENVEC".ROWID="VW_IVPSR_11E7D7DE"."BASE_TABLE_ROWID")
   5 - filter(ROWNUM<=20)
   7 - filter(ROWNUM<=20)
   8 - access("VW_IVCR_B5B87E67"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
  11 - filter(ROWNUM<=6)
```

```
  13 - filter(ROWNUM<=6)

Column Projection Information (identified by operation id):
-----------------------------------------------------------

   1 - "from$_subquery$_002"."NAME"[VARCHAR2,500]

PLAN_TABLE_OUTPUT
____________________________________________________________________________
____________________________________________________________________________
____________
   2 - (#keys=1) "VEC_DIST"[BINARY_DOUBLE,8], "GENVEC"."NAME"[VARCHAR2,500]
   3 - (#keys=1) "GENVEC"."NAME"[VARCHAR2,500],
"VEC_DIST"[BINARY_DOUBLE,8], "GENVEC"."NAME"[VARCHAR2,500]
   4 - "BASE_TABLE_ROWID"[ROWID,10], "VEC_DIST"[BINARY_DOUBLE,8]
   5 - "VW_IVPSJ_578B79F1"."BASE_TABLE_ROWID"[ROWID,10],
"VW_IVPSJ_578B79F1"."VEC_DIST"[BINARY_DOUBLE,8]
   6 - "VW_IVPSJ_578B79F1"."BASE_TABLE_ROWID"[ROWID,10],
"VW_IVPSJ_578B79F1"."VEC_DIST"[BINARY_DOUBLE,8]
   7 - (#keys=1) VECTOR_DISTANCE("VTIX_CNPART"."DATA_VECTOR" /*+
LOB_BY_VALUE */ , VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ),
       EUCLIDEAN)[BINARY_DOUBLE,8],
"VTIX_CNPART"."BASE_TABLE_ROWID"[ROWID,10]
   8 - (#keys=1) "VTIX_CNPART"."BASE_TABLE_ROWID"[ROWID,10],
VECTOR_DISTANCE("VTIX_CNPART"."DATA_VECTOR" /*+ LOB_BY_VALUE */ ,
VECTOR(:QUERY_VECTOR, *, * /*+
       USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)[BINARY_DOUBLE,8]
   9 - "VW_IVCR_B5B87E67"."CENTROID_ID"[NUMBER,22],
"VW_IVCR_B5B87E67"."CENTROID_ID"[NUMBER,22]
  10 - "CENTROID_ID"[NUMBER,22]
  11 - "VW_IVCN_9A1D2119"."CENTROID_ID"[NUMBER,22]
  12 - "VW_IVCN_9A1D2119"."CENTROID_ID"[NUMBER,22]
  13 - (#keys=1)
VECTOR_DISTANCE("VECTOR$GENVEC_IVF_IDX$87355_87370_0$IVF_FLAT_CENTROIDS"."C
ENTROID_VECTOR" /*+ LOB_BY_VALUE */ , VECTOR(:QUERY_VECTOR, *, *

PLAN_TABLE_OUTPUT
____________________________________________________________________________
____________________________________________________________________________
___________
       /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)[BINARY_DOUBLE,8],
"VTIX_CENTRD"."CENTROID_ID"[NUMBER,22]
  14 - "VTIX_CENTRD"."CENTROID_ID"[NUMBER,22],
VECTOR_DISTANCE("VECTOR$GENVEC_IVF_IDX$87355_87370_0$IVF_FLAT_CENTROIDS"."C
ENTROID_VECTOR" /*+ LOB_BY_VALUE */
       , VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)
[BINARY_DOUBLE,8]
  15 - "VTIX_CNPART"."BASE_TABLE_ROWID"[ROWID,10],
"VTIX_CNPART"."CENTROID_ID"[NUMBER,22],
VECTOR_DISTANCE("VTIX_CNPART"."DATA_VECTOR" /*+ LOB_BY_VALUE */ ,
       VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)
[BINARY_DOUBLE,8]
  16 - "VTIX_CNPART"."BASE_TABLE_ROWID"[ROWID,10],
"VTIX_CNPART"."CENTROID_ID"[NUMBER,22],
VECTOR_DISTANCE("VTIX_CNPART"."DATA_VECTOR" /*+ LOB_BY_VALUE */ ,
       VECTOR(:QUERY_VECTOR, *, * /*+  USEBLOBPCW_QVCGMD */ ), EUCLIDEAN)
```

```
[BINARY_DOUBLE,8]
  17 - "GENVEC".ROWID[ROWID,10], "GENVEC"."NAME"[VARCHAR2,500]

Note
-----
   - dynamic statistics used: dynamic sampling (level=2)
   - this is an adaptive plan

83 rows selected.
```

15. Finally, run the similarity search once again.

```
SELECT name
  FROM genvec
  ORDER BY VECTOR_DISTANCE(v, :query_vector, EUCLIDEAN)
  FETCH FIRST 20 ROWS ONLY;
```

Example output:

```
NAME
________
C5
C5-15
C5-13
C5-3
C5-11
C5-1
C5-8
C5-6
C5-7
C5-12
C5-9
C5-4
C5-2
C5-5

NAME
________
C5-14
C5-10
C4-5
C4-12
C4-4
C4-13

20 rows selected.
```

SQL Quick Start Using a BINARY Vector Generator

A set of procedures generate BINARY vectors, providing a simple way to get started with Oracle AI Vector Search without a vector embedding model.

The included procedures allow you to randomly generate binary vectors with a specified number of dimensions and clusters. The output of the generation process is the population of a table called genbvec that you can then use, for example, to experiment with similarity searches.

The following instructions assume you already have access to a database account with sufficient privileges (minimally the DB_DEVELOPER_ROLE role).

> **Note**
>
> Do not use the BINARY vector generator on production databases. This tutorial is made available for testing and demo purposes.

> **Note**
>
> If you already have access to a third-party BINARY vector embedding model, you can perform a real text-to-BINARY-embedding transformation by calling third-party REST APIs using the Vector Utility PL/SQL package DBMS_VECTOR. For more information, refer to the example in Convert Text String to BINARY Embedding Outside Oracle AI Database.

1. Create the genbvec table.

```
DROP TABLE genbvec PURGE;

CREATE TABLE genbvec (
  id NUMBER,              -- id of the generated vector
  v VECTOR(*, BINARY),  -- generated vector
  name VARCHAR2(500),   -- name for the generated vector: C1 to Cn are
centroids, Cx_y is vector number y in cluster number x
  bv VARCHAR2(40),      -- bit version of the generated vector
  ly NUMBER             -- random number you can use to filter out rows in
addiion to similarity search on vectors
);
```

2. Create the procedures used to generate BINARY vectors:

```
CREATE OR REPLACE PROCEDURE generate_random_binary_vector(
dimensions IN NUMBER,
result_int OUT VECTOR,
result_binary OUT VARCHAR2
) IS
    binary_vector VARCHAR2(40);
    int8_value NUMBER;
    number_of_bits NUMBER;
    char_vector VARCHAR2(40);
BEGIN
```

```
  -- Validate dimension is a multiple of 8
  IF MOD(dimensions, 8) != 0 THEN
    RAISE_APPLICATION_ERROR(-20001, 'Number of dimensions must be a 
multiple of 8');
  END IF;

  -- Generate the random binary vector
  binary_vector := '';
  FOR i IN 1 .. dimensions LOOP
    IF DBMS_RANDOM.VALUE(0, 1) < 0.5 THEN
      binary_vector := binary_vector || '0';
    ELSE
      binary_vector := binary_vector || '1';
    END IF;
  END LOOP;

  -- Convert 8-bit packets to their int8 values and build the result string
  number_of_bits := dimensions/8;
  char_vector := '[';
  FOR i IN 0 .. number_of_bits - 1 LOOP
    int8_value := 0;
    FOR j IN 0 .. 7 LOOP
      int8_value := int8_value + TO_NUMBER(SUBSTR(binary_vector, i*8+j+1, 
1)) * POWER(2, j);
    END LOOP;
    char_vector := char_vector || int8_value;
    IF i < number_of_bits - 1 THEN
      char_vector := char_vector || ',';
    END IF;
  END LOOP;
  char_vector := char_vector || ']';

  -- Return the generated vector value
  result_int := to_vector(char_vector, dimensions, BINARY);
  result_binary := binary_vector;
END generate_random_binary_vector;
/


CREATE OR REPLACE PROCEDURE generate_binary_cluster(
  centroid IN VARCHAR2,            -- a string of 1 and 0
  spread IN NUMBER,                -- Maximum Hamming distance between 
centroid and other vectors in the same cluster
  cluster_size IN NUMBER,          -- Number of vectors to generate in 
addition to the centroid
  result_binary OUT SYS_REFCURSOR,
  result_int8 OUT SYS_REFCURSOR
) IS
    dimension NUMBER;
    max_spread NUMBER;
    vector VARCHAR2(40);
    char_vector VARCHAR2(40);
    flip_positions DBMS_SQL.VARCHAR2_TABLE;
    random_position NUMBER;
    tresult_binary DBMS_SQL.VARCHAR2_TABLE;
    tresult_int8 DBMS_SQL.VARCHAR2_TABLE;
```

```
    binary_vector VARCHAR2(40);
    cluster_index NUMBER := 1;
    number_of_bits NUMBER;
    int8_value NUMBER;
BEGIN
  -- Determine the dimension of the centroid vector
  dimension := LENGTH(centroid);

  -- Ensure dimension is a multiple of 8
  IF MOD(dimension, 8) != 0 THEN
    RAISE_APPLICATION_ERROR(-20001, 'Number of dimensions must be a 
multiple of 8');
  END IF;

  -- Generate the cluster of binary vectors
  WHILE cluster_index <= cluster_size LOOP
    binary_vector := centroid;

    -- Randomly flip bits in the centroid vector with a max of spread bits
    max_spread := TRUNC(DBMS_RANDOM.VALUE(1, spread+1));
    flip_positions.DELETE;
    FOR i IN 1 .. max_spread LOOP
      random_position := TRUNC(DBMS_RANDOM.VALUE(1, dimension+1));
      -- Ensure no duplicates
      WHILE flip_positions.EXISTS(random_position) LOOP
        random_position := TRUNC(DBMS_RANDOM.VALUE(1, dimension+1));
      END LOOP;
      flip_positions(random_position) := '1';
    END LOOP;

    -- Apply flips to binary vector
    FOR i IN 1 .. dimension LOOP
      IF flip_positions.EXISTS(i) THEN
        IF SUBSTR(binary_vector, i, 1) = '0' THEN
          binary_vector := SUBSTR(binary_vector, 1, i-1) || '1' || 
SUBSTR(binary_vector, i+1);
        ELSE
          binary_vector := SUBSTR(binary_vector, 1, i-1) || '0' || 
SUBSTR(binary_vector, i+1);
        END IF;
      END IF;
    END LOOP;

    -- Convert binary vector to int8 values
    number_of_bits := dimension/8;
    char_vector := '[';
    FOR i IN 0 .. number_of_bits-1 LOOP
      int8_value := 0;
      FOR j IN 0 .. 7 LOOP
        int8_value := int8_value + TO_NUMBER(SUBSTR(binary_vector, 
i*8+j+1, 1)) * POWER(2, j);
      END LOOP;
      char_vector := char_vector || int8_value;
      IF i < number_of_bits-1 THEN
        char_vector := char_vector || ',';
      END IF;
```

```
    END LOOP;
    char_vector := char_vector || ']';

    -- Add generated vectors to result tables
    tresult_binary(cluster_index) := binary_vector;
    tresult_int8(cluster_index) := char_vector;

    cluster_index := cluster_index + 1;
    END LOOP;

    -- Open cursor for binary result set
    OPEN result_binary FOR
      SELECT COLUMN_VALUE AS binary_vector
      FROM TABLE(tresult_binary);

    -- Open cursor for int8 result set
    OPEN result_int8 FOR
      SELECT COLUMN_VALUE AS int8_vector
      FROM TABLE(tresult_int8);

END generate_binary_cluster;
/


CREATE OR REPLACE PROCEDURE generate_binary_vectors(
  num_vectors NUMBER,   -- If numbers of vector is not a multiple of
num_clusters, remaining vectors are not generated
  num_clusters NUMBER,  -- Must be greater than 0
  dimensions NUMBER,    -- Must be a multiple of 8
  cluster_spread NUMBER -- Maximum Hamming distance between centroid and
other vectors in the same cluster: max number of bits flipped
) IS
    vectors_per_cluster NUMBER;
    remaining_vectors NUMBER;
    i NUMBER := 1;
    j NUMBER := 1;
    idx NUMBER := 1;
    max_id NUMBER;
    ri VECTOR(*, BINARY);
    rb VARCHAR2(40);
    result_binary SYS_REFCURSOR;
    result_int8 SYS_REFCURSOR;
    vb VARCHAR2(40);
    vi VARCHAR2(40);
BEGIN

  IF (num_vectors) <=0 OR (num_clusters < 1) OR (num_vectors <
num_clusters)
        OR (dimensions <= 0) OR (dimensions > 504) OR (cluster_spread <=
0) THEN
    RAISE_APPLICATION_ERROR(-20001, 'Issues with arguments provided');
  END IF;

  SELECT MAX(id) INTO max_id FROM genbvec;
  IF max_id IS NULL THEN max_id := 0;
  END IF;
```

```
  -- Calculate vectors per cluster
  vectors_per_cluster := TRUNC(num_vectors / num_clusters);
  remaining_vectors := num_vectors MOD num_clusters; -- remaining vectors
are not generated

  -- Generate cluster centroids
  FOR i IN 1..num_clusters LOOP
    generate_random_binary_vector(dimensions, ri, rb);
    INSERT INTO genbvec VALUES (max_id + idx, ri, 'C'||i, rb,
DBMS_RANDOM.VALUE(3,600000000));
    idx := idx + 1;

    -- Generate vectors for each cluster
    IF vectors_per_cluster > 1 THEN
      generate_binary_cluster(rb, cluster_spread, vectors_per_cluster,
result_binary, result_int8);

      -- Output the binary result
      j:= 1;
      LOOP
        FETCH result_binary INTO vb;
        FETCH result_int8 INTO vi;
        EXIT WHEN result_binary%NOTFOUND;
        ri := TO_VECTOR(vi, dimensions, BINARY);
        INSERT INTO genbvec VALUES (max_id + idx, ri, 'C'||i||'-'||j, vb,
DBMS_RANDOM.VALUE(3,600000000));
        j := j+1;
        idx := idx + 1;
      END LOOP;
      CLOSE result_binary;
      CLOSE result_int8;
    END IF;
  END LOOP;
  COMMIT;

END generate_binary_vectors;
/
```

3. After you have your vector generator procedures set up, you can run the commands in this step to get started experimenting with `BINARY` vectors in the database.

   This example generates two clusters, each having twenty-one 32-dimension vectors (including the centroid) with a maximum spread of 3 from the centroid:

   a. Start out by generating some `BINARY` vectors using the `generate_binary_vectors` procedure. The results of the generation are inserted into the table, `genbvec`.

   ```
   BEGIN
     generate_binary_vectors(
       num_vectors  =>   40,   -- If numbers of vector is not a multiple
   of num_clusters, remaining vectors are not generated
       num_clusters =>    2,   -- Must be grather than 0
       dimensions   =>   32,   -- Must be a multiple of 8 and less than 504
       cluster_spread =>  3    -- Maximum Hamming distance between
   centroid and other vectors in the same cluster: max number of bits
   flipped
   ```

```
  );
END;
/
```

b. Run a SELECT statement to view the generated BINARY vectors.

```
SET SERVEROUTPUT ON;

SELECT id, v, name, VECTOR_DIMENSION_COUNT(v) DIMS,
VECTOR_DIMENSION_FORMAT(v) FORMAT, bv, ly FROM genbvec;
```

Example output:

```
     ID  V                    NAME       DIMS      FORMAT
BV                                                LY
-------  ------------------  ---------  --------  ----------
-----------------------------------  --------------
      1  [24,153,161,63]      C1         32        BINARY
00011000100110011000010111111100     99789021.1
      2  [24,153,165,63]      C1-1       32        BINARY
00011000100110011010010111111100     60221003.5
      3  [26,152,165,63]      C1-2       32        BINARY
01011000000110011010010111111100     387124796
      4  [24,201,161,62]      C1-3       32        BINARY
00011000100100111000010101111100     291263868
      5  [24,187,161,63]      C1-4       32        BINARY
00011000110111011000010111111100     583827824
      6  [24,153,161,61]      C1-5       32        BINARY
00011000100110011000010110111100     144826451
      7  [24,153,171,55]      C1-6       32        BINARY
00011000100110011101010111101100     113684378
      8  [88,153,161,61]      C1-7       32        BINARY
00011010100110011000010110111100     312081799
      9  [152,217,161,47]     C1-8       32        BINARY
00011001100110111000010111110100     173971628
     10  [24,153,163,59]      C1-9       32        BINARY
00011000100110011100010111011100     500775192
     11  [24,153,160,61]      C1-10      32        BINARY
00011000100110010000010110111100     137309652
     12  [25,185,161,47]      C1-11      32        BINARY
10011000100111011000010111110100     483392712
     13  [89,153,161,63]      C1-12      32        BINARY
10011010100110011000010111111100     458730494
     14  [24,153,229,31]      C1-13      32        BINARY
00011000100110011010011111111000     325738354

     ID  V                    NAME       DIMS      FORMAT
BV                                                LY
-------  ------------------  ---------  --------  ----------
-----------------------------------  --------------
     15  [24,152,161,63]      C1-14      32        BINARY
00011000000110011000010111111100     260267242
     16  [24,153,165,63]      C1-15      32        BINARY
00011000100110011010010111111100     153663322
```

```
     17  [24,137,169,63]       C1-16      32        BINARY
00011000100100011001010111111100     411918780
     18  [24,185,161,63]       C1-17      32        BINARY
00011000100111011000010111111100     53885587.1
     19  [152,137,161,63]      C1-18      32        BINARY
00011001100100011000010111111100     321305137
     20  [25,153,161,63]       C1-19      32        BINARY
10011000100110011000010111111100     180742593
     21  [16,153,161,63]       C1-20      32        BINARY
00001000100110011000010111111100     511768659
     22  [183,107,24,190]      C2         32        BINARY
11101101110101100001100001111101     529205377
     23  [181,251,24,190]      C2-1       32        BINARY
10101101110111110001100001111101     391560729
     24  [191,107,25,186]      C2-2       32        BINARY
11111101110101101001100001011101     191852938
     25  [182,106,24,190]      C2-3       32        BINARY
01101101010101100001100001111101     164088550
     26  [183,107,56,187]      C2-4       32        BINARY
11101101110101100001110011011101     20400437.6
     27  [183,106,16,190]      C2-5       32        BINARY
11101101010101100000100001111101     363725396
     28  [183,107,40,190]      C2-6       32        BINARY
11101101110101100001010001111101     144549103

     ID  V                     NAME       DIMS      FORMAT
BV                                                 LY
-------  ------------------  ---------  --------  ----------
----------------------------------  --------------
     29  [183,107,26,190]      C2-7       32        BINARY
11101101110101100101100001111101     318036129
     30  [183,123,24,188]      C2-8       32        BINARY
11101101110111100001100000111101     309460286
     31  [179,35,24,190]       C2-9       32        BINARY
11001101110001000001100001111101     25042254.7
     32  [182,251,24,190]      C2-10      32        BINARY
01101101110111110001100001111101     355499793
     33  [183,251,24,190]      C2-11      32        BINARY
11101101110111110001100001111101     483002129
     34  [183,107,24,254]      C2-12      32        BINARY
11101101110101100001100001111111     497697267
     35  [183,42,24,158]       C2-13      32        BINARY
11101101010101000001100001111001     64446273.5
     36  [151,107,28,186]      C2-14      32        BINARY
11101001110101100011100001011101     248483969
     37  [167,43,16,190]       C2-15      32        BINARY
11100101110101000000100001111101     513880134
     38  [183,106,24,190]      C2-16      32        BINARY
11101101010101100001100001111101     558247180
     39  [183,123,24,190]      C2-17      32        BINARY
11101101110111100001100001111101     287706546
     40  [151,107,24,190]      C2-18      32        BINARY
11101001110101100001100001111101     309138884
     41  [167,107,28,186]      C2-19      32        BINARY
11100101110101100011100001011101     433932877
```

```
     42  [63,106,24,190]      C2-20      32        BINARY
11111100010101100001100001111101     84539416.7
```

c. Perform a similarity search on the BINARY vectors.

```
SELECT name, v, bv, VECTOR_DISTANCE(v,
  (SELECT v FROM genbvec WHERE name='C1'), HAMMING) DISTANCE
FROM genbvec
ORDER BY VECTOR_DISTANCE(v, (SELECT v FROM genbvec WHERE name='C1'),
HAMMING);
```

Example output:

```
   NAME  V                  BV                                 DISTANCE
-------  -----------------  --------------------------------
----------
     C1  [24,153,161,63]    00011000100110011000010111111100   0
   C1-1  [24,153,165,63]    00011000100110011010010111111100   1.0
  C1-20  [16,153,161,63]    00001000100110011000010111111100   1.0
  C1-19  [25,153,161,63]    10011000100110011000010111111100   1.0
  C1-17  [24,185,161,63]    00011000100111011000010111111100   1.0
  C1-15  [24,153,165,63]    00011000100110011010010111111100   1.0
  C1-14  [24,152,161,63]    00011000000110011000010111111100   1.0
   C1-5  [24,153,161,61]    00011000100110011000010110111100   1.0
   C1-4  [24,187,161,63]    00011000110111011000010111111100   2.0
  C1-18  [152,137,161,63]   00011001100100011000010111111100   2.0
  C1-16  [24,137,169,63]    00011000100100011001010111111100   2.0
  C1-12  [89,153,161,63]    10011010100110011000010111111100   2.0
  C1-10  [24,153,160,61]    00011000100110010000010110111100   2.0
   C1-9  [24,153,163,59]    00011000100110011100010111011100   2.0

   NAME  V                  BV                                 DISTANCE
-------  -----------------  --------------------------------
----------
   C1-7  [88,153,161,61]    00011010100110011000010110111100   2.0
   C1-2  [26,152,165,63]    01011000000110011010010111111100   3.0
   C1-3  [24,201,161,62]    00011000100100111000010101111100   3.0
   C1-6  [24,153,171,55]    00011000100110011101010111101100   3.0
   C1-8  [152,217,161,47]   00011001100110111000010111110100   3.0
  C1-11  [25,185,161,47]    10011000100111011000010111110100   3.0
  C1-13  [24,153,229,31]    00011000100110011010011111111000   3.0
   C2-1  [181,251,24,190]   10101101111011111000110000111110 1  15.0
  C2-10  [182,251,24,190]   01101101111011111000110000111110 1  15.0
   C2-6  [183,107,40,190]   11101101111010110000101000111110 1  16.0
  C2-11  [183,251,24,190]   11101101111011111000110000111110 1  16.0
   C2-2  [191,107,25,186]   11111101111010110100110000101110 1  17.0
  C2-20  [63,106,24,190]    11111100010101100001100001111101   17.0
  C2-18  [151,107,24,190]   11101001111010110000110000111110 1  17.0

   NAME  V                  BV                                 DISTANCE
-------  -----------------  --------------------------------
----------
  C2-17  [183,123,24,190]   11101101111011110000110000111110 1  17.0
  C2-15  [167,43,16,190]    11100101111010100000010000111110 1  17.0
```

```
   C2-9  [179,35,24,190]    11001101111000100000110000111110 1   17.0
   C2-4  [183,107,56,187]   11101101111010110000111001101110 1   17.0
     C2  [183,107,24,190]   11101101111010110000110000111110 1   18.0
   C2-8  [183,123,24,188]   11101101111011110000110000011110 1   18.0
   C2-3  [182,106,24,190]   01101101010101100001100001111101    18.0
   C2-5  [183,106,16,190]   11101101010101100000100001111101    18.0
   C2-7  [183,107,26,190]   11101101111010110010110000111110 1   19.0
  C2-12  [183,107,24,254]   11101101111010110000110000111111 1   19.0
  C2-13  [183,42,24,158]    11101101010101000001100001111001    19.0
  C2-14  [151,107,28,186]   11101001111010110001110000101110 1   19.0
  C2-16  [183,106,24,190]   11101101010101100001100001111101    19.0
  C2-19  [167,107,28,186]   11100101111010110001110000101110 1   21.0

42 rows selected.
```

d. Run another similarity search, this time limiting the results to the first 21 rows. In this example, this means the results include `BINARY` vectors only from cluster 1.

```
SELECT name, v, bv, VECTOR_DISTANCE(v,
    (SELECT v FROM genbvec WHERE name='C1'), HAMMING) DISTANCE
FROM genbvec
ORDER BY VECTOR_DISTANCE(v, (SELECT v FROM genbvec WHERE name='C1'),
HAMMING)
FETCH EXACT FIRST 21 ROWS ONLY;
```

Example output:

```
   NAME  V                   BV                                  DISTANCE
-------  ------------------  ---------------------------------
----------
     C1  [24,153,161,63]     00011000100110011000010111111100    0
   C1-1  [24,153,165,63]     00011000100110011010010111111100    1.0
  C1-20  [16,153,161,63]     00001000100110011000010111111100    1.0
  C1-19  [25,153,161,63]     10011000100110011000010111111100    1.0
  C1-17  [24,185,161,63]     00011000100111011000010111111100    1.0
  C1-15  [24,153,165,63]     00011000100110011010010111111100    1.0
  C1-14  [24,152,161,63]     00011000000110011000010111111100    1.0
   C1-5  [24,153,161,61]     00011000100110011000010110111100    1.0
   C1-4  [24,187,161,63]     00011000110111011000010111111100    2.0
  C1-18  [152,137,161,63]    00011001100100011000010111111100    2.0
  C1-16  [24,137,169,63]     00011000100100011001010111111100    2.0
  C1-12  [89,153,161,63]     10011010100110011000010111111100    2.0
  C1-10  [24,153,160,61]     00011000100110010000010110111100    2.0
   C1-9  [24,153,163,59]     00011000100110011100010111011100    2.0

   NAME  V                   BV                                  DISTANCE
-------  ------------------  ---------------------------------
----------
   C1-7  [88,153,161,61]     00011010100110011000010110111100    2.0
   C1-2  [26,152,165,63]     01011000000110011010010111111100    3.0
   C1-3  [24,201,161,62]     00011000100100111000010101111100    3.0
   C1-6  [24,153,171,55]     00011000100110011101010111101100    3.0
   C1-8  [152,217,161,47]    00011001100110111000010111110100    3.0
  C1-11  [25,185,161,47]     10011000100111011000010111110100    3.0
```

```
  C1-13  [24,153,229,31]     00011000100110011010011111111000   3.0

21 rows selected.
```

e. In this iteration, the similarity search omits the `HAMMING` distance metric. However, because `HAMMING` is the default metric used with `BINARY` vectors, the results are the same as the previous query.

```
SELECT name, v, bv, VECTOR_DISTANCE(v,
    (SELECT v FROM genbvec WHERE name='C1')) DISTANCE
FROM genbvec
ORDER BY VECTOR_DISTANCE(v, (SELECT v FROM genbvec WHERE name='C1'))
FETCH EXACT FIRST 21 ROWS ONLY;
```

Example output:

```
   NAME  V                  BV                                 DISTANCE
-------  ------------------  ---------------------------------
----------
     C1  [24,153,161,63]     00011000100110011000010111111100   0
   C1-1  [24,153,165,63]     00011000100110011010010111111100   1.0
  C1-20  [16,153,161,63]     00001000100110011000010111111100   1.0
  C1-19  [25,153,161,63]     10011000100110011000010111111100   1.0
  C1-17  [24,185,161,63]     00011000100111011000010111111100   1.0
  C1-15  [24,153,165,63]     00011000100110011010010111111100   1.0
  C1-14  [24,152,161,63]     00011000000110011000010111111100   1.0
   C1-5  [24,153,161,61]     00011000100110011000010110111100   1.0
   C1-4  [24,187,161,63]     00011000110111011000010111111100   2.0
  C1-18  [152,137,161,63]    00011001100100011000010111111100   2.0
  C1-16  [24,137,169,63]     00011000100100011001010111111100   2.0
  C1-12  [89,153,161,63]     10011010100110011000010111111100   2.0
  C1-10  [24,153,160,61]     00011000100110010000010110111100   2.0
   C1-9  [24,153,163,59]     00011000100110011100010111011100   2.0

   NAME  V                  BV                                 DISTANCE
-------  ------------------  ---------------------------------
----------
   C1-7  [88,153,161,61]     00011010100110011000010110111100   2.0
   C1-2  [26,152,165,63]     01011000000110011010010111111100   3.0
   C1-3  [24,201,161,62]     00011000100100111000010101111100   3.0
   C1-6  [24,153,171,55]     00011000100110011101010111101100   3.0
   C1-8  [152,217,161,47]    00011001100110111000010111110100   3.0
  C1-11  [25,185,161,47]     10011000100111011000010111110100   3.0
  C1-13  [24,153,229,31]     00011000100110011010011111111000   3.0

21 rows selected.
```

Your Vector Documentation Map to GenAI Prompts

Follow the included steps to increase your chance of getting better and more consistent answers about this document from your preferred Generative AI Chatbot with internet search capabilities.

1. Click on a topic of interest in the AI Vector Search Topic Map section of this page.
2. Copy and paste the corresponding prompt in your preferred Chatbot UI.
3. Enter your question at the end of the prompt.
4. Run your prompt.

> **Warning**
>
> Your use of Generative AI Chatbots is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to the use of any such chatbots. Oracle provides these information map prompts only for convenience and educational purposes. Notwithstanding any other terms and conditions related to the use of Generative AI Chatbots, your use of these prompts constitutes your acceptance of that risk and expresses exclusion of Oracle's responsibility or liability for any damages resulting from such use.

AI Vector Search Topic Map

- AI Vector Search Overview
- Get Started
- Generate Vector Embeddings
  - Understand Vector Embeddings Generation
  - Convert Pretrained Models to ONNX Format
  - Import ONNX Models into Oracle AI Database
  - Access Third-Party Models for Vector Generation
  - Generate Embedding Functions and Examples
  - Perform Chunking with Embedding Examples
  - Configure Chunking Parameters and Chunking Functions
  - Store Vector Embeddings in Relational Tables
- Vector Indexes
  - Size the Vector Pool
  - Hierarchical Navigable Small World Indexes
  - Inverted File Flat Vector Indexes
  - Partition Management Operations on Vector Indexes
  - Guidelines for Vector Indexes
  - Create and Alter Hybrid Vector Index
  - Search Using Hybrid Vector Index
- Vector Operations
  - Distance Metrics
  - Vector Distance Functions and Operators
  - Vector Constructors, Converters, Descriptors, Arithmetic Operators and Aggregate Functions

   - Query Data with Similarity Searches
   - Work with LLM-Powered APIs
   - Retrieval Augmented Generation
- Vector Diagnostics
   - Oracle AI Vector Search Views
      * Text Processing Views
      * Views Related to Vector Indexes and Hybrid Vector Indexes
   - Oracle AI Vector Search Statistics and Initialization Parameters
   - Vector Search PL/SQL Packages

AI Vector Search Topic Map Prompts

AI Vector Search Overview

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/overview-
ai-vector-search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/why-use-
vector-search-instead-other-vector-databases.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-ai-
vector-search-workflow.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
glossary.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supported-
clients-and-languages.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Get Started

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sql-quick-
start-using-vector-embedding-model-uploaded-database.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sql-quick-
start-using-float32-vector-generator.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sql-quick-
start-using-binary-vector-generator.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.
```

```
Question:
<enter your question here>
```

Understand Vector Embeddings Generation

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
understand-stages-data-transformations.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sql-
functions-generate-embeddings.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/chainable-
utility-functions-and-common-use-cases.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
helper-procedures.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supplied-
vector-utility-pl-sql-packages.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supported-
third-party-provider-operations-and-endpoints.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/terms-
using-vector-utility-pl-sql-packages.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/validate-
json-input-parameters.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Convert Pretrained Models to ONNX Format

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-text-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-image-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-multi-modal-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-text-classification.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-reranking.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
pretrained-models-onnx-model-end-end-instructions.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/python-
classes-convert-pretrained-models-onnx-models.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
```

```
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Import ONNX Models into Oracle AI Database

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-text-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-image-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-multi-modal-embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-text-classification.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/onnx-
pipeline-models-reranking.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/import-
onnx-models-oracle-ai-database-end-end-example.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/alternate-
method-import-onnx-models.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Access Third-Party Models for Vector Generation

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/access-
third-party-models-vector-generation-leveraging-third-party-rest-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dbms_vector-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dbms_vector_chain-vecse.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Generate Embedding Functions and Examples

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_embedding.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_chunks.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
text-string-embedding-oracle-ai-database.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
text-string-binary-embedding-oracle-ai-database.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
text-string-embedding-using-public-third-party-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
text-string-embedding-locally-ollama.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
image-embedding-using-public-third-party-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
multi-modal-embeddings-using-clip.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vectorize-
relational-tables-using-oml-feature-extraction-algorithms.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Perform Chunking With Embedding Examples

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
text-chunks-custom-chunking-specifications.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
file-text-chunks-embeddings-oracle-ai-database.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/convert-
file-embeddings-oracle-ai-database.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
and-use-embeddings-end-end-search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_chunks.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Configure Chunking Parameters and Chunking Functions

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/explore-
chunking-techniques-and-examples.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/create-
and-use-custom-vocabulary.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/create-
and-use-custom-language-data.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_chunks.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Store Vector Embeddings in Relational Tables

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/create-
tables-using-vector-data-type.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vectors-
external-tables.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/querying-
inline-external-table.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
performing-semantic-similarity-search-using-external-table.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vectors-
distributed-database-tables.html#GUID-F75FB9CA-E7D1-46BC-847A-7324EF21D829
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/binary-
vectors.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sparse-
vectors.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/insert-
vectors-database-table-using-insert-statement.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/load-
character-vector-data-using-sqlloader-example.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/load-
binary-vector-data-using-sqlloader-example.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/unload-
and-load-vectors-using-oracle-data-pump.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.
```

```
Question:
<enter your question here>
```

Size the Vector Pool

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/size-
vector-pool.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Hierarchical Navigable Small World Indexes

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/overview-
hierarchical-navigable-small-world-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/hnsw-
duplication-rac-nodes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
hierarchical-navigable-small-world-index-syntax-and-parameters.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/pmop-
global-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
experiment-global-ivf-and-hnsw-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
indexes-gdd.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/hnsw-
index-architecture-transaction-support-and-persistence.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/included-
columns-hnsw-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/scalar-
quantized-hnsw-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/hnsw-
distribution-oracle-rac.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
distributed-hnsw-index-availability-and-performance-cluster-instance-
failure.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.
```

```
Question:
<enter your question here>
```

Inverted File Flat Vector Indexes

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
understand-inverted-file-flat-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/inverted-
file-flat-vector-indexes-online-rebuild.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/rebuild-
ivf-index-online.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/inverted-
file-flat-index-creation-syntax-and-parameters.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/inverted-
file-flat-alter-index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/pmop-
global-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
experiment-global-ivf-and-hnsw-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/pmop-
local-ivf-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
experiment-local-ivf-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
indexes-gdd.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/included-
columns-ivf-indexes.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Partition Management Operations on Vector Indexes

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/pmop-
global-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
experiment-global-ivf-and-hnsw-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/pmop-
local-ivf-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
experiment-local-ivf-indexes.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
```

```
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Guidelines for Vector Indexes

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
guidelines-using-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/index-
accuracy-report.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
index-status-checkpoint-and-advisor-procedures.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Create and Alter Hybrid Vector Index

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/create-
vector-indexes-and-hybrid-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
understand-hybrid-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/create-
hybrid-vector-index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/alter-
index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
guidelines-and-restrictions-hybrid-vector-indexes.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Search Using Hybrid Vector Index

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
```

```
understand-hybrid-search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/query-hybrid-vector-indexes-end-end-example.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/searchpipeline.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/get_sql.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and verbatim code examples.
  3. Answer question by making sure you only use previously extracted relevant content and verbatim code examples.

Question:
<enter your question here>
```

Distance Metrics

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/euclidean-and-squared-euclidean-distances.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/cosine-similarity.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/dot-product-similarity.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/manhattan-distance.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/hamming-distance.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/jaccard-similarity.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/custom-distance-function.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and verbatim code examples.
  3. Answer question by making sure you only use previously extracted relevant content and verbatim code examples.

Question:
<enter your question here>
```

Vector Distance Functions and Operators

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector_distance.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/l1_distance.html
```

```
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
l2_distance.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
cosine_distance.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
inner_product.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
hamming_distance-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
jaccard_distance-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/custom-
distance-function.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Vector Constructors, Converters, Descriptors, Arithmetic Operators and Aggregate Functions

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/to_vector-
vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
from_vector-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_serialize-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_norm-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_dimension_count-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_dims-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vector_dimension_format-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
arithmetic-operators.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
avg_vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
sum_vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/json-
compatibility-vector-data-type.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
```

```
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Query Data with Similarity Searches

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/perform-
exact-similarity-search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
understand-approximate-similarity-search-using-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/optimizer-
plans-hnsw-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/optimizer-
plans-ivf-vector-indexes.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
approximate-search-using-hnsw.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
approximate-search-using-ivf.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vector-
index-hints.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/perform-
multi-vector-similarity-search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
terminable-iteration-ivf.html#VECSE-GUID-26BA0FE6-59A7-48B9-A39C-FFF49E5349A1

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Work with LLM-Powered APIs

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
summary-using-public-third-party-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
summary-using-ollama.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
text-using-public-third-party-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/generate-
text-locally-ollama.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/describe-
images-using-public-third-party-apis.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/describe-
images-using-ollama.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supported-
```

```
third-party-provider-operations-and-endpoints.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Retrieval Augmented Generation

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/retrieval-
augmented-generation1.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/sql-rag-
example.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-ai-
vector-search-integration-langchain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-ai-
vector-search-integration-llamaindex.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/use-
reranking-better-rag-results.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supported-
third-party-provider-operations-and-endpoints.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Text Processing Views

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
all_vector_abbrev_tokens.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
all_vector_lang.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dba_vector_hitcounts.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
user_vector_hitcounts.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
user_vector_abbrev_tokens.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
user_vector_lang.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
user_vector_vocab.html
```

```
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
user_vector_vocab_tokens.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
all_vector_vocab.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
all_vector_vocab_tokens.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Views Related to Vector Indexes and Hybrid Vector Indexes

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_memory_pool.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vecsys-
vectorindex.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_graph_index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_graph_index_checkpoints.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_graph_index_snapshots.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
vvector_partitions_index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/vecsys-
vectorindexcheckpoints.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
indexnamevectors.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Oracle AI Vector Search Statistics and Initialization Parameters

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-ai-
vector-search-dictionary-statistics.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-
```

```
machine-learning-static-dictionary-views.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/oracle-ai-
vector-search-parameters.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

Vector Search PL/SQL Packages

```
Provided URLs:
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dbms_vector-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_credential-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
disable_checkpoint.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_credential-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_onnx_model-procedure-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
enable_checkpoint.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
get_index_status.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
index_accuracy_query.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
index_accuracy_report.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
index_vector_memory_advisor.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
inmemory_onnx_model-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
load_onnx_model-procedure.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/json-
metadata-parameters-onnx-models.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
load_onnx_model_cloud.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/query.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
rebuild_index.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_rerank-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_chunks-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
```

```
utl_to_embedding-and-utl_to_embeddings-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_generate_text-dbms_vector.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dbms_vector_chain-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_credential-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_lang_data.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_preference.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
create_vocabulary.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_credential-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_lang_data.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_preference.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
drop_vocabulary.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_rerank-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_chunks-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_embedding-and-utl_to_embeddings-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_generate_text-dbms_vector_chain.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_summary.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
utl_to_text.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/supported-
languages-and-data-file-locations.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
dbms_hybrid_vector-vecse.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
search.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
searchpipeline.html
https://docs.oracle.com/en/database/oracle/oracle-database/26/vecse/
get_sql.html

Instructions:
  1. Set temperature to 0 to answer question.
  2. Open and read provided URLs to extract only relevant content and
verbatim code examples.
  3. Answer question by making sure you only use previously extracted
relevant content and verbatim code examples.

Question:
<enter your question here>
```

4 Generate Vector Embeddings

Oracle AI Vector Search offers Vector Utilities (SQL and PL/SQL tools) to automatically generate vector embeddings from your unstructured data, either within or outside Oracle AI Database.

Embeddings are vector representations that capture the semantic meaning of your data, rather than the actual words in a document or pixels in an image (as explained in Overview of Oracle AI Vector Search). A vector embedding model creates these embeddings by assigning numerical values to each element of your data (such as a word, sentence, or paragraph).

To generate embeddings within the database, you can import and use vector embedding models in ONNX format. To generate embeddings outside the database, you can access third-party vector embedding models (remotely or locally) by calling third-party REST APIs.

- About Vector Generation
  Learn about *Vector Utility SQL functions* and *Vector Utility PL/SQL packages* that help you transform unstructured data into vector embeddings.
- Import Pretrained Models in ONNX Format
  Oracle Database 26ai includes an ONNX runtime engine for running embedding models directly inside the database. This section covers the process of importing existing pretrained embedding models into Oracle Database, including converting those models into the ONNX format if they are not already converted.
- Access Third-Party Models for Vector Generation Leveraging Third-Party REST APIs
  You can access third-party vector embedding models to generate vector embeddings from your data, outside the database by calling third-party REST APIs.
- Vector Generation Examples
  Run these end-to-end examples to see how you can generate vector embeddings, both within and outside the database.

About Vector Generation

Learn about *Vector Utility SQL functions* and *Vector Utility PL/SQL packages* that help you transform unstructured data into vector embeddings.

- Understand the Stages of Data Transformations
  Your input data may travel through different stages before turning into a vector.
- About SQL Functions to Generate Embeddings
  Choose to implement Vector Utility SQL functions to perform parallel or on-the-fly chunking and embedding operations, within the database. The supplied SQL functions for vector generation are `VECTOR_CHUNKS` and `VECTOR_EMBEDDING`.
- About PL/SQL Packages to Generate Embeddings
  Choose to implement Vector Utility PL/SQL packages to perform chunking, embedding, and text generation operations along with text processing and similarity search, both within and outside the database. You can schedule these operations as end-to-end pipelines. The supplied PL/SQL packages for vector generation are `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN`.

Understand the Stages of Data Transformations

Your input data may travel through different stages before turning into a vector.

Depending on the size of your input data, which can range from small strings to very large documents, the data passes through a pipeline of optional transformation stages from *Plain Text* to *Chunks* to *Tokens* to *Vectors*, with *Vector Index* as the endpoint.

Prepare: Plain Text and Chunks

This stage prepares your unstructured data to ensure that it is in a format that can be processed by vector embedding models.

To prepare large unstructured text, such as a PDF or Word document, you may start by transforming the document into plain text. The text can then be split into smaller, more manageable segments called "chunks" through a process referred to as "chunking". Chunks can be sets of words (to capture specific words or word pieces), sentences (to capture a specific context), or paragraphs (to capture broader themes) and they can help with simplifying the task of transforming the text into vectors.

Later, you will learn about several chunking parameters and techniques that help you define your own chunking specifications and strategies, so that you can generate relevant and meaningful chunks according to your use case.

Splitting your data into chunks is not needed for small documents, phrases, text strings, or short summaries. Embedding models can directly process such content into a vector representation. This is explained in the following section.

Embed: Tokens and Vectors

You now pass the text or extracted chunks as input through a declared vector embedding model for generating vector embeddings on each text string or chunk. Embeddings are vector representations that capture the semantic meaning or context of your data. A vector embedding model creates these embeddings by assigning numerical values to each element of your data, that is, to each word, sentence, or paragraph.

The chunks are first passed through **Tokenizer** associated with your embedding model. The tokenizer further splits the chunks into individual words or word pieces known as **tokens**. An embedding model then embeds each token into a vector representation.

Tokenizers used by embedding models usually have limits on the size of the input text (number of tokens) they can deal with, so it is important that you chunk your data beforehand into appropriate-sized segments to avoid loss of text when generating embeddings. If the number of tokens are larger than the maximum input limit imposed by the model, then some tokens get truncated into a defined input length. Note that chunking is not needed if your text meets this maximum input limit.

The chunker must select a text size that approximates the maximum input limit of your model. The actual number of tokens depends on the specified tokenizer for an embedding model, which typically uses a **vocabulary** list of words, numbers, punctuations, and pieces of tokens. A vocabulary contains a set of tokens that are collected during a statistical training process. Each tokenizer uses a different vocabulary format to process text into tokens.

For example, the BERT multilingual model uses Word-Piece encoding or the GPT model uses Byte-Pair encoding.

The BERT model may tokenize the following sentence (containing four words):

```
Embedding usecase for chunking
```

into the following eight tokens and also include ## (number signs) to indicate non-initial pieces of words:

```
Em ##bedd ##ing use ##case for chunk ##ing
```

Vocabulary files are included as part of a model's distribution. You can supply a vocabulary file (recognized by your model's tokenizer) to the chunker beforehand, so that it can correctly estimate the token count of your input data at the time of chunking.

Populate and Query: Vector Indexes

Finally, you store the extracted vectors in a vector index to implement combined similarity and relational searches on those vectors.

Instead of creating a vector index alone, you can also choose to create a hybrid vector index, which is a combination of both vector index and Oracle Text search index. This lets you implement hybrid searches to retrieve more relevant search results by performing both vector-based similarity searches and text-based keyword searches on the same data, simultaneously.

About SQL Functions to Generate Embeddings

Choose to implement Vector Utility SQL functions to perform parallel or on-the-fly chunking and embedding operations, within the database. The supplied SQL functions for vector generation are `VECTOR_CHUNKS` and `VECTOR_EMBEDDING`.

Vector Utility SQL functions are intended for a direct and quick interaction with data, within pure SQL.

VECTOR_CHUNKS

Use the `VECTOR_CHUNKS` SQL function if you want to split plain text into chunks (pieces of words, sentences, or paragraphs) in preparation for the generation of embeddings, to be used with a vector index.

For example, you can use this function to build a standalone Text Chunking system that lets you break down a large PDF document into smaller yet semantically meaningful chunk texts. You can experiment with your chunks by running parallel chunking operations, where you can inspect each chunk text, accordingly amend the chunking results, and then proceed further with other data transformation stages.

To generate chunks, this function uses the in-house implementation with Oracle AI Database.

For detailed information on this function, see VECTOR_CHUNKS.

VECTOR_EMBEDDING

Use the `VECTOR_EMBEDDING` function if you want to generate a single vector embedding for different data types.

For example, you can use this function in information-retrieval applications or chatbots, where you want to generate a query vector on the fly from a user's natural language text input string. You can then query a vector field with this query vector for a fast similarity search.

To generate an embedding, this function uses a vector embedding model (in ONNX format) that you load into the database.

Note

If you want to generate embeddings by using third-party vector embedding models, then use Vector Utility PL/SQL packages. These packages let you work with both embedding models (in ONNX format) stored in the database and third-party embedding models (by calling third-party REST APIs).

For detailed information on this function, see VECTOR_EMBEDDING.

Related Topics

- Import Pretrained Models in ONNX Format
  Oracle Database 26ai includes an ONNX runtime engine for running embedding models directly inside the database. This section covers the process of importing existing pretrained embedding models into Oracle Database, including converting those models into the ONNX format if they are not already converted.
- Generate Embeddings
  In these examples, you can see how to use the `VECTOR_EMBEDDING` SQL function or the `UTL_TO_EMBEDDING` PL/SQL function to generate a vector embedding from input text strings and images.

About PL/SQL Packages to Generate Embeddings

Choose to implement Vector Utility PL/SQL packages to perform chunking, embedding, and text generation operations along with text processing and similarity search, both within and outside the database. You can schedule these operations as end-to-end pipelines. The supplied PL/SQL packages for vector generation are `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN`.

These packages can work with both vector embedding models in ONNX format (by importing these models into the database) and third-party vector embedding models (by calling third-party REST APIs). Each package is made up of subprograms, such as *chainable utility functions* and *vector helper procedures*.

- About Chainable Utility Functions and Common Use Cases
  These are intended to be a set of chainable and flexible "stages" through which you pass your input data to transform into a different representation, including vectors.
- About Vector Helper Procedures
  Vector helper procedures let you configure authentication credentials, preferences, and language-specific data for use in chainable utility functions.
- Supplied Vector Utility PL/SQL Packages
  Use either a lightweight `DBMS_VECTOR` package or a more advanced `DBMS_VECTOR_CHAIN` package with full capabilities.
- Terms of Using Vector Utility PL/SQL Packages
  You must understand the terms of using REST APIs that are part of Vector Utility PL/SQL packages.
- Validate JSON Input Parameters
  You can optionally validate the structure of your JSON input to the `DBMS_VECTOR.UTL` and `DBMS_VECTOR_CHAIN.UTL` functions, which use JSON to define their input parameters.

Related Topics

- Vector Generation Examples
  Run these end-to-end examples to see how you can generate vector embeddings, both within and outside the database.

About Chainable Utility Functions and Common Use Cases

These are intended to be a set of chainable and flexible "stages" through which you pass your input data to transform into a different representation, including vectors.

Supplied Chainable Utility Functions

You can combine a set of chainable utility (`UTL`) functions together in an end-to-end pipeline.

Each pipeline or **transformation chain** can include a single function or a combination of functions, which are applied to source documents as they are transformed into other representations (text, chunks, summary, or vector). These functions are chained together, such that the output from one function is used as an input for the next.

Each chainable utility function performs a specific task of transforming data into other representations, such as converting data to text, converting text to chunks, or converting the extracted chunks to embeddings.

At a high level, the supplied chainable utility functions include:

Function	Description	Input and Return Value
`UTL_TO_TEXT()`	Converts data (for example, Word, HTML, or PDF documents) to plain text.	Accepts the input as `CLOB` or `BLOB`. Returns a plain text version of the document as `CLOB`.
`UTL_TO_CHUNKS()`	Converts data to chunks.	Accepts the input as plain text (`CLOB` or `VARCHAR2`). Splits the data to return an array of chunks (`CLOB`).
`UTL_TO_EMBEDDING()`	Converts data to a single embedding.	Accepts the input as plain text (`CLOB`) or image (`BLOB`). Returns a single embedding (`VECTOR`).
`UTL_TO_EMBEDDINGS()`	Converts an array of chunks to an array of embeddings.	Accepts the input as an array of chunks (`VECTOR_ARRAY_T`). Returns an array of embeddings (`VECTOR_ARRAY_T`).
`UTL_TO_SUMMARY()`	Generates a concise summary for data, such as large or complex documents.	Accepts the input as plain text (`CLOB`). Returns a summary in plain text as `CLOB`.
`UTL_TO_GENERATE_TEXT()`	Generates text for prompts and images.	Accepts the input as text data (`CLOB`) for prompts, or as media data (`BLOB`) for media files such as images. Processes this information to return `CLOB` containing the generated text.
`UTL_TO_RERANK()`	Reassesses and reorders an initial list of documents based on their similarity score.	Accepts the input as a query (`CLOB`) and a list of documents (`JSON`). Processes this information to return a `JSON` object containing a reranked list of documents, sorted by score.

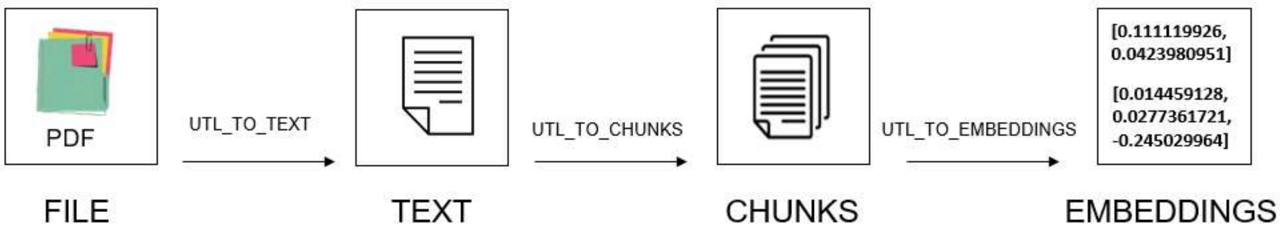
Sequence of Chains

Chainable utility functions are designed to be flexible and modular. You can create transformation chains in various sequences, depending on your use case.

For example, you can directly extract vectors from a large PDF file by creating a chain of the `UTL_TO_TEXT`, `UTL_TO_CHUNKS`, and `UTL_TO_EMBEDDINGS` chainable utility functions.

As shown in the following diagram, a *file-to-text-to-chunks-to-embeddings* chain performs a set of operations in this order:

1. Converts a PDF file to a plain text file by calling `UTL_TO_TEXT`.
2. Splits the resulting text into many appropriate-sized chunks by calling `UTL_TO_CHUNKS`.
3. Generates vector embeddings on each chunk by calling `UTL_TO_EMBEDDINGS`.



Common Use Cases

Let us look at some common use cases to understand how you can customize and apply these transformation chains:

Single-Step or Direct Transformation:

- **Document to vectors**:

  As discussed earlier, a common use case is to automatically generate vector embeddings from a set of documents. Embeddings are created by vector embedding models, and are numerical representations that capture the semantic meaning of your input data (as explained in Overview of Oracle AI Vector Search). The resulting embeddings are high-dimensional vectors, which you can use for text classification, question-answering, information retrieval, or text processing applications.

  – You can convert a set of documents to plain text, split the resulting text into smaller chunks to finally generate embeddings on each chunk, in a single *file-to-text-to-chunks-to-embeddings* chain. See Perform Chunking With Embedding.

  – For smaller documents or text strings, you can eliminate chunking (as explained in Understand the Stages of Data Transformations) and directly generate embeddings in a single *text-embeddings* chain. See Generate Embeddings.

- **Document to vectors, with chunking and summarization**:

  Another use case might be to generate a short summary of a document and then automatically extract vectors from that summary.

  After generating the summary, you can either generate a single vector (using `UTL_TO_EMBEDDING`) or chunk it and then generate multiple vectors (using `UTL_TO_EMBEDDINGS`).

  – You can convert the document to plain text, summarize the text into a concise gist to finally create a single embedding on the summarized text, in a *file-to-text-to-summary-to-embedding* chain.

  - You can convert the document to plain text, summarize the text into a gist, split the gist into chunks to finally create multiple embeddings on each summarized chunk, in a *file-to-text-to-summary-to-chunks-to-embeddings* chain.

  While both the chunking and summarization techniques make text smaller, they do so in different ways. Chunking just breaks the text into smaller pieces, whereas summarization extracts a salient meaning and context of that text into free-form paragraphs or bullet points.

  By summarizing the entire document and appending the summary to each chunk, you get the best of both worlds, that is, an individual piece that also has a high-level understanding of the overall document.

- **Image to vectors**:

  You can directly generate a vector embedding based on an image, which can be used for image classification, object detection, comparison of large datasets of images, or for a more effective similarity search on documents that include images. Embeddings are created by image embedding models or multimodal embedding models that extract each visual element of an image (such as shape, color, pattern, texture, action, or object) and accordingly assign a numerical value to each element.

  See Convert Image to Embedding Using Public REST Providers.

Step-by-Step or Parallel Transformation:

- **Text to vector**:

  A common use case might be information retrieval applications or chatbots, where you can convert a user's natural language text query string to a query vector on the fly. You can then compare that query vector against the existing vectors for a fast similarity search.

  You can vectorize a query vector and then run that query vector against your index vectors, in a *text-to-embedding* chain.

  See Convert Text String to Embedding Within Oracle AI Database, Convert Text String to Embedding Using Public REST Providers, and Convert Text String to Embedding Using the Local REST Provider Ollama.

- **Text to chunks**:

  Another use case might be to build a standalone Text Chunking system to break down a large amount of text into smaller yet semantically meaningful pieces, in a *text-to-chunks* chain.

  This method also gives you more flexibility to experiment with your chunks, where you can create, inspect, and accordingly amend the chunking results and then proceed further.

  See Convert Text to Chunks With Custom Chunking Specifications and Convert File to Text to Chunks to Embeddings Within Oracle AI Database.

- **Prompt or media file to text**:

  You can communicate with Large Language Models (LLMs) to perform several language-related tasks, such as text generation, translation, summarization, or Q&A. You can input text data in natural language (for prompts) or media data (for images) to an LLM-powered chat interface. LLM then processes this information to generate a text response.

  - **Prompt to text**:

    A prompt can be an input text string, such as a question that you ask an LLM. For example, "`What is Oracle Text?`". A prompt can also be a set of instructions or a command, such as "`Summarize the following ...`", "`Draft an email asking for ...`", or "`Rewrite the following ...`". The LLM responds with a textual answer, description, or summary based on the specified task in the prompt.

See Generate Text Using Public REST Providers and Generate Text Using the Local REST Provider Ollama.

- **Image to text**:

  You can also prompt with a media file (such as an image) to generate a textual analysis or description of the contents of the image, which can then be used for image classification, object detection, or similarity search.

  Here, you additionally supply a text question as the prompt along with the image. For example, the prompt can be "`What is this image about?`" or "`How many birds are there in this image?`".

  See Describe Images Using Public REST Providers.

- **Text to summary**:

  You can build a standalone Text Summarization system to generate a summary of large or complex documents, in a *text-to-summary* chain.

  This method also gives you more flexibility to experiment with your summaries, where you can create, inspect, and accordingly amend the summarization results and then proceed further.

  See Generate Summary Using Public REST Providers.

  > **Note**
  >
  > In addition to using external LLMs for summarization, you can use Oracle Text to generate a gist (summary) within the database. See UTL_TO_SUMMARY.

- **RAG implementation**

  A prompt can include results from a search, through the implementation of Retrieval Augmented Generation (RAG). Using RAG can mitigate inaccuracies and hallucinations, which may occur in generated responses when using LLMs.

  See SQL RAG Example.

  You can additionally utilize reranking to enhance the quality of information ingested into an LLM. See Use Reranking for Better RAG Results.

Schedule Vector Utility Packages

Some of the transformation chains may take a long time depending on your workload and implementation, thus you can schedule to run Vector Utility PL/SQL packages in the background.

The `DBMS_SCHEDULER` PL/SQL package helps you effectively schedule these packages, without manual intervention.

For information on how to create, run, and manage jobs with Oracle Scheduler, see *Oracle AI Database Administrator's Guide*.

About Vector Helper Procedures

Vector helper procedures let you configure authentication credentials, preferences, and language-specific data for use in chainable utility functions.

At a high level, the supplied vector helper procedures include:

- **Credential helper procedures** to securely manage authentication credentials, which are used to access third-party providers when making REST API calls.

Function	Description
CREATE_CREDENTIAL	Creates a credential name for securely storing user authentication credentials in Oracle AI Database.
DROP_CREDENTIAL	Drops an existing credential name.

- **Preference helper procedures** to manage vectorizer preferences, which are used when creating or altering hybrid vector indexes.

Function	Description
CREATE_PREFERENCE	Creates a vectorizer preference in the database.
DROP_PREFERENCE	Drops an existing vectorizer preference from the database.

- **Chunker helper procedures** to manage custom vocabulary and language data, which are used when chunking user data.

Function	Description
CREATE_VOCABULARY	Loads your own vocabulary file into the database.
DROP_VOCABULARY	Removes the specified vocabulary data from the database.
CREATE_LANG_DATA	Loads your own language data file (abbreviation tokens) into the database.
DROP_LANG_DATA	Removes abbreviation data for a given language from the database.

Related Topics

- Vector Search PL/SQL Packages
  The DBMS_VECTOR, DBMS_VECTOR_CHAIN, and DBMS_HYBRID_VECTOR PL/SQL APIs are available to support Oracle AI Vector Search capabilities.
- Text Processing Views
  These views display language-specific data (abbreviation token details) and vocabulary data related to the Oracle AI Vector Search SQL and PL/SQL utilities.

Supplied Vector Utility PL/SQL Packages

Use either a lightweight DBMS_VECTOR package or a more advanced DBMS_VECTOR_CHAIN package with full capabilities.

- DBMS_VECTOR:

  This package simplifies common operations with Oracle AI Vector Search, such as chunking text into smaller segments, extracting vector embeddings from user data, or generating text for a given prompt.

Subprogram	Operation	Provider	Implementation
Chainable Utility Functions	UTL_TO_CHUNKS to perform chunking	Oracle AI Database	Calls the VECTOR_CHUNKS SQL function under the hood
	UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS to generate one or more embeddings	Oracle AI Database	Calls the embedding model in ONNX format stored in the database

Subprogram	Operation	Provider	Implementation
		Third-party REST providers	Calls the specified third-party embedding model
	UTL_TO_GENERATE_TEXT to generate text for prompts and images	Third-party REST providers	Calls the specified third-party text generation model
	UTL_TO_RERANK to retrieve more relevant search output	Third-party REST providers	Calls the specified third-party embedding model
Credential Helper Procedures	CREATE_CREDENTIAL and DROP_CREDENTIAL to manage credentials for third-party service providers	Oracle AI Database	Stores credentials securely for use in Chainable Utility Functions

For detailed information on this package, see DBMS_VECTOR.

- DBMS_VECTOR_CHAIN:

  This package provides chunking and embedding functions along with some text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Subprogram	Operation	Provider	Implementation
Chainable Utility Functions	UTL_TO_TEXT to extract plain text data from documents	Oracle AI Database	Uses the Oracle Text component (CONTEXT) of Oracle AI Database
	UTL_TO_CHUNKS to perform chunking	Oracle AI Database	Calls the VECTOR_CHUNKS SQL function under the hood
	UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS to generate one or more embeddings	Oracle AI Database	Calls the embedding model in ONNX format stored in the database
		Third-party REST providers	Calls the specified third-party embedding model
	UTL_TO_SUMMARY to generate summaries	Oracle AI Database	Uses Oracle Text
		Third-party REST providers	Calls the specified third-party text summarization model
	UTL_TO_GENERATE_TEXT to generate text for prompts and images	Third-party REST providers	Calls the specified third-party text generation model
	UTL_TO_RERANK to retrieve more relevant search output	Third-party REST providers	Calls the specified third-party embedding model
Credential Helper Procedures	CREATE_CREDENTIAL and DROP_CREDENTIAL to manage credentials for third-party service providers	Oracle AI Database	Stores credentials securely for use in Chainable Utility Functions
Preference Helper Procedures	CREATE_PREFERENCE and DROP_PREFERENCE to manage preferences for hybrid vector indexes	Oracle AI Database	Creates vectorizer preferences for use in hybrid vector indexing pipelines
Chunker Helper Procedures	CREATE_VOCABULARY and DROP_VOCABULARY to manage custom token vocabularies	Oracle AI Database	Uses Oracle Text

Subprogram	Operation	Provider	Implementation
	CREATE_LANG_DATA and DROP_LANG_DATA to manage language-specific data (abbreviation tokens)	Oracle AI Database	Uses Oracle Text

Note

The DBMS_VECTOR_CHAIN package requires you to install the CONTEXT component of Oracle Text, an Oracle AI Database technology that provides indexing, term extraction, text analysis, text summarization, word and theme searching, and other utilities.

Due to underlying dependance on the text processing capabilities of Oracle Text, note that both the UTL_TO_TEXT and UTL_TO_SUMMARY chainable utility functions and all the chunker helper procedures are available only in this package through Oracle Text.

For detailed information on this package, see DBMS_VECTOR_CHAIN.

Related Topics

- Supported Third-Party Provider Operations and Endpoints
  Review a list of third-party REST providers and REST endpoints that are supported for various vector generation, summarization, text generation, and reranking operations.

Terms of Using Vector Utility PL/SQL Packages

You must understand the terms of using REST APIs that are part of Vector Utility PL/SQL packages.

Some of the Vector Utility PL/SQL APIs enable you to perform embedding, summarization, and text generation operations outside Oracle AI Database, by using third-party REST providers (such as Cohere, Google AI, Hugging Face, Generative AI, OpenAI, or Vertex AI).

Warning

Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.

Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Validate JSON Input Parameters

You can optionally validate the structure of your JSON input to the `DBMS_VECTOR.UTL` and `DBMS_VECTOR_CHAIN.UTL` functions, which use JSON to define their input parameters.

The JSON data is schemaless, so the amount of validation that Vector Utility package APIs do at runtime is minimal for better performance. The APIs validate only the mandatory JSON parameters, that is, the parameters that you supply for the APIs to run (not optional JSON parameters and attributes).

Before calling an API, you can use subprograms in the `DBMS_JSON_SCHEMA` package to test whether the input data to be specified in the `PARAMS` clause is valid with respect to a given JSON schema. This offers more flexibility and also ensures that only schema-valid data is inserted in a JSON column.

Validate JSON input parameters for the `DBMS_VECTOR.UTL` and `DBMS_VECTOR_CHAIN.UTL` functions against the following schema:

- **For the Database Provider**:
  - `SCHEMA_CHUNK`

```
{ "title" : "utl_to_chunks",
  "description" : "Chunk parameters",
  "type" : "object",
  "properties" : {
    "by"           : {"type" : "string", "enum" : [ "chars",
"characters", "words", "vocabulary" ]   },
    "max"          : {"type" : "string", "pattern" : "^[1-9][0-9]*$" },
    "overlap"      : {"type" : "string", "pattern" : "^[0-9]+$" },
    "split"        : {"type" : "string", "enum" : [ "none", "newline",
"blankline", "space", "recursively", "custom" ]  },
    "vocabulary"   : {"type" : "string"  },
    "language"     : {"type" : "string"  },
    "normalize"    : {"type" : "string", "enum" : [ "all", "none",
"options" ]  },
      "norm_options" : {"type" : "array",  "items": {  { "type":
"string", "enum": ["widechar", "whitespace", "punctuation"] }  },
    "custom_list"  : {"type" : "array",  "items": { "type":
"string" }  },
    "extended"     : {"type" : "boolean" } },
  "additionalProperties" : false
}
```

  - `SCHEMA_VOCAB`

```
{ "title" : "create_vocabulary",
  "description" : "Create vocabulary parameters",
  "type" : "object",
  "properties" : {
    "table_name"      : {"type" : "string" },
    "column_name"     : {"type" : "string" },
    "vocabulary_name" : {"type" : "string" },
    "format"          : {"type" : "string", "enum" : [ "BERT", "GPT2",
"XLM" ]  },
    "cased"           : {"type" : "boolean" } },
```

```
  "additionalProperties" : false,
  "required" : [ "table_name", "column_name", "vocabulary_name" ]
}
```

– SCHEMA_LANG

```
{ "title" : "create_lang_data",
  "description" : "Create language data parameters",
  "type" : "object",
  "properties" : {
    "table_name"      : {"type" : "string" },
    "column_name"     : {"type" : "string" },
    "preference_name" : {"type" : "string" },
    "language"        : {"type" : "string" } },
  "additionalProperties" : false,
  "required" : [ "table_name", "column_name", "preference_name",
"language" ]
}
```

– SCHEMA_TEXT

```
{ "title": "utl_to_text",
  "description": "To text parameters",
  "type" : "object",
  "properties" : {
    "plaintext"       : {"type" : "boolean" },
    "charset"         : {"type" : "string", "enum" : [ "UTF8" ] } },
  "additionalProperties": false
}
```

– SCHEMA_DBEMB

```
{ "title" : "utl_to_embedding",
  "description" : "To DB embeddings parameters",
  "type" : "object",
  "properties" : {
    "provider"   : {"type" : "string" },
    "model"      : {"type" : "string" } },
  "additionalProperties": true,
  "required" : [ "provider", "model" ]
}
```

– SCHEMA_SUM

```
{ "title" : "utl_to_summary",
  "description" : "To summary parameters",
  "type" : "object",
  "properties" : {
    "provider"       : {"type" : "string" },
    "numParagraphs"  : {"type" : "number" },
    "language"       : {"type" : "string" },
    "glevel"         : {"type" : "string" }
   },
  "additionalProperties" : true,
```

```
    "required" : [ "provider" ]
  }
```

- **For REST Providers**:

  SCHEMA_REST

  ```
  {  "title" : "REST parameters",
     "description" : "REST versions of utl_to_embedding, utl_to_summary, 
  utl_to_generate_text",
     "type" : "object",
     "properties" : {
       "provider"          : {"type" : "string" },
       "credential_name"   : {"type" : "string" },
       "url"               : {"type" : "string" },
       "model"             : {"type" : "string" }
      },
     "additionalProperties" : true,
     "required" : [ "provider", "credential_name", "url", "model" ] }
  ```

  Note that all the REST calls to third-party service providers share the same schema for their respective embedding, summarization, and text generation operations.

Examples:

- To validate your JSON data against JSON schema, use the PL/SQL function or procedure `DBMS_JSON_SCHEMA.is_valid()`.

  The function returns `1` for valid and `0` for invalid (invalid data can optionally raise an error). The procedure returns `TRUE` for valid and `FALSE` for invalid as the value of an `OUT` parameter.

  ```
  l_valid := sys.DBMS_JSON_SCHEMA.is_valid(params, json(SCHEMA), 
  dbms_json_schema.RAISE_ERROR);
  ```

- If you use the procedure (not function) `is_valid`, then you have access to the validation errors report as an `OUT` parameter. This use of the procedure checks data against schema, providing output in parameters validity (`BOOLEAN`) and errors (`JSON`).

  ```
  sys.DBMS_JSON_SCHEMA.is_valid(params, json(SCHEMA), l_valid, l_errors);
  ```

- To read a detailed validation report of errors, use the PL/SQL procedure `DBMS_JSON_SCHEMA.validate_report`.

  If you use the function (not the procedure) `is_valid`, then you do not have access to such a report. Instead of using the function `is_valid`, you can use the PL/SQL function `DBMS_JSON_SCHEMA.validate_report` in a SQL query to validate and return the same full validation information that the reporting `OUT` parameter of the procedure `is_valid` provides, as a JSON type instance.

  ```
  SELECT JSON_SERIALIZE(DBMS_JSON_SCHEMA.validate_report('json',SCHEMA) 
  returning varchar2 PRETTY);
  ```

Related Topics

- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.
- DBMS_JSON_SCHEMA

Import Pretrained Models in ONNX Format

Oracle Database 26ai includes an ONNX runtime engine for running embedding models directly inside the database. This section covers the process of importing existing pretrained embedding models into Oracle Database, including converting those models into the ONNX format if they are not already converted.

Open Neural Network Exchange or ONNX is an open standard format of machine learning models. Consider the following use cases:

- Using complex media input such as text or image for similarity search
- Perform text classification
- Perform reranking

You need vector embedding of the media input to perform all the tasks mentioned above. ONNX pipeline models allows you to convert text and/or image models into ONNX format if they are not already in ONNX format, import the ONNX format models into Oracle Database, and generate embeddings from your data within the database. The ONNX format models imported to the database could be used for tasks such as search and classification.

> **Note**
>
> You can use the ONNX pipeline utilities directly within the OML4Py Client Container. See Install OML4Py Client Container for more information. The container includes OML4Py client 2.1.1 and the `oml.utils` utility package (including `ONNXPipeline`, `ONNXPipelineConfig`, and `MiningFunction`), so you can list preconfigured models and export a selected model to a file and/or into the database from a Python session in the container.

Pretrained models are models that are already trained on a media data (text, image, etc.) and saved to a storage format for future use. Hugging Face is the most popular platform that hosts pretrained models typically created with PyTorch.

This table lists machine learning models, along with a brief description, model size, and a PAR download link for each.

Model Name	Description	Dimensions	Size (MB)	Last Updated	Download
all_MiniLM_L12_v2	This is a SentenceTransformers model suitable for semantic similarity search and clustering use cases.	384	116.92	7/15/2024	Link
multilingual_e5_base	This is a multilingual sentence-transformer model that supports 100 languages. Supports text embedding, translation, and multilingual understanding. Features 12 layers with 768-dimensional embeddings for multilingual tasks.	768	1040	2/17/2024	Link
multilingual_e5_small	This is a multilingual sentence-transformer model that supports 100. Supports text embedding, translation, or multilingual understanding. Optimized for efficiency, it can handle multiple languages while maintaining a smaller size for faster inference and reduced computational requirements.	384	76.8	9/9/2024	Link

Model Name	Description	Dimensions	Size (MB)	Last Updated	Download
clip_vit_base_patch32_txt	This is a text encoder that produces embeddings compatible with the CLIP image encoder. Enables text-to-image search and similarity matching between text descriptions and images.	512	243.57	1/29/2021	Link
clip_vit_base_patch32_img	This is an image encoder that produces embeddings compatible with the CLIP text encoder. Enables image-to-text search and similarity matching between images and text descriptions.	512	335.38	1/29/2021	Link

The Python package `oml.utils` contains three classes: `ONNXPipeline`, `ONNXPipelineConfig`, and `MiningFunction`. The package handles ONNX pipeline generation and export, while also incorporating the necessary pre- and post-processing steps.

- **ONNXPipeline** : Allows you to import a pretrained model (Your own pretrained model or one of the supported models from Hugging Face)
- **ONNXPipelineConfig** : Allows you to configure the attributes of pretrained model (Your own pretrained model or one of the supported models from Hugging Face)
- **MiningFunction**: Allows you to choose from one of the following mining options:
  - `EMBEDDING` : Corresponds to text and image embedding
  - `CLASSIFICATION` : Corresponds to text classification
  - `REGRESSION` : Corresponds to re-ranking

> **Warning**
>
> EmbeddingModel and EmbeddingModelConfig are deprecated. Instead, please use ONNXPipeline and ONNXPipelineConfig respectively. The details of the deprecated classes can be found in Python Classes to Convert Pretrained Models to ONNX Models (Deprecated). If a you choose to use a deprecated class, a warning message will be shown indicating that the classes will be removed in the future and advising the user to switch to the new class.

The ONNX pipeline models are available for text embedding, image embedding, multi-modal embedding, text classification and re-ranking.

- Convert Pretrained Models to ONNX Model: End-to-End Instructions for Text Embedding
  This section provides end-to-end instructions from installing the OML4Py client to downloading a pretrained embedding model in ONNX-format using the Python utility package offered by Oracle.
- Python Classes to Convert Pretrained Models to ONNX Models
- ONNX Pipeline Models: Text Embedding
  ONNX pipeline models provides text embedding models that accepts text as input and produces embeddings as output. The pipeline models also provide the necessary pre-processing and post-processing needed for the text.
- ONNX Pipeline Models: Image Embedding
  ONNX pipeline models provides image embedding models that accepts image as an input and produces embeddings as output. The pipeline models also provide the necessary pre-processing.
- ONNX Pipeline Models: CLIP Multi-Modal Embedding
  ONNX pipeline models provides multi-modal embedding models that accepts both image and text as an input and produces embeddings as output. The pipeline models also provide the necessary pre-processing needed.
- ONNX Pipeline Models: Text Classification
  ONNX pipeline models provides text classification models that accepts text strings as input and produces the probablity of the input string belonging to a specific label. The pipeline models also provide the necessary pre-processing and post-processing.
- ONNX Pipeline Models: Reranking Pipeline
  ONNX pipeline models provide a reranking pipeline that calculates similarity score for a given pair of texts.
- Load Custom Models from The Local Filesystem
  Custom models that have been fine-tuned from pre-trained Hugging Face models and reside on the local filesystem can be converted to the ONNX format using the new `local_model_path` setting. This setting can also be useful in cases where networking is unavailable and a pre-trained model has previously been downloaded.
- Support For Large ONNX Format Model
  OML4Py supports large ONNX format models outside of the database. A large model is divided into a single ONNX file and one or more external data files. The maximum size limit for an ONNX file is 2GB, so for models larger than 2GB, the ONNX file must be split into smaller parts.
- Model Summary
  A model summary provides a concise overview of the model. To obtain the summary after exporting the model, call the `summary()` function.
- Import ONNX Models into Oracle AI Database End-to-End Example
  Learn to import a pretrained embedding model that is in ONNX format and generate vector embeddings.

Convert Pretrained Models to ONNX Model: End-to-End Instructions for Text Embedding

This section provides end-to-end instructions from installing the OML4Py client to downloading a pretrained embedding model in ONNX-format using the Python utility package offered by Oracle.

> **Note**
>
> This example provides end to end instructions for converting pretrained text models to ONNX models. Steps 1 - 9 are identical for image models and multi-modals. You can use appropriate code/syntax mentioned in the corresponding topics to convert image models and multi models to ONNX pipeline models.

These instructions assume you have configured your Oracle Linux 8 repo in `/etc/yum.repos.d`, configured a Wallet if using an Autonomous AI Database, and set up a proxy if needed.

1. Install Python:

   ```
   sudo yum install libffi-devel openssl openssl-devel tk-devel xz-devel \
   zlib-devel bzip2-devel readline-devel libuuid-devel \
   ncurses-devel libaio tcl-devel openblas


   wget https://www.python.org/ftp/python/3.13.5/Python-3.13.5.tgz
   mkdir -p $HOME/python
   tar -xvzf Python-3.13.5.tgz --strip-components=1 -C $HOME/python
   cd $HOME/python
   ./configure --enable-shared --prefix=$HOME/python
   make clean; make
   make altinstall
   ```

2. Set variables `PYTHONHOME`, `PATH`, and `LD_LIBRARY_PATH`:

   ```
   export PYTHONHOME=$HOME/python
   export PATH=$PYTHONHOME/bin:$PATH
   export LD_LIBRARY_PATH=$PYTHONHOME/lib:$LD_LIBRARY_PATH
   ```

3. Create symlink for python3 and pip3:

   ```
   cd $HOME/python/bin
   ln -s python3.13 python3
   ln -s pip3.13 pip3
   ```

4. Install Oracle Instant Client if you plan to load embedded models from Python directly into a database. If you are exporting models to a file, you can skip this step—see the note about environment variables in Step 5 for further details.

   ```
   sudo yum install -y oracle-instantclient-release-el8 oracle-instantclient-
   basic
   ```

5. Create an environment file, for example, `env.sh`, that defines the Python and Oracle Instant client environment variables and source these environment variables before each OML4Py client session. Alternatively, add the environment variable definitions to `.bashrc` so they are defined when the user logs into their Linux machine.

```
# Environment variables for Python
export PYTHONHOME=$HOME/python
export PATH=$PYTHONHOME/bin:$PATH
export LD_LIBRARY_PATH=$PYTHONHOME/lib:$LD_LIBRARY_PATH
```

> **Note**
>
> Environment variable for Oracle Instant Client - only if the Oracle Instant Client is installed for exporting models to the database.
> `export LD_LIBRARY_PATH=$HOME/instantclient_23_26:$LD_LIBRARY_PATH`
>
> .

6. Create a file named requirements.txt that contains the required third-party packages listed below.

```
--extra-index-url https://download.pytorch.org/whl/cpu
pandas==2.2.3
setuptools==80.8.0
scipy==1.14.1
matplotlib==3.10.0
oracledb==3.3.0
scikit-learn==1.6.1
numpy==2.1.0
pyarrow==19.0.0
onnxruntime==1.20.0
onnxruntime-extensions==0.14.0
onnx==1.18.0
torch==2.9.0
transformers==4.56.1
sentencepiece==0.2.1
```

> **Note**
>
> Alternatively, you can install the required supporting packages by downloading and using the supporting packages bundle: oml4py-supporting-linux-x86_64-2.1.1.zip. See Install the Required Supporting Packages for Linux for On-Premises Databases for more information.

7. Upgrade pip3 and install the packages listed in requirements.txt.

```
pip3 install --upgrade pip
pip3 install -r requirements.txt
```

8. To install the OML4Py client, download the OML4Py 2.1.1 client from the [OML4Py download page](). Accept the Oracle License Agreement, then click on oml4py-client-linux-

x86_64-2.1.1.zip to download the ZIP file. After downloading, upload the ZIP file to your Linux machine.

```
unzip oml4py-client-linux-x86_64-2.1.1.zip
pip3 install client/oml-2.1.1-cp313-cp313-linux_x86_64.whl
```

> **Note**
>
> Install the OML4Py Slim Client Modules if you only need access to the ONNX model conversion feature. This lightweight client is ideal for scenarios where you require ONNX model conversion without the complete set of OML4Py functionalities. See Install OML4Py Slim Client for more information.

9. To get a list of all preconfigured models, start Python and import ONNXPipelineConfig from `oml.utils`.

```
python3
from oml.utils import ONNXPipelineConfig
ONNXPipelineConfig.show_preconfigured()
```

```
['sentence-transformers/all-mpnet-base-v2',
'sentence-transformers/all-MiniLM-L6-v2',
'sentence-transformers/multi-qa-MiniLM-L6-cos-v1',
'sentence-transformers/distiluse-base-multilingual-cased-v2',
'sentence-transformers/all-MiniLM-L12-v2',
'BAAI/bge-small-en-v1.5',
'BAAI/bge-base-en-v1.5',
'taylorAI/bge-micro-v2',
'intfloat/e5-small-v2',
'intfloat/e5-base-v2',
'thenlper/gte-base',
'thenlper/gte-small',
'TaylorAI/gte-tiny',
'sentence-transformers/paraphrase-multilingual-mpnet-base-v2',
'intfloat/multilingual-e5-base',
'intfloat/multilingual-e5-small',
'sentence-transformers/stsb-xlm-r-multilingual',
'Snowflake/snowflake-arctic-embed-xs',
'Snowflake/snowflake-arctic-embed-s',
'Snowflake/snowflake-arctic-embed-m',
'mixedbread-ai/mxbai-embed-large-v1',
'openai/clip-vit-large-patch14',
'google/vit-base-patch16-224',
'microsoft/resnet-18',
'microsoft/resnet-50',
'WinKawaks/vit-tiny-patch16-224',
'Falconsai/nsfw_image_detection',
'WinKawaks/vit-small-patch16-224',
'nateraw/vit-age-classifier',
'rizvandwiki/gender-classification',
'AdamCodd/vit-base-nsfw-detector',
'trpakov/vit-face-expression',
'BAAI/bge-reranker-base',
```

```
'BAAI/bge-large-en-v1.5',
'ibm-granite/granite-embedding-278m-multilingual',
'thenlper/gte-large',
'Snowflake/snowflake-arctic-embed-l',
'WhereIsAI/UAE-Large-V1']
```

10. Choose from:
    - To generate an ONNX file that you can manually upload to the database using the `DBMS_VECTOR.LOAD_ONNX_MODEL`, refer to step 3 of SQL Quick Start and skip step 10.
    - To upload the model directly into the database using the full version of the OML4Py client, skip this step and proceed to step 11.

    Export a preconfigured embedding model to a local file. Import ONNXPipeline and ONNXPipelineConfig from `oml.utils`. This exports the ONNX-format model to your local file system.

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig

# Export to file
pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-
v2")
pipeline.export2file("your_preconfig_file_name",output_dir=".")
```

    Move the ONNX file to a directory on the database server, and create a directory on the file system and in the database for the import.

```
mkdir -p /tmp/models
sqlplus / as sysdba
alter session set container=<name of pluggable database>;
```

    Apply the necessary permissions and grants.

```
-- directory to store ONNX files for import
CREATE DIRECTORY ONNX_IMPORT AS '/tmp/models';
-- grant your OML user read and write permissions on the directory
GRANT READ, WRITE ON DIRECTORY ONNX_IMPORT to OMLUSER;
-- grant to allow user to import the model
GRANT CREATE MINING MODEL TO OMLUSER;
```

    Use the `DBMS_VECTOR.LOAD_ONNX_MODEL` procedure to load the model in your OML user schema. In this example, the procedure loads the ONNX model file named `all-MiniLM-L6.onnx` from the ONNX_IMPORT directory into the database as a model named ALL_MINILM_L6.

```
BEGIN
   DBMS_VECTOR.LOAD_ONNX_MODEL(
     directory => 'ONNX_IMPORT',
     file_name => 'all-MiniLM-L6-v2.onnx',
     model_name => 'ALL_MINILM_L6',
     metadata => JSON('{"function" : "embedding", "embeddingOutput" :
"embedding", "input": {"input": ["DATA"]}}'));
END;
```

11. Export a preconfigured embedding model to the database. If you are using a database connection, make sure to update the connection details to match your credentials and database environment.

   > **Note**
   >
   > To ensure step 11 works properly, complete steps 4 first.

```
# Import oml library and EmbeddingModel from oml.utils
import oml
from oml.utils import ONNXPipeline, ONNXPipelineConfig

# Set embedded mode to false for Oracle AI Database on premises. This is
not supported or required for Oracle Autonomous AI Database.
oml.core.methods.__embed__ = False

# Create a database connection.

# Oracle AI Database on-premises
oml.connect("<user>", "<password>", port=<port number> host="<hostname>",
service_name="<service name>")

# Oracle Autonomous AI Database
oml.connect(user="<user>", password="<password>", dsn="myadb_low")
pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-
v2")
pipeline.export2db("ALL_MINILM_L6")
```

   Query the model and its views, and you can generate embeddings from Python or SQL.

```
import oracledb
cr = oml.cursor()
data = cr.execute("select vector_embedding(ALL_MINILM_L6 using 'RES' as
DATA)AS embedding from dual")
data.fetchall()
```

```
[(array('f', [-0.11642304062843323, 0.015433191321790218,
-0.04692629724740982, 0.007167273201048374,
0.035023435950279236, -0.04029884934425354, 0.018412720412015915,
0.0648672804236412, 0.014992560259997845,
0.012053844518959522, -0.03542192652821541, -0.09510122984647751,
-0.020823240280151367, 0.049922555685043335,
-0.01863112859427929, -0.026279663667082787, -0.03260187804698944,
0.05227319151163101, -0.09848847985267639,
0.0009467907948419452, 0.04395370930433273, 0.01060416642576456,
0.029333725571632385, -0.04252052679657936,
-0.025767622515559196, -0.03291202709078789, -0.053170595318078995,
0.07415973395109177, 0.0082617262378335,
```

```
...
...


SELECT VECTOR_EMBEDDING(ALL_MINILM_L6 USING 'RES' as DATA) AS embedding;


EMBEDDING
---------------------------------------------------------------------------
-----
[-1.16423041E-001,1.54331913E-002,-4.69262972E-002,7.1672732E-003,3.5023436
E-002
,-4.02988493E-002,1.84127204E-002,6.48672804E-002,1.49925603E-002,1.2053844
5E-00
2,-3.54219265E-002,-9.51012298E-002,-2.08232403E-002,4.99225557E-002,-1.863
11286
E-002,-2.62796637E-002,-3.2601878E-002,5.22731915E-002,-9.84884799E-002,9.4
67907
95E-004,4.39537093E-002,1.06041664E-002,2.93337256E-002,-4.25205268E-002,-2
.5767
...
...
```

Note

To suppress Torch warnings, use the following code:

```
import warnings
import torch warnings.filterwarnings("ignore",
category=DeprecationWarning)
warnings.filterwarnings("ignore", category=torch.jit.TracerWarning)
```

12. Verify the model exists using SQL:

```
sqlplus $USER/pass@PDBNAME;


select model_name, algorithm, mining_function from user_mining_models
where  model_name='ALL_MINILM_L6';


---------------------------------------------------------------------------
MODEL_NAME                  ALGORITHM                      MINING_FUNCTION
------------------------------ ------------------------------------------
ALL_MINILM_L6               ONNX                           EMBEDDING
```

Python Classes to Convert Pretrained Models to ONNX Models

This topic provides the functions, attributes and example usage of the python classes `ONNXPipelineConfig` and `ONNXPipeline`.

Functions and Attributes of ONNXPipelineConfig

The `ONNXPipelineConfig` class contains the properties required for the package to perform downloading, exporting, augmenting, validation, and storing of an ONNX model. The class provides access to configuration properties using the dot operator. As a convenience, well-known configurations are provided as templates.

Parameters

This table describes the functions and properties of the `ONNXPipelineConfig` class.

Functions	Parameter Type	Returns	Description
`from_template(name, **kwargs)`	• `name` (`String`): The name of the template • `**kwargs`: template properties to override or add	Instance of `ONNXPipelineConfig`	A static function that creates an `ONNXPipelineConfig` object based on a predefined template given by the name parameter. You can use named arguments to override the template properties.
`show_templates()`	NA	List of existing templates	A static function that returns a list of existing templates by name.
`show_preconfigured()`	• `include_properties` (`bool,optional`): A flag indicating whether properties should be included in the results. Defaults to `False` so only names will be included by default. • `model_name` (`str,optional`): A model name to filter by when including properties. This argument will be ignored if `include_properties` is `False`. Otherwise only the properties of this model will be included in the results.	A list of preconfigured model names or properties.	Shows a list of preconfigured model names, or properties. By default, this function returns a list of names only. If the properties are required, pass the `include_properties` parameter as `True`. The returned list will contain a single dict where each key of the dict is the name of a preconfigured model and the value is the property set for that model. Finally, if only a single set of properties for a specific model is required, pass the name of the model in the `model_name` parameter (the `include_properties` parameter should also be `True`). This will return a list of a single dict with the properties for the specified model.

Template Properties

The text template has configuration properties shown below:

```
"do_lower_case": true,
"post_processors":[{"name":"Pooling","type":"mean"},{"name":"Normalize"}]
```

> **Note**
>
> All other properties in the Properties table will take the default values. Any property without a default value must be provided when creating the `ONNXPipelineConfig` instance.

Properties

This table shows all properties that can be configured. preconfigured models already have these properties set to specific values. Templates will use the default values unless a user overrides it when using the `from_template` function on `ONNXPipelineConfig`.

Property	Description
`post_processors`	An array of post_processors that will be loaded after the model is loaded or initialized. The list of known and supported post_processors is provided later in this section. Templates may define a list of post_processors for the types of models they support. Otherwise, an empty array is the default.
`max_seq_length`	This property is applicable for text-based models only. The maximum length of input to the model as number of tokens. There is no default value. Specify this value for models that are not preconfigured.
`do_lower_case`	Specifies whether or not to lowercase the input when tokenizing. The default value is `True`.
`quantize_model`	Perform quantization on the model. This could greatly reduce the size of the model as well as speed up the process. It may however result in different results for the embedding vector (against the original model) and possibly small reduction in accuracy. The default value is `False`.
`distance_metrics`	An array of names of suitable distance metrics for the model. The names must be name of distance metrics used for Oracle vector distance operator. Only used when exporting the model to the database. Supported list is ["`EUCLIDEAN`","`COSINE`","`MANHATTAN`","`HAMMING`","`DOT`","`EUCLIDEAN_SQUARED`", "`JACCARD`"]. The default value is an empty array.
`languages`	An array of language (Abbreviation) supported in the Database. Only used when exporting the model to the database. For a supported list of languages, see Languages. The default value is an empty array.
`use_float16`	Specifies whether or not to convert the exported onnx model to float16. The default value is `False`.

Properties of post_processors

This table describes the built-in post_processors and their configuration parameters.

post_processor	Parameters	Description
`Pooling`	• `name`: `Pooling`. • `type`: Valid values should be `mean`(Default), `max`, `cls`	The Pooling post_processor summarizes the output of the transformer model into a fixed-length vector.
`Normalize`	• `name`: Specify `Normalize`	The Normalize post_processor bounds the vector values to a range using L2 normalization.
`Dense`	• `name`: Dense • `in_features`: Input feature size • `out_features`: Output feature size • `bias`: Whether to learn an additive bias. The default value is `True`. • `activation_function`: Activation function of the dense layer. Currently only supports `Tanh` as the activation function.	Applies transformation to the incoming data.

Example: Configure post_processors

In this example, you override the post_processors in the sentence-transformers template with a `Max` Pooling post_processor followed by Normalization.

```
config = ONNXPipelineConfig.from_template("text")
config.post_processors = [{"name":"Pooling","type":"max"},
{"name":"Normalize"}]
```

Functions and Attributes of ONNXPipeline

Use the `ONNXPipeline` class to convert transformer models to the ONNX format with post_processing steps embedded into the final model.

Parameters

This table describes the signature and properties of the `ONNXPipeline` class.

Functions	Parameters	Description
`ONNXPipeline(model_name,configuration=None,settings={})`	• `model_name(str)`: Name of the model to be used. For example, `medicalai/ClinicalBERT`. • `configuration (ONNXPipelineConfig, optional)`: An initialized `ONNXPipelineConfig` object. This parameter must be specified when using a template. If not specified, the model is assumed to be a preconfigured model. • `settings (dict, optional)`: A dictionary of global settings that control operations such as logging levels, download behavior, and file locations.	Creates a new instance of the `ONNXPipeline` class.

Settings

The settings object is a dictionary passed to the `ONNXPipeline` class. It provides global properties used for non-model-specific operations, such as logging and model download behavior.

For example:

```
pipe = ONNXPipeline(model_name, config, settings={"ignore_safetensors_error":
True})
```

where: `ignore_safetensors_error=True` allows execution to continue even if the requested model does not provide a `safetensors` version.

Property	Type	Default Value	Description
`cache_dir`	`str`	`$HOME/.cache/OML`	Directory where Hugging Face models are downloaded and cached. Model files will be downloaded to directories relative to the `cache_dir`. If the directory does not exist at execution time, it will be created automatically.

Property	Type	Default Value	Description
`logging_level`	`str`	`ERROR`	The level of severity for log messages. Valid values are `[‘DEBUG’, ‘INFO’, ‘WARNING’, ‘ERROR’, ‘CRITICAL’]`.

Note

This log level is also applied globally to all Python

Property	Type	Default Value	Description
			packages and is also mapped to the ONNXRuntime libraries.
`force_download`	`bool`	`False`	Forces downloading model files from Hugging Face even if they already exist in the local cache.

Property	Type	Default Value	Description
`local_model_path`	`str`	`None`	Path to a local copy of a model to be exported instead of downloading the model from Hugging Face. Must be a valid directory path.
`cleanup`	`bool`	`True`	Controls whether intermediate files generated during model export are removed after execution. If set to `False`, intermediate export files are retained.
`ignore_checksum_error`	`bool`	`False`	Ignores any errors caused by mismatch in checksums when using preconfigured models.
`ignore_safetensors_error`	`bool`	`False`	Ignores errors when the requested model does not have a `safetensors` version available.

Functions

This table describes the function and properties of the `ONNXPipeline` class.

Function	Parameters	Description
`export2file(export_name,output_dir=None)`	• `export_name(string)`: Name of the exported file. The file is saved with the `.onnx` extension. • `output_dir(string)`: Output directory where the file will be saved. If not specified, the file is saved in the current working directory.	Exports the model to a local `.onnx` file.
`export2db(export_name)`	• `export_name(string)`: Name used for the mining model object in the database. The name must comply with database object naming rules.	Exports the model to the database.

Example: Preconfigured Model

This example illustrates the preconfigured embedding model that comes with the Python package. You can use this model without any additional configurations.

```
"sentence-transformers/distiluse-base-multilingual-cased-v2": {
        "max_seq_length": 128,
        "do_lower_case": false,
        "post_processors":[{"name":"Pooling","type":"mean"},
{"name":"Dense","in_features":768, "out_features":512, "bias":true,
```

```
"activation_function":"Tanh"}],
        "quantize_model":true,
        "distance_metrics": ["COSINE"],
        "languages": ["ar", "bg", "ca", "cs", "dk", "d", "us", "el", "et",
"fa", "sf", "f", "frc", "gu", "iw", "hi", "hr", "hu", "hy", "in", "i", "ja",
"ko", "lt",
                      "lv", "mk", "mr", "ms", "n", "nl", "pl", "pt", "ptb",
"ro", "ru", "sk", "sl", "sq", "lsr", "s", "th", "tr", "uk", "ur", "vn",
"zhs", "zht"]
    }
```

ONNX Pipeline Models: Text Embedding

ONNX pipeline models provides text embedding models that accepts text as input and produces embeddings as output. The pipeline models also provide the necessary pre-processing and post-processing needed for the text.

> **Note**
>
> - This API is updated for 23.7. The version used for 23.6 and below used python packages EmbeddingModel and EmbeddingModelConfig. These packages are replaced with ONNXPipeline and ONNXPipelineConfig respectively. Oracle recommends that you use the latest version. You can find the details of the deprecated python classes in Python Classes to Convert Pretrained Models to ONNX Models (Deprecated).
> - This feature will **only** work on OML4Py 2.1 client. It is not supported on the OML4Py server.
> - In-database embedding models must include tokenization and post-processing. Providing only the core ONNX model to `DBMS_VECTOR.LOAD_ONNX_MODEL` is insufficient because you need to handle tokenization externally, pass tensors into the SQL operator, and convert output tensors into vectors.
> - Oracle is providing a Hugging Face all-MiniLM-L12-v2 model in ONNX format, available to download directly to the database using `DBMS_VECTOR.LOAD_ONNX_MODEL`. For more information, see the blog post Now Available! Pre-built Embedding Generation model for Oracle Database 23ai.

If you do not have a pretrained embedding model in ONNX-format to generate embeddings for your data, Oracle offers a Python package that downloads pretrained models from an external source, converts the model to ONNX format augmented with pre-processing and post-processing steps, and imports the resulting ONNX-format model into Oracle Database. Use the `DBMS_VECTOR.LOAD_ONNX_MODEL` procedure or `export2db()` function to import the file as a mining model. (`export2db()` is a method on the ONNXPipeline object). Then leverage the in-database ONNX Runtime with the ONNX model to produce vector embeddings.

At a high level, the Python package performs the following tasks:

- Downloads the pretrained model from external source (example: from Hugging Face repository) to your system
- Augments the model with pre-processing and post-processing steps and creates a new ONNX model
- Validates the augmented ONNX model

- Loads into the database as a data mining model or optionally exports to a file

The Python package can take any of the models in the preconfigured list as input. Alternatively, you can use the built-in template that contains common configurations for certain groups of models such as text or image models. To understand what a preconfigured list, what a built-in template is, and how to use them, read further.

Limitations

This table describes the limitations of the Python package.

> **Note**
>
> This feature is available with the OML4Py 2.1 client only.

Parameter	Description
`Transformer Model Type`	Supports transformer models that supports text embedding.
`Model Size`	Quantization can help reduce the size.
`Tokenizers`	Must be either `BERT`, `GPT2`, `SENTENCEPIECE`, `CLIP`, or `ROBERTA`.

Preconfigured List of Models

Preconfigured list of models are common models from external resource repositories that are provided with the Python package. The preconfigured models have an existing specification. Users can create their own specification using the text template as a starting point. To get a list of all model names in the preconfigured list, you can use the `show_preconfigured` function.

Templates

The Python package provides built-in text template for you to configure the pretrained models with pre-processing and post-processing operations. The template has a default specification for the pretrained models. This specification can be changed or augmented to create custom configurations. The text template uses Mean Pooling and Normalization as post-processing operations by default.

Converting Text Models to ONNX Format and Storing them as a File or in a Database:

To use the Python package `oml.utils`, ensure that you have the following:

- OML4Py 2.1 Client running on Linux X64 for On-Premises Databases
- Python 3.12 (the earlier versions are not compatible)
- For the full end-to-end example, see Convert Pretrained Models to ONNX Model: End-to-End Instructions.

1. Start Python in your work directory.

   ```
   python3
   ```

   ```
   Python 3.12.6 | (main, Feb 27 2024, 17:35:02) [GCC 11.2.0] on linux
   Type "help", "copyright", "credits" or "license" for more information.
   ```

2. On the OML4Py client, load the Python classes:

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig
```

3. You can get a list of all preconfigured models by running the following:

```
ONNXPipelineConfig.show_preconfigured()
```

4. To get a list of available templates:

```
ONNXPipelineConfig.show_templates()
```

5. Choose from:
   - Generating a text pipeline using a preconfigured model "sentence-transformers/all-MiniLM-L6-v2" and save it in a local directory:

```
#generate from preconfigured model "sentence-transformers/all-MiniLM-L6-
v2"
pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-
v2")
pipeline.export2file("your_preconfig_file_name",output_dir=".")
```

   - Generating a text pipeline using a preconfigured model "sentence-transformers/all-MiniLM-L6-v2" and save it in the database:

```
#generate from preconfigured model "sentence-transformers/all-MiniLM-L6-
v2"
pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-
v2")
pipeline.export2db("your_preconfig_file_name")
```

Note

In order for export2db( ) to succeed, a connection is required to the database. It is assumed that you have provided your credentials in the following code and the connection is successful.

```
import oml
oml.connect(user="username",password="password",dsn="dsn");
```

   - Generating a text pipeline with a template and save it in a local directory

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig
config   = ONNXPipelineConfig.from_template("text", max_seq_length=256,
           distance_metrics=["COSINE"], quantize_model=True)
pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-
v2",
           config=config)
pipeline.export2file("your_preconfig_model_name", output_dir=".")
```

Let's understand the code:

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig
```

This line imports two classes, `ONNXPipeline` and `ONNXPipelineConfig`.

In the preconfigured models first example, where the ONNX format model is saved in a particular directory:

- `pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-v2")` creates an instance of the `ONNXPipeline` class, loading a pretrained model specified by the `model_name` parameter. `pipeline` is the text embedding model object. `sentence-transformers/all-MiniLM-L6-v2` is the model name for computing sentence embeddings. This is the model name under Hugging Face. Oracle supports models from Hugging Face.
- The `export2file` command creates an ONNX format model with a user-specified model name in the file. `your_preconfig_file_name` is a user defined ONNX model file name.
- `output_dir="."` specifies the output directory where the file will be saved. The `"."` denotes the current directory (that is, the directory from which the script is running).

In the preconfigured models second example, where the ONNX model is saved in the database:

- `pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-v2")` creates an instance of the `ONNXPipleline` class, loading a pretrained model specified by the `model_name` parameter. `pipeline` is the text embedding model object. `sentence-transformers/all-MiniLM-L6-v2` is the model name for computing sentence embeddings. This is the model name under Hugging Face. Oracle supports models from Hugging Face.
- The `export2db` command creates an ONNX format model with a user defined model name in the database. `your_preconfig_model_name` is a user defined ONNX model name.

In the template example:

- `ONNXPipelineConfig.from_template("text", max_seq_length=256 distance_metrics=["COSINE"], quantize_model=True)`: This line creates a configuration object for an embedding model using a method called `from_template`. The `"text"` argument indicates the name of the template. The `max_seq_length=512` parameter specifies the maximum length of input to the model as number of tokens. There is no default value. Specify this value for models that are not preconfigured.
- `pipeline = ONNXPipeline(model_name="sentence-transformers/all-MiniLM-L6-v2", config=config)` initializes an `ONNXPipeliene` instance with a specific model and the previously defined configuration. The `model_name="all-MiniLM-L6-v2"` argument specifies the name or identifier of the pretrained model to be loaded.
- The `export2file` command creates an ONNX format model with a user defined model name in the file. `your_preconfig_model_name` is a user defined ONNX model name.
- `output_dir="."` specifies the output directory where the file will be saved. The `"."` denotes the current directory (that is, the directory from which the script is running).

> **Note**
>
> - For models larger than 400MB, Oracle recommends quantization.
>
>   **Quantization** reduces the model size by converting model weights from high-precision representation to low-precision format. The quantization option converts the weights to INT8. The smaller model size enables you to cache the model in shared memory further improving the performance.
> - The `.onnx` file is created with opset version 17 and ir version 8. For more information about these version numbers, see https://onnxruntime.ai/docs/reference/compatibility.html#onnx-opset-support.

6. Exit Python.

   ```
   exit()
   ```

7. Inspect if the converted models are present in your directory.

   > **Note**
   >
   > ONNX files are only created when `export2file` is used. If `export2db` is used, local ONNX files will be generated, and the model will be saved to the database.

   List all files in the current directory that have the extension `onnx`.

   ```
   ls -ltr *.onnx
   ```

   The Python utility package validates the embedding text model before you can run them using ONNX Runtime. Oracle supports ONNX embedding models that conform to `string` as input and `float32 [<batch size>,<vector dimension>]` as output.

   If the input or output of the model doesn't conform to the description, you receive an error during the import.

8. Choose from:

   - Load the ONNX model to your database using PL/SQL:

     a. You can load the ONNX model using this syntax:

     ```
     BEGIN
        DBMS_VECTOR.LOAD_ONNX_MODEL(
          directory => 'ONNX_IMPORT',
          file_name => 'all-MiniLM-L6.onnx',
          model_name => 'ALL_MINILM_L6');
     END;
     ```

b. Move the ONNX file named `your_template_file_name` to a directory on the database server, and create a directory on the file system and in the database for the import.

```
mkdir -p /tmp/models
sqlplus / as sysdba
alter session set container=ORCLPDB;
```

Apply the necessary permissions and grants. In this example, we are using a pluggable database named ORCLPDB.

```
-- directory to store ONNX files for import
CREATE DIRECTORY ONNX_IMPORT AS '/tmp/models';
-- grant your OML user read and write permissions on the directory
GRANT READ, WRITE ON DIRECTORY ONNX_IMPORT to OMLUSER;
-- grant to allow user to import the model
GRANT CREATE MINING MODEL TO OMLUSER;
```

- Load the ONNX model to your database using Python:

9. Verify the model is in the database:

At your operating system prompt, start SQL*Plus, connect to it :

```
sqlplus OML_USER/password@pdbname_medium;



SQL*Plus: Release 23.0.0.0.0 - Production on Wed May 1 15:33:29 2024
Version 23.4.0.24.05

Copyright (c) 1982, 2024, Oracle. All rights reserved.

Last Successful login time: Wed May 01 2024 15:27:06 -07:00

Connected to:
Oracle
                                Database 26ai Enterprise Edition Release
23.0.0.0.0 - Production
Version 23.4.0.24.05
```

List the ONNX model in the database to make sure it was loaded:

```
SQL> select MODEL_NAME, ALGORITHM from user_mining_models WHERE
MODEL_NAME='YOUR_PRECONFIG_MODEL_NAME';


MODEL_NAME ALGORITHM
-------------------------------------
YOUR_PRECONFIG_MODEL_NAME ONNX
```

`DBMS_VECTOR.LOAD_ONNX_MODEL` is only needed if `export2file` was used to save the ONNX model file to the local system instead of using `export2db` to save the model in the database. The `DBMS_VECTOR.LOAD_ONNX_MODEL` imports the ONNX format model into the Oracle Database to leverage the in-database ONNX Runtime to produce vector embeddings using the `VECTOR_EMBEDDING` SQL operator.

See Also

- *Oracle AI Database SQL Language Reference* for information about the `VECTOR_EMBEDDING` SQL function
- *Oracle AI Database PL/SQL Packages and Types Reference* for information about the `IMPORT_ONNX_MODEL` procedure
- *Oracle AI Database PL/SQL Packages and Types Reference* for information about the `LOAD_ONNX_MODEL` procedure
- *Oracle Machine Learning for SQL Concepts* for more information about importing pretrained embedding models in ONNX format and generating vector embeddings
- https://onnx.ai/onnx/intro/ for ONNX documentation

ONNX Pipeline Models: Image Embedding

ONNX pipeline models provides image embedding models that accepts image as an input and produces embeddings as output. The pipeline models also provide the necessary pre-processing.

Image Embedding Pipeline

1. **Input** : Input to the pipeline is an image. Each image is represented as an array of bytes. This can be a BLOB when using SQL or a similar in-memory byte representation like `io.ByetIO` in Python.
2. **Pre-Processing** : The image embedding pipeline utilizes an Image Processor in the pre-processing step. The Image Processor is dependent on the model configuration in the Hugging Face repository. When you provide a pretrained image model (e.g. "google/vit-base-patch16-224"), the pipeline builder will look up the Image Processor class from the model's configuration and determine if it's supported by looking up the Image Processor name in pipeline builder templates. If a match is found then the specific Image Processor is loaded and configured according to the template. The following table shows the relationship between pipeline builder templates and their corresponding Image Processor classes. Image Processor classes are provided by the transformer library.

   Note

   You do not need to reference these templates when using preconfigured image models. However any non-preconfigured model is template driven and these templates can be used for such models.

Template	Image Processor Class
Image_ViT	transformers.ViTImageProcessor
Image_ConvNext	transformers.ConvNextImageProcessor

Each Image Processor has several possible operations that will modify the input image in a specific way. Each of these operations is represented by a configuration in the Image Processor template. The modification of these operations and their respective configuration is not supported. You are able to view the configurations and should be aware of them. The following table shows the image operations and their respective configurations:

Table 4-1 Image Processor Configurations

Image Processor Operation	Description
Decode	Converts the compressed image to 3-D raster format.
Resize	Resizes the given image to the new shape.
Rescale	Rescales (element wise multiplication) an image with a numeric value.
Normalize	Normalizes (adjusts the intensity values of pixels to a desired range, often between 0 to 1) an image using the given mean and standard deviation (std) using the following formulation: *(image - mean) / std*
CenterCrop	Crops the image with a fixed size from the center.

> **Note**
>
> The Image pre-processors are considered to be fixed and modifications are not allowed. We are exposing these details for your information and understanding.

3. **Original Model**: Pre-trained pytorch model from Hugging Face repository. The repository also contain preconfigured models which have an existing specification. To get a list of all model names in the preconfigured list, you can use the `show_preconfigured` function. You can also create your own specification using the above mentioned templates as a starting point.
4. **Output** : The pipeline generates embeddings for each input image.

Supported Input for Image Embedding Models

Standalone baseline 8-bit lossy JPEG images are accepted as input for in-database image embedding generation. Image preprocessing or format conversion must be performed before passing the image as input to the pipeline.

The following cases are not supported:

Unsupported Input	Details
Non-JPEG or malformed input	Empty input, invalid magic, input that is too short, or truncated marker segments.
Progressive/interlaced JPEG	SOF2 marker 0xC2
Extended sequential JPEG	SOF1 marker 0xC1
Lossless JPEG	SOF3
Differential JPEG	SOF5-SOF7 and SOF13-SOF15
Arithmetic-coded JPEG	SOF9-SOF11, and arithmetic conditioning marker 0xCC

Unsupported Input	Details
Non-8-bit precision	SOF0 with 12-bit or 16-bit precision
Reserved JPEG extension	SOF marker 0xC8
Oversized decoded output	Width or height above 8192 pixels, or decoded output size above 512MB.
Oversized encoded input	Image size is limited to 20MB.

Unsupported inputs do not produce a successful, decoded image output. At SQL scoring time, this surfaces as a `NULL` embedding.

Image Embedding Examples

1. Exporting a pre-configured image model:

```
from oml.utils import ONNXPipeline
pipeline =ONNXPipeline("google/vit-base-patch16-224")
pipeline.export2file("your_preconfig_model_name")
```

   This example uses the `google/vit-base-patch16-224` model, which is a pre-configured model shipped in OML4Py 2.1. The pipeline is exported to a file which will result in a file **your_preconfig_model_name.onnx** in the current directory. This model can optionally be exported directly to the database using the pipeline.export2db function. Although the exported pipeline will just produce embeddings, not a classification output, it is possible to train an OML classification model to take these embeddings and produce the desired classification.

2. Exporting a non pre-configured model with a template:

   > **Note**
   >
   > Refer to this section for understanding the templates

```
from oml.utils import ONNXPipeline,ONNXPipelineConfig
import oml
config = ONNXPipelineConfig.from_template("Image_ViT")
pipeline = ONNXPipeline("nateraw/food",config=config)
oml.connect("pyquser","pyquser",dsn="pydsn")
oml.export2db("your_preconfig_model_name")
```

   In this example, since the `nateraw/food` model is not included as a preconfigured model in OML4Py 2.1, a template approach was chosen. Since this is a ViT based model, the ViT template is selected.

The process of loading a image model converted to a ONNX format is very similar to the ONNX format model generated from text. Refer to steps 6-9 from ONNX Pipeline Models : Text Embedding for understanding how to load a ONNX model and use it further.

ONNX Pipeline Models: CLIP Multi-Modal Embedding

ONNX pipeline models provides multi-modal embedding models that accepts both image and text as an input and produces embeddings as output. The pipeline models also provide the necessary pre-processing needed.

CLIP Multi-Modal Embedding Pipeline

CLIP models are multi-modal which allows you to perform embedding on both text and image input data. The main advantage of this model is to do image-text similarity. You can compare vectors related to text snippets and a given image, to understand which text can better describe given image. The pipeline generator will generate two ONNX pipeline models for a pretrained CLIP model, distinguished by their suffixes. The pipeline model for images is suffixed with **_img** and the model for text is suffixed with **_txt**. The same models with their suffixes will be loaded to the database when using *export2db*. For performing CLIP related tasks such as image-text similarity, both models will need to be used at inference time.

1. **Input**: CLIP models consist of two pipelines: an image embedding pipeline and a text embedding pipeline. The Image pipeline takes images as described in the Image Embedding Pipeline section of ONNX Pipeline Models: Image Embedding, and the text Pipeline takes text as described in ONNX Pipeline Models: Text Embedding (Text Embedding support was introduced in OML4Py 2.0).
2. **Pre-processing**: The image pipeline for CLIP models utilizes the same pre-processing strategy as described in the Image Embedding Pipeline section of ONNX Pipeline Models: Image Embedding. That is, an image processor that matches the model's configuration is utilized to prepare the images. The text pipeline utilizes a specific tokenizer: the **CLIPTokenizer (transformers.models.clip.CLIPTokenizer)**.
3. **Post-processing**: As a post-processing step, Normalization is added to both the image and text pipelines.
4. **Output**: Both models will produce vectors of the same shape that can then be compared using some similarity measure.

CLIP Multi-Modal Embedding Examples

1. Exporting a pre-configured image model to a file:
   The following example will produce two pipelines called **clip_img.onnx** and **clip_txt.onnx**, which can be used for image and text embeddings respectively.

   ```
   from oml.utils import ONNXPipeline
   pipeline = ONNXPipeline("openai/clip-vit-large-patch14")
   pipeline.export2file("clip")
   ```

2. Exporting a pre-configured image model to the database:
   This example will produce two in-database models called **clip_img** and **clip_txt**, which can be used for image and text embeddings respectively.

   ```
   fromoml.utils import ONNXPipeline
   import oml
   pipeline = ONNXPipeline("openai/clip-vit-large-patch14")
   oml.connect("pyquser","pyquser",dsn="pydsn")
   pipeline.export2db("clip")
   ```

3. Exporting a non pre-configured with a template to a file:

This example will work for clip models that are not preconfigured. It will create two files called **clip_14_img.onnx** and **clip_14_txt.onnx**.

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig
config = ONNXPipelineConfig.from_template("multimodal_clip")
pipeline = ONNXPipeline("openai/clip-vit-base-patch16",config=config)
pipeline.export2file("clip_16")
```

For an end to end example of using CLIP to generate multi-modal embeddings, see Generate Multi-modal Embeddings Using CLIP.

ONNX Pipeline Models: Text Classification

ONNX pipeline models provides text classification models that accepts text strings as input and produces the probablity of the input string belonging to a specific label. The pipeline models also provide the necessary pre-processing and post-processing.

In addition to text embedding models, Hugging Face repository also hosts transformer models that can be used for text classification. These models are typically fine-tuned embedding models on which a classification "head" was appended. The head changes the output of the model from a vector of embeddings to a list of labels and probabilities. For text based classification tasks such as sentiment analysis, you can generate a Text Classification Pipeline using OML4Py 2.1.

Text Classification Pipeline

1. **Input**: Input to the text classification pipeline is provided in the form of a batch, or array, of 1 ore more text strings. Each text string provided in the input will correspond to an output list containing the labels and probabilities.
2. **Pre-Processing**: Similar to text embedding pipeline, the text classification pipeline also configures a tokenizer for tokenizing the text inputs. The following tokenizer classes are supported in OML4Py 2.1:

Table 4-2 Tokenizer Classes available for Text Classifiaction Pipeline

Tokenizer Class	Tokenizer Type
`transformers.models.bert.BertTokenizer`	BERT
`transformers.models.clip.CLIPTokenizer`	CLIP
`transformers.models.distilbert.DistilBertTokenizer`	BERT
`transformers.models.gpt2.GPT2Tokenizer`	GPT2
`transformers.models.mpnet.MPNetTokenizer`	BERT
`transformers.models.roberta.tokenization_roberta.RobertaTokenizer`	ROBERTA
`transformers.models.xlm_roberta.XLMRobertaTokenizer`	SENTENCEPIECE

> **Note**
>
> A tokenizer is automatically configured based on the tokenizer class configured for the model on Hugging Face. Tokenizer classes are provided by the transformer library. If the tokenizer class configured for the model in Hugging Face is not supported by OML4Py 2.1, an error will be raised.

3. **Original Model**: The original model must be a pre-trained PyTorch model in Hugging Face repository or from the local file system. Models on the local file system must match the Hugging Face format.
4. **Post-Processing**: The text classification pipeline provides a softmax function for post-processing by default. The softmax function will normalize the set of scores (logits) produced by the model into a probability distribution where each value is normalized to a range of [0,1]and all values sum to 1. You can choose to not include the softmax post-processing by providing an empty list of post-processors when generating the pipeline in OML4Py 2.1.
5. **Output**: The output of the classification pipeline is a list of probabilities for each input string. The length of the list is equal to the total number of classification targets or labels.

The pipeline generator assigns output labels based on the output indices. Currently, the mapping between model outputs and labels defaults to the order in which the outputs are produced.

> **Note**
>
> Label metadata does not become part of the ONNX model. It is only applied when exporting the model to the db with export2db.

Text Classification Pipeline Examples

In order to generate ONNX pipeline models for image classification, `MiningFunction` class is used from the python utilities. The `MiningFunction` class can take one of the three values: `EMBEDDING`, `CLASSIFICATION`, and `REGRESSION`. You can choose one depending on the task which you would like to perform. The `MiningFunction` enum is defined as follows :

```
class MiningFunction(Enum) :
    EMBEDDING = 1
    CLASSIFCATION = 2
    REGRESSION = 3
```

Since you are working on text classification pipeline, you would choose `CLASSIFICATION` or `2` (Enum) for the class `MiningFunction`

> **Warning**
>
> EmbeddingModel and EmbeddingModelConfig are deprecated. Instead, please use ONNXPipeline and ONNXPipelineConfig respectively. The details of the deprecated classes can be found in Python Classes to Convert Pretrained Models to ONNX Models (Deprecated). If a you choose to use a deprecated class, a warning message will be shown indicating that the classes will be removed in the future and advising the user to switch to the new class.

1. Example for generating a text pipeline with a template:

```
from oml.utils import ONNXPipeline,ONNXPipelineConfig,MiningFunction
config = ONNXPipelineConfig.from_template("text",max_seq_length=512)
pipeline = ONNXPipeline("SamLowe/roberta-base-
go_emotions",config=config,function=MiningFunction.CLASSIFICATION)
pipeline.export2file("emotions","testouput")
```

2. Importing the text classification pipeline generated in the above example in the DB (SQL)

> **Note**
>
> The following code assumes that you have created a directory in PL/SQL named 'ONNX_IMPORT'

```
BEGIN
    DBMS_VECTOR.LOAD_ONNX_MODEL(
    'ONNX_IMPORT',
    'emotions.onnx',
    'emotions',

JSON('{"function":"classification","classificationProbOutput":"logits","inp
ut":{"input":["DATA"]},
        "labels":
["admiration","amusement","anger","annoyance","approval","caring","confusio
n","curiosity",

"desire","disappointment","disapproval","disgust","embarrassment","exciteme
nt","fear","gratitude",

"grief","joy","love","nervousness","optimism","pride","realization","relief
","remorse","sadness","surprise","neut"]}')
    );
END;
```

In this example the label metadata is being applied when invoking the `DBMS_VECTOR.LOAD_ONNX_MODEL` function to import the ONNX pipeline into the database. Please recall that the "labels" property is part of the JSON argument to the function and not a part of the ONNX file. The labels must be provided separately (if not provided only the numerical indices of the output will be used). When exporting directly to the database from OML4Py using the `export2db` function, you can either rely on the default label metadata provided in the model's configuration on Hugging Face, or provide your own label metadata via a template argument.

3. Example for exporting a text classification pipeline with default label metadata:

```
from oml.utils importONNXPipeline,ONNXPipelineConfig,MiningFunction
import oml
config =
ONNXPipelineConfig.from_template("text",max_seq_length=512,labels=["admirat
ion","amusement","anger","annoyance","approval","caring",

"confusion","curiosity","desire","disappointment","disapproval","disgust","
```

```
embarrassment","excitement",

"fear","gratitude","grief","joy","love","nervousness","optimism","pride","r
ealization","relief",

"remorse","sadness","surprise","neut"])
pipeline = ONNXPipeline("SamLowe/roberta-base-
go_emotions",config=config,function=MiningFunction.CLASSIFICATION)
oml.connect("pyquser","pyquser",dsn="pydsn")
pipeline.export2db("emotions")
```

In this example the pipeline is exported with label metadata that you provided in the template. This example is a combination of the examples provided in the previous two steps.

4. Example for scoring a text classification pipeline:

```
select prediction(emotions using 'Today is a good day' as data),
         prediction_probability(emotions using 'Today is a good day'as
data) from dual;
```

ONNX Pipeline Models: Reranking Pipeline

ONNX pipeline models provide a reranking pipeline that calculates similarity score for a given pair of texts.

Reranking Pipeline

Reranking models (also known as cross-encoders or rerankers) calculate a similarity score for given pairs of texts. Instead of encoding the input text into a fixed-length vector as text embedding models do, rerankers encode two input texts simultaneously and produces a similarity score for the pair. By default it outputs a logit which can be converted to a number between 0 and 1 by applying a sigmoid activation function on it. Although you can compute similarity score between two texts through embedding models by first computing the embeddings of the texts and then applying cosine similarity, the re-ranker models usually provide superior performance and they can be used to re-rank the top-k results from an embedding model.

1. **Input**: There are two inputs to the reranking pipeline named `first_input` and `second_input`. Each contains an array of one or more text strings. The two inputs should have equal number of text strings. An array will be produced for each pair of texts from `first_input` and`second_input`. For example, for the following input pairs two arrays are produced, one for the pair`'hi'` and `'hi'` and the other for the pair `'halloween'` and `'hello'`:
   `{'first_input':['hi','halloween'],'second_input':['hi','hello']}`
2. **Pre-Processing**:
   Pre-processing for reranking pipeline includes tokenization. The tokenizer class: `transformers.models.xlm_roberta.tokenization_xlm_roberta.XLMRobertaTokenizer` is supported in OML4Py 2.1 for reranking models.
3. **Output**: The output of a reranking pipeline is an array of similarity scores, one for each input text pair. For the above example in the Input section, the output similarity score is 9.290878 for the pair `'hi'` and `'hi'` and 4.3913193 for the pair `'halloween'` and `'hello'`.

Reranking Pipeline Examples

1. Generating a Reranking Pipeline:

```
from oml.utils import ONNXPipeline, ONNXPipelineConfig
from oml import MiningFunction, ONNXPipeline
mn = 'BAAI/bge-reranker-base'
em = ONNXPipeline(mn, function=MiningFunction.REGRESSION)
mf = 'bge-reranker-base'
em.export2file(mf)
```

2. Importing the reranker model to the database:

```
BEGIN
DBMS_VECTOR.LOAD_ONNX_MODEL('DM_DUMP','bge-reranker-
base.onnx','doc_model', JSON('{"function" : "regression",
                            "regressionOutput" : "output", "input":
{"first_input": ["DATA1"],"second_input": ["DATA2"]}}'));
END;
```

3. Obtain the similarity score in the database:

```
SELECT prediction(doc_model USING 'what is panda?' as DATA1, 'hi' as
DATA2) from dual;
```

The above example produces a similarity score between two text strings using the re-ranker ONNX pipeline imported in step 2. The score range from -infinity to +infinity. Score close to +infinity indicates a high similarity between the two strings, and score closer to -infinity indicate low similarity between the strings. A sigmoid function can map the similarity score to a float value in the range [0,1]. A sample output score corresponding to the above example is shown here:

```
PREDICTION(DOC_MODELUSING'WHATISPANDA?'ASDATA1,'HI'ASDATA2)
-----------------------------------------------------------
                    -7.73E+000
```

Load Custom Models from The Local Filesystem

Custom models that have been fine-tuned from pre-trained Hugging Face models and reside on the local filesystem can be converted to the ONNX format using the new `local_model_path` setting. This setting can also be useful in cases where networking is unavailable and a pre-trained model has previously been downloaded.

The model directory for a custom model must follow the following format:

```
<basedir>/<model_file_subfolders>
```

- basedir: This is the base directory for the model sub folders and is passed as the `local_model_path` settings parameter. This parameter is described in the Generate the Augmented ONNX File section. It can be an absolute or relative path. If relative, it will be relative to where Python was started.

- model_file_subfolders: These nested folders are created based on the model's name. For each '/' character in the model's name, you should create a new subdirectory.

Example 4-1 Local Directory for Custom Model Files

For example, if the model name is `mymodel`, only one directory named `mymodel` is expected. However, if the model name is `user/mymodel`, you should create a parent directory called `user` with a subdirectory named `mymodel`. All model files should be placed in the final subdirectory (`mymodel` in this case).

```
user
    mymodel
```

Generate the Augmented ONNX File

When initializing the EmbeddingModel object, use the settings parameter to pass a new property `local_model_path`, which identifies the `basedir` for the model. For example, if the `basedir` for the model on the local filesystem is `$HOME/sample-model`, and there is a custom model named `my-model`, you can achieve this with the following Python code:

```
from oml.utils import EmbeddingModel
import os

em = EmbeddingModel('my-
model',settings={'local_model_path':f'{os.environ["HOME"]}/sample-model'})
em.export2file('my-onnx-model','.')
```

At the end of this process, you should find a file named `my-onnx-model.onnx` in the current directory.

> **Note**
>
> The above code is the same for exporting to the database except that export2db is used instead of export2file. For more information, see ONNX Pipeline Models: Text Embedding. The code is compatible with all downloaded pretrained models, preconfigured models, and templates.

Error and Warning Conditions

Table 4-3 Error and Warning Conditions

Error (or Warning) Conditions	Error (or Warning) Message	Notes
force_download specified	force_download cannot be true when specifying 'local_model_path'	force_download is mutually exclusive with this feature so it cannot be specified when local_model_path has a value.
local_model_path is not a directory	'{local_path}' specified by 'local_model_path' must be an existing directory.	`{local_path}` is the value specified in `local_model_path` . This message is returned when the value of local_model_path is not an existing directory.

Table 4-3 (Cont.) Error and Warning Conditions

Error (or Warning) Conditions	Error (or Warning) Message	Notes
Missing model config file	Expected local model folder {model_path} to contain the file config.json but it was not found.	`{model_path}` is the full path to the local model folder. This message is returned when the local model folder does not contain a file named config.json.
Missing model files	Expected local model folder {model_path} to at least contain one of model.safetensors or pytorch_model.bin files. Neither were found.	`{model_path}` is the full path to the local model folder. This message is returned when the local model folder does not contain either a pytorch_mode.bin file OR at least one file that does not end in .safetensors.
preconfigured model used (Warning)	Warning: checksum validation is disabled when loading preconfigured models from a local path.	This warning is shown when a preconfigured model is loaded from a local path.

Support For Large ONNX Format Model

OML4Py supports large ONNX format models outside of the database. A large model is divided into a single ONNX file and one or more external data files. The maximum size limit for an ONNX file is 2GB, so for models larger than 2GB, the ONNX file must be split into smaller parts.

You can leverage ONNXPipelines to utilize external data. External data can be applied in two ways:

- If model size is less than 1 GB: If the model size is less than 1 GB, manually activate the use of external data, set the parameter `use_external_data` to True. It can be used with both preconfigured models and non-configured models.
  - By using a preconfigured model.

    ```
    from oml.utils import ONNXPipeline,ONNXPipelineConfig
    config = ONNXPipelineConfig.from_template("text", max_seq_length=512,
    use_external_data=True)
    pipeline = ONNXPipeline("Snowflake/snowflake-arctic-embed-s",config)
    ```

  - If the model is not configured, you can use templates.

    ```
    from oml.utils import ONNXPipeline, ONNXPipelineConfig
    config = ONNXPipelineConfig.from_template("text", max_seq_length=512,
    use_external_data=True)
    pipeline = ONNXPipeline("Snowflake/snowflake-arctic-embed-s", config)
    ```

- If model size exceeds 1 GB: Loading a model of size greter than 1 GB without quantization enabled will default to using external data. For models greater than 2 GB with quantization enabled (`quantize_model=True`), will result in an error, instead of activating the external data. The following example, which uses a large multilingual-e5 model, will default to using the external data:

```
from oml.utils import ONNXPipeline,ONNXPipelineConfig
config = ONNXPipelineConfig.from_template("text", max_seq_length=512)
pipeline = ONNXPipeline("intfloat/multilingual-e5-large", config)
pipeline.export2db("multilingual")
```

> **Note**
>
> You can always manually activate the use of external data by setting the parameter `use_external_data=True` for models of any size.

The `export2file` function will create a zip file instead of a single ONNX file. This zip file will contain:

- An ONNX file: This file corresponds to the model graph.
- One or more .data files: These files will contain the tensors and any other external data associated with the model.
- A JSON file: This file details which tensor belongs to a specific location on each .data file.

An example of such a zip file would be as follows:

- large_model.onnx
- large_model_data_0.data, large_model_data_1.data
- large_model_external_data.json

Model size support in ONNX pipeline API:

- Less than 1 GB: Supported with optimal performance.
- 1 GB to 2 GB: If quantization is enabled, full support is provided; otherwise, the model is exported using external initializers.
- 0.4 GB to 2 GB (without quantization): A warning is displayed, recommending quantization to improve performance.
- Greater than 2 GB: Models over 2 GB are supported only with quantization disabled, using external data. With quantization enabled, the limit is 2 GB.

Model Summary

A model summary provides a concise overview of the model. To obtain the summary after exporting the model, call the `summary()` function.

The model summary can be retrieved from ONNX pipeline by calling the `summary()` function after executing either `export2db()` or `export2file()`. The summary includes details such as:

- Whether the model was quantized
- Whether external data was used
- The tokenizer type, pre-processors and post-processors applied

- Other relevant metadata

```
from oml.utils import ONNXPipeline,ONNXPipelineConfig
model_name="sentence-transformers/all-MiniLM-L6-v2"
em = ONNXPipeline(model_name=model_name,settings={"logging_level":"ERROR"})
em.export2db("allminil6v2")
em.summary()


{'description': "Original model sentence-transformers/all-MiniLM-L6-v2
augmented with pre/post processors. Generated by OML4Py ver. 2.1.
Generated model is not quantized. Generated model doesn't use external data.
Model tokenizer is <class
'transformers.models.bert.tokenization_bert.BertTokenizer'>.
Model post processors include: mean_pooling,mean_pooling.", 'producer':
'OML4Py 2.1'}
```

If the model was exported to the database, the summary is also included in the view storing ONNX metadata information:

```
SQL> select * from DM$VJALLMINIL6V2;

METADATA
------------------------------------------------------------------------------
--
{"function":"embedding","embeddingOutput":"embedding","input":{"input":
["DATA"]}
,"pooling":"mean","normalization":true,"maxSequenceLength":256,"suitableDistan
ce
Metrics":["COSINE","DOT","EUCLIDEAN"],"languages":["us"],"modelDescription":
{"de
scription":"Original model sentence-transformers/all-MiniLM-L6-v2 augmented
with
 pre/post processors. Generated by OML4Py ver. 2.1. Generated model is not
quant
ized. Generated model doesn't use external data. Model tokenizer is <class
'tran
sformers.models.bert.tokenization_bert.BertTokenizer'>. Model post processors
in
clude: mean_pooling,mean_pooling.","producer":"OML4Py 2.1"}}
```

Import ONNX Models into Oracle AI Database End-to-End Example

Learn to import a pretrained embedding model that is in ONNX format and generate vector embeddings.

Follow the steps below to import a pretrained ONNX formatted embedding model into the Oracle AI Database.

Prepare Your Data Dump Directory

Prepare your data dump directory and provide the necessary access and privileges to `dmuser`.

1. Choose from:

   a. If you already have a pretrained ONNX embedding model, store it in your working folder.
   b. If you do not have pretrained embedding model in ONNX format, perform the steps listed in Convert Pretrained Models to ONNX Format.

2. Login to SQL*Plus as SYSDBA in your PDB.

```
CONN sys/<password>@pdb as sysdba;
```

3. Grant the DB_DEVELOPER_ROLE to dmuser.

```
GRANT DB_DEVELOPER_ROLE TO dmuser identified by <password>;
```

4. Grant CREATE MINING MODEL privilege to dmuser.

```
GRANT create mining model TO dmuser;
```

5. Set your working folder as the data dump directory (DM_DUMP) to load the ONNX embedding model.

```
CREATE OR REPLACE DIRECTORY DM_DUMP as '<work directory path>';
```

6. Grant READ permissions on the DM_DUMP directory to dmuser.

```
GRANT READ ON DIRECTORY dm_dump TO dmuser;
```

7. Grant WRITE permissions on the DM_DUMP directory to dmuser.

```
GRANT WRITE ON DIRECTORY dm_dump TO dmuser;
```

8. Drop the model if it already exits.

```
exec DBMS_VECTOR.DROP_ONNX_MODEL(model_name => 'doc_model', force => true);
```

Import ONNX Model Into the Database

You created a data dump directory and now you load the ONNX model into the Database. Use the DBMS_VECTOR.LOAD_ONNX_MODEL procedure to load the model. The DBMS_VECTOR.LOAD_ONNX_MODEL procedure facilitates the process of importing ONNX format model into the Oracle AI Database. In this example, the procedure loads an ONNX model file, named my_embedding_model.onnx from the DM_DUMP directory, into the Database as doc_model, specifying its use for embedding tasks.

1. Connect as dmuser.

```
CONN dmuser/<password>@<pdbname>;
```

2. Load the ONNX model into the Database.

   If the ONNX model to be imported already includes an output tensor named embeddingOutput and an input string tensor named data, JSON metadata is unnecessary.

Embedding models converted from OML4Py follow this convention and can be imported without the JSON metadata.

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
  'DM_DUMP',
  'my_embedding_model.onnx',
  'doc_model');
```

Alternately, you can load the ONNX embedding model by specifying the JSON metadata.

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
  'DM_DUMP',
  'my_embedding_model.onnx',
  'doc_model',
  JSON('{"function" : "embedding", "embeddingOutput" : "embedding", "input": {"input":
["DATA"]}}'));
```

The procedure `LOAD_ONNX_MODEL` declares these parameters:

- `DM_DUMP`: specifies the directory name of the data dump.

> **Note**
>
> Ensure that the `DM_DUMP` directory is defined.

- `my_embedding_model`: is a `VARCHAR2` type parameter that specifies the name of the ONNX model.
- `doc_model`: This parameter is a user-specified name under which the model is stored in the Oracle AI Database.
- The JSON metadata associated with the ONNX model is declared as:

  `"function" : "embedding"`: Indicates the function name for text embedding model.

  `"embeddingOutput" : "embedding"`: Specifies the output variable which contains the embedding results.
- `"input": {"input": ["DATA"]}`: Specifies a JSON object (`"input"`) that describes the input expected by the model. It specifies that there is an input named `"input"`, and its value should be an array with one element, `"DATA"`. This indicates that the model expects a single string input to generate embeddings.

For more information about the `LOAD_ONNX_MODEL` procedure, see *Oracle AI Database PL/SQL Packages and Types Reference*.

Alternatively, if your ONNX embedding model is loaded on cloud object storage, the `LOAD_ONNX_MODEL_CLOUD` procedure can be used. For more information, see *Oracle AI Database PL/SQL Packages and Types Reference*.

Query Model Statistics

You can view model attributes and learn about the model by querying machine learning dictionary views and model detail views.

> **Note**
>
> *DOC_MODEL* is the user-specified name of the embedding text model.

1. Query `USER_MINING_MODEL_ATTRIBUTES` view.

   ```
   SELECT model_name, attribute_name, attribute_type, data_type, vector_info
   FROM user_mining_model_attributes
   WHERE model_name = 'DOC_MODEL'
   ORDER BY ATTRIBUTE_NAME;
   ```

   To learn about `USER_MINING_MODEL_ATTRIBUTES` view, see USER_MINING_MODEL_ATTRIBUTES.

2. Query `USER_MINING_MODELS` view.

   ```
   SELECT MODEL_NAME, MINING_FUNCTION, ALGORITHM,
   ALGORITHM_TYPE, MODEL_SIZE
   FROM user_mining_models
   WHERE model_name = 'DOC_MODEL'
   ORDER BY MODEL_NAME;
   ```

   To learn about `USER_MINING_MODELS` view, see USER_MINING_MODELS.

3. Check model statistics by viewing the model detail views. Query the `DM$VM`*`DOC_MODEL`* view.

   ```
   SELECT * FROM DM$VMDOC_MODEL ORDER BY NAME;
   ```

   To learn about model details views for ONNX embedding models, see Model Details Views for ONNX Models.

4. Query the `DM$VP`*`DOC_MODEL`* model detail view.

   ```
   SELECT * FROM DM$VPDOC_MODEL ORDER BY NAME;
   ```

5. Query the `DM$VJ`*`DOC_MODEL`* model detail view.

   ```
   SELECT * FROM DM$VJDOC_MODEL;
   ```

Generate Embeddings

Apply the model and generate vector embeddings for your input. Here, the input is *hello*.

Generate vector embeddings using the `VECTOR_EMBEDDING` function.

```
SELECT VECTOR_EMBEDDING(doc_model USING 'hello' as data) AS embedding;
```

To learn about the `VECTOR_EMBEDDING` SQL function, see VECTOR_EMBEDDING. You can use the `UTL_TO_EMBEDDING` function in the `DBMS_VECTOR_CHAIN` PL/SQL package to generate vector embeddings generically through REST endpoints. To explore these functions, see the example Convert Text String to Embedding.

Example: Importing a Pretrained ONNX Model to Oracle AI Database

The following presents a comprehensive step-by-step example of importing ONNX embedding and generating vector embeddings.

```
conn sys/<password>@pdbname as sysdba
grant db_developer_role to dmuser identified by <password>;
grant create mining model to dmuser;

create or replace directory DM_DUMP as '<work directory path>';
grant read on directory dm_dump to dmuser;
grant write on directory dm_dump to dmuser;
>conn dmuser/<password>@<pdbname>;

-- Drop the model if it exits
exec DBMS_VECTOR.DROP_ONNX_MODEL(model_name => 'doc_model', force => true);

-- Load Model
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
    'DM_DUMP',
    'my_embedding_model.onnx',
    'doc_model',
    JSON('{"function" : "embedding", "embeddingOutput" : "embedding"}'));
/

--check the attributes view
set linesize 120
col model_name format a20
col algorithm_name format a20
col algorithm format a20
col attribute_name format a20
col attribute_type format a20
col data_type format a20

SQL> SELECT model_name, attribute_name, attribute_type, data_type, vector_info
FROM user_mining_model_attributes
WHERE model_name = 'DOC_MODEL'
ORDER BY ATTRIBUTE_NAME;


OUTPUT:

MODEL_NAME           ATTRIBUTE_NAME       ATTRIBUTE_TYPE       DATA_TYPE
VECTOR_INFO
-------------------- -------------------- -------------------- ----------
---------------
DOC_MODEL                INPUT_STRING         TEXT                 VARCHAR2
DOC_MODEL                ORA$ONNXTARGET       VECTOR               VECTOR
VECTOR(128,FLOA
                                                                          T32)



SQL> SELECT MODEL_NAME, MINING_FUNCTION, ALGORITHM,
ALGORITHM_TYPE, MODEL_SIZE
```

```
FROM user_mining_models
WHERE model_name = 'DOC_MODEL'
ORDER BY MODEL_NAME;

OUTPUT:
MODEL_NAME           MINING_FUNCTION                ALGORITHM
ALGORITHM_ MODEL_SIZE
-------------------- ------------------------------ --------------------
---------- ----------
DOC_MODEL                EMBEDDING                          ONNX
NATIVE       17762137



SQL> select * from DM$VMDOC_MODEL ORDER BY NAME;

OUTPUT:
NAME                                     VALUE
----------------------------------------
----------------------------------------
Graph Description                        Graph combining g_8_torch_jit and
torch_
                                         jit
                                         g_8_torch_jit



                                         torch_jit


Graph Name                               g_8_torch_jit_torch_jit
Input[0]                                 input:string[1]
Output[0]                                embedding:float32[?,128]
Producer Name                            onnx.compose.merge_models
Version                                  1

6 rows selected.


SQL> select * from DM$VPDOC_MODEL ORDER BY NAME;

OUTPUT:
NAME                                     VALUE
----------------------------------------
----------------------------------------
batching                                 False
embeddingOutput                          embedding


SQL> select * from DM$VJDOC_MODEL;

OUTPUT:
METADATA
------------------------------------------------------------------------------
--
{"function":"embedding","embeddingOutput":"embedding","input":{"input":
```

```
["DATA"]}}



--apply the model
SQL> SELECT VECTOR_EMBEDDING(doc_model USING 'hello' as data) AS embedding;

-----------------------------------------------------------------------------
--
[-9.76553112E-002,-9.89954844E-002,7.69771636E-003,-4.16760892E-003,-9.6930563
4E-002,
-3.01141385E-002,-2.63396613E-002,-2.98553891E-002,5.96499592E-002,4.13885899E
-002,
5.32859489E-002,6.57707453E-002,-1.47056757E-002,-4.18472625E-002,4.1588001E-0
02,
-2.86354572E-002,-7.56499246E-002,-4.16395674E-003,-1.52879998E-001,6.60010576
E-002,
-3.9013084E-002,3.15719917E-002,1.2428958E-002,-2.47651711E-002,-1.16851285E-0
01,
-7.82847106E-002,3.34323719E-002,8.03267583E-002,1.70483496E-002,-5.42407483E-
002,
6.54291287E-002,-4.81935125E-003,6.11041225E-002,6.64106477E-003,-5.47
```

Oracle AI Vector Search SQL Scenario

To learn how you can chunk *database-concepts26ai.pdf* and *oracle-ai-vector-search-users-guide.pdf*, generate vector embeddings, and perform similarity search using vector indexes, see Quick Start SQL.

- Alternate Method to Import ONNX Models
  Use the `DBMS_DATA_MINING.IMPORT_ONNX_MODEL` procedure to import the model and declare the input name. A PL/SQL helper block is used to facilitate the process of importing the ONNX format model into the Oracle AI Database in the included example.

Alternate Method to Import ONNX Models

Use the `DBMS_DATA_MINING.IMPORT_ONNX_MODEL` procedure to import the model and declare the input name. A PL/SQL helper block is used to facilitate the process of importing the ONNX format model into the Oracle AI Database in the included example.

Perform the following steps to import ONNX model into the Database using `DBMS_DATA_MINING` PL/SQL package.

- Connect as `dmuser`.

  ```
  CONN dmuser/<password>@<pdbname>;
  ```

- Run the following helper PL/SQL block:

  ```
  DECLARE
      m_blob BLOB default empty_blob();
      m_src_loc BFILE ;
      BEGIN
      DBMS_LOB.createtemporary (m_blob, FALSE);
      m_src_loc := BFILENAME('DM_DUMP', 'my_embedding_model.onnx');
      DBMS_LOB.fileopen (m_src_loc, DBMS_LOB.file_readonly);
  ```

```
    DBMS_LOB.loadfromfile (m_blob, m_src_loc, DBMS_LOB.getlength
(m_src_loc));
    DBMS_LOB.CLOSE(m_src_loc);
    DBMS_DATA_MINING.import_onnx_model ('doc_model', m_blob,
JSON('{"function" : "embedding", "embeddingOutput" : "embedding", "input":
{"input": ["DATA"]}}'));
    DBMS_LOB.freetemporary (m_blob);
    END;
    /
```

The code sets up a BLOB object and a BFILE locator, creates a temporary BLOB for storing the my_embedding_model.onnx file from the DM_DUMP directory, and reads its contents into the BLOB. It then closes the file and uses the content to import an ONNX model into the database with specified metadata, before releasing the temporary BLOB resources.

The schema of the IMPORT_ONNX_MODEL procedure is as follows: DBMS_DATA_MINING.IMPORT_ONNX_MODEL(model_data, model_name, metadata). This procedure loads IMPORT_ONNX_MODEL from the DBMS_DATA_MINING package to import the ONNX model into the Database using the name provided in model_name, the BLOB content in m_blob, and the associated metadata.

- doc_model: This parameter is a user-specified name under which the imported model is stored in the Oracle AI Database.
- m_blob: This is a model data in BLOB that holds the ONNX representation of the model.
- "function" : "embedding": Indicates the function name for text embedding model.
- "embeddingOutput" : "embedding": Specifies the output variable which contains the embedding results.
- "input": {"input": ["DATA"]}: Specifies a JSON object ("input") that describes the input expected by the model. It specifies that there is an input named "input", and its value should be an array with one element, "DATA". This indicates that the model expects a single string input to generate embeddings.

Alternately, the DBMS_DATA_MINING.IMPORT_ONNX_MODEL procedure can also accept a BLOB argument representing an ONNX file stored and loaded from OCI Object Storage. The following is an example to load an ONNX model stored in an OCI Object Storage.

```
DECLARE
  model_source BLOB := NULL;
BEGIN
  -- get BLOB holding onnx model
  model_source := DBMS_CLOUD.GET_OBJECT(
    credential_name => 'myCredential',
    object_uri => 'https://objectstorage.us-phoenix -1.oraclecloud.com/' ||
      'n/namespace -string/b/bucketname/o/myONNXmodel.onnx');

  DBMS_DATA_MINING.IMPORT_ONNX_MODEL(
    "myonnxmodel",
    model_source,
    JSON('{ function : "embedding" })
  );
END;
/
```

This PL/SQL block starts by initializing a `model_source` variable as a `BLOB` type, initially set to NULL. It then retrieves an ONNX model from Oracle Cloud Object Storage using the `DBMS_CLOUD.GET_OBJECT` procedure, specifying the credentials (`OBJ_STORE_CRED`) and the URI of the model. The ONNX model resides in a specific bucket named `bucketname` in this case, and is accessible through the provided URL. Then, the script loads the ONNX model into the `model_source` BLOB. The `DBMS_DATA_MINING.IMPORT_ONNX_MODEL` procedure then imports this model into the Oracle AI Database as `myonnxmodel`. During the import, a JSON metadata specifies the model's function as `embedding`, for embedding operations.

See IMPORT_ONNX_MODEL Procedure and GET_OBJECT Procedure and Function to learn about the PL/SQL procedure.

Example: Importing a Pretrained ONNX Model to Oracle AI Database

The following presents a comprehensive step-by-step example of importing ONNX embedding and generating vector embeddings.

```
conn sys/<password>@pdb as sysdba
grant db_developer_role to dmuser identified by dmuser;
grant create mining model to dmuser;

create or replace directory DM_DUMP as '<work directory path>';
grant read on directory dm_dump to dmuser;
grant write on directory dm_dump to dmuser;
>conn dmuser/<password>@<pdbname>;

-- Drop the model if it exits
exec DBMS_VECTOR.DROP_ONNX_MODEL(model_name => 'doc_model', force => true);

-- Load Model
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
    'DM_DUMP',
    'my_embedding_model.onnx',
    'doc_model',
    JSON('{"function" : "embedding", "embeddingOutput" : "embedding"}'));
/
--Alternately, load the model
EXECUTE DBMS_DATA_MINING.IMPORT_ONNX_MODEL(
       'my_embedding_model.onnx',
    'doc_model',
    JSON('{"function" : "embedding",
    "embeddingOutput" : "embedding",
    "input": {"input": ["DATA"]}}')
    );

--check the attributes view
set linesize 120
col model_name format a20
col algorithm_name format a20
col algorithm format a20
col attribute_name format a20
col attribute_type format a20
col data_type format a20

SQL> SELECT model_name, attribute_name, attribute_type, data_type, vector_info
FROM user_mining_model_attributes
```

```
WHERE model_name = 'DOC_MODEL'
ORDER BY ATTRIBUTE_NAME;


OUTPUT:

MODEL_NAME           ATTRIBUTE_NAME       ATTRIBUTE_TYPE       DATA_TYPE
VECTOR_INFO
-------------------- -------------------- -------------------- ----------
---------------
DOC_MODEL                INPUT_STRING         TEXT                 VARCHAR2
DOC_MODEL                ORA$ONNXTARGET       VECTOR               VECTOR
VECTOR(128,FLOA
                                                                          T32)



SQL> SELECT MODEL_NAME, MINING_FUNCTION, ALGORITHM,
ALGORITHM_TYPE, MODEL_SIZE
FROM user_mining_models
WHERE model_name = 'DOC_MODEL'
ORDER BY MODEL_NAME;

OUTPUT:
MODEL_NAME           MINING_FUNCTION                ALGORITHM
ALGORITHM_ MODEL_SIZE
-------------------- ------------------------------ --------------------
---------- ----------
DOC_MODEL                EMBEDDING                          ONNX
NATIVE       17762137



SQL> select * from DM$VMDOC_MODEL ORDER BY NAME;

OUTPUT:
NAME                                     VALUE
----------------------------------------
----------------------------------------
Graph Description                        Graph combining g_8_torch_jit and
torch_
                                         jit
                                         g_8_torch_jit



                                         torch_jit


Graph Name                               g_8_torch_jit_torch_jit
Input[0]                                 input:string[1]
Output[0]                                embedding:float32[?,128]
Producer Name                            onnx.compose.merge_models
Version                                  1

6 rows selected.
```

```
SQL> select * from DM$VPDOC_MODEL ORDER BY NAME;

OUTPUT:
NAME                                       VALUE
----------------------------------------
----------------------------------------
batching                                   False
embeddingOutput                            embedding


SQL> select * from DM$VJDOC_MODEL;

OUTPUT:
METADATA
------------------------------------------------------------------------------
--
{"function":"embedding","embeddingOutput":"embedding","input":{"input":
["DATA"]}}



--apply the model
SQL> SELECT TO_VECTOR(VECTOR_EMBEDDING(doc_model USING 'hello' as data)) AS
embedding;

------------------------------------------------------------------------------
--
[-9.76553112E-002,-9.89954844E-002,7.69771636E-003,-4.16760892E-003,-9.6930563
4E-002,
-3.01141385E-002,-2.63396613E-002,-2.98553891E-002,5.96499592E-002,4.13885899E
-002,
5.32859489E-002,6.57707453E-002,-1.47056757E-002,-4.18472625E-002,4.1588001E-0
02,
-2.86354572E-002,-7.56499246E-002,-4.16395674E-003,-1.52879998E-001,6.60010576
E-002,
-3.9013084E-002,3.15719917E-002,1.2428958E-002,-2.47651711E-002,-1.16851285E-0
01,
-7.82847106E-002,3.34323719E-002,8.03267583E-002,1.70483496E-002,-5.42407483E-
002,
6.54291287E-002,-4.81935125E-003,6.11041225E-002,6.64106477E-003,-5.47
```

Access Third-Party Models for Vector Generation Leveraging Third-Party REST APIs

You can access third-party vector embedding models to generate vector embeddings from your data, outside the database by calling third-party REST APIs.

The Vector Utility PL/SQL packages `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN` (outlined in Supplied Vector Utility PL/SQL Packages) provide the third-party REST APIs that let you interact with external embedding models.

You can remotely access third-party service providers, such as Cohere, Google AI, Hugging Face, Generative AI, OpenAI, and Vertex AI. Alternatively, you can install Ollama on your local host to access open LLMs, such as Llama 3, Phi 3, Mistral, and Gemma 2. All the supported providers and corresponding REST operations allowed for each provider are listed in Supported Third-Party Provider Operations and Endpoints.

Review these high-level steps involved in generating embeddings by calling third-party REST APIs:

1. **Understand the terms of using third-party embedding models**.

   > **Warning**
   >
   > Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
   >
   > Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

2. **Configure your REST API connection**.

   By configuring your REST API connection, you enable Oracle AI Database to communicate with the REST endpoint URL where you want to send requests to access data from the third-party embedding service. The requested embedding service then processes the data and returns a vector representation.

   This involves the following tasks:

   a. **Create a user, set up storage, and grant necessary privileges**.

      Create a tablespace and a user, and then grant the `DB_DEVELOPER_ROLE` to that user. This role assigns all basic roles and privileges that are necessary for a database developer.

      You can pass the input directly as a text string, or prepare a data dump directory to upload your data for vector generation. Ensure to grant the user access to read and write your data dump directory.

   b. **Grant the `HTTP` or `CONNECT` privilege to allow connection to the third-party host**.

      Grant the either the `HTTP` or `CONNECT` privilege to the user for connecting to your third-party host. The `HTTP` privilege provides a limited set of permissions, while the `CONNECT` privilege is broader and grants additional capabilities. You use the `DBMS_NETWORK_ACL_ADMIN` PL/SQL procedure (as described in DBMS_NETWORK_ACL_ADMIN) to specify the users and their privilege assignments that can access external network services from within the database.

      This procedure appends an access control entry (ACE) to the network access control list (ACL) for the specified host. The ACE grants a privilege to a principal (user name), which enables the user to connect to the external host that has been authorized in the database's networking ACL. This increases security by controlling the users that can connect to the specified hosts, and thus prevents unauthorized connections.

   c. **Set the proxy server, if configured**.

If you have configured a proxy server on your network, then you must specify the proxy server's host name and port number. This helps you access external web resources or embedding services by routing requests through your proxy server.

You use the `UTL_HTTP.SET_PROXY` PL/SQL procedure (as described in UTL_HTTP.SET_PROXY) to set the proxy server.

d. **Set up credentials to enable access to the REST provider**.

   You require authentication credentials to enable access to your chosen third-party service provider. A credential name holds authentication parameters, such as user name, password, access token, private key, or fingerprint.

   You use the credential helper procedures `CREATE_CREDENTIAL` and `DROP_CREDENTIAL` to securely manage credentials in the database. These procedures are available with both the `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN` PL/SQL packages.

3. **Generate vector embeddings**.

   You use chainable utility functions `UTL_TO_EMBEDDING` and `UTL_TO_EMBEDDINGS` to convert input data to one or more vector embeddings. These APIs are available with both the `DBMS_VECTOR` and the `DBMS_VECTOR_CHAIN` PL/SQL packages.

   Determine which API to use:

   - `UTL_TO_EMBEDDING` converts plain text (`CLOB`) or image (`BLOB`) to a single embedding (`VECTOR`).
   - `UTL_TO_EMBEDDINGS` converts an array of chunks (`VECTOR_ARRAY_T`) to an array of embeddings (`VECTOR_ARRAY_T`).

   For example, a typical `DBMS_VECTOR.UTL_TO_EMBEDDING` call specifies the REST provider, credential name, REST endpoint URL for an embedding service, and embedding model name parameters in a JSON object, as follows:

```
var params clob;
exec :params := '
{
 "provider": "cohere",
  "credential_name": "COHERE_CRED",
  "url": "https://api.cohere.example.com/embed",
  "model": "embed-model"
}';

select DBMS_VECTOR.UTL_TO_EMBEDDING('hello', json(:params)) from dual;
```

   In this example, the input for generating embedding is `hello`.

Run end-to-end embedding examples:

To see how to apply these steps for performing various embedding use cases, you can run some end-to-end example scenarios listed in Vector Generation Examples.

Related Topics

- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.

- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Vector Generation Examples

Run these end-to-end examples to see how you can generate vector embeddings, both within and outside the database.

- Generate Embeddings
  In these examples, you can see how to use the `VECTOR_EMBEDDING` SQL function or the `UTL_TO_EMBEDDING` PL/SQL function to generate a vector embedding from input text strings and images.
- Perform Chunking With Embedding
  In these examples, you can see how to explore the `VECTOR_CHUNKS` SQL function along with chainable utility PL/SQL functions to split large textual extracts and documents into chunks and then represent each chunk as a vector embedding.
- Configure Chunking Parameters
  Oracle AI Vector Search provides many parameters for chunking text data, such as `SPLIT [BY]`, `OVERLAP`, or `NORMALIZE`. In these examples, you can see how to configure these parameters to define your own chunking specifications and strategies, so that you can create meaningful chunks.

Generate Embeddings

In these examples, you can see how to use the `VECTOR_EMBEDDING` SQL function or the `UTL_TO_EMBEDDING` PL/SQL function to generate a vector embedding from input text strings and images.

- Convert Text String to Embedding Within Oracle AI Database
  Perform a text-to-embedding transformation by accessing a vector embedding model stored in the database.
- Convert Text String to BINARY Embedding Outside Oracle AI Database
  Perform a text-to-`BINARY`-embedding transformation by accessing a third-party `BINARY` vector embedding model.
- Convert Text String to Embedding Using Public REST Providers
  Perform a text-to-embedding transformation, using publicly hosted third-party embedding models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI.
- Convert Text String to Embedding Using the Local REST Provider Ollama
  Perform a text-to-embedding transformation by accessing open embedding models, using the local host REST endpoint provider Ollama.
- Convert Image to Embedding Using Public REST Providers
  Perform an image-to-embedding transformation by making a REST call to the third-party service provider, Vertex AI. In this example, you can see how to vectorize both image and text inputs using a multimodal embedding model and then query a vector space containing vectors from both content types.
- Generate Multi-modal Embeddings Using CLIP
  This section provides end-to-end instructions from installing the OML4Py client to generating multi-modal embeddings using CLIP.

- Vectorize Relational Tables Using OML Feature Extraction Algorithms

Convert Text String to Embedding Within Oracle AI Database

Perform a text-to-embedding transformation by accessing a vector embedding model stored in the database.

You can download an embedding machine learning model, convert it into ONNX format (if not already in ONNX format), and load the model into Oracle AI Database. You can then access that model to vectorize your data that is used to populate a vector index. Note that you must use the same embedding model on both the data to be indexed and the user's input query. In this example, you can see how to vectorize a user's input query on the fly.

Here, you can call either the `VECTOR_EMBEDDING` SQL function or the `UTL_TO_EMBEDDING` PL/SQL function (note the singular "embedding"). Both `VECTOR_EMBEDDING` and `UTL_TO_EMBEDDING` directly return a `VECTOR` type (not an array).

To convert a user's input text string "`hello`" to a vector embedding, using an embedding model in ONNX format:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

      ```
      conn sys/password as sysdba
      ```

      ```
      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      EXTENT MANAGEMENT LOCAL
      SEGMENT SPACE MANAGEMENT AUTO;
      ```

      ```
      SET ECHO ON
      SET FEEDBACK 1
      SET NUMWIDTH 10
      SET LINESIZE 80
      SET TRIMSPOOL ON
      SET TAB OFF
      SET PAGESIZE 10000
      SET LONG 10000
      ```

   b. Create a local user (`docuser`) and grant necessary privileges:

      ```
      DROP USER docuser cascade;
      ```

      ```
      CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
      unlimited on tbs1;
      ```

      ```
      GRANT DB_DEVELOPER_ROLE, create credential to docuser;
      ```

   c. Connect as the local user (`docuser`):

      ```
      CONN docuser/password
      ```

2. Call either VECTOR_EMBEDDING or UTL_TO_EMBEDDING.

   a. Load your ONNX format embedding model into Oracle AI Database.

      For detailed information on how to perform this step, see Import ONNX Models into Oracle AI Database End-to-End Example.

   b. Call VECTOR_EMBEDDING or UTL_TO_EMBEDDING. You can use UTL_TO_EMBEDDING either from the DBMS_VECTOR or the DBMS_VECTOR_CHAIN package, depending on your use case.

      - VECTOR_EMBEDDING:

```
SELECT VECTOR_EMBEDDING(doc_model USING 'hello' as data) AS 
embedding;
```

      - DBMS_VECTOR.UTL_TO_EMBEDDING:

```
var params clob;
exec :params := '{"provider":"database", "model":"doc_model"}';

select dbms_vector.utl_to_embedding('hello', json(:params)) from 
dual;
```

      Here, doc_model specifies the name under which your embedding model is stored in the database.

   The generated embedding appears as follows:

```
EMBEDDING
---------------------------------------------------------------------------
----------------------------------------------------
[8.78423732E-003,-4.29633334E-002,-5.93001908E-003,-4.65480909E-002,2.14333
013E-002,6.53376281E-002,-5.93746938E-002,2.10403297E-002,
4.38376889E-002,5.22960871E-002,1.25104953E-002,6.49512559E-002,-9.26998071
E-003,-6.97442219E-002,-3.02916039E-002,-4.74979728E-003,
-1.08755399E-002,-4.63751052E-003,3.62781435E-002,-9.35919806E-002,-1.13934
642E-002,-5.74270077E-002,-1.36667723E-002,2.42995787E-002,
-6.96804151E-002,4.93822657E-002,1.01460628E-002,-1.56464987E-002,-2.394105
68E-002,-4.27529104E-002,-5.65665103E-002,-1.74160264E-002,
5.05326502E-002,4.31500375E-002,-2.6994409E-002,-1.72731467E-002,9.30535868
E-002,6.85951149E-004,5.61876409E-003,-9.0233935E-003,
-2.55788807E-002,-2.04174276E-002,3.74175981E-002,-1.67872179E-002,1.074793
04E-001,-6.64602639E-003,-7.65537247E-002,-9.71965566E-002,
-3.99636962E-002,-2.57076006E-002,-5.62455431E-002,-1.3583754E-001,3.459460
29E-002,1.85191762E-002,3.01524661E-002,-2.62163244E-002,
-4.05582506E-003,1.72979087E-002,-3.66434865E-002,-1.72491539E-002,3.952284
16E-002,-1.05518714E-001,-1.27463877E-001,1.42578809E-002
```

Related Topics

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the DBMS_VECTOR.UTL_TO_EMBEDDING and DBMS_VECTOR.UTL_TO_EMBEDDINGS chainable utility functions to generate one or more vector embeddings from textual documents and images.
- VECTOR_EMBEDDING
  Use VECTOR_EMBEDDING to generate a single vector embedding for different data types using embedding or feature extraction machine learning models.

Convert Text String to BINARY Embedding Outside Oracle AI Database

Perform a text-to-BINARY-embedding transformation by accessing a third-party BINARY vector embedding model.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To generate a vector embedding with "hello" as the input using Cohere ubinary embed-english-v3.0 model:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the SYS user, connecting as SYSDBA:

      ```
      conn sys/password as sysdba


      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      EXTENT MANAGEMENT LOCAL
      SEGMENT SPACE MANAGEMENT AUTO;


      SET ECHO ON
      SET FEEDBACK 1
      SET NUMWIDTH 10
      SET LINESIZE 80
      SET TRIMSPOOL ON
      SET TAB OFF
      SET PAGESIZE 10000
      SET LONG 10000
      ```

   b. Create a local user (docuser) and grant necessary privileges:

      ```
      DROP USER docuser cascade;


      CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
      unlimited on tbs1;


      GRANT DB_DEVELOPER_ROLE, create credential to docuser;
      ```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to allow connection to the host.

   Grant connect privilege to docuser for connecting to the host, using the DBMS_NETWORK_ACL_ADMIN procedure. This example uses * to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Set up your credentials for the REST provider (in this case, Cohere) and then call UTL_TO_EMBEDDING.

   a. Run DBMS_VECTOR.CREATE_CREDENTIAL to create and store a credential.

   Cohere requires the following authentication parameter:

```
{ "access_token": "<access token>" }
```

   Replace <access token> with your own values. You will later refer to this credential name when declaring JSON parameters for the UTL_TO_EMBEDDING call.

```
EXEC DBMS_VECTOR.DROP_CREDENTIAL('COHERE_CRED');

DECLARE
  jo json_object_t;
BEGIN
  jo := json_object_t();
  jo.put('access_token', '<access token>');
  DBMS_VECTOR.CREATE_CREDENTIAL(
    credential_name   => 'COHERE_CRED',
    params            => json(jo.to_string));
END;
/
```

   b. Call DBMS_VECTOR.UTL_TO_EMBEDDING to generate the BINARY embedding.

> **Note**
>
> For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

```
var params clob;

BEGIN
:params := '
             {
              "provider": "cohere",
              "credential_name": "COHERE_CRED",
              "url": "https://api.cohere.ai/v1/embed",
              "model": "embed-english-v3.0",
              "input_type": "search_query",
              "embedding_types": ["ubinary"]
             }';
END;
/

SELECT TO_VECTOR(FROM_VECTOR(DBMS_VECTOR.UTL_TO_EMBEDDING('hello',
JSON(:params))),*,BINARY);
```

The generated BINARY embedding appears as follows:

```
TO_VECTOR(FROM_VECTOR(DBMS_VECTOR.UTL_TO_EMBEDDING('HELLO',JSON(:PARAMS))),
*,BIN
---------------------------------------------------------------------------
-----
[137,218,245,195,211,132,169,63,43,22,12,93,112,93,85,208,145,27,76,245,99,
222,1
21,63,1,161,200,24,1,30,202,233,208,2,113,27,119,78,123,192,132,115,187,146
,58,1
36,40,63,221,52,68,241,53,88,20,99,85,248,114,177,100,248,100,158,94,53,57,
97,18
2,129,14,64,173,236,107,109,37,195,173,49,128,113,204,183,158,55,139,10,205
,65,4
0,53,243,247,134,63,125,133,55,230,129,64,165,103,102,46,251,164,213,139,22
7,225
,66,98,112,100,64,145,98,80,97,192,149,77,43,114,146,197]
```

Related Topics

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the DBMS_VECTOR.UTL_TO_EMBEDDING and DBMS_VECTOR.UTL_TO_EMBEDDINGS chainable utility functions to generate one or more vector embeddings from textual documents and images.

Convert Text String to Embedding Using Public REST Providers

Perform a text-to-embedding transformation, using publicly hosted third-party embedding models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI.

You can use third-party embedding models to vectorize your data that is used to populate a vector index. Note that you must use the same embedding model on both the data to be indexed and the user's input query. In this example, you can see how to vectorize a user's input query on the fly.

Here, you can call the chainable utility function `UTL_TO_EMBEDDING` (note the singular "embedding") from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case. This example uses the `DBMS_VECTOR.UTL_TO_EMBEDDING` API.

`UTL_TO_EMBEDDING` directly returns a `VECTOR` type (not an array).

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To convert a user's input text "`hello`" to a vector embedding, using a public third-party embedding model:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `sys` user, connecting as `sysdba`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to docuser for allowing connection to the host, using the DBMS_NETWORK_ACL_ADMIN procedure.

   This example uses * to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Set up your credentials for the REST provider that you want to access and then call UTL_TO_EMBEDDING.

   - **Using Generative AI**:

     a. Run DBMS_VECTOR.CREATE_CREDENTIAL to create and store an OCI credential (OCI_CRED).

     Generative AI requires the following authentication parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
"fingerprint"     : "<fingerprint>"
}
```

     You will later refer to this credential name when declaring JSON parameters for the UTL_to_EMBEDDING call.

> **Note**
>
> The generated private key may appear as:
>
> ```
> -----BEGIN RSA PRIVATE KEY-----
> <private key string>
> -----END RSA PRIVATE KEY-----
> ```
>
> You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

```
exec dbms_vector.drop_credential('OCI_CRED');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('user_ocid','<user ocid>');
  jo.put('tenancy_ocid','<tenancy ocid>');
  jo.put('compartment_ocid','<compartment ocid>');
  jo.put('private_key','<private key>');
  jo.put('fingerprint','<fingerprint>');
  dbms_vector.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

Replace all the authentication parameter values. For example:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();

jo.put('user_ocid','ocid1.user.oc1..aabbalbbaa1112233aabbaabb1111222
aa1111bb');

jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaaaalbbbb1112233aaaabbaa1
111222aaa111a');

jo.put('compartment_ocid','ocid1.compartment.oc1..ababalabab1112233a
bababab1111222aba11ab');
  jo.put('private_key','AAAaaaBBB11112222333...AAA111AAABBB222aaa1a/
+');

jo.put('fingerprint','01:1a:a1:aa:12:a1:12:1a:ab:12:01:ab:a1:12:ab:1
a');
  dbms_vector.create_credential(
    credential_name   => 'OCI_CRED',
    parameters        => json(jo.to_string));
```

```
end;
/
```

b. Call `DBMS_VECTOR.UTL_TO_EMBEDDING`:

Here, the cohere.embed-english-v3.0 model is used. You can replace `model` with your own value, as required.

> **Note**
>
> For a list of all REST endpoint URLs and models that are supported to use with Generative AI, see Supported Third-Party Provider Operations and Endpoints.

```
-- select example

var params clob;
exec :params := '
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/embedText",
  "model": "cohere.embed-english-v3.0",
  "batch_size": 10
}';

select dbms_vector.utl_to_embedding('hello', json(:params)) from
dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  v vector;
begin
  input := 'hello';
  params := '
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/embedText",
  "model": "cohere.embed-english-v3.0",
  "batch_size": 10
}';

  v := dbms_vector.utl_to_embedding(input, json(params));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
```

```
end;
/
```

- **Using Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI**:

  a. Run `DBMS_VECTOR.CREATE_CREDENTIAL` to create and store a credential.

     Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

     ```
     { "access_token": "<access token>" }
     ```

     You will later refer to this credential name when declaring JSON parameters for the `UTL_to_EMBEDDING` call.

     ```
     exec dbms_vector.drop_credential('<credential name>');


     declare
       jo json_object_t;
     begin
       jo := json_object_t();
       jo.put('access_token', '<access token>');
       dbms_vector.create_credential(
         credential_name   => '<credential name>',
         params            => json(jo.to_string));
     end;
     /
     ```

     Replace the `access_token` and `credential_name` values. For example:

     ```
     declare
       jo json_object_t;
     begin
       jo := json_object_t();
       jo.put('access_token', 'AbabA1B123aBc123AbabAb123a1a2ab');
       dbms_vector.create_credential(
         credential_name   => 'HF_CRED',
         params            => json(jo.to_string));
     end;
     /
     ```

  b. Call `DBMS_VECTOR.UTL_TO_EMBEDDING`:

     ```
     -- select example

     var params clob;
     exec :params := '
     {
       "provider": "<REST provider>",
       "credential_name": "<credential name>",
       "url": "<REST endpoint URL for embedding service>",
       "model": "<embedding model name>"
     }';

     select dbms_vector.utl_to_embedding('hello', json(:params)) from
     dual;
     ```

```
-- PL/SQL example

declare
  input clob;
  params clob;
  v vector;
begin
  input := 'hello';

  params := '
{
  "provider": "<REST provider>",
  "credential_name": "<credential name>",
  "url": "<REST endpoint URL for embedding service>",
  "model": "<embedding model name>"
}';

  v := dbms_vector.utl_to_embedding(input, json(params));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

> **Note**
>
> For a complete list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Replace `provider`, `credential_name`, `url`, and `model` with your own values. Optionally, you can specify additional REST provider parameters. This is shown in the following examples:

Cohere example:

```
{
  "provider"       : "cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.ai/v1/embed",
  "model"          : "embed-english-light-v2.0",
  "input_type"     : "search_query"
}
```

Google AI example:

```
{
  "provider"       : "googleai",
  "credential_name": "GOOGLEAI_CRED",
  "url"            : "https://generativelanguage.googleapis.com/
v1beta/models/",
```

```
  "model"          : "embedding-001"
}
```

Hugging Face example:

```
{
  "provider"       : "huggingface",
  "credential_name": "HF_CRED",
  "url"            : "https://api-inference.huggingface.co/pipeline/
feature-extraction/",
  "model"          : "sentence-transformers/all-MiniLM-L6-v2"
}
```

OpenAI example:

```
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/embeddings",
  "model"          : "text-embedding-3-small"
}
```

Vertex AI example:

```
{
  "provider"       : "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url"            : "https://LOCATION-aiplatform.googleapis.com/v1/
projects/PROJECT/locations/LOCATION/publishers/google/models/",
  "model"          : "textembedding-gecko:predict"
}
```

The generated embedding may appear as:

```
EMBEDDING
---------------------------------------------------------------------------
----------------------------------------------------------
[8.78423732E-003,-4.29633334E-002,-5.93001908E-003,-4.65480909E-002,2.14333
013E-002,6.53376281E-002,-5.93746938E-002,2.10403297E-002,
4.38376889E-002,5.22960871E-002,1.25104953E-002,6.49512559E-002,-9.26998071
E-003,-6.97442219E-002,-3.02916039E-002,-4.74979728E-003,
-1.08755399E-002,-4.63751052E-003,3.62781435E-002,-9.35919806E-002,-1.13934
642E-002,-5.74270077E-002,-1.36667723E-002,2.42995787E-002,
-6.96804151E-002,4.93822657E-002,1.01460628E-002,-1.56464987E-002,-2.394105
68E-002,-4.27529104E-002,-5.65665103E-002,-1.74160264E-002,
5.05326502E-002,4.31500375E-002,-2.6994409E-002,-1.72731467E-002,9.30535868
E-002,6.85951149E-004,5.61876409E-003,-9.0233935E-003,
-2.55788807E-002,-2.04174276E-002,3.74175981E-002,-1.67872179E-002,1.074793
04E-001,-6.64602639E-003,-7.65537247E-002,-9.71965566E-002,
-3.99636962E-002,-2.57076006E-002,-5.62455431E-002,-1.3583754E-001,3.459460
29E-002,1.85191762E-002,3.01524661E-002,-2.62163244E-002,
-4.05582506E-003,1.72979087E-002,-3.66434865E-002,-1.72491539E-002,3.952284
16E-002,-1.05518714E-001,-1.27463877E-001,1.42578809E-002
```

This example uses the default settings for each provider. For detailed information on additional parameters, refer to your third-party provider's documentation.

Related Topics

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the `DBMS_VECTOR.UTL_TO_EMBEDDING` and `DBMS_VECTOR.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.
- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Convert Text String to Embedding Using the Local REST Provider Ollama

Perform a text-to-embedding transformation by accessing open embedding models, using the local host REST endpoint provider Ollama.

Ollama is a free and open-source command-line interface tool that allows you to run open embedding models and LLMs locally and privately on your Linux, Windows, or macOS systems. You can access Ollama as a service using SQL and PL/SQL commands.

You can call embedding models, such as all-minilm, mxbai-embed-large, or nomic-embed-text to vectorize your data. Note that you must use the same embedding model on both the data to be indexed and the user's input query. In this example, you can see how to vectorize a user's input query on the fly.

Here, you can call the chainable utility function `UTL_TO_EMBEDDING` (note the singular "embedding") from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case. This example uses the `DBMS_VECTOR.UTL_TO_EMBEDDING` API. `UTL_TO_EMBEDDING` directly returns a `VECTOR` type (not an array).

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To convert a user's input text string "`hello`" to a query vector, by calling a local embedding model using Ollama:

1. Connect to Oracle AI Database as a local user.

a. Log in to SQL*Plus as the SYS user, connecting as SYSDBA:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Install Ollama and run an embedding model locally.

   a. Download and run the Ollama application from https://ollama.com/download.

      You can either install Ollama as a service that runs in the background or as a standalone binary with a manual install. For detailed installation-specific steps, see Quick Start in the Ollama Documentation.

      Note the following:

      - The Ollama server needs to be able to connect to the internet so that it can download the models. If you require a proxy server to access the internet, remember to set the appropriate environment variables before running the Ollama server. For example, to set for Linux:

```
-- set a proxy if you require one

export https_proxy=<proxy-hostname>:<proxy-port>
export http_proxy=<proxy-hostname>:<proxy-port>
export no_proxy=localhost,127.0.0.1,.example.com
export ftp_proxy=<proxy-hostname>:<proxy-port>
```

- If you are running Ollama and the database on different machines, then on the database machine, you must change the `URL` to refer to the host name or IP address that is running Ollama instead of the local host.
- You may need to change your firewall settings on the machine that is running Ollama to allow the port through.

b. If running Ollama as a standalone binary from a manual install, then start the server:

```
ollama serve
```

c. Run a model using the `ollama pull <embedding_model_name>` command.

For example, to call the all-minilm model:

```
ollama pull all-minilm
```

For detailed information on this step, see Ollama Readme.

d. Verify that Ollama is running locally by using a cURL command.

For example:

```
-- get embeddings

curl http://localhost:11434/api/embeddings -d '{
  "model" : "all-minilm",
  "prompt": "What is Oracle AI Vector Search?"}'
```

3. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

4. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

5. Call `UTL_TO_EMBEDDING`.

The Ollama service has a REST API endpoint for generating embedding. Specify the URL and other configuration parameters in a JSON object.

```
var embed_ollama_params clob;
exec :embed_ollama_params := '{
     "provider": "ollama",
     "host"    : "local",
     "url"     : "http://localhost:11434/api/embeddings",
```

```
     "model"    : "all-minilm"
}';

select dbms_vector.utl_to_embedding('hello', json(:embed_ollama_params))
ollama_output from dual;
```

You can replace the `url` and `model` with your own values, as required.

> **Note**
>
> For a complete list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

An excerpt from the generated embedding output is as follows:

```
OLLAMA_OUTPUT
---------------------------------------------------------------------------
-----
[-2.31221581E+000,-3.26045007E-001,2.48111725E-001,-1.65610778E+000,1.10871
601E+
000,-1.78519666E-001,-2.44365096E+000,-3.32534742E+000,-1.3771069E+000,1.88
42382
4E+000,1.26494539E+000,-2.05359578E+000,-1.78593469E+000,-3.16150457E-001,-
5.362
42545E-001,6.42113638E+000,-2.36518741E+000,2.21405053E+000,6.52316332E-001
,-7.8
1692028E-001,2.32031775E+000,5.31627655E-001,-5.02781868E-001,7.03743398E-0
01,5.
48391223E-001,-3.16579014E-001,5.28999329E+000,1.63369191E+000,1.34206653E-
001,9
.54429448E-001,-1.94197679E+000,2.39797616E+000,3.5270077E-001,-1.6536833E+
000,-
5.74707508E-001,1.60994816E+000,3.80332375E+000,-6.30351126E-001,-1.5865227
E+000
,-2.48650503E+000,-1.42142653E+000,-2.79453158E+000,1.76355612E+000,-2.4869
0337E
-001,1.5521245E+000,-1.95240334E-001,1.42441893E+000,-3.57098508E+000,4.020
83158
E+000,-2.38530707E+000,2.34579134E+000,-2.79158998E+000,-5.92314243E-001,-9
.7153

OLLAMA_OUTPUT
---------------------------------------------------------------------------
-----
9557E-001,1.6514441E-002,1.03710043E+000,1.96799666E-001,-2.18394065E+000,-
2.786
71598E+000,-1.1549623E+000,1.92903787E-001,7.72498465E+000,-1.63462329E+000
,3.33
839393E+000,-7.17389703E-001,-3.99817854E-001,7.7395606E-001,6.43829286E-00
1,1.8
5182285E+000,2.95272923E+000,-8.72635692E-002,1.77895337E-001,-1.19716954E+
000,9
.43063736E-001,1.51703429E+000,-7.93301344E-001,1.92319798E+000,3.33290434E
+000,
```

```
-1.29852867E+000,2.02961493E+000,-4.35824203E+000,1.30186975E+000,-1.097941
4E+00
0,-7.07521975E-001,-4.50183332E-001,3.77224755E+000,-2.49138021E+000,-1.125
75901
E+000,1.15370192E-001,1.66767395E+000,3.52229095E+000,-3.78049397E+000,-6.1
71851
75E-001,1.71992481E+000,2.33226371E+000,-3.35983014E+000,-3.78417182E+000,3
.8380
127E+000,2.59293962E+000,-1.33574629E+000,-1.76218402E+000,3.53816301E-001,
-1.80
47899E+000,9.85167921E-001,1.93026745E+000,2.15959966E-001,9.94020045E-001,
6.189
```

Related Topics

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the `DBMS_VECTOR.UTL_TO_EMBEDDING` and `DBMS_VECTOR.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.
- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Convert Image to Embedding Using Public REST Providers

Perform an image-to-embedding transformation by making a REST call to the third-party service provider, Vertex AI. In this example, you can see how to vectorize both image and text inputs using a multimodal embedding model and then query a vector space containing vectors from both content types.

You can directly generate a vector embedding based on an image, which can be used for classifying or detecting images, comparing large datasets of images, or for performing a more effective similarity search on documents that include images. To get image embeddings, you can use any image embedding model or multimodal embedding model supported by Vertex AI. By analyzing an image, a model generates image embedding that encodes each visual element of the image (shape, color, pattern, texture, action, or object) as a vector representation.

Multimodal embedding is a technique that vectorizes data from different modalities such as text and images. This lets you use the same embedding model to generate embeddings for both types of content. By doing so, the resulting embeddings are compatible and situated in the same vector space, which allows for effective comparison between the two modalities (text and image) during similarity searches.

Here, you can use the `UTL_TO_EMBEDDING` function from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To find a bird that is similar to a given image or text input, using similarity search:

1. Connect as a local user and prepare your data dump directory.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

      ```
      conn sys/password as sysdba


      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      EXTENT MANAGEMENT LOCAL
      SEGMENT SPACE MANAGEMENT AUTO;


      SET ECHO ON
      SET FEEDBACK 1
      SET NUMWIDTH 10
      SET LINESIZE 80
      SET TRIMSPOOL ON
      SET TAB OFF
      SET PAGESIZE 10000
      SET LONG 10000
      ```

   b. Create a local user (`docuser`) and grant necessary privileges:

      ```
      drop user docuser cascade;


      create user docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
      unlimited on tbs1;


      grant DB_DEVELOPER_ROLE to docuser;
      ```

c. Create a local directory (`VEC_DUMP`) to store image files. Grant necessary privileges:

```
create or replace directory VEC_DUMP as '/my_local_dir/';


grant read, write on directory VEC_DUMP to docuser;

commit;
```

d. Connect as the local user (`docuser`):

```
conn docuser/password
```

2. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

   This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Set up credentials for Vertex AI.

   Vertex AI requires the following authentication parameter:

```
{ "access_token": "<access token>" }
```

```
begin
  DBMS_VECTOR_CHAIN.DROP_CREDENTIAL(credential_name  => 'VERTEXAI_CRED');
exception
  when others then null;
end;
/
```

   Here, replace `<access token>` with your own value:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', '<access token>');
  DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL(
    credential_name   =>  'VERTEXAI_CRED',
    params            => json(jo.to_string));
```

```
end;
/
```

5. Create a relational table (`docs`) and insert some images in it.

   You will later compare your query image to this database of images.

   Here, you first create a `docs` table with `id` and `content` columns. Then, using the `to_blob` function, you convert the contents of your image files into `BLOB` for storing into `content` (blob column), reading the files from the `VEC_DUMP` directory.

   > **Note**
   >
   > For image `BLOB` input, standalone baseline 8-bit lossy JPEG images are accepted. Unsupported or malformed image input does not produce a successful image decode. At SQL scoring time, this can result in a `NULL` embedding result. Perform any required image preprocessing or format conversion externally before embedding creation.
   >
   > For more information about image input restrictions, see ONNX Pipeline Models: Image Embedding.

```
drop table docs;


create table docs(id number primary key, content blob);

insert into docs(id, content)
values(1, to_blob(bfilename('VEC_DUMP', 'cat.jpg')));
insert into docs (id, content)
values(2, to_blob(bfilename('VEC_DUMP', 'eagle.jpg')));
```

6. Get image embedding for your query image (`parrots.jpg`):

   First, upload your query image to the `VEC_DUMP` directory.

   Then, use `UTL_TO_EMBEDDING` to vectorize that image. Here, the input is specified as `parrots.jpg` (with a pointer to the `VEC_DUMP` directory), the modality is specified as `image`, and the provider-specific embedding parameters for Vertex AI are passed as JSON.

   > **Note**
   >
   > For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

```
-- declare embedding parameters

var params clob;

begin
  :params := '
{
  "provider": "vertexai",
```

```
  "credential_name": "VERTEXAI_CRED",
  "url": "https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/
locations/LOCATION/publishers/google/models/",
  "model": "multimodalembedding:predict"
}';
end;
/

-- get image embedding: PL/SQL example

declare
  v vector;
  output clob;
begin
  v := dbms_vector_chain.utl_to_embedding(
    to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
  output := vector_serialize(v);
  dbms_output.put_line('vector data=' || dbms_lob.substr(output, 100) ||
'...');
end;
/

-- get image embedding: select example

select dbms_vector_chain.utl_to_embedding(
  to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
```

7. Search using an image input. Here, input is the generated embedding for your query image (`parrots.jpg`):

```
select docs.id, vector_distance(
  dbms_vector_chain.utl_to_embedding(to_blob(bfilename('VEC_DUMP',
'parrots.jpg')), 'image', json(:params)),
  dbms_vector_chain.utl_to_embedding(docs.content, 'image', json(:params)),
  cosine) dist
from docs
order by dist asc;
```

An output appears as:

```
ID       DIST
-------
-----------
2
5.847E-001
1        6.295E-001
```

This query shows that `ID 2` (`eagle.jpg`) is the closest match to the given image input, while `ID 1` (`cat.jpg`) is less similar.

8. Search using a text input. Here, input is the image's description as "`bald eagle`":

```
select docs.id, vector_distance(
  dbms_vector_chain.utl_to_embedding('bald eagle', json(:params)),
  dbms_vector_chain.utl_to_embedding(docs.content, 'image', json(:params)),
```

```
  cosine) dist
from docs
order by dist asc;
```

An output appears as:

```
ID      DIST
------- -----------
2       8.449E-001
1       9.87E-001
```

This query also implies that `ID 2` is the closest match to your text input, while `ID 1` is less similar. However, both values are lower than those in the previous query where you specified the image as input.

Related Topics

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING` and `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.
- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Generate Multi-modal Embeddings Using CLIP

This section provides end-to-end instructions from installing the OML4Py client to generating multi-modal embeddings using CLIP.

These instructions assume you have configured your Oracle Linux 8 repo in `/etc/yum.repos.d`, configured a Wallet if using an Autonomous AI Database, and set up a proxy if needed.

1. Install Python:

```
sudo yum install libffi-devel openssl openssl-devel tk-devel xz-devel zlib-
devel bzip2-devel readline-devel libuuid-devel ncurses-devel libaio
mkdir -p $HOME/python
wget https://www.python.org/ftp/python/3.12.3/Python-3.12.3.tgz
tar -xvzf Python-3.12.3.tgz --strip-components=1 -C $HOME/python
cd $HOME/python
./configure --prefix=$HOME/python
make clean; make
make altinstall
```

2. Set variables PYTHONHOME, PATH, and LD_LIBRARY_PATH:

```
export PYTHONHOME=$HOME/python
export PATH=$PYTHONHOME/bin:$PATH
export LD_LIBRARY_PATH=$PYTHONHOME/lib:$LD_LIBRARY_PATH
```

3. Create symlink for python3 and pip3:

```
cd $HOME/python/bin
ln -s python3.12 python3
ln -s pip3.12 pip3
```

4. Install Oracle Instant client if you will be exporting embedded models to the database from Python. If you will be exporting to a file, skip steps 4 and 5 and see the note under environment variables in step 6:

```
cd $HOME
wget https://download.oracle.com/otn_software/linux/instantclient/2340000/
instantclient-basic-linux.x64-23.4.0.24.05.zip
unzip instantclient-basic-linux.x64-23.4.0.24.05.zip
```

5. Set variable LD_LIBRARY_PATH:

```
export LD_LIBRARY_PATH=$HOME/instantclient_23_4:$LD_LIBRARY_PATH
```

6. Create an environment file, for example, env.sh, that defines the Python and Oracle Instant client environment variables and source these environment variables before each OML4Py client session. Alternatively, add the environment variable definitions to .bashrc so they are defined when the user logs into their Linux machine.

```
# Environment variables for Python
export PYTHONHOME=$HOME/python
export PATH=$PYTHONHOME/bin:$PATH
export LD_LIBRARY_PATH=$PYTHONHOME/lib:$LD_LIBRARY_PATH
```

> **Note**
>
> Environment variable for Oracle Instant Client - only if the Oracle Instant Client is installed for exporting models to the database.
> `export LD_LIBRARY_PATH=$HOME/instantclient_23_4:$LD_LIBRARY_PATH`
>
> .

7. Create a file named requirements.txt that contains the required third-party packages listed below.

```
--extra-index-url https://download.pytorch.org/whl/cpu
pandas==2.1.1
setuptools==68.0.0
scipy==1.12.0
matplotlib==3.8.4
oracledb==2.2.0
scikit-learn==1.4.1.post1
numpy==1.26.4
```

```
onnxruntime==1.17.0
onnxruntime-extensions==0.10.1
onnx==1.16.0
torch==2.2.0+cpu
transformers==4.38.1
sentencepiece==0.2.0
```

8. Upgrade pip3 and install the packages listed in requirements.txt.

```
pip3 install --upgrade pip
pip3 install -r requirements.txt
```

9. Install OML4Py client. Download OML4Py 2.0.1 client from OML4Py download page and upload it to the Linux machine.

```
unzip oml4py-client-linux-x86_64-2.0.1.zip
pip3 install client/oml-2.0-cp312-cp312-linux_x86_64.whl
```

10. Get a list of all preconfigured models. Start Python and import ONNXPipelineConfig from `oml.utils`.

```
python3

from oml.utils import ONNXPipelineConfig

ONNXPipelineConfig.show_preconfigured()


['sentence-transformers/all-mpnet-base-v2',
'sentence-transformers/all-MiniLM-L6-v2',
'sentence-transformers/multi-qa-MiniLM-L6-cos-v1',
'sentence-transformers/distiluse-base-multilingual-cased-v2',
'sentence-transformers/all-MiniLM-L12-v2',
'BAAI/bge-small-en-v1.5',
'BAAI/bge-base-en-v1.5',
'taylorAI/bge-micro-v2',
'intfloat/e5-small-v2',
'intfloat/e5-base-v2',
'thenlper/gte-base',
'thenlper/gte-small',
'TaylorAI/gte-tiny',
'sentence-transformers/paraphrase-multilingual-mpnet-base-v2',
'intfloat/multilingual-e5-base',
'intfloat/multilingual-e5-small',
'sentence-transformers/stsb-xlm-r-multilingual',
'Snowflake/snowflake-arctic-embed-xs',
'Snowflake/snowflake-arctic-embed-s',
'Snowflake/snowflake-arctic-embed-m',
'mixedbread-ai/mxbai-embed-large-v1',
'openai/clip-vit-large-patch14',
'google/vit-base-patch16-224',
'microsoft/resnet-18',
'microsoft/resnet-50',
'WinKawaks/vit-tiny-patch16-224',
'Falconsai/nsfw_image_detection',
'WinKawaks/vit-small-patch16-224',
```

```
'nateraw/vit-age-classifier',
'rizvandwiki/gender-classification',
'AdamCodd/vit-base-nsfw-detector',
'trpakov/vit-face-expression',
'BAAI/bge-reranker-base']
```

11. Use OML4Py to load a multi-modal model on the database.

    - To use an alternate method other than OML4Py, skip this step and proceed to step 12.

    Export a preconfigured embedding model to the database. Import the oml library and import `ONNXPipeline` and `ONNXPipelineConfig` from `oml.utils`. This exports the ONNX-format model to your local file system. In the following steps, replace the placeholders with your own credentials.

    ```
    import oml
    from oml.utils import ONNXPipeline, ONNXPipelineConfig
    ```

    If your Oracle AI Database is on premises, set embedded mode to false. This step is not supported or required for Oracle Autonomous AI Database.

    ```
    oml.core.methods.__embed__ = False
    ```

    Create a database connection.

    - Using Oracle AI Database on premises:

      ```
      oml.connect("<user>", "<password>", port=<port number>
      host="<hostname>",
      service_name="<service name>")

      pipeline = ONNXPipeline(model_name="openai/clip-vit-large-patch14")
      pipeline.export2db("CLIP")
      ```

    - Using Oracle Autonomous AI Database:

      ```
      oml.connect(user="<user>", password="<password>", dsn="myadb_low")

      pipeline = ONNXPipeline(model_name="openai/clip-vit-large-patch14")
      pipeline.export2db("CLIP")
      ```

    Once this step is complete, there will be two models loaded on the database called "CLIP_TXT" and "CLIP_IMG".

12. Export a preconfigured embedding model to a local file.

    This exports the ONNX-format model to your local file system:

    ```
    # Export to file
    pipeline = ONNXPipeline(model_name="openai/clip-vit-large-patch14")
    pipeline.export2file("clip",output_dir="/tmp/models")
    ```

Move the ONNX file to a directory on the database server and create a directory on the file system and in the database for the import.

```
mkdir -p /tmp/models
sqlplus / as sysdba
alter session set container=<name of pluggable database>;
```

Apply the necessary permissions and grants.

```
-- directory to store ONNX files for import
CREATE DIRECTORY ONNX_IMPORT AS '/tmp/models';
-- grant your OML user read and write permissions on the directory
GRANT READ, WRITE ON DIRECTORY ONNX_IMPORT to OMLUSER;
-- grant to allow user to import the model
GRANT CREATE MINING MODEL TO OMLUSER;
```

Use the `DBMS_VECTOR.LOAD_ONNX_MODEL` procedure to load the models in your OML user schema. In this example, the procedure loads the ONNX model files named `clip_txt.onnx` and `clip_img.onnx` from the `ONNX_IMPORT` directory into the database as models named `CLIP_TXT` and `CLIP_IMG`, respectively.

```
BEGIN
    DBMS_VECTOR.LOAD_ONNX_MODEL(
    directory => 'ONNX_IMPORT',
    file_name => 'clip_txt.onnx',
    model_name => 'CLIP_TXT',
    metadata => JSON('{"function" : "embedding", "embeddingOutput" :
"embedding", "input": {"input": ["DATA"]}}'));
END;
BEGIN
    DBMS_VECTOR.LOAD_ONNX_MODEL(
    directory => 'ONNX_IMPORT',
    file_name => 'clip_img.onnx',
    model_name => 'CLIP_IMG',
    metadata => JSON('{"function" : "embedding", "embeddingOutput" :
"embedding", "input": {"input": ["DATA"]}}'));
END;
```

13. Verify the models exist using SQL.

```
sqlplus $USER/pass@PDBNAME;


SELECT model_name, algorithm, mining_function
FROM user_mining_models
WHERE model_name='CLIP_TXT' OR model_name='CLIP_IMG';


--------------------------------------------------------
MODEL_NAME          ALGORITHM            MINING_FUNCTION
--------------------------------------------------------
CLIP_TXT            ONNX                 EMBEDDING
CLIP_IMG            ONNX                 EMBEDDING
```

14. Generate embeddings with exported models using Python.

> **Note**
>
> For image BLOB input, standalone baseline 8-bit lossy JPEG images are accepted. Unsupported or malformed image input does not produce a successful image decode. At SQL scoring time, this can result in a NULL embedding result. Perform any required image preprocessing or format conversion externally before embedding creation.
>
> For more information about image input restrictions, see ONNX Pipeline Models: Image Embedding.

```
from oracledb import DB_TYPE_BLOB
with open('cat.jpg', 'rb') as f:
    img = f.read()
cr = oml.cursor()
blob = cr. var(DB_TYPE_BLOB)
blob.setvalue(0, img)
data = cr.execute("select vector_embedding(CLIP_TXT using 'RES' as DATA)
from dual")
txt_embed = data.fetchall()
data = cr.execute("select vector_embedding(CLIP_IMG using to_blob(:1) as
DATA) from dual", [blob])
img_embed = data.fetchall()
```

Calculate similarity between an image and text using Python:

```
from oracledb import DB_TYPE_BLOB
with open('cat.jpg', 'rb') as f:
    img = f.read()
cr = oml.cursor()
blob = cr.var(DB_TYPE_BLOB)
blob.setvalue(0, img)
data = cr.execute("""select 1-vector_distance(vector_embedding(CLIP_TXT
using 'RES' as DATA),
                    vector_embedding(CLIP_IMG using to_blob(:1) as DATA))
from dual""", [blob])
data.fetchall()
```

Result:

```
[(0.1637756726800217,)]
```

Generate embeddings with the exported models using SQL:

- `SELECT VECTOR_EMBEDDING(CLIP_TXT USING 'RES' as DATA) AS embedding;`

An example of results are shown in the following excerpt:

```
EMBEDDING
------------------------------------------------------------------------
--------
[2.86132172E-002,-5.59654366E-003,8.66401661E-003,-1.4299524E-002,1.0201
2949E-00
2,6.00034464E-003,1.86244473E-002,-7.81036681E-003, ...
```

- ```
  SELECT VECTOR_EMBEDDING(CLIP_IMG USING TO_BLOB(BFILENAME('ONNX_IMPORT',
  'cat.jpg')) as DATA) AS embedding;
  ```

  An example of results are shown in the following excerpt:

  ```
  EMBEDDING
  ------------------------------------------------------------------------
  --------
  [-2.2028232E-002,1.29058748E-003,-2.0222881E-004,4.58140159E-003,1.98919
  605E-002
  ,-9.51210782E-003,8.22519697E-003,1.06151737E-002, ...
  ```

Calculate the similarity between an image and text using SQL:

```
SELECT 1-VECTOR_DISTANCE(VECTOR_EMBEDDING(
            CLIP_IMG USING TO_BLOB(BFILENAME('ONNX_IMPORT', 'cat.jpg')) as
DATA),
            VECTOR_EMBEDDING(CLIP_TXT USING 'RES' as DATA)) AS similarity;
```

Example result:

```
SIMILARITY
----------
1.638E-001
```

Vectorize Relational Tables Using OML Feature Extraction Algorithms

This example shows you how to use OML's Feature Extraction algorithms in conjunction with the `VECTOR_EMBEDDING()` operator to vectorize sets of relational data, build similarity indexes, and perform similarity searches on the resulting vectors.

Feature Extraction algorithms help in extracting the most informative features/columns from the data and aim to reduce the dimensionality of large data sets by identifying the principal components that capture the most variance in the data. This reduction simplifies the data set while retaining the most important information, making it easier to analyze correlations and redundancies in the data.

The Principal Component Analysis (PCA) algorithm, a widely used dimensionality reduction technique in machine learning, is used in this tutorial.
```

Note

This example uses customer bank marketing data available at https://archive.ics.uci.edu/dataset/222/bank+marketing.

The relational data table includes a mix of numeric and categorical columns. It has more than 4000 records.

```
SELECT column_name, data_type
FROM user_tab_columns
WHERE table_name = 'BANK'
ORDER BY data_type, column_name;
```

Output:

```
COLUMN_NAME          DATA_TYPE
-------------------- --------------------
AGE                  NUMBER
CAMPAIGN             NUMBER
CONS_CONF_IDX        NUMBER
CONS_PRICE_IDX       NUMBER
DURATION             NUMBER
EMP_VAR_RATE         NUMBER
EURIBOR3M            NUMBER
ID                   NUMBER
NR_EMPLOYED          NUMBER
PDAYS                NUMBER
PREVIOUS             NUMBER
CONTACT              VARCHAR2
CREDIT_DEFAULT       VARCHAR2
DAY_OF_WEEK          VARCHAR2
EDUCATION            VARCHAR2
HOUSING              VARCHAR2
JOB                  VARCHAR2
LOAN                 VARCHAR2
MARITAL              VARCHAR2
MONTH                VARCHAR2
POUTCOME             VARCHAR2
Y                    VARCHAR2
```

To perform a similarity search, you need to vectorize the relational data. To do so, you can first use the OML Feature Extraction algorithm to project the data onto a more compact numeric space. In this example, you configure the SVD algorithm to perform a Principal Component Analysis (PCA) projection of the original data table. The number of features/columns (5 in this case) is specified in the setting table. The input number determines the number of principal features or columns that will be retained after the dimensionality reduction process. Each of these columns represent a direction in the feature space along which the data varies the most.

Because you need to use the `DBMS_DATA_MINING` package to create the model, you need the `CREATE MINING MODEL` privilege in addition to the other privileges relevant to vector indexes and similarity search. For more information about the `CREATE MINING MODEL` privilege, see *Oracle Machine Learning for SQL User's Guide*.

1. Create a setting table, insert values, and then create a model.

   Use the `DBMS_DATA_MINING` package to create a model, using `mod_sett` as the setting table:

```
CREATE TABLE mod_sett(
  setting_name  VARCHAR2(30),
  setting_value VARCHAR2(30)
);


BEGIN
  INSERT INTO mod_sett (setting_name, setting_value) VALUES
        (dbms_data_mining.algo_name,
dbms_data_mining.algo_singular_value_decomp);
  INSERT INTO mod_sett (setting_name, setting_value) VALUES
        (dbms_data_mining.prep_auto, dbms_data_mining.prep_auto_on);
  INSERT INTO mod_sett (setting_name, setting_value) VALUES
        (dbms_data_mining.svds_scoring_mode,
dbms_data_mining.svds_scoring_pca);
  INSERT INTO mod_sett (setting_name, setting_value) VALUES
        (dbms_data_mining.feat_num_features, 5);
  commit;
END;
/


BEGIN
  DBMS_DATA_MINING.CREATE_MODEL(
    model_name          => 'pcamod',
    mining_function     => dbms_data_mining.feature_extraction,
    data_table_name     => 'bank',
    case_id_column_name => 'id',
    settings_table_name => 'mod_sett');
END;
/
```

2. Use the `VECTOR_EMBEDDING()` function to output the SVD projection results as vectors.

   The dimension of the vector column is the same as the number of features in the PCA model and the value of the vector represents the PCA projection results of the original row data.

   > **Note**
   >
   > `USING *` results in the use of all the relevant features present in the input row.

```
SELECT id, vector_embedding(pcamod USING *) embedding
FROM bank
WHERE id=10000;
```

Output:

```
          ID      EMBEDDING
--------------    ---------------------------------------------------
       10000      [-2.3551013972411354E+002,2.8160084506788273E+001,
                    5.2821278275005774E+001,-1.8960922352439308E-002,
                   -2.5441143639048378E+000]
```

3. Create a table to hold the vector output results for all the data records.

   This represents a vectorization of your original relational data for the top-5 most important columns.

   ```
   CREATE TABLE pca_output AS
   (SELECT id, vector_embedding(pcamod USING *) embedding
    FROM bank);
   ```

4. Build a vector index based on the vector table.

   For example, you can build an IVF index with cosine distance:

   ```
   CREATE VECTOR INDEX my_ivf_idx ON pca_output(embedding)
   ORGANIZATION NEIGHBOR PARTITIONS
   DISTANCE COSINE WITH TARGET ACCURACY 95;
   ```

5. Perform a similarity search using the `my_ivf_idx` index.

   In this example, you search for the top-3 results that are closest to `id=10000` based on the cosine distance and join the vector table with the original bank table to retrieve the most impactful attributes from the original table. To identify the most impactful columns for this row, use the `FEATURE_DETAILS()` function.

   ```
   SELECT feature_details(pcamod, 5 USING *) features
   FROM bank
   WHERE id=10000;
   ```

   Output:

   ```
   FEATURES
   ---------------------------------------------------------------------------
   --------------------
   <Details algorithm="Singular Value Decomposition" feature="5">
   <Attribute name="PDAYS" actualValue="999" weight=".041" rank="1"/>
   <Attribute name="EURIBOR3M" actualValue="4.959" weight=".028" rank="2"/>
   <Attribute name="CONTACT" actualValue="telephone" weight=".016" rank="3"/>
   <Attribute name="EMP_VAR_RATE" actualValue="1.4" weight=".014" rank="4"/>
   <Attribute name="DAY_OF_WEEK" actualValue="wed" weight=".002" rank="5"/>
   </Details>
   ```

   Join the original data table to retrieve the most impactful information:

   ```
   SELECT p.id id, b.PDAYS PDAYS, b.EURIBOR3M EURIBOR3M, b.CONTACT CONTACT,
       b.EMP_VAR_RATE EMP_VAR_RATE, b.DAY_OF_WEEK DAY_OF_WEEK
   FROM pca_output p, bank b
   ```

```
WHERE p.id <> 10000 AND p.id=b.id
ORDER BY VECTOR_DISTANCE(embedding, (select embedding from pca_output
where id=10000), COSINE)
FETCH FIRST 3 ROWS ONLY;
```

Output:

```
        ID      PDAYS  EURIBOR3M   CONTACT EMP_VAR_RATE DAY_OF_WEEK
---------- ---------- ---------- --------- ------------ -----------
      9416        999      4.967 telephone          1.4         fri
     13485        999      4.963 telephone          1.4         thu
      9800        999      4.959 telephone          1.4         wed
```

The results of the previous query illustrate that the closest 3 records are very similar. In contrast, the distributions of these features across the data set are dispersed as shown in the following queries:

```
SELECT avg(PDAYS) avg, stddev(PDAYS) std, min(PDAYS) min, max(PDAYS) max
FROM bank;
```

Output:

```
       AVG        STD        MIN        MAX
---------- ---------- ---------- ----------
962.475454 186.910907          0        999
```

```
SELECT avg(EURIBOR3M) avg, stddev(EURIBOR3M) std, min(EURIBOR3M) min,
max(EURIBOR3M) max
FROM bank;
```

Output:

```
       AVG       STD        MIN        MAX
---------- --------- ---------- ----------

3.62129081 1.7344474       .634      5.045
```

This tutorial demonstrates how you can vectorize relational data very efficiently and achieve significant compression while maintaining a high quality similarity search.

See Also

Oracle Machine Learning for SQL Concepts for more information about Feature Extraction algorithms

Perform Chunking With Embedding

In these examples, you can see how to explore the `VECTOR_CHUNKS` SQL function along with chainable utility PL/SQL functions to split large textual extracts and documents into chunks and then represent each chunk as a vector embedding.

To embed large textual data, you first need to prepare it in a format that can be processed by embedding models. You first transform the data into plain text, split the resulting text into smaller chunks of text, and then transform each chunk into a vector. This is done to comply with the input limits set by embedding models. Chunks can be words (to capture specific words or word pieces), sentences (to capture a specific context), or paragraphs (to capture broader themes).

- Convert Text to Chunks With Custom Chunking Specifications
  A chunked output, especially for long and complex documents, sometimes loses contextual meaning or coherence with its parent content. In this example, you can see how to refine your chunks by applying custom chunking specifications.
- Convert File to Text to Chunks to Embeddings Within Oracle AI Database
  First convert a PDF file to text, split the text into chunks, and then create vector embeddings on each chunk by accessing a vector embedding model stored in the database.
- Convert File to Embeddings Within Oracle AI Database
  Directly extract vector embeddings from a PDF document, using a single-step statement, by accessing a vector embedding model stored in the database.
- Generate and Use Embeddings for an End-to-End Search
  First generate vector embeddings from textual content by using a vector embedding model stored in the database, and then populate and query a vector index. At query time, you also vectorize the query criteria on the fly.

Convert Text to Chunks With Custom Chunking Specifications

A chunked output, especially for long and complex documents, sometimes loses contextual meaning or coherence with its parent content. In this example, you can see how to refine your chunks by applying custom chunking specifications.

Here, you use the `VECTOR_CHUNKS` SQL function or the `UTL_TO_CHUNKS()` PL/SQL function from the `DBMS_VECTOR_CHAIN` package.

> **Note**
>
> This example shows how to apply some of the custom chunking parameters. To further explore all the supported chunking parameters with more detailed examples, see Explore Chunking Techniques and Examples.

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
```

```
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local test user (`docuser`) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE to docuser;
```

c. Connect as the local user (`docuser`):

```
CONN docuser/password
```

2. Create a relational table (`documentation_tab`) to store unstructured text chunks in it:

```
DROP TABLE IF EXISTS documentation_tab;

CREATE TABLE documentation_tab (
    id NUMBER,
    text VARCHAR2(2000));


INSERT INTO documentation_tab VALUES(1,
'Oracle AI Vector Search stores and indexes vector embeddings'||
' for fast retrieval and similarity search.'||CHR(10)||CHR(10)||
'    About Oracle AI Vector Search'||CHR(10)||
'    Vector Indexes are a new classification of specialized indexes'||
' that are designed for Artificial Intelligence (AI) workloads that
allow'||
' you to query data based on semantics, rather than keywords.'||CHR(10)||
CHR(10)||
'    Why Use Oracle AI Vector Search?'||CHR(10)||
' The biggest benefit of Oracle AI Vector Search is that semantic search'||
' on unstructured data can be combined with relational search on
business'||
' data in one single system.'||CHR(10));
```

```
COMMIT;


SET LINESIZE 1000;
SET PAGESIZE 200;
COLUMN doc FORMAT 999;
COLUMN id  FORMAT 999;
COLUMN pos FORMAT 999;
COLUMN siz FORMAT 999;
COLUMN txt FORMAT a60;
COLUMN data FORMAT a80;
```

3. Call the `VECTOR_CHUNKS` SQL function and specify the following custom chunking parameters. Setting the `NORMALIZE` parameter to `all` ensures that the formatting of the results is easier to read, which is especially helpful when the data includes PDF documents:

```
SELECT D.id doc, C.chunk_offset pos, C.chunk_length siz, C.chunk_text txt
FROM documentation_tab D, VECTOR_CHUNKS(D.text
    BY words
    MAX 50
    OVERLAP 0
    SPLIT BY recursively
    LANGUAGE american
    NORMALIZE all) C;
```

To call the same operation using the `UTL_TO_CHUNKS` function from the `DBMS_VECTOR_CHAIN` package, run:

```
SELECT D.id doc,
    JSON_VALUE(C.column_value, '$.chunk_id' RETURNING NUMBER) AS id,
    JSON_VALUE(C.column_value, '$.chunk_offset' RETURNING NUMBER) AS pos,
    JSON_VALUE(C.column_value, '$.chunk_length' RETURNING NUMBER) AS siz,
    JSON_VALUE(C.column_value, '$.chunk_data') AS txt
FROM documentation_tab D,
   dbms_vector_chain.utl_to_chunks(D.text,
   JSON('{"by":"words",
          "max":"50",
          "overlap":"0",
          "split":"recursively",
          "language":"american",
          "normalize":"all"}')) C;
```

This returns a set of three chunks, which are split by words recursively using blank lines, new lines, and spaces:

```
DOC  POS  SIZ  TXT
---- ---- ---- ------------------------------------------------------------
1   1    108   Oracle AI Vector Search stores and indexes vector embeddings
         for fast retrieval and similarity search.

1   109  234   About Oracle AI Vector Search
            Vector Indexes are a new classification of specialized index
```

```
                es that are designed for Artificial Intelligence (AI) worklo
                ads that allow you to query data based on semantics, rather
                than keywords.

1    343  204   Why Use Oracle AI Vector Search?
                The biggest benefit of Oracle AI Vector Search is that seman
                tic search on unstructured data can be combined with relatio
                nal search on business data in one single system.
```

The chunking results contain:

- `chunk_id` as `DOC`: ID for each chunk
- `chunk_offset` as `POS`: Original position of each chunk in the source document, relative to the start of document (which has a position of `1`)
- `chunk_length` as `SIZ`: Character length of each chunk
- `chunk_data` as `TXT`: Textual content from each chunk

Related Topics

- VECTOR_CHUNKS
  Use `VECTOR_CHUNKS` to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.
- UTL_TO_CHUNKS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS` chainable utility function to split a large plain text document into smaller chunks of text.

Convert File to Text to Chunks to Embeddings Within Oracle AI Database

First convert a PDF file to text, split the text into chunks, and then create vector embeddings on each chunk by accessing a vector embedding model stored in the database.

You can run parallel or step-by-step transformations like this for standalone applications where you want to review, inspect, and accordingly amend results at each stage and then proceed further.

Here, you use a set of functions from the `DBMS_VECTOR_CHAIN` package, such as `UTL_TO_TEXT`, `UTL_TO_CHUNKS`, and `UTL_TO_EMBEDDINGS`.

To generate embeddings using a vector embedding model stored in the database, through step-by-step transformation chains:

1. **Connect as a local user and prepare your data dump directory**.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

   ```
   conn sys/password as sysdba


   CREATE TABLESPACE tbs1
   DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
   EXTENT MANAGEMENT LOCAL
   SEGMENT SPACE MANAGEMENT AUTO;


   SET ECHO ON
   SET FEEDBACK 1
   ```

```
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

   b. Create a local user (`docuser`) and grant necessary privileges:

```
drop user docuser cascade;


create user docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


grant DB_DEVELOPER_ROLE to docuser;
```

   c. Create a local directory (`VEC_DUMP`) to store your input data and model files. Grant necessary privileges:

```
create or replace directory VEC_DUMP as '/my_local_dir/';


grant read, write on directory VEC_DUMP to docuser;

commit;
```

   d. Connect as the local user (`docuser`):

```
conn docuser/password
```

2. **Convert file to text**.
   a. Create a relational table (`documentation_tab`) and store your PDF document (*Oracle AI Database Concepts*) in it:

```
drop table documentation_tab purge;


CREATE TABLE documentation_tab (id number, data blob);

INSERT INTO documentation_tab values(1, to_blob(bfilename('VEC_DUMP',
'database-concepts26ai.pdf')));

commit;


SELECT dbms_lob.getlength(t.data) from documentation_tab t;
```

   b. Call `UTL_TO_TEXT` to convert the PDF document into text format:

```
SELECT dbms_vector_chain.utl_to_text(dt.data) from documentation_tab dt;
```

An excerpt from the output is as follows:

```
DBMS_VECTOR_CHAIN.UTL_TO_TEXT(DT.DATA)
--------------------------------------------------------------------------
Database Concepts
26ai
Oracle AI Database Database Concepts, 26ai

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1 row selected.
```

3. **Convert text to chunks.**
   a. Call `UTL_TO_CHUNKS` to chunk the text document:

Here, you create the chunks using the default chunking parameters.

```
SELECT ct.*
  from documentation_tab dt,

dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(dt.data))
ct;
```

An excerpt from the output is as follows:

```
{"chunk_id":1,"chunk_offset":1508024,"chunk_length":579,"chunk_data":"In
ventory
\n\n\n\nAnalysis \n\n\n\nReporting \n\n\n\nMining\n\n\n\nSummary
\n\n\n\nData
 \n\n\n\nRaw Data\n\n\n\nMetadata\n\n\n\nSee Also:\n\n\n\nOracle AI
Database Data
Warehousing Guide to learn about transformation
\n\n\n\nmechanisms\n\n\n\nOvervie
w of Extraction, Transformation, and Loading (ETL) \n\n\n\nThe process
of extrac
ting data from source systems and bringing it into the warehouse is
\n\n\n\ncomm
only called ETL: extraction, transformation, and loading. ETL refers to
a broad
process \n\n\n\nrather than three well-defined steps.\n\n\n\nIn a
typical scenar
io, data from one or more operational systems is extracted and then"}

{"chunk_id":2,"chunk_offset":1508603,"chunk_length":607,"chunk_data":"ph
ysica
lly transported to the target system or an intermediate system for
processing. \
n\n\n\nDepending on the method of transportation, some transformations
can occur
 during this \n\n\n\nprocess. For example, a SQL statement that
directly accesse
s a remote target through a \n\n\n\ngateway can concatenate two columns
as part
of the \n\n\n\nSELECT\n\n\n\nstatement. \n\n\n\nOracle AI Database is
not itself an
 ETL tool. However, Oracle AI Database provides a rich set of
\n\n\n\ncapabilities
usable by ETL tools and customized ETL solutions. ETL capabilities
provided by \
n\n\n\nOracle AI Database include:\n\n\n\n? \n\n\n\nTransportable
tablespaces"}

3728 rows selected.
```

Notice the extra spaces and newline characters (`\n\n`) in the chunked output. This is because normalization is not applied by default. As shown in the next step, you can apply custom chunking specifications, such as `normalize` to omit duplicate characters, extra spaces, or newline characters from the output. You can further refine your chunks by applying other chunking specifications, such as split conditions or maximum size limits.

b. Apply the following JSON parameters to use normalization and some of the custom chunking specifications (described in Explore Chunking Techniques and Examples):

```
SELECT ct.*
  from documentation_tab dt,
  dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(
    dt.data),
    JSON('{
           "by" : "words",
           "max" : "100",
           "overlap" : "0",
           "split" : "recursively",
           "language" : "american",
           "normalize" : "all"
          }')) ct;
```

The output may now appear as:

```
{"chunk_id":2536,"chunk_offset":1372527,"chunk_length":633,"chunk_data":
"The dat
abase maps granules to parallel execution servers at execution time.
When a para
llel execution server finishes reading the rows corresponding to a
granule, and
when granules remain, it obtains another granule from the query
coordinator. This
operation continues until the table has been read. The execution
servers send
results back to the coordinator, which assembles the pieces into the
desired full
table scan. Oracle AI Database VLDB and Partitioning Guide to learn how
to use
parallel execution. Oracle AI Database Data Warehousing Guide to learn
about
recommended"}

{"chunk_id":2537,"chunk_offset":1373160,"chunk_length":701,"chunk_data":
"initial
ization parameters for parallelism\n\nChapter 18\n\nOverview of
Background Proce
sses\n\nApplication and Oracle Net Services Architecture\n\nThis
chapter defines
application architecture and describes how an Oracle AI database and
database appli
cations work in a distributed processing environment. This material
applies to
almost every type of Oracle AI Database environment. Overview of Oracle
Application
Architecture In the context of this chapter, application architecture
refers to
the computing environment in which a database application connects to
an Oracle
AI database."}

3728 rows selected.
```

The chunking results contain:

- `chunk_id`: Chunk ID for each chunk
- `chunk_offset`: Original position of each chunk in the source document, relative to the start of document (which has a position of 1)
- `chunk_length`: Character length of each chunk
- `chunk_data`: Text pieces from each chunk

4. **Convert chunks to embeddings.**

   a. Load your embedding model into Oracle AI Database by calling the `load_onnx_model` procedure.

```
EXECUTE dbms_vector.drop_onnx_model(model_name => 'doc_model', force =>
true);


EXECUTE dbms_vector.load_onnx_model('VEC_DUMP',
'my_embedding_model.onnx', 'doc_model', JSON('{"function" :
"embedding", "embeddingOutput" : "embedding" , "input": {"input":
["DATA"]}}'));
```

   In this example, the procedure loads an ONNX model file, named `my_embedding_model.onnx` from the `VEC_DUMP` directory, into the database as `doc_model`. You must replace `my_embedding_model.onnx` with an ONNX export of your embedding model and `doc_model` with the name under which the imported model is stored in the database.

   > **Note**
   >
   > If you do not have an embedding model in ONNX format, then perform the steps listed in ONNX Pipeline Models: Text Embedding.

   b. Call `UTL_TO_EMBEDDINGS` to generate a set of vector embeddings corresponding to the chunks:

```
var embed_params clob;
exec :embed_params := '{"provider":"database", "model":"doc_model"}';

SELECT et.* from
  documentation_tab dt,
  dbms_vector_chain.utl_to_embeddings(

dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(dt.data)),
    json(:embed_params)) et;
```

An excerpt from the output is as follows:

```
{"embed_id":"1","embed_data":"Introduction to Oracle AI Database\n\n\nThis
chapter provides an overview of Oracle AI Database. Every organization has
informat
ion that it must store and manage to meet its requirements. For example, a
corpo
ration must collect and maintain human resources records for its
```

```
employees. About
Relational Databases and Schema Objects\n\n\nData Access tables with
parent keys,
base, upgrade, UROWID data type, user global area (UGA), user program
interface,
","embed_vector":"[0.111119926,0.0423980951,-0.00929224491,-0.0352411047,-0
.0144
591287,0.0277361721,0.183199733,-0.0245029964,-0.137614027,0.0730137378,0.0
17934
6036,0.0788726509,0.0176453814,0.100403085,-0.0518687107,-0.0152645027,0.02
83792
187,-0.114087239,0.0139923804,0.0747490972,-0.181839675,-0.130034953,0.1012
07718
,0.117135495,-0.0682030097,-0.217743069,0.0613380745,0.0150767341,0.0361393
057,-
0.113082513,-0.0550440662,-0.000983044971,-0.00719357422,0.1590323,-0.02204
14512
,-0.0723528489,-0.0126240514,-0.175765082,0.168952227,0.0466451086,-0.12136
507,-
0.0442310236,0.0139067639,0.054659389,-0.29653421,-0.0988782048,0.079434923
8,-0.
0758788213,0.0152856084,-0.0260562375,0.0652966872,-0.0782724097,-0.0226081
386,0
.0909011662,-0.184569761,0.159565002,-0.15350005,-0.0108382348,0.101788878,
1.919
59683E-002,2.54665539E-002,2.50248201E-002,5.29858321E-002,1.42359538E-002,
5.655
82886E-002,3.41602638E-002,3.18607911E-002,-3.07250433E-002,-3.60006578E-00
2,-3.
26940455E-002,-5.13980947E-002,-9.18597169E-003,-2.40122043E-002,2.15246622
E-002
,-3.89301814E-002,1.09825116E-002,-8.59739035E-002,-3.34327705E-002,-6.5231
0252E
-002,2.46418975E-002,6.27725571E-003,6.54156879E-002,-2.97986511E-003,-1.48
5541E
-003,-9.00155635E-003]"}

{"embed_id":2,"embed_data":"1-18 Database, 19-6 write-ahead,18-17 Learn
session
memory in a large pool. The multitenant architecture enables an Oracle AI
database
to function as a multitenant container database (CDB). XStream,20-38\nZero
Data
Loss Recovery Appliance See Recovery Appliance zone maps, An application
contai
ner consists of exactly one application root, and PDBs plugged in to this
root.
Index-21\n D:20231208125114-08'00' D:20231208125114-08'00'
D:20231208125114-08'0
0'
19-6","embed_vector":"[6.30229115E-002,6.80943206E-002,-6.94655553E-002,-2.
58
157589E-002,-1.89648587E-002,-9.02655348E-002,1.97774544E-002,-9.39233322E-
003,-
5.06882742E-002,2.0078931E-002,-1.28898735E-003,-4.10606936E-002,2.09831214
E-003
```

```
,-4.53372523E-002,-7.09890276E-002,5.38906306E-002,-5.81014939E-002,-1.3959
175E-
004,-1.08725717E-002,-3.79145369E-002,-4.39973129E-003,3.25602405E-002,6.58
87302
2E-002,-4.27212603E-002,-3.00925318E-002,3.07144262E-002,-2.26370787E-004,-
4.623
15865E-002,1.11807801E-001,7.36674219E-002,-1.61244173E-003,7.35205635E-002
,4.16
726843E-002,-5.08309156E-002,-3.55720241E-003,4.49763797E-003,5.03803678E-0
02,2.
32542045E-002,-2.58533042E-002,9.06257033E-002,8.49585086E-002,8.65213498E-
002,5
.84013164E-002,-7.72946924E-002,6.65430725E-002,-1.64568517E-002,3.23978886
E-002
,2.24988302E-003,3.02566844E-003,-2.43405364E-002,9.75424573E-002,4.1463053
8E-00
3,1.89351812E-002,-1.10467218E-001,-1.24333188E-001,-2.36738548E-002,7.5427
7706E
-002,-1.64660662E-002,-1.38906585E-002,3.42438952E-003,-1.88432514E-005,-2.
47511
379E-002,-3.42802797E-003,3.23110656E-003,4.24311385E-002,6.59448802E-002,-
3.311
67318E-002,-5.14010936E-002,2.38897409E-002,-9.00154635E-002]"}

3728 rows selected.
```

The embedding results contain:

- `embed_id`: ID number of each vector embedding
- `embed_data`: Input text that is transformed into embeddings
- `embed_vector`: Generated vector representations

Related Topics

- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Convert File to Embeddings Within Oracle AI Database

Directly extract vector embeddings from a PDF document, using a single-step statement, by accessing a vector embedding model stored in the database.

Here, you perform a file-to-text-to-chunks-to-embeddings transformation by calling a set of `DBMS_VECTOR_CHAIN.UTL` functions in a single `CREATE TABLE` statement.

This statement creates a relational table (`doc_chunks`) from unstructured text chunks and the corresponding vector embeddings:

```
CREATE TABLE doc_chunks as
(select dt.id doc_id, et.embed_id, et.embed_data, to_vector(et.embed_vector)
embed_vector
 from
   documentation_tab dt,
   dbms_vector_chain.utl_to_embeddings(

dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(dt.data),
json('{"normalize":"all"}')),
       json('{"provider":"database", "model":"doc_model"}')) t,
   JSON_TABLE(t.column_value, '$[*]' COLUMNS (embed_id NUMBER PATH
'$.embed_id', embed_data VARCHAR2(4000) PATH '$.embed_data', embed_vector
CLOB PATH '$.embed_vector')) et
);
```

Note that each successive function depends on the output of the previous function, so the order of chains is important here. First, the output from `utl_to_text` (`dt.data` column) is passed as an input for `utl_to_chunks` and then the output from `utl_to_chunks` is passed as an input for `utl_to_embeddings`.

For complete example, run SQL Quick Start Using a Vector Embedding Model Uploaded into the Database, where you can see how to embed Oracle AI Database Documentation books in the `doc_chunks` table and perform similarity searches using vector indexes.

Generate and Use Embeddings for an End-to-End Search

First generate vector embeddings from textual content by using a vector embedding model stored in the database, and then populate and query a vector index. At query time, you also vectorize the query criteria on the fly.

This example covers the entire workflow of Oracle AI Vector Search (as explained in Understand the Stages of Data Transformations). If you are not yet familiar with the concepts beyond generating embeddings (such as creating and querying vector indexes), review the remaining sections before running this scenario.
To run an end-to-end similarity search workflow by accessing a vector embedding model stored in the database:

1. Start SQL*Plus and connect to Oracle AI Database as a local user:
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

      ```
      conn sys/password as sysdba


      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      EXTENT MANAGEMENT LOCAL
      SEGMENT SPACE MANAGEMENT AUTO;


      SET ECHO ON
      SET FEEDBACK 1
      SET NUMWIDTH 10
      SET LINESIZE 80
      ```

```
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (`docuser`) and grant necessary privileges:

```
drop user docuser cascade;


create user docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


grant DB_DEVELOPER_ROLE to docuser;
```

c. Create a local directory (`VEC_DUMP`) to store your input data and model files. Grant necessary privileges:

```
create or replace directory VEC_DUMP as '/my_local_dir/';


grant read, write on directory VEC_DUMP to docuser;

commit;
```

d. Connect as the local user (`docuser`):

```
conn docuser/password
```

2. Create a relational table (`documentation_tab`) and store your textual content in it.

```
drop table documentation_tab purge;


create table documentation_tab (id number, text clob);


insert into documentation_tab values (1,
    'Analytics empowers business analysts and consumers with modern, AI-
powered, self-service analytics capabilities for data preparation,
visualization, enterprise reporting, augmented analysis, and natural
language processing.
     Oracle Analytics Cloud is a scalable and secure public cloud service
that provides capabilities to explore and perform collaborative analytics
for you, your workgroup, and your enterprise.

     Oracle Analytics Cloud is available on Oracle Cloud Infrastructure
Gen 2 in several regions in North America, EMEA, APAC, and LAD when you
subscribe through Universal Credits. You can subscribe to Professional
Edition or Enterprise Edition.');

insert into documentation_tab values (3,
    'Generative AI Data Science is a fully managed and serverless platform
for data science teams to build, train, and manage machine learning models
```

```
in the Oracle Cloud Infrastructure.');

insert into documentation_tab values (4,
    'Language allows you to perform sophisticated text analysis at scale.
Using the pretrained and custom models, you can process unstructured text
to extract insights without data science expertise.
     Pretrained models include sentiment analysis, key phrase extraction,
text classification, and named entity recognition. You can also train
custom models for named entity recognition and text
     classification with domain specific datasets. Additionally, you can
translate text across numerous languages.');

insert into documentation_tab values (5,
    'When you work with Oracle Cloud Infrastructure, one of the first
steps is to set up a virtual cloud network (VCN) for your cloud resources.
This topic gives you an overview of Oracle Cloud
     Infrastructure Networking components and typical scenarios for using
a VCN. A virtual, private network that you set up in Oracle data centers.
It closely resembles a traditional network, with
     firewall rules and specific types of communication gateways that you
can choose to use. A VCN resides in a single Oracle Cloud Infrastructure
region and covers one or more CIDR blocks
     (IPv4 and IPv6, if enabled). See Allowed VCN Size and Address Ranges.
The terms virtual cloud network, VCN, and cloud network are used
interchangeably in this documentation.
     For more information, see VCNs and Subnets.');

insert into documentation_tab values (6,
    'NetSuite banking offers several processing options to accurately
track your income. You can record deposits to your bank accounts to
capture customer payments and other monies received in the
     course of doing business. For a deposit, you can select payments
received for existing transactions, add funds not related to transaction
payments, and record any cash received back from the bank.');

commit;
```

3. Load your embedding model by calling the `load_onnx_model` procedure.

```
EXECUTE dbms_vector.drop_onnx_model(model_name => 'doc_model', force =>
true);


EXECUTE dbms_vector.load_onnx_model(
   'VEC_DUMP',
   'my_embedding_model.onnx',
   'doc_model',
   json('{"function" : "embedding", "embeddingOutput" : "embedding" ,
"input": {"input": ["DATA"]}}')
);
```

In this example, the procedure loads an ONNX model file, named `my_embedding_model.onnx` from the `VEC_DUMP` directory, into the database as `doc_model`. You must replace `my_embedding_model.onnx` with an ONNX export of your embedding model and `doc_model` with the name under which the imported model is stored in the database.

> **Note**
>
> If you do not have an embedding model in ONNX format, then perform the steps listed in ONNX Pipeline Models: Text Embedding.

4. Create a relational table (`doc_chunks`) to store unstructured data chunks and associated vector embeddings, by using `doc_model`.

```
create table doc_chunks as (
  SELECT d.id id,
      l.chunk_id chunk_id,
         l.chunk_offset chunk_offset,
         l.chunk_length chunk_length,
         l.chunk_text chunk,
         vector_embedding(doc_model using l.chunk_text as data) vector
  FROM documentation_tab d,
       lateral( select vc.*,
               rownum vc.chunk_id
                from vector_chunks(d.text by words max 100 overlap 10
split RECURSIVELY) vc
               ) l
);
```

The `CREATE TABLE` statement reads the text from the `DOCUMENTATION_TAB` table, first applies the `VECTOR_CHUNKS` SQL function to split the text into chunks based on the specified chunking parameters, and then applies the `VECTOR_EMBEDDING` SQL function to generate corresponding vector embedding on each resulting chunk text.

5. Explore the `doc_chunks` table by selecting rows from it to see the chunked output.

```
desc doc_chunks;
set linesize 100
set long 1000
col id for 999
col chunk_id for 99999
col chunk_offset for 99999
col chunk_length for 99999
col chunk for a30
col vector for a100


select id, chunk_id, chunk_offset, chunk_length, chunk from doc_chunks;
```

The chunking output returns a set of seven chunks, which are split recursively, that is, using the `BLANKLINE`, `NEWLINE`, `SPACE`, `NONE` sequence. Note that Document 5 produces two chunks when the maximum word limit of `100` is reached.

You can see that the first chunk ends at a blank line. The text from the first chunk overlaps onto the second chunk, that is, 10 words (including comma and period; underlined below) are overlapping. Similarly, there is an overlap of 10 (also underlined below) between the fifth and sixth chunks.

```
  ID CHUNK_ID CHUNK_OFFSET CHUNK_LENGTH CHUNK
---- -------- ------------ ------------ ------------------------------
   1      1         1        418  Analytics empowers business an
                              alysts and consumers with mode
                              rn, AI-powered, self-service a
                              nalytics capabilities for data
                              preparation, visualization, e
                              nterprise reporting, augmented
                              analysis, and natural languag
                              e processing.
                              Oracle Analytics Cloud is
                              a scalable and secure public
                              cloud service that provides ca
                              pabilities to explore and perf
                              orm collaborative analytics for
                              you, your workgroup, and your
                              enterprise.

   1      2         373      291 for you, your workgroup, and
                              your enterprise.
                              Oracle Analytics Cloud is
                              available on Oracle Cloud Inf
                              rastructure Gen 2 in several r
                              egions in North America, EMEA,
                              APAC, and LAD when you subscr
                              ibe through Universal Credits.
                              You can subscribe to Professi
                              onal Edition or Enterprise Edi
                              tion.

   3      1        1      180  Generative AI Data Science is
                              a fully managed and serverless
                              platform for data science tea
                              ms to build, train, and manage
                              machine learning models in th
                              e Oracle Cloud Infrastructure.

   4      1        1      505  Language allows you to perform
                              sophisticated text analysis a
                              t scale. Using the pretrained
                              and custom models, you can pro
                              cess unstructured text to extr
                              act insights without data scie
                              nce expertise.
                              Pretrained models include
                              sentiment analysis, key phras
                              e extraction, text classificat
                              ion, and named entity recognit
                              ion. You can also train custom
                              models for named entity recog
                              nition and text
                              classification with domai
```

```
                                       n specific datasets. Additiona
                                       lly, you can translate text ac
                                       ross numerous languages.

         5     1        1      386  When you work with Oracle Clou
                                       d Infrastructure, one of the f
                                       irst steps is to set up a virt
                                       ual cloud network (VCN) for yo
                                       ur cloud resources. This topic
                                       gives you an overview of Orac
                                       le Cloud
                                       Infrastructure Networking
                                       components and typical scenar
                                       ios for using a VCN. A virtual
                                       , private network that you set
                                       up in Oracle data centers. It
                                       closely resembles a tradition
                                       al network, with

         5     2          329     474  centers. It closely resembles
                                       a traditional network, with
                                       firewall rules and specif
                                       ic types of communication gate
                                       ways that you can choose to us
                                       e. A VCN resides in a single O
                                       racle Cloud Infrastructure reg
                                       ion and covers one or more CID
                                       R blocks
                                       (IPv4 and IPv6, if enable
                                       d). See Allowed VCN Size and A
                                       ddress Ranges. The terms virtu
                                       al cloud network, VCN, and clo
                                       ud network are used interchang
                                       eably in this documentation.
                                       For more information, see
                                       VCNs and Subnets.

         6     1        1      393  NetSuite banking offers severa
                                       l processing options to accura
                                       tely track your income. You ca
                                       n record deposits to your bank
                                       accounts to capture customer
                                       payments and other monies rece
                                       ived in the
                                       course of doing business.
                                       For a deposit, you can select
                                       payments received for existin
                                       g transactions, add funds not
                                       related to transaction payment
                                       s, and record any cash receive
                                       d back from the bank.

7 rows selected.
```

6. Explore the first vector result by selecting rows from the `doc_chunks` table to see the embedding output.

```
select vector from doc_chunks where rownum <= 1;
```

An excerpt from the output is as follows:

```
[1.18813422E-002,2.53968383E-003,-5.33896387E-002,1.46877998E-003,5.7720981
5E-002,-1.58939194E-002,3
.12595293E-002,-1.13087103E-001,8.5138239E-002,1.10731693E-002,3.70671228E-
002,4.03710492E-002,1.503
95066E-001,3.31836529E-002,-1.98343433E-002,6.16453104E-002,4.2827677E-002,
-4.02921103E-002,-7.84291
551E-002,-4.79201972E-002,-5.06678E-002,-1.36317732E-002,-3.7761624E-003,-2
.3332756E-002,1.42400926E
-002,-1.11553416E-001,-3.70503664E-002,-2.60722954E-002,-1.2074843E-002,-3.
55089158E-002,-1.03518805
E-002,-7.05051869E-002,5.63110895E-002,4.79055084E-002,-1.46315445E-003,8.8
3129537E-002,5.12795225E-
002,7.5858552E-003,-4.13030013E-002,-5.2099824E-002,5.75958602E-002,3.72097
567E-002,6.11167103E-002,
,-1.23207876E-003,-5.46219759E-003,3.04734893E-002,1.80617068E-002,-2.85708
476E-002,-1.01670986E-002
,6.49402961E-002,-9.76506807E-003,6.15146831E-002,5.27246818E-002,7.4499443
2E-002,-5.86469211E-002,8
.84285953E-004,2.77456306E-002,1.99283361E-002,2.37570312E-002,2.33389344E-
002,-4.07911092E-002,-7.6
1070028E-002,1.23929314E-001,6.65794984E-002,-6.15389943E-002,2.62510721E-0
02,-2.48490628E-002]
```

7. Create an index on top of the `doc_chunks` table's `vector` column.

```
create vector index vidx on doc_chunks (vector)
   organization neighbor partitions
   with target accuracy 95
   distance EUCLIDEAN parameters (
    type IVF,
    neighbor partitions 2);
```

8. Run queries using the vector index.

   - Query about Machine Learning:

```
select id, vector_distance(
    vector,
    vector_embedding(doc_model using 'machine learning models' as data),
    EUCLIDEAN) results
FROM doc_chunks order by results;



  ID    RESULTS
---- ----------
   3 1.074E+000
   4 1.086E+000
```

```
   5 1.212E+000
   5 1.296E+000
   1 1.304E+000
   6 1.309E+000
   1 1.365E+000

7 rows selected.
```

- Query about Generative AI:

```
select id, vector_distance(
    vector,
    vector_embedding(doc_model using 'gen ai' as data),
    EUCLIDEAN) results
FROM doc_chunks order by results;


  ID    RESULTS
---- ----------
   4 1.271E+000
   3 1.297E+000
   1 1.309E+000
   5  1.32E+000
   1 1.352E+000
   5 1.388E+000
   6 1.424E+000

7 rows selected.
```

- Query about Networks:

```
select id, vector_distance(
  vector,
  vector_embedding(doc_model using 'computing networks' as data),
  MANHATTAN) results
FROM doc_chunks order by results;


  ID    RESULTS
---- ----------
   5 1.387E+001
   5 1.441E+001
   3 1.636E+001
   1 1.707E+001
   4 1.758E+001
   1 1.795E+001
   6 1.902E+001

7 rows selected.
```

- Query about Banking:

```
select id, vector_distance(
    vector,
    vector_embedding(doc_model using 'banking, money' as data),
```

```
    MANHATTAN) results
FROM doc_chunks order by results;


  ID    RESULTS
---- ----------
   6 1.363E+001
   1 1.969E+001
   5 1.978E+001
   5 1.997E+001
   3 1.999E+001
   1 2.058E+001
   4 2.079E+001

7 rows selected.
```

Related Topics

- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Configure Chunking Parameters

Oracle AI Vector Search provides many parameters for chunking text data, such as `SPLIT [BY]`, `OVERLAP`, or `NORMALIZE`. In these examples, you can see how to configure these parameters to define your own chunking specifications and strategies, so that you can create meaningful chunks.

- Explore Chunking Techniques and Examples
  Review these examples of all the supported chunking parameters. These examples can provide an idea on what are good and bad chunking techniques, and thus help you define a strategy when chunking your data.
- Create and Use Custom Vocabulary
  Create and use your own vocabulary of tokens when chunking data.
- Create and Use Custom Language Data
  Create and use your own language-specific conditions (such as common abbreviations) when chunking data.

Explore Chunking Techniques and Examples

Review these examples of all the supported chunking parameters. These examples can provide an idea on what are good and bad chunking techniques, and thus help you define a strategy when chunking your data.

Here, you can see how the following sample text of five lines is split when you apply various chunking parameters to it:

1[15]Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
[blank line 1]
2[05] About Oracle AI Vector Search
3[33] Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
[blank line 2]
4[07] Why Use Oracle AI Vector Search?
5[29]The biggest benefit of Oracle AI Vector Search is that semantic search on unstructured data
can be combined with relational search on business data in one single system.

Note the following:

- The lines are numbered as a reference for the explanations and include the word count in square brackets (for example, **`1[15]`**). The blank lines are also noted.
- The start and end boundaries of chunks are represented with colored markers.
- To perform examples with the `BY VOCABULARY` mode, you must create custom vocabulary beforehand (for example, `DOC_VOCAB`). See [Create and Use Custom Vocabulary](#).

Example 4-2 BY chars MAX 200 OVERLAP 0 SPLIT BY none

This example shows the simplest form of chunking, where you split the text by a fixed number of characters (including the end-of-line characters), at whatever point that occurs in the text.

The text from the first chunk is split at an absolute maximum character of `200`, which divides the word `indexes` between the first two chunks. Similarly, you can see the word `Oracle` splitting between second and third chunks.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search on unstructured data
can be combined with relational search on business data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY chars MAX 200 OVERLAP 0
                                       SPLIT BY none LANGUAGE american NORMALIZE none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
-------------------------------------------------------------------------
1         200          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized ind

201         200          exes that are designed for Artificial Intelligence
(AI) workloads that allow you to query data based on semantics, rather than
keywords.

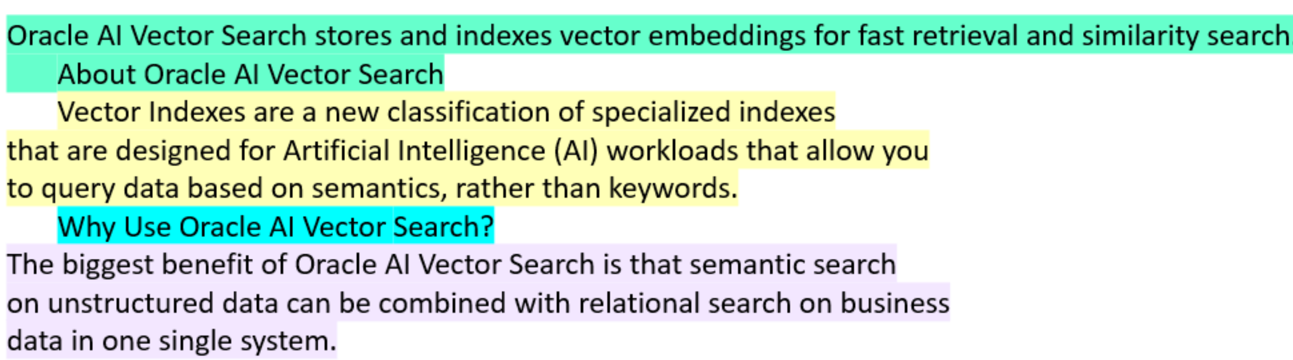
                          Why Use Oracle AI Vector Search?
                          The biggest benefit of O

401         146          racle AI Vector Search is that semantic search on
unstructured data can be combined with relational search on business data in
one single system.
```

Example 4-3 BY chars MAX 200 OVERLAP 0 SPLIT BY newline

In this example, the text is split into four chunks at new lines, if possible, within the maximum limit of `200` characters.

The text from the first chunk is split after the second line because the third line would exceed the maximum. The third line fits within the maximum perfectly. The fourth and fifth line would also exceed the maximum, so it produces two chunks.


Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY chars MAX 200 OVERLAP 0
                           SPLIT BY newline LANGUAGE american NORMALIZE none)
C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1          138         Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search

143          196         Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads that allow you to query data based on semantics, rather than
keywords.

343          33          Why Use Oracle AI Vector Search?

377          170         The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with relational
search on business data in one single system.
```

Example 4-4 BY chars MAX 200 OVERLAP 0 SPLIT BY recursively

In this example, the text is split into five chunks recursively using blank lines, newlines and then spaces, if possible, within the maximum of `200` characters.

The first chunk is split after the first blank line because including the text after the second blank line would exceed the maximum. The second passage exceeds the maximum on its own, so it is broken into two chunks at the new lines. Similarly, the third section is also broken into two chunks at the new lines.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY chars MAX 200 OVERLAP 0
                           SPLIT BY recursively LANGUAGE american NORMALIZE
none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1            104          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

109         30           About Oracle AI Vector Search

143         196          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads that allow you to query data based on semantics, rather than
keywords.

343         33           Why Use Oracle AI Vector Search?

377         170          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with relational
search on business data in one single system.
```

Example 4-5 BY words MAX 40 OVERLAP 0 SPLIT BY none

In this example, the text is split into three chunks at an absolute maximum word of `40`, the third line after `wordloads` and the fifth line after `with`.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
    About Oracle AI Vector Search
    Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
    Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 0
                           SPLIT BY none LANGUAGE american NORMALIZE none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1          266          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search
```

```
                              Vector Indexes are a new classification of
specialized indexes that are designed for Artificial
                              Intelligence (AI) workloads

267         223          that allow you to query data based on semantics,
rather than keywords.

                              Why Use Oracle AI Vector Search?
                              The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with


490         57           relational search on business data in one single
system.
```

Example 4-6   BY words MAX 40 OVERLAP 0 SPLIT BY newline

In this example, the text is split into chunks at new lines, if possible, within the maximum of `40` words.

The first chunk (of 21 words) is split after the second line, as the third line would exceed the maximum number of words (21+33 words). The third and fourth lines fit within the maximum. The fifth line is 29 words, so fits in the last chunk.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
    About Oracle AI Vector Search
    Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
    Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 0
                               SPLIT BY newline LANGUAGE american NORMALIZE none)
C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1          138          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                              About Oracle AI Vector Search
```

```
143         233          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads that allow you to query data based on semantics, rather than
keywords.

                          Why Use Oracle AI Vector Search?

377          170          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with relational
search on business data in one single system.
```

Example 4-7 BY words MAX 40 OVERLAP 0 SPLIT BY recursively

In this example, the chunks are split by words recursively using blank lines, newlines, and spaces.

The text after the second blank line exceeds the maximum words, so the first chunk ends at the first blank line. The second chunk (of 38 words) ends at the next blank line. The final chunk (of 35 words) consists of the rest of the input.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
    About Oracle AI Vector Search
    Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
    Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 0
                           SPLIT BY recursively LANGUAGE american NORMALIZE
none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1         104          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

109         230          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial
                          Intelligence (AI) workloads that allow you to query
data based on semantics, rather than keywords.
```

```
343         204          Why Use Oracle AI Vector Search?
                          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be
                          combined with relational search on business data in
one single system.
```

Example 4-8 BY vocabulary MAX 40 OVERLAP 0 SPLIT BY none

In this example, the text is split into four chunks at an absolute maximum vocabulary token of 40, which contrasts with the three chunks produced in Example 4-5. This is because vocabulary tokens include pieces of words, so the chunk text is generally smaller than simple word splitting.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY vocabulary doc_vocab MAX 40
OVERLAP 0
                           SPLIT BY none LANGUAGE american NORMALIZE none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1         157          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search
                          Vector Indexes
158         156          are a new classification of specialized indexes that
are designed for Artificial Intelligence (AI) workloads that allow you to
query data based on semantics

314          150          , rather than keywords.

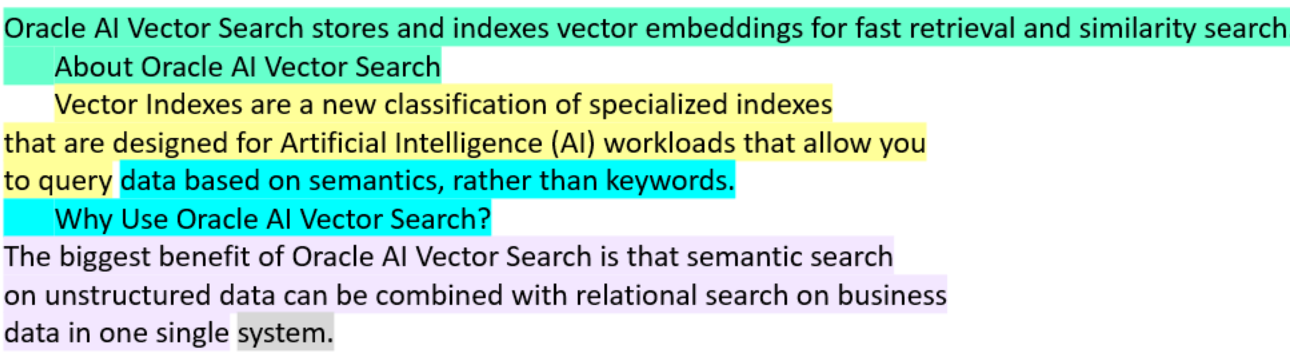
                          Why Use Oracle AI Vector Search?
                          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured
```

```
464         83          data can be combined with relational search on
business data in one single system.
```

Example 4-9 BY vocabulary MAX 40 OVERLAP 0 SPLIT BY newline

In this example, the text is split into five chunks with newlines, using an absolute maximum vocabulary token of 40, which contrasts with Example 4-6.

Vocabulary tokens include pieces of words, so the chunk text is generally smaller than simple word splitting. This example produces five chunks rather than three in Example 4-6, with the middle passage split into two, and the final word unable to fit into the fourth chunk.



Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY vocabulary doc_vocab MAX 40
OVERLAP 0
                           SPLIT BY newline LANGUAGE american NORMALIZE none)
C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1         138          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search

143         148          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads that allow you to query

291         85           data based on semantics, rather than keywords.

                          Why Use Oracle AI Vector Search?

377         162          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with relational
search on business data in one single
```

```
539         8            system.
```

Example 4-10 BY vocabulary MAX 40 OVERLAP 0 SPLIT BY recursively

In this example, the text is split into seven chunks recursively using blank lines, new lines, and spaces and an absolute maximum vocabulary token of 40, which contrasts with the three chunks produced in Example 4-7.

Vocabulary tokens include pieces of words, so the chunk text is generally smaller than simple word splitting. This example produces seven chunks with the middle passage split into three and the final passage split into three.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY vocabulary doc_vocab MAX 40
OVERLAP 0
                           SPLIT BY recursively LANGUAGE american NORMALIZE
none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1         104          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

109         30           About Oracle AI Vector Search

143         148          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads that allow you to query

291         48           data based on semantics, rather than keywords.

343         33           Why Use Oracle AI Vector Search?

377         162          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be combined with relational
```

```
search on business data in one single

539         8            system.
```

Example 4-11   BY words MAX 40 OVERLAP 5 SPLIT BY none

This example is similar to Example 4-5, except that an overlap of 5 is used here.

The first chunk ends at the maximum 40 words (after `workloads`). The second chunk overlaps with the last 5 words including parentheses of the first chunk, and ends after `unstructured`. The overlapping words are underlined below. The third chunk overlaps with the last 5 words, which are also underlined.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial **Intelligence (AI) workloads** that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is **that semantic search**
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 5
                           SPLIT BY none LANGUAGE american NORMALIZE none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
-------------------
1         266         Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                         About Oracle AI Vector Search
                         Vector Indexes are a new classification of
specialized indexes that are designed for Artificial Intelligence (AI)
workloads

239         225         Intelligence (AI) workloads that allow you to query
data based on semantics, rather than keywords.

                         Why Use Oracle AI Vector Search?
                         The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured

427         120         that semantic search on unstructured data can be
combined with relational search on business data in one single system.
```

Example 4-12 BY words MAX 40 OVERLAP 5 SPLIT BY newline

This example is similar to Example 4-6, except that an overlap of 5 is used here. The overlapping portion of a chunk must obey the same split condition, in this case must begin on a new line.

The first chunk ends at the second line, as the third line would exceed the maximum 40 words. The second chunk starts with the second line of 5 words of the first chunk (underlined below) and ends at the third line. The third chunk has no overlap because the preceding line exceeds the maximum of 5.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
    About Oracle AI Vector Search
    Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
    Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 5
                           SPLIT BY newline LANGUAGE american NORMALIZE none)
C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
-------------------------------------------------------------------------------
---------------------
1         138          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search

109         230          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial
                          Intelligence (AI) workloads that allow you to query
data based on semantics, rather than keywords.

343         204          Why Use Oracle AI Vector Search?
                          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be
                          combined with relational search on business data in
one single system.
```

Example 4-13 BY words MAX 40 OVERLAP 5 SPLIT BY recursively

This example is similar to Example 4-7, except that an overlap of 5 is used here. The overlapping portion of a chunk must obey the same split condition, in this case must begin at either a blank line, new line, or space.

The text after the second blank line exceeds the maximum words, so the first chunk ends at the first blank line. The second chunk overlaps with 5 words (beginning on a space; underlined below) and includes the second line, but excludes the third line of 33 words. The third chunk overlaps 5 words and ends on the second blank line. The fourth chunk consumes the rest of the input.

Oracle AI Vector Search stores and indexes vector embeddings for fast **retrieval and similarity search.**
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, **rather than keywords.**
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 5
                           SPLIT BY recursively LANGUAGE american NORMALIZE
none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
-------------------------------------------------------------------------------
--------------------
1         104         Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

71          68          retrieval and similarity search.

                         About Oracle AI Vector Search

109         230         About Oracle AI Vector Search
                         Vector Indexes are a new classification of
specialized indexes that are designed for Artificial
                         Intelligence (AI) workloads that allow you to query
data based on semantics, rather than keywords.

316          231         rather than keywords.

                         Why Use Oracle AI Vector Search?
                         The biggest benefit of Oracle AI Vector Search is
```

```
that semantic search on unstructured data can be
                         combined with relational search on business data in
one single system.
```

Example 4-14 BY chars MAX 200 OVERLAP 0 SPLIT BY none NORMALIZE none

This example is the same as the first one (Example 4-2), to contrast with the next example (Example 4-15) with normalization.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
    About Oracle AI Vector Search
    Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
    Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY chars MAX 200 OVERLAP 0
                           SPLIT BY none LANGUAGE american NORMALIZE none) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1         200          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized ind

201         200           exes that are designed for Artificial Intelligence
(AI) workloads that allow you to query data based
                          on semantics, rather than keywords.

                          Why Use Oracle AI Vector Search?
                          The biggest benefit of O

401         146           racle AI Vector Search is that semantic search on
unstructured data can be combined with relational
                          search on business data in one single system.
```

Example 4-15 BY chars MAX 200 OVERLAP 0 SPLIT BY none NORMALIZE whitespace

This example enables `whitespace` normalization, which collapses redundant white space to produce more content within a chunk maximum.

The first chunk extends 8 more characters due to the two indented lines of 4 spaces each (marked with underscores _ below). The second chunk extends 4 more characters due to the one indented line of 4 total spaces. The third chunk has the remaining input.

This example shows that the chunk length (normally in bytes) can differ from the chunk text's size. The `CHUNK_OFFSET` and `CHUNK_LENGTH` represent the original source location of the chunk.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
____About Oracle AI Vector Search
____Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
____Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY chars MAX 200 OVERLAP 0
                           SPLIT BY none LANGUAGE american NORMALIZE
whitespace) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1          208          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

                          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized indexes tha

209          205          t are designed for Artificial Intelligence (AI)
workloads that allow you to query data based on sema
                          ntics, rather than keywords.

                          Why Use Oracle AI Vector Search?
                          The biggest benefit of Oracle AI Vect

414          133          or Search is that semantic search on unstructured
data can be combined with relational search on business data in one single
system.
```

Example 4-16 BY words MAX 40 OVERLAP 0 SPLIT BY sentence LANGUAGE American

This example uses end-of-sentence splitting which uses language-specific data and heuristics (such as sentence punctuations, contextual rules, or common abbreviations) to determine likely sentence boundaries. Three chunks are produced, each ending at the periods.

You can use this technique to keep your text intact for chunks that contain many split sentences. Otherwise, the text may lose semantic context and may not be useful for queries that target specific information.

Oracle AI Vector Search stores and indexes vector embeddings for fast retrieval and similarity search.
About Oracle AI Vector Search
Vector Indexes are a new classification of specialized indexes
that are designed for Artificial Intelligence (AI) workloads that allow you
to query data based on semantics, rather than keywords.
Why Use Oracle AI Vector Search?
The biggest benefit of Oracle AI Vector Search is that semantic search
on unstructured data can be combined with relational search on business
data in one single system.

Syntax:

```
SELECT C.*
FROM documentation_tab D, VECTOR_CHUNKS(D.text BY words MAX 40 OVERLAP 0
                           SPLIT BY sentence LANGUAGE american NORMALIZE
NONE) C;
```

Output:

```
CHUNK_OFFSET CHUNK_LENGTH CHUNK_TEXT
------------------------------------------------------------------------------
--------------------
1            102          Oracle AI Vector Search stores and indexes vector
embeddings for fast retrieval and similarity search.

109          228          About Oracle AI Vector Search
                          Vector Indexes are a new classification of
specialized indexes that are designed for Artificial
                          Intelligence (AI) workloads that allow you to query
data based on semantics, rather than keywords.

343         203          Why Use Oracle AI Vector Search?
                          The biggest benefit of Oracle AI Vector Search is
that semantic search on unstructured data can be
                          combined with relational search on business data in
one single system.
```

Example 4-17 BY words MAX 40 OVERLAP 0 SPLIT BY sentence LANGUAGE Simplified Chinese

In continuation with the preceding example, this example uses a Simplified Chinese text as the input to specify language-specific sentence chunking.

The output contains four chunks, each ending at the periods:

使用 My Oracle Support 之前，您的用户概要信息中必须至少具有一个客户服务号。客户服务号是标识您所在组织的唯一参考号。使用 My Oracle Support 门户向您的概要信息中添加一个客户服务号。有关详细信息，请参阅 My Oracle Support 帮助的“如何将客户服务号添加到概要信息？

For the purpose of clarity, in this example, `documentation_tab` is a `CLOB` inserted with the following `ChineseDoc.txt` document:

```
使用 My Oracle Support 之前，您的用⊠概要信息中必⊠至少具有一个客⊠服⊠号。客⊠服⊠号是
⊠⊠您所在⊠⊠的唯一参考号。使用 My Oracle Support ⊠⊠向您的概要信息中添加一个客⊠服⊠
号。有关⊠⊠信息，⊠参⊠ My Oracle Support 帮助的“如何将客⊠服⊠号添加到概要信息？
```

Perform the chunking operation as follows:

```
-- create a relational table

DROP TABLE IF EXISTS documentation_tab;
CREATE TABLE documentation_tab (
    id   NUMBER,
    text CLOB);

-- create a local directory and store the document into the table

CREATE OR REPLACE DIRECTORY VEC_DUMP AS '/my_local_dir/';
CREATE OR REPLACE PROCEDURE my_clob_from_file(
    p_dir in varchar2,
    p_file in varchar2,
    p_id in number
  ) AS
  dest_loc CLOB;
  v_bfile bfile := null;
  v_lang_context number := dbms_lob.default_lang_ctx;
  v_dest_offset integer := 1;
  v_src_offset integer := 1;
  v_warning number;
BEGIN
        insert into documentation_tab values(p_id,empty_clob()) returning text
        into dest_loc;

                v_bfile := BFileName(p_dir, p_file);

                dbms_lob.open(v_bfile, dbms_lob.lob_readonly);
                dbms_lob.loadClobFromFile(
                                dest_loc,
                                v_bfile,
                                dbms_lob.lobmaxsize,
                                v_dest_offset,
                                v_src_offset,
                                873,
                                v_lang_context,
                                v_warning);
```

```
        dbms_lob.close(v_bfile);
END my_clob_from_file;
/

show errors;

-- transform clob into chunks

exec my_clob_from_file('VEC_DUMP', 'ChineseDoc.txt', 1);

SELECT rownum as id, C.chunk_offset pos, C.chunk_length as siz,
       REPLACE(SUBSTR(C.chunk_text,1,15),CHR(10),'_') as beg,
       '...' as rng,
       REPLACE(SUBSTR(C.chunk_text,-15),CHR(10),'_') as end
FROM documentation_tab D, VECTOR_CHUNKS(to_char(D.text) BY words
                                 MAX 40
                                 OVERLAP 0
                                 SPLIT BY sentence
                                 LANGUAGE "simplified chinese"
                                 NORMALIZE none) C;
```

Output:

```
ID   POS  SIZ  BEG                          RNG END
---- ---- ---- ---------------------------  --- ------------------------
 1    1   103  使用 My Oracle Su             ... 中必☒至少具有一个客☒服☒号。
 2  104    60  客☒服☒号是☒☒您所在☒☒的唯     ... 是☒☒您所在☒☒的唯一参考号。
 3  164    85  使用 My Oracle Su             ... 概要信息中添加一个客☒服☒号。
 4  249   109  有关☒☒信息，☒参☒ My O        ... 何将客☒服☒号添加到概要信息？
```

Related Topics

- VECTOR_CHUNKS
  Use VECTOR_CHUNKS to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.
- UTL_TO_CHUNKS
  Use the DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS chainable utility function to split a large plain text document into smaller chunks of text.

Create and Use Custom Vocabulary

Create and use your own vocabulary of tokens when chunking data.

Here, you use the chunker helper function CREATE_VOCABULARY from the DBMS_VECTOR_CHAIN package to load custom vocabulary. This vocabulary file contains a list of tokens, recognized by your vector embedding model's tokenizer.

1. Connect as a local user and prepare your data dump directory.
   a. Log in to SQL*Plus as the SYS user, connecting as SYSDBA:

      ```
      conn sys/password as sysdba


      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      ```

```
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (`docuser`) and grant necessary privileges:

```
drop user docuser cascade;


create user docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


grant DB_DEVELOPER_ROLE to docuser;
```

c. Create a local directory (`VEC_DUMP`) to store your vocabulary file. Grant necessary privileges:

```
create or replace directory VEC_DUMP as '/my_local_dir/';


grant read, write on directory VEC_DUMP to docuser;

commit;
```

d. Transfer the vocabulary file for your required model to the `VEC_DUMP` directory.

For example, if using the WordPiece tokenization, you can download and transfer the `vocab.txt` vocabulary file for "bert-base-uncased".

e. Connect as the local user (`docuser`):

```
conn docuser/password
```

2. Create a relational table (`doc_vocabtab`) to store your vocabulary tokens in it:

```
CREATE TABLE doc_vocabtab(token nvarchar2(64))
  ORGANIZATION EXTERNAL
  (default directory VEC_DUMP
   ACCESS PARAMETERS (RECORDS DELIMITED BY NEWLINE)
   location ('bert-vocabulary-uncased.txt'));
```

3. Create a vocabulary (`doc_vocab`) by calling `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY`:

```
DECLARE
  vocab_params clob := '{
                        "table_name"      : "doc_vocabtab",
```

```
                                   "column_name"     : "token",
                                   "vocabulary_name" : "doc_vocab",
                                   "format"          : "bert",
                                   "cased"           : false
                                  }';

BEGIN
  dbms_vector_chain.create_vocabulary(json(vocab_params));
END;
/
```

After loading the token vocabulary, you can now use the `BY VOCABULARY` chunking mode (with `VECTOR_CHUNKS` or `UTL_TO_CHUNKS`) to split data by counting the number of tokens.

Related Topics

- CREATE_VOCABULARY
  Use the `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY` chunker helper procedure to load your own token vocabulary file into the database.
- VECTOR_CHUNKS
  Use `VECTOR_CHUNKS` to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.
- UTL_TO_CHUNKS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS` chainable utility function to split a large plain text document into smaller chunks of text.

Create and Use Custom Language Data

Create and use your own language-specific conditions (such as common abbreviations) when chunking data.

Here, you use the chunker helper function `CREATE_LANG_DATA` from the `DBMS_VECTOR_CHAIN` package to load the data file for Simplified Chinese. This data file contains abbreviation tokens for your chosen language.

1. Connect as a local user and prepare your data dump directory.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

      ```
      conn sys/password as sysdba


      CREATE TABLESPACE tbs1
      DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
      EXTENT MANAGEMENT LOCAL
      SEGMENT SPACE MANAGEMENT AUTO;


      SET ECHO ON
      SET FEEDBACK 1
      SET NUMWIDTH 10
      SET LINESIZE 80
      SET TRIMSPOOL ON
      SET TAB OFF
      SET PAGESIZE 10000
      SET LONG 10000
      ```

b. Create a local user (docuser) and grant necessary privileges:

```
drop user docuser cascade;


create user docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


grant DB_DEVELOPER_ROLE to docuser;
```

c. Create a local directory (VEC_DUMP) to store your language data file. Grant necessary privileges:

```
create or replace directory VEC_DUMP as '/my_local_dir/';


grant read, write on directory VEC_DUMP to docuser;

commit;
```

d. Transfer the data file for your required language to the VEC_DUMP directory. For example, dreoszhs.txt for Simplified Chinese.

   To know the data file location for your language, see Supported Languages and Data File Locations.

e. Connect as the local user (docuser):

```
conn docuser/password
```

2. Create a relational table (doc_langtab) to store your abbreviation tokens in it:

```
CREATE TABLE doc_langtab(token nvarchar2(64))
  ORGANIZATION EXTERNAL
  (default directory VEC_DUMP
   ACCESS PARAMETERS (RECORDS DELIMITED BY NEWLINE CHARACTERSET AL32UTF8)
   location ('dreoszhs.txt'));
```

3. Create language data (doc_lang_data) by calling DBMS_VECTOR_CHAIN.CREATE_LANG_DATA:

```
DECLARE
  lang_params clob := '{
                         "table_name"      : "doc_langtab",
                         "column_name"     : "token",
                         "language"        : "simplified chinese",
                         "preference_name" : "doc_lang_data"
                       }';
BEGIN
  dbms_vector_chain.create_lang_data(json(lang_params));
END;
/
```

After loading the language data, you can now use language-specific chunking by specifying the LANGUAGE chunking parameter with VECTOR_CHUNKS or UTL_TO_CHUNKS.

Related Topics

- CREATE_LANG_DATA
  Use the `DBMS_VECTOR_CHAIN.CREATE_LANG_DATA` chunker helper procedure to load your own language data file into the database.
- VECTOR_CHUNKS
  Use `VECTOR_CHUNKS` to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.
- UTL_TO_CHUNKS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS` chainable utility function to split a large plain text document into smaller chunks of text.

5

Store Vector Embeddings

Store vector embeddings and associated unstructured data with your relational business data in Oracle AI Database.

- Create Tables Using the VECTOR Data Type
  You can declare a table's column as a `VECTOR` data type.
- Insert Vectors in a Database Table Using the INSERT Statement
  Once you create a table with a `VECTOR` data type column, you can directly insert vectors into the table using the `INSERT` statement.
- Load Vector Data Using SQL*Loader
  Use these examples to understand how you can load character and binary vector data.
- Unload and Load Vectors Using Oracle Data Pump
  Starting with Oracle Database 26ai, Oracle Data Pump enables you to use multiple components to load and unload vectors to databases.

Create Tables Using the VECTOR Data Type

You can declare a table's column as a `VECTOR` data type.

The following command shows a simple example:

```
CREATE TABLE my_vectors (id NUMBER, embedding VECTOR);
```

In this example, you don't have to specify the number of dimensions or their format, which are both optional. If you don't specify any of them, you can enter vectors of different dimensions with different formats. This is a simplification to help you get started with using vectors in Oracle AI Database.

> **Note**
>
> Such vectors typically arise from different embedding models. Vectors from different models (providing a different semantic landscape) are not comparable for use in similarity search.

Here's a more complex example that imposes more constraints on what you can store:

```
CREATE TABLE my_vectors (id NUMBER, embedding VECTOR(768, INT8)) ;
```

In this example, each vector that is stored:

- Must have 768 dimensions, and
- Each dimension will be formatted as an `INT8`.

The number of dimensions must be strictly greater than zero with a maximum of 65535 for non-`BINARY` vectors and 65528 for `BINARY` vectors. If you attempt to use larger values, the following error is raised:

```
ORA-51801: Invalid VECTOR type specification: Invalid dimension
count ('...'). Valid values can either be * (i.e. flexible) or an
integer between 1 and 65535.
```

`BINARY` vectors must have a dimension that is a multiple of 8. If the specified dimension is not a multiple of 8, the following error is raised:

```
ORA-51813: Vector of BINARY format should have a dimension count
that is a multiple of 8.
```

The possible dimension formats are:

- `INT8` (8-bit integers)
- `FLOAT32` (32-bit IEEE floating-point numbers)
- `FLOAT64` (64-bit IEEE floating-point numbers)
- `BINARY` (packed `UINT8` bytes where each dimension is a single bit)

Oracle AI Database automatically casts the values as needed.

Here is some simple code that shows you how to define and insert a `VECTOR` in a table:

```
DROP TABLE my_vect_tab PURGE;
CREATE TABLE my_vect_tab (v01 VECTOR(3, INT8));
INSERT INTO my_vect_tab VALUES ('[10, 20, 30]');

SELECT * FROM my_vect_tab;

V01
----------
[10,20,30]
```

You use a textual form to represent a vector in SQL statements. The `DENSE` textual form as shown in the preceding `INSERT` statement is a basic example: coordinate 1 has a value of 10, coordinate 2 has a value of 20, and coordinate 3 has a value of 30. You separate each coordinate with a comma and you enclose the list with square brackets.

So far, the vector types shown are, by default, `DENSE` vectors where each dimension is physically stored. All the definitions seen are equivalent to the following form:

```
VECTOR(number_of_dimensions, dimension_element_format, DENSE) or
VECTOR(number_of_dimensions, dimension_element_format, *)
```

However, you also have the option to create `SPARSE` vectors. In contrast to `DENSE` vectors, a sparse vector is a vector whose dimension values are expected to be mostly zero. When using `SPARSE` vectors, only the non-zero values are physically stored. You define sparse vectors using the following form:

```
VECTOR(number_of_dimensions, dimension_element_format, SPARSE)
```

> **Note**
>
> - It is not supported to have SPARSE storage in one row and DENSE storage in another row for the same vector column as the coding of the two representations are very different.
> - VECTOR(…, …, *) is always interpreted as DENSE.
> - Inverted File Flat (IVF) vector indexes cannot be created on top of SPARSE vectors. This restriction does not apply to Hierarchical Navigable Small World (HNSW) vector indexes.
> - Depending on the compiler used to compile the Oracle AI Database code, values are rounded to 7 or 8 significant digits, meaning values greater than the maximum allowed for FLOAT32 and FLOAT64 can be inserted without error when the number rounds down to the maximum. Keep this in mind when considering the necessary precision for the dimensions of the vectors being used for your use case.

The following table guides you through the possible declaration format for a VECTOR data type with a DENSE storage format:

Possible Declaration Format	Explanation
VECTOR	Vectors can have an arbitrary number of dimensions and formats.
VECTOR(*, *) VECTOR(*, *, *) VECTOR(*, *, DENSE)	Vectors can have an arbitrary number of dimensions and formats. VECTOR, VECTOR(*, *), VECTOR(*,*,*), and VECTOR(*, *, DENSE) are equivalent.
VECTOR(number_of_dimensions) VECTOR(number_of_dimensions, *) VECTOR(number_of_dimensions, *, *) VECTOR(number_of_dimensions, *, DENSE)	Vectors must all have the specified number of dimensions or an error is thrown. Every vector will have its dimensions stored without format modification. VECTOR(number_of_dimensions), VECTOR(number_of_dimensions, *), VECTOR(number_of_dimensions, *, *), and VECTOR(number_of_dimensions, *, DENSE) are equivalent.
VECTOR(*, dimension_element_format) VECTOR(*, dimension_element_format, *) VECTOR(*, dimension_element_format, DENSE)	Vectors can have an arbitrary number of dimensions, but their format will be up-converted or down-converted to the specified dimension_element_format (INT8, FLOAT32, FLOAT64, or BINARY). VECTOR(*, dimension_element_format), VECTOR(*, dimension_element_format, *), and VECTOR(*, dimension_element_format, DENSE) are equivalent.

The following table guides you through the possible declaration format for a VECTOR data type for sparse vectors.

Possible Declaration Format	Explanation
`VECTOR(*, *, SPARSE)`	Vectors can have an arbitrary number of dimensions and formats.
`VECTOR(number_of_dimensions, *, SPARSE)`	Vectors must all have the specified number of dimensions or an error is thrown. Every vector will have its dimensions stored without format modification.
`VECTOR(*, dimension_element_format, SPARSE)`	Vectors can have an arbitrary number of dimensions but their format will be up-converted or down-converted to the specified `dimension_element_format` (`INT8`, `FLOAT32`, `FLOAT64`, or `BINARY`)

The following SQL*Plus code example shows how the system interprets various vector definitions:

```
CREATE TABLE my_vect_tab (
     v1  VECTOR(3, FLOAT32),
     v2  VECTOR(2, FLOAT64),
     v3  VECTOR(1, INT8),
     v4  VECTOR(1024, BINARY),
     v5  VECTOR(1, *),
     v6  VECTOR(*, FLOAT32),
     v7  VECTOR(*, *),
     v8  VECTOR,
     v9  VECTOR(10),
     v10 VECTOR(*, *, DENSE),
     v11 VECTOR(1024, FLOAT32, DENSE),
     v12 VECTOR(1000, INT8, SPARSE),
     v13 VECTOR(*, INT8, SPARSE),
     v14 VECTOR(*, *, SPARSE),
     v15 VECTOR(2048, FLOAT32, *)
   );

Table created.

DESC my_vect_tab;
 Name                           Null?    Type
 --------------------------- -------- ---------------------------
 V1                                      VECTOR(3 , FLOAT32, DENSE)
 V2                                      VECTOR(2 , FLOAT64, DENSE)
 V3                                      VECTOR(1 , INT8, DENSE)
 V4                                      VECTOR(1024, BINARY, DENSE)
 V5                                      VECTOR(1 , *, DENSE)
 V6                                      VECTOR(* , FLOAT32, DENSE)
 V7                                      VECTOR(* , *, DENSE)
 V8                                      VECTOR(* , *, DENSE)
 v9                                      VECTOR(10, *, DENSE)
 v10                                     VECTOR(*, *, DENSE)
 v11                                     VECTOR(1024, FLOAT32, DENSE)
 v12                                     VECTOR(1000, INT8, SPARSE)
 v13                                     VECTOR(*, INT8, SPARSE)
 v14                                     VECTOR(*, *, SPARSE)
 v15                                     VECTOR(2048, FLOAT32, DENSE)
```

A vector can be `NULL` but its dimensions cannot (for example, you cannot have a `VECTOR` with a `NULL` dimension such as `[1.1, NULL, 2.2]`).

Note

Vectors (`DENSE` and `SPARSE`) are internally stored as Securefile BLOBs and most popular embedding model vector sizes are between 1.5KB and 12KB in size. You can use the following formula to determine the size of your vectors on disk:

- For `DENSE` vectors:
  *number of vectors * number of dimensions * size of your vector dimension type* (for example, a `FLOAT32` is equivalent to `BINARY_FLOAT` and is 4 bytes in size).
- For `SPARSE` vectors:
  *number of vectors * ((average number of sparse dimensions * 4 bytes) + (number of sparse dimensions * size of your vector dimension type))*.

Restrictions

You currently cannot define `VECTOR` columns in/as:

- IOTs (neither as Primary Key nor as non-Key column)
- Clusters/Cluster Tables
- Global Temp Tables
- (Sub)Partitioning Key
- Primary Key
- Foreign Key
- Unique Constraint
- Check Constraint (`NULL` and `NOT NULL` check constraints are supported)
- Default Value
- Modify Column
- Manual Segment Space Management (MSSM) tablespace (only SYS user can create `VECTOR`s as Basicfiles in MSSM tablespace)
- Continuous Query Notification (CQN) queries
- Non-vector indexes such as B-tree, Bitmap, Reverse Key, Text, Spatial indexes, etc

Oracle AI Database does not support the following SQL constructs with `VECTOR` columns:

- Distinct, Count Distinct
- Order By, Group By
- Join condition
- Comparison operators (e.g. >, <, =) etc

Oracle's Globally Distributed Database has the following restrictions on vector data types:

- Sharding keys: A distributed database only supports sharding keys on non-vector columns. The vector data can be distributed across shards using a primary key on any other non-vector column identified as a sharding key.

- Raft replication: A distributed database using the Raft replication method for high availability does not support vector columns.

Topics

- [Vectors in External Tables](#)
  External tables can be created with columns of type `VECTOR`, allowing vector embeddings represented in text or binary format stored in external files to be rendered as the `VECTOR` data type in Oracle AI Database.
- [Vectors in Distributed Database Tables](#)
  There is no new SQL syntax or keyword when creating sharded tables and duplicated tables with vector columns in a Globally Distributed Database; however, there are some requirements and restrictions to consider.
- [BINARY Vectors](#)
  In addition to `FLOAT32` (the default format for comma-separated string representations of vectors), `FLOAT64`, and `INT8` dimension formats, you can also use the `BINARY` dimension format.
- [SPARSE Vectors](#)
  The storage format of a vector can be specified as `SPARSE` or `DENSE`. Sparse vectors are vectors that typically have a large number of dimensions but with very few non-zero dimension values, while Dense vectors are vectors where every dimension stores a value, including zero-values.

Vectors in External Tables

External tables can be created with columns of type `VECTOR`, allowing vector embeddings represented in text or binary format stored in external files to be rendered as the `VECTOR` data type in Oracle AI Database.

The ability to store `VECTOR` data in external tables can be beneficial when considering the vast amount of data that is involved in AI workflows. Using external tables provides a convenient option to store vector embeddings and contextual information outside of the database while still using the database as a semantic search engine.

The following types of external files support `VECTOR` columns in external tables:

- CSV
- Parquet
- Avro
- ORC
- Dmp

External tables with `VECTOR` columns can be accessed by using the drivers `ORACLE_LOADER`, `ORACLE_DATAPUMP`, and `ORACLE_BIGDATA`. Once the chosen driver has loaded the file data into the database, you can interact with the external table rows in any SQL operation (supported by your preferred SQL interface), such as joins, SQL functions, aggregation, and so on.

Vectors of any dimension format and storage format are supported. If a vector is `SPARSE`, the data must be provided as text as opposed to an array or list format.

Columns of type `VECTOR` can be included in both explicitly created external tables as well as inline external tables created as part of a `SELECT` statement. The benefit of this approach is that there is no need to predefine a static table to access the vectors in the external table before loading them into the database. The external tables mapping is persisted only while the external table is in use by the SQL query. For an example, see Querying an Inline External Table, which shows how the external table mappings are created as part of the SQL query operation. Once the query has completed, the external table mapping is discarded from the database.

Additionally, the `row_limiting_clause` can be used in `SELECT` statements that reference external tables. Internal and external tables can be referenced in the same query. You can use the `CREATE TABLE AS SELECT` statement to create an internal table by selecting from an external table with `VECTOR` column(s). Similarly, you can use the `INSERT INTO SELECT` statement to insert values into an internal table from an external table called in the `SELECT` subquery.

> **Note**
>
> - External tables are not currently supported in multi-vector similarity searches.
> - HNSW indexes cannot currently be created on `VECTOR` columns stored in external tables.
> - IVF indexes are supported only for external Iceberg tables. For more information, see IVF Indexing on External Iceberg Tables.

Vector embeddings in external tables can be accessed for use in similarity searches in the same way as you would use an internal table, as in the following query:

```
SELECT id, embedding
FROM external_table
ORDER BY VECTOR_DISTANCE(embedding, '[1,1]', COSINE)
FETCH FIRST 3 ROWS ONLY WITH TARGET ACCURACY 90;
```

The following examples demonstrate the syntax used to create external tables with `VECTOR` columns depending on the access driver:

- Using `ORACLE_LOADER`:

  ```
  CREATE TABLE ext_vec_tab1(
    v1 VECTOR,
    v2 VECTOR
  )
    ORGANIZATION EXTERNAL
    (
      TYPE ORACLE_LOADER
      DEFAULT DIRECTORY my_dir
      ACCESS PARAMETERS
      (
        RECORDS DELIMITED BY NEWLINE
        FIELDS TERMINATED BY ":"
        MISSING FIELD VALUES ARE NULL
      )
      LOCATION('my_ext_vec_embeddings.csv')
    )
  REJECT LIMIT UNLIMITED;
  ```

- Using ORACLE_DATAPUMP:

```
-- First create the table with the loader
CREATE TABLE dp_ext_tab(
  country_code         VARCHAR2(5),
  country_name         VARCHAR2(50),
  country_language     VARCHAR2(50),
  country_vector       VECTOR(*,*)
)
  ORGANIZATION EXTERNAL
  (
    TYPE ORACLE_LOADER
    DEFAULT DIRECTORY my_dir
    ACCESS PARAMETERS
    (
      RECORDS DELIMITED BY NEWLINE
      FIELDS TERMINATED BY ":"
      MISSING FIELD VALUES ARE NULL
      (
        country_code         CHAR(5),
        country_name         CHAR(50),
        country_language     CHAR(50),
        country_vector       CHAR(10000)
      )
    )
    LOCATION ('ext_vectorcountries.dat')
    )
  PARALLEL 5
  REJECT LIMIT UNLIMITED;

-- Then generate the dmp file
CREATE TABLE ext_export_table
ORGANIZATION EXTERNAL
(
  TYPE ORACLE_DATAPUMP
  DEFAULT DIRECTORY my_dir
  LOCATION ('ext.dmp')
)
  AS SELECT * FROM dp_ext_tab;

-- Finally, create an external table with the datapump driver
CREATE TABLE dp_ext_tab_final
(
  country_code         VARCHAR2(5),
  country_name         VARCHAR2(50),
  country_language     VARCHAR2(50),
  country_vector       VECTOR(3, INT8)
)
  ORGANIZATION EXTERNAL
  (
    TYPE ORACLE_DATAPUMP
    DEFAULT DIRECTORY my_dir
    LOCATION ('ext.dmp')
  )
```

```
  PARALLEL 5
  REJECT LIMIT UNLIMITED;
```

- Using `ORACLE_BIGDATA`:

```
CREATE TABLE bd_ext_tab
(
  COL1 vector(5,INT8),
  COL2 vector(5,INT8),
  COL3 vector(5,INT8),
  COL4 vector(5,INT8)
)
  ORGANIZATION external
  (
    TYPE ORACLE_BIGDATA
    DEFAULT DIRECTORY my_dir
    ACCESS PARAMETERS
  (
    com.oracle.bigdata.credential.name\=OCI_CRED
    com.oracle.bigdata.credential.schema\=PDB_ADMIN
    com.oracle.bigdata.fileformat=parquet
    com.oracle.bigdata.debug=true
  )
  LOCATION ( 'https://swiftobjectstorage.<region>.oraclecloud.com/v1/
<namespace>/<filepath>/basic_vec_data.parquet' )
  )
REJECT LIMIT UNLIMITED PARALLEL 2;
```

- Querying an Inline External Table
  In this example, vectors in an external table of type `ORACLE_BIGDATA` are queried as part of a vector search.
- Performing a Semantic Similarity Search Using External Table
  See a SQL example plan that illustrates how you can use external tables as the data set for semantic similarity searches

See Also

- IVF Indexing on External Iceberg Tables
- *Oracle AI Database Utilities* for more information about external tables

Querying an Inline External Table

In this example, vectors in an external table of type `ORACLE_BIGDATA` are queried as part of a vector search.

```
select * from external (
    (
      COL1 vector,
      COL2 vector,
      COL3 vector,
      COL4 vector
    )
```

```
    TYPE ORACLE_BIGDATA
    DEFAULT DIRECTORY DEF_DIR1
    ACCESS PARAMETERS
    (
      com.oracle.bigdata.credential.name\=OCI_CRED
      com.oracle.bigdata.credential.schema\=PDB_ADMIN
      com.oracle.bigdata.fileformat=parquet
      com.oracle.bigdata.debug=true
    )
    location ( 'https://swiftobjectstorage.us-phoenix-1.oraclecloud.com/v1/
myvdataproject/BIGDATA_PARQUET/vector_data/basic_vector_data.parquet' )
    REJECT LIMIT UNLIMITED
  ) tkexobd_bd_vector_inline;
```

Performing a Semantic Similarity Search Using External Table

See a SQL example plan that illustrates how you can use external tables as the data set for semantic similarity searches

The following is an example of an explain plan for `select id`, embedding from `ext_table_3`, and using `order by vector_distance('[1,1]', embedding, cosine)` to return approximately only the first three rows of data with a target accuracy of 90 percent:

```
SQL> select * from table(dbms_xplan.display('plan_table', null, 'advanced
predicate'));

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
Plan hash value: 1784440045

------------------------------------------------------------------------------
--
---------------------

| Id  | Operation                     | Name         | Rows   | Bytes |TempSpc|
Co
st (%CPU)| Time      |

------------------------------------------------------------------------------
--
---------------------


PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
|   0 | SELECT STATEMENT              |              |     3 | 48945 |
|  1
466K  (2)| 00:00:58 |

|*  1 |  COUNT STOPKEY                |              |       |       |       |
         |           |

|   2 |   VIEW                        |              |  102K|  1588M|
```

```
|  1
466K  (2)| 00:00:58 |

|*  3 |    SORT ORDER BY STOPKEY      |              |   102K|  1589M|
798M|  1
466K  (2)| 00:00:58 |

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--

|   4 |     EXTERNAL TABLE ACCESS FULL| EXT_TABLE_3 |   102K|  1589M|       |
362   (7)| 00:00:01 |

------------------------------------------------------------------------------
--
---------------------


Query Block Name / Object Alias (identified by operation id):
-------------------------------------------------------------


PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
   1 - SEL$2
   2 - SEL$1 / "from$_subquery$_002"@"SEL$2"
   3 - SEL$1
   4 - SEL$1 / "EXT_TABLE_3"@"SEL$1"

Outline Data
-------------

  /*+
      BEGIN_OUTLINE_DATA
      FULL(@"SEL$1" "EXT_TABLE_3"@"SEL$1")

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
      NO_ACCESS(@"SEL$2" "from$_subquery$_002"@"SEL$2")
      OUTLINE_LEAF(@"SEL$2")
      OUTLINE_LEAF(@"SEL$1")
      ALL_ROWS
      OPT_PARAM('_fix_control' '6670551:0')
      OPT_PARAM('_optimizer_cost_model' 'fixed')
      DB_VERSION('26.1.0')
      OPTIMIZER_FEATURES_ENABLE('26.1.0')
      IGNORE_OPTIM_EMBEDDED_HINTS
      END_OUTLINE_DATA
  */

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
```

```
Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=3)
   3 - filter(ROWNUM<=3)

Column Projection Information (identified by operation id):
-----------------------------------------------------------

   1 - "from$_subquery$_002"."ID"[NUMBER,22],
"from$_subquery$_002"."EMBEDDING"[

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
VECTOR,32600]

   2 - "from$_subquery$_002"."ID"[NUMBER,22],
"from$_subquery$_002"."EMBEDDING"[
VECTOR,32600]

   3 - (#keys=1) VECTOR_DISTANCE(VECTOR('[1,1]', *, *, * /*+
USEBLOBPCW_QVCGMD
*/ ),

       "EMBEDDING" /*+ LOB_BY_VALUE */ , COSINE)[BINARY_DOUBLE,8],
"ID"[NUMBER,2
2], "EMBEDDING" /*+


PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
       LOB_BY_VALUE */ [VECTOR,32600]
   4 - "ID"[NUMBER,22], "EMBEDDING" /*+ LOB_BY_VALUE */ [VECTOR,32600],
       VECTOR_DISTANCE(VECTOR('[1,1]', *, *, * /*+  USEBLOBPCW_QVCGMD */ ),
"EMB
EDDING" /*+

       LOB_BY_VALUE */ , COSINE)[BINARY_DOUBLE,8]

Query Block Registry:
---------------------

  SEL$1 (PARSER) [FINAL]

PLAN_TABLE_OUTPUT
------------------------------------------------------------------------------
--
  SEL$2 (PARSER) [FINAL]
```

Vectors in Distributed Database Tables

There is no new SQL syntax or keyword when creating sharded tables and duplicated tables with vector columns in a Globally Distributed Database; however, there are some requirements and restrictions to consider.

User Permissions

Only an all-shards user can create sharded and duplicated tables. You must connect to the shard catalog as an all-shards user. Connecting to the shard catalog as an all-shards user automatically enables `SHARD DDL`, and the DDL to create the tables is propagated to all the shards in the distributed database.

Creating Sharded Tables with a Vector Column

- Sharded tables must be created on the catalog database with `SHARD DDL` enabled.
- A vector column cannot be part of the sharding key or the partitionset key.
- The `CREATE SHARDED TABLE` command is propagated to all of the shards by the shard coordinator.

The syntax to create a sharded table with a vector column is same as the syntax to create a non-sharded table with a vector column. The only difference is to include the `SHARDED` keyword in the `CREATE TABLE` statement.

```
CREATE SHARDED TABLE REALTORS(
     REALTOR_ID NUMBER PRIMARY KEY,
     NAME VARCHAR2(20),
     IMAGE VECTOR,
     ZIPCODE VARCHAR2(40))
PARTITION BY CONSISTENT HASH(REALTOR_ID)
TABLESPACE SET TS1;
```

Creating Duplicated Tables with a Vector Column

- Duplicated tables must be created on the shard catalog database with `SHARD DDL` enabled.

The syntax to create a duplicated table with a vector column is same as the syntax to create a non-sharded table with a vector column. The only difference is to include the `DUPLICATED` keyword in the `CREATE TABLE` statement.

```
CREATE DUPLICATED TABLE PRODUCT_DESCRIPTIONS
     (
     PRODUCT_ID           NUMBER(6,0) NOT NULL,
     ORDER_ID             NUMBER(6,0) NOT NULL,
     LANGUAGE_ID          VARCHAR2(6 BYTE),
     TRANSLATED_NAME      NVARCHAR2(50),
     TRANSLATED_DESCRIPTION NVARCHAR2(2000),
     VECT4 VECTOR,
     VECT5 VECTOR,
     CONSTRAINT  PRODUCT_DESCRIPTIONS_PK primary key (PRODUCT_ID)
     ) tablespace products
     STORAGE (INITIAL 1M NEXT 1M);
```

BINARY Vectors

In addition to FLOAT32 (the default format for comma-separated string representations of vectors), FLOAT64, and INT8 dimension formats, you can also use the BINARY dimension format.

A BINARY vector represents each dimension as a single bit (0 or 1). The following statement is an example of declaring a 1024 dimension vector using the BINARY format:

```
VECTOR(1024, BINARY)
```

The main advantages of using the BINARY format are:

- The storage footprint of vectors can be reduced by a factor of 32 compared to the default FLOAT32 format.
- Distance computations between two vectors are up to 40 times faster.

The downside of using the BINARY format is the potential loss of accuracy. However, the loss is often not very substantial. BINARY vector accuracy is often greater than 90% compared to that of FLOAT32 vectors. Several third-party providers have added embedding models that have the ability to generate binary embeddings, including Cohere, Hugging Face, and Jina AI.

BINARY vectors are stored as packed UINT8 bytes (unsigned integer). This means that a single byte represents exactly 8 BINARY dimensions and no less.

> **Note**
>
> - The default distance metric for BINARY vectors is HAMMING.
> - Conversions between BINARY vectors and any other dimension format are not currently supported.
> - A column can be declared as VECTOR(*, BINARY). In this case, '*' means that vectors can have an arbitrary number of dimensions. However, because the maximum possible number of dimensions supported is 65535 for other formats, you cannot exceed a UINT8 array of size 8191 for a BINARY vector, which represents 8191 * 8 = 65528 dimensions, the greatest multiple of 8 less than 65535.
> - Oracle does not currently support using OML4Py to export BINARY models to ONNX format and import them in the Oracle AI Database.
> - Oracle AI Database server does not currently provide quantization techniques. When applicable, this should be handled by the client.

The following is an example of a Cohere INT8 embedding and a UBINARY embedding. Note when considering this example that Oracle AI Database server works with the data provided in the VECTOR constructor as it is received. Any quantization logic necessary for processing the data into binary format should be handled by the client. The example is provided to help in understanding the concept.

```
INT8 Embedding of 1024 dimensions from Cohere embed-english-v3.0:
[25, 11, -99, -114, 13, -17, -59, 44, 65, 33, -50, -2, 28, -16, -6, -20, -33,
49, -59, -50, 0, -82, -67, 10, 82, -2, -126, -28, -32, -69, -13, 120, 54, 4,
```

```
-71, 24, 4, -37, -57, 34, 16, -7, 27, -74, -12, 13, 1, -24, 65, -24, 28, 46,
25, -33, -25, 36, 3, -47, 12, -49, -17, 11, 53, 70, -18, 10, -8, 4, 0, -33,
10, -3, 27, -24, -35, -24, 23, -32, 0, -4, -21, -7, -29, -48, -7, -28, -25,
-8, 54, -7, 14, -8, 39, 78, 0, -13, 26, 2, 40, 27, -35, -26, 5, -23, 15, 72,
-4, -5, 33, 14, 18, 11, 0, -6, 6, -16, -53, 56, -35, 15, -1, -8, 83, 28, -2,
27, -34, -60, 36, 4, 14, 21, -69, 17, -22, 0, 16, -77, 29, 27, 26, 0, 81, 15,
-90, 7, 22, -2, -26, -39, -31, -10, 2, 32, -30, 40, -71, 29, 2, 36, -72, -6,
42, -16, -16, 6, 40, 30, 1, -31, -42, 31, 56, 18, 0, 9, 27, 59, 11, 38, 28,
-30, 73, -10, -56, 6, 17, 87, 15, 1, 49, -33, -68, 0, 10, -49, 18, -10, 8,
12,
52, -31, 7, -37, -25, -53, 9, -5, 72, 14, -37, -41, 30, -54, -60, 30, -62,
20,
3, 7, 64, -7, 48, 16, 19, 1, -43, -18, -91, -6, -113, 104, 42, 61, -24, -15,
20, -9, 4, 36, 27, 46, -30, -39, 43, -14, 53, -36, -4, 35, 74, 37, 1, -19,
62,
12, -13, 8, -11, 21, -4, 96, 29, 17, -99, 2, -67, -32, -55, -8, 55, 16, -29,
28, 47, 47, -77, 0, -24, 1, 1, 38, 28, -11, 2, -55, 4, 18, 42, 99, 98, 1, 17,
18, -21, 4, 89, 66, -32, 17, 56, 14, -2, -45, 19, -30, 26, 14, 34, -36, 5,
74,
50, 33, 47, -37, 34, 61, -8, -62, 46, 56, -55, 0, 33, 5, -72, -29, -48, 21,
40,
22, 3, 39, -1, 10, 32, -47, 28, 19, 92, -5, -13, 2, 12, -21, -33, -9, 31, -2,
-25, -20, -14, 1, 53, -34, -26, 17, 72, -35, -36, -26, -86, -20, 55, -4, -53,
-14, 47, 26, 82, -3, -41, -18, -40, -94, 87, 3, -17, 38, 54, 17, 62, -23, 61,
20, -4, 18, 37, 21, -37, -10, -43, -32, -40, -29, 43, 75, -44, -3, 47, 9,
-10,
29, -26, 55, 35, -17, 43, 37, -8, 19, 0, -32, -49, 43, -27, 16, -81, 34, 56,
15, -33, -13, -30, -13, -28, 54, -61, -90, -45, -101, -52, -101, 5, 22, 7,
72,
-30, 31, 27, 42, -47, -6, -30, -30, 42, 13, -23, 63, -84, -20, -17, 61, -40,
35, 37, 21, -8, 110, 108, 26, -49, -1, -31, 8, 10, 7, 29, -67, -29, 72, 15,
11,
4, -34, 12, 28, -48, -21, -81, 38, -29, 26, 4, 10, 29, -11, 26, -78, -51,
-52,
27, -92, -23, -5, -11, 31, 18, -33, -49, 7, -51, -35, -57, -14, 121, -8, 29,
25,
70, -19, 29, 48, -41, 48, -18, 19, -18, -13, 46, 27, 47, 42, 1, -33, 20, -27,
8,
-31, 31, 1, 0, 11, -4, 32, -65, -7, 9, -11, 15, 3, -34, 42, -15, -71, -5, 3,
8,
-8, 22, -7, -70, 10, 21, -127, -114, 13, -11, 46, -13, -10, -10, 29, -59, 43,
-1, -17, -21, 8, -15, 12, 1, -73, -26, -5, 6, 37, 23, 46, 73, 14, -74, 84,
-2,
-22, -6, 5, -7, -26, 28, -39, -23, -22, 14, 38, 0, -2, 41, 27, -65, 30, 3,
-23,
53, 86, 35, -32, -48, -15, 32, 21, -26, -48, -26, 32, 32, 4, -70, -72, -62,
-28,
-14, -86, -10, -63, 44, -68, -41, 27, -52, 33, -56, -30, 5, 84, -54, 16, -22,
-20, 16, 34, 14, -25, 8, -14, -13, -28, -40, 16, 41, -5, -88, -35, 55, -82,
55,
74, -55, -12, 58, 57, -83, -26, 55, 32, -6, 42, -14, 35, -5, -36, 84, -40,
-29,
7, -20, -17, 23, -20, -49, -48, 22, 49, -30, 35, 48, 5, 34, 17, 13, 30, 33,
-38,
-37, 10, -52, -24, 67, -15, -12, -3, -11, -46, -7, 32, 10, -46, 3, 18, -7,
-26,
```

```
0, -40, 23, -46, 89, 37, 3, -29, -51, -32, 49, -51, 9, 16, -47, -26, 14, 10,
14,
-13, 11, 16, -18, 54, -24, 18, -14, -51, -89, -24, 20, 12, 2, 62, 13, 53,
-22,
2, 22, -14, 29, -9, 51, -42, -97, 28, 49, -4, -93, -17, -26, 46, 47, 33, -33,
25, 81, -29, 5, 17, 24, 54, -10, -14, -2, 29, 17, -4, -47, 56, 4, 9, 30, -87,
39, -16, 39, 67, -13, 37, 13, 67, 50, -16, -55, 8, 24, -50, -1, -36, -51,
-20,
-58, 11, -28, -22, -26, 16, 7, -17, 39, -9, -21, -9, -8, -18, 37, -47, -19,
36,
-8, 6, -39, 58, -26, -37, 11, 86, 33, 67, -35, 25, -11, -7, -22, 20, 14, 8,
8,
7, -30, -58, 37, -1, 16, -13, 89, -6, 81, -46, -37, -7, 9, -23, -11, -41,
-13,
18, -17, -4, -42, 0, 91, -128, 33, -18, -88, -84, -11, -62, 79, -34, -39, 54,
-17, -14, 15, 79, -33, -4, 30, 5, 8, -55, -9, -38, 10, -41, 37, -5, 2, 62, 3,
-5, -42, 17, -50, 14, -58, -16, 26, -20, -49, 52, 73, -42, 9, 7, -50, 14,
-11,
39, 0, -45, -90, -30, -16, -19, -6, -1, 43, -7, -47, -4, 40, -6, 5, 2, 2,
-20,
-40, 39, 10, -16, 64, -11, -36, -5, 37, -16, 49, 24, -20, 17, 27, -21, -49,
-49,
-38, -19, -31, -2, 15, 52, -68, -14, 20, 38, 10, -48, -2, -52, -60, -55, -30,
37, -32, -80, 1, -1, -12, -45, 15, 29, 8, -46, -42, -28, -38, 11, 4, 19, 2,
67,
-44, -5, -28, 21, 17, -16, -34, 16, -6, 10, -11, 15, 2, 33, -25, -13, 8, -7,
2,
-22, 21, -41, 10, -29, -36, 46, 19, -41, 36, -39, 10, -23, -13, -2, -53, 39,
-25, -4]


UBINARY Embedding from the same model (1024 dimensions = 128 packed UINT8
bytes)
[201, 200, 65, 129, 217, 166, 185, 167, 90, 138, 0, 172, 242, 207, 165, 52,
245,
187, 96, 215, 39, 159, 250, 126, 107, 162, 201, 123, 193, 203, 202, 123, 87,
67,
113, 235, 253, 220, 187, 236, 220, 125, 185, 136, 102, 8, 224, 222, 220, 12,
214,
217, 92, 16, 61, 195, 69, 220, 121, 236, 94, 136, 100, 46, 212, 250, 189, 45,
26,
101, 20, 88, 253, 18, 51, 110, 49, 192, 37, 52, 232, 98, 204, 212, 146, 55,
249,
32, 108, 174, 44, 237, 67, 246, 166, 29, 188, 103, 173, 230, 4, 104, 37, 79,
71,
202, 162, 16, 160, 147, 56, 174, 82, 109, 96, 34, 230, 139, 96, 51, 129, 35,
135,
198, 87, 42, 154, 132]
```

The `BINARY` vector is generated through a binary quantization mechanism using the following rule:

- If the `INT8` dimension value > 0, the `BINARY` dimension value is 1
- If the `INT8` dimension value <= 0, the `BINARY` dimension format is 0

Consider the first 8 INT8 dimensions from the preceding example:

```
[25, 11, -99, -114, 13, -17, -59, 44]
```

In BINARY, this translates to:

```
[1, 1, 0, 0, 1, 0, 0, 1]
```

Representing this as a UINT8 byte makes it 201, which is the first byte value of the packed UINT8 representation. So, each BINARY vector can therefore be inserted as a UINT8 array whose size is: *number of BINARY vector dimensions/8*.

Note

BINARY vectors are only supported with a number of dimensions that is a multiple of 8.

The following is an example of an invalid declaration of a BINARY vector column, due to the fact that the vector dimension, 12, is not divisible by 8:

```
CREATE TABLE vectab (id NUMBER, data VECTOR(12, BINARY));
```

Result:

```
CREATE TABLE vectab (id NUMBER, data VECTOR(12, BINARY))
                                                 *
ERROR at line 1:
ORA-51813: Vector of BINARY format should have a dimension count that is a
multiple of 8.
```

The following statements are an example of a valid table creation with a BINARY vector column and a valid insert (string representation):

```
CREATE TABLE vectab(id NUMBER, data VECTOR(16, BINARY));
INSERT INTO vectab VALUES (1, '[201, 15]');
SELECT data FROM vectab;
```

Result:

```
DATA
---------
[201,15]
```

These next statements are examples of invalid inserts (string representation):

```
SQL> INSERT INTO vectab VALUES (1, '[201]');
INSERT INTO vectab VALUES (1, '[201]')
                               *
ERROR at line 1:
ORA-51803: Vector dimension count must match the dimension count specified in
the column definition (actual: 8, required: 16).
```

```
SQL> INSERT INTO vectab VALUES (1, '[201, 15, 123]');
INSERT INTO vectab VALUES (1, '[201, 15, 123]')
                              *
ERROR at line 1:
ORA-51803: Vector dimension count must match the dimension count specified in
the column definition (actual: 24, required: 16).

SQL> INSERT INTO vectab VALUES (1, '[256, 15]');
INSERT INTO vectab VALUES (1, '[256, 15]')
                              *
ERROR at line 1:
ORA-51806: Vector column is not properly formatted (dimension value 1 is
outside the allowed precision range).
```

SPARSE Vectors

The storage format of a vector can be specified as `SPARSE` or `DENSE`. Sparse vectors are vectors that typically have a large number of dimensions but with very few non-zero dimension values, while Dense vectors are vectors where every dimension stores a value, including zero-values.

Sparse vectors can be generated by Sparse Encoding models such as SPLADE or BM25. Generally speaking, sparse models such as SPLADE outperform dense models, such as BERT and All-MiniLM, in keyword awareness search. They are also widely used for Hybrid Vector Search by combining sparse and dense vectors.

Conceptually, a sparse vector can be thought of as a vector where every dimension corresponds to a keyword in a certain vocabulary. For a given document, the sparse vector contains non-zero dimension values representing the number of occurrences for the keywords within that document. For example, BERT has a vocabulary size of 30,522 and several sparse encoders generate vectors of this dimensionality.

Representing a dense vector with 30,522 dimensions with only 100 non-zero `FLOAT32` dimension values would still require *30,522 * 4 = ~120KB* of storage. Such a format takes up a lot of space for no reason as most of the dimension values are 0. This would cause a huge performance deficit compared to the `SPARSE` representation of such vectors.

That is why when using `SPARSE` vectors, only the non-zero dimension values are physically stored.

Here is an example of creating and inserting a `SPARSE` vector:

```
DROP TABLE my_sparse_tab PURGE;
CREATE TABLE my_sparse_tab (v01 VECTOR(5, INT8, SPARSE));

INSERT INTO my_sparse_tab VALUES('[5,[2,4],[10,20]]');
INSERT INTO my_sparse_tab VALUES('[[2,4],[10,20]]');

SELECT * FROM my_sparse_tab;

V01
--------------------
[5,[2,4],[10,20]]
[5,[2,4],[10,20]]
```

You can see the difference between the SPARSE textual form within the INSERT statement with the one used for a DENSE vector. The SPARSE textual form looks like:

```
'[Total Dimension Count, [Dimension Index Array], [Dimension Value Array]]'
```

The example uses the number of dimensions in total (5 here but it is optional to specify it in this case as it is defined in the column's declaration), then gives the list of coordinates that have non-zero values, then the list of the corresponding values. In this example, coordinate 2 has the value 10 and coordinate 4 has the value 20. Coordinates 1, 3, and 5 have the value 0.

It is not permitted to use a DENSE textual form for SPARSE vectors and vice versa. However, it is possible to use vector functions to transform one into the other as illustrated in the following sample code using the table my_sparse_tab (created in the previous snippet):

The following INSERT statement fails:

```
INSERT INTO my_sparse_tab VALUES('[0, 10, 0, 20, 0]');

Error starting at line : 1 in command -
INSERT INTO my_sparse_tab VALUES ('[0,10,0,20,0]')
Error at Command Line : 1 Column : 33
Error report -
SQL Error: ORA-51833: Textual input conversion between sparse and dense
vector is not
supported.
```

However, this insertion works:

```
INSERT INTO my_sparse_tab VALUES (TO_VECTOR('[0,0,10,0,20]', 5, INT8, DENSE));

SELECT * FROM my_sparse_tab;

V01
____________________
[5,[2,4],[10,20]]
[5,[2,4],[10,20]]
[5,[3,5],[10,20]]
```

You can also transform a SPARSE vector into a DENSE textual form if needed and vice versa:

```
SELECT FROM_VECTOR(v01 RETURNING CLOB FORMAT DENSE)
FROM my_sparse_tab
WHERE ROWNUM<2;

FROM_VECTOR(V01RETURNINGCLOBFORMATDENSE)
____________________________________________
[0,10,0,20,0]
```

> **Note**
>
> The RETURNING clause used in the preceding example can also return a VARCHAR2 or a BLOB.

Insert Vectors in a Database Table Using the INSERT Statement

Once you create a table with a VECTOR data type column, you can directly insert vectors into the table using the INSERT statement.

The following examples assume you have already created vectors and know their values.

Here is a simple example:

```
DROP TABLE galaxies PURGE;
CREATE TABLE galaxies (id NUMBER, name VARCHAR2(50), doc VARCHAR2(500),
embedding VECTOR);

INSERT INTO galaxies VALUES (1, 'M31', 'Messier 31 is a barred spiral galaxy
in the Andromeda constellation which has a lot of barred spiral galaxies.',
'[0,2,2,0,0]');
INSERT INTO galaxies VALUES (2, 'M33', 'Messier 33 is a spiral galaxy in the
Triangulum constellation.', '[0,0,1,0,0]');
INSERT INTO galaxies VALUES (3, 'M58', 'Messier 58 is an intermediate barred
spiral galaxy in the Virgo constellation.', '[1,1,1,0,0]');
INSERT INTO galaxies VALUES (4, 'M63', 'Messier 63 is a spiral galaxy in the
Canes Venatici constellation.', '[0,0,1,0,0]');
INSERT INTO galaxies VALUES (5, 'M77', 'Messier 77 is a barred spiral galaxy
in the Cetus constellation.', '[0,1,1,0,0]');
INSERT INTO galaxies VALUES (6, 'M91', 'Messier 91 is a barred spiral galaxy
in the Coma Berenices constellation.', '[0,1,1,0,0]');
INSERT INTO galaxies VALUES (7, 'M49', 'Messier 49 is a giant elliptical
galaxy in the Virgo constellation.', '[0,0,0,1,1]');
INSERT INTO galaxies VALUES (8, 'M60', 'Messier 60 is an elliptical galaxy in
the Virgo constellation.', '[0,0,0,0,1]');
INSERT INTO galaxies VALUES (9, 'NGC1073', 'NGC 1073 is a barred spiral
galaxy in Cetus constellation.', '[0,1,1,0,0]');
COMMIT;
```

Here is a more sophisticated example:

```
DROP TABLE doc_queries PURGE;
CREATE TABLE doc_queries (id NUMBER, query VARCHAR2(500), embedding VECTOR);

DECLARE
  e CLOB;
BEGIN
e:=
'[-7.73346797E-002,1.09683955E-002,4.68435362E-002,2.57333983E-002,6.95586428E
-00'||
'2,-2.43412293E-002,-7.25011379E-002,6.66433945E-002,3.78751606E-002,-2.223544
75E'||
'-002,3.02388351E-002,9.36625451E-002,-1.65204913E-003,3.50606232E-003,-5.4773
859'||
'7E-002,-7.5879097E-002,-2.72218436E-002,7.01764375E-002,-1.32512336E-003,3.14
728'||
'022E-002,-1.39147148E-001,-7.52705336E-002,2.62449421E-002,1.91645715E-002,4.
055'||
'73137E-002,5.83701171E-002,-3.26474942E-002,2.0509012E-002,-3.81141738E-003,-
```

```
7.1'||
'8656182E-002,-1.95893757E-002,-2.56917924E-002,-6.57705888E-002,-4.39117625E-
002'||
',-6.82357177E-002,1.26592368E-001,-3.46683599E-002,1.07687116E-001,-3.9695449
2E-'||
'002,-9.06721968E-003,-2.4109887E-002,-1.29214963E-002,-4.82468568E-002,-3.872
307'||
'76E-002,5.13443872E-002,-1.40985977E-002,-1.87066793E-002,-1.11725368E-002,9.
367'||
'76772E-002,-6.39425665E-002,3.13162468E-002,8.61801133E-002,-5.5481784E-002,4
.13'||
'125418E-002,2.0447813E-002,5.03717586E-002,-1.73418857E-002,3.94522659E-002,-
7.2'||
'6833269E-002,3.13266069E-002,1.2377765E-002,7.64856935E-002,-3.77447419E-002,
-6.'||
'41075056E-003,1.1455299E-001,1.75497644E-002,4.64923214E-003,1.89623125E-002,
9.1'||
'3506597E-002,-8.22509527E-002,-1.28537193E-002,1.495138E-002,-3.22528258E-002
,-4'||
'.71280375E-003,-3.15563753E-003,2.20409594E-002,7.77796134E-002,-1.927099E-00
2,-'||
'1.24283969E-001,4.69769612E-002,1.78121701E-002,1.67152807E-002,-3.83916795E-
002'||
',-1.51029453E-002,2.10864041E-002,6.86845928E-002,-7.4719809E-002,1.17681816E
-00'||
'3,3.93113159E-002,6.04066066E-002,8.55340436E-002,3.68878953E-002,2.41579115E
-00'||
'2,-5.92489541E-002,-1.21883564E-002,-1.77226216E-002,-1.96259264E-002,8.51236
377'||
'E-003,1.43039867E-001,2.62829307E-002,2.96348184E-002,1.92485824E-002,7.66567
141'||
'E-002,-1.18600562E-001,3.01779062E-002,-5.88010997E-002,7.07774982E-002,-6.60
426'||
'617E-002,6.44619241E-002,1.29240509E-002,-2.51785964E-002,2.20869959E-004,-2.
514'||
'38171E-002,5.52265197E-002,8.65883753E-002,-1.83726232E-002,-8.13263431E-002,
1.1'||
'6624301E-002,1.63392909E-002,-3.54643688E-002,2.05128491E-002,4.67337575E-003
,1.'||
'20488718E-001,-4.89500947E-002,-3.80397178E-002,6.06209273E-003,-1.37961926E-
002'||
',4.68355882E-031,3.35873142E-002,6.20040558E-002,2.13472452E-002,-1.87379227E
-00'||
'3,-5.83158981E-004,-4.04266678E-002,2.40761992E-002,-1.93725452E-002,9.376372
4E-'||
'002,-3.02913114E-002,7.67844869E-003,6.11112304E-002,6.02455214E-002,-6.38855
845'||
'E-002,-8.03523697E-003,2.08786246E-003,-7.45898336E-002,8.74964818E-002,-5.02
371'||
'937E-002,-4.99385223E-003,3.37120108E-002,8.99377018E-002,1.09540671E-001,5.8
501'||
'102E-002,1.71627291E-002,-3.26152593E-002,8.36912021E-002,5.05600758E-002,-9.
737'||
'63615E-002,-1.40264994E-002,-2.07926836E-002,-4.20163684E-002,-5.97197041E-00
2,1'||
'.32461395E-002,2.26361351E-003,8.1473738E-002,-4.29272018E-002,-3.86809185E-0
```

```
02,'||
'-8.24682564E-002,-3.89646105E-002,1.9992901E-002,2.07321253E-002,-1.74706057E
-00'||
'2,4.50415723E-003,4.43851873E-002,-9.86309871E-002,-7.68082142E-002,-4.538143
05E'||
'-003,-8.90906602E-002,-4.54972908E-002,-5.71065396E-002,2.10020249E-003,1.224
947'||
'07E-002,-6.70659095E-002,-6.52298108E-002,3.92126441E-002,4.33384106E-002,4.3
899'||
'6181E-002,5.78813367E-002,2.95345876E-002,4.68395352E-002,9.15119275E-002,-9.
629'||
'58392E-003,-5.96637605E-003,1.58674959E-002,-6.74034096E-003,-6.00510836E-002
,2.'||
'67188111E-003,-1.10706768E-003,-6.34015873E-002,-4.80389707E-002,6.84534572E-
003'||
',-1.1547043E-002,-3.44865513E-003,1.18979132E-002,-4.31232266E-002,-5.9022788
E-0'||
'02,4.87607308E-002,3.95954074E-003,-7.95252472E-002,-1.82770658E-002,1.182642
49E'||
'-002,-3.79164703E-002,3.87993976E-002,1.09805465E-002,2.29136664E-002,-7.2278
082'||
'4E-002,-5.31538352E-002,6.38669729E-002,-2.47980515E-003,-9.6999377E-002,-3.7
566'||
'7699E-002,4.06541862E-002,-1.69874367E-003,5.58868013E-002,-1.80723771E-033,-
6.6'||
'5985467E-003,-4.45010923E-002,1.77929532E-002,-4.8369132E-002,-1.49722975E-00
2,-'||
'3.97582203E-002,-7.05247298E-002,3.89178023E-002,-8.26886389E-003,-3.91006246
E-0'||
'02,-7.02963024E-002,-3.91333885E-002,1.76661201E-002,-5.09723537E-002,2.37749
107'||
'E-002,-1.83419678E-002,-1.2693027E-002,-1.14232123E-001,-6.68751821E-002,7.52
167'||
'869E-003,-9.94713791E-003,6.03599809E-002,6.61353692E-002,3.70595567E-002,-2.
019'||
'52495E-002,-2.40410417E-002,-3.36526595E-002,6.20064288E-002,5.50279953E-002,
-2.'||
'68641673E-002,4.35859226E-002,-4.57317568E-002,2.76936609E-002,7.88119733E-00
2,-'||
'4.78852056E-002,1.08523415E-002,-6.43479973E-002,2.0192951E-002,-2.09538229E-
002'||
',-2.2202393E-002,-1.0728148E-003,-3.09607089E-002,-1.67067181E-002,-6.0357227
9E-'||
'002,-1.58187654E-002,3.45828459E-002,-3.45360823E-002,-4.4002533E-003,1.77463
517'||
'E-002,6.68234832E-004,6.14458732E-002,-5.07084019E-002,-1.21073434E-002,4.195
981'||
'85E-002,3.69152687E-002,1.09461844E-002,1.83413982E-001,-3.89185362E-002,-5.1
846'||
'0497E-002,-8.71620141E-003,-1.17692262E-001,4.04785499E-002,1.07505821E-001,1
.41'||
'624091E-002,-2.57720836E-002,2.6652012E-002,-4.50568087E-002,-3.34110335E-002
,-1'||
'.11387551E-001,-1.29796984E-003,-6.51671961E-002,5.36890551E-002,1.0702607E-0
01,'||
'-2.34011523E-002,3.97406481E-002,-1.01149324E-002,-9.95831117E-002,-4.4019784
```

```
8E-'||
'002,6.88989647E-003,4.85475454E-003,-3.94048765E-002,-3.6099229E-002,-5.40755
13E'||
'-002,8.58292207E-002,1.0697281E-002,-4.70785573E-002,-2.96272673E-002,-9.4919
120'||
'9E-003,1.57316476E-002,-5.4926388E-002,6.49022609E-002,2.55531631E-002,-1.839
057'||
'17E-002,4.06873561E-002,4.74951901E-002,-1.22502812E-033,-4.6441108E-002,3.74
079'||
'868E-002,9.14599106E-004,6.09740615E-002,-7.67391697E-002,-6.32521287E-002,-2
.17'||
'353106E-002,2.45231949E-003,1.50869079E-002,-4.96984981E-002,-3.40828523E-002
,8.'||
'09691194E-003,3.31339166E-002,5.41345142E-002,-1.16213948E-001,-2.49572527E-0
02,'||
'5.00682592E-002,5.90037219E-002,-2.89178211E-002,8.01460445E-003,-3.41945067E
-00'||
'2,-8.60121697E-002,-6.20261126E-004,2.26721354E-002,1.28968194E-001,2.8765536
8E-'||
'002,-2.20255274E-002,2.7228903E-002,-1.12029864E-002,-3.20301466E-002,4.98079
099'||
'E-002,2.89051589E-002,2.413591E-002,3.64605561E-002,6.26017479E-003,6.5463289
6E-'||
'002,1.11282602E-001,-3.60428065E-004,1.95987038E-002,6.16615731E-003,5.935930
46E'||
'-002,1.50377362E-003,2.95319762E-002,2.56325547E-002,-1.72190219E-002,-6.5816
819'||
'7E-002,-4.08149995E-002,2.7983617E-002,-6.80195764E-002,-3.52494679E-002,-2.9
840'||
'0577E-002,-3.04043181E-002,-1.9352382E-002,5.49411364E-002,8.74160081E-002,5.
614'||
'25127E-002,-5.60747795E-002,-3.43311466E-002,9.83581021E-002,2.01142877E-002,
1.3'||
'193069E-002,-3.22583504E-002,8.54402035E-002,-2.20514946E-002]';

INSERT INTO doc_queries VALUES (13, 'different methods of backup and
recovery', e);
COMMIT;
END;
/
```

You can also create a table that holds `BINARY` vectors:

```
DROP TABLE my_bin_tab PURGE;
CREATE TABLE my_bin_tab(id NUMBER, data VECTOR(16, BINARY));

INSERT INTO my_bin_tab VALUES (1, '[201, 15]');
```

Here's an additional example demonstrating the insertion of `SPARSE` vectors into a table:

```
DROP TABLE my_sparse_tab PURGE;
CREATE TABLE my_sparse_tab(v01 VECTOR(5, INT8, SPARSE));
```

```
INSERT INTO my_sparse_tab VALUES ('[5,[2,4],[10,20]]');
INSERT INTO my_sparse_tab VALUES ('[[2,4],[10,20]]');
```

You can also generate vectors by calling services outside the database or generate vectors directly from within the database after you have imported pretrained embedding models.

If you have already loaded an embedding model into the database, called, let's say, `MyModel`, you can insert vectors using the `VECTOR_EMBEDDING` function. For example:

```
INSERT INTO doc_queries VALUES (13, 'different methods of backup and
recovery',
  vector_embedding(MyModel using 'different methods of backup and recovery'
as data));"
```

> **See Also**
>
> - Import Pretrained Models in ONNX Format
> - Convert Text String to Embedding Within Oracle AI Database

Load Vector Data Using SQL*Loader

Use these examples to understand how you can load character and binary vector data.

SQL*Loader supports loading `VECTOR` columns from character data and binary floating point array `fvec` files. The format for `fvec` files is that each binary 32-bit floating point array is preceded by a four (4) byte value, which is the number of elements in the vector. There can be multiple vectors in the file, possibly with different dimensions. Export and import of a table with vector datatype columns is supported in all the modes (FULL, SCHEMA, TABLES) using all the available methods (access_method=direct_path, access_method=external_table, access_method=automatic) for unloading/loading data.

- Load Character Vector Data Using SQL*Loader Example
  In this example, you can see how to use SQL*Loader to load vector data into a five-dimension vector space.
- Load Binary Vector Data Using SQL*Loader Example
  In this example, you can see how to use SQL*Loader to load binary vector data files.

Load Character Vector Data Using SQL*Loader Example

In this example, you can see how to use SQL*Loader to load vector data into a five-dimension vector space.

Let's imagine we have the following text documents classifying galaxies by their types:

- **DOC1**: "Messier 31 is a barred spiral galaxy in the Andromeda constellation which has a lot of barred spiral galaxies."
- **DOC2**: "Messier 33 is a spiral galaxy in the Triangulum constellation."
- **DOC3**: "Messier 58 is an intermediate barred spiral galaxy in the Virgo constellation."
- **DOC4**: "Messier 63 is a spiral galaxy in the Canes Venatici constellation."

- **DOC5**: "Messier 77 is a barred spiral galaxy in the Cetus constellation."
- **DOC6**: "Messier 91 is a barred spiral galaxy in the Coma Berenices constellation."
- **DOC7**: "NGC 1073 is a barred spiral galaxy in Cetus constellation."
- **DOC8**: "Messier 49 is a giant elliptical galaxy in the Virgo constellation."
- **DOC9**: "Messier 60 is an elliptical galaxy in the Virgo constellation."

You can create vectors representing the preceding galaxy's classes using the following five-dimension vector space based on the count of important words appearing in each document:

Table 5-1 Five dimension vector space

Galaxy Classes	Intermediate	Barred	Spiral	Giant	Elliptical
M31	0	2	2	0	0
M33	0	0	1	0	0
M58	1	1	1	0	0
M63	0	0	1	0	0
M77	0	1	1	0	0
M91	0	1	1	0	0
M49	0	0	0	1	1
M60	0	0	0	0	1
NGC1073	0	1	1	0	0

This naturally gives you the following vectors:

- **M31**: `[0,2,2,0,0]`
- **M33**: `[0,0,1,0,0]`
- **M58**: `[1,1,1,0,0]`
- **M63**: `[0,0,1,0,0]`
- **M77**: `[0,1,1,0,0]`
- **M91**: `[0,1,1,0,0]`
- **M49**: `[0,0,0,1,1]`
- **M60**: `[0,0,0,0,1]`
- **NGC1073**: `[0,1,1,0,0]`

You can use SQL*Loader to load this data into the `GALAXIES` database table defined as:

```
drop table galaxies purge;
create table galaxies (id number, name varchar2(50), doc varchar2(500),
embedding vector);
```

Based on the data described previously, you can create the following `galaxies_vec.csv` file:

```
1:M31:Messier 31 is a barred spiral galaxy in the Andromeda constellation
which has a lot of barred spiral galaxies.:[0,2,2,0,0]:
2:M33:Messier 33 is a spiral galaxy in the Triangulum constellation.:
[0,0,1,0,0]:
3:M58:Messier 58 is an intermediate barred spiral galaxy in the Virgo
```

```
constellation.:[1,1,1,0,0]:
4:M63:Messier 63 is a spiral galaxy in the Canes Venatici constellation.:
[0,0,1,0,0]:
5:M77:Messier 77 is a barred spiral galaxy in the Cetus constellation.:
[0,1,1,0,0]:
6:M91:Messier 91 is a barred spiral galaxy in the Coma Berenices
constellation.:[0,1,1,0,0]:
7:M49:Messier 49 is a giant elliptical galaxy in the Virgo constellation.:
[0,0,0,1,1]:
8:M60:Messier 60 is an elliptical galaxy in the Virgo constellation.:
[0,0,0,0,1]:
9:NGC1073:NGC 1073 is a barred spiral galaxy in Cetus constellation.:
[0,1,1,0,0]:
```

Here is a possible SQL*Loader control file `galaxies_vec.ctl`:

```
recoverable
LOAD DATA
infile 'galaxies_vec.csv'
INTO TABLE galaxies
fields terminated by ':'
trailing nullcols
(
id,
name char (50),
doc  char (500),
embedding char (32000)
)
```

> **Note**
>
> You cannot use comma-delimited vectors (vectors separated by commas) as the field terminator in your CSV file. You must use another deliminator. In these examples the deliminator is a colon (:).

After you have created the two files `galaxies_vec.csv` and `galaxies_vec.ctl`, you can run the following sequence of instructions directly from your favorite SQL command line tool:

```
host sqlldr vector/vector@CDB1_PDB1 control=galaxies_vec.ctl
log=galaxies_vec.log

SQL*Loader: Release 23.0.0.0.0 - Development on Thu Jan 11 19:46:21 2024
Version 23.4.0.23.00

Copyright (c) 1982, 2024, Oracle and/or its affiliates.  All rights reserved.

Path used:      Conventional
Commit point reached - logical record count 10

Table GALAXIES2:
  9 Rows successfully loaded.

Check the log file:
```

```
  galaxies_vec.log
for more information about the load.

SQL>

select * from galaxies;

 ID NAME    DOC
EMBEDDING
--- ------  ----------------------------------------------------------
---------------------------------
  1 M31     Messier 31 is a barred spiral galaxy in the Andromeda ...
[0,2.0E+000,2.0E+000,0,0]
  2 M33     Messier 33 is a spiral galaxy in the Triangulum ...
[0,0,1.0E+000,0,0]
  3 M58     Messier 58 is an intermediate barred spiral galaxy ...
[1.0E+000,1.0E+000,1.0E+000,0,0]
  4 M63     Messier 63 is a spiral galaxy in the Canes Venatici ...
[0,0,1.0E+000,0,0]
  5 M77     Messier 77 is a barred spiral galaxy in the Cetus ...
[0,1.0E+000,1.0E+000,0,0]
  6 M91     Messier 91 is a barred spiral galaxy in the Coma ...
[0,1.0E+000,1.0E+000,0,0]
  7 M49     Messier 49 is a giant elliptical galaxy in the Virgo ...
[0,0,0,1.0E+000,1.0E+000]
  8 M60     Messier 60 is an elliptical galaxy in the Virgo ...
[0,0,0,0,1.0E+000]
  9 NGC1073 NGC 1073 is a barred spiral galaxy in Cetus ...
[0,1.0E+000,1.0E+000,0,0]

9 rows selected.

SQL>
```

Here is the resulting log file for this load (`galaxies_vec.log`):

```
cat galaxies_vec.log

SQL*Loader: Release 23.0.0.0.0 - Development on Thu Jan 11 19:46:21 2024
Version 23.4.0.23.00

Copyright (c) 1982, 2024, Oracle and/or its affiliates.  All rights reserved.

Control File:   galaxies_vec.ctl
Data File:      galaxies_vec.csv
  Bad File:     galaxies_vec.bad
  Discard File:  none specified

 (Allow all discards)

Number to load: ALL
Number to skip: 0
Errors allowed: 50
Bind array:     250 rows, maximum of 1048576 bytes
Continuation:    none specified
```

```
Path used:      Conventional

Table GALAXIES, loaded from every logical record.
Insert option in effect for this table: INSERT
TRAILING NULLCOLS option in effect

Column Name   Position   Len  Term Encl Datatype
----------- ---------- ----- ---- ---- ----------
ID               FIRST     *   :       CHARACTER
NAME              NEXT    50   :       CHARACTER
DOC               NEXT   500   :       CHARACTER
EMBEDDING         NEXT 32000   :       CHARACTER

value used for ROWS parameter changed from 250 to 31


Table GALAXIES2:
  9 Rows successfully loaded.
  0 Rows not loaded due to data errors.
  0 Rows not loaded because all WHEN clauses were failed.


Space allocated for bind array:               1017234 bytes(31 rows)
Read   buffer bytes: 1048576

Total logical records skipped:          0
Total logical records read:             9
Total logical records rejected:         0
Total logical records discarded:        1

Run began on Thu Jan 11 19:46:21 2024
Run ended on Thu Jan 11 19:46:24 2024

Elapsed time was:     00:00:02.43
CPU time was:         00:00:00.03
$
```

> **Note**
>
> This example uses `embedding char (32000)` vectors. For very large vectors, you can use the `LOBFILE` feature

Related Topics

- Loading LOB Data from LOBFILEs

Load Binary Vector Data Using SQL*Loader Example

In this example, you can see how to use SQL*Loader to load binary vector data files.

The vectors in a binary (`fvec`) file are stored in raw 32-bit Little Endian format.

Each vector takes 4+*d**4 bytes for the `.fvecs` file where the first 4 bytes indicate the dimensionality (*d*) of the vector (that is, the number of dimensions in the vector) followed by *d**4 bytes representing the vector data, as described in the following table:

Table 5-2 Fields for Vector Dimensions and Components

Field	Field Type	Description
d	int	The vector dimension
components	array of floats	The vector components

For binary `fvec` files, they must be defined as follows:

- You must specify `LOBFILE`.
- You must specify the syntax format `fvecs` to indicate that the dafafile contains binary dimensions.
- You must specify that the datafile contains raw binary data (`raw`).

The following is an example of a control file used to load `VECTOR` columns from binary floating point arrays using the galaxies vector example described in Understand Hierarchical Navigable Small World Indexes, but in this case importing `fvecs` data, using the control file syntax `format "fvecs"`:

> **Note**
>
> SQL*Loader supports loading `VECTOR` columns from character data and binary floating point array `fvec` files. The format for `fvec` files is that each binary 32-bit floating point array is preceded by a four (4) byte value, which is the number of elements in the vector. There can be multiple vectors in the file, possibly with different dimensions.

```
LOAD DATA
infile 'galaxies_vec.csv'
INTO TABLE galaxies
fields terminated by ':'
trailing nullcols
(
id,
name char (50),
doc  char (500),
embedding lobfile (constant '/u01/data/vector/embedding.fvecs' format
"fvecs")  raw
)
```

The data contained in `galaxies_vec.csv` in this case does not have the vector data. Instead, the vector data will be read from the secondary `LOBFILE` in the `/u01/data/vector` directory (`/u01/data/vector/embedding.fvecs`), which contains the same information in `float32` floating point binary numbers, but is in `fvecs` format:

```
1:M31:Messier 31 is a barred spiral galaxy in the Andromeda constellation
which has a lot of barred spiral galaxies.:
2:M33:Messier 33 is a spiral galaxy in the Triangulum constellation.:
```

```
3:M58:Messier 58 is an intermediate barred spiral galaxy in the Virgo
constellation.:
4:M63:Messier 63 is a spiral galaxy in the Canes Venatici constellation.:
5:M77:Messier 77 is a barred spiral galaxy in the Cetus constellation.:
6:M91:Messier 91 is a barred spiral galaxy in the Coma Berenices
constellation.:
7:M49:Messier 49 is a giant elliptical galaxy in the Virgo constellation.:
8:M60:Messier 60 is an elliptical galaxy in the Virgo constellation.:
9:NGC1073:NGC 1073 is a barred spiral galaxy in Cetus constellation.:
```

Unload and Load Vectors Using Oracle Data Pump

Starting with Oracle Database 26ai, Oracle Data Pump enables you to use multiple components to load and unload vectors to databases.

Oracle Data Pump technology enables very high-speed movement of data and metadata from one database to another. Oracle Data Pump is made up of three distinct components: Command-line clients, expdp and impdp; the DBMS_DATAPUMP PL/SQL package (also known as the Data Pump API); and the DBMS_METADATA PL/SQL package (also known as the Metadata API).

Unloading and Loading a table with vector datatype columns is supported in all modes (`FULL`, `SCHEMA`, `TABLES`) using all the available access methods (`DIRECT_PATH`, `EXTERNAL_TABLE`, `AUTOMATIC`, `INSERT_AS_SELECT`).

Examples Vector Export and Import Syntax

```
expdp <username>/<password>@<Database-instance-TNS-alias>  dumpfile=<dumpfile-
name>.dmp directory=<directory-name> full=y metrics=y
access_method=direct_path

expdp <username>/<password>@<Database-instance-TNS-alias>  dumpfile=<dumpfile-
name>.dmp directory=<directory-name> schemas=<schema-name> metrics=y
access_method=external_table

expdp <username>/<password>@<Database-instance-TNS-alias>  dumpfile=<dumpfile-
name>.dmp directory=<directory-name> tables=<schema-name>.<table-name>
metrics=y access_method=direct_path

impdp <username>/<password>@<Database-instance-TNS-alias>  dumpfile=<dumpfile-
name>.dmp directory=<directory-name> metrics=y access_method=direct_path
```

> **Note**
>
> - `TABLE_EXISTS_ACTION=APPEND | TRUNCATE` can only be used with the `EXTERNAL_TABLE` access method.
> - `TABLE_EXISTS_ACTION=APPEND | TRUNCATE` can load `VECTOR` column data into a `VARCHAR2` column if the conversion can fit into that `VARCHAR2`.
> - `TABLE_EXISTS_ACTION=APPEND | TRUNCATE` can only load a `VECTOR` column with the source `VECTOR` data dimasion that matches that loaded `VECTOR` column's dimension. If the dimension does not match, then an error is raised.
> - `TABLE_EXISTS_ACTION=REPLACE` supports any access method.
> - It is not possible to use a the transportable tablespace mode with vector indexes. However, this mode supports tables with the `VECTOR` datatype.

Related Topics

- Overview of Oracle Data Pump
- DBMS_DATAPUMP
- DBMS_METADATA

6

Create Vector Indexes and Hybrid Vector Indexes

Depending on whether to use similarity search or hybrid search, you can create either a vector index or a hybrid vector index.

Vector indexes are a class of specialized indexing data structures that are designed to accelerate similarity searches using high-dimensional vectors. They use techniques such as clustering, partitioning, and neighbor graphs to group vectors representing similar items, which drastically reduces the search space, thereby making the search process extremely efficient. Create vector indexes on your vector embeddings to use these indexes for running similarity searches over huge vector spaces.

Hybrid vector indexes leverage the existing Oracle Text indexing data structures and vector indexing data structures. Such indexes can provide more relevant search results by integrating the keyword matching capabilities of text search with the semantic precision of vector search. Create hybrid vector indexes directly on your input data or on your vector embeddings to use these indexes for performing a combination of full-text search and similarity search.

- Size the Vector Pool
  To allow vector index creation, you must enable a new memory area stored in the SGA called the **Vector Pool**.
- Manage the Different Categories of Vector Indexes
- Manage Hybrid Vector Indexes
  Learn how to manage a hybrid vector index, which is a single index for searching by *similarity* and *keywords*, to enhance the accuracy of your search results.
- Vector Indexes in a Globally Distributed Database
  Inverted File Flat (IVF) index and Hierarchical Navigable Small World (HNSW) index are supported on sharded tables in a distributed database; however there are some considerations.
- IVF Indexing on External Iceberg Tables

Size the Vector Pool

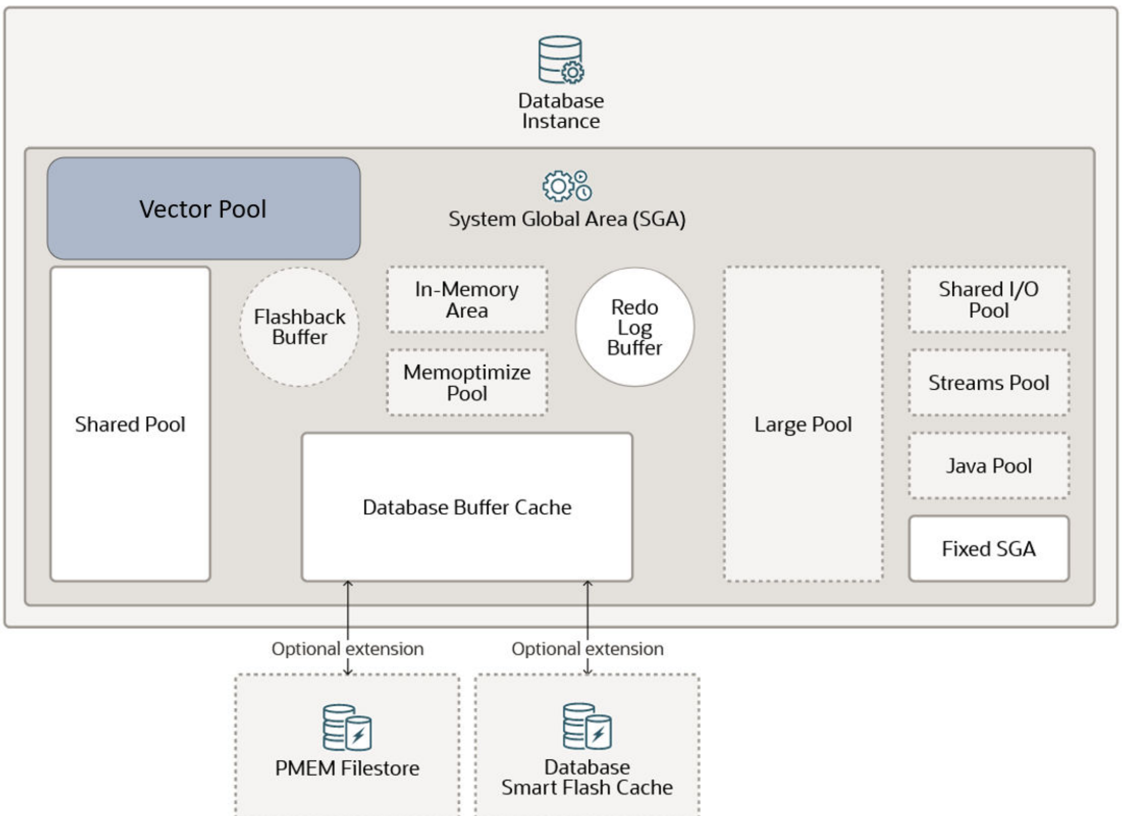
To allow vector index creation, you must enable a new memory area stored in the SGA called the **Vector Pool**.

The Vector Pool is a memory allocated in SGA to store Hierarchical Navigable Small World (HNSW) vector indexes and all associated metadata. It is also used to speed up Inverted File Flat (IVF) index creation as well as DML operations on base tables with IVF indexes.

> **Note**
>
> IVF centroid vectors are stored in the vector pool to improve IVF index creation and index maintenance performance. If space runs out in the vector pool, the centroid vectors are cached in the large pool. If space runs out in the large pool, the centroid vectors are not cached inmemory. This rule applies to both unpartitioned and partitioned local and global IVF indexes.

Enabling a Vector Pool is illustrated in the following diagram:

Figure 6-1 Vector Pool



To size the Vector Pool in an on-premises environment, use the `VECTOR_MEMORY_SIZE` initialization parameter. You can dynamically modify this parameter at the following levels:

- At the CDB level `VECTOR_MEMORY_SIZE` specifies the current size of the Vector Pool. Reducing the parameter value will fail if there is current vector usage.
- At the PDB level `VECTOR_MEMORY_SIZE` specifies the maximum Vector Pool usage allowed by a PDB. Reducing the parameter value will be allowed even if current vector usage exceeds the new quota.

You can change the value of a parameter in a parameter file in the following ways:

- By editing an initialization parameter file. In most cases, the new value takes effect the next time you start an instance of the database.

- By issuing an `ALTER SYSTEM SET ... SCOPE=SPFILE` statement to update a server parameter file.
- By issuing an `ALTER SYSTEM RESET` statement to clear an initialization parameter value and set it back to its default value.

Here is an example of how to change the value for `VECTOR_MEMORY_SIZE` at the PDB level if you are using an SPFILE:

```
SQL> show con_name
CON_NAME
------------------------------
MYPDB1

SQL> show user
USER is "SYS"

SQL> show parameter vector_memory_size
NAME               TYPE        VALUE
------------------ ----------- -----
vector_memory_size big integer 500M

SQL> SELECT ISPDB_MODIFIABLE
  2  FROM   V$SYSTEM_PARAMETER
  3* WHERE  NAME='vector_memory_size';

ISPDB_MODIFIABLE
___________________
TRUE

SQL> ALTER SYSTEM SET vector_memory_size=1G SCOPE=BOTH;

System altered.

SQL> show parameter vector_memory_size

NAME                TYPE        VALUE
------------------- ----------- -------
vector_memory_size  big integer 1G
SQL>
```

For more information about changing initialization parameter values, see Managing Initialization Parameters Using a Server Parameter File.

If `VECTOR_MEMORY_SIZE` is set to 1 and the `sga_target` is greater than 0 at CDB initialization, HNSW index creation will automatically grow the vector memory pool to satisfy the new index. The maximum PDB `VECTOR_MEMORY_SIZE` value is limited to 70% of the PDB `sga_target`. Dropping an HNSW index shrinks the vector memory pool accordingly.

In this configuration, where the vector memory pool grows automatically, the PDB `VECTOR_MEMORY_SIZE` value will default to 0 and cannot be changed using the `ALTER SYSTEM` command. Changes in vector pool size as a result of automatic growth are not persisted to the spfile, so when the database is restarted, the vector pool size is reset.

You can query the `V$VECTOR_MEMORY_POOL` view to monitor the Vector Pool.

Note

To roughly determine the memory size needed to store an HNSW index, use the following formula: *1.3 * number of vectors * number of dimensions * size of your vector dimension type* (for example, a `FLOAT32` is equivalent to `BINARY_FLOAT` and is 4 bytes in size).

See Also

- Vector Memory Pool Views

Vector Pool Size with Autonomous AI Database Serverless Services

When using Autonomous AI Database Serverless services (ADB-S), you cannot explicitly set any SGA-related memory parameters. This includes direct modification of the Vector Pool size.

With ADB-S, the Vector Pool can dynamically grow and shrink:

- An increase to the Vector Pool size is automatically triggered by HNSW index creation and when reopening a PDB containing HNSW indexes.
- A reduction to the Vector Pool size is automatically triggered by an HNSW index drop, PDB close, and CPU reduction.

Note

Vector Pool size is limited to a maximum of 70% of the PDB SGA size. The SGA size can be retrieved by using the following query:

```
SELECT value FROM V$PARAMETER WHERE name='sga_target';
```

Note

Changing instance configuration can cause a previously created HNSW index to be evicted due to insufficient Vector Pool memory. For example, this could be the case if one instance with 15 OCPUs is split into two instances, each with 8 OCPUs.

You can check how much memory is allocated to the Vector Pool after HNSW index creation using a `SELECT` statement similar to the following:

```
SELECT sum(alloc_bytes) FROM V$VECTOR_MEMORY_POOL;
```

Note

Until an HNSW index is created, `V$VECTOR_MEMORY_POOL` shows `0` in the `ALLOC_BYTES` column.

Manage the Different Categories of Vector Indexes

There are two ways to make vector searches faster:

- Reduce the search scope by clustering vectors (nearest neighbors) into structures based on certain attributes and restricting the search to closest clusters.
- Reduce the vector size by reducing the number of bits representing vectors values.

Oracle AI Vector Search supports the following categories of vector indexing methods based on approximate nearest-neighbors (ANN) search:

- In-Memory Neighbor Graph Vector Index
- Neighbor Partition Vector Index

The distance function used to create and search the index should be the one recommended by the embedding model used to create the vectors. You can specify this distance function at the time of index creation or when you perform a similarity search using the `VECTOR_DISTANCE()` function. If you use a different distance function than the one used to create the index, an exact match is triggered because you cannot use the index in this case.

> **Note**
>
> - Oracle AI Vector Search indexes supports the same distance metrics as the `VECTOR_DISTANCE()` function. `COSINE` is the default metric if you do not specify any metric at the time of index creation or during a similarity search using the `VECTOR_DISTANCE()` function.
> - You should always define the distance metric in an index based on the distance metric used by the embedding model you are using.

- In-Memory Neighbor Graph Vector Index
  Hierarchical Navigable Small World (HNSW) is the only type of In-Memory Neighbor Graph vector index supported. HNSW graphs are very efficient indexes for vector approximate similarity search. HNSW graphs are structured using principles from small world networks along with layered hierarchical organization.
- Neighbor Partition Vector Index
  Inverted File Flat (IVF) index is the only type of Neighbor Partition vector index supported. Inverted File Flat Index (IVF Flat or simply IVF) is a partitioned-based index that lets you balance high-search quality with reasonable speed.
- Partition Maintenance Operations and Vector indexes
  Partition administration is an important task to consider when working with partitioned tables. Partition Maintenance Operations are supported with tables that have either global IVF or HNSW indexes.
- Guidelines for Using Vector Indexes
  Use these guidelines to create and use Hierarchical Navigable Small World (HNSW) or Inverted File Flat (IVF) vector indexes.
- Index Accuracy Report
  The index accuracy reporting feature lets you determine the accuracy of your vector indexes.

- Vector Index Status, Checkpoint, and Advisor Procedures
  Review these high-level details on the `GET_INDEX_STATUS`, `ENABLE_CHECKPOINT`, `DISABLE_CHECKPOINT`, and `INDEX_VECTOR_MEMORY_ADVISOR` procedures that are available with the `DBMS_VECTOR` PL/SQL package.

In-Memory Neighbor Graph Vector Index

Hierarchical Navigable Small World (HNSW) is the only type of In-Memory Neighbor Graph vector index supported. HNSW graphs are very efficient indexes for vector approximate similarity search. HNSW graphs are structured using principles from small world networks along with layered hierarchical organization.

- About In-Memory Neighbor Graph Vector Index
  The default type of index created for an In-Memory Neighbor Graph vector index is Hierarchical Navigable Small World.
- Local Hierarchical Navigable Small World Indexes
  A local HNSW index is an index created for each partition or sub partition of a partitioned table. Instead of building a single, global HNSW (Hierarchical Navigable Small World) graph across all vectors in a large, partitioned table, Oracle enables the creation of individual HNSW graphs for each (sub)partition. This local partitioned indexing approach improves scalability, performance, and manageability essential for enterprise workloads involving high-dimensional similarity search applications, such as semantic search, recommendation systems, and AI-driven analytics.
- Included Columns with HNSW Indexes
  Included columns in vector indexes facilitate faster searches with attribute filters by incorporating non-vector columns within an In-Memory Neighbor Graph Vector Index.
- Scalar Quantized HNSW Indexes
  Scalar quantized HNSW indexes can be used to reduce memory requirements and accelerate similarity search.
- Hierarchical Navigable Small World Index Syntax and Parameters
  Syntax and examples for creating Hierarchical Navigable Small World vector indexes.

About In-Memory Neighbor Graph Vector Index

The default type of index created for an In-Memory Neighbor Graph vector index is Hierarchical Navigable Small World.

- Overview of Hierarchical Navigable Small World Indexes
  Use these examples to understand how to create HNSW indexes for vector approximate similarity searches.
- HNSW Index Architecture: Transaction Support and Persistence
  Hierarchical Navigable Small World (HNSW) indexes are specialized, memory-only structures designed for efficient vector search. Once an HNSW index is created, any subsequent inserts, updates, or deletes (DML operations) performed on the base table will not be reflected in the index.
- HNSW RAC Duplication
  Learn how Hierarchical Navigable Small World (HNSW) indexes are populated during index creation, index repopulation, or instance startup in an Oracle Real Application Clusters (Oracle RAC) or a non-RAC environment.
- HNSW Distribution on Oracle RAC
  Learn how Hierarchical Navigable Small World (HNSW) indexes can be distributed across multiple Oracle Real Application Clusters (Oracle RAC) instances.

Overview of Hierarchical Navigable Small World Indexes

Use these examples to understand how to create HNSW indexes for vector approximate similarity searches.

With Navigable Small World (NSW), the idea is to build a proximity graph where each vector in the graph connects to several others based on three characteristics:

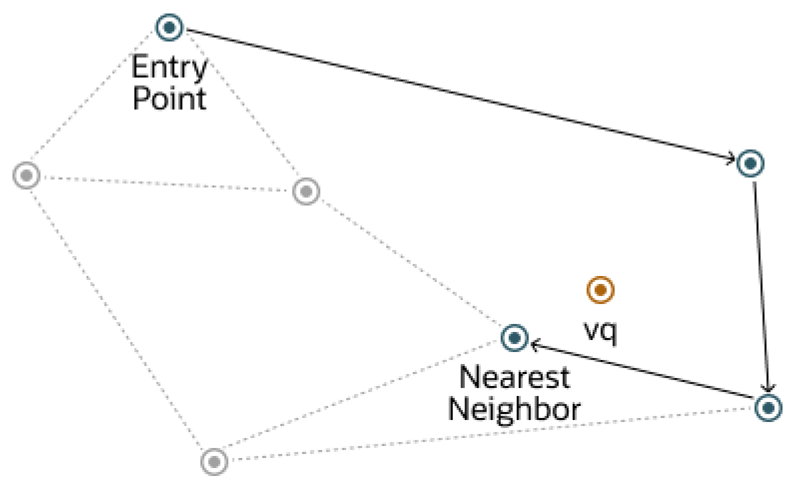
- The distance between vectors
- The maximum number of closest vector candidates considered at each step of the search during insertion (`EFCONSTRUCTION`)
- The maximum number of connections (`NEIGHBORS`) permitted per vector

If the combination of the above thresholds is too high, then you may end up with a densely connected graph, which can slow down the search process. On the other hand, if the combination of those thresholds is too low, then the graph may become too sparse and/or disconnected, which makes it challenging to find a path between certain vectors during the search.

Navigable Small World (NSW) graph traversal for vector search begins with a predefined entry point in the graph, accessing a cluster of closely related vectors. The search algorithm employs two key lists: Candidates, a dynamically updated list of vectors that we encounter while traversing the graph, and Results, which contains the vectors closest to the query vector found thus far. As the search progresses, the algorithm navigates through the graph, continually refining the Candidates by exploring and evaluating vectors that might be closer than those in the Results. The process concludes once there are no vectors in the Candidates closer than the farthest in the Results, indicating a local minimum has been reached and the closest vectors to the query vector have been identified.

This is illustrated in the following graphic:

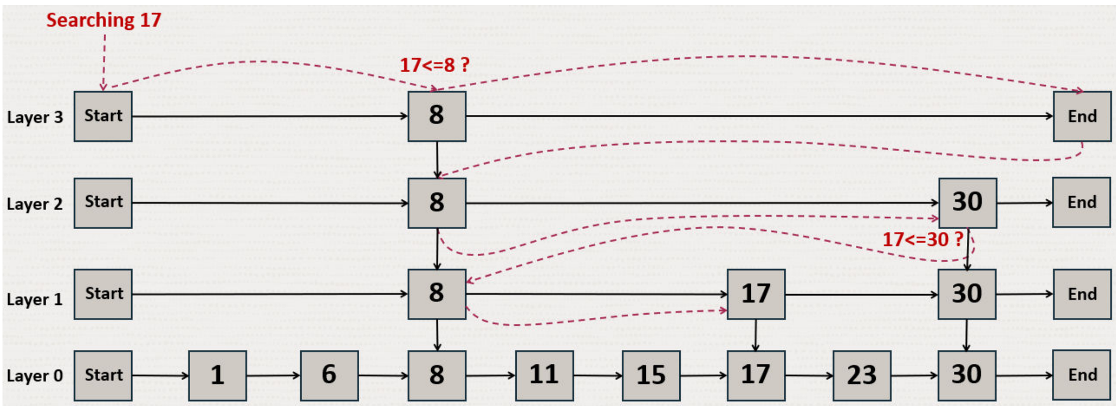
Figure 6-2 Navigable Small World Graph



The described method demonstrates robust performance up to a certain scale of vector insertion into the graph. Beyond this threshold, the Hierarchical Navigable Small World (HNSW) approach enhances the NSW model by introducing a multi-layered hierarchy, akin to the structure observed in Probabilistic Skip Lists. This hierarchical architecture is implemented by distributing the graph's connections across several layers, organizing them in a manner where each subsequent layer contains a subset of the vectors from the layer below. This stratification ensures that the top layers capture long-distance links, effectively serving as express pathways across the graph, while the lower layers focus on shorter links, facilitating fine-grained, local navigation. As a result, searches begin at the higher layers to quickly approximate the region of the target vector, progressively moving to lower layers for a more precise search, significantly improving search efficiency and accuracy by leveraging shorter links (smaller distances) between vectors as one moves from the top layer to the bottom.

To better understand how this works for HNSW, let's look at how this hierarchy is used for the Probability Skip List structure:

Figure 6-3 Probability Skip List Structure

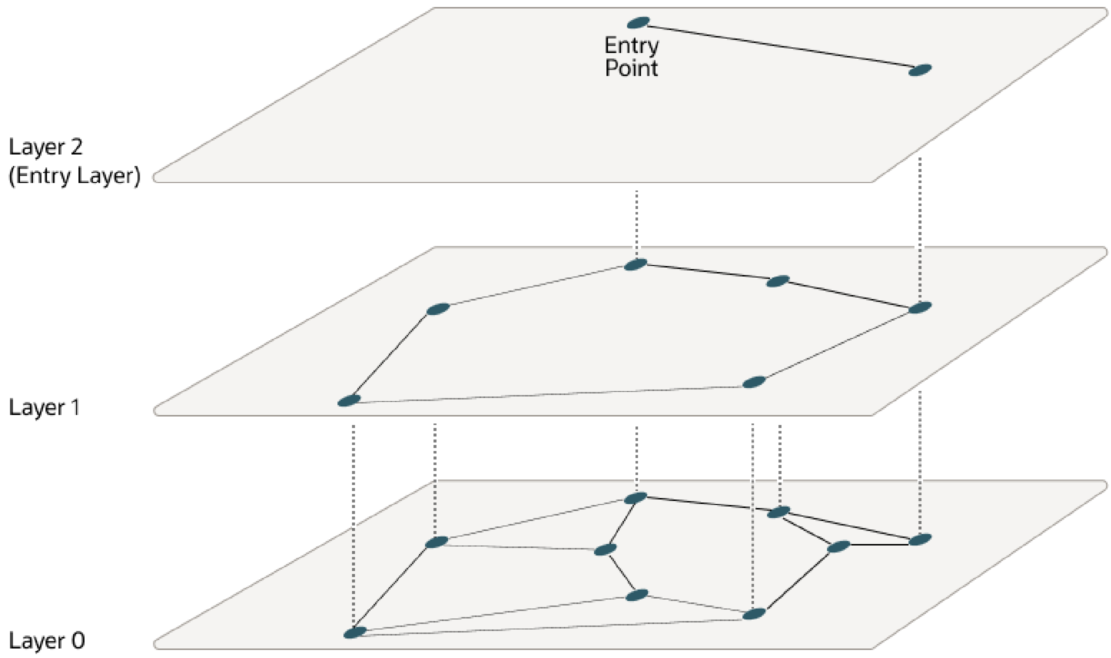


The Probability Skip List structure uses multiple layers of linked lists where the above layers are skipping more numbers than the lower ones. In this example, you are trying to search for number 17. You start with the top layer and jump to the next element until you either find 17, reach the end of the list, or you find a number that is greater than 17. When you reach the end of a list or you find a number greater than 17, then you start in the previous layer from the latest number less than 17.

HNSW uses the same principle with NSW layers where you find greater distances between vectors in the higher layers. This is illustrated by the following diagrams in a 2D space:

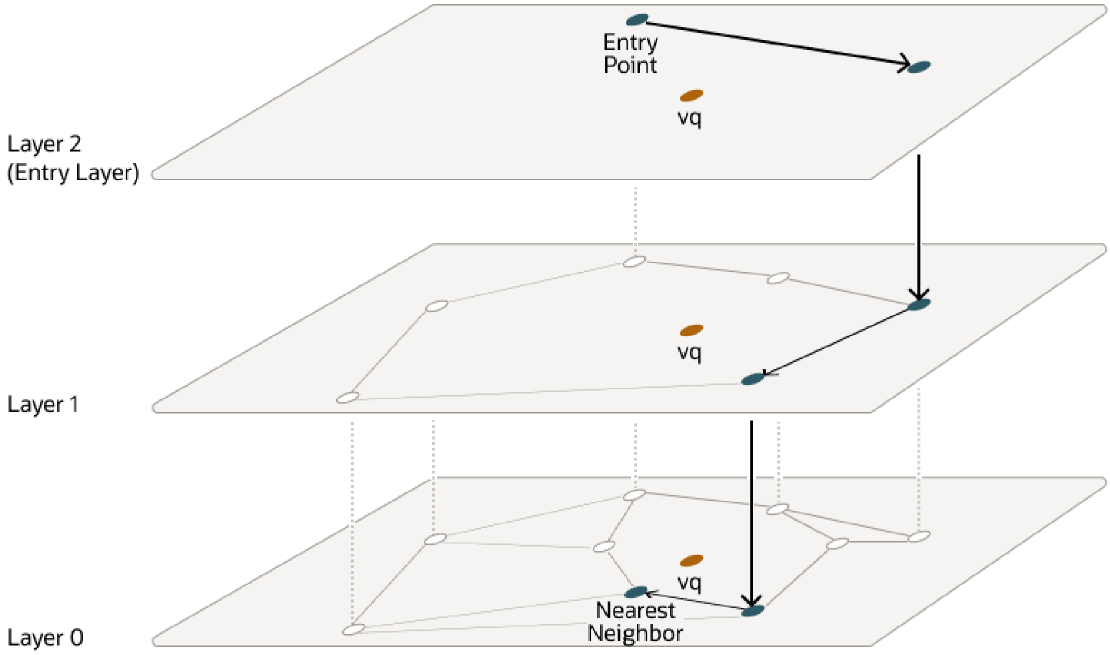
At the top layer are the longest edges and at the bottom layer are the shortest ones.

Figure 6-4   Hierarchical Navigable Small World Graphs



Starting from the top layer, the search in one layer starts at the entry vector. Then for each node, if there is a neighbor that is closer to the query vector than the current node, it jumps to that neighbor. The algorithm keeps doing this until it finds a local minimum for the query vector. When a local minimum is found in one layer, the search goes to the next layer by using the same vector in that new layer and the search continues in that layer. This process repeats itself until the local minimum of the bottom layer is found, which contains all the vectors. At this point, the search is transformed into an approximate similarity search using the NSW algorithm around that latest local minimum found to extract the top `k` most similar vectors to your query vector. While the upper layers can have a maximum of connections for each vector set by the `NEIGHBORS` parameter, layer `0` can have twice as much. This process is illustrated in the following graphic:

Figure 6-5 Hierarchical Navigable Small World Graphs Search



Layers are implemented using in-memory graphs (not Oracle Inmemory graph). Each layer uses a separate in-memory graph. As already seen, when creating an HNSW index, you can fine tune the maximum number of connections per vector in the upper layers using the `NEIGHBORS` parameter as well as the maximum number of closest vector candidates considered at each step of the search during insertion using the `EFCONSTRUCTION` parameter, where EF stands for Enter Factor.

As explained earlier, when using Oracle AI Vector Search to run an approximate search query using HNSW indexes, you have the possibility to specify a target accuracy at which the approximate search should be performed.

In the case of an HNSW approximate search, you can specify a target accuracy percentage value to influence the number of candidates considered to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose a similar result as an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the `EFSEARCH` parameter to impose a certain maximum number of candidates to be considered while probing the index. The higher that number, the higher the accuracy.

> **Note**
>
> - If you do not specify any target accuracy in your approximate search query, then you will inherit the one set when the index was created. You will see that at index creation, you can specify a target accuracy either using a percentage value or parameters values depending on the index type you are creating.
> - It is possible to specify a different target accuracy at index search compared to the one set at index creation. For HNSW indexes, you may look at more neighbors using the `EFSEARCH` parameter (higher than the `EFCONSTRUCTION` value specified at index creation) to get more accurate results. The target accuracy that you give during index creation decides the index creation parameters and also acts as the default accuracy values for vector index searches.

HNSW Index Architecture: Transaction Support and Persistence

Hierarchical Navigable Small World (HNSW) indexes are specialized, memory-only structures designed for efficient vector search. Once an HNSW index is created, any subsequent inserts, updates, or deletes (DML operations) performed on the base table will not be reflected in the index.

To maintain accuracy, performance, and consistency with transactional data operations, Oracle employs several supporting data structures. This page outlines the key internal components used in maintaining transactional vector search and persistence for HNSW indexes.

Transaction Support Structures:

- A **private journal** is an in-memory, per-transaction structure that resides in the vector memory pool. It captures all vectors inserted or deleted by an active transaction. This is comparable to transaction journals that are used to maintain the in-memory column store data (explained in *Oracle AI Database In-Memory Guide*).
- The **shared journal** is an on-disk, table-backed component created alongside each HNSW index. It holds the history of committed transactions that modified the indexed vector columns. Each journal entry is associated with a commit SCN, and includes metadata such as inserted and deleted vector identifiers.
- Every HNSW index includes a dedicated **ROWID-to-VID mapping table** that links each base table row ID (ROWID) to the corresponding vector ID (VID) used in the HNSW graph.

Graph Refresh and Persistence Structures:

Queries that come after an index is created would need to lookup the index as well as the DMLs that occurred after it to get the top-K result. With accumulating DMLs, queries become slower as the exact search on the shared journal vectors becomes more expensive than the approximate search on the currently indexed HNSW graph. To maintain performance and accuracy, Oracle provides automated graph refresh and persistence mechanisms.

- An **incremental snapshot** represents an updated in-memory version of the HNSW graph associated with a specific SCN. It adds newly inserted vectors into the graph and tracks deletions using compact bitmaps. Snapshots reduce shared journal overhead, improve query performance, and minimize memory usage. They are especially effective when frequent small DML operations occur.

> **Note**
>
> – Any query that comes below the build SCN for the current latest snapshot runs into error `ORA 51815 "INMEMORY NEIGHBOR GRAPH HNSW vector index snapshot is too old."`

- When incremental refresh becomes inefficient, usually due to the accumulation of deleted vectors, a **full repopulation** is triggered to rebuild the HNSW graph from scratch using the current base table. Unlike incremental snapshots, which updates an existing graph, full repopulation builds a fresh graph while keeping the old one active for ongoing queries.

  > **Note**
  >
  > At the time of full repopulation (until a new HNSW graph becomes available), if a query tries to access an older version of the HNSW graph that no longer exists, then the read consistency error `ORA-51815 "INMEMORY NEIGHBOR GRAPH HNSW vector index snapshot is too old."` is raised.

- A **checkpoint** is a disk-based, serialized copy of the HNSW graph's topology and metadata (not the actual vectors). It is created automatically during key events like index creation or repopulation.

  A distributed HNSW index creates multiple checkpoints, one for each HNSW slice graph hosted by each RAC instance. Checkpointed graphs are not associated with specific physical instances. Instead, they are associated with the instances that owns the HNSW slice graph. The system tracks which HNSW slice graph belongs to which instance to enable easy redistribution of index during instance failures or removals. On instance failure, any RAC instance mapped to the same HNSW slice graph as the failed instance directly reuses the existing checkpointed graph, leading to faster recovery and less downtime.

  You can disable or re-enable full HNSW checkpoints by using the ENABLE_CHECKPOINT and DISABLE_CHECKPOINT procedures of the `DBMS_VECTOR` PL/SQL package.

Explicitly Specifying a Graph Refresh

You can use the `idx_rebuild_mode` parameter of the DBMS_VECTOR.REBUILD_INDEX procedure to specify how the HNSW graph should be refreshed. This parameter accepts values : `FULL` or `INCREMENTAL`. This enables a full graph rebuild. By default, `idx_rebuild_mode` is set to `NULL`. In this case, the system follows the existing behavior: drop and recreates the index. The `idx_rebuild_mode` allows for a fine-grained control when managing index refresh operations.

An example which triggers a full repopulation:

```
execute dbms_vector.rebuild_index('galaxies_hnsw_idx',
                                   'galaxies',
                                   'embedding',
                                    NULL,
                                    NULL,
                                   'INMEMORY NEIGHBOR GRAPH',
                                   'EUCLIDEAN',
                                    95,
                                    FULL,
```

```
                                   '{"type" : "HNSW", "neighbors" :
3, "efConstruction" : 4 }') ;
```

> **Note**
>
> This code assumes that the index `galaxies_hnsw_idx` is already created on the `galaxies` table. See the Oracle documentation on Hierarchical Navigable Small World (HNSW) index syntax and parameters for guidance on creating an HNSW index.

HNSW indexes rely on these structures to balance performance and accuracy while retrieving transactionally consistent top-K results. Together, they ensure scalable, low-latency vector search even as tables evolve through frequent updates and deletes.

HNSW RAC Duplication

Learn how Hierarchical Navigable Small World (HNSW) indexes are populated during index creation, index repopulation, or instance startup in an Oracle Real Application Clusters (Oracle RAC) or a non-RAC environment.

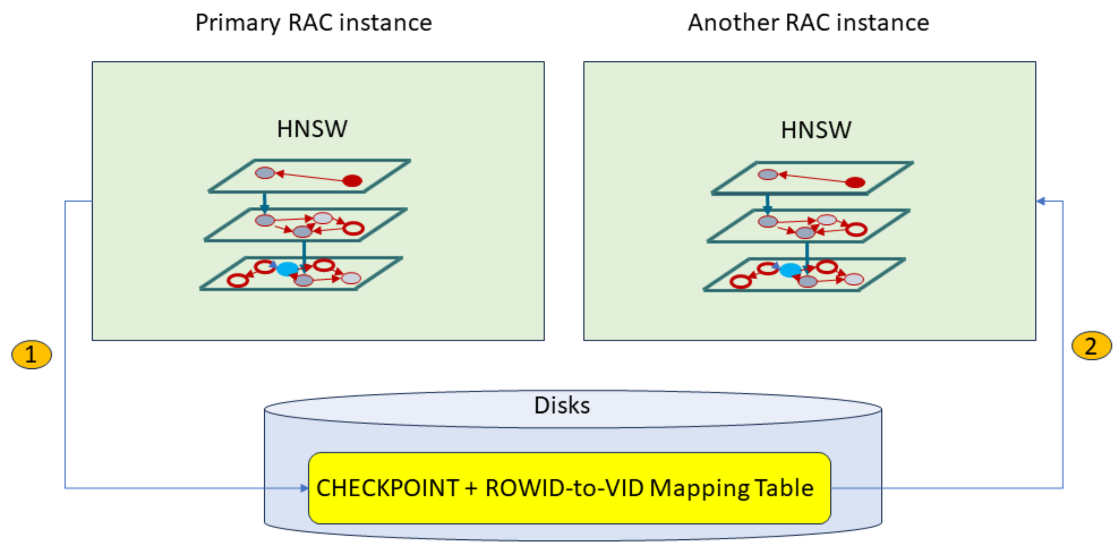
HNSW Index Creation in RAC Environment

Currently HNSW index in an Oracle RAC environment is created in **centralized mode**. The diagram below depicts the workflow for HNSW index creation in an Oracle RAC environment:

1. The RAC instance that creates the HNSW index is responsible for creating the HNSW ROWID-to-VID mapping table on the disk. As explained in Optimizer Plans for HNSW Vector Indexes, this table is needed by the optimizer to run certain optimization plans.
2. By default, once the first HNSW slice graph is created, all other RAC instances are informed to start their own HNSW In-Memory slice graph creation concurrently. This operation is called an **HNSW duplication** operation. Duplication is achieved in two steps: The primary RAC instance creates a disk checkpoint of the HNSW slice graph that it created and then all the other secondary RAC nodes use that disk checkpoint to load the HNSW slice graph. Creating HNSW index in the centralized mode ensures that all nodes share an identical graph structure.

   HNSW slice graphs constructed on the RAC nodes are categorized into the following two categories, based on whether the base table is partitioned. In this context, a slice refers to the subset of vector data used to build an index. A slice can represent either the entire table or an individual partition or sub partition of a table. Each HNSW graph is constructed over a single slice:

   - HNSW index table-slice graph
   - HNSW index partition-slice graph



> **Note**
>
> Instances that do not have enough vector memory cannot participate in the parallel RAC-wide HNSW index population.

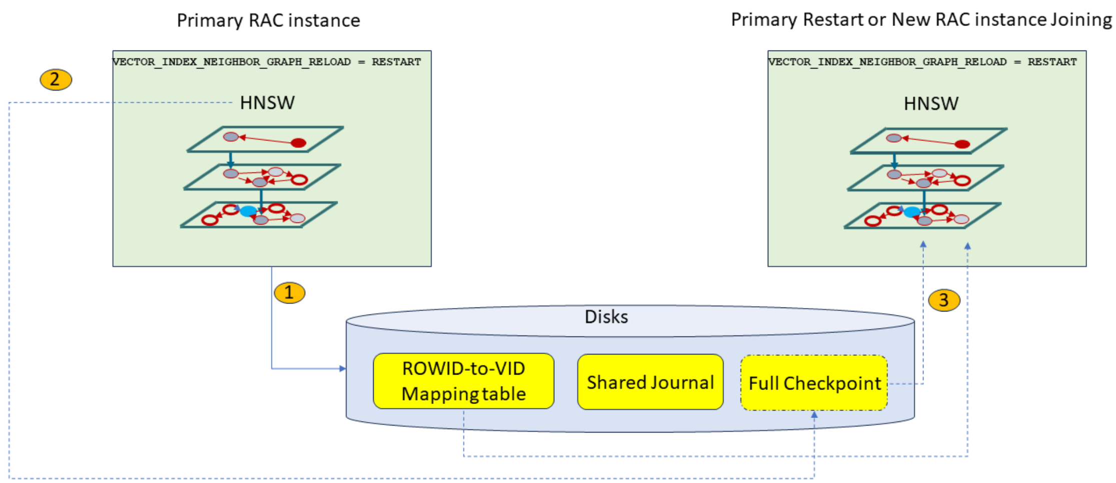
In addition to the **duplication mechanism** that creates the HNSW index across the RAC nodes, full repopulation is used to perform a **HNSW slice graph refresh duplication**. See HNSW Index Architecture: Transaction Support and Persistence for more information about why and when HNSW index full repopulation operation is triggered. The **HNSW slice graph refresh duplication** process is identical to that used during index creation: The primary instance initiates the HNSW slice graph refresh and creates a new full disk checkpoint. The other instances reload the HNSW slice graph from the latest checkpoint.

HNSW Index Reload at Instance Restart or New Node Joining Cluster

Because HNSW indexes reside in memory, the in-memory HNSW slice graphs representing your HNSW indexes are lost if an instance shuts down. Upon restart or when a new node joins the cluster, Oracle triggers a **reload** mechanism to quickly restore the HNSW slice graphs in memory. This reload mechanism is enabled by default for both Oracle RAC and non-RAC single instance environments. As explained in HNSW Index Creation in RAC Environment, during index creation time, the in-memory HNSW slice graph and the ROWID-to-VID mapping table on disks are created. If enabled, a full checkpoint is also created on disks. The way the reload mechanism is processed depends on the existence of a full HNSW slice graph checkpoint on disk. Read HNSW Index Architecture: Transaction Support and Persistence to understand more about checkpoints.

- **If a recent full checkpoint exists**: The instance loads the HNSW slice graph directly from the checkpoint, ensuring rapid recovery.
- **If no valid checkpoint is available**: HNSW slice graph is recreated from the scratch and the new HNSW slice graph will not be identical to the older HNSW slice graph.

The reload behavior is governed by the `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` parameter. If the `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` instance parameter is set to `RESTART` (default setting) at the time the instance joins the cluster or when restarting, and if a full checkpoint exists and is not deemed too old compared to the current instance SCN, then it is used by the starting instance to create its HNSW slice graph in memory. If these two conditions are not met, then the starting instance uses the duplication mechanism to create the HNSW slice graph in memory from scratch. If the `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` instance parameter is set to `OFF` at the time the instance joins the cluster or when restarting, then the HNSW slice graph is not reloaded.



HNSW Distribution on Oracle RAC

Learn how Hierarchical Navigable Small World (HNSW) indexes can be distributed across multiple Oracle Real Application Clusters (Oracle RAC) instances.

A distributed HNSW index on Oracle RAC enables scalable similarity search across multiple instances in a RAC (Real Application Clusters) environment. In a distributed HNSW index:

- The vector data is segmented and distributed across RAC instances in a memory-efficient manner.
- Each instance builds and manages an HNSW slice graph on its assigned subset of the vector data. The complete HNSW index is the logical union of these HNSW slice graphs.
- During query execution, each instance searches its HNSW slice graph, and the partial results are merged to produce the final query result.

Why use a distributed HNSW index?

The following section summarize the advantages of using a distributed HNSW index.

- Horizontal Scalability

  Distributing vector data across multiple RAC instances in a memory-efficient manner enables the total available memory to scale with cluster size. HNSW indexes are no longer constrained by the memory or processing resources of a single instance. Each RAC instance independently builds and maintains its HNSW slice graph, allowing a single distributed HNSW index to scale with increasing data volume and vector dimensionality.

- Parallelization

  Enables parallel index operations and query execution across multiple RAC instances. Each RAC instance handles its own HNSW index maintenance and query processing, preventing a single instance from becoming a bottleneck, resulting in improved query throughput.

- Efficiency

  Eliminates redundant graph storage and build-time computations across instances. This significantly improves memory utilization and reduces overall index build and maintenance costs.
- Resilience and Flexibility

  Automatically adapts to cluster changes by redistributing the index data as needed. This improves fault isolation and supports cluster re-balancing when instances fail or are removed from the cluster. For more information, refer to About HNSW Index Availability and Performance During Cluster Node Failure

Usage Notes

- No Support for Parallel DML Operations

  Distributed HNSW index does not support parallel DML operations.
- No Support for Snapshots (Full Repopulation Required)

  Distributed HNSW indexes do not support snapshots. Full repopulation is the only method used to refresh a distributed HNSW index. See HNSW Index Architecture: Transaction Support and Persistence for more information about why and when HNSW index full repopulation operation is triggered.
- No Support for Covering Columns

  Including covering columns (extra columns) in the HNSW index is not supported. If your query needs columns other than the indexed vector column, Oracle Database automatically uses the base table while processing the query.
- Index Rebuild on Certain Cluster Changes

  If the cluster size decreases such that there are not enough available instances to host all HNSW slice graphs, Oracle rebuilds the index.
- No Support for Online Index Build

  Online creation of distributed HNSW indexes with concurrent DML operations is not supported.
- Index Unavailability During Cluster Reconfiguration

  When an instance fails or is removed, affected HNSW slice graphs are temporarily disabled. The index remains unavailable for queries until graphs are redistributed or rebuilt.
- Scalar Quantization Support
  Scalar quantization is supported with distributed HNSW indexes and can be specified during index creation. For more information, see Scalar Quantized HNSW Indexes.

Distributed HNSW Index: High-Level Workflow

- Index Creation with the `DISTRIBUTE` Clause

  You create a distributed HNSW index by specifying the `DISTRIBUTE` clause in the `CREATE VECTOR INDEX` statement. You can choose one of the following distribution strategies:
  - `BY ROWID RANGE`
  - `BY PARTITION`
  - `BY SUBPARTITION`

If no distribution strategy is specified along with the `DISTRIBUTE` clause, the Oracle Database defaults to `DISTRIBUTE AUTO`. In `DISTRIBUTE AUTO` mode, non-partitioned tables use `ROWID RANGE` distribution and partitioned tables use `[SUB] PARTITION` distribution.

Distributed HNSW index creation syntax:

```
CREATE VECTOR INDEX index_name ON table_name(column_name)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
WITH <distance_and_target_metrics>
[QUANTIZATION SCALAR COMPRESSION RATIO {2|4|8}]
[PARAMETERS (TYPE HNSW ,
             {NEIGHBORS max_closest_vectors_connected | M
max_closest_vectors_connected},
             EFCONSTRUCTION max_candidates_to_consider,
             RESCORE FACTOR rescore_factor,
             ALGORITHM UNIFORM_QUANTIZATION
            )]
DISTRIBUTE [AUTO | BY ROWID RANGE | BY PARTITION | BY SUBPARTITION ];
```

> **See Also**
>
> CREATE VECTOR INDEX

- RAC Instance Memory Assessment

  The coordinating instance initiates a memory check (a cross-instance coordination call) to estimate available vector pool memory across all participating RAC instances. Each participating instance reports its projected available memory. This information is used to assign vector data proportionally and efficiently across instances.

- Partitioning the Vector Dataset

  Oracle Database divides the vector dataset into logical units called vector distribution units using the selected distribution strategy. The mapping between distribution units and HNSW slice graphs is recorded in auxiliary metadata structures for maintenance and recovery.

- Initiation of Distributed Graph Build

  The coordinating instance triggers all participating instances (including itself) to start building their HNSW slice graphs for their assigned vector data. Graph construction is centrally coordinated and monitored to ensure all instances complete their HNSW slice graph construction successfully.

- Graph Construction on RAC

  Each RAC instance reads its assigned vector data from the base table and builds its HNSW slice graph. It then creates a checkpoint of the completed graph.

  The HNSW slice graph constructed on assigned vector data is categorized according to the distribution strategy used to partition the vector dataset:

  – **HNSW rowid-range-slice graph**: An HNSW graph built over a subset of vector data partitioned using the `ROWID RANGE` distribution strategy. In this context, “slice” refers to the subset of vector data, while “rowid-range” specifies the distribution strategy.

  – **HNSW partition-slice graph**: An HNSW graph built over a subset of vector data partitioned using the `PARTITION` or `SUBPARTITION` distribution strategy. In this context, “slice” refers to the subset of vector data, while “partition” specifies the distribution strategy.

- Index Finalization

  Once all HNSW slice graphs of the distributed HNSW index are built and checkpointed, the coordinating instance marks the index as usable. The index is now available for distributed, parallel similarity search queries, each routed efficiently to the relevant instances.

- Distributed Query Execution

  When a vector search query is issued, the Oracle Database identifies all instances that host an HNSW slice graph. The query is executed in parallel across these instances.

  Each query performs an approximate nearest neighbor (ANN) search on its HNSW slice graph using the configured search parameters such as `ef_search` and `k` (number of neighbors).

  To ensure transactional consistency, each instance scans the shared journal for unindexed changes (committed inserts and deletes that are not yet indexed) and includes them in the similarity search.

  The query optimizer selects distributed index plans only when sufficient parallelism is available (at least one PQ process per index-hosting instance). If not, Oracle Database falls back to a no-index plan.

  For information on optimizer plans, refer to Optimizer Plans for HNSW Vector Indexes.

- Index Maintenance and Persistence

  – HNSW multiple checkpoints: If checkpoint creation is enabled, multiple checkpoints are created for faster recovery after instance failures. One checkpoint for each HNSW slice graph hosted by each RAC instance. For more information refer to Checkpointing .

  – Cluster reconfiguration: A leader instance monitors cluster changes such as instance failures, or removals to maintain index consistency by dynamically adapting to these changes. For more information refer to About HNSW Index Availability and Performance During Cluster instance Failure

HNSW RAC Duplication vs. HNSW Index Distribution

Table 6-1 Duplicated vs. Distributed

HNSW RAC Duplication	HNSW Index Distribution
Data access Each instance reads the entire vector dataset to build the full HNSW graph.	Data access The vector dataset is split into vector distribution units (`by ROWID RANGE`, `by [SUB]PARTITION`). Each instance only reads a subset of vector data assigned to it.
Index structure Each RAC instance has an identical, full copy of the HNSW graph for the entire vector dataset.	Index structure Each instance holds only a portion (HNSW slice graph) of the full HNSW index.
Memory usage High: Every instance must hold the entire index in memory.	Memory usage Efficient: Memory load is split across instances, and each instance holds only a portion (HNSW slice graph) of the full HNSW index.

Table 6-1 (Cont.) Duplicated vs. Distributed

HNSW RAC Duplication	HNSW Index Distribution
Index build process The coordinating RAC instance creates a disk checkpoint of the HNSW graph that it created and then all the other participating RAC instances use that disk checkpoint to load the graph. Every instance independently loads the entire HNSW graph using the full vector dataset.	Index build process Each instance builds an HNSW slice graph for a subset of vector data assigned to the instance.
Query execution Queries execute on a single instance (no parallelism across RAC instances).	Query execution Queries are distributed and parallelized across all instances holding HNSW slice graphs.
Resource utilization Poor: Redundant storage and build compute on all instances.	Resource utilization Good: Storage is not redundant and workload is balanced across all instances.
Scalability Limited by the memory and processing power of a single instance.	Scalability Scales with cluster size; can handle much larger datasets.
Cluster reconfiguration All instances must reload or rebuild full HNSW index if cluster changes.	Cluster reconfiguration Only affected HNSW slice graphs are redistributed or rebuilt.
Best use case Small datasets	Best use case Large or growing datasets

- About Distributed HNSW Index Availability and Performance During Cluster instance Failure

About Distributed HNSW Index Availability and Performance During Cluster instance Failure

Distributed HNSW (Hierarchical Navigable Small World) indexes in a Real Application Clusters (RAC) environment are designed to stay reliable even when the cluster undergoes changes, such as instance failures or removals. The system continuously monitors cluster changes such as instance failures, additions, and removals. It tracks HNSW slice graph status, assignments, recovery tasks, and ownership changes, and ensures this information is consistently visible across all instances.

The following steps are automatically performed to handle cluster reconfiguration and recovery:

- Disable index

  If an instance hosting a HNSW slice graph fails or becomes unreachable, the Oracle Database quickly detects which HNSW indexes are impacted. Affected HNSW indexes are temporarily disabled to prevent incomplete or incorrect query results.

- Temporarily switch to a no-index plan

  Any open queries or processes using disabled indexes are invalidated. The affected queries are automatically switched to no-index plans such as full scans or alternative indexes. Until HNSW slice graphs from failed instances are reloaded on available instances, queries execute without using the distributed HNSW index. Oracle Database invalidates the cursors for queries that rely on disabled indexes and automatically re-compiles them once the HNSW slice graphs are available.

- Redistribution of HNSW slice graphs and vector distribution units

  Once the cluster stabilizes, HNSW slice graphs from the failed instances are reassigned to available healthy instances based on memory availability.

  Vector distribution units are recomputed only when required. Since HNSW slice graphs are checkpointed, the Oracle Database can quickly reload the affected graphs on healthy instances without reprocessing the entire vector dataset.

> **Note**
>
> In case of multiple concurrent instance failures or permanently reduced cluster size, the Oracle Database may not be able to reassign the HNSW slice graphs directly. If the available healthy instances cannot accommodate all HNSW slice graphs for an index, the Oracle Database falls back to rebuilding the index.

Local Hierarchical Navigable Small World Indexes

A local HNSW index is an index created for each partition or sub partition of a partitioned table. Instead of building a single, global HNSW (Hierarchical Navigable Small World) graph across all vectors in a large, partitioned table, Oracle enables the creation of individual HNSW graphs for each (sub)partition. This local partitioned indexing approach improves scalability, performance, and manageability essential for enterprise workloads involving high-dimensional similarity search applications, such as semantic search, recommendation systems, and AI-driven analytics.

In addition, partition pruning and parallel execution further optimize query efficiency.

Using partition pruning Oracle Database restricts query processing to only those partitions that satisfy the query's filtering predicates. The optimizer analyzes `FROM` and `WHERE` clauses in SQL statements to eliminate unneeded partitions thereby minimizing unnecessary data access and computation.

For example, if a houses table is partitioned by `state` column using list or hash partitioning, and a query specifies `WHERE state = CA`, then the Oracle Database performs partition pruning. It searches only the `CA` partition and its local HNSW index to process the query:

```
SELECT id
FROM houses

WHERE state = 'CA'
ORDER BY vector_distance(data_vector, :query_vector)
FETCH FIRST 10 ROWS ONLY;

--With a local HNSW index, only the CA partition and its local HNSW graphs
are searched.
```

For more information on partition pruning, see Partition Pruning.

Parallel execution further amplifies efficiency by allowing multiple partitions and their corresponding local HNSW indexes to be searched simultaneously across multiple CPU threads.

Usage Notes

- The base table must be partitioned. Local HNSW indexes can only be created on partitioned tables.
- The base table must be loaded with vector data before creating local HNSW indexes. After creating local HNSW indexes, any DML operations (INSERT, UPDATE, DELETE) on the indexed base table results in errors. Therefore, it is essential to load the table with vector data before creating the local HNSW indexes.
- Adequate sizing of the SGA "Vector Memory Pool" is needed for in-memory index operations.
- Including column clause (covering columns) is not supported.
- Local HNSW on Sparse vector columns is not supported.
- Online creation of local HNSW vector indexes is not supported.
- Parameters, such as Target Accuracy, Distance metric, HNSW graph parameters, and table space clause are globally applied to all HNSW index partitions.
- DML operations (insert, update, delete) are not integrated for local HNSW indexes. This means if you insert, update, or delete rows in the underlying partitioned table, the corresponding partitioned local HNSW index will not reflect those changes.
- In case of HNSW RAC Duplication in an Oracle RAC environment, partitioned local HNSW indexes are built by constructing an independent HNSW graph for each partition of the table, with index creation and loading distributed and coordinated across RAC nodes. This can be achieved in two modes: centralized mode, where one node builds the HNSW graph for a partition and writes a checkpoint that other nodes then load to ensure consistency; or decentralized mode, where each node independently builds its assigned partition graphs in parallel without relying on shared checkpoints.

Syntax

```
CREATE VECTOR INDEX VIDX_HNSW ON HOUSES(VEC) ORGANIZATION INMEMORY NEIGHBOR
GRAPH [WITH TARGET ACCURACY 95] [DISTANCE EUCLIDEAN] [PARAMETERS (type HNSW,
neighbors 32, efConstruction 500)] LOCAL;
```

Example

```
CREATE VECTOR INDEX VIDX_HNSW ON HOUSES(VEC)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  [WITH TARGET ACCURACY 95]
  [DISTANCE EUCLIDEAN]
  [PARAMETERS (type HNSW, neighbors 32, efConstruction 500)]
  LOCAL;
```

Included Columns with HNSW Indexes

Included columns in vector indexes facilitate faster searches with attribute filters by incorporating non-vector columns within an In-Memory Neighbor Graph Vector Index.

Consider a classic example where you are analyzing the home price data. You want only the top 5 homes from your description where the price is less than 2 million dollars and/or from the specific zip code. Or to a finer granularity, you just need to retrieve only the top home that matches the description and your filter attribute. Included columns make this possible by incorporating filterable attributes, such as price and zip code, directly into the vector index. This integration allows the index to evaluate and return results based on both the vector similarity and the attribute filters, without requiring additional steps to cross-reference the base table.

By bridging the gap between traditional database filters and AI-powered vector searches, included columns deliver faster, more efficient results. This synergy enhances search performance, especially in use cases requiring fine-grained filtering alongside advanced similarity computations.

Syntax and Specifications

Create a base table, which contains house sales data with prices, descriptions, and vectors:

```
CREATE TABLE houses (id NUMBER, zip_code NUMBER,
    price NUMBER, description CLOB, data_vector VECTOR);
```

Then, you can create an HNSW vector index **with an included column on `price`** as follows:

```
CREATE VECTOR INDEX vidx_hnsw
ON houses (data_vector)
INCLUDE (price)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
WITH TARGET ACCURACY 95;
```

The above syntax creates a vector index, `VIDX_HNSW`, with `ORGANIZATION` as `INMEMORY NEIGHBOR GRAPH` on the `DATA_VECTOR` column where `PRICE` is an included column. The `INCLUDE` keyword allows a user to specify the attributes they wish to include in the HNSW index.

You can check the included columns by querying the `V$VECTOR_GRAPH_INDEX` view:

```
SELECT OWNER, INDEX_NAME, COVERING_COLS
FROM V$VECTOR_GRAPH_INDEX;



OWNER   INDEX_NAME   COVERING_COLS
------- ------------ --------------------------------
SYS     VIDX_HNSW    PRICE
```

Note

Restrictions on `INCLUDE` :
The `INCLUDE` list can contain:

- Limited to a maximum of 31 columns - Oracle indexes support 32 key columns.
- Supports only the following data types : `NUMBER`, `CHAR`, `VARCHAR2`, `DATE`, `TIMESTAMP`, `RAW`, `JSON`, and reference-based LOB types such as `BLOB` and `CLOB`.
- Does not support NLS types and `LONG` types.

Benefits of Using Included Columns

1. **HNSW No-Filter with and without Included Columns**:
   If you consider a non-filter query that fetches the five homes nearest to a specified vector, the resulting query will result in an execution plan with multiple joins.

```
SELECT /*+ VECTOR_INDEX_TRANSFORM(t) */ price
FROM houses t
ORDER BY VECTOR_DISTANCE(data_vector, :query_vector)
FETCH FIRST 5 ROWS ONLY;
```

> **Note**
>
> `query_vector` contains the actual input vector. You can use the instructions mentioned in [SQL Quick Start Using a FLOAT32 Vector Generator](#) to generate query_vector.

The multiple joins are displayed in the following execution plan:

```
PLAN_TABLE_OUTPUT
---------------------------------------------------------------------------
-----------------
Plan hash value: 2484688059

---------------------------------------------------------------------------
-----------------
| Id | Operation                      | Name         | Rows | Bytes | Cost
(%CPU)| Time     |
---------------------------------------------------------------------------
-----------------
|  0 | SELECT STATEMENT               |              |    5 |    20 |
3 (67)| 00:00:01 |
|* 1 |  COUNT STOPKEY                 |              |      |
|            |          |
|  2 |   VIEW                         |              |    5 |    20 |
3 (67)| 00:00:01 |
|* 3 |    SORT ORDER BY STOPKEY       |              |    5 |  7965 |
3 (67)| 00:00:01 |
|  4 |     TABLE ACCESS BY INDEX ROWID| HOUSES       |    5 |  7965 |
2 (50)| 00:00:01 |
|  5 |      VECTOR INDEX HNSW SCAN    | VIDX_HNSW    |    5 |  7965 |
2 (50)| 00:00:01 |
---------------------------------------------------------------------------
-----------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=5)
   3 - filter(ROWNUM<=5)
```

You can see in the execution plan that the `VIDX_HNSW` index is scanned for the closest vectors and then the base table `HOUSES` is accessed by the `INDEX ROWID` returned. This is because after we have located the vectors in the HNSW Index, we return to the `HOUSES` table to retrieve the non-indexed columns from the base table.

The following execution plan displays the same operation when using an In-Memory Neighbor Graph vector index (HNSW) created with included columns:

```
PLAN_TABLE_OUTPUT
---------------------------------------------------------------------------
------------
Plan hash value: 656317923

---------------------------------------------------------------------------
------------
| Id | Operation                  | Name         | Rows  | Bytes | Cost
(%CPU)| Time     |
---------------------------------------------------------------------------
------------
|  0 | SELECT STATEMENT           |              |     5 |    20 |      3
(67)| 00:00:01 |
|* 1 |  COUNT STOPKEY             |              |       |       |
|          |
|  2 |   VIEW                     |              |     5 |    20 |      3
(67)| 00:00:01 |
|* 3 |    SORT ORDER BY STOPKEY   |              |     5 |  7965 |      3
(67)| 00:00:01 |
|  4 |     VECTOR INDEX HNSW SCAN| VIDX_HNSW   |     5 |  7965 |      2
(50)| 00:00:01 |
---------------------------------------------------------------------------
------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=5)
   3 - filter(ROWNUM<=5)
```

As you can see in the execution plan above, adding included columns to a vector index ensures there is no base table pre-filter evaluation required. The query can be resolved by scanning only the HNSW vector index as the columns queried can be sourced directly from the index.

2. **HNSW Filtering with Included Columns**:
One of the more commonly used plans for neighbor graph vector indexes is the pre-filter plan. In this case you are filtering the contents of the base table to eliminate non-relevant rows. But evaluating a query without using a filter attribute can be very expensive as the operation involves several joins. The following query uses a neighbor partition vector index without included columns:

```
SELECT /*+ vector_index_transform(houses vidx_hnsw
in_filter_with_join_back) */
price
FROM houses
WHERE price < 2000000
```

```
ORDER BY vector_distance(data_vector, :query_vector)
FETCH FIRST 5 ROWS ONLY;
```

> **Note**
>
> `query_vector` contains the actual input vector. You can use the instructions mentioned in [SQL Quick Start Using a FLOAT32 Vector Generator] to generate query_vector.

On the other hand, if you use the same in-filter query on the HNSW index created with `price` as the `included` column, the execution plan shows fewer joins, as the join with the base table is avoided. This is shown in the plan below:

```
PLAN_TABLE_OUTPUT
---------------------------------------------------------------------------
---------------------------
Plan hash value: 942384497

---------------------------------------------------------------------------
---------------------------
| Id | Operation                           | Name             | Rows   |
Bytes | Cost (%CPU)| Time     |
---------------------------------------------------------------------------
---------------------------
|  0 | SELECT STATEMENT                    |                  |     5 |
20 |  1272   (1)| 00:00:01 |
|* 1 |  COUNT STOPKEY                      |                  |
|       |             |          |
|  2 |   VIEW                              |                  |     5 |
20 |  1272   (1)| 00:00:01 |
|* 3 |    SORT ORDER BY STOPKEY            |                  |     5 |
8025 |  1272   (1)| 00:00:01 |
|  4 |     VECTOR INDEX HNSW SCAN IN-FILTER| VIDX_HNSW        |     5 |
8025 |  1271   (1)| 00:00:01 |
|  5 |      VIEW                           | VW_HIJ_3C2A15A9 |     1
|       |     2   (0)| 00:00:01 |
|* 6 |       FILTER                        |                  |
|       |             |          |
|  7 |        FAST DUAL                    |                  |     1
|       |     2   (0)| 00:00:01 |
---------------------------------------------------------------------------
---------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=5)
   3 - filter(ROWNUM<=5)
   6 - filter("HOUSES"."PRICE"<2000000)
```

Scalar Quantized HNSW Indexes

Scalar quantized HNSW indexes can be used to reduce memory requirements and accelerate similarity search.

In the context of vector databases, scalar quantization is a method of compressing or approximating high-dimensional data vectors by independently quantizing each dimension (or scalar component) of the vector. This approach is used to reduce the storage requirements of a vector database while preserving the essential structure of the data for tasks such as similarity search. Scalar quantization can be particularly beneficial when simplicity and low storage requirements are high priorities.

High-dimensional vectors stored as floating-point numbers require significant storage space. Scalar quantization saves memory by compressing the vectors. For instance, a 128-dimensional vector stored in 8 bits (`INT8`) per dimension requires only 128 bytes of memory, while 32-bit floating-point storage (`Float32`) would require 512 bytes and 64-bit double storage (`Float64`) would require 1024 bytes. Because of this reduced storage requirement, scalar quantization can enable vector databases to handle very large datasets, comprised of even billions of vectors, that would otherwise be impractical due to storage constraints.

Another benefit of scalar quantization is improved search speed. Approximate nearest neighbor (ANN) search methods can leverage quantized data for faster comparisons, as operations on integers or binary codes are faster than floating-point calculations.

With Oracle AI Database, uniform scalar quantization is the implemented compression method, meaning the input range is divided into equal-size intervals. Each interval is represented by a single quantization level, usually its midpoint. The range of each dimension is divided into 2^b equal intervals, where b is the number of bits (for example, 8 bits for int8). Each float value is mapped to the nearest quantization level.

> **Note**
>
> With scalar quantization, the amount of memory allocated in the vector pool is determined based on the quantization scale, not on the actual data type size.

Scalar quantization is supported with HNSW indexes and can be specified during index creation using the `CREATE VECTOR INDEX` DDL. For syntax and additional information, see Hierarchical Navigable Small World Index Syntax and Parameters and *Oracle AI Database SQL Language Reference*.

When performing inference on scalar quantized vectors, a `rescore factor` parameter is available as an option of the `SELECT` statement and can be used to refine your top-K results during query execution. For information and syntax, see *Oracle AI Database SQL Language Reference*.

Hierarchical Navigable Small World Index Syntax and Parameters

Syntax and examples for creating Hierarchical Navigable Small World vector indexes.

Syntax

```
CREATE VECTOR INDEX vector_index_name
ON table_name (vector_column)
[GLOBAL] ORGANIZATION INMEMORY [NEIGHBOR] GRAPH
```

```
[WITH] [DISTANCE [CUSTOM [<schema>.][<package>.]] metric name]
[WITH TARGET ACCURACY percentage_value]
[QUANTIZATION SCALAR COMPRESSION RATIO {2|4|8}]
[PARAMETERS (TYPE HNSW ,
             {NEIGHBORS max_closest_vectors_connected | M
max_closest_vectors_connected},
             EFCONSTRUCTION max_candidates_to_consider,
             [OFFLOAD_CREDENTIAL_NAME
credential_name_containing_gpu_access_token,
              OFFLOAD_URL https_endpoint_for_gpu_index_construction]
             RESCORE FACTOR rescore_factor,
             ALGORITHM UNIFORM_QUANTIZATION
            )]
[DUPLICATE ALL | DISTRIBUTE [AUTO | BY ROWID RANGE | BY PARTITION | BY
SUBPARTITION  ]]
[PARALLEL degree_of_parallelism]
```

HNSW Parameters

`NEIGHBORS` and `M` are equivalent and represent the maximum number of neighbors a vector can have on any layer, except the last layer, where a vector can have up to 2M neighbors.

`EFCONSTRUCTION` represents the maximum number of closest vector candidates considered at each step of the search during insertion.

`OFFLOAD_CREDENTIAL_NAME` is the identifier of the credential that should be used when authenticating with the Private AI Services Container GPU offloading service. The referenced credential name should be provided as a JSON object with an `access_token` field containing the API key to access the container. If `OFFLOAD_CREDENTIAL_NAME` is provided, `OFFLOAD_URL` must also be provided (and vice versa).

`OFFLOAD_URL` is the Private AI Services Container's service endpoint for index creation. It should be provided as an `https://` URL with the hostname or IP address of the remote server and the `/v1/index` path. For example,

```
https://<my_privateai_service_hostname>/v1/index
```

where `<my_privateai_service_hostname>` is the URL that the Oracle AI Database can use to connect to the Private AI Services Container. For more information about the Private AI Services Container, see *Oracle Private AI Services Container User's Guide*

`RESCORE FACTOR` is an optional integer parameter that defines the rescoring multiplier for semantic search queries. It has a default value of 1 and is applied only when the index is quantized (when `QUANTIZATION` is specified).

The valid range for HNSW vector index parameters are:

`ACCURACY`	> 0 and <= 100

`DISTANCE`	`COSINE` (default)
	`EUCLIDEAN`
	`L2_SQUARED` (as in `EUCLIDEAN_SQUARED`)
	`DOT`

	MANHATTAN HAMMING JACCARD CUSTOM <custom metric name>
QUANTIZATION	SCALAR
TYPE	HNSW
NEIGHBORS	> 0 and <= 2048
EFCONSTRUCTION	> 0 and <= 65535
RESCORE FACTOR	>=1 and <= 100 (default 1)
ALGORITHM	UNIFORM_QUANTIZATION

Usage Notes

You can specify either `DUPLICATE ALL` or `DISTRIBUTE`, but not both. These clauses are optional. If they are omitted, the HNSW vector index is created using the default behavior, consistent with the default behavior of a standard `CREATE INDEX` DDL execution.

You can optionally use the `QUANTIZATION` keyword together with the `COMPRESSION RATIO` parameter in order to apply scalar quantization to a new vector index.

The `COMPRESSION RATIO` specifies the level of compression for a vector column when creating a scalar-quantized index. Higher compression ratios yield smaller storage requirements but increased quantization. The possible values for `COMPRESSION RATIO` are 2, 4, and 8.

The following table shows the resulting data type stored in the index for each compression ratio, depending on the original vector column format.

Compression Ratio	FLOAT64 Input	FLOAT32 Input	FLOAT16 Input
2	Float32	Float16	Int8
4	Float16	Int8	Not Supported
6	Int8	Not Supported	Not Supported

Note that "not supported" means that scalar quantization is not available for the given combination of compression ratio and vector column format.

> **Note**
>
> Any parallel DML executed against a table with an HNSW index is automatically converted to a serial DML. HNSW indexes are not currently supported with parallel DML operations.

Examples

```
CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies (embedding)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
```

```
  DISTANCE COSINE
  WITH TARGET ACCURACY 95;

CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies (embedding)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  DISTANCE COSINE
  WITH TARGET ACCURACY 90
  PARAMETERS (
    TYPE             HNSW,
    NEIGHBORS        40,
    EFCONSTRUCTION   500
  ) PARALLEL 8;

CREATE VECTOR INDEX galaxies_quantized_idx ON galaxies (embedding)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  DISTANCE EUCLIDEAN
  WITH TARGET ACCURACY 90
  QUANTIZATION SCALAR COMPRESSION RATIO 2
  PARAMETERS (
    TYPE             HNSW,
    NEIGHBORS        32,
    EFCONSTRUCTION   200,
    RESCORE FACTOR   2,
    ALGORITHM        UNIFORM_QUANTIZATION
  );

CREATE VECTOR INDEX gist_idx ON gist(embedding)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  DISTANCE EUCLIDEAN
  PARAMETERS (
    TYPE                       HNSW,
    NEIGHBORS                  32,
    EFCONSTRUCTION             200,
    OFFLOAD_CREDENTIAL_NAME    mycredname,
    OFFLOAD_URL                'https://<container_url>:<port>/v1/index'
  ) PARALLEL 4;
```

For detailed information, see CREATE VECTOR INDEX in *Oracle AI Database SQL Language Reference*.

- Online Hierarchical Navigable Small World Index
  In situations where a base table needs to be available for ongoing updates and cannot be locked during index creation, you can create your HNSW (Hierarchical Navigable Small World) index online with the `ONLINE` clause of `CREATE VECTOR INDEX`. This way an application with heavy DML need not stop updating the base table for indexing.
- Function-Based Hierarchical Navigable Small World Index
  Oracle AI Database has long supported function-based indexes, allowing indexing expressions and computed values. The function-based HNSW index is an extension of this capability which applies the function-based indexing paradigm to the vector data type.

Online Hierarchical Navigable Small World Index

In situations where a base table needs to be available for ongoing updates and cannot be locked during index creation, you can create your HNSW (Hierarchical Navigable Small World) index online with the `ONLINE` clause of `CREATE VECTOR INDEX`. This way an application with heavy DML need not stop updating the base table for indexing.

Syntax

```
CREATE VECTOR INDEX index_name ON table_name(vector_column)
  ORGANIZATION INMEMORY NEIGHBOR GRAPH
  [WITH TARGET ACCURACY 95]
  [DISTANCE EUCLIDEAN]
  PARAMETERS (type HNSW, neighbors 32, efConstruction 500)
  ONLINE;
```

Usage Notes

- All vectors indexed need to have the same dimension and storage type throughout the index creation. Any changes or inconsistencies during index creation result in errors (for example, ORA-51902, ORA-51934).
- If the base table experiences a very high rate of DML during index build, then index creation time may increase.
- Online index creation with included columns (columns in addition to the vector column) is not currently supported.
- Online distributed HNSW index creation is not currently supported.

Example

```
CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies(embedding)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
DISTANCE EUCLIDEAN
WITH TARGET ACCURACY 95
PARAMETERS (type HNSW, neighbors 32, efConstruction 500)
ONLINE;
```

Function-Based Hierarchical Navigable Small World Index

Oracle AI Database has long supported function-based indexes, allowing indexing expressions and computed values. The function-based HNSW index is an extension of this capability which applies the function-based indexing paradigm to the vector data type.

This index is created on an expression instead of a vector column. The index expression can be an arithmetic, conditional expression or any other expression that produces a vector. Oracle AI Database computes the value of the expression and stores the resulting vectors in the index. The base table does not get an extra vector column; the index holds the vector values resulting from the expression.

The index is automatically maintained and remains consistent whenever a DML operation is performed.

Function-based HNSW indexes are especially useful to extract vector values from JSON documents for indexing. When creating the index, you provide an expression to extract vectors from the JSON document. At the time of indexing, the vector data is extracted from JSON documents by evaluating the specified expression.

When a query includes an expression that matches a function-based vector index expression, the optimizer automatically rewrites and routes queries to leverage the function-based index. This boosts query performance for similarity searches.

Usage Notes

- Online index creation is required; offline mode is not permitted.
- The expression in the index must return a single vector value per row.
- Only one expression per function-based vector index; multi- vector expression or hybrid vector indexes are not supported.
- The expression length is limited (currently 1000 characters).
- If a column that is used within the expression of the index is dropped, the associated index is also automatically dropped.
- Scalar quantization is not currently supported.
- RAC Distributed HNSW Indexes are not supported.
- There are currently the following two syntax types supported for extracting the vectors from JSON documents:
  - `Dot-path` notation:
    Example:`t.doc.v.vector()`
  - `json_value` application:
    Example: `json_value(doc, '$.v' returning vector(512, float32))`

    It is often beneficial to specify the dimension and element type of the vectors for best performance and correctness. However, with the dot-path notation, you cannot currently specify these attributes. The only way to do so is using the `json_value` syntax.

Syntax

```
CREATE VECTOR INDEX index_name ON table_name (expression)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
ONLINE;
```

Examples

Example 1

Indexing a vector extracted from JSON document. Indexes the vector in the field v inside the JSON column `doc`.

Example JSON document:

In this JSON document, the field v is a JSON array representing the vector you want to index.

```
{
  "name": "Paul",
  "age": 60,
  "v": [1, 2, 3],
  "salary": 1000000
}
```

Index creation statement:

```
CREATE VECTOR INDEX vecidx ON mytable t (t.doc.v.vector())
ORGANIZATION INMEMORY NEIGHBOR GRAPH
ONLINE;
```

Example 2

Indexing a vector extracted from JSON document. Indexes the vector in the field `street_embedding` from the `address` object inside the JSON column `doc`. In this example, the`json_value` SQL function instead of `dot-path` notation to specify the vector dimension and the vector element types. This is enables to improve performance when building the index.

Example JSON document:

In this JSON document, `street_embedding` is an array of numbers (a vector) inside the nested `address` object of the JSON column `doc`. This example JSON document is in the extended JSON format. The extended JSON format enables Oracle Database to recognize vectors within textual JSON data. When this text is ingested as extended JSON, Oracle Database can store the resulting vectors in an optimized binary format, enabling faster extraction and processing. For more information about how Oracle Database handles extended JSON data, see JSON Data Type.

```
{
  "id": 1,
  "doc": {
    "address": {
      "street": "500 Oracle Pkwy",
      "street_embedding": {
        "$vector": [0.12, 0.19, 0.75, 0.44, 0.08, 0.63, 0.27, 0.91],
        "$vectorElementType": "float32"
      },
      "zip_code": "94065"
    }
  }
}
```

Index creation statement:

```
CREATE VECTOR INDEX addr_street_ejson_jv_ix
ON addr_docs t (JSON_VALUE(t.doc, '$.address.street_embedding' RETURNING
VECTOR(8, float32)))
ORGANIZATION INMEMORY NEIGHBOR GRAPH
WITH TARGET ACCURACY 95
DISTANCE EUCLIDEAN
ONLINE;
```

Example 3

Indexing a Computed Vector Expression. Indexes the sum of two vector columns, `(v + u)`

```
CREATE VECTOR INDEX vecidx ON mytable (v + u)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
ONLINE;
```

Example 4

Conditional vector indexing. Indexes the vector only when the condition (`country_code = 'US'`) is satisfied; otherwise, stores `NULL` in the index.

```
CREATE VECTOR INDEX emp_resume_us_ix
ON employees_resume e (DECODE(e.country_code, 'US', e.resume_embedding, NULL))
ORGANIZATION INMEMORY NEIGHBOR GRAPH
WITH TARGET ACCURACY 95
DISTANCE EUCLIDEAN
ONLINE;
```

Neighbor Partition Vector Index

Inverted File Flat (IVF) index is the only type of Neighbor Partition vector index supported. Inverted File Flat Index (IVF Flat or simply IVF) is a partitioned-based index that lets you balance high-search quality with reasonable speed.

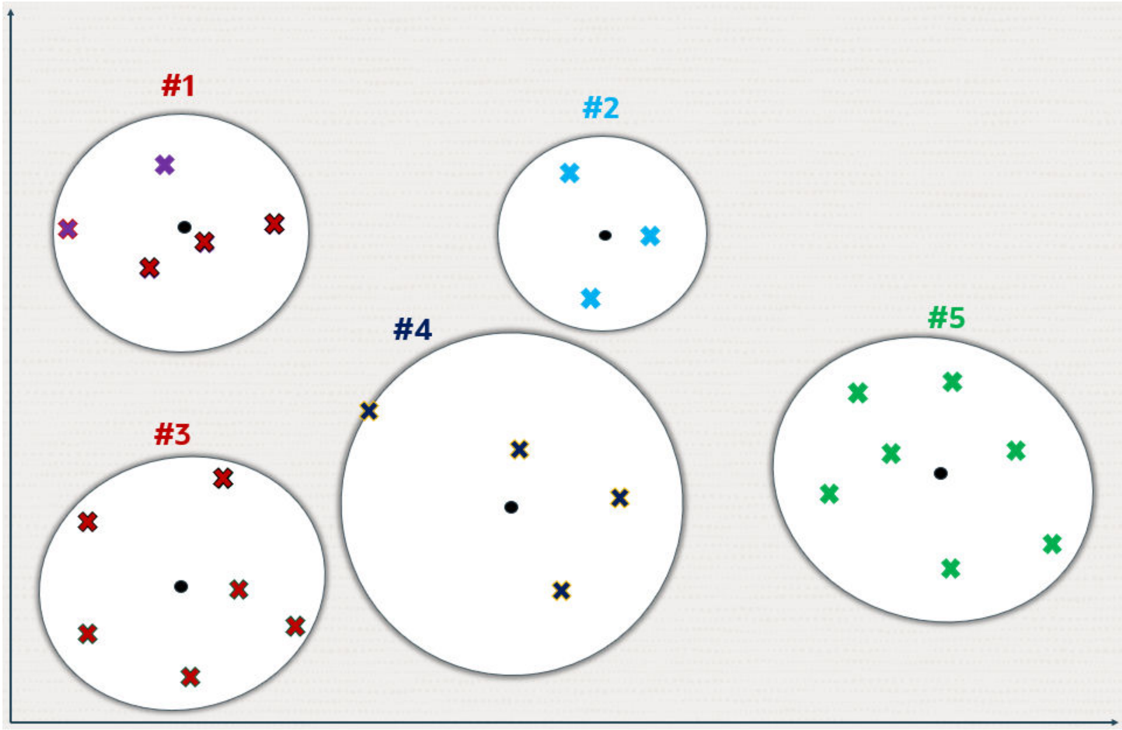
- Understand Inverted File Flat Vector Indexes
  The Inverted File Flat vector index is a technique designed to enhance search efficiency by narrowing the search area through the use of neighbor partitions or clusters.
- Understand Local Inverted File Flat Vector Indexes
  A local IVF index can be used to accelerate search queries. With local IVF indexes, there is a one-to-one relationship between the base table partitions or subpartitions and the index partitions.
- Included Columns with IVF Indexes
  Included columns in vector indexes facilitate faster searches with attribute filters by incorporating non-vector columns within a Neighbor Partition Vector Index.
- Inverted File Flat CREATE INDEX
  Syntax and examples for creating Inverted File Flat vector indexes.
- Inverted File Flat Vector Indexes Online Rebuild
  Data distribution shifts caused by new DML operations (inserts, updates, or deletes) that occur after an IVF (Inverted File Flat) vector index has been created can significantly influence accuracy of the results. These changes require careful maintenance strategies to ensure sustained retrieval accuracy.

Understand Inverted File Flat Vector Indexes

The Inverted File Flat vector index is a technique designed to enhance search efficiency by narrowing the search area through the use of neighbor partitions or clusters.

The following diagrams depict how partitions or clusters are created in an approximate search done using a 2D space representation. But this can be generalized to much higher dimensional spaces.

Figure 6-6 Inverted File Flat Index Using 2D



Crosses represent the vector data points in this space.

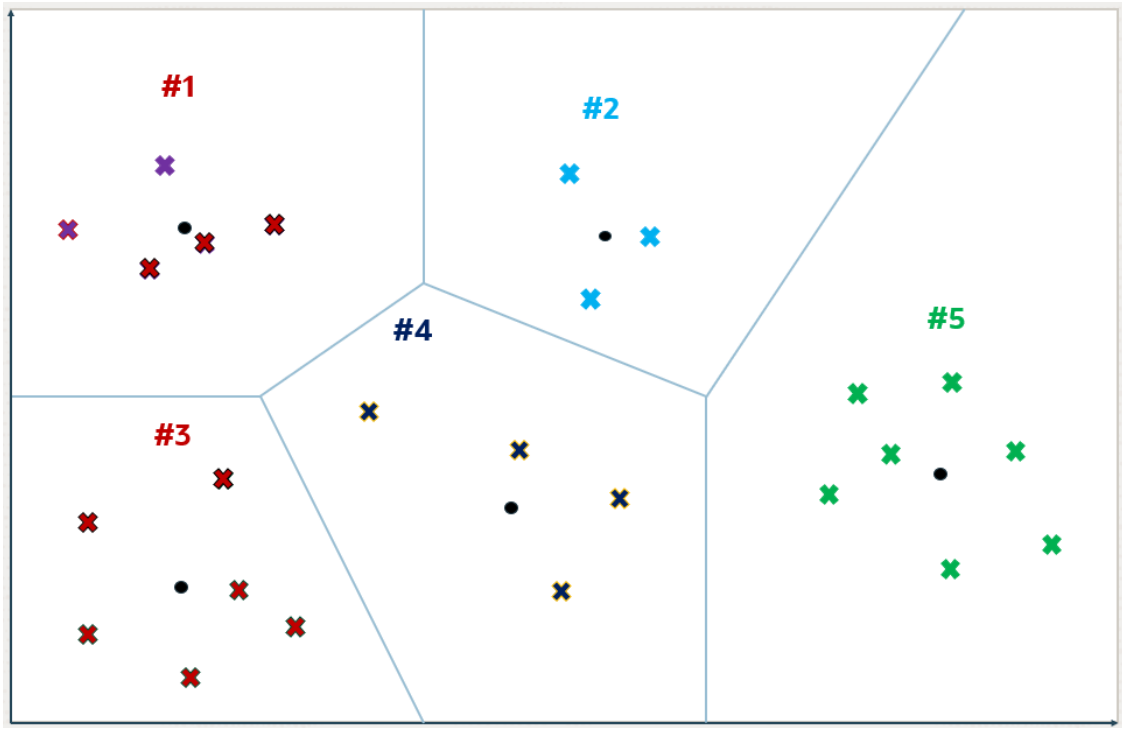
New data points, shown as small plain circles, are added to identify `k` partition centroids, where the number of centroids (`k`) is determined by the size of the dataset (`n`). Typically `k` is set to the square root of `n`, though it can be adjusted by specifying the `NEIGHBOR PARTITIONS` parameter during index creation.

Each centroid represents the average vector (center of gravity) of the corresponding partition.

The centroids are calculated by a training pass over the vectors whose goal is to minimize the total distance of each vector from the closest centroid.

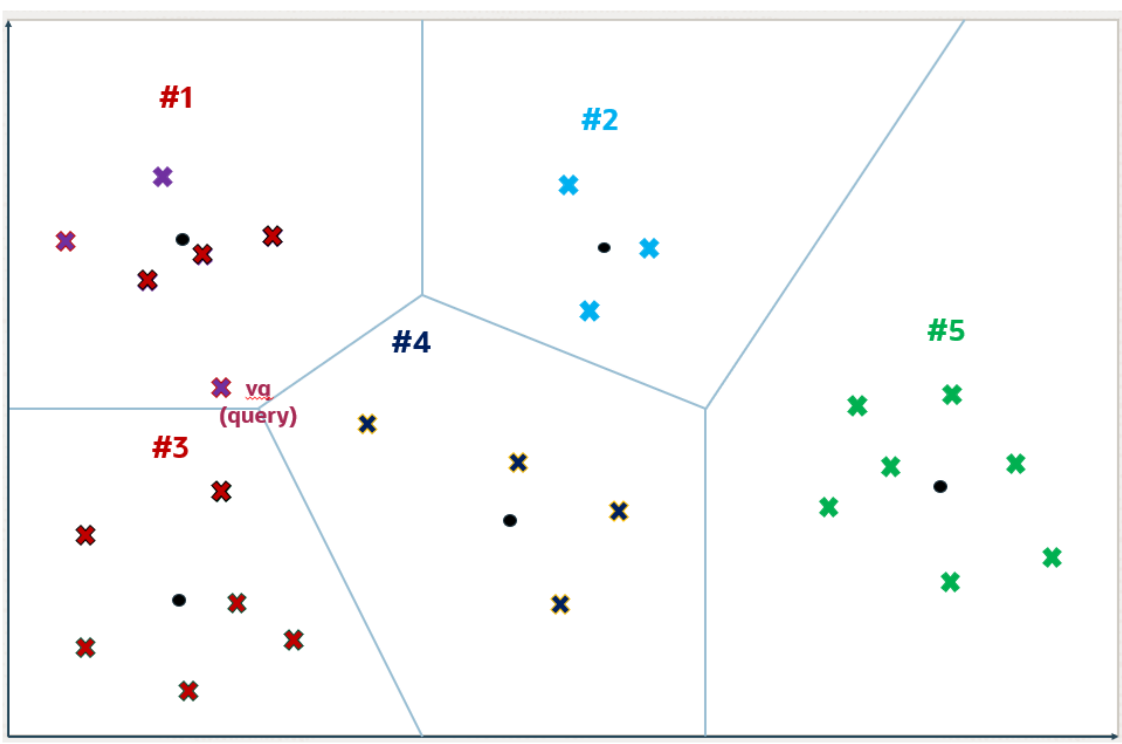
The centroids ends up partitioning the vector space into `k` partitions. This division is conceptually illustrated as expanding circles from the centroids that stop growing as they meet, forming distinct partitions.

Figure 6-7 Inverted File Flat Index



Except for those on the periphery, each vector falls within a specific partition associated with a centroid.

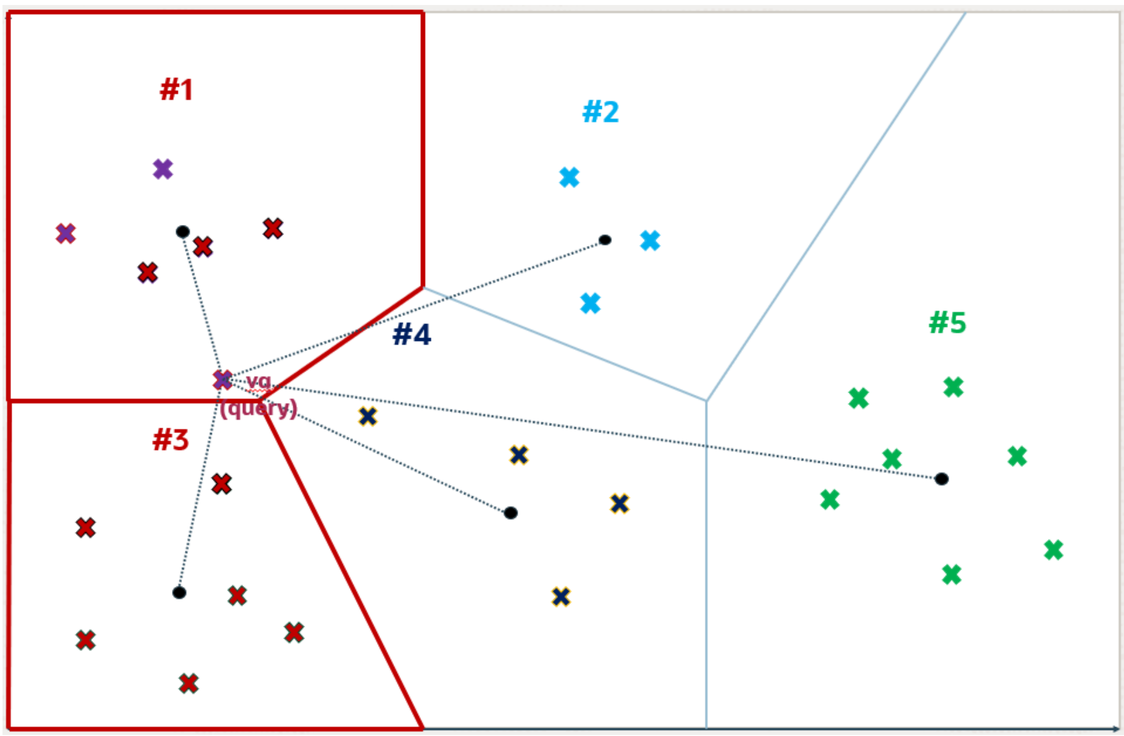
Figure 6-8 Inverted File Flat Index



For a query vector vq, the search algorithm identifies the nearest `i` centroids, where `i` defaults to the square root of `k` but can be adjusted for a specific query by setting the `NEIGHBOR PARTITION PROBES` parameter. This adjustment allows for a trade-off between search speed and accuracy.

Higher numbers for this parameter will result in higher accuracy. In this example, `i` is set to 2 and the two identified partitions are partitions number 1 and 3.

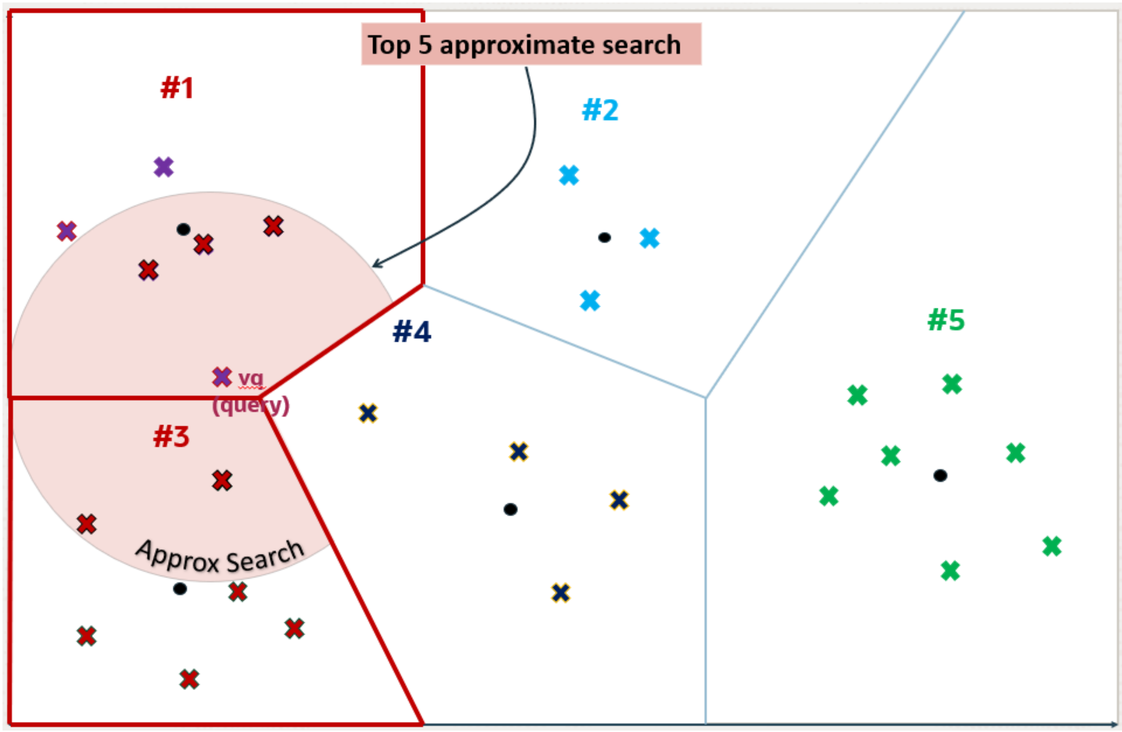
Figure 6-9 Inverted File Flat Index



Once the `i` partitions are determined, they are fully scanned to identify, in this example, the top 5 nearest vectors. This number 5 can be different from `k` and you specify this number in your query. The five nearest vectors to `vq` found in partitions number 1 and 3 are highlighted in the following diagram.

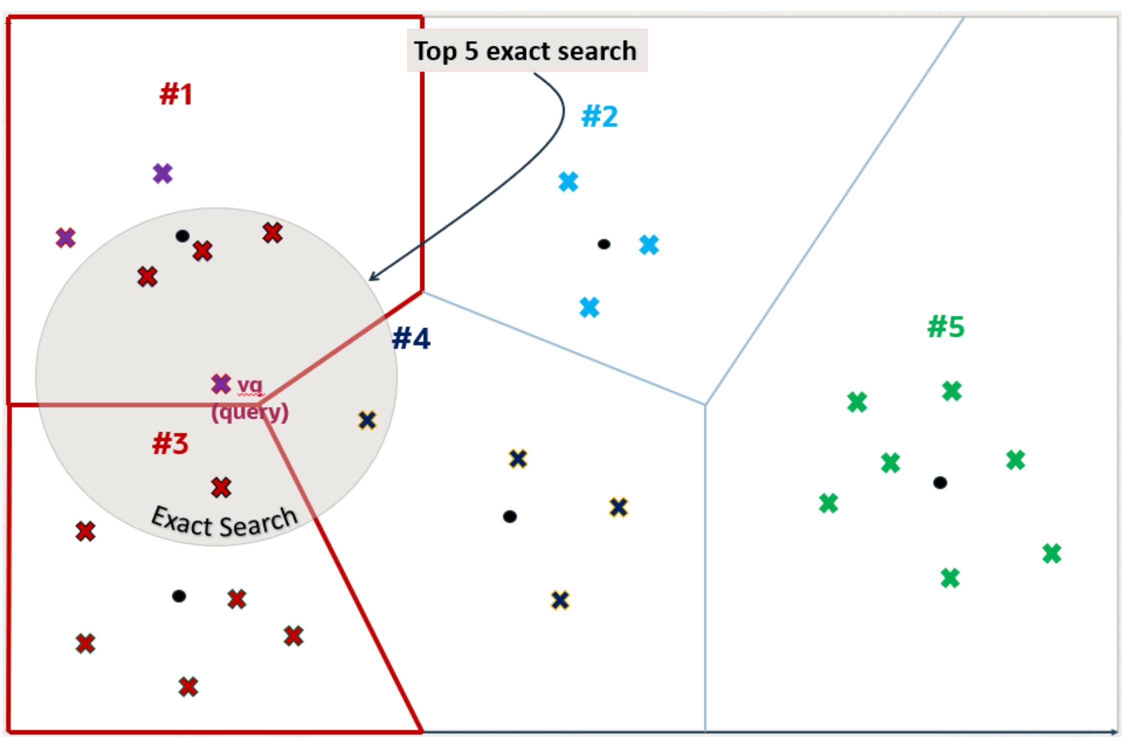
This method constitutes an approximate search as it limits the search to a subset of partitions, thereby accelerating the process but potentially missing closer vectors in unexamined partitions. This example illustrates that an approximate search might not yield the exact nearest vectors to `vq`, demonstrating the inherent trade-off between search efficiency and accuracy.

Figure 6-10 Inverted File Flat Index



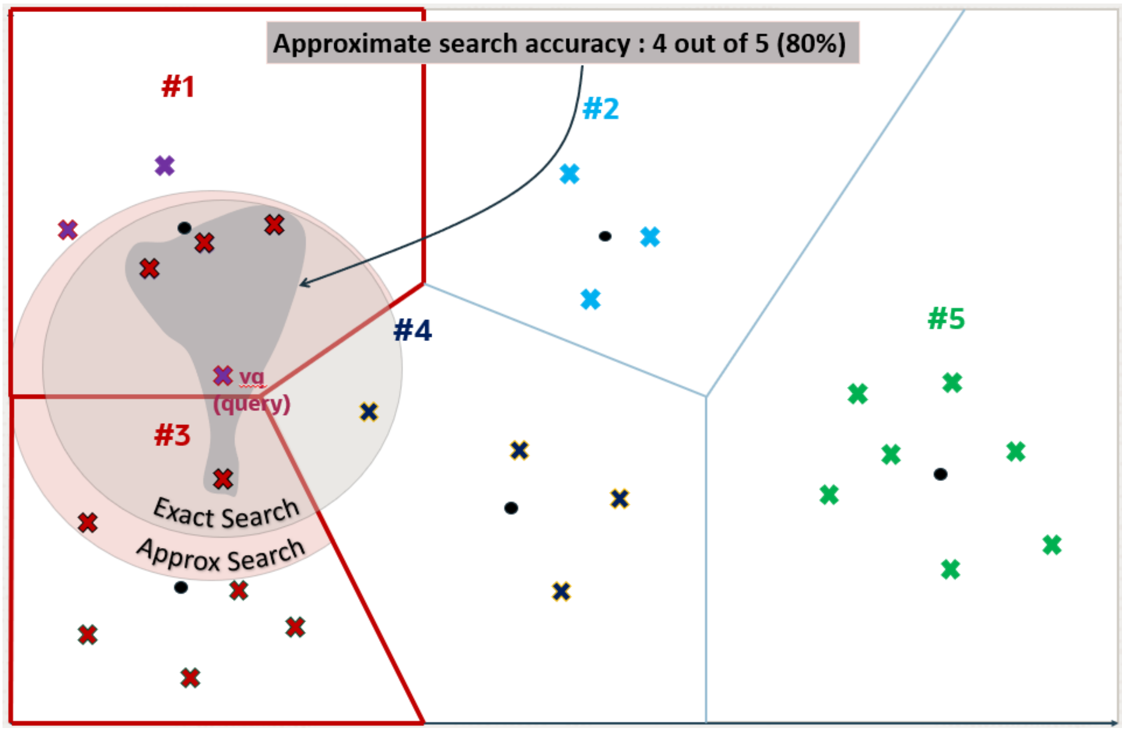
However, the five exact nearest vectors from vq are not the ones found by the approximate search. You can see that one of the vectors in partition number 4 is closer to vq than one of the retrieved vectors in partition number 3.

Figure 6-11 Inverted File Flat Index



You can now see why using vector index searches is not always an exact search and is called an approximate search instead. In this example, the approximate search accuracy is only 80% as it has retrieved only 4 out of 5 of the exact search vectors' result.

Figure 6-12 Inverted File Flat Index



When using Oracle AI Vector Search to run an approximate search query using vector indexes, you have the possibility to specify a target accuracy at which the approximate search should be performed.

In the case of an IVF approximate search, you can specify a target accuracy percentage value to influence the number of partitions used to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the `NEIGHBOR PARTITION PROBES` parameter to impose a certain maximum number of partitions to be probed by the search. The higher that number, the higher the accuracy.

> **Note**
>
> - If you do not specify any target accuracy in your approximate search query, then you will inherit the one set when the index was created. You will see that at index creation time, you can specify a target accuracy either using a percentage value or parameters values depending on the type of index you are creating.
> - It is possible to specify a different target accuracy at index search, compared to the one set at index creation. For IVF indexes, you may probe more centroid partitions using the `NEIGHBOR PARTITION PROBES` parameter to get more accurate results. The target accuracy that you provide during index creation decides the index creation parameters and also acts as the default accuracy value for vector index searches.

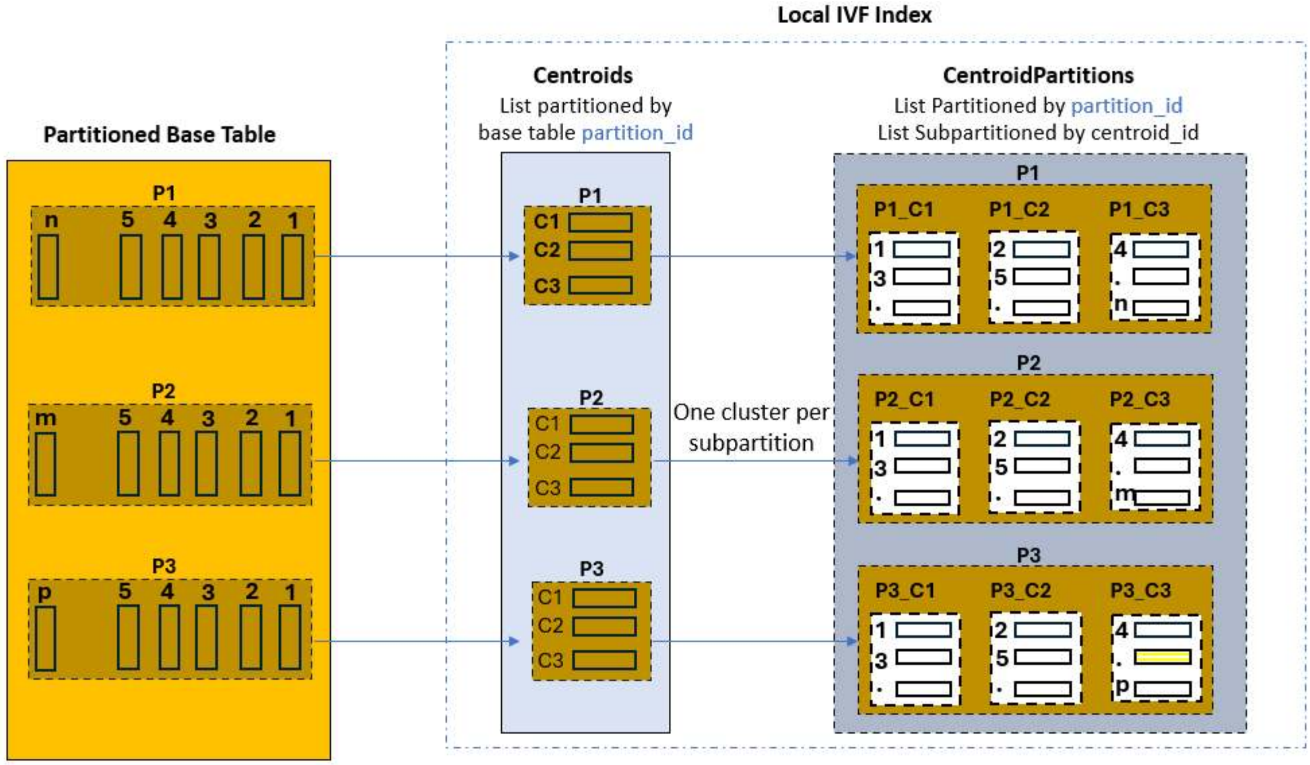
Understand Local Inverted File Flat Vector Indexes

A local IVF index can be used to accelerate search queries. With local IVF indexes, there is a one-to-one relationship between the base table partitions or subpartitions and the index partitions.

Local IVF index creation and DML operations on base tables with IVF indexes can be accelerated if **Vector Pool** is enabled. **Vector Pool** is a new memory area stored in the SGA. For more information related to **Vector Pool**, see Size the Vector Pool.

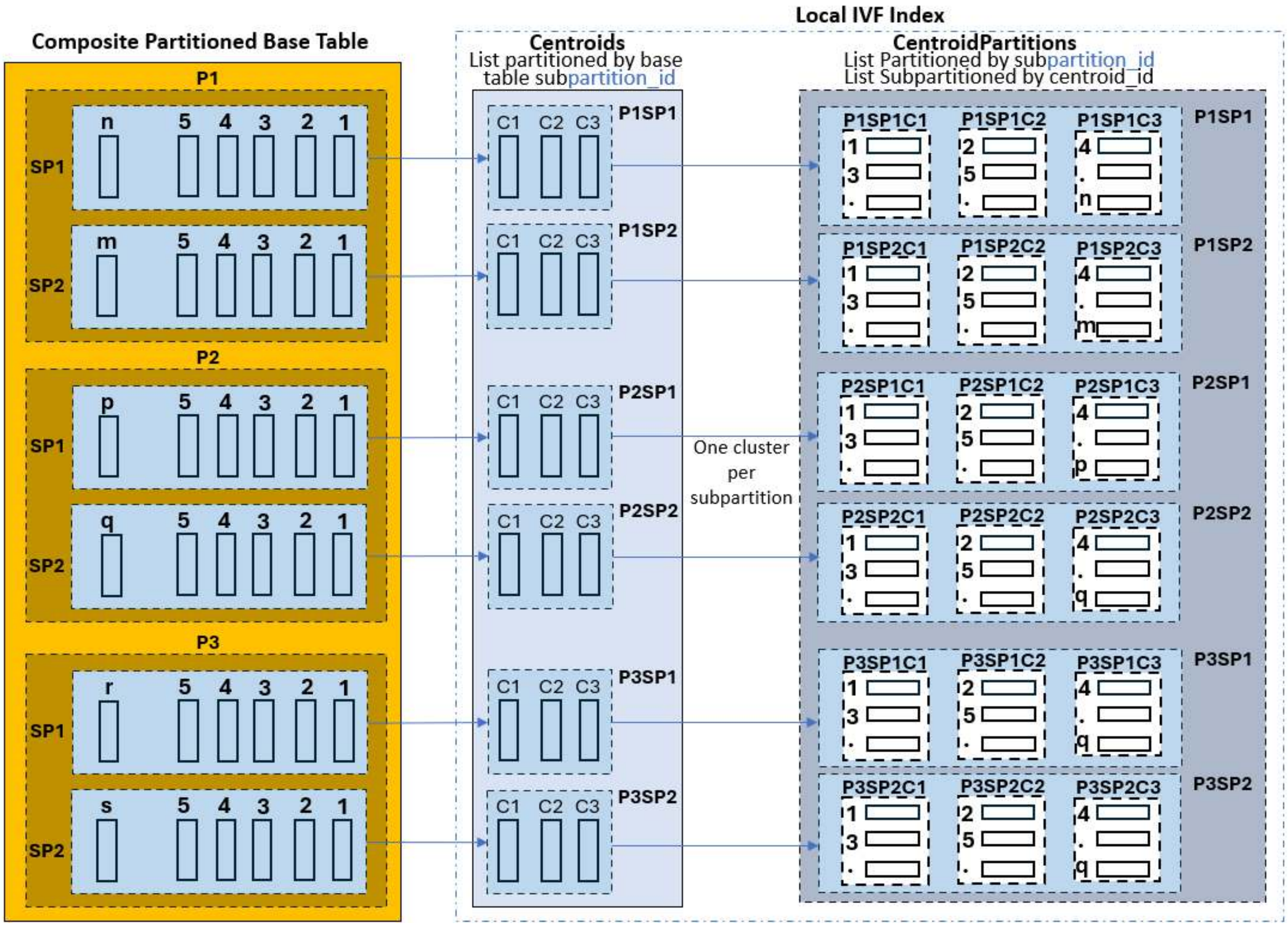
This is illustrated by the following graphic, where the base table has three partitions. The created local IVF index is constituted by two internal tables:

- One called `VECTOR$<base table name>_IVF_IDX$<object info>$IVF_FLAT_CENTROIDS`, which is list-partitioned by base table partition ids, and is thus equi-partitioned with the base table. Each partition contains the list of corresponding identified centroid vectors and associated ids.
- The second called `VECTOR$<base table name>_IVF_IDX$<object info>$IVF_FLAT_CENTROID_PARTITIONS`, which is list-partitioned by base table partition id and list-subpartitioned by centroid id. This table is also equi-partitioned with the base table, and each subpartition contains the base table vectors closely related (cluster) to the corresponding centroid id for that subpartition.



If, for example, you search the top-10 houses in California similar to a vectorized picture, your query benefits from partition pruning on the base table and Centroids table (California) as they are both partitioned by state. Once the closest centroid is identified in that partition, the query simply needs to scan the corresponding centroid cluster subpartition in the Centroid Partitions table without having to scan other centroid subpartitions.

Another possibility is for the base table to be composite partitioned. Here is a graphical representation corresponding to that case. The Centroids table is list-partitioned according to base table subpartitions. Each partition in the Centroids table contains all centroid vectors found in the corresponding base table subpartition. The Centroid Partitions table is list-partitioned by base table subpartition id, and is further subpartitioned by centroid id:



Note

- You can create local IVF indexes only on a partitioned base table.
- Local IVF indexes inherit all system catalog tables and views used by regular local indexes. A flag (`idx_spare2`) in the `vecsys.vector$index` table indicates if an index is a local or global vector index.
- If a Partition Maintenance Operation (PMOP), such as `ALTER TABLE SPLIT/MERGE/MOVE/EXCHANGE/COALECSE`, is performed on the base table, all corresponding IVF index partitions are marked as `UNUSABLE` after the operation.
- Partition Maintenance Operations are not supported with local IVF indexes.

Using local IVF indexes brings additional advantages:

- Simplified Partition Maintenance Operations:

  For example, dropping a table partition just involves dropping the corresponding index partition.
- Flexible Indexing schemes:

  For example, marking certain index partitions `UNUSABLE` to avoid indexing certain table partitions through partial indexing.

> **Note**
>
> If you want your user queries to use the full potential of local IVF indexes by taking advantage of the benefit of partition pruning, user queries must have partition pruning conditions.

Included Columns with IVF Indexes

Included columns in vector indexes facilitate faster searches with attribute filters by incorporating non-vector columns within a Neighbor Partition Vector Index.

Consider a classic example where you are analyzing the home price data. You want only the top 10 homes from your description where the price is less than 2 million dollars and/or from the specific zip code. Or to a finer granularity, you just need to retrieve only the top home that matches the description and your filter attribute. Included columns make this possible by incorporating filterable attributes, such as price and zip code, directly into the vector index. This integration allows the index to evaluate and return results based on both the vector similarity and the attribute filters, without requiring additional steps to cross-reference the base table.

By bridging the gap between traditional database filters and AI-powered vector searches, included columns deliver faster, more efficient results. This synergy enhances search performance, especially in use cases requiring fine-grained filtering alongside advanced similarity computations.

Syntax and Specifications

Create a partitioned base table, which contains the sales data for all products sold in the current year:

```
CREATE TABLE houses (
    id           NUMBER,
    zip_code     NUMBER,
    price        NUMBER,
    description  CLOB,
    data_vector  VECTOR
)
PARTITION BY LIST (zip_code) (
    PARTITION p_zip_10000_19999 VALUES (10000, 10001, 10002, 10003, 10004),
    PARTITION p_zip_20000_29999 VALUES (20000, 20001, 20002, 20003, 20004),
    PARTITION p_zip_30000_39999 VALUES (30000, 30001, 30002, 30003, 30004),
    PARTITION p_zip_others VALUES (DEFAULT)
);
```

> **Note**
>
> To demonstrate the creation of local and global indexes with included columns, the example uses a partitioned base table here. The choice between a partitioned and a non-partitioned table should be made based on business requirements.

Then, you can create an IVF index **with included column on price** as follows:

```
create vector index vidx_ivf
on houses(data_vector)
include (price)
organization neighbor partitions
with target accuracy 95
distance EUCLIDEAN
parameters(type IVF, neighbor partitions 2);
```

The above syntax creates a vector index `VIDX_IVF` with `organization` as `NEIGHBOR PARTITIONS` on the `DATA_VECTOR` column where the `PRICE` column is an included column. The `INCLUDE` keyword allows a user to specify the attributes they wish to include in the IVF index.

By default, an IVF index is global. To create a local IVF index **with included column on price** for the partitioned base table, add the `LOCAL` clause during index creation:

```
CREATE VECTOR INDEX vidx_ivf_local
ON houses(data_vector)
INCLUDE (price)
ORGANIZATION NEIGHBOR PARTITIONS
WITH TARGET ACCURACY 95
DISTANCE EUCLIDEAN
PARAMETERS (type IVF, neighbor partitions 2)
LOCAL;
```

The above syntax creates a local IVF vector index `vidx_ivf_local` with `organization` as `NEIGHBOR PARTITIONS` on the `DATA_VECTOR` column where the `PRICE` column is an included column. The `INCLUDE` keyword allows a user to specify the attributes they wish to include in the IVF index.

> **See Also**
>
> Understand Local Inverted File Flat Vector Indexes

Recall that the IVF index has two auxiliary tables : Centroids table - `IVF_FLAT_CENTROIDS` and the Centroid Partitions table - `IVF_FLAT_CENTROID_PARTITIONS`. The following code block describes the Centroid Partitions table without the included columns:

```
  Name                           Null?    Type
 ----------------------------------------- --------
---------------------------
 BASE_TABLE_ROWID                  NOT NULL ROWID
 CENTROID_ID                      NOT NULL NUMBER
 DATA_VECTOR                           VECTOR(384, FLOAT32, DENSE)
------------------------------------------------------------------------------
--
```

In both global and local IVF indexes, the list of included columns are augmented in the `IVF_FLAT_CENTROID_PARTITIONS` table of IVF index. The BASE_TABLE_ROWID column can be thought of as an implicit included column. The following table describes the

IVF_FLAT_CENTROID_PARTITIONS table with the included column augmented as the last row. In this case, the price.

```
 Name                          Null?    Type
 ----------------------------------------- --------
---------------------------
 BASE_TABLE_ROWID                 NOT NULL ROWID
 CENTROID_ID                     NOT NULL NUMBER
 DATA_VECTOR                          VECTOR(384, FLOAT32, DENSE)
 PRICE                              NUMBER
------------------------------------------------------------------------------
--
```

DML operations on the base table also maintain the included columns. Updating any row on the base table for the included columns will all also update the centroid partitions for the IVF index.

> **Note**
>
> **Restrictions on INCLUDE :**
> The INCLUDE list can contain:
>
> - Limited to a maximum of 31 columns - Oracle indexes support 32 key columns.
> - Supports only the following data types : NUMBER, CHAR, VARCHAR2, DATE, TIMESTAMP, RAW, JSON, and reference-based LOB types such as BLOB and CLOB.
> - Does not support NLS types and LONG types.

> **Note**
>
> If a serial direct load is attempted on a table that has an included column, the operation will be automatically converted to a conventional insert.

Benefits of Using Included Columns

1. **IVF No-Filter with and without Included Columns**:
   If you consider a non-filter query that fetches the two homes nearest to a specified vector, the resulting query will result in an execution plan with multiple joins.

   ```
   select /*+ VECTOR_INDEX_TRANSFORM(t) */ price
   from houses t
   order by VECTOR_DISTANCE(data_vector, :query_vector)
   fetch approx first 2 rows only;
   ```

   > **Note**
   >
   > query_vector contains the actual input vector. You can use the instructions mentioned in SQL Quick Start Using a FLOAT32 Vector Generator to generate query_vector.

The multiple joins are displayed in the execution plan below:

```
Execution Plan
----------------------------------------------------------
Plan hash value: 466477707

---------------------------------------------------------------------------
---------------------------
| Id  | Operation            | Name                                 |
---------------------------------------------------------------------------
---------------------------
|   0 | SELECT STATEMENT        |
|
|   1 |  VIEW                |
|
|   2 |   NESTED LOOPS            |
|
|   3 |    VIEW              | VW_IVPSR_11E7D7DE
|
|*  4 |     COUNT STOPKEY        |
|
|   5 |      VIEW             | VW_IVPSJ_578B79F1
|
|*  6 |       SORT ORDER BY STOPKEY    |
|
|   7 |        NESTED LOOPS        |
|
|   8 |     VIEW              | VW_IVCR_B5B87E67
|
|*  9 |      COUNT STOPKEY        |
|
|  10 |       VIEW             | VW_IVCN_9A1D2119
|
|* 11 |        SORT ORDER BY STOPKEY|
|
|  12 |         TABLE ACCESS FULL    |
VECTOR$VIDX_IVF$74478_74488_0$IVF_FLAT_CENTROIDS          |
|  13 |     PARTITION LIST ITERATOR |
|
|* 14 |      TABLE ACCESS FULL    |
VECTOR$VIDX_IVF$74478_74488_0$IVF_FLAT_CENTROID_PARTITIONS |
|  15 |    TABLE ACCESS BY USER ROWID    |
HOUSES                              |
---------------------------------------------------------------------------
---------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   4 - filter(ROWNUM<=2)
   6 - filter(ROWNUM<=2)
   9 - filter(ROWNUM<=1)
  11 - filter(ROWNUM<=1)
  14 - filter("VW_IVCR_B5B87E67"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
```

You can see in the execution plan that the IVF centroids are joined with the centroid partition table to scan for the centroids in the nearest neighbor partitions fetched from the centroid table. This operation is expected. However, you are also performing a join with the base table: HOUSES to apply the attribute filters. This second join against the base table could be avoided if the columns were included in the centroid partitions table.

The following execution plan displays the same operation when using a neighbor partition vector index (IVF) created with included columns:

```
Execution Plan
----------------------------------------------------------
Plan hash value: 504484986

---------------------------------------------------------------------------
----------------------------
| Id  | Operation               | Name                                  |
---------------------------------------------------------------------------
----------------------------
|   0 | SELECT STATEMENT           |                                   |
|   1 |  VIEW                   |                                   |
|   2 |   VIEW                   |
VW_IVPSR_11E7D7DE                         |
|*  3 |    COUNT STOPKEY           |
|
|   4 |     VIEW                |
VW_IVPSJ_578B79F1                         |
|*  5 |      SORT ORDER BY STOPKEY     |
|
|   6 |       NESTED LOOPS           |
|
|   7 |        VIEW                |
VW_IVCR_B5B87E67                         |
|*  8 |     COUNT STOPKEY           |                                   |
|   9 |      VIEW                |
VW_IVCN_9A1D2119                         |
|* 10 |       SORT ORDER BY STOPKEY|                                   |
|  11 |        TABLE ACCESS FULL    |
VECTOR$VIDX_IVF_1$74483_74508_0$IVF_FLAT_CENTROIDS           |
|  12 |        PARTITION LIST ITERATOR |
|
|* 13 |     TABLE ACCESS FULL       |
VECTOR$VIDX_IVF_1$74483_74508_0$IVF_FLAT_CENTROID_PARTITIONS |
---------------------------------------------------------------------------
----------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   3 - filter(ROWNUM<=2)
   5 - filter(ROWNUM<=2)
   8 - filter(ROWNUM<=1)
  10 - filter(ROWNUM<=1)
  13 - filter("VW_IVCR_B5B87E67"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
```

As you can see in the execution plan above, adding included columns to a vector index ensures there is no base table pre-filter evaluation required. The query can be resolved by scanning the centroids partitions table for the neighbor partitioned vector index.

2. **IVF Filtering with Included Columns**:
One of the more commonly used plans for neighbor graph vector indexes is the pre-filter plan. In this case you are filtering the contents of the base table to eliminate non-relevant rows. But evaluating a query without using a filter attribute can be very expensive as the operation involves several joins. The following query uses a neighbor partition vector index without included columns:

```
SELECT /*+ VECTOR_INDEX_TRANSFORM(houses, vidx_ivf) */
price
FROM houses
WHERE price = 1400000
ORDER BY vector_distance(data_vector, :query_vector)
FETCH APPROX FIRST 4 ROWS ONLY;
```

> **Note**
>
> `query_vector` contains the actual input vector. You can use the instructions mentioned in [SQL Quick Start Using a FLOAT32 Vector Generator](#) to generate query_vector.

The execution plan shows multiple joins and the mandatory join with the base table `HOUSES`.

```
Execution Plan
----------------------------------------------------------
Plan hash value: 3359903466

---------------------------------------------------------------------------
----------------------------------
| Id  | Operation                   | Name
|
---------------------------------------------------------------------------
----------------------------------
|   0 | SELECT STATEMENT                |
|
|*  1 |  COUNT STOPKEY                    |
|
|   2 |   VIEW                         |                                 |
|*  3 |    SORT ORDER BY STOPKEY
|                                 |
|   4 |     NESTED LOOPS                |
|
|   5 |      MERGE JOIN CARTESIAN
|                                 |
|*  6 |       TABLE ACCESS FULL             |
HOUSES                                |
|   7 |       BUFFER SORT                |
|
|   8 |        VIEW                    |
VW_IVCR_2D77159E                        |
```

```
|*  9 |     COUNT STOPKEY               |
|
|  10 |      VIEW                   |
VW_IVCN_9A1D2119                        |
|* 11 |       SORT ORDER BY STOPKEY
|                             |
|  12 |        TABLE ACCESS FULL           |
VECTOR$VIDX_IVF$74478_74488_0$IVF_FLAT_CENTROIDS        |
|* 13 |      TABLE ACCESS BY GLOBAL INDEX ROWID|
VECTOR$VIDX_IVF$74478_74488_0$IVF_FLAT_CENTROID_PARTITIONS |
|* 14 |       INDEX UNIQUE SCAN            |
SYS_C008791                            |
---------------------------------------------------------------------------
---------------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=4)
   3 - filter(ROWNUM<=4)
   6 - filter("HOUSES"."PRICE"=1400000)
   9 - filter(ROWNUM<=3)
  11 - filter(ROWNUM<=3)
  13 - filter("VW_IVCR_2D77159E"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
  14 - access("HOUSES".ROWID="VTIX_CNPART"."BASE_TABLE_ROWID")
```

On the other hand, if you use the same in-filter query on the IVF index created with `price` as the `included` column, the execution plan would show less joins, the join with the base table is avoided. This is shown in the plan below:

```
Execution Plan
----------------------------------------------------------
Plan hash value: 4183240211

---------------------------------------------------------------------------
---------------------------
| Id  | Operation               | Name                                  |
---------------------------------------------------------------------------
---------------------------
|   0 | SELECT STATEMENT           |                                  |
|*  1 |  COUNT STOPKEY               |                                  |
|   2 |   VIEW                   |                                  |
|*  3 |    SORT ORDER BY STOPKEY       |
|
|*  4 |     HASH JOIN                |                                  |
|   5 |      PART JOIN FILTER CREATE
| :BF0000                            |
|   6 |       VIEW                   |
VW_IVCR_2D77159E                           |
|*  7 |        COUNT STOPKEY           |
|
|   8 |     VIEW                 | VW_IVCN_9A1D2119
|
|*  9 |      SORT ORDER BY STOPKEY |                                  |
|  10 |       TABLE ACCESS FULL    |
```

```
VECTOR$VIDX_IVF_1$74483_74508_0$IVF_FLAT_CENTROIDS          |
|  11 |      PARTITION LIST JOIN-FILTER|
|
|* 12 |       TABLE ACCESS FULL        |
VECTOR$VIDX_IVF_1$74483_74508_0$IVF_FLAT_CENTROID_PARTITIONS |
---------------------------------------------------------------------------
----------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=4)
   3 - filter(ROWNUM<=4)
   4 - access("VW_IVCR_2D77159E"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
   7 - filter(ROWNUM<=3)
   9 - filter(ROWNUM<=3)
  12 - filter("VTIX_CNPART"."PRICE"=1400000)
```

Inverted File Flat CREATE INDEX

Syntax and examples for creating Inverted File Flat vector indexes.

Syntax

```
CREATE VECTOR INDEX <vector index name>
ON <table name> ( <vector column> )
[GLOBAL] ORGANIZATION [NEIGHBOR] PARTITIONS
[WITH] [DISTANCE <metric name>]
[WITH TARGET ACCURACY <percentage value>
[PARAMETERS ( TYPE IVF,
              { NEIGHBOR PARTITIONS <number of partitions> |
                SAMPLES_PER_PARTITION <number of samples> |
                MIN_VECTORS_PER_PARTITION <minimum number of vectors per
partition> |
                NEIGHBOR PARTITION GROUPING <ON | OFF> }
            )]]
[PARALLEL <degree of parallelism>]
[LOCAL];
```

The `GLOBAL` clause specifies a global IVF index. By default, IVF vector indexes are globally partitioned by centroid.

Specify the `LOCAL` clause to create a local IVF index.

IVF PARAMETERS

Parameter	Purpose
`NEIGHBOR PARTITIONS`	Determines the target number of centroid partitions that are created by the index.

Parameter	Purpose
SAMPLES_PER_PARTITION	Decides the total number of vectors that are passed to the clustering algorithm (samples_per_partition * neighbor_partitions). Passing all the vectors could significantly increase the total index creation time. The goal is to pass in a subset of vectors that can capture the data distribution.
MIN_VECTORS_PER_PARTITION	Represents the target minimum number of vectors per partition. The goal is to trim out any partition that can end up with few vectors (<= 100).
NEIGHBOR PARTITION GROUPING	Enables, in addition to partitioning, further data grouping in the storage layer for centroids in the IVF index. This can result in query performance benefits in serial, parallel, and multi-user (concurrent) queries due to additional pruning during the scan. When this storage optimization is enabled, the default number of centroids may increase, which can lead to a longer IVF index creation time. To disable this feature, you must drop the index and recreate it. Neighbor partition grouping will continue to be enabled on an index that is rebuilt. Online index rebuild is supported on IVF indexes that have this storage option enabled. Neighbor partition grouping can only be specified upon index creation and cannot be enabled or disabled using ALTER INDEX or REBUILD. This parameter is only applicable to global IVF indexes. It is not supported with HNSW indexes or local IVF indexes.

The valid range for IVF vector index parameters are:

ACCURACY	> 0 and <= 100
DISTANCE	COSINE (default) EUCLIDEAN L2_SQUARED (as in EUCLIDEAN_SQUARED) DOT MANHATTAN HAMMING
TYPE	IVF
NEIGHBOR PARTITIONS	>0 and <= 10000000
SAMPLES_PER_PARTITION	From 1 to (num_vectors/neighbor_partitions)
MIN_VECTORS_PER_PARTITION	From 0 (no trimming of centroid partitions) to total number of vectors (which would result in 1 centroid partition)

NEIGHBOR PARTITION GROUPING	ON | OFF. The default value is OFF.

Examples

- To create a global IVF index:

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 95;
```

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 90
PARAMETERS (
    TYPE IVF,
    NEIGHBOR PARTITIONS 10
);
```

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE EUCLIDEAN
WITH TARGET ACCURACY 95
PARAMETERS (
    TYPE IVF,
    NEIGHBOR PARTITIONS 100,
    NEIGHBOR PARTITION GROUPING ON
) TABLESPACE my_tablespace;
```

- To create a local IVF index:

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 90
PARAMETERS (
    TYPE IVF,
    NEIGHBOR PARTITIONS 10
) LOCAL;
```

For detailed information on the syntax, see CREATE VECTOR INDEX in *Oracle AI Database SQL Language Reference*.

- Create IVF Index Online
  When the base table must remain available for ongoing updates and cannot be locked during index creation, use the `ONLINE` clause with `CREATE VECTOR INDEX` to build the IVF index online. This allows applications with heavy DML to continue updating the table while the index is being created.

Related Topics

- Partition Maintenance Operations with GLOBAL Vector Indexes
  The table that a Partition Maintenance Operation is performed on must be a partitioned table. It can be partitioned using various methods, including `RANGE`, `LIST`, `HASH`, and `COMPOSITE`, and can have one or more global IVF and HNSW indexes.

Create IVF Index Online

When the base table must remain available for ongoing updates and cannot be locked during index creation, use the `ONLINE` clause with `CREATE VECTOR INDEX` to build the IVF index online. This allows applications with heavy DML to continue updating the table while the index is being created.

Syntax

```
CREATE VECTOR INDEX index_name ON table_name (vector_column)
  ORGANIZATION NEIGHBOR PARTITIONS
  DISTANCE distance_metric
  WITH TARGET ACCURACY target_accuracy
  PARAMETERS (type IVF, neighbor partitions num_partitions)
  ONLINE;
```

Usage Notes

- All vectors indexed need to have the same dimension and storage type throughout the index creation. Any changes or inconsistencies during index creation result in errors (for example, ORA-51902, ORA-51934).
- If the base table experiences a very high rate of DML during index build, then index creation time may increase.
- Online index creation with included columns (columns in addition to the vector column) is not currently supported.
- Online creation of local IVF vector indexes is not currently supported.

Example

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
  ORGANIZATION NEIGHBOR PARTITIONS
  DISTANCE COSINE
  WITH TARGET ACCURACY 90
  PARAMETERS (type IVF, neighbor partitions 10)
  ONLINE;
```

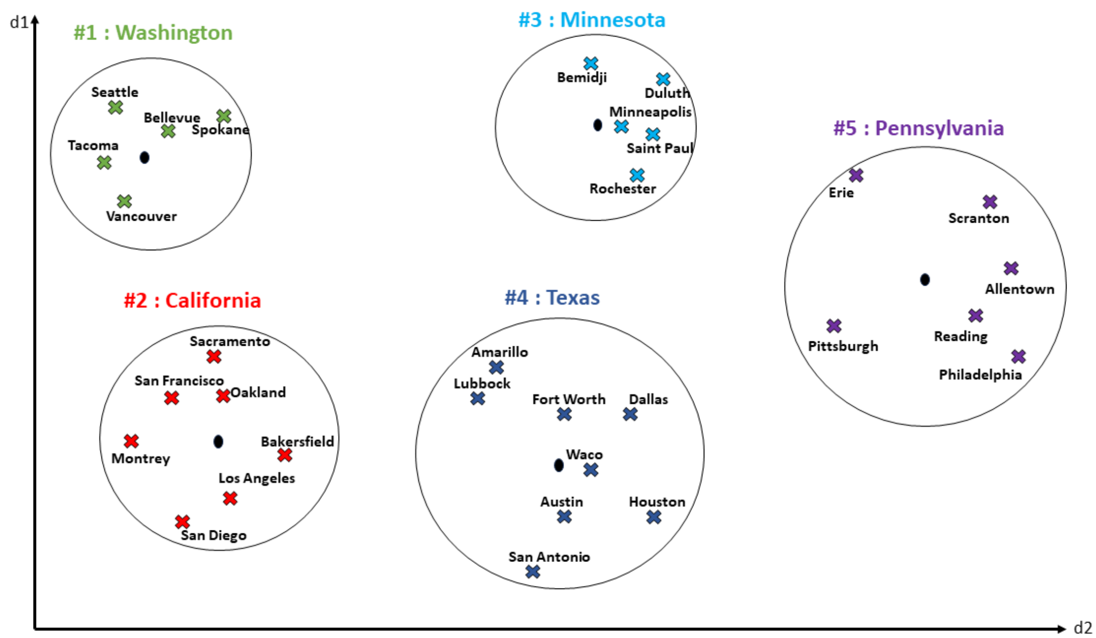
Inverted File Flat Vector Indexes Online Rebuild

Data distribution shifts caused by new DML operations (inserts, updates, or deletes) that occur after an IVF (Inverted File Flat) vector index has been created can significantly influence accuracy of the results. These changes require careful maintenance strategies to ensure sustained retrieval accuracy.

Before we explore the impact of data changes on IVF index quality, let’s briefly recall how an IVF index is constructed. The process begins by clustering a sample set of vectors from the base table. These vectors are defined as CENTROIDS and are stored in an underlying Centroids table. The base table is then scanned, and all remaining vectors are then also clustered and mapped to the centroids. The vector assignment is stored into a Centroids Partition table.

The effectiveness of the IVF index is largely dependent on the choices made during the index creation. Parameters provided during IVF index creation determine how well the index partitions the data and, ultimately, how efficiently it can support high-quality nearest neighbor searches. For example, consider a dataset representing US cities. Based on the parameters set during the index creation, the data would be divided into several partitions as shown in the following figure:

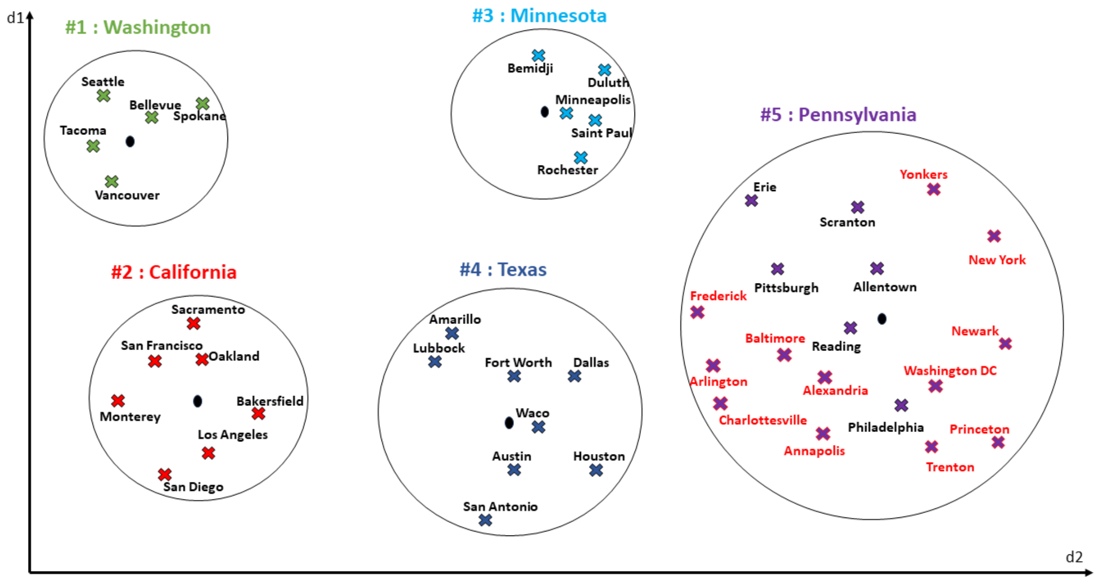
Figure 6-13 IVF Clusters Corresponding to US Cities



After the IVF index is created, post-creation DML operations such as inserts, updates, and deletes can significantly alter the underlying data distribution. This often leads to partition skew, where some neighbor partitions become disproportionately large while others remain underpopulated. Since the centroids used during index creation remain static, they may no longer accurately represent the evolving dataset. This misalignment can cause newly added vectors to be misclassified into incorrect partitions, further degrading recall rates and overall index effectiveness.

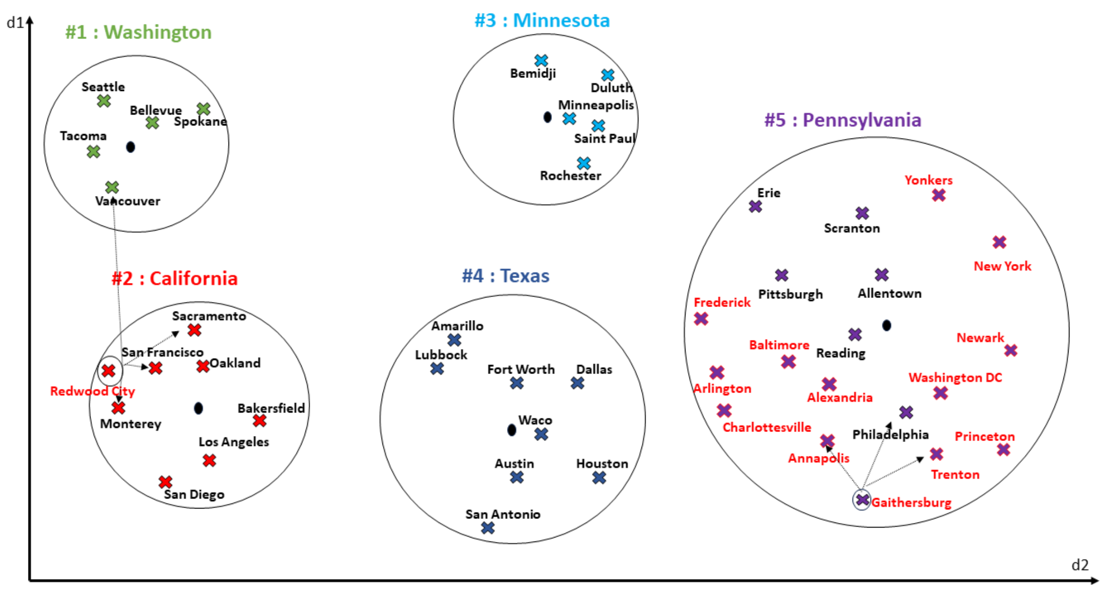
For the example mentioned above, consider that new cities are added (by DML inserts) to the shared journal table after the index is created. The new inserts are added to the nearest cluster. In this case, you can see that all the new inserts are added to the same cluster, making one cluster significantly larger than the other clusters. New inserts are represented in the following figure:

Figure 6-14 DML Updates added to the Existing Index



Consider a similarity search corresponding to the search query vectors: "Redwood City" and "Gaithersburg". The top-3 matches corresponding to "Redwood City" would be San Francisco, Monterey, and Sacramento. And the top-3 matches corresponding to "Gaithersburg" would be Philadelphia, Trenton, and Annapolis. Note, these search results are from the original index represented in the figure above.

Figure 6-15 Sample Similarity Search Prior to Index Reorganization



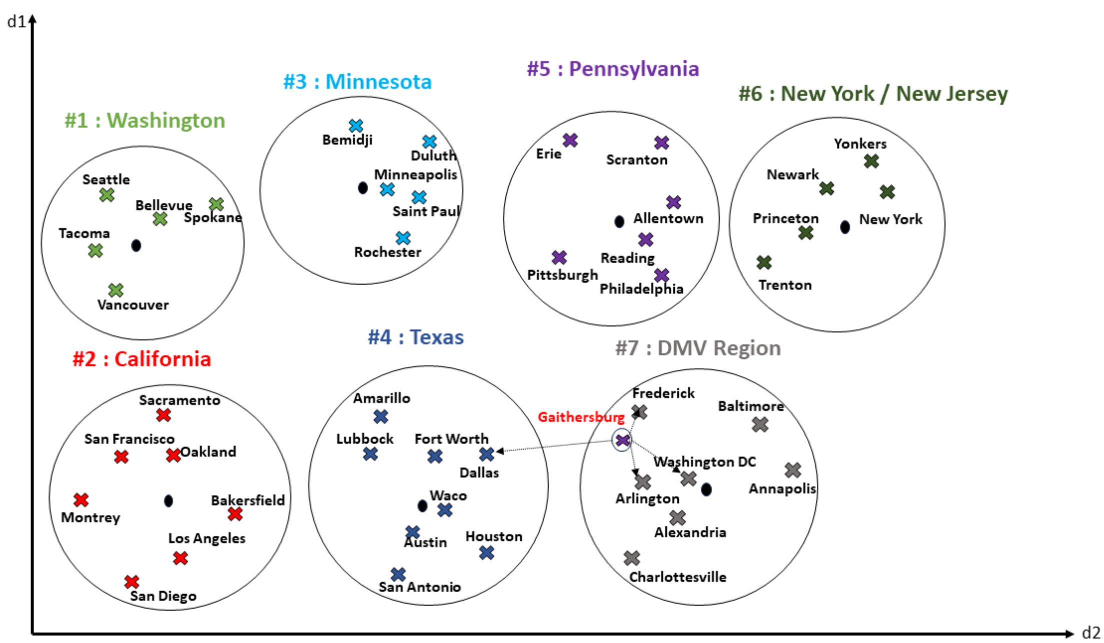
While the search results corresponding to "Redwood City" matches the query, the search results corresponding to "Gaithersburg" are not accurately representing the query. One of the key reason is that the DML updates (shown in the table) are made to the centroid partitions table and the new data may or may not be accurately represented in the index. Also, the retrieval time could be higher due to the large cluster size.

To maintain optimal accuracy, it is essential to implement appropriate **index reorganization** that address these issues as the data evolves.

Index reorganization is achieved by triggering either the global IVF index reorganization or the local IVF index reorganization. Full rebuild is done in multiple phases. The changes that come in to the journal table during the static rebuild phase are applied to the index through a catchup phase. The local IVF index reorganization allows for rebuilding only the selected partition, rather than the entire index. For more information, see Inverted File Flat Vector Indexes Online Rebuild .

After reorganizing the index, the same US cities index would include more clusters representing the new data added. You could also note that the search corresponding to "Gaithersburg" now returns Frederick, Washington DC and Arlington as the top 3 results, which are more accurate in comparison to the results prior to the index reorganization. The following plot represents the new clusters and also note that the size of the clusters being uniform after the reorganization.

Figure 6-16 Similarity Search After Index Reorganization



- Rebuild IVF Indexes Online

Rebuild IVF Indexes Online

The Inverted File Flat (IVF) index may require a rebuild when significant DML activity, such as inserts, updates, or deletes occurs on the underlying base table. Heavy modifications can cause the index structure to become suboptimal, resulting in reduced search efficiency or accuracy. To ensure optimal accuracy, it is important to perform suitable **index reorganization** as the data changes over time. Internal testing has demonstrated that queries per second (QPS) increases significantly following index reorganization. The IVF indexes can be rebuilt either manually or automatically.

Consider an IVF index created using syntax and parameters specified in this Inverted File Flat CREATE INDEX section.

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 95;
```

As the underlying table data changes over time, you can rebuild the IVF index online to reorganize it and help maintain search accuracy and performance.

> **Note**
>
> If a parallel DML is executed on a table that has an IVF index that is in the process of being rebuilt, the operation is automatically converted to a serial DML.

Manual IVF Index Rebuild

Manual reorganization gives the user control to initiate an index rebuild when needed. This can be particularly useful to maintain index performance and ensure data consistency as your data evolves. The manual IVF index reorganization supports both Global and Local index rebuilding:

- **Global IVF Index Reorganization**
  Global IVF index reorganization rebuilds the entire index. You can initiate a global IVF index rebuild using the following syntax:

  ```
  ALTER INDEX <ivf_index_name> REBUILD ONLINE [PARALLEL
        <degree_of_parallelism>];
  ```

  Starting with Oracle AI Database 26ai, Release Update 23.26.1, users can also specify parameters when rebuilding an IVF index. This allows users to select a different set of parameters from those used during the initial index creation, without needing to drop and recreate the index. This flexibility makes it easier to tune index performance or optimize for new data patterns by simply running the index rebuild command with updated parameter values. For more information on parameters, see Inverted File Flat CREATE INDEX.

  ```
  ALTER INDEX <ivf_index_name> REBUILD
  [WITH TARGET ACCURACY]
  [PARAMETERS (
      NEIGHBOR PARTITIONS <number of partitions>,
      SAMPLES_PER_PARTITION <number of samples>,
      MIN_VECTORS_PER_PARTITION <minimum number of vectors per partition>
  )]
  ONLINE
  [PARALLEL <degree of parallelism>];
  ```

- **Local IVF Index Reorganization**
  Local IVF indexing organizes both your data and its search index into matching partitions. Each partition of the data table has a corresponding partition in the index, and related data within each partition is grouped together. This structure improves search performance and simplifies management as data grows or changes. For more information, see Understand Local Inverted File Flat Vector Indexes. Consequently, it allows you to rebuild only the selected partition rather than the entire index, enhancing efficiency and reducing maintenance overhead.

To rebuild a specific partition of the index, use the following syntax:

```
ALTER INDEX <ivf_index_name> REBUILD PARTITION <part_name>
[WITH TARGET ACCURACY]
[PARAMETERS (
    NEIGHBOR PARTITIONS <number of partitions>,
    SAMPLES_PER_PARTITION <number of samples>,
    MIN_VECTORS_PER_PARTITION <minimum number of vectors per partition>
)]
ONLINE
[PARALLEL <degree of parallelism>];
```

For detailed information on the syntax, see ALTER INDEX in Oracle AI Database SQL Language Reference

Auto IVF Index Reorganization

Automatic index optimization enhances index performance and minimizes manual maintenance by continuously monitoring index usage and health. When necessary, it automatically rebuilds indexes to ensure optimal efficiency. A background process periodically scans all IVF indexes at fixed intervals, assessing each for potential optimization. It tracks DML activity, such as inserts, updates, or deletes on corresponding tables. When DML changes surpass a configurable threshold (configured parameters), the system automatically schedules a rebuild for the affected index. This proactive approach ensures that IVF indexes remain efficient and performant without user intervention.

The Automatic Optimization framework is fully configurable, offering several parameters to customize its operation. Users can control how often the background process runs, set the sensitivity to DML changes, and define how rebuild operations are scheduled. These configurable options enable users to fine-tune the automation to match their workload characteristics and performance requirements.

> **Note**
>
> Automatic IVF index reorganization is currently not enabled for IVF indexes that have included columns.

Table 6-2 Configuration Parameters

Parameter	Description
`VECTOR_INDEX_OPTIMIZATION_BACKGROUND`	Enables or disables the vector index background coordinator process that manages and optimizes vector indexes in the background for both global and local IVF indexes. **Valid values**: • `VECTOR_INDEX_OPTIMIZATION_BACKGROUND_DISABLED`: disables the vector index background coordinator process. • `VECTOR_INDEX_OPTIMIZATION_BACKGROUND_ENABLED`: enables the vector index background coordinator process for both global and local IVF indexes. • `VECTOR_INDEX_OPTIMIZATION_BACKGROUND_ENABLED_GLOBAL`: enables the vector index background coordinator process for global IVF indexes only. **Default value**: `VECTOR_INDEX_OPTIMIZATION_BACKGROUND_ENABLED`
`VECTOR_INDEX_OPTIMIZATION_BACKGROUND_INTERVAL`	Specifies the frequency, in seconds, at which the background coordinator process optimizes the index. **Valid values**: 600 to 2^{31} - 1 **Default value**: 1800 seconds
`VECTOR_INDEX_OPTIMIZATION_BACKGROUND_MAXPROCS`	Specifies the maximum number of processes scheduled for vector index optimization by the background coordinator process. **Valid values**: 1 to 2^{31} - 1 **Default value**: 1
`VECTOR_INDEX_OPTIMIZATION_DML_THRESHOLD`	Specifies the percentage threshold of DML (insert, update, or delete) operations executed on the base table that will trigger index optimization actions. For example, if the threshold is set to 20%, index optimization is initiated each time the number of DML operations equals 20% of the base table's current row count. This ensures timely optimization in response to significant data changes. **Valid values**: 1 to 2^{31} - 1 **Default value**: 20
`VECTOR_INDEX_OPTIMIZATION_MIN_ROWS`	Specifies the minimum row count threshold that a base table must meet before index optimization actions are initiated. **Valid values**: 1 to 2^{31} - 1 **Default value**: 100,000

Table 6-2 (Cont.) Configuration Parameters

Parameter	Description
`VECTOR_INDEX_OPTIMIZATION_TASK_BACKOFF_MINS`	Sets a backoff time (in minutes) to be observed after a failed automatic index optimization action. During this backoff period, the vector index optimization background coordinator will not schedule any further optimization tasks for the affected index until the specified time has elapsed since the most recent failure. **Valid values**: 0 to 2^{31} - 1 **Default value**: 60 minutes
`VECTOR_INDEX_OPTIMIZATION_TASK_JOB_CLASS`	Allows users to specify a job class for index optimization tasks that are scheduled to run in the background (as part of PL/SQL procedure call as well as auto-optimization background coordinator actions). **Valid values**: • `VECTOR_INDEX_OPTIMIZATION_JOB_CLASS_DEFAULT`: assigns background jobs scheduled with `DBMS_SCHEDULER` a default priority level. These jobs are managed with standard resource allocation and scheduling. • `VECTOR_INDEX_OPTIMIZATION_JOB_CLASS_HIGH`: assigns background jobs scheduled with `DBMS_SCHEDULER` a high priority level. These jobs are given preference in resource allocation and scheduling, allowing critical or time-sensitive tasks to run ahead of lower-priority jobs. **Default value**: `VECTOR_INDEX_OPTIMIZATION_JOB_CLASS_DEFAULT`

> **Note**
>
> The configurable parameters listed above must be prefixed with `DBMS_VECTOR_ADMIN`, as they are PL/SQL constants defined within the package.

Configuring Parameters using the DBMS_VECTOR_ADMIN Package

The **DBMS_VECTOR_ADMIN** package provides procedures to configure and retrieving these parameter values. For more information, see DBMS_VECTOR_ADMIN. The following procedures are available:

Table 6-3 DBMS_VECTOR_ADMIN PACKAGE

Subprogram	Description

SET_PARAMETER	Sets a value for the specified input parameter.
	The following is an example of setting the background optimization interval for vector indexes to 1800 seconds:
	
	`DECLARE`
	`BEGIN`
	
	`DBMS_VECTOR_ADMIN.SET_PARAMETER(parameter =>`
	`DBMS_VECTOR_ADMIN.VECTOR_INDEX_OPTIMIZATION_BACKGROUND_INTERVAL,`
	`   value => 1800);`
	`END;`
	`/`

GET_PARAMETER	Gets the current value for the specified input parameter.
	The following is an example of retrieving and displaying the current background optimization interval setting for vector indexes:
	
	`DECLARE`
	`  v_param_val NUMBER;`
	`BEGIN`
	
	`DBMS_VECTOR_ADMIN.GET_PARAMETER(parameter =>`
	`DBMS_VECTOR_ADMIN.VECTOR_INDEX_OPTIMIZATION_BACKGROUND_INTERVAL,`
	`  value => v_param_val);`
	`  DBMS_OUTPUT.PUT_LINE('parameter value = '|| v_param_val);`
	`END;`
	`/`

Partition Maintenance Operations and Vector indexes

Partition administration is an important task to consider when working with partitioned tables. Partition Maintenance Operations are supported with tables that have either global IVF or HNSW indexes.

- Partition Maintenance Operations with GLOBAL Vector Indexes
  The table that a Partition Maintenance Operation is performed on must be a partitioned table. It can be partitioned using various methods, including `RANGE`, `LIST`, `HASH`, and `COMPOSITE`, and can have one or more global IVF and HNSW indexes.

Partition Maintenance Operations with GLOBAL Vector Indexes

The table that a Partition Maintenance Operation is performed on must be a partitioned table. It can be partitioned using various methods, including `RANGE`, `LIST`, `HASH`, and `COMPOSITE`, and can have one or more global IVF and HNSW indexes.

The following Partition Maintenance Operations are supported on global IVF and HNSW indexes:

- `ADD [SUB]PARTITION`
- `DROP [SUB]PARTITION`
- `MERGE [SUB]PARTITION`
- `SPLIT [SUB]PARTITION`
- `EXCHANGE [SUB]PARTITION`
- `MOVE [SUB]PARTITION`
- `RENAME [SUB]PARTITION`
- `COALESCE [SUB]PARTITION`

In general, a Partition Maintenance Operation does not maintain its global vector indexes by default, unless the clause `UPDATE GLOBAL INDEX` or `UPDATE INDEXES` has been specified in the `ALTER TABLE` statement. One exception to this rule is when a new partition is added to a table that is partitioned on range or list. In this case, the clause `UPDATE GLOBAL INDEX` or `UPDATE INDEXES` is not required because the new partition has no data yet and thus no data needs to be maintained during the Partition Maintenance Operation.

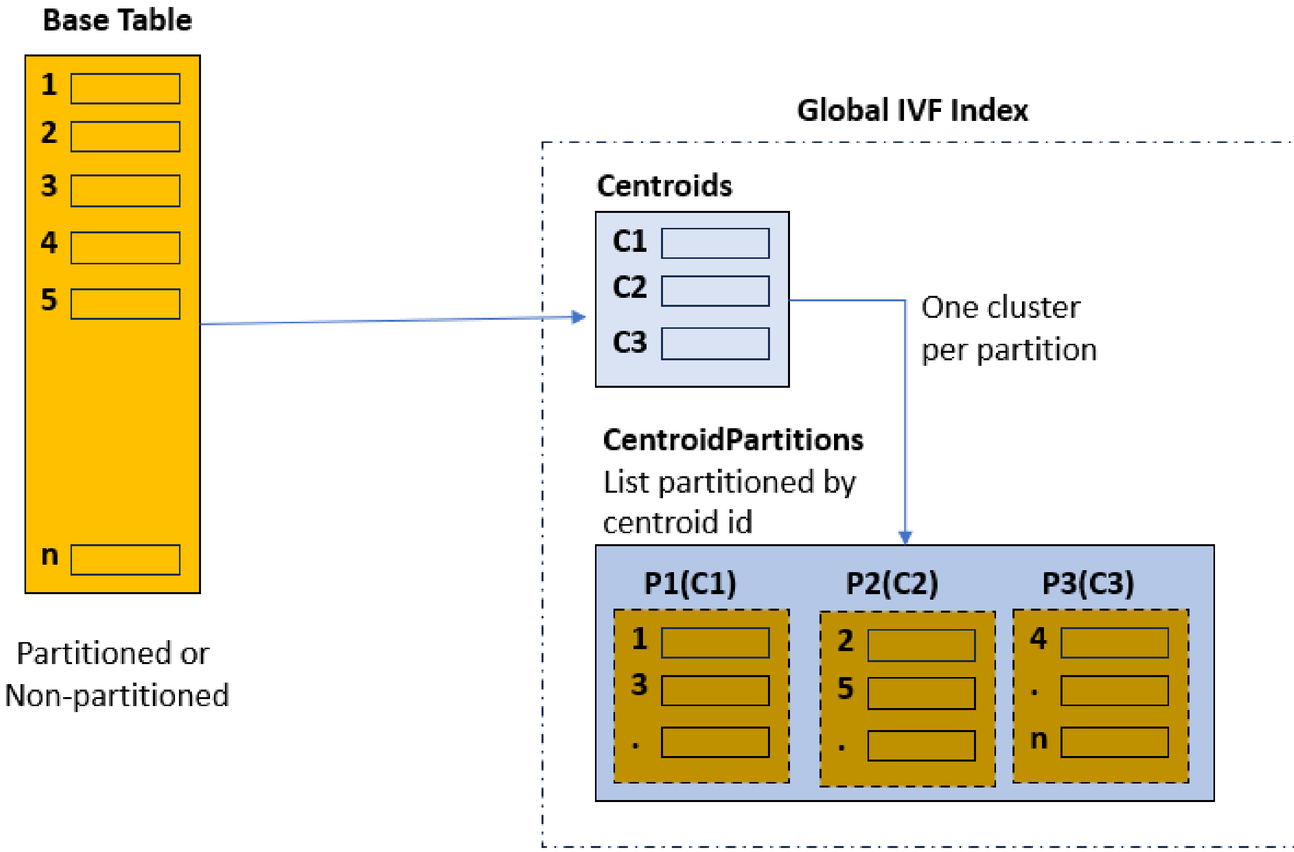
If its status is `VALID` before a Partition Maintenance Operation, a global IVF or HNSW index is maintained automatically during the processing of the Partition Maintenance Operation when the clause `UPDATE GLOBAL INDEXES` or `UPDATE INDEXES` has been specified. After the Partition Maintenance Operation is complete, its status will still be `VALID`.

> **Note**
>
> The use of the `ONLINE` clause in the `ALTER TABLE` statement is not currently supported. Executing a statement such as `ALTER ABLE MERGE PARTITION ONLINE` fails with an `ORA-14808: table does not support ONLINE MERGE PARTITION due to unsupported vector index.`

A **global IVF index** is composed of two tables:

- One called `VECTOR$`<*`base table name`*>`_IVF_IDX$`<*`object info`*>`$IVF_FLAT_CENTROIDS`, containing the list of identified centroid vectors and associated ids.
- The second called `VECTOR$`<*`base table name`*>`_IVF_IDX$`<*`object info`*>`$IVF_FLAT_CENTROID_PARTITIONS`, which is list-partitioned on the centroid ids. Each partition contains the base table vectors closely related (cluster) to the corresponding centroid id for that partition.

This is illustrated by the following diagram:



This structure is used to accelerate searches in the index by identifying first the centroid that is the closest to your query vector, and then using the corresponding centroid id to prune unnecessary partitions.

However, if the base table is partitioned on some relational data and your query is filtering on the base table partition key, then global IVF indexes are not optimal because they are completely independent of the base table partition key. For example, if you search the top-10 houses in California similar to a vectorized picture, the picture itself most likely has no relationship with the state of California. While your query benefits from the fact that your base table is partitioned by state, so you can search only the partition corresponding to California, the query still must look at pictures that may not be in California.

To further accelerate such type of queries, you have the possibility to create a **local IVF index**. The term **local** for an index refers to a one-to-one relationship between the base table partitions or subpartitions and the index partitions. For more information about local IVF indexes, see Understand Local Inverted File Flat Vector Indexes.

- Experiment with Global IVF and HNSW Indexes
  You can start experimenting with `GLOBAL` IVF and HNSW indexes using the included code. These are not complete scenarios but rather a series of SQL commands to help you get started on your own testing.

Experiment with Global IVF and HNSW Indexes

You can start experimenting with `GLOBAL` IVF and HNSW indexes using the included code. These are not complete scenarios but rather a series of SQL commands to help you get started on your own testing.

1. **Create a table partitioned on `RANGE`:**

```
CREATE TABLE sales_data (product_id NUMBER,  sale_date DATE, vec vector(2))
PARTITION BY RANGE (sale_date)
(
    PARTITION sales_2021 VALUES LESS THAN (TO_DATE('01-JAN-2022','DD-MON-
YYYY')),
    PARTITION sales_2022 VALUES LESS THAN (TO_DATE('01-JAN-2023','DD-MON-
YYYY')),
    PARTITION sales_2023 VALUES LESS THAN (TO_DATE('01-JAN-2024','DD-MON-
YYYY')),
    PARTITION sales_2024 VALUES LESS THAN (TO_DATE('01-JAN-2025','DD-MON-
YYYY'))
);
```

2. **Populate the table with data for all 4 partitions:**

```
INSERT INTO sales_data (product_id, sale_date, vec)
VALUES
    (101, TO_DATE('20-DEC-2021', 'DD-MON-YYYY'), '[0.1, 0.1]'),
    (102, TO_DATE('15-NOV-2021', 'DD-MON-YYYY'), '[0.2, 0.2]'),
    (103, TO_DATE('05-OCT-2021', 'DD-MON-YYYY'), '[0.3, 0.3]'),
    (104, TO_DATE('20-SEP-2021', 'DD-MON-YYYY'), '[0.4, 0.4]'),
    (105, TO_DATE('10-AUG-2021', 'DD-MON-YYYY'), '[0.5, 0.5]');

INSERT INTO sales_data (product_id, sale_date, vec)
VALUES
    (201, TO_DATE('20-DEC-2022', 'DD-MON-YYYY'), '[10.1, 10.1]'),
    (202, TO_DATE('15-NOV-2022', 'DD-MON-YYYY'), '[10.2, 10.2]'),
    (203, TO_DATE('05-OCT-2022', 'DD-MON-YYYY'), '[10.3, 10.3]'),
    (204, TO_DATE('20-SEP-2022', 'DD-MON-YYYY'), '[10.4, 10.4]'),
    (205, TO_DATE('10-AUG-2022', 'DD-MON-YYYY'), '[10.5, 10.5]');

INSERT INTO sales_data (product_id, sale_date, vec)
VALUES
    (301, TO_DATE('20-DEC-2023', 'DD-MON-YYYY'), '[0.1, 0.1]'),
    (302, TO_DATE('15-NOV-2023', 'DD-MON-YYYY'), '[0.2, 0.2]'),
    (303, TO_DATE('05-OCT-2023', 'DD-MON-YYYY'), '[0.3, 0.3]'),
    (304, TO_DATE('20-SEP-2023', 'DD-MON-YYYY'), '[0.4, 0.4]'),
    (305, TO_DATE('10-AUG-2023', 'DD-MON-YYYY'), '[0.5, 0.5]');

INSERT INTO sales_data (product_id, sale_date, vec)
VALUES
    (401, TO_DATE('20-DEC-2024', 'DD-MON-YYYY'), '[10.1, 10.1]'),
    (402, TO_DATE('15-NOV-2024', 'DD-MON-YYYY'), '[10.2, 10.2]'),
    (403, TO_DATE('05-OCT-2024', 'DD-MON-YYYY'), '[10.3, 10.3]'),
    (404, TO_DATE('20-SEP-2024', 'DD-MON-YYYY'), '[10.4, 10.4]'),
    (405, TO_DATE('10-AUG-2024', 'DD-MON-YYYY'), '[10.5, 10.5]');
```

3. **Create a `GLOBAL` vector index:**

   a. Create a `GLOBAL` IVF index on column vec

   ```
   CREATE VECTOR INDEX VIDXP ON sales_data(vec)
   ORGANIZATION NEIGHBOR PARTITIONS
   WITH TARGET ACCURACY 95 DISTANCE EUCLIDEAN
   PARAMETERS(type IVF, NEIGHBOR PARTITION 2);
   ```

   b. Create a `GLOBAL` HNSW index on column `vec`:

   ```
   CREATE VECTOR INDEX VIDXP ON sales_data(vec)
   ORGANIZATION INMEMORY NEIGHBOR GRAPH
   WITH TARGET ACCURACY 95 DISTANCE EUCLIDEAN
   PARAMETERS (type HNSW, efConstruction 500);
   ```

   c. Check the partitions of the base table:

   ```
   SELECT
     object_name,
     subobject_name,
     object_id,
     data_object_id
   FROM user_objects
   WHERE object_name = 'SALES_DATA'
   ORDER BY subobject_name;
   ```

   d. Check the status of the IVF index:

   ```
   SELECT
     index_name,
     status
   FROM user_indexes
   WHERE index_type = 'VECTOR' AND table_name = 'SALES_DATA';
   ```

4. **Add a partition:**

   ```
   ALTER TABLE sales_data
   ADD PARTITION sales_2025
   VALUES LESS THAN (TO_DATE('01-JAN-2026','DD-MON-YYYY'));

   commit;
   ```

5. **Drop a partition:**

   ```
   ALTER TABLE sales_data
   DROP PARTITION sales_2022
   UPDATE GLOBAL INDEXES;

   commit;
   ```

6. **Merge two partitions into a new one:**

   ```
   ALTER TABLE sales_data
   MERGE PARTITIONS sales_2023, sales_2024
   ```

```
INTO PARTITION sales_2023_2024
UPDATE GLOBAL INDEXES;

commit;
```

7. **Split a partition:**

```
ALTER TABLE sales_data
SPLIT PARTITION sales_2025
AT (TO_DATE('01-NOV-2025', 'DD-MON-YYYY'))
UPDATE GLOBAL INDEXES;

commit;
```

8. **Exchange a partition with a source table:**

```
--create another table to exchange data with the table sales_data
CREATE TABLE sales_data2 (product_id NUMBER, sale_date DATE, vec
VECTOR(2));

INSERT INTO sales_data2 (product_id, sale_date, vec)
VALUES
  (111, TO_DATE('22-DEC-2021', 'DD-MON-YYYY'), '[110.1, 20.1]'),
  (112, TO_DATE('22-NOV-2021', 'DD-MON-YYYY'), '[110.1, 20.1]'),
  (113, TO_DATE('22-OCT-2021', 'DD-MON-YYYY'), '[110.1, 20.1]'),
  (114, TO_DATE('22-SEP-2021', 'DD-MON-YYYY'), '[110.1, 20.1]'),
  (115, TO_DATE('22-AUG-2021', 'DD-MON-YYYY'), '[110.1, 20.1]');

--exchange the data between sales_data and sales_data2
ALTER TABLE sales_data
EXCHANGE PARTITION sales_2021
WITH TABLE sales_data2
UPDATE GLOBAL INDEXES;

commit;
```

9. **Truncate a partition:**

```
ALTER TABLE sales_data
TRUNCATE PARTITION sales_2021
UPDATE GLOBAL INDEXES;

commit;
```

10. **Create a table partitioned on `HASH` to test the `COALESCE` operation:**

```
CREATE TABLE sales
(
  part_no     NUMBER,
  sale_date   DATE,
  vec         VECTOR(2),
  pcount      NUMBER
)
PARTITION BY HASH (pcount) (PARTITION P1, PARTITION P2, PARTITION P3);
```

```
--populate data into the table
INSERT INTO sales (part_no, sale_date, vec, pcount)
VALUES
  (101, TO_DATE('20-DEC-2021', 'DD-MON-YYYY'), '[1.01, 1.01]', 101),
  (102, TO_DATE('15-NOV-2021', 'DD-MON-YYYY'), '[1.02, 1.02]', 102),
  (103, TO_DATE('05-OCT-2021', 'DD-MON-YYYY'), '[1.03, 1.03]', 103),
  (104, TO_DATE('20-SEP-2021', 'DD-MON-YYYY'), '[1.04, 1.04]', 104),
  (105, TO_DATE('10-AUG-2021', 'DD-MON-YYYY'), '[1.05, 1.05]', 105),
  (201, TO_DATE('20-DEC-2022', 'DD-MON-YYYY'), '[2.01, 2.01]', 201),
  (202, TO_DATE('15-NOV-2022', 'DD-MON-YYYY'), '[2.02, 2.02]', 202),
  (203, TO_DATE('05-OCT-2022', 'DD-MON-YYYY'), '[2.03, 2.03]', 203),
  (204, TO_DATE('20-SEP-2022', 'DD-MON-YYYY'), '[2.04, 2.04]', 204),
  (205, TO_DATE('10-AUG-2022', 'DD-MON-YYYY'), '[2.05, 2.05]', 205);
```

11. **Create a `GLOBAL` vector index on column vec:**
    a. Create a `GLOBAL` IVF index:

    ```
    CREATE VECTOR INDEX VIDXP ON sales(vec)
    ORGANIZATION NEIGHBOR PARTITIONS WITH TARGET ACCURACY 95
    DISTANCE EUCLIDEAN PARAMETERS(type IVF, NEIGHBOR PARTITION 2);
    ```

    b. Create a `GLOBAL` HNSW index:

    ```
    CREATE VECTOR INDEX VIDXP ON sales(vec)
    ORGANIZATION INMEMORY NEIGHBOR GRAPH WITH TARGET ACCURACY 95
    DISTANCE EUCLIDEAN PARAMETERS(type HNSW, efConstruction 500);
    ```

12. **`RENAME` a partition:**

    ```
    ALTER TABLE sales RENAME PARTITION P1 TO NewP1;

    commit;
    ```

13. **`COALESCE` partitions**:

    ```
    ALTER TABLE sales COALESCE PARTITION UPDATE INDEXES:

    commit;
    ```

Guidelines for Using Vector Indexes

Use these guidelines to create and use Hierarchical Navigable Small World (HNSW) or Inverted File Flat (IVF) vector indexes.

Create Index Guidelines

The minimum information required to create a vector index is to specify one `VECTOR` data type table column and a vector index type: `INMEMORY NEIGHBOR GRAPH` for HNSW and `NEIGHBOR PARTITIONS` for IVF. However, you also have the possibility to specify more information, such as the following:

- You can optionally provide more information including the distance metric to use. Supported metrics are `EUCLIDEAN SQUARED`, `EUCLIDEAN`, `COSINE`, `DOT`, `MANHATTAN`, and `HAMMING`. If not specified, `COSINE` is used by default.

- Specific parameters that impact the accuracy of index creation and approximate searches. A target accuracy percentage value and, `NEIGHBORS` (or M) and `EFCONSTRUCTION` for HNSW and `NEIGHBOR PARTITIONS` for IVF.
- You can create globally partitioned vector indexes.
- You can also specify the degree of parallelism to use for index creation.

You cannot currently define a vector index on:

- IOTs
- Clusters/Cluster tables
- Global Temp tables
- Blockchain tables
- Immutable tables
- Materialized views
- Function-based vector index

You can find information about your vector indexes by looking at `ALL_INDEXES`, `DBA_INDEXES`, and `USER_INDEXES` family of views. The columns of interest are `INDEX_TYPE` (`VECTOR`) and `INDEX_SUBTYPE` (`INMEMORY_NEIGHBOR_GRAPH_HNSW` or `NEIGHBOR_PARTITIONS_IVF`). In the case the index is not a vector index, `INDEX_SUBTYPE` is `NULL`.

See VECSYS.VECTOR$INDEX for detailed information about vector indexes.

Note

- The `VECTOR` column is designed to be extremely flexible to support vectors of any number of dimensions and any format for the vector dimensions. However, you can create a vector index only on a `VECTOR` column containing vectors that all have the same number of dimensions. This is required as you can't compute distances over vectors with different dimensions. For example, if a `VECTOR` column is defined as `VECTOR(*, FLOAT32)`, and two vectors with different dimensions (128 and 256 respectively) are inserted in that column. When you try to create the vector index on that column, you will get an error.
- Once an IVF index is created on a vector column of an empty table and non-null vectors with a particular dimension have been inserted into it, new vectors with a different dimension cannot be inserted into the same vector column due to the existence of the IVF index.
- If a table is truncated, any IVF index created on the table is marked as `UNUSABLE`. Thus, the IVF index will not be maintained on any subsequent `INSERT` statements with vectors of the same or different dimension count. You must rebuild or recreate the IVF index before using the index in a query again.
- You can only create one type of vector index per vector column.
- Oracle recommends that you allocate larger, temporary tablespaces for proper IVF vector index creation with big vector spaces and vector sizes. In such cases, the system internally makes extensive use of temporary space.
- On a RAC environment you can set up a vector pool on each instance for the best performance of vector indexes.

Use Index Guidelines

For the Oracle AI Database Optimizer to consider a vector index, you must ensure these conditions in your SQL statements:

- The vector index must exist.
- The distance function for the index must be the same as the distance function used in the `vector_distance()` function.
- If the vector index DDL does not specify the distance function and the `vector_distance()` function uses `EUCLIDEAN`, `DOT`, `MANHATTAN` or `HAMMING`, then the vector index is not used.
- If the vector index DDL uses the `DOT` distance function and the `vector_distance()` function uses the default distance function `COSINE`, then the vector index is not used.
- The `vector_distance()` must not be encased in another SQL function.
- If using the partition row-limiting clause, then the vector index is not used.
- Index accuracy with an IVF index may diminish over time due to DML operations being performed on the underlying table. You can check for this by using the `INDEX_ACCURACY_QUERY` function provided by the `DBMS_VECTOR` package. In such a case, the index can be rebuilt using the `REBUILD_INDEX` function also provided by the `DBMS_VECTOR` package. See DBMS_VECTOR for more information about `DBMS_VECTOR` and its subprograms.

> **Note**
>
> Except for `ALTER TABLE tab SUBPARTITION/PARTITION` (`RANGE/LIST`) with or without Update Global Indexes, global vector indexes are marked as unusable as part of other Partition Management Operations on a partitioned table. In those cases you must manually rebuild the global vector indexes.

Index Accuracy Report

The index accuracy reporting feature lets you determine the accuracy of your vector indexes.

Accuracy of One Specific Query Vector

After a vector index is created, you may be interested to know how accurate your vector searches are. One possibility might be to run two queries using the same query vector, that is, one performing an approximate search using a vector index and the other performing an exact search without an index. Then, you need to manually compare the results to determine the real accuracy of your index.

Instead, you can use an index accuracy report provided by the `DBMS_VECTOR.INDEX_ACCURACY_QUERY` procedure. This procedure provides an accuracy report for a top-K index search for a specific query vector and a specific target accuracy. For syntax and parameter details, see INDEX_ACCURACY_QUERY.

Here is a usage example of this procedure using our galaxies scenario:

```
declare
    q_v VECTOR;
    report varchar2(128);
begin
```

```
    q_v := to_vector('[0,1,1,0,0]');
    report := dbms_vector.index_accuracy_query(
        OWNER_NAME => 'COSMOS',
        INDEX_NAME => 'GALAXIES_HNSW_IDX',
        qv => q_v, top_K =>10,
        target_accuracy =>90 );
    dbms_output.put_line(report);
end;
/
```

The preceding example computes the top-10 accuracy of the `GALAXIES_HNSW_IDX` vector index using the embedding corresponding to the NGC 1073 galaxy and a 90% accuracy requested.

The index accuracy report for this may look like:

```
Accuracy achieved (100%) is 10% higher than the Target Accuracy requested
(90%)
```

The possible parameters are:

- `owner_name`: Index owner name
- `index_name`: Index name
- `qv`: Query vector
- `top_K`: Top K value for accuracy computation
- `target_accuracy`: Target accuracy for the index

Accuracy of Automatically Captured Query Vectors

An overloaded version of the `DBMS_VECTOR.INDEX_ACCURACY_REPORT` function allows you to capture from your past workloads, accuracy values achieved by your approximate searches using a particular vector index for a certain period of time. Query vectors used for approximate searches are captured automatically in memory and persisted to a catalog table every hour.

The `INDEX_ACCURACY_REPORT` function computes the achieved accuracy using the captured query vectors for a given index. To compute the achieved accuracy for each query vector, the function compares the result set of approximate similarity searches with exact similarity searches for the same query vectors.

The accuracy findings are stored in dictionary and exposed using the `DBA_VECTOR_INDEX_ACCURACY_REPORT` dictionary view.

Here is a usage example of this function using the galaxies scenario:

```
VARIABLE t_id NUMBER;
BEGIN
  :t_id := DBMS_VECTOR.INDEX_ACCURACY_REPORT('VECTOR', 'GALAXIES_HNSW_IDX');
END;
/
```

You can also run the following statement to get the corresponding task identifier:

```
SELECT DBMS_VECTOR.INDEX_ACCURACY_REPORT('VECTOR', 'GALAXIES_HNSW_IDX');
```

The following are possible parameters for the INDEX_ACCURACY_REPORT function:

- owner_name (IN): Index owner name
- ind_name (IN): Index name
- start_time (IN): Query vectors captured from this time are considered for the accuracy computation. A NULL start_time uses query vectors captured in the last 24 hours.
- end_time (IN): Query vectors captured until this time are considered for accuracy computation. A NULL end_time uses query vectors captured from start_time until the current time.
- Return Values: A numeric task ID if the accuracy for the given index was successfully computed. Otherwise, a NULL task ID is returned.

> **Note**
>
> - If both start_time and end_time are NULL, accuracy is computed using query vectors captured in the last 24 hours.
> - If start_time is NULL and end_time is not NULL, accuracy is computed using query vectors captured between 24 hours before end_time until end_time.
> - If start_time is not NULL and end_time is NULL, accuracy is computed using query vectors captured between start_time and the current time.

You can see the analysis results using the DBA_VECTOR_INDEX_ACCURACY_REPORT view:

```
desc DBA_VECTOR_INDEX_ACCURACY_REPORT


 Name                                      Null?    Type
 ----------------------------------------- --------
----------------------------
 TASK_ID                                            NUMBER
 TASK_TIME                                          TIMESTAMP(6)
 OWNER_NAME                                         VARCHAR2(128)
 INDEX_NAME                                         VARCHAR2(128)
 INDEX_TYPE                                         VARCHAR2(16)
 MIN_TARGET_ACCURACY                                NUMBER
 MAX_TARGET_ACCURACY                                NUMBER
 NUM_VECTORS                                        NUMBER
 MEDIAN_ACHIEVED_ACCURACY                           NUMBER
 MIN_ACHIEVED_ACCURACY                              NUMBER
 MAX_ACHIEVED_ACCURACY                              NUMBER
```

Select target accuracy values in the following statement:

```
SELECT MIN_TARGET_ACCURACY, MAX_TARGET_ACCURACY, num_vectors,
MIN_ACHIEVED_ACCURACY, MEDIAN_ACHIEVED_ACCURACY, MAX_ACHIEVED_ACCURACY
FROM DBA_VECTOR_INDEX_ACCURACY_REPORT WHERE task_id = 1;



MIN_TARGET_ACCURACY MAX_TARGET_ACCURACY NUM_VECTORS MIN_ACHIEVED_ACCURACY
```

```
MEDIAN_ACHIEVED_ACCURACY MAX_ACHIEVED_ACCURACY
------------------ ------------------ ----------- --------------------
----------------------- --------------------
                 1                 10           2
49                       57                    65
                11                 20           3
60                       73                    83
                21                 30           3
44                       64                    84
                31                 40           2
63                     76.5                    90
                41                 50           3
63                       81                    90
                61                 70           2
57                       68                    79
                71                 80           3
79                       87                    89
                81                 90           3
70                       71                    78
                91                100           4
67                     79.5                    88
```

Each row in the output represents a bucket of 10 target accuracy values: 1-10, 11-20, 21-30, … , 91-100.

Consider the following partial statement that runs an approximate similarity search for a particular query vector and a particular target accuracy:

```
SELECT ...
FROM ...
WHERE ...
ORDER BY VECTOR_DISTANCE( embedding, :my_query_vector, COSINE )
FETCH FIRST 3 ROWS ONLY WITH TARGET ACCURACY 65;
```

Once the preceding approximate similarity search is run and captured by your accuracy report task, the value of `NUM_VECTORS` would be increased by one in row 6 (bucket values between 61 and 70) of the results of the select statement on the `DBA_VECTOR_INDEX_ACCURACY_REPORT` view for your task. `NUM_VECTORS` represents the number of query vectors that fall in a particular target accuracy bucket.

`MIN_ACHIEVED_ACCURACY`, `MEDIAN_ACHIEVED_ACCURACY`, and `MAX_ACHIEVED_ACCURACY` are the actual achieved accuracy values for the given target accuracy bucket.

> **Note**
>
> The initialization parameter `VECTOR_QUERY_CAPTURE` is used to enable and disable capture of query vectors. The parameter value is set to `ON` by default. You can turn off this background functionality by setting `VECTOR_QUERY_CAPTURE` to `OFF`. When `VECTOR_QUERY_CAPTURE` is `ON`, the database captures some of the query vectors through sampling. The captured query vectors are retrained for a week and then purged automatically.

Vector Index Status, Checkpoint, and Advisor Procedures

Review these high-level details on the `GET_INDEX_STATUS`, `ENABLE_CHECKPOINT`, `DISABLE_CHECKPOINT`, and `INDEX_VECTOR_MEMORY_ADVISOR` procedures that are available with the `DBMS_VECTOR` PL/SQL package.

GET_INDEX_STATUS

Purpose: To query the status of a vector index creation.

Syntax:

```
DBMS_VECTOR.GET_INDEX_STATUS ('USER_NAME','INDEX_NAME');
```

Usage Notes:

- You can use the `GET_INDEX_STATUS` procedure only during a vector index creation.
- The `Percentage` value is shown in the output only for Hierarchical Navigable Small World (HNSW) indexes (and not for Inverted File Flat (IVF) indexes).
- Along with the `DB_DEVELOPER_ROLE` privilege, you must have read access to the `VECSYS.VECTOR$INDEX$BUILD$` table.
- You can use the following query to view all auxiliary tables:

  ```
  select IDX_AUXILIARY_TABLES from vecsys.vector$index;
  ```

  – For HNSW indexes:

    `rowid_vid_map` stores the mapping between a row ID and vector ID. `shared_journal_change_log` stores the DML changes that are yet to be incorporated into an HNSW graph.

  – For IVF indexes:

    `centroids` stores the location for each centroid. `centroid_partitions` stores the closest centroid for each vector.

- The possible values of `Stage` for HNSW vector indexes are:

Value	Description
`HNSW Index Initialization`	Initialization phase for the HNSW vector index creation
`HNSW Index Auxiliary Tables Creation`	Creation of the internal auxiliary tables for the HNSW Neighbor Graph vector index
`HNSW Index Graph Allocation`	Allocation of memory from the vector memory pool for the HNSW graph
`HNSW Index Loading Vectors`	Loading of the base table vectors into the vector pool memory
`HNSW Index Graph Construction`	Creation of the multi-layered HNSW graph with the previously loaded vectors
`HNSW Index Creation Completed`	HNSW vector index creation finished

- The possible values of `Stage` for IVF vector indexes are:

Value	Description
`IVF Index Initialization`	Initialization phase for the IVF vector index creation
`IVF Index Centroids Creation`	The K-means clustering phase that computes the cluster centroids on a sample of base table vectors
`IVF Index Centroid Partitions Creation`	Centroids assignment phase for the base table vectors
`IVF Index Creation Completed`	IVF vector index creation completed

Example:

```
exec DBMS_VECTOR.GET_INDEX_STATUS('VECTOR_USER','VIDX_HNSW');

Index objn: 74745
Stage: HNSW Index Loading Vectors
Percentage: 80%
```

ENABLE_CHECKPOINT

Purpose: To enable the Checkpoint feature for a given HNSW index user and HNSW index name.

> **Note**
>
> This procedure only allows the index to create checkpoints. The checkpoint is created as part of the next HNSW graph refresh.

The `INDEX_NAME` clause is optional. If you do not specify the index name, then this procedure enables the Checkpoint feature for all HNSW indexes under the given user.

Syntax:

```
DBMS_VECTOR.ENABLE_CHECKPOINT('INDEX_USER',['INDEX_NAME']);
```

Example 1: Using index name and index user:

```
DBMS_VECTOR.ENABLE_CHECKPOINT('VECTOR_USER','VIDX1');
```

Example 2: Using index user:

```
DBMS_VECTOR.ENABLE_CHECKPOINT('VECTOR_USER');
```

> **Note**
>
> By default, HNSW checkpointing is enabled. You can disable it using the `DBMS_VECTOR.DISABLE_CHECKPOINT` procedure.

DISABLE_CHECKPOINT

Purpose: To purge all older checkpoint data. This procedure disables the Checkpoint feature for a given HNSW index user and HNSW index name. It also disables the creation of future checkpoints as part of the HNSW graph refresh.

The *INDEX_NAME* clause is optional. If you do not specify the index name, then this procedure disables the Checkpoint feature for all HNSW indexes under the given user.

Syntax:

```
DBMS_VECTOR.DISABLE_CHECKPOINT('INDEX_USER',['INDEX_NAME']);
```

Example 1: Using index name and index user:

```
DBMS_VECTOR.DISABLE_CHECKPOINT('VECTOR_USER','VIDX1');
```

Example 2: Using index user:

```
DBMS_VECTOR.DISABLE_CHECKPOINT('VECTOR_USER');
```

INDEX_VECTOR_MEMORY_ADVISOR

Purpose: To determine the vector memory size needed for a particular vector index. This helps you evaluate the number of indexes (HNSW or IVF) that can fit for each simulated vector memory size.

Syntax:

- Using the number and type of vector dimensions that you want to store in your vector index.

  ```
  DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
      INDEX_TYPE,
      NUM_VECTORS,
      DIM_COUNT,
      DIM_TYPE,
      PARAMETER_JSON,
      RESPONSE_JSON);
  ```

- Using the table and vector column on which you want to create your vector index:

  ```
  DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
      TABLE_OWNER,
      TABLE_NAME,
      COLUMN_NAME,
      INDEX_TYPE,
      PARAMETER_JSON,
      RESPONSE_JSON);
  ```

`INDEX_TYPE` can be one of the following:

- `IVF` for IVF vector indexes
- `HNSW` for HNSW vector indexes

PARAMETER_JSON can have only one of the following form:

- `PARAMETER_JSON=>{"accuracy":value}`
- `INDEX_TYPE=>IVF, parameter_json=>{"neighbor_partitions":value}`
- `INDEX_TYPE=>HNSW, parameter_json=>{"neighbors":value}`

> **Note**
>
> You cannot specify values for accuracy along with neighbor_partitions or neighbors.

Example 1: Using neighbors in the parameters list:

```
exec DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
    INDEX_TYPE=>'HNSW',
    NUM_VECTORS=>10000,
    DIM_COUNT=>768,
    DIM_TYPE=>'FLOAT32',
    PARAMETER_JSON=>'{"neighbors":128}',
    RESPONSE_JSON=>:response_json);

Suggested vector memory pool size: 59918628 Bytes
```

Example 2: Using accuracy in the parameters list:

```
exec DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
    INDEX_TYPE=>'HNSW',
    NUM_VECTORS=>10000,
    DIM_COUNT=>768,
    DIM_TYPE=>'FLOAT32',
    PARAMETER_JSON=>'{"accuracy":90}',
    RESPONSE_JSON=>:response_json);

Suggested vector memory pool size: 53926765 Bytes
```

Example 3: Using the table and vector column on which you want to create the vector index:

```
exec DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
    'VECTOR_USER',
    'VECTAB',
    'DATA_VECTOR',
    'HNSW',
    RESPONSE_JSON=>:response_json);

Using default accuracy: 90%
Suggested vector memory pool size: 76396251 Bytes
```

Related Topics

- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.

Manage Hybrid Vector Indexes

Learn how to manage a hybrid vector index, which is a single index for searching by *similarity* and *keywords*, to enhance the accuracy of your search results.

- Understand Hybrid Vector Indexes
  A hybrid vector index inherits all the information retrieval capabilities of Oracle Text search indexes and leverages the semantic search capabilities of Oracle AI Vector Search vector indexes.
- Guidelines and Restrictions for Hybrid Vector Indexes
  Review these guidelines and the current set of restrictions when working with hybrid vector indexes.
- CREATE HYBRID VECTOR INDEX
  Use the `CREATE HYBRID VECTOR INDEX` SQL statement to create a hybrid vector index, which allows you to index and query documents using a combination of full-text search and vector similarity search.
- Creating Hybrid Vector Indexes using the DBMS_SEARCH API
  The `DBMS_SEARCH` package provides procedures and functions for creating, managing, and querying hybrid vector indexes when multiple sources are used for the search. Sources includes tables and views as described in the DBMS_SEARCH documentation.
- ALTER INDEX
  Use the `ALTER INDEX` SQL statement to modify an existing hybrid vector index.

Understand Hybrid Vector Indexes

A hybrid vector index inherits all the information retrieval capabilities of Oracle Text search indexes and leverages the semantic search capabilities of Oracle AI Vector Search vector indexes.

Hybrid vector indexes allow you to index and query documents using a combination of full-text search and semantic vector search. A hybrid vector index is a class of specialized Domain Index that combines the existing Oracle Text indexing data structures and vector indexing data structures into *one unified structure*. A single index contains both textual and vector fields for a document, enabling you to perform a combination of keyword search and vector search simultaneously.

The purpose of a hybrid vector index is to enhance search relevance of an Oracle Text index by allowing users to search by both *vectors* and *keywords* in various combinations, using out-of-the-box and custom scoring techniques. By integrating traditional keyword-based text search with vector-based similarity search, you can improve the overall search experience and provide users with more accurate information.

When to Use a Hybrid Vector Index

Consider using a hybrid vector index for hybrid search scenarios where your query requires information that is semantically similar but pertains to a specific focus area, that is, involves a particular organization, user name, product code, technical term, date, or time. For example, a typical hybrid search query can be to find "`top 10 instances of stock fraud for ABC Corporation`".

Such a query involves two separate components:

- One where you want to identify the notion of "`stock fraud`"
- The second where you want to narrow the results to focus only on "`ABC Corporation`"

Pure keyword search may return results that specifically contain the query words like "stock", "fraud", "ABC", or "Corporation" because it focuses on matching the exact keywords or surface-level representation of words or phrases with tokenized terms in a text index. Therefore, keyword search alone may not be suitable here because it can overlook the semantic meaning behind the words in our query, especially if the exact terms are not present in the content.

Pure vector search focuses on understanding the meaning and context of words or phrases rather than just matching keywords. Vector search considers semantic relationship between the query words, so it may include more contextually-relevant results like "corporate fraud", "stock market manipulation", "stock misconduct", "financial irregularities", or "lawsuits in the financial sector". Vector search also may not be suitable here because it can include results about the broader topic of stock fraud involving ABC or similar organizations, especially if the exact phrase "stock fraud for ABC Corporation" is not present in the content.

Hybrid search can address both components of such a query by running keyword search and vector search on the same data and then combining the two search results into a single result set. In this way, you can utilize the strengths of both text indexes and vector indexes to retrieve the most relevant results.

Why Choose a Hybrid Vector Index?

Let us summarize the advantages of a hybrid vector index.

- Higher recall compared to pure vector search or keyword search:

  As discussed earlier, hybrid search lets you combine the power of Oracle AI Vector Search and Oracle Text Search to provide more accurate and personalized information. Keyword search or vector search alone may not be relevant in complex search scenarios and may lead to a lot of spurious results.

- Mitigates the downsides of chunking:

  Vector embedding models usually impose limits on the size of input text, which forces large documents to be split into smaller chunks of data for semantic search (as explained in Understand the Stages of Data Transformations). A chunk can lose broader context of the original document due to truncation, which may lead to missed results. A hybrid vector index helps to restore the entire context of each document by performing textual search at the document level.

- Simple to deploy and manage compared to maintaining independent indexes:

  A hybrid vector index is a single domain index that can maintain text and vectors with DML. Both keyword search and vector search are performed on all documents, and then the two search results are combined and scored to return a unified result set. It provides an end-to-end indexing pipeline that automatically transforms your input data for vector search alongside keyword search, thereby enhancing the indexing performance. This index exposes optional preferences to configure indexing parameters, but it is not required.

Here is an example of the hybrid vector index DDL:

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  DOCS(file_name)
  PARAMETERS ('MODEL MY_INDB_MODEL');
```

Here, a hybrid vector index named `my_hybrid_idx` is created on the `file_name` column of the `DOCS` table. The embedding model used for vector generation is an in-database ONNX format model named `MY_INDB_MODEL`.

The preceding example shows the minimum input requirements for index creation. For complete syntax, see [CREATE HYBRID VECTOR INDEX](#).

- Unified query API to search by vectors and keywords:

  A single `SEARCH` API (available with the `DBMS_HYBRID_VECTOR` PL/SQL package) lets you specify both a traditional `CONTAINS` query on document text indexes and a `VECTOR_DISTANCE` query on vectorized chunk indexes. You can switch between keyword-only, vector-only, and hybrid search modes to retrieve the best documents or chunks.

  Here is an example of hybrid search using the `SEARCH` API for our earlier "`top 10 instances of stock fraud for ABC Corporation`" scenario:

```
select DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name"     : "my_hybrid_idx",
            "vector":
                    { "search_text" : "stock fraud" },
            "text"  :
                    { "contains"    : "$ABC AND $Corporation" },
            "return":
                    { "topN"        : 10 }
          }'))
from dual;
```

  This query specifies the search text for vector search (using the vector distance function) as `stock fraud`, the search text for keyword search (using the `CONTAINS` Oracle Text index operator) as `$ABC AND $Corporation`, and the maximum number of rows to return as the top 10. To understand how to use stem ($) and other `CONTAINS` operators, see *Oracle Text Reference*.

  For complete syntax, see [Understand Hybrid Search](#).

Use Case Examples of a Hybrid Vector Index

Here are some use case scenarios to understand how you can implement a hybrid vector index.

- Fraud detection:

  You can identify fraudulent transactions that do not follow the exact patterns of known fraud cases but are similar in behavior or context. In this case, a hybrid vector index can combine traditional rule-based filters (such as transactions above a certain threshold or in specific locations) with a vector index that compares transaction embeddings (which capture features like timing, merchant type, and transaction history) to identify semantically similar but not identical fraudulent patterns.

- Legal document analysis:

You need to search large volumes of legal documents for relevant cases and precedents. You might first filter documents using traditional legal jargon and keyword searches. Then, a vector index built from document embeddings can further filter previously identified documents that are semantically similar, even if they use different terminology or legal reasoning, ensuring comprehensive research.

You can perform the search in a different order if, for example, you are looking for cases that are contextually similar to a landmark case but need to ensure that the cases also include specific legal terminologies or citations. Here, the search can start with vector-based search to retrieve cases that are semantically similar to the landmark case using embeddings. Then, you can apply a traditional keyword-based filter to ensure that the results include specific legal terms, statutes, or citations.

- Medical document analysis:

  Medical research papers or articles either directly reference the terms of interest or do not mention those keywords at all because these documents are written by researchers or scientists in a distinct field of specialization. Here, you can switch between keyword-only and vector-only searches to quickly locate documents that mention specific medical terms of interest and specialization areas, respectively. You can also combine semantic search with keyword search in a different order, starting with a vector-based semantic search using embeddings to first identify papers that are contextually similar to your research area. Then, apply a keyword-based filter to further refine the results to include only specific references or medical terms that are critical to your research.

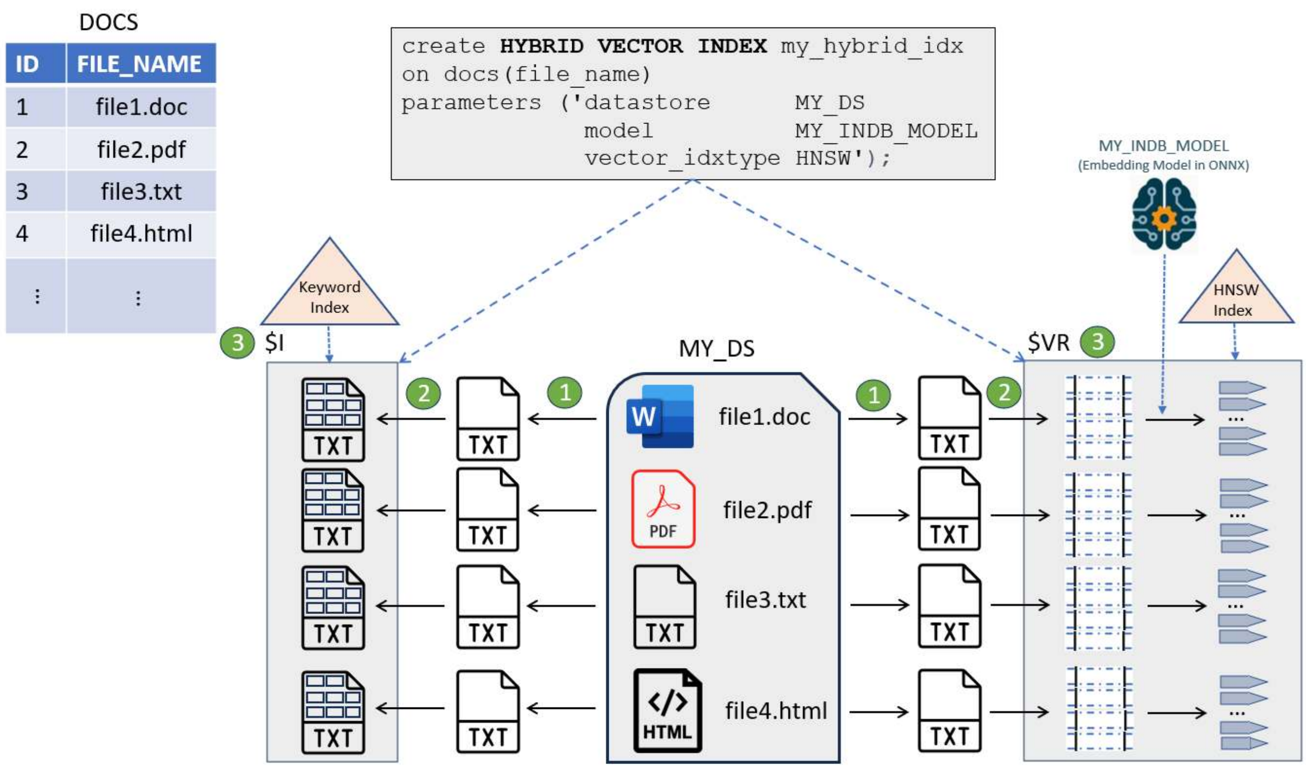
- HR recruitment:

  You want to recruit new employees with strong technical skills in programming languages but who also display certain personality and interpersonal traits. Here, you can first apply a keyword-based filter to best match the technical skills (such as "Java" and "Database"). Then, you can perform a semantic search to identify the personality and interpersonal traits (such as "team work" and "leadership experience"). A hybrid search approach can help recruiters to effectively shortlist candidates by scanning and comparing resumes or portfolios with a dual focus.

Hybrid Vector Index Creation Overview

You create a hybrid vector index by simply specifying on which table and column to create it along with some details, such as the local or remote location where all source documents are stored (datastore), the ONNX in-database embedding model to use for generating embeddings, and the type of vector index to create. You can specify additional parameters that are discussed later in this chapter.

As illustrated in the following diagram, the hybrid vector index DDL creates a single index that contains both textual fields (with derived text tokens) and vector fields (with extracted chunks and corresponding embeddings) for each indexed document.



As you can see, the implementation of a hybrid vector index leverages the existing capabilities of the Oracle Text search index and the Oracle AI Vector Search vector index. You can define PL/SQL preferences to customize all these indexing pipeline stages for both the index types.

A document table `DOCS` contains IDs and corresponding document names or file names stored in a location called `MY_DS` datastore.

The indexing pipeline starts with reading the documents from `MY_DS` (datastore), and then passes the documents through a series of processing stages:

1. Filter (conversion of binary documents such as PDF, Word, or Excel to plain text)
2. Tokenizer (tokenization of data for keyword search) and Vectorizer (chunking and embedding generation for vector search)
3. Indexing Engine (creation of secondary tables)

   As the system passes documents through this indexing engine, it populates and indexes a set of secondary tables that are collectively part of a hybrid vector index.

   The two main secondary tables created are:

   - `$I` is the same structure as the existing Oracle Text index, which contains the inverted indexed data with tokenized terms.

     An Oracle Text index is created over a textual column based on your specifications. For a deeper understanding of the Oracle Text indexing process, see *Oracle Text Application Developer's Guide*.

   - `$VR` contains the indexed data with generated chunks and corresponding embeddings.

     A vector index is also created over the `VECTOR` column based on your specifications. `$VR` also contains a `ROWID` column that maps back to the document table's row IDs, and a DOCID column that links back to Oracle Text document-level IDs. This creates a link between chunks, tokens, and documents.

You can directly examine the $VR table using the dictionary view <*index name*>$VECTORS, which lets you query all row ids, chunks, and embeddings. See <index name>$VECTORS.

Temporarily, the $D secondary table (not shown in the diagram) is also created to save a copy of the document if the original document is not already in the database or needs filtering. This table gets truncated after chunks are obtained.

Hybrid Vector Index Maintenance Operations

A hybrid vector index supports mostly all the traditional operations of an Oracle Text index, such as MAINTENANCE AUTO , SYNC, or OPTIMIZE. For details, see Guidelines and Restrictions for Hybrid Vector Indexes.

Related Topics

- Creating and Querying Oracle Text Indexes
- Perform Hybrid Search
  Hybrid search is an advanced information retrieval technique that lets you search documents by *keywords* and *vectors*, to achieve more relevant search results.

Guidelines and Restrictions for Hybrid Vector Indexes

Review these guidelines and the current set of restrictions when working with hybrid vector indexes.

Index Creation Guidelines

A hybrid vector index is a single domain index that includes both Oracle Text search index and Oracle AI Vector Search vector index. Any failure during the construction of one index type can cause the entire index creation to fail as a whole. Therefore, you must adhere to all the indexing guidelines that are individually applicable to a vector index and an Oracle Text search index.

For example, if there is insufficient vector memory pool for the HNSW vector index or if there is inadequate tablespace, then the vector index-related error can lead to a failure in creating a hybrid vector index.

- If a destructive operation occurs, such as running Ctrl+C or shutting down the database, then you must check the status of the index. If the index status does not display as valid, it indicates that the index may be corrupted or inconsistent, making it unsafe for use. In such cases, you must recreate the index.
- If there are any errors in the hybrid vector index DDL or if there is any DDL failure, then you must manually drop the index and create it again. Before using the index, ensure that it is safe to use by checking the index status.

To check the status of your index, you can query the regular Oracle Text data dictionary views, such as CTX_INDEXES and CTX_USER_INDEXES. See *Oracle Text Reference*.

Hybrid Vector Index Maintenance Operations

You can maintain your hybrid vector indexes just like an Oracle Text index. A hybrid vector index supports mostly all the traditional operations of Oracle Text indexes, such as Synchronization, Optimization, and Automatic Maintenance.

Here are some guidelines on maintaining indexes after DML operations on base tables:

- **Default index maintenance**:

  A hybrid vector index runs in an automatic maintenance mode (MAINTENANCE AUTO), which means that your DMLs are automatically synchronized into the index in the background at optimal intervals.

  You do not need to manually configure a maintenance type or synchronization type for maintaining a hybrid vector index. However, if required, you can modify the default settings to set any synchronization interval, specify a SYNC type (such as MANUAL, EVERY, or ON COMMIT), or schedule any background synchronization job using the DBMS_SCHEDULER.

  In an automatic maintenance mode, indexes are asynchronously maintained without any user intervention. Oracle recommends that you periodically examine regular Oracle Text views to know the status of all background maintenance events.

  For information on these views, see *Oracle Text Application Developer's Guide*.

- **Default index optimization**:

  A hybrid vector index runs with an automatic background Optimize Full job every midnight "local" time. This job optimizes your index to defragment it and clean up any lazy deletes from secondary tables such as the $I, $D, and $VR.

- **Optimizing the $VR Table**

  For a hybrid vector index, you can also optimize only the $VR table. This optimization is run in “section mode” for the new “section” (semantic index). This will compact the $VR table only by removing any deleted ROWIDs.

  ```
  CTX_DDL.OPTIMIZE_INDEX('index_name,'token_type',
  section_type=>ctx_ddl.section_semantic_index,parallel_degree=>3);
  ```

  Oracle recommends you to explicitly run index optimization more frequently, especially if your hybrid vector index involves a large number of inserts, updates, or deletes to base tables. This is because frequent index synchronization can cause fragmentation of your index, which can adversely affect query response time. You can reduce fragmentation and index size by optimizing the index with CTX_DDL.OPTIMIZE_INDEX.

  For information on how to configure optimization methods, see the index maintenance preferences in CREATE HYBRID VECTOR INDEX.

Guidelines to Run Direct SQL Queries on $VR

If you query secondary tables such as $VR directly before an index optimization job is run, you may still see old data or deleted rows because these are ignored at query time by Oracle provided search API (DBMS_HYBRID_VECTOR.SEARCH). For example, you may see lazy deletes that mark the row IDs as deleted but are only removed from secondary tables after the index is optimized.

Instead of examining the $VR table directly, you can use the data dictionary view <*index name*>$VECTORS to work with the contents of the $VR table, and accordingly run direct SQL queries on the view. This view lets you query the document table's row IDs by excluding all lazy deletes (along with corresponding chunks and embeddings that are generated).

Note that the <*index name*>$VECTORS view resides in a user's schema. Therefore, if a hybrid vector index is named IDX, then the view name is IDX$VECTORS.

For information on this view, see <index name>$VECTORS.

Guidelines for VPD Policy Protection

If you have applied an Oracle Virtual Private Database (VPD) policy to the base table, then you must:

- Protect the entire hybrid vector index with the same VPD policy, including the `$I`, `$VR`, and `$D` secondary tables.
- Protect the `<index name>$VECTORS` view (which is created on the `$VR` table) with the same VPD policy.
- Ensure that any SQL queries run on the hybrid vector index and any direct SQL queries run on secondary tables that are created within the schema also respect that VPD policy.

For information on how to use VPD to control data access in general, see *Oracle AI Database Security Guide*.

Hybrid Vector Index DDL Restrictions

- The supported data types are `CLOB`, `VARCHAR2`, or `BLOB`. Textual columns with the `IS JSON` check constraint are not supported.
- You currently cannot import externally created chunks or vectors directly into the `$VR` table of a hybrid vector index.
- Currently, only in-database ONNX embedding models are supported (not third-party embedding models) to generate vector embeddings for a hybrid vector index.
- The `HAMMING` and `JACCARD` distance metrics are currently not supported with hybrid vector indexes.
- The DML tracking support aligns with the behavior of the IVF and HNSW vector indexes for the vector index part of a hybrid vector index.
- Creating materialized views based on the outputs produced by the `VECTOR_CHUNKS`, `UTL_TO_CHUNKS`, `UTL_TO_TEXT`, and `UTL_TO_SUMMARY` functions is currently not supported.

> **Note**
>
> With the current set of restrictions, Oracle recommends that you create a hybrid vector index in the following scenarios:
>
> - If you want to search through textual documents (with `CLOB`, `VARCHAR2`, `JSON`, or `BLOB` data types) in a hybrid search mode.
> - If you want to alternatively switch between keyword search and semantic search when querying textual data.
>
> If you have the data that will only be searched semantically or if you want to manually create your own third-party vector embeddings (using the `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN` third-party REST APIs), then use a vector index instead of creating a hybrid vector index.

Related Topics

- Understand Hybrid Vector Indexes
  A hybrid vector index inherits all the information retrieval capabilities of Oracle Text search indexes and leverages the semantic search capabilities of Oracle AI Vector Search vector indexes.

CREATE HYBRID VECTOR INDEX

Use the `CREATE HYBRID VECTOR INDEX` SQL statement to create a hybrid vector index, which allows you to index and query documents using a combination of full-text search and vector similarity search.

Purpose

To create a class of specialized Domain Index called a hybrid vector index.

A **hybrid vector index** is an Oracle Text `SEARCH INDEX` type that combines the existing Oracle Text indexing data structures and vector indexing data structures into one unified structure. It is a single domain index that stores both text fields and vector fields for a document. Both text search and similarity search are performed on tokenized terms and vectors respectively. The two search results are combined and scored to return a unified result set.

The purpose of a hybrid vector index is to enhance search relevance of an Oracle Text index by allowing users to search by both *vectors* and *keywords*. By integrating traditional keyword-based text search with vector-based similarity search, you can improve the overall search experience and provide users with more accurate information.

Usage Notes

To create a hybrid vector index, you can provide minimal information such as:

- table or column on which you want to create the index
- in-database ONNX embedding model, or the REST endpoint of your preferred vector embedding model service provider, for generating embeddings. See Supported Third-Party Provider Operations and Endpoints.

For cases where multiple columns or tables need to be indexed together, you can specify the `MULTI_COLUMN_DATASTORE` or `USER_DATASTORE` preference.

All other indexing parameters are predefined to facilitate the indexing of documents without requiring you to be an expert in any text processing, chunking, or embedding strategies. If required, you can modify the predefined parameters using:

- Vector search preferences for the *vector index* part of the index
- Text search preferences for the *text index* part of the index
- Index maintenance preferences for DML operations on the combined index

During HVI (Hybrid View Index) creation, any rows that fail to be indexed are logged in the CTX_USER_INDEX_ERRORS table. You can query this table to review details about any rows that could not be indexed, including error messages and relevant row information. This is helpful for easier identification of issues encountered during the index creation.

When your database is configured with `MAX_STRING_SIZE=EXTENDED`, you can leverage this database setting to index significantly larger text chunks (up to 32,767 bytes) for hybrid vector indexing using the DBMS_VECTOR_CHAIN.CREATE_PREFERENCE PL/SQL function.

When performing IMPDP (Oracle Data Pump Import) of a schema containing a Hybrid Vector Index (IVF) ensure that index preferences (such as datastore, lexer, wordlist, storage, and section group) exist in the target schema prior to import. Always manually create all text index preferences (datastore, lexer, wordlist, storage, section group) in the target schema before running Data Pump Import or index creation DDL for hybrid vector indexes. Failing to do so will cause errors about missing preferences. Review the export SQL file, prepare corresponding `ctx_ddl.create_preference` statements, and execute them before import.

For detailed information on the creation process of a hybrid vector index or in general about what hybrid vector indexes are, see Understand Hybrid Vector Indexes.

> **Note**
>
> There are some key points to note when creating and using hybrid vector indexes. See Guidelines and Restrictions for Hybrid Vector Indexes.

Syntax

```
CREATE HYBRID VECTOR INDEX [schema.]index_name ON
  [schema.]table_name(column_name)
  [FILTER BY filter_column[, filter_column]...]
  [ORDER BY oby_column[desc|asc][, oby_column[desc|asc]]...]
  PARAMETERS ('paramstring')
  [LOCAL [PARTITION [partition] ][, PARTITION [partition] ]]
  [PARALLEL n];
```

Here is an example DDL specified with only the minimum required parameters.

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
  PARAMETERS('MODEL my_embed_model');
```

Setting an explicit memory and parallel degree is highly recommended as the default memory is low and the index will take longer to create.

Here is an example DDL which specifies the memory and the parallel degree.

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
  PARAMETERS('MODEL my_embed_model MEMORY 1G') PARALLEL 4;
```

> **Note**
>
> The total PGA memory used will be the value of the `MEMORY` parameter multiplied by the `PARALLEL` degree. In this example, the total PGA memory used would be 1G * 4 = 4GB. This means that up to 4GB of PGA memory can be used by the hybrid vector index.

More comprehensive examples are given at the end of this section.

Let us explore all the required and optional indexing parameters:

[*schema*.]*index_name*
Specify the name of the hybrid vector index to create.

[*schema*.]*table_name*(*column_name*)
Specify the name of the table and column on which you want to create the hybrid vector index. You can create a hybrid vector index on one or more text columns with `VARCHAR2`, `CLOB`, and `BLOB` data types.

> **Note**
>
> You cannot create hybrid vector indexes on a text column that uses the `IS JSON` check constraint.

Because the system can index most document formats, including HTML, PDF, Microsoft Word, and plain text, you can load a supported type into the text column. For a complete list, see Supported Document Formats.
For cases where multiple columns or tables need to be indexed together, specify a datastore preference (described later in Text search preferences).

PARAMETERS (paramstring)
Specify preferences in `paramstring`:

- **Vector Search Preferences**:

  Configures the "vector index" part of a hybrid vector index, pertaining to processing input for vector search.

  > **Note**
  >
  > You can either pass a minimal set of parameters (the required `MODEL` and the optional `VECTOR_IDXTYPE` parameters) directly in the `PARAMETERS` clause or use a vectorizer preference to specify a complete set of vector search parameters. You cannot use both (directly set parameters along with vectorizer) in the `PARAMETERS` clause.

  – With `MODEL` and `VECTOR_IDXTYPE` directly specified:

```
CREATE HYBRID VECTOR INDEX [schema.]index_name ON
  [schema.]table_name(column_name)
   PARAMETERS ('MODEL <model_name>
               [VECTOR_IDXTYPE <vector_index_type>]')
  [FILTER BY filter_column[, filter_column]...]
  [ORDER BY oby_column[desc|asc][, oby_column[desc|asc]]...]
  [PARALLEL n];
```

  Here, `MODEL` specifies the vector embedding model that you import into the database for generating vector embeddings on your input data.

> **Note**
>
> Currently, only ONNX in-database embedding models are supported.

`VECTOR_IDXTYPE` specifies the type of vector index to create, such as `IVF` (default) for the Inverted File Flat (IVF) vector index and `HNSW` for the Hierarchical Navigable Small World (HNSW) vector index. If you omit this parameter, then the IVF vector index is created by default.

Creating a `LOCAL` index on an Hybrid Vector Index is supported when the underlying `index_type` is `IVF`. An example is shown below:

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
doc_table(text_column)
parameters('MODEL my_doc_model
            VECTOR_IDXTYPE IVF')
LOCAL PARALLEL;
```

> **Caution**
>
> Creating a `LOCAL` index on Hybrid Vector Index when the underlying `index_type` is `HNSW`, would throw an error before starting any document processing (early failure).

– With the vectorizer preference:

  A vectorizer preference is a JSON object that collectively holds all indexing parameters related to chunking (`UTL_TO_CHUNKS` or `VECTOR_CHUNKS`), embedding (`UTL_TO_EMBEDDING`, `UTL_TO_EMBEDDINGS`, or `VECTOR_EMBEDDING`), and vector index (`distance`, `accuracy`, or `vector_idxtype`).

  You use the `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` PL/SQL function to create a vectorizer preference. To create a vectorizer preference, see DBMS_VECTOR_CHAIN.CREATE_PREFERENCE.

  After creating a vectorizer preference, you can use the `VECTORIZER` parameter to pass the preference name here. For example:

```
begin
  DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
    'my_vectorizer_pref',
     dbms_vector_chain.vectorizer,
    json('{
            "vector_idxtype":  "hnsw",
            "model"         :  "my_doc_model",
            "by"            :  "words",
            "max"           :  100,
            "overlap"       :  10,
            "split"         :  "recursively"
          }'
```

```
        ));
end;
/

CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
    parameters('VECTORIZER my_vectorizer_pref');
```

A vectorizer preference with externally hosted vector embedding models:

`DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` also lets you use vector embedding models hosted externally in third-party services like Open AI and Google, or other external services, rather than importing them into the database.

After creating a vectorizer preference with externally hosted vector embedding model , you can use the `VECTORIZER` parameter to pass the preference name here. For example:

privateai example:

```
begin
  dbms_vector_chain.create_preference('hvi_pref_restapi_e',
      DBMS_VECTOR_CHAIN.VECTORIZER,
      json('{
              "embedder_spec":
               {
                 "provider": "privateai",
                 "url": http://....com:port/omlmodels/all_mini_l12/
score,
                 "host": "local",
                 "model": "all_minilm_l12"
               }
            }'));
end;
/

CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
    parameters('VECTORIZER hvi_pref_restapi_e');
```

ocigenai example:

```
begin
  jo := json_object_t();
  jo.put('user_ocid','ocid1.user.oc1… gca');
  jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaa…….4ma');
  jo.put('compartment_ocid','ocid1.compartment.oc1..aaaaaa………cxea');
  jo.put('private_key', ' ..private key here …’ );
  jo.put('fingerprint','09:…….:4f');
  dbms_vector_chain.create_credential(
    credential_name => 'OCI_CRED',
```

```
    params => json(jo.to_string));
end;
/
begin
    dbms_vector_chain.create_preference('docs_vec',
      dbms_vector_chain.vectorizer,
      json('{
        "chunker_spec": {
          "mode": "textual",
          "by": "words",
          "max": 100
        },
        "embedder_spec":{
          "provider": "ocigenai",
          "credential_name": "OCI_CRED",
          "url": https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/embedText,
          "model": "cohere.embed-english-v3.0",
          "proxy": "www-proxy-…..com:80"
        }
      }'));
  end;
end;
/

CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
    parameters('VECTORIZER docs_vec');
```

See Embedding Parameters from CREATE_PREFERENCE for detailed description of each parameter.

> **Note**
>
> `"host": "local"` is valid only if the embedding service is running locally and accessible from the database server. If it's not running locally, then do not provide this parameter. If you set `"host": "local"` for the embedding service that is not running locally, the index creation fails to index vectors.

A vectorizer preference to enable extended string data type capability for a Hybrid Vector Index:

When your database is configured with `MAX_STRING_SIZE=EXTENDED`, you can leverage this database setting to index significantly larger text chunks (up to 32,767 bytes) for hybrid vector indexing. This enables you to store and index text segments up to 32,767 bytes or 8,000 words. `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` lets you to leverage the extended data type for a hybrid vector index using `extended": 1` attribute:

```
begin
  DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
    'my_vectorizer_pref',
     dbms_vector_chain.vectorizer,
    json('{"by": "WORDS", "max": 2000, "extended": 1}')
);
```

> **Note**
>
> If you do not specify `"extended": 1` or your database is not configured with `MAX_STRING_SIZE=EXTENDED`, the previous limits (1,000 words or 4,000 bytes) will apply, and inserting larger chunks will result in errors.

- **Text Search Preferences**:

  Configures the "Oracle Text index" part of a hybrid vector index, pertaining to processing input for keyword search.

  These parameters define the text processing and tokenization stages of a hybrid vector indexing pipeline. All these are the same set of parameters that you provide when working with Oracle Text indexes alone.

  ```
  [DATASTORE datastore_pref]
  [STORAGE storage_pref]
  [MEMORY memsize]
  [STOPLIST stoplist]
  [LEXER lexer_pref]
  [FILTER filter_pref]
  [WORDLIST wordlist_pref]
  [SECTION GROUP section_group]
  ```

  DATASTORE datastore_pref
  Specify the name of your datastore preference. Use the datastore preference to specify the local or remote location where your source files are stored.

  If you want to index multiple columns or tables together, see MULTI_COLUMN_DATASTORE and USER_DATASTORE.

  For a complete list of all datastore preferences, see Datastore Types.

  Default: `DIRECT_DATASTORE`

  STORAGE storage_pref
  Specify the name of your storage preference for an Oracle Text search index. Use the storage preference to specify how the index tables are stored. See Storage Types.

MEMORY memsize
Specify the amount of run-time memory to use for indexing.

```
memsize = number[K|M|G]
```

K is for kilobytes, M is for megabytes, and G is for gigabytes.

The value you specify for `memsize` must be between 1M and the value of `MAX_INDEX_MEMORY` in the `CTX_PARAMETERS` view. To specify a memory size larger than the `MAX_INDEX_MEMORY`, you must reset this parameter with `CTX_ADM.SET_PARAMETER` to be larger than or equal to `memsize`. See CTX_ADM.SET_PARAMETER.

The default for Oracle Text search index is 500 MB.

The `memsize` parameter specifies the amount of memory Oracle Text uses for indexing before flushing the index to disk. Specifying a large amount memory improves indexing performance because there are fewer I/O operations and improves query performance and maintenance, because there is less fragmentation.

Specifying smaller amounts of memory increases disk I/O and index fragmentation, but might be useful when run-time memory is scarce.

STOPLIST stoplist
Specify the name of your stoplist. Use stoplist to identify words that are not to be indexed. See CTX_DDL.CREATE_STOPLIST.

Default: `CTXSYS.DEFAULT_STOPLIST`

LEXER lexer_pref
Specify the name of your lexer or multilexer preference. Use the lexer preference to identify the language of your text and how text is tokenized for indexing. See Lexer Types.

Default: `CTXSYS.DEFAULT_LEXER`

FILTER filter_pref
Specify the name of your filter preference. Use the filter preference to specify how to filter formatted documents to plain text. See Filter Types.

The default for binary text columns is `NULL_FILTER`. The default for other text columns is `AUTO_FILTER`.

WORDLIST wordlist_pref
Specify the name of your wordlist preference. Use the wordlist preference to enable features such as fuzzy, stemming, and prefix indexing for better wildcard searching. See Wordlist Type.

SECTION GROUP section_group
Specify the name of your section group. Use section groups to create sections in structured documents. See CTX_DDL.CREATE_SECTION_GROUP.

Default: `NULL_SECTION_GROUP`

- **Index Maintenance Preferences**:

  Configures the DML operations on the entire hybrid vector index, that is, how to synchronize and optimize the index.

Because a hybrid vector index is basically an Oracle Text search index type, so all maintenance-specific capabilities of an Oracle Text index are applicable.

```
[MAINTENANCE AUTO | MAINTENANCE MANUAL]
[SYNC (MANUAL | EVERY "interval-string" | ON COMMIT)]
[OPTIMIZE (MANUAL | AUTO_DAILY | EVERY "interval-string")]
```

MAINTENANCE AUTO | MAINTENANCE MANUAL

Specify the maintenance type for synchronization of a hybrid vector index when there are inserts, updates, or deletes to the base table. The maintenance type specified for an index applies to all index partitions.

You can specify one of the following maintenance types:

Maintenance Type	Description
`MAINTENANCE AUTO`	This method sets your index to automatic maintenance, that is, the index is automatically synchronized in the background at optimal intervals. You do not need to manually configure a `SYNC` type or set any synchronization interval. The background mechanism automatically determines the synchronization interval and schedules background `SYNC.INDEX` operations by tracking the DML queue.
`MAINTENANCE MANUAL`	This method sets your index to manual maintenance. This is a non-automatic maintenance (synchronization) mode in which you can specify `SYNC` types, such as `MANUAL`, `EVERY`, or `ON COMMIT`.

SYNC (MANUAL | EVERY "interval-string" | ON COMMIT)

Specify the `SYNC` type for synchronization of a hybrid vector index when there are inserts, updates, or deletes to the base table. These `SYNC` settings are applicable only to the indexes that are set to manual maintenance.

> **Note**
>
> By default, a hybrid vector index runs in an automatic maintenance mode (`MAINTENANCE AUTO`), which means that your DMLs are automatically synchronized into the index in the background at optimal intervals. Therefore, you do not need to manually configure a `SYNC` type for maintaining a hybrid vector index. However, if required, you can do so if you want to modify the default settings for an index.

You can specify one of the `SYNC` methods:

SYNC Type	Description
`MANUAL`	With this method, automatic synchronization is not provided. You must manually synchronize the index using `CTX_DDL.SYNC_INDEX`.

SYNC Type	Description
`EVERY` *interval-string*	Automatically synchronize the index at a regular interval specified by the value of *interval-string*, which takes the same syntax as that for scheduler jobs. Automatic synchronization using `EVERY` requires that the index creator have `CREATE JOB` privileges. Ensure that *interval-string* is set to a considerable time period so that any previous synchronization jobs will have completed. Otherwise, the synchronization job may stop responding. The *interval-string* argument must be enclosed in double quotation marks (" ").
`ON COMMIT`	Synchronize the index immediately after a commit. The commit does not return until the sync is complete. The operation uses the memory specified with the *memory* parameter. This sync type works best when the `STAGE_ITAB` index option is enabled, otherwise it causes significant fragmentation of the main index, requiring frequent `OPTIMIZE` calls.

With automatic (`EVERY`) synchronization, you can specify memory size and parallel synchronization. You can define repeating schedules in the *interval-string* argument using calendaring syntax values. These values are described in *Oracle AI Database PL/SQL Packages and Types Reference*.

Syntax:

```
SYNC [EVERY "interval-string"] MEMORY mem_size PARALLEL paradegree
```

For example, to sync the index at an interval of 20 seconds:

```
SYNC [EVERY "freq=secondly;interval=20"] MEMORY 500M PARALLEL 2
```

OPTIMIZE (MANUAL | AUTO_DAILY | EVERY "interval-string)

Specify `OPTIMIZE` to enable automatic background index optimization of a hybrid vector index. You can specify any one of the following `OPTIMIZE` methods:

OPTIMIZE Type	Description
`MANUAL`	Provides no automatic optimization. You must manually optimize the index with `CTX_DDL.OPTIMIZE_INDEX`.
`AUTO_DAILY`	This is the default setting. With `OPTIMIZE (AUTO_DAILY)`, the optimize `FULL` job is scheduled to run midnight from 12 A.M. local time everyday.

OPTIMIZE Type	Description
EVERY "*interval-string*"	Automatically runs optimize token at a regular interval specified by the value *interval-string*, which takes the same syntax as the scheduler jobs. Ensure that *interval-string* is set to a considerable time period so that the previous optimize jobs are complete; otherwise, the optimize job might stop responding. *interval-string* must be enclosed in double quotes, and any single quote within *interval-string* must be preceded by the escape character with another single quote.

With AUTO_DAILY | EVERY "*interval-string*" setting, you can specify parallel optimization.

Syntax:

```
OPTIMIZE [AUTO_DAILY | EVERY "interval-string"] PARALLEL paradegree ...
```

For example, to optimize the index at an interval of 20 minutes:

```
OPTIMIZE [EVERY "freq=minutely;interval=20"] PARALLEL 2
```

FILTER BY filter_column
Specify the structured indexed column on which a range or equality predicate in the WHERE clause of a mixed query will operate. You can specify one or more structured columns for filter_column, on which the relational predicates are expected to be specified along with the CONTAINS() predicate in a query.

- You can use these relational operators:

  <, <=, =, >=, >, between, and LIKE (for VARCHAR2)

- These columns can only be of CHAR, NUMBER, DATE, VARCHAR2, or RAW type. Additionally, CHAR, VARCHAR2 and VARCHAR2 types are supported only if the maximum length is specified and does not exceed 249 bytes.

  If the maximum length of a CHAR or VARCHAR2 column is specified in characters, for example, VARCHAR2 (50 CHAR), then it cannot exceed FLOOR (249/max_char_width), where max_char_width is the maximum width of any character in the database character set.

  For example, the maximum specified column length cannot exceed 62 characters, if the database character set is AL32UTF8. The ADT attributes of supported types (CHAR, NUMBER, DATE, VARCHAR2, or RAW) are also allowed.

  An error is raised for all other data types. Expressions, for example, func(cola), and virtual columns are not allowed.

- txt_column is allowed in the FILTER BY column list.

- DML operations on FILTER BY columns are always transactional.

ORDER BY oby_column[desc|asc]
Specify one or more structured indexed columns by which you want to sort query results. You can specify a list of structured *oby_columns*. These columns can only be of CHAR, NUMBER, DATE, VARCHAR2, or RAW type. VARCHAR2 and RAW columns longer than 249 bytes are truncated to the first 249 bytes. Expressions, for example func(cola), and virtual columns are not allowed. The order of the specified columns matters. The ORDER BY clause in a query can contain:

- The entire ordered `ORDER BY` columns
- Only the prefix of the ordered `ORDER BY` columns
- The score followed by the prefix of the ordered `ORDER BY` columns

`DESC` sorts the results in a descending order (from highest to lowest), while `ASC` (default) sorts the results in an ascending order (from lowest to highest).

[PARALLEL n]
Parallel indexing can improve index performance when you have multiple CPUs. To create an index in parallel, use the `PARALLEL` clause with a parallel degree.
Optionally specifies the parallel degree for parallel indexing. The actual degree of parallelism might be smaller depending on your resources. You can use this parameter on nonpartitioned tables. However, creating a nonpartitioned index in parallel does not turn on parallel query processing. Parallel indexing is supported for creating a local partitioned index.
The indexing memory size specified in the parameter clause applies to each parallel worker. For example, if indexing memory size is specified in the parameter clause as 500M and parallel degree is specified as 2, then you must ensure that there is at least 1GB of memory available for indexing.

Examples

- **With vector search preferences directly specified**:

  In this example, only the required parameter `model` is specified in the `PARAMETERS` clause:

  ```
  CREATE HYBRID VECTOR INDEX my_hybrid_idx on
    doc_table(text_column)
    parameters('MODEL my_doc_model');
  ```

  In this example, both the parameters `model` and `vector_idxtype` are specified:

  ```
  CREATE HYBRID VECTOR INDEX my_hybrid_idx on
    doc_table(text_column)
    parameters('MODEL my_doc_model
                VECTOR_IDXTYPE HNSW');
  ```

- **With vector search preferences specified using VECTORIZER**:

  In this example, the `vectorizer` parameter is used in the `PARAMETERS` clause to specify the `my_vectorizer_spec` preference:

  ```
  begin
    DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
      'my_vectorizer_spec',
       dbms_vector_chain.vectorizer,
      json('{"vector_idxtype" :  "hnsw",
             "model"          :  "my_doc_model",
             "by"             :  "words",
             "max"            :  100,
             "overlap"        :  10,
             "split"          :  "recursively"}'));
  end;
  /

  CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  ```

```
  doc_table(text_column)
  parameters('VECTORIZER my_vectorizer_spec');
```

- **With text search and vector search preferences directly specified**:

  In this example, only the required Vector Search parameter `MODEL` is specified in the `PARAMETERS` clause. Text Search parameters are also specified:

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
  parameters('MODEL my_doc_model
              DATASTORE my_datastore
              STORAGE my_storage
              STOPLIST my_stoplist
              LEXER my_lexer')
  ORDER BY docid asc;
```

- **With text search and index maintenance preferences directly specified and vector search preferences specified using VECTORIZER**:

  In this example, the `VECTORIZER` parameter is used to specify the `my_vectorizer_spec` preference that holds vector search parameters. All the Text Search and Index Maintenance preferences are directly specified.

```
begin
  DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
    'my_vectorizer_spec',
     dbms_vector_chain.vectorizer,
    json('{
            "vector_idxtype" :  "hnsw",
            "model"          :  "my_doc_model",
            "by"             :  "words",
            "max"            :  100,
            "overlap"        :  10,
            "split"          :  "recursively"
         }'
        ));
end;
/

CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_table(text_column)
  parameters('VECTORIZER my_vectorizer_spec
              DATASTORE my_datastore
              STORAGE my_storage
              MEMORY 128M
              MAINTENANCE AUTO
              OPTIMIZE AUTO_DAILY
              STOPLIST my_stoplist
              LEXER my_lexer
              FILTER my_filter
              WORDLIST my_wordlist
              SECTION GROUP my_section_group')
  FILTER BY category, author
  ORDER BY score(10), score(20) desc
  PARALLEL 3;
```

Related Topics

- Perform Hybrid Search
- Query Hybrid Vector Indexes End-to-End Example

Creating Hybrid Vector Indexes using the DBMS_SEARCH API

The `DBMS_SEARCH` package provides procedures and functions for creating, managing, and querying hybrid vector indexes when multiple sources are used for the search. Sources includes tables and views as described in the DBMS_SEARCH documentation.

> **Note**
>
> Starting with Oracle AI Database 26ai, Release Update 23.10, you are required to have the following user privileges to ensure proper functionality of `DBMS_SEARCH` package:
>
> - `CREATE SEQUENCE`
> - `CREATE TRIGGER`
> - `CREATE JOB`

The CREATE_INDEX procedure can be used to create a hybrid vector index.

Syntax:

```
DBMS_SEARCH.CREATE_INDEX(
    index_name        VARCHAR2,
    tablespace        VARCHAR2 DEFAULT NULL,
    datatype          VARCHAR2 DEFAULT NULL,
    lexer             VARCHAR2 DEFAULT NULL,
    stoplist          VARCHAR2 DEFAULT NULL,
    wordlist          VARCHAR2 DEFAULT NULL,
    vectorizer        VARCHAR2 DEFAULT NULL
);
```

The syntax contains parameters that can define both text as well as vector search preferences. The parameters `lexer`, `stoplist`, and `wordlist` allow you to specify the text preferences for the hybrid vector index and the parameter `vectorizer` can be used to specify the vector preference for the hybrid search. When using the `vectorizer` parameter, you must first use the `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` PL/SQL function to create a vectorizer preference. For more information, see CREATE_PREFERENCE. After creating a vectorizer preference, you can use the vectorizer parameter to pass the preference name. You can define the `lexer`, `stoplist` and `wordlist` using CREATE_PREFERENCE and CREATE_STOPLIST functions.

Example:

The following example also displays the sample code to define lexer, stoplist, wordlist and vectorizer.

```
 exec CTX_DDL.CREATE_PREFERENCE('my_lexer','BASIC_LEXER');
 exec CTX_DDL.CREATE_STOPLIST('my_stoplist','BASIC_STOPLIST');
 exec CTX_DDL.CREATE_PREFERENCE('my_wordlist','BASIC_WORDLIST');
```

```
begin
    DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
       PREF_NAME => 'my_vectorizer_spec',
       PREF_TYPE =>  DBMS_VECTOR_CHAIN.VECTORIZER,
       PARAMS    => json(
                         '{"vector_idxtype" : "ivf",
                           "model"          : "ALL_MINILM_L12",
                           "by"             : "words",
                           "max"            : "100",
                           "overlap"        : "10",
                           "split"          : "recursively",
                           "language"       : "english"
                          }'));
 end;
 /

 begin
    DBMS_SEARCH.CREATE_INDEX(
       index_name => 'my_hybrid_idx',
       datatype   => 'JSON',
       lexer      => 'my_lexer',
       stoplist   => 'my_stoplist',
       wordlist   => 'my_wordlist',
       vectorizer => 'my_vectorizer_spec');
 end;
/
```

The `DBMS_SEARCH` package also contains a procedure, ADD_SOURCE, that allows you to add one or more data sources (tables, views, or duality views) from different schemas to the `DBMS_SEARCH` index.

Syntax:

```
DBMS_SEARCH.ADD_SOURCE (
index_name      VARCHAR2,
source_name     VARCHAR2,
memory          VARCHAR2 DEFAULT NULL,
parallel_degree NUMBER DEFAULT NULL);
```

Example:

```
exec DBMS_SEARCH.ADD_SOURCE('my_hybrid_idx','MYVIEW');
```

You can also specify more than one source. Oracle recommends that the sources are added rather serially than concurrently. Here is an example where you can add two sources to the same index (`my_hybrid_idx`) created above:

```
exec DBMS_SEARCH.ADD_SOURCE('my_hybrid_idx','MYVIEW1');
exec DBMS_SEARCH.ADD_SOURCE('my_hybrid_idx','MYVIEW2');
```

You can query the `ctx_user_index_partitions` view to check if the source is added to the index and is ready to be queried. The query

```
select ixp_index_partition_name, ixp_status from ctx_user_index_partitions
where IXP_INDEX_NAME = 'my_hybrid_idx' and IXP_PARTITION_NAME LIKE ‘DSR$’;
```

would display the status of the sources as "INDEXED" indicating that they are indexed and ready to be queried. Sample output is shown below:

```
IXP_INDEX_PARTITION_NAME                              IXP_STATUS
-------------------------------------------------------------------
DSR$1                                                 INDEXED
DSR$2                                                 INDEXED
```

When you use IVF index to create the hybrid vector index, Oracle recommends to run a `rebuild` of the index after the sources are added. Since our example uses IVF indexes, you would run the following command once all sources have been indexed:

```
alter index DR$ARTICLES_IDX$VI rebuild online;
```

The `DBMS_SEARCH` package also contains procedures that allow you to remove a data source and all its associated data from the index, remove the index and all its associated data from the database, return a virtual indexed JSON document for the specified source metadata, retrieve a hitlist, and produce an aggregation of JSON documents based on the specified filter conditions. Refer to DBMS_SEARCH Package for more details about the procedure, syntax, usage, restrictions, and so on.

ALTER INDEX

Use the `ALTER INDEX` SQL statement to modify an existing hybrid vector index.

Purpose

To make changes to hybrid vector indexes.

Syntax

```
ALTER INDEX [schema.]index_name REBUILD
  PARAMETERS(
      ['UPDATE VECTOR INDEX [VECTOR_IDXTYPE HNSW/IVF]']
      ['REPLACE vectorizer vectorizer_pref_name']
      )
  [PARALLEL n];
```

> **Note**
>
> - If you do not specify the `PARAMETERS` clause, then all parts of the hybrid vector index (both Oracle Text index and vector index) are recreated with existing preference settings.
> - Renaming hybrid vector indexes using the `ALTER INDEX RENAME` syntax is not supported.
> - The `ALTER INDEX` parameter `UDPATE VECTOR INDEX` is not supported for Local HVI and HNSW vector indexes.

[*schema*.]*index_name*
Specifies name of the hybrid vector index that you want to modify.

PARAMETERS(UPDATE VECTOR INDEX)
Rebuilds both text part and vector part of the hybrid vector index. For text part, the rebuild uses the preferences which are specified during the index creation or the default preferences if not specified. For the vector part (chunking, embedding and vector index creation), rebuild uses the new vectorizer preference specified in the `ALTER INDEX REBUILD` syntax. See CREATE HYBRID VECTOR INDEX for detailed information on the preferences set during the index creation.

PARAMETERS(REPLACE vectorizer *vectorizer_pref_name*)
Recreates only the vector index part of a hybrid vector index with the specified vectorizer preference settings.

> **Note**
>
> For non HVI index, replace operation would throw an error, as it cannot replace something that was not present.

PARALLEL
Specifies parallel indexing, as described for the `CREATE HYBRID VECTOR INDEX` statement. For detailed information on the `PARALLEL` clause, see CREATE HYBRID VECTOR INDEX.

Examples

Here are some examples on how you can modify existing hybrid vector indexes:

- **To rebuild all parts of a hybrid vector index**:

  Use the following syntax to rebuild all parts of a hybrid vector index (both Oracle Text index and vector index) with the original preference settings:

  Syntax:

  ```
  ALTER INDEX index_name REBUILD [PARALLEL n];
  ```

  Note that you do not need to specify any `PARAMETERS` clause when rebuilding both parts of a hybrid vector index.

Example:

```
ALTER INDEX my_hybrid_idx REBUILD;

SELECT (select id from doc_table where rowid = jt.doc_rowid) as doc,
       jt.chunk
FROM JSON_TABLE(
         DBMS_HYBRID_VECTOR.SEARCH(
           json(
             '{ "hybrid_index_name" : "my_hybrid_idx",
                "vector" :
                 { "search_text"    : "vector based search capabilities",
                   "search_mode"    : "CHUNK"
                 },
                "return" :
                 { "topN"           : 10 }
             }')
           ),
           '$[*]' COLUMNS doc_rowid  PATH '$.rowid',
                          chunk      PATH '$.chunk_text') jt;
```

- **To rebuild only the vector index part**:

  Use the following syntax to rebuild only the vector index part of a hybrid vector index with the original preference settings:

  Syntax:

```
ALTER INDEX index_name REBUILD
  PARAMETERS('UPDATE VECTOR INDEX') [PARALLEL n];
```

  Example:

```
ALTER INDEX my_hybrid_idx REBUILD
  PARAMETERS('UPDATE VECTOR INDEX') PARALLEL 3;

SELECT (select id from doc_table where rowid = jt.doc_rowid) as doc,
       jt.chunk
FROM JSON_TABLE(
         DBMS_HYBRID_VECTOR.SEARCH(
           json(
             '{ "hybrid_index_name" : "my_hybrid_idx",
                "vector" :
                 { "search_text"    : "vector based search capabilities",
                   "search_mode"    : "CHUNK"
                 },
                "return" :
                 { "topN"           : 10 }
             }')
           ),
           '$[*]' COLUMNS doc_rowid  PATH '$.rowid',
                          chunk      PATH '$.chunk_text') jt;
```

- **To recreate indexes with a vectorizer preference**:

  You can create a vectorizer preference using the `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` PL/SQL function. For detailed information on how to create a vectorizer preference, see

CREATE_PREFERENCE. After creating the preference, use the `REPLACE vectorizer` parameter to pass the preference name here.

Syntax:

```
ALTER INDEX index_name REBUILD
  parameters('REPLACE vectorizer vectorizer_pref_name') [PARALLEL n];
```

> **Note**
>
> For non-HVI index, the `REPLACE` operation would throw an error.

Example:

```
ALTER INDEX my_hybrid_idx REBUILD
  parameters('REPLACE vectorizer my_vectorizer_pref') [PARALLEL n];
```

- **To replace only the model and/or vector index type**

  For an existing HVI index, you can replace the model and/or the index type without specifying the full vectorizer preference using the following syntax.

  Syntax:

  ```
  ALTER INDEX schema.index_name REBUILD[
  parameters('REPLACE MODEL model_name VECTOR_IDXTYPE hnsw/ivf')];
  ```

  Example:

  ```
  ALTER INDEX schema.my_hybrid_idx REBUILD[
  parameters('REPLACE MODEL my_model_name VECTOR_IDXTYPE ivf')];
  ```

  > **Note**
  >
  > For non HVI indexes, this syntax would throw an error. If a vectorizer is also specified alongside the model and/or vector_idxtype, it would lead to an error, as only one of either vectorizer or model/vector_idxtype is allowed.

Related Topics

- *Oracle Text Reference*

Vector Indexes in a Globally Distributed Database

Inverted File Flat (IVF) index and Hierarchical Navigable Small World (HNSW) index are supported on sharded tables in a distributed database; however there are some considerations.

> **Note**
>
> - Global indexes are not supported on sharded tables; however, this limitation does not exist for the global HNSW and IVF index.
> - Hybrid Vector Indexes (HVI) are not currently supported on sharded tables.
> - GDSCTL commands `MOVE CHUNK`, `ADD CDB`, and `ADD SHARD`, and incremental deployment, will raise ORA-05118 if there are global vector indexes on sharded tables. Drop the global vector indexes before performing these operations.

Inverted File Flat Index

Inverted File Flat Index (IVF Flat or simply IVF) is a partitioned-based index that lets you balance high-search quality with reasonable speed.

You can create a local IVF index on vector columns in a sharded table. There is no syntax change required.

- IVF indexes and HNSW indexes on a sharded table must be created on the shard catalog database with `SHARD DDL` enabled.
- The `CREATE INDEX` command is propagated as is to all of the shards by the shard coordinator. The `CREATE INDEX` clause scope is the shard.

There is no syntax change to create an IVF index on a sharded table, when compared to the syntax to create an IVF index on a non-sharded table.

```
CREATE VECTOR INDEX ivf_image
     ON houses (image)
 ORGANIZATION NEIGHBOR PARTITIONS WITH TARGET ACCURACY 95
 DISTANCE EUCLIDEAN PARAMETERS
 (type IVF, NEIGHBOR PARTITIONS 1000) PARALLEL 16;
```

Hierarchical Navigable Small World Index

There is no syntax change to create a Hierarchical Navigable Small World (HNSW) index on a sharded table, when compared to the syntax to create an HNSW index on a non-sharded table.

```
CREATE VECTOR INDEX hnsw_image
     ON houses (image)
 ORGANIZATION INMEMORY NEIGHBOR GRAPH
 WITH TARGET ACCURACY 95;
```

IVF Indexing on External Iceberg Tables

This feature enables vector search capabilities on Iceberg external tables stored in cloud object storage systems such as Oracle Cloud Infrastructure Object Storage or Amazon S3.

Iceberg tables are often extremely large and consist of many data files. A single global index can take too long to refresh even when only a small number of data files change because the refresh may still require scanning and rebuilding large portions of the overall index.

To address this, you can create an IVF vector index within the database for an Iceberg table stored outside the Oracle AI Database. This design uses a local, per-file indexing approach, enabling targeted, incremental refresh aligned with Iceberg's snapshot-based change patterns. As a result, it delivers fast index refresh, more predictable performance, and more efficient index maintenance at scale.

External table support for Iceberg has the following requirements:

- The Iceberg table must be non-partitioned
- The Iceberg table must use the Copy-on-Write table property

This section introduces how to set up, create, and validate an IVF vector index on external Iceberg tables. It covers system prerequisites, required configurations, index creation, post-creation validation, and index maintenance operations.

- File-Based Local IVF Index Architecture
  Unlike the original partition-local IVF index, where the partition key is derived from the base table fragment object ID, the Iceberg external table implementation uses the underlying data file.
- Prerequisites
  Before creating an IVF vector index on an Iceberg external table, configure the following prerequisites:
- Create External Iceberg Table for IVF Indexing
  There are two main approaches to create external tables for use with IVF indexing on Iceberg tables. Each approach determines how your table and vector index interact with data changes on the source Iceberg table.
- Create IVF Index for an External Iceberg Table
  The IVF index support for external tables over Iceberg enables scalable and efficient vector search on large datasets stored in external object stores.
- Vector Search with IVF Index on Iceberg External Tables
  After you create an IVF vector index on an Iceberg external table, you can run approximate k-Approximate Nearest Neighbor (k-ANN) queries that use the index to accelerate semantic or similarity search.
- IVF Index Maintenance
  Index maintenance ensures that the IVF index stays synchronized with changes to the underlying Iceberg table so that queries can utilize the most current data.

File-Based Local IVF Index Architecture

Unlike the original partition-local IVF index, where the partition key is derived from the base table fragment object ID, the Iceberg external table implementation uses the underlying data file.

In this design, the partition key is the data file ID associated with each Iceberg/Parquet file. Oracle assigns and tracks this identifier in the `XT_FILE` auxiliary table, which stores file metadata such as file path, name, and last-modified timestamp. Although the external base table itself is not physically partitioned, it can be treated as virtually partitioned by data file ID for indexing purposes. This enables the system to build and maintain file-local IVF index structures, improving refresh and maintenance efficiency because changes typically affect only a subset of files.

FILE_ID	FILE_PATH	FILE_NAME	FILE_LASTMOD	PART_ID
1	path/to/file	file1.parquet	24-MAR-25 10.23.52.000 PM	1
2	path/to/file	file2.parquet	24-MAR-25 10.23.54.000 PM	2
3	path/to/file	file3.parquet	24-MAR-25 10.23.54.000 PM	3

Prerequisites

Before creating an IVF vector index on an Iceberg external table, configure the following prerequisites:

Enable hidden columns

Vector indexing relies on hidden columns to relate each row to its underlying storage file. These must be enabled before index creation:

```
-- Session level
ALTER SESSION SET "_xt_table_hidden_column" = TRUE;      -- Enable FILE$PATH,
FILE$NAME

-- System level
ALTER SYSTEM SET "_xt_table_hidden_column" = TRUE;      -- Enable FILE$PATH,
FILE$NAME

-- Flush plan cache
ALTER SYSTEM FLUSH SHARED_POOL;                            -- Recommended
after changes
```

Create External Iceberg Table for IVF Indexing

There are two main approaches to create external tables for use with IVF indexing on Iceberg tables. Each approach determines how your table and vector index interact with data changes on the source Iceberg table.

Metadata-Based External Table Creation (Snapshot-Reference)

To enable IVF indexing, you must first create an external table that references a specific Iceberg snapshot (typically through a `metadata.json` URI). This table is bound to the snapshot described by the chosen `metadata.json` file.

Because external tables lack reliable `ROWID` semantics, you must define a single column primary key, which is used for precise row identification during vector search. The primary key should be declared with the `PRIMARY KEY RELY DISABLE` clause and the corresponding column in the underlying Iceberg table must be **UNIQUE** and **NOT NULL**.

```
DEFINE metadata_uri = 'https://objectstorage.<region>.oraclecloud.com/n/
<tenancy>/b/<bucket>/o/<pathtometadata>/metadata.json';

CREATE TABLE IF NOT EXISTS vdb_ext_ivf01
(
  id   VARCHAR(10)   PRIMARY KEY RELY DISABLE,
```

```
  c1   NUMBER,
  n1   VARCHAR2(10),
  emb  VECTOR(1024, float32)
)
ORGANIZATION EXTERNAL
(
  TYPE ORACLE_BIGDATA
  DEFAULT DIRECTORY DATA_PUMP_DIR
  ACCESS PARAMETERS
  (
    com.oracle.bigdata.credential.name = OCI_CRED
    com.oracle.bigdata.fileformat      = parquet
    com.oracle.bigdata.access_protocol = iceberg
  )
  LOCATION ('iceberg:&metadata_uri')
)
REJECT LIMIT UNLIMITED;
```

If you change or update the source Iceberg table (for example, adding new files), a new `metadata.json` will be generated, but the base table and its index will not see these changes. To see new data, you must recreate or manually repoint the external table to the latest `metadata.json` and rebuild the index.

Catalog-Based Table Creation (Dynamic Reference)

This method creates the external table using Iceberg's catalog capabilities, referencing the catalog instead of referencing to a specific snapshot. Using this method, the table always reflects the most up-to-date state.

```
CREATE TABLE IF NOT EXISTS vdb_ext_ivf01
(
  id   VARCHAR(10)   PRIMARY KEY RELY DISABLE,
  c1   NUMBER,
  n1   VARCHAR2(10),
  emb  VECTOR(1024, float32)
)
ORGANIZATION EXTERNAL
(
  TYPE ORACLE_BIGDATA
  DEFAULT DIRECTORY DATA_PUMP_DIR
  ACCESS PARAMETERS
  (
    com.oracle.bigdata.fileformat               = parquet
    com.oracle.bigdata.credential.name          = OCI_CRED
    com.oracle.bigdata.access_protocol          = iceberg
    com.oracle.bigdata.access_protocol.config   =
    {
      "iceberg_catalog_type": "aws_glue",
      "iceberg_glue_region": "us-west-2",
      "iceberg_table_path": "<database>.<table>"
    }
  )
  LOCATION ('iceberg:')
)
REJECT LIMIT UNLIMITED;
```

Access is configured by specifying catalog details in `ACCESS PARAMETERS`, rather than specifying a specific metadata file. When the table is queried, Oracle automatically fetches the latest snapshot and underlying data files from the catalog. Any changes to the source Iceberg table (such as newly added or modified partitions) are automatically reflected after a short interval.

Create IVF Index for an External Iceberg Table

The IVF index support for external tables over Iceberg enables scalable and efficient vector search on large datasets stored in external object stores.

The approach leverages a file-local IVF partition mechanism, in which each external data file (such as a Parquet file) is associated with its own dedicated IVF index partition. The indexing process is tightly integrated with Iceberg snapshot management, ensuring data consistency and operational efficiency.

IVF index creation on Iceberg tables uses the same syntax as on regular tables. The index build mechanism automatically detects the underlying files of the declared Iceberg snapshot and constructs a file-local IVF index for each. Internal partition-local IVF logic is leveraged to optimize both index build and query performance.

You can specify any distance metric for similarity calculations. In this example, cosine distance is used.

```
CREATE VECTOR INDEX vdb_ext_ivf01_idx
ON vdb_ext_ivf01 (emb)
ORGANIZATION NEIGHBOR PARTITIONS DISTANCE COSINE;
```

Upon a successful IVF index creation, the following auxiliary tables are generated:

- `VECTOR$VDB_EXT_IVF01_IDX$76013_76015_0$IVF_FLAT_CENTROIDS`
- `VECTOR$VDB_EXT_IVF01_IDX$76013_76015_0$IVF_FLAT_CENTROID_PARTITIONS`
- `VECTOR$VDB_EXT_IVF01_IDX$76013_76015_0$XT_FILE`
- `VECTOR$VDB_EXT_IVF01_IDX$76013_76015_0$XT_FILE_MAINT_TASK`

It is important to validate the associated auxiliary tables to ensure the index is correctly built and aligned with the underlying data. Consider the following validation points:

- `XT_FILE Auxiliary` Table
  The `XT_FILE auxiliary` table should list all external data files that are part of the Iceberg snapshot used during index creation. This ensures the index covers the complete set of files referenced by that snapshot.
- `ID` and `PART_ID` Consistency
  Within the `XT_FILE` table, the `ID` and `PART_ID` columns must have identical values. This reflects the one-to-one mapping between each external file and its corresponding IVF index partition.
- `IVF_FLAT_CENTROID_PARTITIONS` Table Size
  The `IVF_FLAT_CENTROID_PARTITIONS` table must contain the same number of records as the base table. This guarantees that every row in your data has a corresponding entry within the index partition, ensuring completeness for vector search operations.

Vector Search with IVF Index on Iceberg External Tables

After you create an IVF vector index on an Iceberg external table, you can run approximate k-Approximate Nearest Neighbor (k-ANN) queries that use the index to accelerate semantic or similarity search.

The following example shows a k-ANN query that computes cosine distance and returns the approximate top 5 closest vectors:

```
SELECT   id, vector_distance(emb, '[0,1,7]', COSINE) AS dist
FROM     vdb_ext_ivf01
ORDER BY dist
FETCH FIRST 5 ROWS ONLY;
```

Snapshot-referenced external table: k-ANN results are based on the specific Iceberg snapshot pointed to by the table's `metadata.json`. Newer Iceberg commits are not visible until the external table is updated to reference a newer snapshot.

Catalog-referenced external table: k-ANN results reflect the latest committed Iceberg snapshot available at query time. As new snapshots are committed, k-ANN query results are always evaluated as of the latest snapshot regardless of whether the vector index has been maintained/refreshed. The k-ANN query is internally transformed to use the correct data files as of the latest snapshot. Since the vector index is file-local, the transformed query can perform k-ANN using a partial vector index along with data-file scans, always returning results consistent with the latest snapshot.

Supported k-ANN Query Patterns
IVF indexing on an Iceberg external table supports k-ANN queries with no filter and with pre-filtering.

> **Note**
>
> IVF indexes with included columns are not currently supported on Iceberg external tables.

> **See Also**
>
> - Optimizer Plans for IVF Vector Indexes

IVF Index Maintenance

Index maintenance ensures that the IVF index stays synchronized with changes to the underlying Iceberg table so that queries can utilize the most current data.

When the underlying Iceberg table changes, the index may require maintenance to reflect the latest snapshot. During maintenance, the index maintenance process builds indexes for newly added data files.

To achieve this, the system uses coordinated background tasks to advance the IVF index from an older Iceberg snapshot to the latest one while preserving query consistency during ongoing maintenance.

Key Components

Index maintenance uses the following internal auxiliary tables to track external data files and maintenance tasks:

- **`XT_FILE`**: Tracks each external data file and whether its file-local IVF index is usable.

  ```
  CREATE TABLE VECTOR$<index_name>$<index_objn>$XT_FILE (
    id     NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  -- File ID
    path   VARCHAR2(4000)          NOT NULL,                 -- File path
    name   VARCHAR2(4000)          NOT NULL,                 -- File name
    status NUMBER        DEFAULT 1 NOT NULL                  -- 1: Index
  usable
                                                             -- 0: Index not
  usable
  );
  ```

  – `status`: `1` (index is usable) or `0` (index is not usable)

- **`XT_FILE_MAINT_TASK`**: Records maintenance operations required to align the index with a newer Iceberg snapshot.

  ```
  CREATE TABLE VECTOR$<index_name>$<index_objn>$XT_FILE_MAINT_TASK (
    task_id      NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,   -- Task
  ID
    file_id      NUMBER,                                            -- File
  ID
    task_type    CHAR(1)        NOT NULL,                           -- Task
  type, A: Add index partition, D: Drop index partition
    task_status  CHAR(1)        NOT NULL,                           -- Task
  status, R: Ready, I: In progress, C: Completed, F: Failed
    snapshot_id  VARCHAR2(4000) NOT NULL                            --
  Target iceberg snapshot ID
  );
  ```

  – `task_type`: `A` (add index partition) or `D` (drop index partition)

  – `tast_status`: `R` (Ready), `I` (In Progress), `C` (Completed), `F` (Failed)

  – `snapshot_id`: identifies the Iceberg snapshot that introduced the change

Updates to the vector index catalog

The vector index catalog (`VECSYS.VECTOR$INDEX`) stores additional information required for Iceberg-based indexing, including:

- Iceberg access information (credential schema/name) and the current/next metadata_uri (snapshot metadata).
- References to the auxiliary tables created for the index (object numbers and generated table names).

```
// SPARE2 JSON column
{
  "iceberg": {
    "credential_schema": "<CREDENTIAL_SCHEMA>",
    "credential_name"  : "<CREDENTIAL_NAME>",
```

```
    "metadata_uri"     : "https://oci_bucket/iceberg/vector/metadata/
v1.metadata.json",
    "new_metadata_uri" : "https://oci_bucket/iceberg/vector/metadata/
v2.metadata.json"
  }
}

// IDX_AUXILIARY_TABLES JSON column
{
  "xt_file_objn"           : 74315,
  "xt_file_name"           : "VECTOR$IDX_NAME$74315_74318_0$XT_FILE",
  "xt_file_maint_task_objn": 74320,
  "xt_file_maint_task_name":
"VECTOR$IDX_NAME$74320_74321_0$XT_FILE_MAINT_TASK"
}
```

This catalog allows background tasks to locate the correct Iceberg table version and the corresponding maintenance state tables for a given IVF index.

Background (Autotask++) tasks

Automated maintenance is implemented using the following background tasks:

Table Change Detection Autotask

Periodically checks the Iceberg catalog for a newer snapshot than the one currently indexed. When changes are detected, it will do the following:

1. Retrieves the list of files referenced by the new snapshot.
2. Compares it with files recorded in `XT_FILE`.
3. Computes a file-level diff. The file-list diffs consist of an `add-file` list, containing files newly introduced in the newer snapshot, and a `drop-file` list, containing files no longer present in the new snapshot.
4. Registers the corresponding maintenance tasks in the `XT_FILE_MAINT_TASK` table.

Index Creation Autotask

Builds file-local IVF index structures for files in the add-file list. For each new file, it performs the following:

1. Allocates a new file identifier (`FILE_ID`) using `XT_FILE`.
2. Adds the corresponding partitions to the IVF index tables. For example, `CENTROIDS` and `CENTROID_PARTITIONS` using the file's ID as the partition key.
3. Runs k-means clustering to populate `CENTROIDS`.
4. Populates `CENTROID_PARTITIONS` based on the clustering output.
5. Updates the `XT_FILE_MAINT_TASK` and `XT_FILE` tables to mark the index creation as completed.

7

Use SQL Functions for Vector Operations

There are a number of SQL functions and operators that you can use with vectors in Oracle AI Vector Search.

> **Note**
>
> You can also use PL/SQL packages to perform similar operations and additional tasks. See Vector Search PL/SQL Packages.

- Vector Distance Functions and Operators
  A vector distance function takes in two vector operands and a distance metric to compute a mathematical distance between those two vectors, based on the distance metric provided. You can optionally use shorthand distance functions and operators instead of their corresponding distance functions.
- Chunking and Vector Generation Functions
  Oracle AI Vector Search offers Vector Utilities, which provide the `VECTOR_CHUNKS` and `VECTOR_EMBEDDING` SQL functions for chunking data and generating vector embedding, respectively.
- Constructors, Converters, Descriptors, and Arithmetic Operators
  Other basic vector operations for Oracle AI Vector Search involve creating, converting, and describing vectors.
- JSON Compatibility with the VECTOR Data Type
  The JSON data type supports the `VECTOR` type as a JSON scalar type. A `VECTOR` instance is convertible to a JSON type instance and vice versa using the JSON constructor and the `VECTOR` constructor, respectively.

Vector Distance Functions and Operators

A vector distance function takes in two vector operands and a distance metric to compute a mathematical distance between those two vectors, based on the distance metric provided. You can optionally use shorthand distance functions and operators instead of their corresponding distance functions.

Distances determine similarity or dissimilarity between vectors.

- Vector Distance Metrics
  Measuring distances in a vector space is at the heart of identifying the most relevant results for a given query vector. That process is very different from the well-known keyword filtering in the relational database world.
- VECTOR_DISTANCE
  `VECTOR_DISTANCE` is the main function that you can use to calculate the distance between two vectors.

- L1_DISTANCE
  `L1_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the Manhattan distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.
- L2_DISTANCE
  `L2_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the Euclidean distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.
- COSINE_DISTANCE
  `COSINE_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the cosine distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.
- INNER_PRODUCT
  `INNER_PRODUCT` calculates the inner product of two vectors. It takes two vectors as input and returns the inner product as a `BINARY_DOUBLE`. `INNER_PRODUCT(<expr1>, <expr2>)` is equivalent to `-1 * VECTOR_DISTANCE(<expr1>, <expr2>, DOT)`.
- HAMMING_DISTANCE
  `HAMMING_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the hamming distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.
- JACCARD_DISTANCE
  `JACCARD_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the jaccard distance between two vectors. It takes two `BINARY` vectors as input and returns the distance between them as a `BINARY_DOUBLE`.

Related Topics

- Perform Exact Similarity Search
  A similarity search looks for the relative order of vectors compared to a query vector. Naturally, the comparison is done using a particular distance metric but what is important is the result set of your top closest vectors, not the distance between them.
- Perform Approximate Similarity Search Using Vector Indexes
  For a vector search to be useful, it needs to be fast and accurate. Approximate similarity searches seek a balance between these goals.

Vector Distance Metrics

Measuring distances in a vector space is at the heart of identifying the most relevant results for a given query vector. That process is very different from the well-known keyword filtering in the relational database world.

When working with vectors, there are several ways you can calculate distances to determine how similar, or dissimilar, two vectors are. Each distance metric is computed using different mathematical formulas. The time it takes to calculate the distance between two vectors depends on many factors, including the distance metric used as well as the format of the vectors themselves, such as the number of vector dimensions and the vector dimension formats. Generally it's best to match the distance metric you use to the one that was used to train the vector embedding model that generated the vectors.

- Euclidean and Euclidean Squared Distances
  Euclidean distance reflects the distance between each of the vectors' coordinates being compared—basically the straight-line distance between two vectors. This is calculated using the Pythagorean theorem applied to the vector's coordinates (SQRT(SUM((xi-yi)^2))).

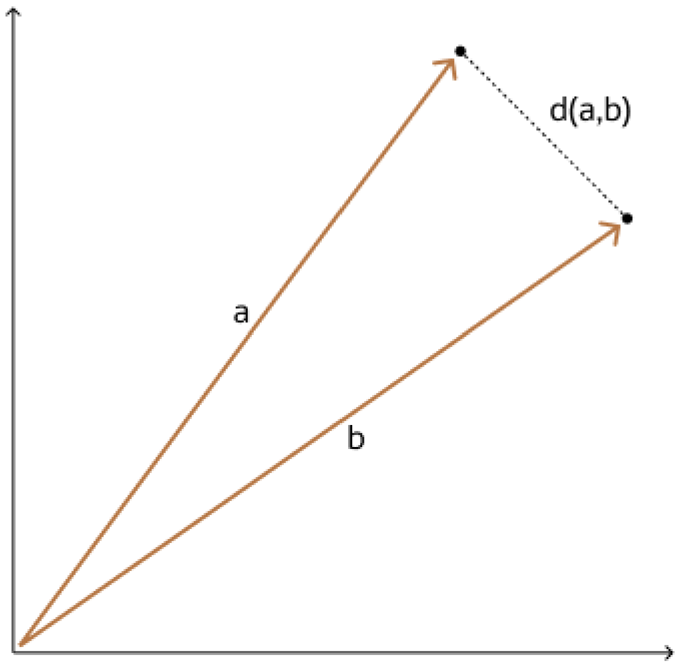
- Cosine Similarity
  One of the most widely used similarity metrics, especially in natural language processing (NLP), is cosine similarity, which measures the cosine of the angle between two vectors.
- Dot Product Similarity
  The dot product similarity of two vectors can be viewed as multiplying the size of each vector by the cosine of their angle. The corresponding geometrical interpretation of this definition is equivalent to multiplying the size of one of the vectors by the size of the projection of the second vector onto the first one, or vice versa.
- Manhattan Distance
  This metric is calculated by summing the distance between the dimensions of the two vectors that you want to compare.
- Hamming Distance
  The Hamming distance between two vectors represents the number of dimensions where they differ.
- Jaccard Similarity
  The Jaccard similarity is used to determine the share of significant (non-zero) dimensions (bit's position) common between two `BINARY` vectors.
- Custom Distance Function
  JavaScript user-defined functions can be used to define a custom vector distance. This provides greater flexibility in the types of distance equations that can be employed, extending vector search functionality to a broader range of use cases.

Euclidean and Euclidean Squared Distances

Euclidean distance reflects the distance between each of the vectors' coordinates being compared—basically the straight-line distance between two vectors. This is calculated using the Pythagorean theorem applied to the vector's coordinates (`SQRT(SUM((xi-yi)`^{2}`)))`.

This metric is sensitive to both the vector's size and it's direction.

With Euclidean distances, comparing squared distances is equivalent to comparing distances. So, when ordering is more important than the distance values themselves, the Squared Euclidean distance is very useful as it is faster to calculate than the Euclidean distance (avoiding the square-root calculation).



Cosine Similarity

One of the most widely used similarity metrics, especially in natural language processing (NLP), is cosine similarity, which measures the cosine of the angle between two vectors.

The smaller the angle, the more similar the two vectors. Cosine similarity measures the similarity in the direction or angle of the vectors, ignoring differences in their size (also called *magnitude*). The smaller the angle, the bigger its cosine. So, the cosine distance and the cosine similarity have an inverse relationship. While cosine distance measures how different two vectors are, cosine similarity measures how similar two vectors are.

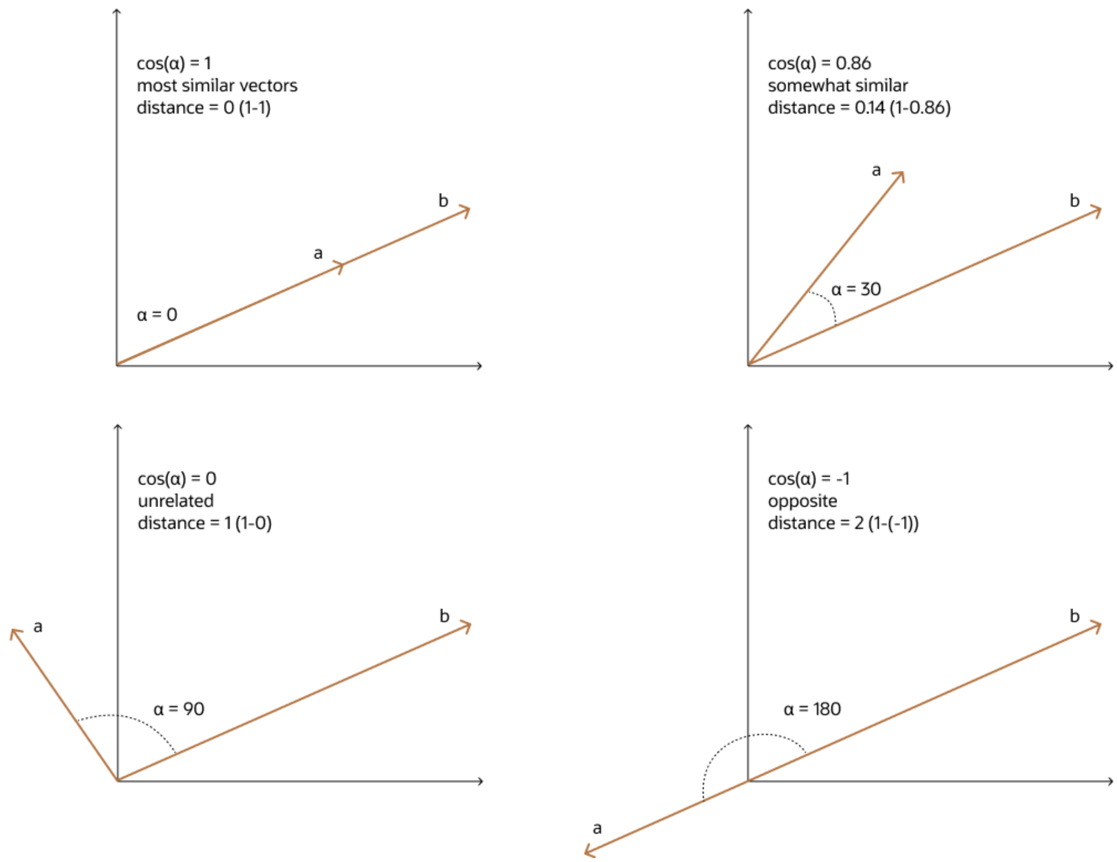
The cosine similarity is calculated between two vectors, A and B, using the following formula:

```
cos(α) = (A · B) ÷ ( ||A|| * ||B||)
```

- `A · B` is the dot product of vectors A and B
- `||A||` is the magnitude (Euclidean norm) of vector A
- `||B||` is the magnitude (Euclidean norm) of vector B

The cosine distance is calculated by subtracting the cosine similarity from 1:

```
Distance(A, B) = 1 - cos(α)
```



Dot Product Similarity

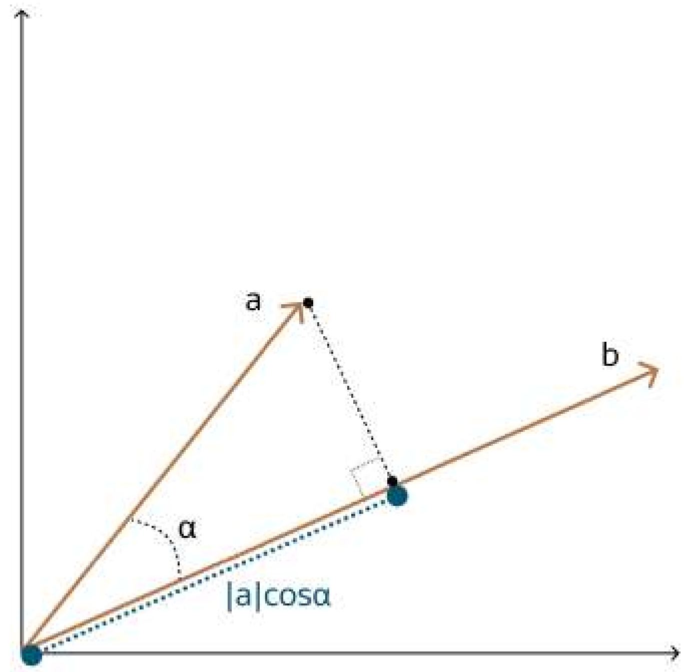
The dot product similarity of two vectors can be viewed as multiplying the size of each vector by the cosine of their angle. The corresponding geometrical interpretation of this definition is equivalent to multiplying the size of one of the vectors by the size of the projection of the second vector onto the first one, or vice versa.

As illustrated in the following diagram, you project one vector on the other and multiply the resulting vector sizes.

Incidentally, this is equivalent to the sum of the products of each vector's coordinate. Often, you do not have access to the cosine of the two vector's angle, hence this calculation is easier.

Larger dot product values imply that the vectors are more similar, while smaller values imply that they are less similar. Compared to using Euclidean distance, using the dot product similarity is especially useful for high-dimensional vectors.

Note that normalizing vectors and using the dot product similarity is equivalent to using cosine similarity. There are cases where dot product similarity is faster to evaluate than cosine similarity, and conversely where cosine similarity is faster than dot product similarity. A normalized vector is created by dividing each dimension by the norm (or length) of the vector, such that the norm of the normalized vector is equal to 1.

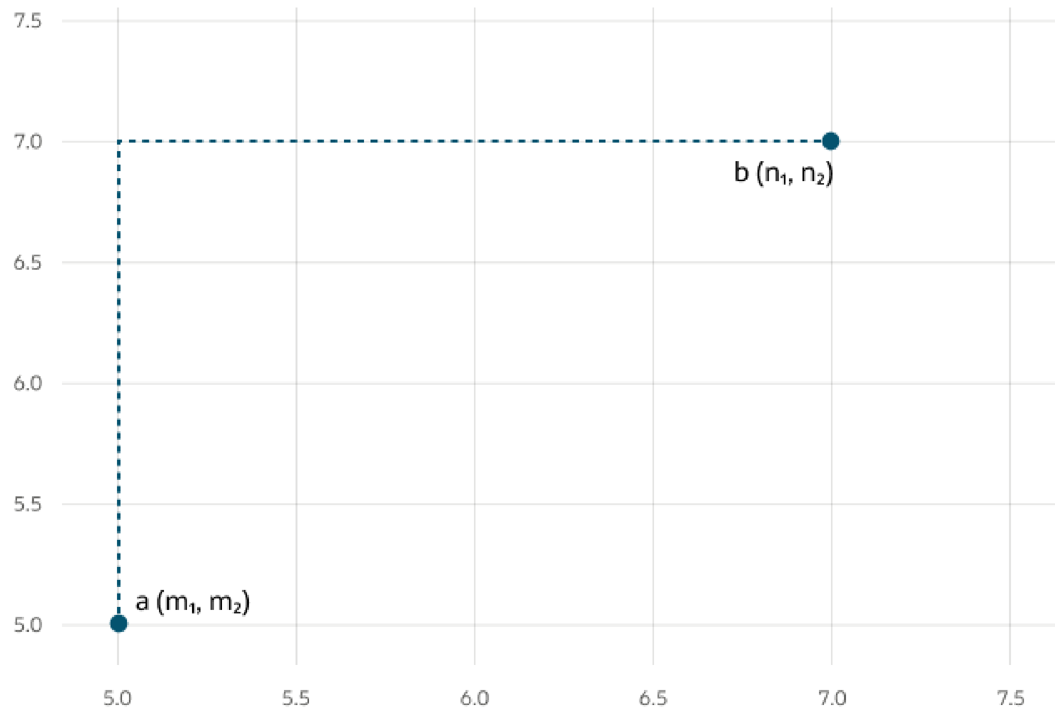


Manhattan Distance

This metric is calculated by summing the distance between the dimensions of the two vectors that you want to compare.

Imagine yourself in the streets of Manhattan trying to go from point A to point B. A straight line is not possible.

This metric is most useful for vectors describing objects on a uniform grid, such as city blocks, power grids, or a chessboard. It can be useful for higher dimensional vector spaces too. Compared to the Euclidean metric, the Manhattan metric is faster for calculations and you can use it advantageously for higher dimensional vector spaces.



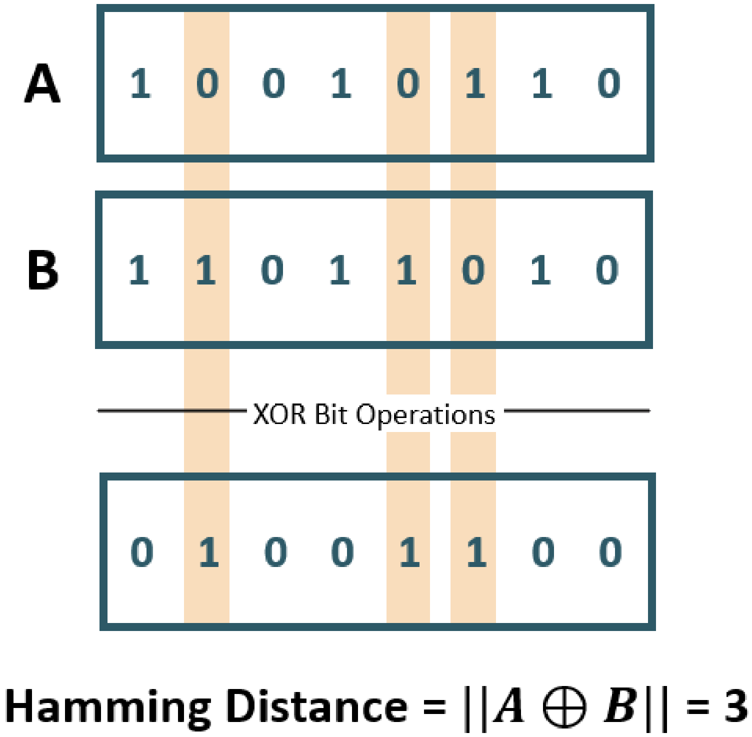
Hamming Distance

The Hamming distance between two vectors represents the number of dimensions where they differ.

For example, when using binary vectors, the Hamming distance between two vectors is the number of bits you must change to change one vector into the other. To compute the Hamming distance between two vectors *A* and *B*, you need to:

- Compare the position of each bit in the sequence. You do this by using an `exclusive or` (also called the XOR bit operation) between *A* and *B*. This operation outputs 1 if the bits in the sequence do not match, and 0 otherwise.
- Count the number of '1's in the resulting vector, the outcome of which is called the Hamming weight or norm of that vector.

It's important to note that the bit strings need to be of equal length for the comparison to make sense. The Hamming metric is mainly used with binary vectors for error detection over networks.



Jaccard Similarity

The Jaccard similarity is used to determine the share of significant (non-zero) dimensions (bit's position) common between two `BINARY` vectors.

The Jaccard similarity is only applicable to `BINARY` vectors and only the non-zero bits of each vector are considered.

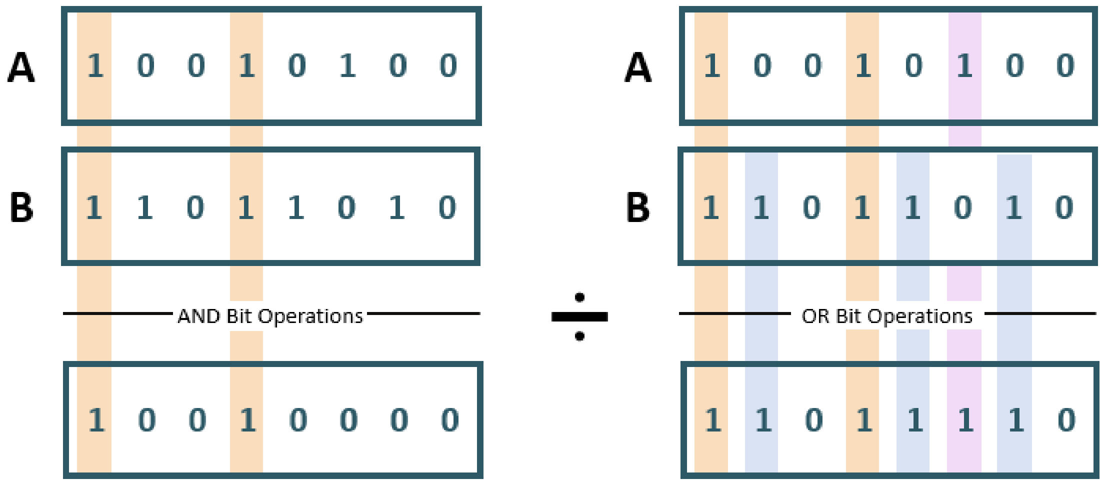
The Jaccard similarity between vectors *A* and *B* is the calculation of the Hamming weight (norm, or the number of '1's in the resulting vector) of the result of an `AND` bit operation between *A* and *B*, divided by the Hamming weight of the result of an `OR` bit operation between *A* and *B*.

As shown in the included diagram of vectors *A* and *B*:

- The `AND` bit operation outputs a 1 if the bits in the sequence match a 1, and 0 otherwise.
- The `OR` bit operation outputs a 1 if at least one of the bits in the sequence matches a 1, and 0 otherwise.

The result of the calculation is from 0 to 1, where results approaching 1 are more similar. A result of 0 means that the two vectors share no non-zero attributes while a result of 1 indicates the two vectors share identical sets of non-zero attributes. In the included diagram, the two vectors share 33% of the significant attributes.

While the Jaccard similarity indicates how similar two vectors are, the Jaccard distance indicates the *dissimilarity* between the vectors. The Jaccard distance can be found by subtracting the Jaccard similarity from 1. For example, two vectors with a Jaccard similarity of *0.25* have a Jaccard distance of *0.75*. When determining distance, the meaning of the result is opposite to that of a similarity calculation. A result of 0 indicates that the two vectors are identical while a result of 1 indicates the vectors are completely disjoint, sharing no common elements.



$$\text{Jaccard Similarity} = \frac{\|A \cap B\|}{\|A \cup B\|} = \frac{2}{6} = 0.33$$

$$\text{Jaccard Distance} = 1 - \text{Jaccard Similarity} = 1 - 0.33 = 0.67$$

Custom Distance Function

JavaScript user-defined functions can be used to define a custom vector distance. This provides greater flexibility in the types of distance equations that can be employed, extending vector search functionality to a broader range of use cases.

A custom distance function is created by a user-defined JavaScript function defined in a Multilingual Engine (MLE) inline call specification. The signature of the function must match the signature of existing built-in distance functions. As in, it must accept exactly two arguments of type `VECTOR` and return a `BINARY_DOUBLE`. The function signature must also include the `DETERMINISTIC` keyword. The following function definition provides an example of a custom distance function, in this case implementing the Euclidean Squared distance:

```
CREATE OR REPLACE FUNCTION euclidean_sq_vector_distance("a" VECTOR, "b"
VECTOR)
RETURN BINARY_DOUBLE
DETERMINISTIC PARALLEL_ENABLE
AS MLE LANGUAGE JAVASCRIPT PURE
{{
    let len = a.length;
    let sum = 0;
    for(let i = 0; i < len; i++) {
        const tmp = a[i] - b[i];
        sum += tmp * tmp;
    }
    return sum;
}};
/
```

Custom distance functions can be used with HNSW vector indexes. If the degree of parallelism in the vector index is greater than 1, the custom distance function must include the `PARALLEL_ENABLE` clause. Upon index creation, a custom distance can be specified by name in the `DISTANCE` clause. In queries, the custom distance can be used in the `ORDER BY` clause and in the `SELECT` list. The distance function tied to a vector index can be viewed by querying the `VECSYS.VECTOR$INDEX` view.

When used in the creation of an HNSW index, the `PURE` keyword must be specified in the MLE call specification. The `PURE` clause indicates that the JavaScript program should be run in a restricted execution context, which guarantees that the code will not modify stateful objects, such as database tables or PL/SQL packages, regardless of database privileges currently in effect. A user-defined function used to create a custom distance metric only handles computations on function inputs, which do not require access to database state. Restricted contexts provide an extra layer of security by prohibiting unwanted database modifications. For more information on restricted execution contexts and the `PURE` keyword, see *Oracle AI Database JavaScript Developer's Guide*.

In order to use a vector index dependent on a custom distance function, you must have `EXECUTE` privileges on the function specified during index creation. You must also have `EXECUTE` privileges on `JAVASCRIPT`. For vector indexes, only definer's rights are supported.

If the distance function is modified, the associated vector index will be in an `UNUSABLE` state.

> **Note**
>
> The use of custom distance functions with IVF indexes is not currently supported.

Use the previously created custom distance function, `euclidean_sq_custom_distance`, to first create a vector index:

```
CREATE TABLE custom_dist_tab( id NUMBER, data_vector VECTOR(2, FLOAT32));

INSERT INTO custom_dist_tab VALUES (1, vector('[1.1,2.2]', 2, float32));
INSERT INTO custom_dist_tab VALUES (2, vector('[2.2,3.3]', 2, float32));
INSERT INTO custom_dist_tab VALUES (3, vector('[3.3,4.4]', 2, float32));
INSERT INTO custom_dist_tab VALUES (4, vector('[4.4,5.5]', 2, float32));
INSERT INTO custom_dist_tab VALUES (5, vector('[5.5,6.6]', 2, float32));

CREATE VECTOR INDEX cust_dist_idx_hnsw ON custom_dist_tab (data_vector)
ORGANIZATION INMEMORY
NEIGHBOR GRAPH WITH TARGET ACCURACY 95
DISTANCE CUSTOM EUCLIDEAN_SQ_VECTOR_DISTANCE
PARALLEL 3;
```

The custom distance function can be referenced in similarity search queries in the `ORDER BY` clause or in the `SELECT` list:

```
SELECT data_vector
FROM custom_dist_tab
ORDER BY euclidean_sq_vector_distance(data_vector, VECTOR('[1, 2]'))
FETCH FIRST 5 ROWS ONLY;

SELECT
  data_vector,
```

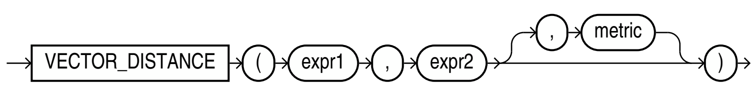
```
  euclidean_sq_vector_distance(data_vector, VECTOR('[1, 2]')) edist
FROM custom_dist_tab
ORDER BY edist
FETCH FIRST 5 ROWS ONLY;
```

> **See Also**
>
> - *Oracle AI Database JavaScript Developer's Guide* for information about using inline call specifications to publish JavaScript functions with the Multilingual Engine (MLE)

VECTOR_DISTANCE

`VECTOR_DISTANCE` is the main function that you can use to calculate the distance between two vectors.

Syntax



Purpose

`VECTOR_DISTANCE` takes two vectors as parameters. You can optionally specify a distance metric to calculate the distance. If you do not specify a distance metric, then the default distance metric is cosine. If the input vectors are `BINARY` vectors, the default metric is hamming.

You can optionally use the following shorthand vector distance functions:

- `L1_DISTANCE`
- `L2_DISTANCE`
- `COSINE_DISTANCE`
- `INNER_PRODUCT`
- `HAMMING_DISTANCE`
- `JACCARD_DISTANCE`

All the vector distance functions take two vectors as input and return the distance between them as a `BINARY_DOUBLE`.

Note the following caveats:

- If you specify a metric as the third argument, then that metric is used.
- If you do not specify a metric, then the following rules apply:
  – If there is a single column referenced in *`expr1`* and *`expr2`* as in: `VECTOR_DISTANCE(vec1, :bind)`, and if there is a vector index defined on *`vec1`*, then the metric used when defining the vector index is used.

    If no vector index is defined on *`vec1`*, then the `COSINE` metric is used.

  - If there are multiple columns referenced in *expr1* and *expr2* as in: `VECTOR_DISTANCE(vec1, vec2)`, or `VECTOR_DISTANCE(vec1+vec2, :bind)`, then for all indexed columns, if their metrics used in the definitions of the indexes are the same, then that metric is used.

    On the other hand, if the indexed columns do not have a common metric, or none of the columns have an index defined, then the `COSINE` metric is used.
- In a similarity search query, if *expr1* or *expr2* reference an indexed column and you specify a distance metric that conflicts with the metric specified in the vector index, then the vector index is not used and the metric you specified is used to perform an exact search.
- Approximate (index-based) searches can be done if only one column is referenced by either *expr1* or *expr2*, and this column has a vector index defined, and the metric that is specified in the vector_distance matches the metric used in the definition of the vector index.

Parameters

- *expr1* and *expr2* must evaluate to vectors and have the same dimension format, storage format, and number of dimensions.

  If you use `JACCARD_DISTANCE` or the `JACCARD` metric, then *expr1* and *expr2* must evaluate to `BINARY` vectors.
- This function returns NULL if either *expr1* or *expr2* is NULL.
- *metric* must be one of the following tokens :
  - `COSINE` metric is the default metric. It calculates the cosine distance between two vectors.
  - `DOT` metric calculates the negated dot product of two vectors. The `INNER_PRODUCT` function calculates the dot product, as in the negation of this metric.
  - `EUCLIDEAN` metric, also known as L2 distance, calculates the Euclidean distance between two vectors.
  - `EUCLIDEAN_SQUARED` metric, also called `L2_SQUARED`, is the Euclidean distance without taking the square root.
  - `HAMMING` metric calculates the hamming distance between two vectors by counting the number dimensions that differ between the two vectors.
  - `MANHATTAN` metric, also known as L1 distance or taxicab distance, calculates the Manhattan distance between two vectors.
  - `JACCARD` metric calculates the Jaccard distance. The two vectors used in the query must be `BINARY` vectors.

Shorthand Operators for Distances

Syntax

- `expr1 <-> expr2`

  <-> is the **Euclidean distance operator**: `expr1 <-> expr2` is equivalent to `L2_DISTANCE(expr1, expr2)` or `VECTOR_DISTANCE(expr1, expr2, EUCLIDEAN)`
- `expr1 <=> expr2`

  <=> is the **cosine distance operator**: `expr1 <=> expr2` is equivalent to `COSINE_DISTANCE(expr1, expr2)` or `VECTOR_DISTANCE(expr1, expr2, COSINE)`
- `expr1 <#> expr2`

<#> is the **negative dot product operator**: `expr1 <#> expr2` is equivalent to `-1*INNER_PRODUCT(expr1, expr2)` or `VECTOR_DISTANCE(expr1, expr2, DOT)`

Examples Using Shorthand Operators for Distances

```
'[1, 2]' <-> '[0,1]'
```

```
v1 <-> '[' || '1,2,3' || ']' is equivalent to v1 <-> '[1, 2, 3]'
```

```
v1 <-> '[1,2]' is equivalent to L2_DISTANCE(v1, '[1,2]')
```

```
v1 <=> v2 is equivalent to COSINE_DISTANCE(v1, v2)
```

```
v1 <#> v2 is equivalent to -1*INNER_PRODUCT(v1, v2)
```

Examples

`VECTOR_DISTANCE` with metric `EUCLIDEAN` is equivalent to `L2_DISTANCE`:

```
VECTOR_DISTANCE(expr1, expr2, EUCLIDEAN);
```

```
L2_DISTANCE(expr1, expr2);
```

`VECTOR_DISTANCE` with metric `COSINE` is equivalent to `COSINE_DISTANCE`:

```
VECTOR_DISTANCE(expr1, expr2, COSINE);
```

```
COSINE_DISTANCE(expr1, expr2);
```

`VECTOR_DISTANCE` with metric `DOT` is equivalent to -1 * `INNER_PRODUCT`:

```
VECTOR_DISTANCE(expr1, expr2, DOT);
```

```
-1*INNER_PRODUCT(expr1, expr2);
```

`VECTOR_DISTANCE` with metric `MANHATTAN` is equivalent to `L1_DISTANCE`:

```
VECTOR_DISTANCE(expr1, expr2, MANHATTAN);
```

```
L1_DISTANCE(expr1, expr2);
```

`VECTOR_DISTANCE` with metric `HAMMING` is equivalent to `HAMMING_DISTANCE`:

```
VECTOR_DISTANCE(expr1, expr2, HAMMING);
```

```
HAMMING_DISTANCE(expr1, expr2);
```

`VECTOR_DISTANCE` with metric `JACCARD` is equivalent to `JACCARD_DISTANCE`:

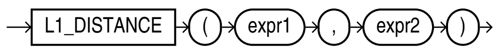
```
VECTOR_DISTANCE(expr1, expr2, JACCARD);
```

```
JACCARD_DISTANCE(expr1, expr2);
```

L1_DISTANCE

`L1_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the Manhattan distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.

Syntax



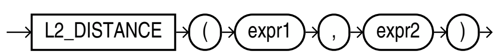
Parameters

- *`expr1`* and *`expr2`* must evaluate to vectors and have the same format and number of dimensions.
- `L1_DISTANCE` returns NULL, if either *`expr1`* or *`expr2`* is NULL.

L2_DISTANCE

`L2_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the Euclidean distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.

Syntax



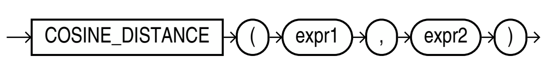
Parameters

- *`expr1`* and *`expr2`* must evaluate to vectors that have the same format and number of dimensions.
- `L2_DISTANCE` returns NULL, if either *`expr1`* or *`expr2`* is NULL.

COSINE_DISTANCE

`COSINE_DISTANCE` is a shorthand version of the `VECTOR_DISTANCE` function that calculates the cosine distance between two vectors. It takes two vectors as input and returns the distance between them as a `BINARY_DOUBLE`.

Syntax



Parameters

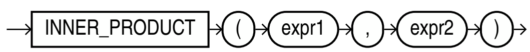
- *`expr1`* and *`expr2`* must evaluate to vectors that have the same format and number of dimensions.

- COSINE_DISTANCE returns NULL, if either *expr1* or *expr2* is NULL.

INNER_PRODUCT

INNER_PRODUCT calculates the inner product of two vectors. It takes two vectors as input and returns the inner product as a BINARY_DOUBLE. INNER_PRODUCT(<*expr1*>, <*expr2*>) is equivalent to -1 * VECTOR_DISTANCE(<*expr1*>, <*expr2*>, DOT).

Syntax



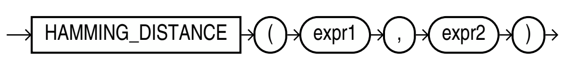
Parameters

- *expr1* and *expr2* must evaluate to vectors that have the same format and number of dimensions.
- INNER_PRODUCT returns NULL, if either *expr1* or *expr2* is NULL.

HAMMING_DISTANCE

HAMMING_DISTANCE is a shorthand version of the VECTOR_DISTANCE function that calculates the hamming distance between two vectors. It takes two vectors as input and returns the distance between them as a BINARY_DOUBLE.

Syntax



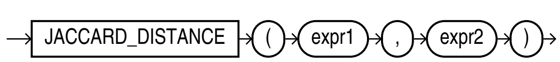
Parameters

- *expr1* and *expr2* must evaluate to vectors that have the same format and number of dimensions.
- HAMMING_DISTANCE returns NULL if either *expr1* or *expr2* is NULL.

JACCARD_DISTANCE

JACCARD_DISTANCE is a shorthand version of the VECTOR_DISTANCE function that calculates the jaccard distance between two vectors. It takes two BINARY vectors as input and returns the distance between them as a BINARY_DOUBLE.

Syntax



Parameters

- *expr1* and *expr2* must evaluate to `BINARY` vectors and have the same number of dimensions. If either expression is not a `BINARY` vector, an error is raised.
- `JACCARD_DISTANCE` returns NULL if either *expr1* or *expr2* is NULL.

Chunking and Vector Generation Functions

Oracle AI Vector Search offers Vector Utilities, which provide the `VECTOR_CHUNKS` and `VECTOR_EMBEDDING` SQL functions for chunking data and generating vector embedding, respectively.

- VECTOR_CHUNKS
  Use `VECTOR_CHUNKS` to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.
- VECTOR_EMBEDDING
  Use `VECTOR_EMBEDDING` to generate a single vector embedding for different data types using embedding or feature extraction machine learning models.

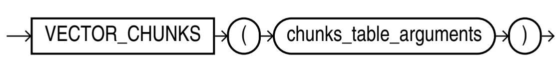
Related Topics

- Vector Generation Examples
  Run these end-to-end examples to see how you can generate vector embeddings, both within and outside the database.

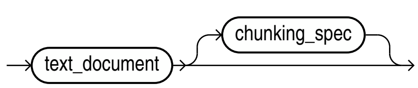
VECTOR_CHUNKS

Use `VECTOR_CHUNKS` to split plain text into smaller chunks to generate vector embeddings that can be used with vector indexes or hybrid vector indexes.

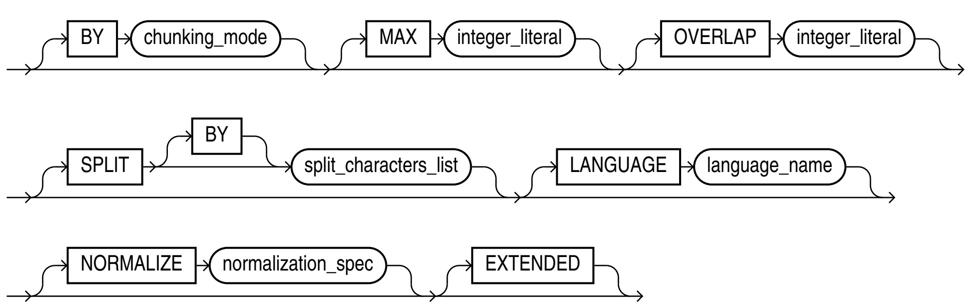
Syntax



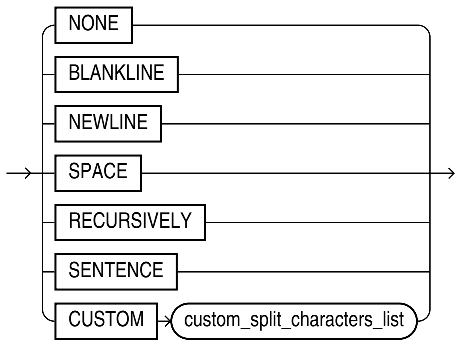
***chunks_table_arguments*::=**



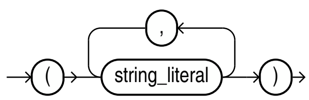
***chunking_spec*::=**



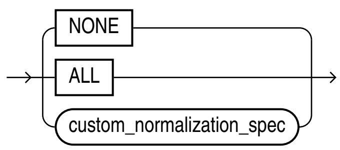
***split_characters_list*::=**



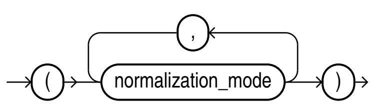
custom_split_characters_list



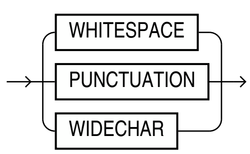
normalization_spec



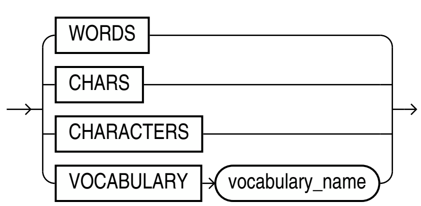
custom_normalization_spec



normalization_mode



***chunking_mode*::=**



Purpose

VECTOR_CHUNKS takes a character value as the *text_document* argument and splits it into chunks using a process controlled by the chunking parameters given in the optional *chunking_spec*. The chunks are returned as rows of a virtual relational table. Therefore, VECTOR_CHUNKS can only appear in the FROM clause of a subquery.

The returned virtual table has the following columns:

- CHUNK_OFFSET of data type NUMBER is the position of each chunk in the source document, relative to the start of the document, which has a position of 1.
- CHUNK_LENGTH of data type NUMBER is the length of each chunk.
- CHUNK_TEXT is a segment of text that has been split off from *text_document*.

The data type of the CHUNK_TEXT column and the length unit used by the values of CHUNK_OFFSET and CHUNK_LENGTH depend on the data type of *text_document* as listed in the following table:

Table 7-1 Input and Output Data Type Details

Input Data Type	Output Data Type	Offset and Length Unit
VARCHAR2	VARCHAR2	byte
CHAR	VARCHAR2	byte
CLOB	VARCHAR2	character
NVARCHAR2	NVARCHAR2	byte
NCHAR	NVARCHAR2	byte
NCLOB	NVARCHAR2	character

> **Note**
>
> - For more information about data types, see *Data Types* in the SQL Reference Manual.
> - The VARCHAR2 input data type is limited to 4000 bytes unless the MAX_STRING_SIZE parameter is set to EXTENDED, which increases the limit to 32767.

Parameters

All chunking parameters are optional, and the default chunking specifications are automatically applied to your chunk data.

When specifying chunking parameters for this API, ensure that you provide these parameters only in the listed order.

Table 7-2 Chunking Parameters Table

Parameter	Description and Acceptable Values
BY	Specifies the mode for splitting your data, that is, to split by counting the number of characters, words, or vocabulary tokens. **Valid values**: • CHARACTERS (or CHARS): Splits by counting the number of characters. • WORDS: Splits by counting the number of words. Words are defined as sequences of alphabetic characters, sequences of digits, individual punctuation marks, or symbols. For segmented languages without whitespace word boundaries (such as Chinese, Japanese, or Thai), each native character is considered a word (that is, unigram). • VOCABULARY: Splits by counting the number of vocabulary tokens. Vocabulary tokens are words or word pieces, recognized by the vocabulary of the tokenizer that your embedding model uses. You can load your vocabulary file using the VECTOR_CHUNKS helper API DBMS_VECTOR_CHAIN.CREATE_VOCABULARY. **Note**: For accurate results, ensure that the chosen model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. **Default value**: WORDS
MAX	Specifies a limit on the maximum size of each chunk. This setting splits the input text at a fixed point where the maximum limit occurs in the larger text. The units of MAX correspond to the BY mode, that is, to split data when it reaches the maximum size limit of a certain number of characters, words, numbers, punctuation marks, or vocabulary tokens. **Valid values**: • BY CHARACTERS: 50 to 4000 characters • BY WORDS: 10 to 1000 words • BY VOCABULARY: 10 to 1000 tokens **Default value**: 100

Table 7-2 (Cont.) Chunking Parameters Table

Parameter	Description and Acceptable Values
SPLIT [BY]	Specifies where to split the input text when it reaches the maximum size limit. This helps to keep related data together by defining appropriate boundaries for chunks. **Valid values**: • NONE: Splits at the MAX limit of characters, words, or vocabulary tokens. • NEWLINE, BLANKLINE, and SPACE: These are single-split character conditions that split at the last split character before the MAX value. Use NEWLINE to split at the end of a line of text. Use BLANKLINE to split at the end of a blank line (sequence of characters, such as two newlines). Use SPACE to split at the end of a blank space. • RECURSIVELY: This is a multiple-split character condition that breaks the input text using an ordered list of characters (or sequences). RECURSIVELY is predefined as BLANKLINE, NEWLINE, SPACE, NONE in this order: 1. If the input text is more than the MAX value, then split by the first split character. 2. If that fails, then split by the second split character. 3. And so on. 4. If no split characters exist, then split by MAX wherever it appears in the text. • SENTENCE: This is an end-of-sentence split condition that breaks the input text at a sentence boundary. This condition automatically determines sentence boundaries by using knowledge of the input language's sentence punctuation and contextual rules. This language-specific condition relies mostly on end-of-sentence (EOS) punctuations and common abbreviations. Contextual rules are based on word information, so this condition is only valid when splitting the text by words or vocabulary (not by characters). **Note**: This condition obeys the BY WORD and MAX settings, and thus may not determine accurate sentence boundaries in some cases. For example, when a sentence is larger than the MAX value, it splits the sentence at MAX. Similarly, it includes multiple sentences in the text only when they fit within the MAX limit. • CUSTOM: Splits based on a custom list of characters strings, for example, markup tags. You can provide custom sequences up to a limit of 16 split character strings, with a maximum length of 10 bytes each. Provide valid text literals as follows: VECTOR_CHUNKS(c. doc, BY character SPLIT CUSTOM ('*<html>*' , '*</html>*')) vc **Default value**: RECURSIVELY
OVERLAP	Specifies the amount (as a positive integer literal or zero) of the preceding text that the chunk should contain, if any. This helps in logically splitting up related text (such as a sentence) by including some amount of the preceding chunk text. The amount of overlap depends on how the maximum size of the chunk is measured (in characters, words, or vocabulary tokens). The overlap begins at the specified SPLIT condition (for example, at NEWLINE). **Valid value**: 5% to 20% of MAX **Default value**: 0

Table 7-2 (Cont.) Chunking Parameters Table

Parameter	Description and Acceptable Values

LANGUAGE	Specifies the language of your input data.
	This clause is important, especially when your text contains certain characters (for example, punctuations or abbreviations) that may be interpreted differently in another language.
	Valid values:
	• NLS-supported language name or its abbreviation, as listed in *Oracle AI Database Globalization Support Guide*.
	• Custom language name or its abbreviation, as listed in Supported Languages and Data File Locations. You use the `DBMS_VECTOR_CHAIN.CREATE_LANG_DATA` chunker helper API to load language-specific data (abbreviation tokens) into the database, for your specified language.
	You must use double quotation marks (") for any language name with spaces. For example:
	`LANGUAGE "simplified chinese"`
	For one-word language names, quotation marks are not needed. For example:
	`LANGUAGE american`
	Default value: `NLS_LANGUAGE` from session

NORMALIZE	Automatically pre-processes or post-processes issues (such as multiple consecutive spaces and smart quotes) that may arise when documents are converted into text. Oracle recommends you to use a normalization mode to extract high-quality chunks. **Valid values**: • `NONE`: Applies no normalization. • `ALL`: Normalizes multi-byte (Unicode) punctuation to standard single-byte. • Applies all supported normalization modes: `PUNCTUATION`, `WHITESPACE`, and `WIDECHAR`. – `PUNCTUATION`: Converts quotes, dashes, and other punctuation characters supported in the character set of the text to their common ASCII form. For example: * U+2013 (En Dash) maps to U+002D (Hyphen-Minus) * U+2018 (Left Single Quotation Mark) maps to U+0027 (Apostrophe) * U+2019 (Right Single Quotation Mark) maps to U+0027 (Apostrophe) * U+201B (Single High-Reversed-9 Quotation Mark) maps to U+0027 (Apostrophe) – `WHITESPACE`: Minimizes whitespace by eliminating unnecessary characters. For example, retain blanklines, but remove any extra newlines and interspersed spaces or tabs: `" \n \n " => "\n\n"` – `WIDECHAR`: Normalizes wide, multi-byte digits and (a-z) letters to single-byte. These are multi-byte equivalents for `0-9` and `a-z A-Z`, which can show up in Chinese, Japanese, or Korean text. **Default value**: `NONE`
EXTENDED	Increases the output limit of a `VARCHAR2` string to `32767` bytes, without requiring you to set the `MAX_STRING_SIZE` parameter to `EXTENDED`. If `EXTENDED` is present in *`chunking_spec`*, the maximum length of a `CHUNK_TEXT` column value is `32767` bytes. If it is absent, the maximum length is `4000` bytes if `MAX_STRING_SIZE` is set to `STANDARD` and `32767` bytes if `MAX_STRING_SIZE` is set to `EXTENDED`.

Examples

VECTOR_CHUNKS can be called for a single character value provided in a character literal or a bind variable as shown in the following example:

```
COLUMN chunk_offset HEADING Offset FORMAT 999
COLUMN chunk_length HEADING Len    FORMAT 999
COLUMN chunk_text   HEADING Text   FORMAT a60

VARIABLE txt VARCHAR2(4000)
EXECUTE :txt := 'An example text value to split with VECTOR_CHUNKS, having over 10 words
because the minimum MAX value is 10';

SELECT * FROM VECTOR_CHUNKS(:txt BY WORDS MAX 10);

SELECT * FROM VECTOR_CHUNKS('Another example text value to split with VECTOR_CHUNKS,
having over 10 words because the minimum MAX value is 10' BY WORDS MAX 10);
```

To chunk values of a table column, the table needs to be joined with the VECTOR_CHUNKS call using left correlation as shown in the following example:

```
CREATE TABLE documentation_tab (
  id   NUMBER,
  text VARCHAR2(2000));

INSERT INTO documentation_tab
   VALUES(1, 'sample');

COMMIT;

SET LINESIZE 100;
SET PAGESIZE 20;
COLUMN pos FORMAT 999;
COLUMN siz FORMAT 999;
COLUMN txt FORMAT a60;

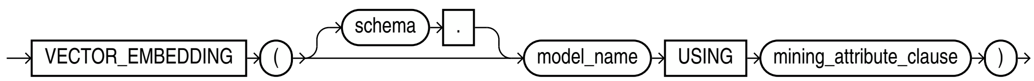
PROMPT SQL VECTOR_CHUNKS
SELECT D.id id, C.chunk_offset pos, C.chunk_length siz, C.chunk_text txt
FROM documentation_tab D, VECTOR_CHUNKS(D.text
                                  BY words
                                  MAX 200
                                  OVERLAP 10
                                  SPLIT BY recursively
                                  LANGUAGE american
                                  NORMALIZE all) C;
```

> **See Also**
>
> - For a complete set of examples on each of the chunking parameters listed in the preceding table, see *Explore Chunking Techniques and Examples of the AI Vector Search User's Guide.*
> - To run an end-to-end example scenario using this function, see *Convert Text to Chunks With Custom Chunking Specifications of the AI Vector Search User's Guide.*

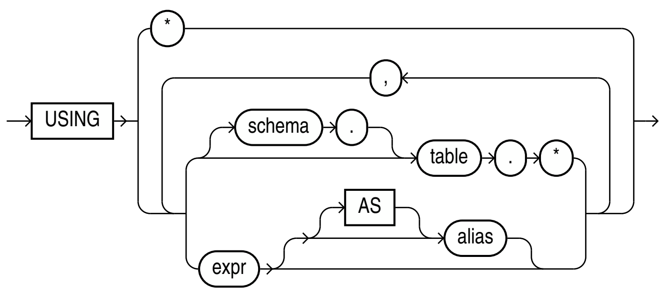
VECTOR_EMBEDDING

Use `VECTOR_EMBEDDING` to generate a single vector embedding for different data types using embedding or feature extraction machine learning models.

Syntax



***mining_attribute_clause*::=**



Purpose

The function accepts the following types as input:

`VARCHAR2` for text embedding models. Oracle automatically converts any other type to `VARCHAR2` except for `NCLOB`, which is automatically converted to `NVARCHAR2`. Oracle does not expect values whose textual representation exceeds the maximum size of a `VARCHAR2`, since embedding models support only text that translates to a couple of thousand tokens. An attribute with a type that has no conversion to `VARCHAR2` results in a SQL compilation error.

For feature extraction models Oracle Machine Learning for SQL supports standard Oracle data types except `DATE`, `TIMESTAMP`, `RAW`, and `LONG`. Oracle Machine Learning supports date type (datetime, date, timestamp) for case_id, `CLOB`/`BLOB`/`FILE` that are interpreted as text columns, and the following collection types as well:

- `DM_NESTED_CATEGORICALS`
- `DM_NESTED_NUMERICALS`
- `DM_NESTED_BINARY_DOUBLES`
- `DM_NESTED_BINARY_FLOATS`

> **Note**
>
> For image `BLOB` input, standalone baseline 8-bit lossy JPEG images are accepted. Unsupported or malformed image input does not produce a successful image decode. At SQL scoring time, this can result in a `NULL` embedding result. Perform any required image preprocessing or format conversion externally before embedding creation.
>
> For more information about image input restrictions, see ONNX Pipeline Models: Image Embedding.

The function always returns a `VECTOR` type, whose dimension is dictated by the model itself. The model stores the dimension information in metadata within the data dictionary.

You can use `VECTOR_EMBEDDING` in `SELECT` clauses, in predicates, and as an operand for SQL operations accepting a `VECTOR` type.

Parameters

`model_name` refers to the name of the imported embedding model that implements the embedding machine learning function.

mining_attribute_clause

- The *`mining_attribute_clause`* argument identifies the column attributes to use as predictors for scoring. This is used as a convenience, as the embedding operator only accepts single input value.
- `USING *` : all the relevant attributes present in the input (supplied in JSON metadata) are used. This is used as a convenience. For an embedding model, the operator only takes one input value as embedding models have only one column.
- `USING` *`expr`* [`AS` *`alias`*] [, *`expr`* [`AS` *`alias`*]] : all the relevant attributes present in the comma-separated list of column expressions are used. This syntax is consistent with the syntax of other machine learning operators. You may specify more than one attribute, however, the embedding model only takes one relevant input. Therefore, you must specify a single mining attribute.

Example

The following example generates vector embeddings with "hello" as the input, utilizing the pretrained ONNX format model `my_embedding_model.onnx` imported into the Database. For complete example, see Import ONNX Models and Generate Embeddings

```
SELECT VECTOR_EMBEDDING(model USING 'hello' as data) AS embedding;
--------------------------------------------------------------------------------
[-9.76553112E-002,-9.89954844E-002,7.69771636E-003,-4.16760892E-003,-9.69305634E-002,
-3.01141385E-002,-2.63396613E-002,-2.98553891E-002,5.96499592E-002,4.13885899E-002,
5.32859489E-002,6.57707453E-002,-1.47056757E-002,-4.18472625E-002,4.1588001E-002,
-2.86354572E-002,-7.56499246E-002,-4.16395674E-003,-1.52879998E-001,6.60010576E-002,
-3.9013084E-002,3.15719917E-002,1.2428958E-002,-2.47651711E-002,-1.16851285E-001,
-7.82847106E-002,3.34323719E-002,8.03267583E-002,1.70483496E-002,-5.42407483E-002,
6.54291287E-002,-4.81935125E-003,6.11041225E-002,6.64106477E-003,-5.47
```

See Also

- *Data Requirements for Machine Learning*
- *Vector Distance Metrics*

Constructors, Converters, Descriptors, and Arithmetic Operators

Other basic vector operations for Oracle AI Vector Search involve creating, converting, and describing vectors.

- Vector Constructors
  `TO_VECTOR()` and `VECTOR()` are synonymous constructors of vectors. The functions take a string of type `VARCHAR2` or `CLOB` as input and return a vector as output.
- Vector Serializers
  `FROM_VECTOR()` and `VECTOR_SERIALIZE()` are synonymous serializers of vectors. The functions take a vector as input and return a string of type `VARCHAR2` or `CLOB` as output.
- VECTOR_NORM
  `VECTOR_NORM` returns the Euclidean norm of a vector as a `BINARY_DOUBLE`. This value is also called the magnitude or size and represents the Euclidean distance between the vector and the origin. For a vector `v = (v1, v2, ..., vn)`, the Euclidean norm is given by $||v|| = SQRT(v1^2 + v2^2 + ... + vn^2)$.
- VECTOR_DIMENSION_COUNT
  `VECTOR_DIMENSION_COUNT` returns the number of dimensions of a vector as a `NUMBER`.
- VECTOR_DIMS
  `VECTOR_DIMS` returns the number of dimensions of a vector as a `NUMBER`. `VECTOR_DIMS` is synonymous with `VECTOR_DIMENSION_COUNT`.
- VECTOR_DIMENSION_FORMAT
  `VECTOR_DIMENSION_FORMAT` returns the storage format of the vector. It returns a `VARCHAR2`, which can be one of the following values: `INT8`, `FLOAT32`, `FLOAT64`, or `BINARY`.
- Arithmetic Operators
  Addition, subtraction, and multiplication can be applied to vectors dimension-wise in SQL and PL/SQL.
- Aggregate Functions
  The aggregate functions `SUM` and `AVG` can be applied to a series of vectors dimension-wise.

Vector Constructors

`TO_VECTOR()` and `VECTOR()` are synonymous constructors of vectors. The functions take a string of type `VARCHAR2` or `CLOB` as input and return a vector as output.

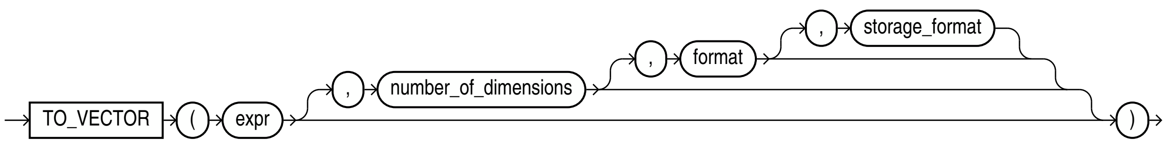
- TO_VECTOR
  `TO_VECTOR` is a constructor that takes a string of type `VARCHAR2`, `CLOB`, `BLOB`, or `JSON` as input, converts it to a vector, and returns a vector as output. `TO_VECTOR` also takes another vector as input, adjusts its format, and returns the adjusted vector as output. `TO_VECTOR` is synonymous with `VECTOR`.
- VECTOR
  `VECTOR` is synonymous with `TO_VECTOR`.

- [VECTOR_ORIGIN]
  The VECTOR_ORIGIN operator generates a vector in which all dimensions are zero, representing the origin point in an n-dimensional graph.
- [VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW]
  The VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW operators generate a vector with all dimension values as some random value, representing a random point in an n-dimensional graph.

TO_VECTOR

TO_VECTOR is a constructor that takes a string of type VARCHAR2, CLOB, BLOB, or JSON as input, converts it to a vector, and returns a vector as output. TO_VECTOR also takes another vector as input, adjusts its format, and returns the adjusted vector as output. TO_VECTOR is synonymous with VECTOR.

Syntax



Parameters

- *expr* must evaluate to one of:
  – A string (of character types or CLOB) that represents a vector.
  – A VECTOR.
  – A BLOB. The BLOB must represent the vector's binary bytes.
  – A JSON array. All elements in the array must be numeric.

  If *expr* is NULL, the result is NULL.

  The string representation of the vector must be in the form of an array of non-null numbers enclosed with a bracket and separated by commas, such as `[1, 3.4, -05.60, 3e+4]`. TO_VECTOR converts a valid string representation of a vector to a vector in the format specified. If no format is specified the default format is used.
- *number_of_dimensions* must be a numeric value that describes the number of dimensions of the vector to construct. The number of dimensions may also be specified as an asterisk (*), in which case the dimension is determined by *expr*.
- *format* must be one of the following tokens: INT8, FLOAT32, FLOAT64, BINARY, or *. This is the target internal storage format of the vector. If * is used, the format will be FLOAT32.

  Note that this behavior is different from declaring a vector column. When you declare a column of type `VECTOR(3, *)`, then all inserted vectors will be stored as is without a change in format.
- *storage_format* must be one of the following tokens: DENSE, SPARSE, or *. If no storage format is specified or if * is used, the following will be observed depending on the input type:
  – Textual input: the storage format will default to DENSE.

- JSON input: the storage format will default to DENSE.
- VECTOR input: there is no default and the storage format is not changed.
- BLOB input: there is no default and the storage format is not changed.

Examples

```
SELECT TO_VECTOR('[34.6, 77.8]');

TO_VECTOR('[34.6,77.8]')
----------------------------------------------------------
[3.45999985E+001,7.78000031E+001]
```

```
SELECT TO_VECTOR('[34.6, 77.8]', 2, FLOAT32);

TO_VECTOR('[34.6,77.8]',2,FLOAT32)
----------------------------------------------------------
[3.45999985E+001,7.78000031E+001]
```

```
SELECT TO_VECTOR('[34.6, 77.8, -89.34]', 3, FLOAT32);

TO_VECTOR('[34.6,77.8,-89.34]',3,FLOAT32)
------------------------------------------------------------
[3.45999985E+001,7.78000031E+001,-8.93399963E+001]
```

```
SELECT TO_VECTOR('[34.6, 77.8, -89.34]', 3, FLOAT32, DENSE);

TO_VECTOR('[34.6,77.8,-89.34]',3,FLOAT32,DENSE)
----------------------------------------------------------------------
[3.45999985E+001,7.78000031E+001,-8.93399963E+001]
```

Note

- For applications using Oracle Client libraries prior to 26ai connected to Oracle AI Database 26ai, use the TO_VECTOR function to insert vector data. For example:

  ```
  INSERT INTO vecTab VALUES(TO_VECTOR('[1.1, 2.9, 3.14]'));
  ```

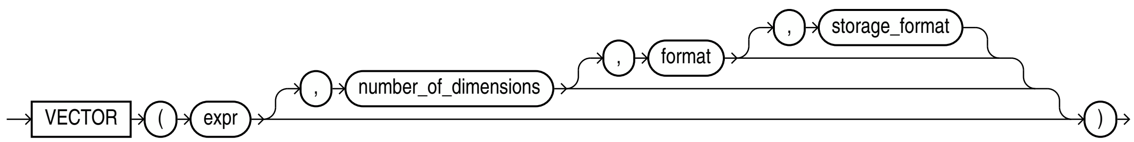
- Applications using Oracle Client 26ai libraries or Thin mode drivers can insert vector data directly as a string or a CLOB. For example:

  ```
  INSERT INTO vecTab VALUES ('[1.1, 2.9, 3.14]');
  ```

VECTOR

`VECTOR` is synonymous with `TO_VECTOR`.

Syntax



Purpose

See TO_VECTOR for semantics and examples.

> **Note**
>
> Applications using Oracle Client 26ai libraries or Thin mode drivers can insert vector data directly as a string or a `CLOB`. For example:
>
> ```
> INSERT INTO vecTab VALUES ('[1.1, 2.9, 3.14]');
> ```

Examples

```
SELECT VECTOR('[34.6, 77.8]');

VECTOR('[34.6,77.8]')
----------------------------------------------------------
[3.45999985E+001,7.78000031E+001]
```

```
SELECT VECTOR('[34.6, 77.8]', 2, FLOAT32);

VECTOR('[34.6,77.8]',2,FLOAT32)
----------------------------------------------------------
[3.45999985E+001,7.78000031E+001]
```

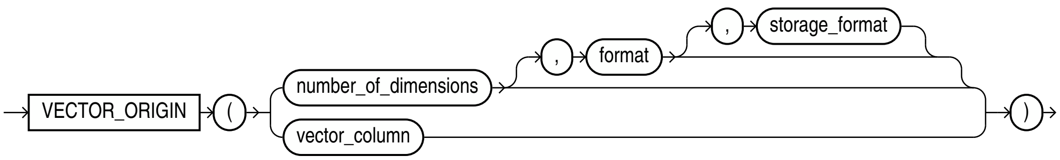
```
SELECT VECTOR('[34.6, 77.8, -89.34]', 3, FLOAT32);

VECTOR('[34.6,77.8,-89.34]',3,FLOAT32)
------------------------------------------------------------
[3.45999985E+001,7.78000031E+001,-8.93399963E+001]
```

VECTOR_ORIGIN

The `VECTOR_ORIGIN` operator generates a vector in which all dimensions are zero, representing the origin point in an n-dimensional graph.

Syntax



Parameters

`VECTOR_ORIGIN` can be used in two different ways. In the first case, a dimension count, element format, and or storage format are provided as parameters. In the second case, you provide only a vector column, from which the dimension and formats are derived.

Table 7-3 VECTOR_ORIGIN Parameters Using Dimension and Formats

Parameter	Accepted Values	Description
`number_of_dimensions`	1 to 65535 (inclusive)	An integer value that describes the number of dimensions of the vector to construct. If the number of dimensions is not provided, or if it is provided as `*`, an error is thrown.
`format`	`INT8`, `FLOAT32`, `FLOAT64`, `BINARY`, or `*`	Optionally provide the element format. If it is not provided, or specified as `*`, the format will default to `FLOAT32`.
`storage_format`	`DENSE` or `SPARSE`	Optionally provide the storage format. If not specified, or provided as `*`, the storage format will default to `DENSE`.

Table 7-4 VECTOR_ORIGIN Parameters Using a Vector Column

Parameter	Accepted Values	Description
`vector_column`	The name of an existing vector column.	The dimension count, format, and storage format are derived from the provided vector column. In the case that the vector column is defined with a flexible dimension count (*), and there is no index on top of the vector column, an error is thrown. If the vector column has a flexible dimension count but *does* have an associated index, the dimension count from the index is used. Otherwise, in the case that the vector column is defined with a valid dimension count, that number is used. If the provided vector column has a flexible format, FLOAT32 is used by default. Otherwise, the format used to define the vector column is used. The vector column's storage format, either DENSE or SPARSE is used.

Examples

```
-- Optional cleanup so the script can be rerun
DROP TABLE IF EXISTS vec_d_tab;
DROP TABLE IF EXISTS vec_s_tab;

-- vec_d_tab: dense vector column A with schema vector(3, float32)
CREATE TABLE vec_d_tab (
  a VECTOR(3, FLOAT32)
);

-- Insert two dense vectors.
-- Using vector_origin here creates a dense zero vector: [0,0,0]
INSERT INTO vec_d_tab VALUES (vector_origin(3, FLOAT32));
INSERT INTO vec_d_tab VALUES (vector_origin(3, FLOAT32));


-- vector_origin using the column metadata
SELECT vector_origin(a) AS dense_origin_from_column
FROM vec_d_tab;

-- vector_origin using explicit dense vector metadata
SELECT vector_origin(3, FLOAT32) AS dense_origin_from_type
FROM vec_d_tab;


-- vec_s_tab: sparse vector column A with schema vector(32000, float32,
sparse)
CREATE TABLE vec_s_tab (
  a VECTOR(32000, FLOAT32, SPARSE)
);
```

```
-- Insert two sparse vectors.
-- Using vector_origin here creates an empty sparse zero vector: [32000,[],[]]
INSERT INTO vec_s_tab VALUES (vector_origin(32000, FLOAT32, SPARSE));
INSERT INTO vec_s_tab VALUES (vector_origin(32000, FLOAT32, SPARSE));


-- vector_origin using the sparse column metadata
SELECT vector_origin(a) AS sparse_origin_from_column
FROM vec_s_tab;

-- vector_origin using explicit sparse vector metadata
SELECT vector_origin(32000, FLOAT32, SPARSE) AS sparse_origin_from_type
FROM vec_s_tab;
```

Example results:

```
DENSE_ORIGIN_FROM_COLUMN
---------------------------------------------------------------------
[0,0,0]
[0,0,0]

DENSE_ORIGIN_FROM_TYPE
---------------------------------------------------------------------
[0,0,0]
[0,0,0]


SPARSE_ORIGIN_FROM_COLUMN
---------------------------------------------------------------------
[32000,[],[]]
[32000,[],[]]

SPARSE_ORIGIN_FROM_TYPE
---------------------------------------------------------------------
[32000,[],[]]
[32000,[],[]]
```

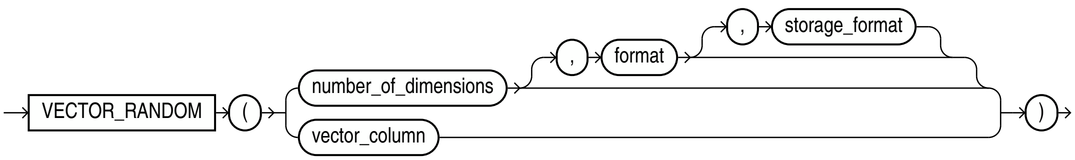
VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW

The `VECTOR_RANDOM` and `VECTOR_RANDOM_PER_ROW` operators generate a vector with all dimension values as some random value, representing a random point in an n-dimensional graph.

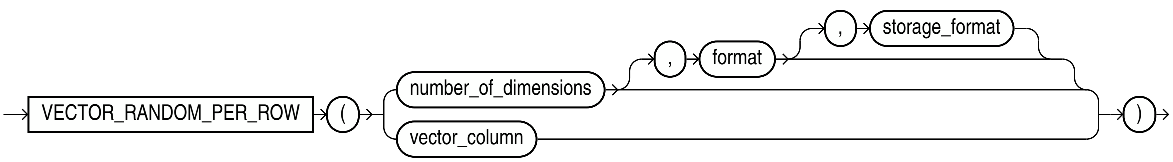
Depending on the operation, you may need the same random vector for all rows in the statement, or you may need different unique random vectors for different rows. For example, when a random vector is used on top of `VECTOR_DISTANCE`, the same random vector must be used as the query vector to compare against each vector in the source table. Alternately, if you want to insert synthetic data into a vector column across multiple rows, you would expect different values for each vector. This is the benefit of these two operators, with `VECTOR_RANDOM` generating the same random vector for all rows in the statement and `VECTOR_RANDOM_PER_ROW` generating unique random vectors for different rows.

Syntax

`VECTOR_RANDOM` Syntax:



`VECTOR_RANDOM_PER_ROW` Syntax:



Parameters

`VECTOR_RANDOM` and `VECTOR_RANDOM_PER_ROW` can be used in two different ways. In the first case, a dimension count, element format, and or storage format are provided as parameters. In the second case, you provide only a vector column, from which the dimension and formats are derived.

Table 7-5 VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW Parameters Using Dimension and Formats

Parameter	Accepted Values	Description
`number_of_dimensions`	1 to 65535 (inclusive)	An integer value that describes the number of dimensions of the vector to construct. If the number of dimensions is not provided, or if it is provided as `*`, an error is thrown. The potential dimension values, depending on element format, are as follows: • `INT8`: -128 to 127 • `FLOAT32` and `FLOAT64`: -SQRT(<float_max_value>) to SQRT(<float_max_value>), where the float max value is $3.4028235 * 10^{38}$. • `BINARY`: 0 to 255 for `DENSE` vectors where each value represents 8 dimensions. No dimension values for `SPARSE` vectors, as sparse indices alone suffice.

Table 7-5 (Cont.) VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW Parameters Using Dimension and Formats

Parameter	Accepted Values	Description
`format`	`INT8`, `FLOAT32`, `FLOAT64`, `BINARY`, or `*`	Optionally provide the element format. If it is not provided, or specified as `*`, the format will default to `FLOAT32`.
`storage_format`	`DENSE` or `SPARSE`	Optionally provide the storage format. If not specified, or provided as `*`, the storage format will default to `DENSE`. If specified as `SPARSE`, all components of the generated `SPARSE` vector will be random. This includes the total sparse dimension count, sparse index array, and sparse index values.

Table 7-6 VECTOR_RANDOM and VECTOR_RANDOM_PER_ROW Parameters Using a Vector Column

Parameter	Accepted Values	Description
`vector_column`	The name of an existing vector column.	The dimension count, format, and storage format are derived from the provided vector column. In the case that the vector column is defined with a flexible dimension count (`*`), and there is no index on top of the vector column, an error is thrown. If the vector column has a flexible dimension count but *does* have an associated index, the dimension count from the index is used. Otherwise, in the case that the vector column is defined with a valid dimension count, that number is used. If the provided vector column has a flexible format, `FLOAT32` is used by default. Otherwise, the format used to define the vector column is used. The vector column's storage format, either `DENSE` or `SPARSE` is used. If the vector column is `SPARSE`, all components of the generated `SPARSE` vector will be random. This includes the total sparse dimension count, sparse index array, and sparse index values.

Examples

Examples using `DENSE` vectors:

```
-- Optional cleanup so the script can be rerun
DROP TABLE IF EXISTS vec_rand_d_tab;
```

```
-- vec_rand_d_tab: dense vector column A with schema vector(3, float32)
CREATE TABLE vec_rand_d_tab (
  a VECTOR(3, FLOAT32)
);

-- Insert three dense vectors.
-- The actual values do not matter for vector_random(a);
-- the column metadata is used to produce a random vector
-- with the same vector format.
INSERT INTO vec_rand_d_tab VALUES ('[1,2,3]');
INSERT INTO vec_rand_d_tab VALUES ('[4,5,6]');
INSERT INTO vec_rand_d_tab VALUES ('[7,8,9]');

-- vector_random using the column metadata.
-- This returns the same random vector for every row in the query result.
SELECT vector_random(a) AS random_from_column
FROM vec_rand_d_tab;

-- vector_random_per_row using the column metadata.
-- This returns a different random vector for each row in the query result.
SELECT vector_random_per_row(a) AS random_per_row_from_column
FROM vec_rand_d_tab;
```

Example results:

```
RANDOM_FROM_COLUMN
---------------------------------------------------------------------
[-4.55028462E+018,-1.52599174E+019,-2.17310863E+017]
[-4.55028462E+018,-1.52599174E+019,-2.17310863E+017]
[-4.55028462E+018,-1.52599174E+019,-2.17310863E+017]

RANDOM_PER_ROW_FROM_COLUMN
---------------------------------------------------------------------
[-1.19015746E+019,1.58563222E+019,-7.81389379E+018]
[-1.44516304E+018,1.29408571E+019,9.86734995E+017]
[3.46542865E+018,8.62223824E+018,-1.08430978E+018]
```

Examples using `SPARSE` vectors:

```
-- Optional cleanup so the script can be rerun
DROP TABLE IF EXISTS vec_rand_s_tab;

-- vec_rand_s_tab: sparse vector column A with schema vector(64, float32,
sparse)
CREATE TABLE vec_rand_s_tab (
  a VECTOR(64, FLOAT32, SPARSE)
);

-- Insert three sparse vectors.
-- The actual values do not matter for vector_random(a);
-- the column metadata is used to produce a random sparse vector
-- with the same vector format.
INSERT INTO vec_rand_s_tab VALUES ('[64,[1,10],[1.0,2.0]]');
INSERT INTO vec_rand_s_tab VALUES ('[64,[20,30],[3.0,4.0]]');
```

```
INSERT INTO vec_rand_s_tab VALUES ('[64,[40,50],[5.0,6.0]]');

-- vector_random using the sparse column metadata.
-- This returns the same random sparse vector for every row in the query
result.
SELECT vector_random(a) AS random_from_sparse_column
FROM vec_rand_s_tab;

-- vector_random_per_row using the sparse column metadata.
-- This returns a different random sparse vector for each row in the query
result.
SELECT vector_random_per_row(a) AS random_per_row_from_sparse_column
FROM vec_rand_s_tab;

-- vector_random using explicit sparse vector metadata.
-- This returns the same random sparse vector for every row in the query
result.
SELECT vector_random(64, FLOAT32, SPARSE) AS random_from_sparse_type
FROM vec_rand_s_tab;

-- vector_random_per_row using explicit sparse vector metadata.
-- This returns a different random sparse vector for each row in the query
result.
SELECT vector_random_per_row(64, FLOAT32, SPARSE) AS
random_per_row_from_sparse_type
FROM vec_rand_s_tab;
```

Example results:

```
RANDOM_FROM_SPARSE_COLUMN
------------------------------------------------------------------------------
--
[64,[35,39],[-1.44835412E+019,1.14759327E+019]]
[64,[35,39],[-1.44835412E+019,1.14759327E+019]]
[64,[35,39],[-1.44835412E+019,1.14759327E+019]]

RANDOM_PER_ROW_FROM_SPARSE_COLUMN
------------------------------------------------------------------------------
--
[64,[54],[4.25011217E+018]]
[64,[54,55,59],[4.07524144E+018,1.5152826E+019,-1.03148396E+019]]
[64,[13,16,17,26,28],
[-1.17318276E+019,-1.27177409E+019,1.80867442E+019,-2.57617389E+018,-4.0359413
2E+018]]

RANDOM_FROM_SPARSE_TYPE
------------------------------------------------------------------------------
--
[64,[2,7,23,53],
[-1.35317193E+019,1.50075707E+019,-1.14627595E+019,1.18382108E+019]]
[64,[2,7,23,53],
[-1.35317193E+019,1.50075707E+019,-1.14627595E+019,1.18382108E+019]]
[64,[2,7,23,53],
[-1.35317193E+019,1.50075707E+019,-1.14627595E+019,1.18382108E+019]]
```

```
RANDOM_PER_ROW_FROM_SPARSE_TYPE
--------------------------------------------------------------------------------
--
[64,[21,22,27,31,34,51],
[4.60698918E+018,-1.49234635E+019,2.2355166E+018,1.6858328E+019,4.37048918E+01
8,1.79236877E+019]]
[64,[41,63],[-3.73847917E+018,4.69414293E+017]]
[64,[3,10,25,34],
[2.04608682E+018,1.7566474E+019,6.20288385E+018,1.00214295E+019]]
```

Vector Serializers

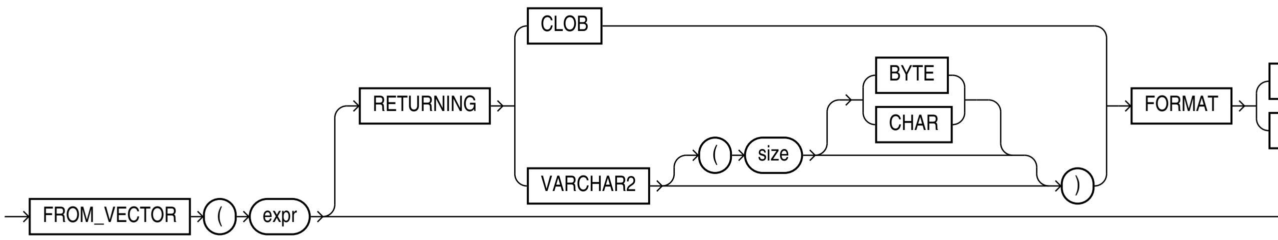
`FROM_VECTOR()` and `VECTOR_SERIALIZE()` are synonymous serializers of vectors. The functions take a vector as input and return a string of type `VARCHAR2` or `CLOB` as output.

- FROM_VECTOR
  `FROM_VECTOR` takes a vector as input and returns a string of type `VARCHAR2` or `CLOB` as output.
- VECTOR_SERIALIZE
  `VECTOR_SERIALIZE` is synonymous with `FROM_VECTOR`.

FROM_VECTOR

`FROM_VECTOR` takes a vector as input and returns a string of type `VARCHAR2` or `CLOB` as output.

Syntax



Purpose

`FROM_VECTOR` optionally takes a `RETURNING` clause to specify the data type of the returned value.

If `VARCHAR2` is specified without size, the size of the returned value size is 32767.

In SQL, you can optionally specify the storage format of the input vector in the `FORMAT` clause, using the tokens `SPARSE` or `DENSE`. The return storage format cannot be specified in PL/SQL; rather, the output format is determined by the storage format of the input vector.

There is no support to convert to `CHAR`, `NCHAR`, and `NVARCHAR2`.

`FROM_VECTOR` is synonymous with `VECTOR_SERIALIZE`.

Parameters

`expr` must evaluate to a vector. The function returns NULL if *`expr`* is NULL.

Examples

- ```
  SELECT FROM_VECTOR(TO_VECTOR('[1, 2, 3]') );
  ```

  Result:

  ```
  FROM_VECTOR(TO_VECTOR('[1,2,3]'))

  ----------------------------------------------------------------
  [1.0E+000,2.0E+000,3.0E+000]

  1 row selected.
  ```

- ```
  SELECT FROM_VECTOR(TO_VECTOR('[1.1, 2.2, 3.3]', 3, FLOAT32) );
  ```

  Result:

  ```
  FROM_VECTOR(TO_VECTOR('[1.1,2.2,3.3]',3,FLOAT32))
  ------------------------------------------------------------------

  [1.10000002E+000,2.20000005E+000,3.29999995E+000]

  1 row selected.
  ```

- ```
  SELECT FROM_VECTOR( TO_VECTOR('[1.1, 2.2, 3.3]', 3, FLOAT32) RETURNING
  VARCHAR2(1000));
  ```

  Result:

  ```
  FROM_VECTOR(TO_VECTOR('[1.1,2.2,3.3]',3,FLOAT32)RETURNINGVARCHAR2(1000))
  ---------------------------------------------------------------------------
  -----

  [1.10000002E+000,2.20000005E+000,3.29999995E+000]

  1 row selected.
  ```

- ```
  SELECT FROM_VECTOR(TO_VECTOR('[1.1, 2.2, 3.3]', 3, FLOAT32) RETURNING
  CLOB );
  ```

  Result:

  ```
  FROM_VECTOR(TO_VECTOR('[1.1,2.2,3.3]',3,FLOAT32)RETURNINGCLOB)

  ---------------------------------------------------------------------------
  -----

  [1.10000002E+000,2.20000005E+000,3.29999995E+000]

  1 row selected.
  ```

Note

- Applications using Oracle Client 26ai libraries or Thin mode drivers can fetch vector data directly, as shown in the following example:

  ```
  SELECT dataVec FROM vecTab;
  ```

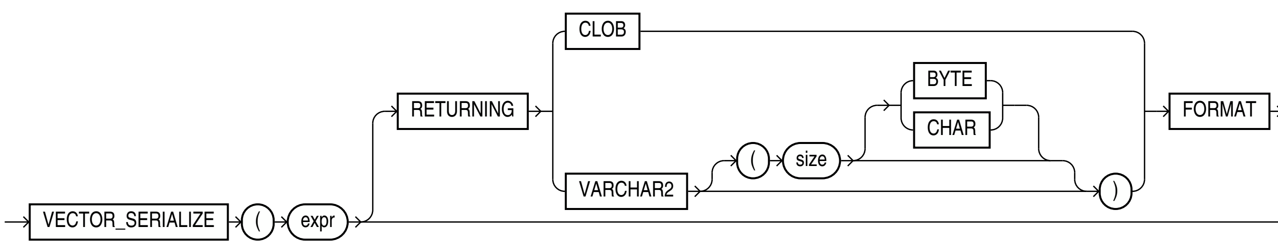
- For applications using Oracle Client 26ai libraries prior to 26ai connected to Oracle AI Database 26ai, use the `FROM_VECTOR` to fetch vector data, as shown by the following example:

  ```
  SELECT FROM_VECTOR(dataVec) FROM vecTab;
  ```

VECTOR_SERIALIZE

`VECTOR_SERIALIZE` is synonymous with `FROM_VECTOR`.

Syntax



Purpose

See FROM_VECTOR for semantics and examples.

Examples

- ```
  SELECT VECTOR_SERIALIZE(VECTOR('[1.1,2.2,3.3]',3,FLOAT32));
  ```

  Result:

  ```
  VECTOR_SERIALIZE(VECTOR('[1.1,2.2,3.3]',3,FLOAT32))
  ---------------------------------------------------------------
  [1.10000002E+000,2.20000005E+000,3.29999995E+000]

  1 row selected.
  ```

- ```
  SELECT VECTOR_SERIALIZE(VECTOR('[1.1, 2.2, 3.3]',3,FLOAT32) RETURNING
  VARCHAR2(1000));
  ```

  Result:

  ```
  VECTOR_SERIALIZE(VECTOR('[...]',3,FLOAT32)RETURNINGVARCHAR2(1000))
  ------------------------------------------------------------------
  [1.10000002E+000,2.20000005E+000,3.29999995E+000]
  ```

```
1 row selected.
```

- SELECT VECTOR_SERIALIZE(VECTOR('[1.1, 2.2, 3.3]',3,FLOAT32) RETURNING CLOB);

Result:

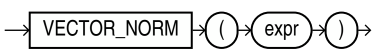
```
VECTOR_SERIALIZE(VECTOR('[1.1, 2.2, 3.3]',3,FLOAT32)RETURNINGCLOB)
--------------------------------------------------------
[1.10000002E+000,2.20000005E+000,3.29999995E+000]

1 row selected.
```

VECTOR_NORM

`VECTOR_NORM` returns the Euclidean norm of a vector as a `BINARY_DOUBLE`. This value is also called the magnitude or size and represents the Euclidean distance between the vector and the origin. For a vector `v = (v1, v2, ..., vn)`, the Euclidean norm is given by $||v|| = SQRT(v1^2 + v2^2 + ... + vn^2)$.

Syntax



Parameters

`expr` must evaluate to a vector.

If *`expr`* is NULL, NULL is returned.

Example

```
SELECT VECTOR_NORM( TO_VECTOR('[4, 3]', 2, FLOAT32) );

VECTOR_NORM(TO_VECTOR('[4,3]',2,FLOAT32))
_________________________________________
5.0
```

VECTOR_DIMENSION_COUNT

`VECTOR_DIMENSION_COUNT` returns the number of dimensions of a vector as a `NUMBER`.

Syntax

Purpose

`VECTOR_DIMENSION_COUNT` is synonymous with [VECTOR_DIMS](#).

Parameters

`expr` must evaluate to a vector.

If *`expr`* is NULL, NULL is returned.

Example

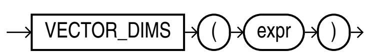
```
SELECT VECTOR_DIMENSION_COUNT( TO_VECTOR('[34.6, 77.8]', 2, FLOAT64) );

VECTOR_DIMENSION_COUNT(TO_VECTOR('[34.6,77.8]',2,FLOAT64))
----------------------------------------------------------
2
```

VECTOR_DIMS

`VECTOR_DIMS` returns the number of dimensions of a vector as a `NUMBER`. `VECTOR_DIMS` is synonymous with `VECTOR_DIMENSION_COUNT`.

Syntax



Purpose

Refer to [VECTOR_DIMENSION_COUNT](#) for full semantics.

VECTOR_DIMENSION_FORMAT

`VECTOR_DIMENSION_FORMAT` returns the storage format of the vector. It returns a `VARCHAR2`, which can be one of the following values: `INT8`, `FLOAT32`, `FLOAT64`, or `BINARY`.

Syntax

Parameters

`expr` must evaluate to a vector.

If *`expr`* is NULL, NULL is returned.

Examples

```
SELECT VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6, 77.8]', 2, FLOAT64));
FLOAT64
```

```
SELECT VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6, 77.8, 9]', 3, FLOAT32));
FLOAT32

SELECT VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6, 77.8, 9, 10]', 3, INT8));
SELECT VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6, 77.8, 9, 10]', 3,
INT8))
                                                                       *
ERROR at line 1:
ORA-51803: Vector dimension count must match the dimension count specified in the column
definition (actual: 4, required: 3).

SELECT VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6, 77.8, 9.10]', 3, INT8));
VECTOR_DIMENSION_FORMAT(TO_VECTOR('[34.6,77.8,9.10
--------------------------------------------------
INT8

SELECT TO_VECTOR('[34.6, 77.8, 9.10]', 3, INT8);
TO_VECTOR('[34.6,77.8,9.10]',3,INT8)
---------------------------------------------------
[3.5E+001,7.8E+001,9.0E+000]
```

Arithmetic Operators

Addition, subtraction, and multiplication can be applied to vectors dimension-wise in SQL and PL/SQL.

Addition

Vector addition is often used in Natural Language Processing (NLP) where words are often represented as vectors that capture their meaning in a numerical format. Vector addition is used to combine these meanings and understand relationships between words, a task also referred to as word analogy.

Given two vectors `A=(a1, a2, a3)` and `B=(b1, b2, b3)`, `C = A + B` is computed as `C = (a1+b1, a2+b2, a3+b3)`.

Subtraction

Vector subtraction can be used in word analogy scenarios but is also useful in the context of facial recognition. Each face can be represented by a vector of facial features (distance between eyes, eye color, and so on). Subtracting one vector from another gives you the main differences between the two faces, giving you the information needed to recognize whether they are similar or not.

Given two vectors `A=(a1, a2, a3)` and `B=(b1, b2, b3)`, `C = A - B` is computed as `C = (a1-b1, a2-b2, a3-b3)`.

Multiplication

Vector multiplication of each corresponding coordinate of two vectors, or element-wise product, is called the Hadamard product. The Hadamard product is often used in neural networks and computer vision.

Given two vectors `A=(a1, a2, a3)` and `B=(b1, b2, b3)`, the Hadamard product `A*B` is computed as `A*B = (a1*b1, a2*b2, a3*b3)`.

Semantics

Both sides of the operation must evaluate to vectors with matching dimensions and must not be `BINARY` or `SPARSE` vectors. The resulting vector has the same number of dimensions as the operands and the format is determined based on the formats of the inputs. If one side of the operation is not a vector, an attempt is made automatically to convert the value to a vector. If the conversion fails, an error is raised.

The format used for the result is ranked in the following order: flexible, `FLOAT64`, `FLOAT32`, then `INT8`. As in, if either side of the operation has a flexible format, the result will be flexible, otherwise, if either side has the format `FLOAT64`, the result will be `FLOAT64`, and so on.

Consider two vectors with the following values:

```
v1 = [1, 2, 3]
v2 = [10, 20, 30]
```

Using arithmetic operators on `v1` and `v2` would, for example, result in the following:

- `v1 + v2` is `[11, 22, 33]`
- `v1 - v2` is `[-9, -18, -27]`
- `v1 * v2` is `[10, 40, 90]`
- `v1 + NULL` is `NULL`

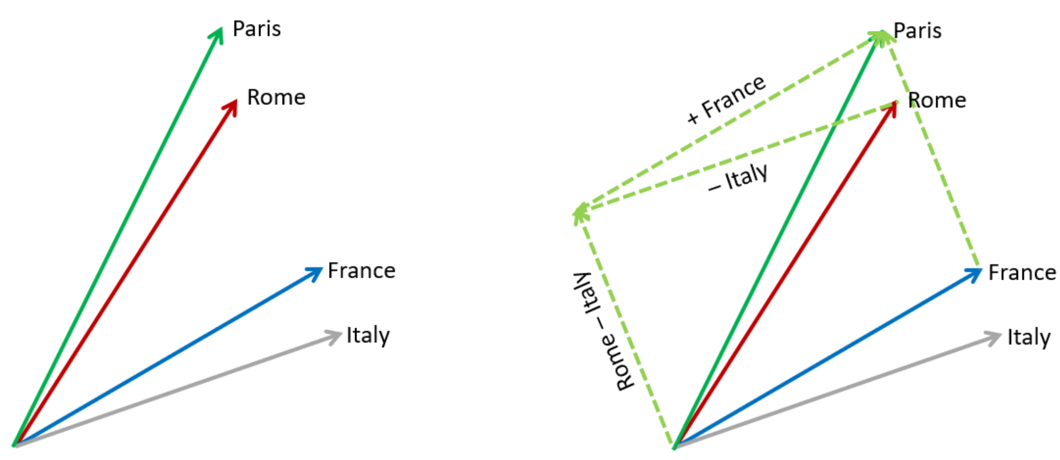
If either side of the arithmetic operation is `NULL`, the result is `NULL`. In the case of dimension overflow, an error is raised. For example, adding `VECTOR('[1, 127]', 2, INT8)` to `VECTOR('[1, 1]', 2, INT8)` results in an error because `127+1=128`, which overflows the `INT8` format.

The use of division operators on vectors is not supported.

Examples

Word Analogy

Using word embeddings, suppose you want to find the relationship between "Rome" and "Paris". You can take the vector for "Rome", subtract the vector for "Italy", and then add the vector for "France". This results in a new vector that approximates the meaning of the word "Paris". The calculation should be "Rome - Italy + France = Paris" using an ideal embedding model.



Rome – Italy + France = Paris

Basic Vector Arithmetic

```
SELECT VECTOR('[5, 10, 15]') - VECTOR('[2, 4, 6]');

VECTOR('[5,10,15]')-VECTOR('[2,4,6]')
------------------------------------------------------------
[3.0E+000,6.0E+000,9.0E+000]


SELECT VECTOR('[1, 2, 3]', 3, FLOAT64) + VECTOR('[4, 5, 6]', 3, FLOAT32) *
'[2, 2, 2]';

VECTOR('[1,2,3]',3,FLOAT64)+VECTOR('[4,5,6]',3,FLOAT32)*'[2,2,2]'
------------------------------------------------------------------------------
--
[9.0E+000,1.2E+001,1.5E+001]


DECLARE
  v1 VECTOR := VECTOR('[10, 20, 30]', 3, INT8);
  v2 VECTOR := VECTOR('[6, 4, 2]', 3, INT8);
BEGIN
  DBMS_OUTPUT.PUT_LINE(TO_CHAR(v1 + v2));
  DBMS_OUTPUT.PUT_LINE(TO_CHAR(v1 - v2));
  DBMS_OUTPUT.PUT_LINE(TO_CHAR(v1 * v2));
END;
/
```

Result:

```
[16,24,32]
[4,16,28]
[60,80,60]
```

Aggregate Functions

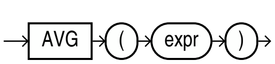
The aggregate functions `SUM` and `AVG` can be applied to a series of vectors dimension-wise.

- AVG
  The `AVG` function takes a vector expression as input and returns the average as a vector with format `FLOAT64`.
- SUM
  The `SUM` function takes a vector expression as input and returns the sum as a vector with format `FLOAT64`.

AVG

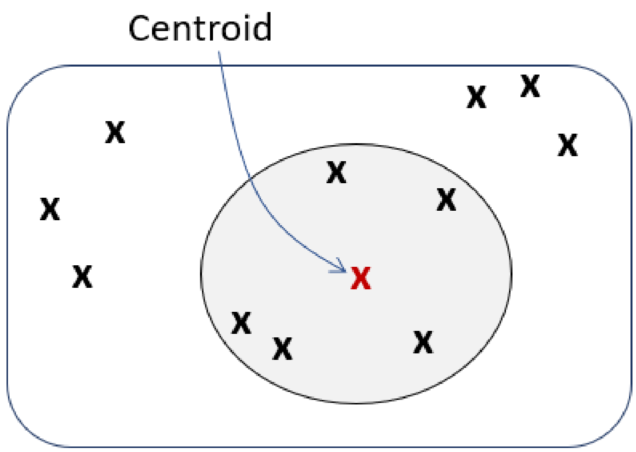
The `AVG` function takes a vector expression as input and returns the average as a vector with format `FLOAT64`.

Syntax



Purpose

AVG is mainly used to create an overall representation (as in a centroid) for a vector set. In applications like Natural Language Processing (NLP), you can compute the average of several vectors to create a single centroid or overall representation. For example, to represent a sentence, you might average the word embeddings of each word in the sentence. This can be used for tasks like text classification, document similarity, or clustering.



The result of AVG with a vector expression is the equivalent of consecutively performing vector addition operations on all non-NULL inputs and then dividing by the total number of non-NULL inputs. The returned vector has the same number of dimensions as the input and has the format FLOAT64. When the expression has a flexible number of dimensions, all inputs must have the same number of dimensions within each aggregate group.

The AVG function with vector expressions as input can be used as a single set aggregate or in the GROUP BY clause. Using ROLLUP is also supported. The AVG function accepts vector expressions as input for aggregate operations but cannot currently be applied to analytic operations.

NULL vectors are ignored and are not counted when calculating the average vector. If all inputs within an aggregate group are NULL, the result is NULL for that group. If the result overflows the FLOAT64 maximum value, an error is raised, regardless of the format of the input vector type.

With vector inputs, using DISTINCT, CUBE, and GROUPING SETS is not supported. Also, BINARY and SPARSE vectors cannot be supplied as input.

For the full definition and implementation of the AVG function, see *Oracle AI Database SQL Language Reference*.

```
CREATE TABLE avg_t (v VECTOR, k1 NUMBER, k2 VARCHAR2(100));
INSERT INTO avg_t VALUES ('[2, 4, 6]', 2, 'even');
INSERT INTO avg_t VALUES ('[8, 10, 12]', 2, 'even');
INSERT INTO avg_t VALUES ('[1, 3, 5]', 3, 'odd');
INSERT INTO avg_t VALUES ('[7, 9, 11]', 3, 'odd');


SELECT AVG(v) v_avg FROM avg_t;

V_AVG
---------------------------------------------------
```

```
[4.5E+000,6.5E+000,8.5E+000]


SELECT AVG(v) v_avg, k1 FROM avg_t GROUP BY k1;

V_AVG                                     K1
---------------------------------------  ----------
[5.0E+000,7.0E+000,9.0E+000]             2
[4.0E+000,6.0E+000,8.0E+000]             3

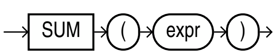

SELECT AVG(v) v_avg FROM avg_t GROUP BY ROLLUP(k1, k2);

V_AVG
---------------------------------------------------
[5.0E+000,7.0E+000,9.0E+000]
[5.0E+000,7.0E+000,9.0E+000]
[4.0E+000,6.0E+000,8.0E+000]
[4.0E+000,6.0E+000,8.0E+000]
[4.5E+000,6.5E+000,8.5E+000]
```

SUM

The SUM function takes a vector expression as input and returns the sum as a vector with format FLOAT64.

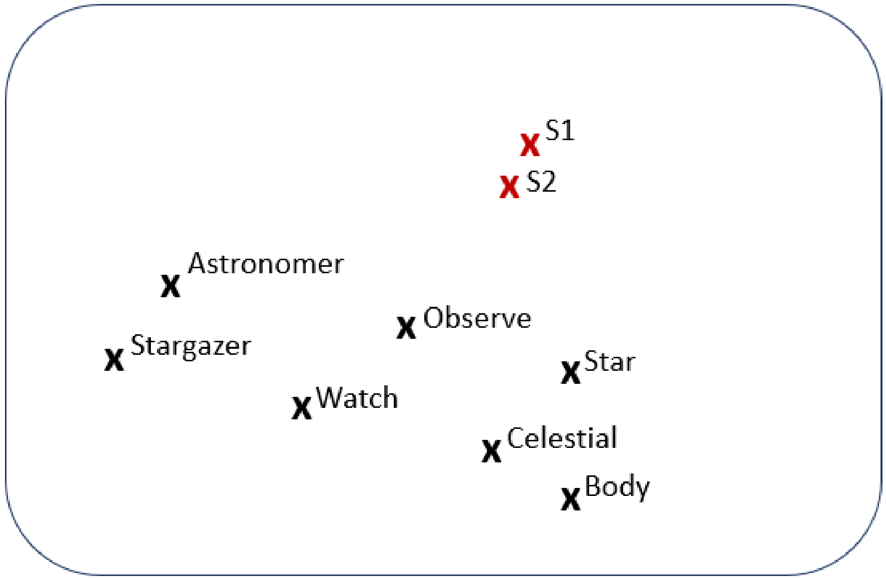
Syntax



Purpose

SUM is mainly used, in Natural Language Processing (NLP), to compute a representation of a sentence or a document. It is common to sum the word embeddings of all words. The resulting vector represents the entire text, allowing models to work with sentences instead of just words.

The following diagram illustrates the vector representation, in a 2D space, of word embeddings. If you take S1, "The astronomer observed the stars", and S2, "The stargazer watched the celestial bodies", you can observe that the corresponding vector sum of each constituting word of each sentence (S1 and S2 in the diagram) are very close, for example, in terms of cosine similarity. This means that the two sentences are very similar in meaning.



S1 = Astronomer + Observe + Star
S2 = Stargazer + Watch + Celestial + Body

The result of `SUM` with a vector expression is the equivalent of consecutively performing vector addition operations on all non-`NULL` inputs. The returned vector has the same number of dimensions as the input and has the format `FLOAT64`. When the expression has a flexible number of dimensions, all inputs must have the same number of dimensions within each aggregate group.

The `SUM` function with vector expressions as input can be used as a single set aggregate or in the `GROUP BY` clause. Using `ROLLUP` is also supported. The `SUM` function accepts vector expressions as input for aggregate operations but cannot currently be applied to analytic operations.

`NULL` vectors are ignored and are not counted when calculating the sum vector. If all inputs within an aggregate group are `NULL`, the result is `NULL` for that group. If the result overflows the `FLOAT64` maximum value, an error is raised, regardless of the format of the input vector type.

With vector inputs, using `DISTINCT`, `CUBE`, and `GROUPING SETS` is not supported. Also, `BINARY` and `SPARSE` vectors cannot be supplied as input.

For the full definition and implementation of the `SUM` function, see *Oracle AI Database SQL Language Reference*.

```
CREATE TABLE sum_t (v VECTOR, k1 NUMBER, k2 VARCHAR2(100));
INSERT INTO sum_t VALUES ('[2, 4, 6]', 2, 'even');
INSERT INTO sum_t VALUES ('[8, 10, 12]', 2, 'even');
INSERT INTO sum_t VALUES ('[1, 3, 5]', 3, 'odd');
INSERT INTO sum_t VALUES ('[7, 9, 11]', 3, 'odd');


SELECT SUM(v) v_sum FROM sum_t;

V_SUM
---------------------------------------------------
```

```
[1.8E+001,2.6E+001,3.4E+001]


SELECT SUM(v) v_sum, k1 FROM sum_t GROUP BY k1;

V_SUM                                     K1
---------------------------------------  ----------
[1.0E+001,1.4E+001,1.8E+001]              2
[8.0E+000,1.2E+001,1.6E+001]              3


SELECT SUM(v) v_sum FROM sum_t GROUP BY ROLLUP(k1, k2);

V_SUM
---------------------------------------------------
[1.0E+001,1.4E+001,1.8E+001]
[1.0E+001,1.4E+001,1.8E+001]
[8.0E+000,1.2E+001,1.6E+001]
[8.0E+000,1.2E+001,1.6E+001]
[1.8E+001,2.6E+001,3.4E+001]


CREATE TABLE sum_diff_dim_t (v VECTOR, k1 NUMBER, k2 VARCHAR2(100));
INSERT INTO sum_diff_dim_t VALUES ('[2, 4, 6]', 2, 'even');
INSERT INTO sum_diff_dim_t VALUES ('[8, 10, 12]', 2, 'even');
INSERT INTO sum_diff_dim_t VALUES ('[1, 3, 5, 7]', 3, 'odd');
INSERT INTO sum_diff_dim_t VALUES ('[9, 11, 13, 15]', 3, 'odd');


SELECT SUM(v) v_sum, k2 FROM sum_diff_dim_t GROUP BY k2;

V_SUM                                     K2
---------------------------------------  ----------
[1.0E+001,1.4E+001,1.8E+001]              even
[1.0E+001,1.4E+001,1.8E+001,2.2E+001]     odd


SELECT SUM(v) v_sum FROM sum_diff_dim_t;

ERROR:
ORA-51808: SUM(vector) requires all vectors to have the same dimension count.
Encountered (3, 4).
```

JSON Compatibility with the VECTOR Data Type

The JSON data type supports the `VECTOR` type as a JSON scalar type. A `VECTOR` instance is convertible to a JSON type instance and vice versa using the JSON constructor and the `VECTOR` constructor, respectively.

Note that when converting a JSON array to a `VECTOR` type instance, the numbers in the JSON array do not have to be of the same numeric type. The input data types do not change the default numeric type for the vector and the `VECTOR` constructor includes an argument to set the format. Using the `VECTOR` constructor (or `TO_VECTOR`) with a JSON data type instance will succeed as long as there is no conversion error.

The following SQL/JSON functions and item methods are compatible for use with the `VECTOR` data type:

- `JSON_VALUE`
- `JSON_OBJECT`
- `JSON_ARRAY`
- `JSON_TABLE`
- `JSON_SCALAR`
- `JSON_TRANSFORM`
- `JSON_SERIALIZE`
- `type()`
- `.vector()`
- `.string()`
- `.stringify()`

See Also

- *Oracle AI Database JSON Developer’s Guide* for more information about `VECTOR` as a data type for JSON data
- *Oracle AI Database JSON Developer’s Guide* for information about SQL/JSON path expression item methods, including `vector()`

8 Query Data With Similarity and Hybrid Searches

Use Oracle AI Vector Search native SQL operations from your development environment to combine similarity with relational searches.

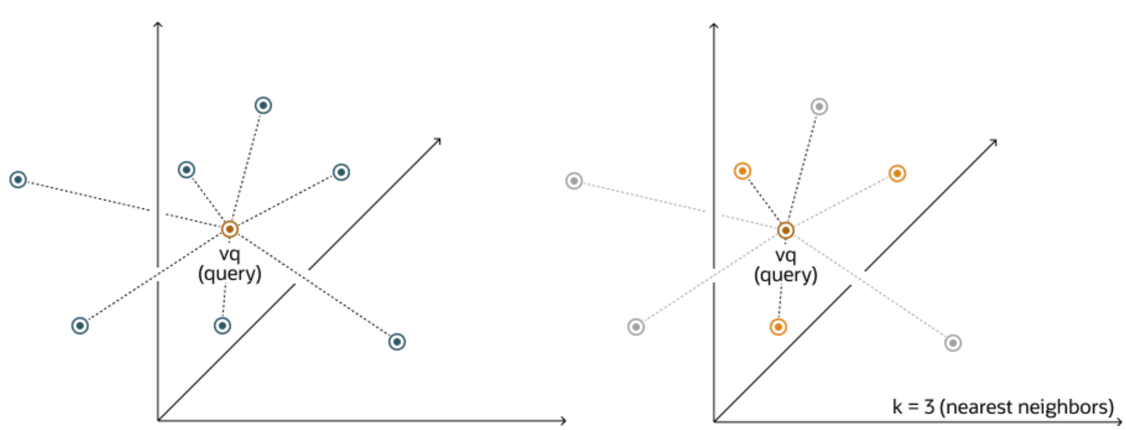
- Perform Exact Similarity Search
  A similarity search looks for the relative order of vectors compared to a query vector. Naturally, the comparison is done using a particular distance metric but what is important is the result set of your top closest vectors, not the distance between them.
- Perform Approximate Similarity Search Using Vector Indexes
  For a vector search to be useful, it needs to be fast and accurate. Approximate similarity searches seek a balance between these goals.
- Perform Multi-Vector Similarity Search
  Another major use-case of vector search is multi-vector search. Multi-vector search is typically associated with a multi-document search, where documents are split into chunks that are individually embedded into vectors.
- Perform Hybrid Search
  Hybrid search is an advanced information retrieval technique that lets you search documents by *keywords* and *vectors*, to achieve more relevant search results.

Perform Exact Similarity Search

A similarity search looks for the relative order of vectors compared to a query vector. Naturally, the comparison is done using a particular distance metric but what is important is the result set of your top closest vectors, not the distance between them.

As an example, and given a certain query vector, you can calculate its distance to all other vectors in your data set. This type of search, also called flat search, or exact search, produces the most accurate results with perfect search quality. However, this comes at the cost of significant search times. This is illustrated by the following diagrams:

Figure 8-1 Exact Search



With an exact search, you compare the query vector `vq` against every other vector in your space by calculating its distance to each vector. After calculating all of these distances, the search returns the nearest `k` of those as the nearest matches. This is called a k-nearest neighbors (kNN) search.

For example, the Euclidean similarity search involves retrieving the top-k nearest vectors in your space relative to the Euclidean distance metric and a query vector. Here's an example that retrieves the top 10 vectors from the `vector_tab` table that are the nearest to `query_vector` using the following exact similarity search query:

```
SELECT docID
FROM vector_tab
ORDER BY VECTOR_DISTANCE( embedding, :query_vector, EUCLIDEAN )
FETCH EXACT FIRST 10 ROWS ONLY;
```

In this example, `docID` and `embedding` are columns defined in the `vector_tab` table and `embedding` has the `VECTOR` data type.

In the case of Euclidean distances, comparing squared distances is equivalent to comparing distances. So, when ordering is more important than the distance values themselves, the Euclidean Squared distance is very useful as it is faster to calculate than the Euclidean distance (avoiding the square-root calculation). Consequently, it is simpler and faster to rewrite the query like this:

```
SELECT docID
FROM vector_tab
ORDER BY VECTOR_DISTANCE( embedding, :query_vector, EUCLIDEAN_SQUARED)
FETCH FIRST 10 ROWS ONLY;
```

Note

The `EXACT` keyword is optional. If omitted while connected to an ADB-S instance, an approximate search using a vector index is attempted if one exists. For more information, see Perform Approximate Similarity Search Using Vector Indexes.

Note

Ensure that you use the distance function that was used to train your embedding model.

See Also

Oracle AI Database SQL Language Reference for the full syntax of the `ROW_LIMITING_CLAUSE`

Perform Approximate Similarity Search Using Vector Indexes

For a vector search to be useful, it needs to be fast and accurate. Approximate similarity searches seek a balance between these goals.

- Understand Approximate Similarity Search Using Vector Indexes
  For faster search speeds with large vector spaces, you can use approximate similarity search using vector indexes.
- Optimizer Plans for Vector Indexes
  Optimizer plans for HNSW and IVF indexes are described in the following sections.
- Approximate Similarity Search Examples
  Review these examples to see how you can perform an approximate similarity search using vector indexes.

Understand Approximate Similarity Search Using Vector Indexes

For faster search speeds with large vector spaces, you can use approximate similarity search using vector indexes.

Using vector indexes for a similarity search is called an **approximate search**. Approximate searches use **vector indexes**, which trade off accuracy for performance.

Approximate search for large vector spaces

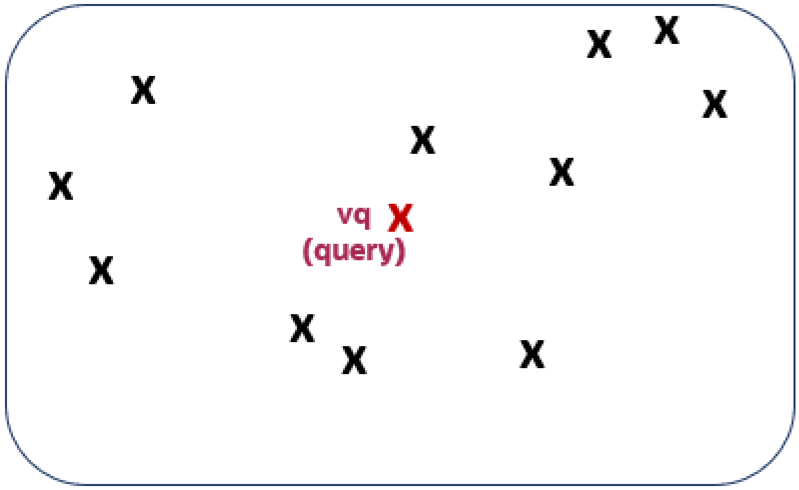
When search quality is your high priority and search speed is less important, Exact Similarity search is a good option. Search speed can be irrelevant for smaller vector spaces, or when you perform searches with high performance servers. However, ML algorithms often perform similarity searches on vector spaces with billions of embeddings. For example, the Deep1B data-set contains 1B images generated by a Convolutional Neural Network (CNN). Computing vector distances with every vector in the corpus to find Top-K matches at 100 percent accuracy is very slow.

Fortunately, there are many types of approximate searches that you can perform using vector indexes. Vector indexes can be less accurate, but they can consume less resources, and can be more efficient. Unlike traditional database indexes, vector indexes are constructed and perform searches using heuristic-based algorithms.

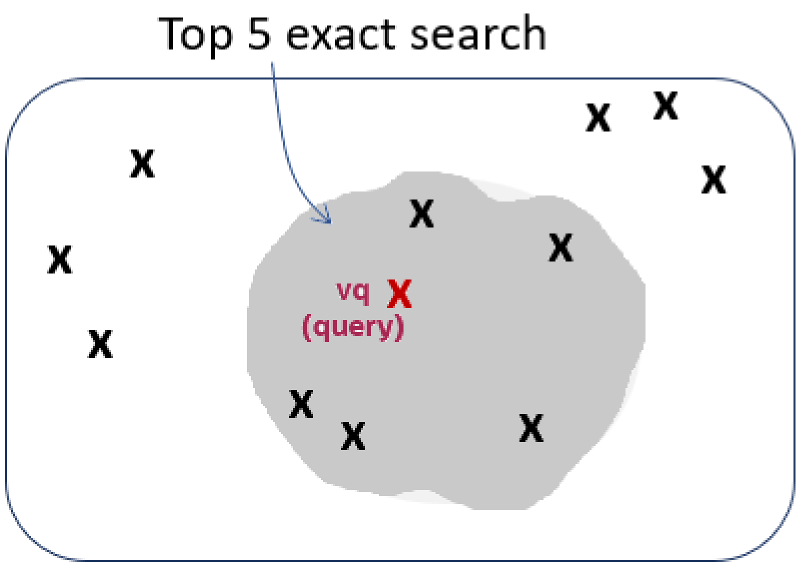
Because 100 percent accuracy cannot be guaranteed by heuristics, vector index searches use **target accuracy**. Internally, the algorithms used for both index creation and index search are doing their best to be as accurate as possible. However, you have the possibility to influence those algorithms by specifying a target accuracy. When creating the index or searching it, you can specify non-default target accuracy values either by specifying a percentage value, or by specifying internal parameters values, depending on the index type you are using.

Target accuracy example

To better understand what is meant by target accuracy look at the following diagrams. The first diagram illustrate a vector space where each vector is represented by a small cross. The one in red represents your query vector.

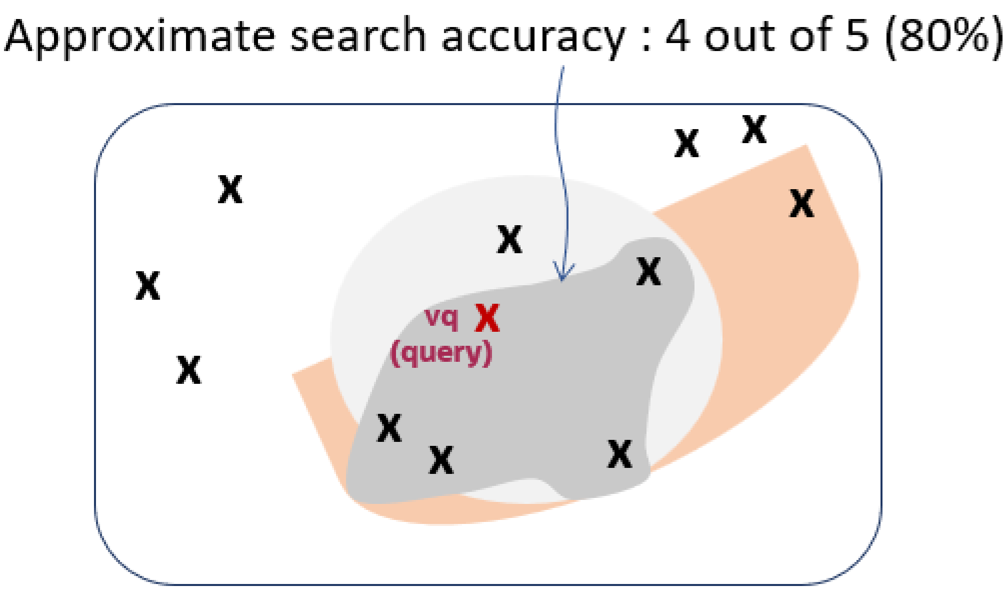


Running a top-5 exact similarity search in that context would return the five vectors shown on the second diagram:



Depending on how your vector index was constructed, running a top-5 approximate similarity search in that context could return the five vectors shown on the third diagram. This is because the index creation process is using heuristics. So searching through the vector index may lead to different results compared to an exact search:

As you can see, the retrieved vectors are different and, in this case, they differ by one vector. This means that, compared to the exact search, the similarity search retrieved 4 out of 5 vectors correctly. The similarity search has 80% accuracy compared to the exact similarity search. This is illustrated on the fourth diagram:



Due to the nature of vector indexes being approximate search structures, it's possible that fewer than K rows are returned in a top-K similarity search.

For information on how to set up your vector indexes, see Create Vector Indexes.

Optimizer Plans for Vector Indexes

Optimizer plans for HNSW and IVF indexes are described in the following sections.

- Optimizer Plans for HNSW Vector Indexes
  A Hierarchical Navigable Small World Graph (HNSW) is a form of In-Memory Neighbor Graph vector index. It is a very efficient index for vector approximate similarity search.

- Optimizer Plans for Local HNSW Vector Indexes
  Oracle Database uses partition pruning to narrow the search and search only the local HNSW graphs for qualifying partitions or sub partitions. Partition pruning narrows the set of local HNSW graphs that need to be searched, and the optimizer plan determines the most efficient execution plan for accessing narrowed graph set during similarity search. When using local HNSW vector indexes, you may see different optimizer plans for similarity searches.
- Optimizer Plans for IVF Vector Indexes
  Inverted File Flat (IVF) is a form of Neighbor Partition Vector index. It is a partition-based index that achieves search efficiency by narrowing the search area through the use of neighbor partitions or clusters.
- Vector Index Hints
  If the optimizer does not choose your existing index while running your query and you still want that index to be used, you can rewrite your SQL statement to include vector index hints.

Optimizer Plans for HNSW Vector Indexes

A Hierarchical Navigable Small World Graph (HNSW) is a form of In-Memory Neighbor Graph vector index. It is a very efficient index for vector approximate similarity search.

In the simplest case, a query has a single table and it does not contain any relational filter predicates or subqueries. The following query example illustrates this situation:

```
SELECT chunk_id, chunk_data
FROM doc_chunks
ORDER BY VECTOR_DISTANCE( chunk_embedding, :query_vector, COSINE )
FETCH FIRST 4 ROWS ONLY WITH TARGET ACCURACY 80;
```

The corresponding execution plan should look like the following, if the optimizer decides to use the index (start from operation id 5):

Top-K similar vectors are first identified using the HNSW vector index. For each of them, corresponding rows are identified in the base table and required selected columns are extracted.

```
---------------------------------------------------------
| Id  | Operation                         | Name          |
---------------------------------------------------------
|   0 | SELECT STATEMENT                  |               |
|*  1 |  COUNT STOPKEY                    |               |
|   2 |   VIEW                            |               |
|*  3 |    SORT ORDER BY STOPKEY          |               |
|   4 |     TABLE ACCESS BY INDEX ROWID| DOC_CHUNKS       |
|   5 |      VECTOR INDEX HNSW SCAN       | DOCS_HNSW_IDX5 |
---------------------------------------------------------
```

> **Note**
>
> The Hierarchical Navigable Small World (HNSW) vector index structure contains `rowids` of corresponding base table rows for each vector in the index.

However, your query may contain traditional relational data filters, as illustrated in the following example:

```
SELECT chunk_id, chunk_data
FROM doc_chunks
WHERE doc_id=1
ORDER BY VECTOR_DISTANCE( chunk_embedding, :query_vector, COSINE )
FETCH FIRST 4 ROWS ONLY WITH TARGET ACCURACY 80;
```

In that case, there are essentially two main alternatives for the optimizer to generate an execution plan using an HNSW vector index. These two alternatives are called **pre-filtering** and **in-filtering**.

The main difference between pre-filtering and in-filtering are:

- *Pre-filtering* runs the filtering evaluation on the base table first and only traverses the HNSW vector index for corresponding vectors. This can be very fast if the filter predicates are selective (that is, most rows filtered out).
- *In-filtering*, on the other hand, starts by traversing the HNSW vector index and invokes the filtering only for each vector matching candidate. This can be better than pre-filtering when more rows pass the filter predicates. In this case, the number of vector candidates to consider, while traversing the HNSW vector index, might be many fewer than the number of rows that pass the filters.

For both in-filtering and pre-filtering, the optimizer may choose to process projected columns from your select list before or after the similarity search operation. If it does so after, this is called a join-back operation. If it does so before, it is called a no-join-back operation. The tradeoff between the two depends on the number of rows returned by the similarity search.

To illustrate these four possibilities, here are some typical execution plans:

> **Note**
>
> Depending on the version you are using, you can find variations of these plans. However, the idea stays the same, so the following plans are just for illustration purposes.

> **Note**
>
> The following sample optimizer plans use an HNSW vector index called `DOCS_HNSW_IDX3`.

Pre-filter with Join-back (Start from Operation id 9)

Filtered `rowids` are first identified in the base table and joined to the auxiliary mapping table before the HNSW vector index identifies the relevant vectors. Relevant base table rows are then identified and required selected columns are extracted.

```
------------------------------------------------------------------------------
--------------------------
| Id  | Operation                             |
Name                                                   |
------------------------------------------------------------------------------
```

```
-------------------------
|   0 | SELECT STATEMENT
|                                                         |
|*  1 |  COUNT STOPKEY
|                                                         |
|   2 |   VIEW
|                                                         |
|*  3 |    SORT ORDER BY STOPKEY
|                                                         |
|*  4 |     TABLE ACCESS BY INDEX ROWID       |
DOC_CHUNKS                                              |
|   5 |      VECTOR INDEX HNSW SCAN PRE-FILTER|
DOCS_HNSW_IDX3                                          |
|   6 |       VIEW                            |
VW_HPJ_56DFF779                                         |
|*  7 |        HASH JOIN RIGHT OUTER
|                                                         |
|   8 |         TABLE ACCESS FULL             |
VECTOR$DOCS_HNSW_IDX3$81357_82726_0$HNSW_ROWID_VID_MAP |
|*  9 |         TABLE ACCESS FULL             |
DOC_CHUNKS                                              |
------------------------------------------------------------------------------
-------------------------
```

Note

- HNSW indexes use an auxiliary table and associated index to store index information like mapping between vector `ids` and `rowids`. Mainly `VECTOR$<`*`index name`*`>$HNSW_ROWID_VID_MAP`.
- For the join, you can also see plans like the following:

  ```
  NESTED LOOPS OUTER
    TABLE ACCESS FULL  DOC_CHUNKS
    TABLE ACCESS BY INDEX ROWID
  VECTOR$DOCS_HNSW_IDX3$81357_82726_0$HNSW_ROWID_VID_MAP
      INDEX UNIQUE SCAN SYS_C008586
  ```

Pre-filter without Join-back (Start from Operation id 8)

Filtered `rowids` with relevant columns are first identified in the base table and buffered. The result is then joined to the auxiliary mapping table before the HNSW vector index identifies the top-K vectors. For those identified vectors, associated base table columns are shown.

```
------------------------------------------------------------------------------
------------------------
| Id  | Operation                            |
Name                                                    |
------------------------------------------------------------------------------
------------------------
|   0 | SELECT STATEMENT
|                                                         |
|*  1 |  COUNT STOPKEY
```

```
|                                                         |
|   2 |   VIEW
|                                                         |
|*  3 |    SORT ORDER BY STOPKEY
|                                                         |
|   4 |     VECTOR INDEX HNSW SCAN PRE-FILTER|
DOCS_HNSW_IDX3                                          |
|   5 |      VIEW                           |
VW_HPF_B919B0A0                                         |
|*  6 |       HASH JOIN RIGHT OUTER
|                                                         |
|   7 |        TABLE ACCESS FULL            |
VECTOR$DOCS_HNSW_IDX3$81357_82726_0$HNSW_ROWID_VID_MAP |
|*  8 |        TABLE ACCESS FULL            |
DOC_CHUNKS                                              |
------------------------------------------------------------------------------
-------------------------
```

In-filter With Join-back (Start from Operation id 5)

HNSW vector index is traversed first, and for each identified vector, filters on the base table for the corresponding `rowid` are applied. Once the top-K `rowids` passing the filters are identified, base table columns are extracted.

```
----------------------------------------------------------------
| Id  | Operation                           | Name             |
----------------------------------------------------------------
|   0 | SELECT STATEMENT                    |                  |
|*  1 |  COUNT STOPKEY                      |                  |
|   2 |   VIEW                              |                  |
|*  3 |    SORT ORDER BY STOPKEY            |                  |
|*  4 |     TABLE ACCESS BY INDEX ROWID     | DOC_CHUNKS       |
|   5 |      VECTOR INDEX HNSW SCAN IN-FILTER| DOCS_HNSW_IDX3  |
|   6 |       VIEW                          | VW_HIJ_B919B0A0 |
|*  7 |        TABLE ACCESS BY USER ROWID   | DOC_CHUNKS       |
----------------------------------------------------------------
```

In-filter without Join-back (Start from Operation id 4)

HNSW vector index is traversed first, and for each identified vector, filters on the base table for the corresponding `rowid` are applied and relevant columns extracted. Once the top-K vectors are identified, corresponding base table columns are shown.

```
---------------------------------------------------------------
| Id  | Operation                          | Name             |
---------------------------------------------------------------
|   0 | SELECT STATEMENT                   |                  |
|*  1 |  COUNT STOPKEY                     |                  |
|   2 |   VIEW                             |                  |
|*  3 |    SORT ORDER BY STOPKEY           |                  |
|   4 |     VECTOR INDEX HNSW SCAN IN-FILTER| DOCS_HNSW_IDX3  |
|   5 |      VIEW                          | VW_HIF_B919B0A0 |
|*  6 |       TABLE ACCESS BY USER ROWID   | DOC_CHUNKS       |
---------------------------------------------------------------
```

HNSW Indexes in the Optimizer Plan

HNSW indexes plans may use an internal table and associated index to store index information like mapping between vector `ids` and `rowids`. Mainly `VECTOR$<`*`index name`*`>$HNSW_ROWID_VID_MAP`.

Table 8-1 HNSW Options

Operation	Options	Object_name
TABLE ACCESS	FULL	`VECTOR$<`*`vector-index-name`*`>$<`*`id`*`>$HNSW_ROW_VID_MAP`
TABLE ACCESS	STORAGE FULL	`VECTOR$<`*`vector-index-name`*`>$<`*`id`*`>$HNSW_ROW_VID_MAP`
TABLE ACCESS	INMEMORY FULL	`VECTOR$<`*`vector-index-name`*`>$<`*`id`*`>$HNSW_ROW_VID_MAP`

Optimizer Plans for Local HNSW Vector Indexes

Oracle Database uses partition pruning to narrow the search and search only the local HNSW graphs for qualifying partitions or sub partitions. Partition pruning narrows the set of local HNSW graphs that need to be searched, and the optimizer plan determines the most efficient execution plan for accessing narrowed graph set during similarity search. When using local HNSW vector indexes, you may see different optimizer plans for similarity searches.

Local HNSW indexes use the same general HNSW vector indexes optimizer plan concepts described in Optimizer Plans for HNSW Vector Indexes topic, such as pre-filtering, in-filtering, and join-back. However, a local HNSW index adds partition pruning to the optimizer decision.

- ***No-filtering*** is used in the following two scenarios:
  - The query does not contain any filtering predicates. In this case, Oracle Database searches all applicable partition-local HNSW graphs.
  - The query contains only partition-pruning predicates and no other filters. In this case, Oracle Database uses the partition-pruning predicates to identify the qualifying partitions or sub-partitions, and searches only the corresponding local HNSW graphs.

  In a no-filtering plan, Oracle Database does not need to access the base table before identifying the approximate nearest rows. This avoids unnecessary base table I/O resources for evaluating those predicates and limits the vector search to the qualifying partition-local HNSW graphs. The approximate nearest rows are computed within the local HNSW index access.

  When the query does not contain a filtering predicate that can restrict the search to a subset of partitions. Oracle Database searches across the applicable local HNSW graphs and returns the approximate nearest rows. The following query example illustrates this situation:

  ```
  SELECT product_id,
         vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
  FROM   TT
  ```

```
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

Oracle Database cannot prune the local HNSW graphs because there is no `WHERE` predicate such as `product_id = ...`. The optimizer plan therefore uses `PARTITION LIST ALL`, meaning all applicable partitions/local graphs are considered.

```
| Id | Operation                               | Name|
|----|----------------------------------------|-----|
|  0 | SELECT STATEMENT                        |     |
|* 1 |  COUNT STOPKEY                          |     |
|  2 |   VIEW                                  |     |
|* 3 |    SORT ORDER BY STOPKEY                |     |
|  4 |     PARTITION LIST ALL                  |     |
|  5 |      TABLE ACCESS BY GLOBAL INDEX ROWID| TT  |
|  6 |       VECTOR INDEX HNSW SCAN            |VIDXH|
```

- ***Parallel no-filtering*** is used when Oracle Database needs to access more than one partitions, parallel plan normally can speed up query response by using parallel execution to distribute the search work across partitions. In a RAC environment, parallel execution plans for local HNSW indexes do not extend to remote nodes.
  Parallel no-filtering plan is used in the following scenarios.
  - The query does not contain any filtering predicates. In this case, Oracle Database searches all applicable partition-local HNSW graphs.
  - The query contains only partition-pruning predicates and no other filters. In this case, Oracle Database uses the partition-pruning predicates to identify the qualifying partitions or sub-partitions, and searches only the corresponding local HNSW graphs.

  Oracle Database searches across all applicable local HNSW graphs. Since there is no partition-pruning predicate is available, the optimizer plan shows `PARTITION LIST ALL`. The following query example illustrates this situation:

```
SELECT product_id,
       vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
FROM   TT
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

  Oracle Database cannot prune the local HNSW graphs because there is no `WHERE` predicate such as `product_id = ...`. The optimizer plan therefore uses `PARTITION LIST ALL`. The scan and sort work are distributed through PX operations:

```
| Id | Operation                                      | Name     |
|----|-----------------------------------------------|---------|
|  0 | SELECT STATEMENT                               |          |
|* 1 |  COUNT STOPKEY                                 |          |
|  2 |   PX COORDINATOR                               |          |
|  3 |    PX SEND QC (ORDER)                          | :TQ10002|
|  4 |     VIEW                                       |          |
|* 5 |      SORT ORDER BY STOPKEY                     |          |
|  6 |       PX RECEIVE                               |          |
```

```
|  7 |         PX SEND RANGE                       | :TQ10001|
|  8 |          PARTITION LIST ALL                 |         |
|* 9 |           SORT ORDER BY STOPKEY             |         |
| 10 |            TABLE ACCESS BY GLOBAL INDEX ROWID  | TT      |
| 11 |             BUFFER SORT                     |         |
| 12 |              PX RECEIVE                     |         |
| 13 |               PX SEND HASH (BLOCK ADDRESS)  | :TQ10000|
| 14 |                PX SELECTOR                  |         |
| 15 |                 VECTOR INDEX HNSW SCAN      | VIDXH   |
```

- ***In-filtering*** evaluates eligible filters as part of local HNSW index access. If the filter identifies one or more partitions, Oracle Database prunes the remaining partitions and searches only the corresponding local HNSW graphs. The following query example illustrates this situation:

  ```
  SELECT product_id,
         vector_distance(vec, vector('[7,0,2,... 0,2]')) AS distance
  FROM   TT
  WHERE  product_id = 2
  ORDER BY distance
  FETCH APPROX FIRST 8 ROWS ONLY;
  ```

  The predicate `product_id = 2` identifies a single partition. Oracle Database prunes the other partitions and searches only the local HNSW graph for the matching partition. The optimizer plan therefore uses `PARTITION LIST SINGLE`.

  ```
  | Id | Operation                              | Name           |
  |----|----------------------------------------|----------------|
  |  0 | SELECT STATEMENT                       |                |
  |* 1 |  COUNT STOPKEY                         |                |
  |  2 |   VIEW                                 |                |
  |* 3 |    SORT ORDER BY STOPKEY               |                |
  |  4 |     PARTITION LIST SINGLE              |                |
  |  5 |      VECTOR INDEX HNSW SCAN IN-FILTER  | VIDXH          |
  |  6 |       VIEW                             |VW_HIF_0F5A0514|
  |* 7 |        TABLE ACCESS BY USER ROWID      | TT             |
  ```

  In-filter plans are most effective when filters are highly selective or inexpensive to evaluate. For costlier filters, such as joins, the optimizer may choose a pre-filter or post-filter plan instead.

- ***In-filtering with binds*** evaluates eligible bind filters as part of local HNSW index access. If the bind values identify one or more partitions, Oracle Database prunes the remaining partitions and searches only the corresponding local HNSW graphs. The following query example illustrates this situation:

  ```
  SELECT product_id,
         vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
  FROM   TT
  WHERE  product_id IN (:prod_id, 3)
  AND    customer_id > 100
  ```

```
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

The `product_id IN (:prod_id, 3)` filter predicate identifies a list of partitions. The optimizer plan shows `PARTITION LIST INLIST`, meaning Oracle Database searches only the local HNSW graphs for the qualifying partition list.

```
| Id | Operation                                 | Name            |
|----|-------------------------------------------|-----------------|
|  0 | SELECT STATEMENT                          |                 |
|* 1 |  COUNT STOPKEY                            |                 |
|  2 |   VIEW                                    |                 |
|* 3 |    SORT ORDER BY STOPKEY                  |                 |
|  4 |     PARTITION LIST INLIST                 |                 |
|  5 |      VECTOR INDEX HNSW SCAN IN-FILTER     | VIDXH           |
|  6 |       VIEW                                | VW_HIF_0F5A0514|
|* 7 |        TABLE ACCESS BY USER ROWID         | TT              |
```

- ***In-filtering with join-back and a join*** evaluates eligible filters as part of local HNSW index access. Oracle Database prunes the remaining partitions and searches only the corresponding local HNSW graphs. After local HNSW index access identifies candidate rows, Oracle Database joins back to the base table or joined tables to evaluate additional predicates, return projected columns, or complete the join. The plan shows `PARTITION LIST INLIST`, and join operations such as `HASH JOIN` or `NESTED LOOPS`. The following query example illustrates this situation:

```
SELECT TT_ORDER.amount_sold,
       product_id,
       vector_distance(vec, vector('[7,0,2,2,..., 2]')) AS distance
FROM   TT, TT_ORDER
WHERE  product_id IN (:prod_id, 3)
AND    customer_id > 100
AND    TT.product_id = TT_ORDER.pid
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

The predicate `product_id IN (:prod_id, 3)` enables partition-list pruning. Oracle Database uses the local HNSW index to identify candidate rows, then joins back to the base table and joins with `TT_ORDER`. The corresponding optimizer plan should look like the following:

```
| Id | Operation                                          | Name            |
|----|----------------------------------------------------|-----------------|
|  0 | SELECT STATEMENT                                   |                 |
|* 1 |  COUNT STOPKEY                                     |                 |
|  2 |   VIEW                                             |                 |
|* 3 |    SORT ORDER BY STOPKEY                           |                 |
|* 4 |     HASH JOIN                                      |                 |
|  5 |      PARTITION LIST INLIST                         |                 |
|  6 |       TABLE ACCESS FULL                            | TT_ORDER        |
|  7 |        TABLE ACCESS BY GLOBAL INDEX ROWID          | TT              |
```

```
|  8 |         PARTITION LIST INLIST                  |                |
|  9 |          VECTOR INDEX HNSW SCAN IN-FILTER      | VIDXH          |
| 10 |           VIEW                                 | VW_HI3_837F1853|
| 11 |            NESTED LOOPS                        |                |
| 12 |             TABLE ACCESS BY USER ROWID         | TT             |
| 13 |              PARTITION LIST ALL                |                |
| 14 |               TABLE ACCESS FULL                | TT_ORDER       |
```

A join-back plan can help reduce memory pressure when caching intermediate results, especially when the query selects many columns or includes large values.

- ***In-filtering with join-back and binds*** evaluates eligible bind filters as part of local HNSW index access. If the bind values identify one or more partitions, Oracle Database prunes the remaining partitions and searches only the corresponding local HNSW graphs. After the local HNSW index identifies candidate rows, Oracle Database joins back to the base table to evaluate the bind filter, access row data, or return projected columns. The following query example illustrates this situation:

```
SELECT product_id,
       vector_distance(vec, vector('[7,0,2,2,..., 2]')) AS distance
FROM   TT
WHERE  product_id IN (:prod_id, :prod_id1)
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

  The predicate `product_id IN (:prod_id, :prod_id1)` enables partition pruning. Oracle Database prunes the remaining partitions and searches only the corresponding local HNSW graphs. The optimizer plan shows `PARTITION LIST INLIST`:

```
| Id | Operation                                   | Name            |
|----|---------------------------------------------|-----------------|
|  0 | SELECT STATEMENT                            |                 |
|* 1 |  COUNT STOPKEY                              |                 |
|  2 |   VIEW                                      |                 |
|* 3 |    SORT ORDER BY STOPKEY                    |                 |
|  4 |     TABLE ACCESS BY GLOBAL INDEX ROWID      | TT              |
|  5 |      PARTITION LIST INLIST                  |                 |
|  6 |       VECTOR INDEX HNSW SCAN POST-FILTER    | VIDXH           |
|  7 |        VIEW                                 | VW_HPST_0F5A0514|
|* 8 |         TABLE ACCESS BY USER ROWID          | TT              |
```

- ***Parallel in-filtering with binds*** evaluates eligible bind filters as part of local HNSW index access. If the bind values identify one or more partitions, Oracle Database prunes the remaining partitions and uses parallel execution to search only the corresponding local HNSW graphs. The following query example illustrates this situation:

```
SELECT product_id,
       vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
FROM   TT
WHERE  product_id IN (:prod_id, 3)
AND    customer_id > 100
```

```
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

The predicate `product_id IN (:prod_id, 3)` enables partition pruning. The optimizer plan shows `PX PARTITION LIST INLIST`, which means the selected partition list is processed through parallel execution.

```
| Id | Operation                                       | Name            |
|----|-----------------------------------------------|-----------------|
|  0 | SELECT STATEMENT                                |                 |
|* 1 |  COUNT STOPKEY                                  |                 |
|  2 |   PX COORDINATOR                                |                 |
|  3 |    PX SEND QC (ORDER)                           | :TQ10001        |
|  4 |     VIEW                                        |                 |
|* 5 |      SORT ORDER BY STOPKEY                      |                 |
|  6 |       PX RECEIVE                                |                 |
|  7 |        PX SEND RANGE                            | :TQ10000        |
|  8 |         PX PARTITION LIST INLIST                |                 |
|* 9 |          SORT ORDER BY STOPKEY                  |                 |
| 10 |           VECTOR INDEX HNSW SCAN IN-FILTER      | VIDXH           |
| 11 |            VIEW                                 | VW_HIF_0F5A0514|
| 12 |             TABLE ACCESS BY USER ROWID          | TT              |
```

- ***Post-filtering*** evaluates filters after local HNSW index access. Oracle Database first uses the local HNSW index to find approximate nearest-neighbor candidates, then evaluates remaining filters on those candidate rows. The following query example illustrates this situation:

```
SELECT product_id,
       vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
FROM   TT
WHERE  product_id IN (:prod_id, :prod_id1)
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

The predicate `product_id IN (:prod_id, :prod_id1)` enables partition pruning. Oracle Database uses the local HNSW index to get vector candidates, then applies the remaining filter afterward. The optimizer plan shows `PARTITION LIST INLIST`:

```
| Id | Operation                                   | Name            |
|----|-------------------------------------------|-----------------|
|  0 | SELECT STATEMENT                            |                 |
|* 1 |  COUNT STOPKEY                              |                 |
|  2 |   VIEW                                      |                 |
|* 3 |    SORT ORDER BY STOPKEY                    |                 |
|  4 |     TABLE ACCESS BY GLOBAL INDEX ROWID      | TT              |
|  5 |      PARTITION LIST INLIST                  |                 |
|  6 |       VECTOR INDEX HNSW SCAN POST-FILTER    | VIDXH           |
|  7 |        VIEW                                 |VW_HPST_0F5A0514|
|* 8 |         TABLE ACCESS BY USER ROWID          | TT              |
```

Post-filter plans work well for expensive filters, such as joins or JSON filters, because the expensive filter is evaluated only for candidate rows returned by the local HNSW index scan. However, after applying the post-filter, the query may not be able to return the requested approximate nearest rows. This can happen if the filter removes many or all rows returned from the local HNSW index evaluation.

- ***Pre-filtering*** evaluates eligible filters before local HNSW vector search. When the filters enable partition pruning, Oracle Database can first identify the qualifying partitions or sub partitions, and then search only the local HNSW graphs for those partitions. This can improve performance for selective filtered similarity searches.
  The following query example illustrates this situation:

```
SELECT product_id,
       vector_distance(vec, vector('[7,0,2,..., 2]')) AS distance
FROM   TT
WHERE  product_id = 2
ORDER BY distance
FETCH APPROX FIRST 8 ROWS ONLY;
```

  The predicate `product_id = 2` enables partition pruning. Oracle Database uses the local HNSW index to get vector candidates. The optimizer plan shows `PARTITION LIST SINGLE`:

```
| Id | Operation                                 | Name  |
|----|-------------------------------------------|-------|
|  0 | SELECT STATEMENT                          |       |
|* 1 |  COUNT STOPKEY                            |       |
|  2 |   VIEW                                    |       |
|* 3 |    SORT ORDER BY STOPKEY                  |       |
|  4 |     PARTITION LIST SINGLE                 |       |
|  5 |      VECTOR INDEX HNSW SCAN PRE-FILTER    | VIDXH |
```

A pre-filter plan evaluates the filter before the local HNSW index scan. Like post-filter plans, pre-filter plans are generally useful for expensive filters. Unlike post-filter plans, pre-filter plans can still return the requested approximate nearest rows because filtering happens before vector index evaluation. The tradeoff is that the filter evaluation cost can be higher than with a post-filter plan. A post-filter plan evaluates the filter only for candidate rows returned by the local HNSW index scan, while a pre-filter plan may need to evaluate the filter earlier in the execution process.

Comparison of In-Filter, Pre-filter, and Post-Filter Plans

Plan type	When the optimizer typically chooses it	How filtering is applied	TOP N behavior	Filter evaluation cost	Partition-wise processing
In-filter (with or with join back)	Typically chosen when filters are inexpensive or the filters are not highly selective (many rows pass the filters).	The filter is evaluated as part of the local HNSW partition graph scan.	Can return TOP N rows during local HNSW index evaluation.	Usually lower when the filter is inexpensive to evaluate.	The optimizer first determines which partitions to access, then evaluates the filter during each selected partition's local HNSW graph scan.

Plan type	When the optimizer typically chooses it	How filtering is applied	TOP N behavior	Filter evaluation cost	Partition-wise processing
Pre-filter (with or with join back)	Typically chosen for expensive filters where preserving TOP N results is important.	The filter is evaluated before the local HNSW partition graph scan.	Can return the requested TOP N rows because filtering happens before vector index evaluation.	Can be higher than a post-filter plan because filtering may occur before candidate rows are reduced by the HNSW scan.	The optimizer first determines which partitions to access, then evaluates the filter before scanning each selected partition's local HNSW graph.
Post-filter	Typically chosen for expensive filters when evaluating the filter only on candidate rows is more efficient.	The filter is evaluated after the local HNSW partition graph scan.	May not return the requested TOP N rows if the filter removes too many candidate rows returned by the HNSW scan.	Usually lower than a pre-filter plan because the filter is evaluated only for candidate rows returned by the local HNSW index scan.	The optimizer first determines which partitions to access, then applies the filter after each selected partition's local HNSW graph scan.

Optimizer Plans for IVF Vector Indexes

Inverted File Flat (IVF) is a form of Neighbor Partition Vector index. It is a partition-based index that achieves search efficiency by narrowing the search area through the use of neighbor partitions or clusters.

When using IVF vector indexes, you can see two possible plans for your similarity searches:

- *Pre-filtering* evaluates the filters first, before using the IVF vector index. Once the optimizer chooses which centroid partitions to scan, it evaluates the filters for all the rows in those partitions. Then it computes the distance to the query vector for the rows that passed the filters.
- *Post-filtering* evaluates the filters after using the IVF vector index. Once the optimizer chooses which centroid partitions to scan, it computes the distance to the query vector for all the rows in those partitions. Once the optimizer finds the closest rows, it then evaluates the filter for those rows and returns only the ones that pass the filters.

> **Note**
>
> Depending on the version you are using, you can find variations of these plans. However, the idea stays the same, so the following plans are just examples for illustration purposes.

Pre-filter plan

- Plan line ids 7 to 12: This part happens first. The optimizer chooses the centroid `ids` that are closest to the query vector.
- Plan line ids 13 to 14: With these lines, centroid `ids` are joined with the centroid partitions table to get the rows only from the passing centroid partitions identified from the centroid `ids` in the first step.

- Plan line id 4: All the rows from 5-14 are joined to the rows from 15 with a `NESTED LOOPS JOIN`. That is, the rows are reduced to only the rows that pass the filter.
- Plan line id 3: This step computes the vector distance and sorts on this value, only keeping the top-K rows.

```
------------------------------------------------------------------------------
------------------------------------------------------------------------------
---------------------
| Id  | Operation                        |
Name                                                                      | Rows
| Bytes |TempSpc| Cost (%CPU)| Time     | Pstart| Pstop |
------------------------------------------------------------------------------
------------------------------------------------------------------------------
---------------------
|   0 | SELECT STATEMENT
|                                                                           |
4 |  8112 |       |  4260   (3)| 00:00:01 |       |       |
|*  1 |  COUNT STOPKEY
|
|       |       |       |            |          |       |       |
|   2 |   VIEW
|                                                                           |
66 |   130K|       |  4260   (3)| 00:00:01 |       |       |
|*  3 |    SORT ORDER BY STOPKEY
|                                                                           |
66 |   127K|   144K|  4260   (3)| 00:00:01 |       |       |
|   4 |     NESTED LOOPS
|                                                                           |
66 |   127K|       |  4217   (3)| 00:00:01 |       |       |
|   5 |      VIEW                        |
VW_IVENJ_47D4581B                                                         |    88
|   880 |       |    89   (4)| 00:00:01 |       |       |
|*  6 |       HASH JOIN
|                                                                           |
88 |   138K|       |    89   (4)| 00:00:01 |       |       |
|   7 |        PART JOIN FILTER CREATE
| :BF0000                                                                   |
3 |    39 |       |     4  (25)| 00:00:01 |       |       |
|   8 |         VIEW                     |
VW_IVCR_B5B87E67                                                          |     3
|    39 |       |     4  (25)| 00:00:01 |       |       |
|*  9 |          COUNT STOPKEY
|
|       |       |       |            |          |       |       |
|  10 |           VIEW                   |
VW_IVCN_9A1D2119                                                          |    16
|   208 |       |     4  (25)| 00:00:01 |       |       |
|* 11 |            SORT ORDER BY STOPKEY
|                                                                           |
16 |   144 |       |     4  (25)| 00:00:01 |       |       |
|  12 |             TABLE ACCESS FULL    |
VECTOR$IDX_IVF_MY_DATA$75132_75138_0$IVF_FLAT_CENTROIDS                   |    16
|   144 |       |     3   (0)| 00:00:01 |       |       |
|  13 |        PARTITION LIST JOIN-
FILTER|
```

```
|   469 |   733K|       |    85   (3)| 00:00:01 |:BF0000|:BF0000|
|  14 |         TABLE ACCESS FULL         |
VECTOR$IDX_IVF_MY_DATA$75132_75138_0$IVF_FLAT_CENTROID_PARTITIONS    |   469
|   733K|       |    85   (3)| 00:00:01 |:BF0000|:BF0000|
|* 15 |      TABLE ACCESS FULL            |
MY_DATA                                                              |     1
|  1965 |       |    47   (3)| 00:00:01 |       |       |
------------------------------------------------------------------------------
------------------------------------------------------------------------------
--------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   1 - filter(ROWNUM<=4)
   3 - filter(ROWNUM<=4)
   6 - access("VW_IVCR_B5B87E67"."CENTROID_ID"="VTIX_CNPART"."CENTROID_ID")
   9 - filter(ROWNUM<=3)
  11 - filter(ROWNUM<=3)
  15 - filter("MY_DATA".ROWID="VW_IVENJ_47D4581B"."BASE_TABLE_ROWID" AND
"MY_DATA"."CATEGORY"<>'Generic')
```

> **Note**
>
> IVF indexes use auxiliary partition tables and associated indexes to store index information. Mainly `VECTOR$<index name>$<ids>$IVF_FLAT_CENTROIDS` and `VECTOR$<index name>$<ids>$IVF_FLAT_CENTROIDS_PARTITIONS`.

Post-filter Plan

- Plan line ids 11 to 15: Here the optimizer chooses the closest centroids (similar to lines 5 through 9 of the pre-filter plan).
- Plan line id 10: This creates a special filter with information about which centroid `ids` were chosen.
- Plan line ids 16 and 17: These lines use the special filter from plan line 10 to scan only the corresponding centroid partitions.
- Plan line id 9: This `HASH JOIN` makes sure that only the rows with the closest centroid `ids` are returned.
- Plan line id 8: This computes the distance for those rows with the closest centroid `ids` and finds the top-K rows.
- Plan line id 4: This `NESTED LOOPS` applies the base table filter to the previously identified top-K rows.

```
------------------------------------------------------------------------------
--------------------------------
| Id  | Operation                           |
Name                                                    |
------------------------------------------------------------------------------
--------------------------------
|   0 | SELECT STATEMENT
|                                                                  |
```

```
|*  1 |  COUNT STOPKEY
|                                                                  |
|   2 |   VIEW
|                                                                  |
|*  3 |    SORT ORDER BY STOPKEY
|                                                                  |
|   4 |     NESTED LOOPS
|                                                                  |
|   5 |      VIEW                          |
VW_IVPSR_11E7D7DE                                                |
|*  6 |       COUNT STOPKEY
|                                                                  |
|   7 |        VIEW                        |
VW_IVPSJ_578B79F1                                                |
|*  8 |         SORT ORDER BY STOPKEY
|                                                                  |
|*  9 |          HASH JOIN
|                                                                  |
|  10 |           PART JOIN FILTER CREATE
| :BF0000                                                          |
|  11 |            VIEW                    |
VW_IVCR_B5B87E67                                                 |
|* 12 |             COUNT STOPKEY
|                                                                  |
|  13 |              VIEW                  |
VW_IVCN_9A1D2119                                                 |
|* 14 |               SORT ORDER BY STOPKEY
|                                                                  |
|  15 |                TABLE ACCESS FULL    |
VECTOR$DOCS_IVF_IDX4$81357_83292_0$IVF_FLAT_CENTROIDS            |
|  16 |           PARTITION LIST JOIN-
FILTER|                                                                  |
|  17 |            TABLE ACCESS FULL        |
VECTOR$DOCS_IVF_IDX4$81357_83292_0$IVF_FLAT_CENTROID_PARTITIONS |
|* 18 |      TABLE ACCESS BY USER ROWID     |
DOC_CHUNKS                                                       |
------------------------------------------------------------------------------
---------------------------------
```

IVF Indexes in the Optimizer Plan

If the IVF index is used by the optimizer, the plan contains the names of the centroids and centroid partitions tables accessed as well as corresponding indexes.

The value displayed in the `Options` column for tables accessed via IVF indexes depends upon whether the table scan is for a regular table or Exadata table.

Table 8-2 Centroids and Centroid Partition Table Options

Operation	Options	Object_name
TABLE ACCESS	FULL	VECTOR$<*vector-index-name*>$<*id*>$IVF_FLAT_CENTROIDS VECTOR$DOCS_IVF_IDX2$<*id*>$IVF_FLAT_CENTROID_PARTITIONS
TABLE ACCESS	STORAGE FULL	VECTOR$<*vector-index-name*>$<*id*>$IVF_FLAT_CENTROIDS VECTOR$DOCS_IVF_IDX2$<*id*>$IVF_FLAT_CENTROID_PARTITIONS
TABLE ACCESS	INMEMORY FULL	VECTOR$<*vector-index-name*>$<*id*>$IVF_FLAT_CENTROIDS VECTOR$DOCS_IVF_IDX2$<*id*>$IVF_FLAT_CENTROID_PARTITIONS

- Terminable Iteration for IVF Index
  Use this feature if you want to ensure that the desired number of rows are returned from a search.

Terminable Iteration for IVF Index

Use this feature if you want to ensure that the desired number of rows are returned from a search.

When using an Inverted File Flat (IVF) index, the optimizer estimates the number of probes (`nProbe`) or clusters to consider in the underlying centroid tables. For example, if the `nProbe = 5`, the five clusters nearest to the query vector are identified and the approximate search is exclusively made within these clusters and the results returned are based on vector proximity within these clusters.

Consider the following query where an IVF index is created on `chunk_embedding` (vector column) within the `doc_chunks` table. The complete example can be found here : SQL Quick Start Using a Vector Embedding Model Uploaded into the Database

```
SELECT chunk_id, chunk_data,
FROM doc_chunks
WHERE doc_id=1
ORDER BY VECTOR_DISTANCE( chunk_embedding, :query_vector, COSINE )
FETCH FIRST 10 ROWS ONLY WITH TARGET ACCURACY 80;
```

One of the existing problems is that the optimizer can under-estimate the number of probes. In this case, the query may return fewer rows than anticipated. In situations where the number of rows fetched is critical to an application, using terminable iteration with IVF indexes makes it possible to set a number (or range of rows) to be returned at a minimum. The underlying method would extend the search to more centroids ensuring that the needed K rows are returned. In order to use terminable iteration, you must specify the operation. If its optimizer cost is higher with a vector index, then the optimizer chooses not to use a vector index.

The terminable IVF index can be enabled by using one of the following methods:

Using Terminable IVF Index with Hint:

The terminable IVF index could be used by specifying a `IVF_ITERATION` as a hint in the query. For example, the above query could be rewritten using the hint as follows :

```
SELECT /*+ IVF_ITERATION*/ chunk_id, chunk_data,
FROM doc_chunks
WHERE doc_id=1
ORDER BY VECTOR_DISTANCE( chunk_embedding, :query_vector, COSINE )
FETCH FIRST 10 ROWS ONLY WITH TARGET ACCURACY 80;
```

Using Terminable IVF Index with Syntax:

The terminable iteration could also be specified by using the new syntax which explicitly sets the bounds for the number of rows that needs to be returned. The same example could be rewritten with the new syntax for terminable IVF index as follows:

```
SELECT chunk_id, chunk_data,
FROM doc_chunks
WHERE doc_id=1
ORDER BY VECTOR_DISTANCE( chunk_embedding, :query_vector, COSINE )
FETCH FIRST 10 to 20 ROWS ONLY WITH TARGET ACCURACY 80;
```

In the above example, **- 10 to 20 -** enables terminable iteration. The lower bound (in this example, 10) indicates that at least 10 rows are returned by performing terminable iteration. The upper bound (in this example, 20) indicates that the query will return at most 20 rows. Lower bound and upper bound can be identical, this would still enable the terminable iteration. No rows are returned if the lower bound is larger than the upper bound or if the upper bound is zero or negative. If there are not enough centroids available to extend the search, it is possible that the query would return less than 10 rows.

Vector Index Hints

If the optimizer does not choose your existing index while running your query and you still want that index to be used, you can rewrite your SQL statement to include vector index hints.

There are two types of vector index hints:

- *Access path hints* apply when running a simple query block with a single table with no predicates and a valid HNSW vector index. This only applies to HNSW vector indexes.
- *Query transformation hints* cover all other cases, where query transformations are used.

Syntax for Access Path Hints

The access path hints are modeled on the existing `INDEX` hints. For example:

```
VECTOR_INDEX_SCAN ( [ @ queryblock ] tablespec [ indexspec ])

NO_VECTOR_INDEX_SCAN ( [ @ queryblock ] tablespec [ indexspec ])
```

The VECTOR_INDEX_SCAN hint instructs the optimizer to use a vector index scan for the specified table. The index is optionally specified. The optimizer does not consider a full table scan.

NO_VECTOR_INDEX_SCAN disables the use of the optionally specified vector index scan.

> **See Also**
>
> - *Oracle AI Database SQL Language Reference* for details about the INDEX hint

Access Path Hint Examples

```
SELECT /*+ VECTOR_INDEX_SCAN(galaxies) */ name
FROM galaxies
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 3 ROWS ONLY;
```

```
SELECT /*+ NO_VECTOR_INDEX_SCAN(galaxies) */ name
FROM galaxies
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 3 ROWS ONLY;
```

Note that even though the second example query is expected not to use a vector index, it is just a hint that the optimizer may or may not follow. To be certain that the index will not be used, it is better to use EXACT rather than the hint, as in the following:

```
SELECT name
FROM galaxies
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH EXACT FIRST 3 ROWS ONLY;
```

Syntax for Query Transformation Hints

The syntax for query transformation hints is similar to the INDEX hints model. For example:

```
VECTOR_INDEX_TRANSFORM ( [ @ queryblock ] tablespec [ indexspec
[ filtertype ]] )

NO_VECTOR_INDEX_TRANSFORM ( [ @ queryblock ] tablespec [ indexspec ] )
```

VECTOR_INDEX_TRANSFORM forces the transformation and uses the optionally specified index, whereas NO_VECTOR_INDEX_TRANSFORM disables the transformation and the use of the optionally specified index. For more information about the potential query transformations, see Optimizer Plans for HNSW Vector Indexes and Optimizer Plans for IVF Vector Indexes.

The optional filtertype specification for the VECTOR_INDEX_TRANSFORM hint can have the following values:

- PRE_FILTER_WITH_JOIN_BACK (only applies to HNSW indexes)
- PRE_FILTER_WITHOUT_JOIN_BACK (applies to both HNSW and IVF indexes)
- IN_FILTER_WITH_JOIN_BACK (only applies to HNSW indexes)

- `IN_FILTER_WITHOUT_JOIN_BACK` (only applies to HNSW indexes)
- `POST_FILTER_WITHOUT_JOIN_BACK` (applies to both HNSW and IVF indexes)

Each `filtertype` value corresponds to exactly one state in the query transformation. If no `filtertype` value is specified in the `VECTOR_INDEX_TRANSFORM` hint, then all of the valid filtering types are considered and the best one is chosen.

> **Note**
>
> If you specify the `filtertype` value, you must also specify the `indexspec` value.

> **See Also**
>
> - *Oracle AI Database SQL Language Reference* for details about the `INDEX` hint

Query Transformation Hint Examples

```
SELECT /*+ vector_index_transform(galaxies galaxies_ivf_idx
pre_filter_without_join_back) */ name
FROM galaxies
WHERE id<5
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 3 ROWS ONLY WITH TARGET ACCURACY 90;


SELECT /*+ vector_index_transform(galaxies) */ name
FROM galaxies
WHERE id<5
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 10 ROWS ONLY WITH TARGET ACCURACY 90;


SELECT /*+ no_vector_index_transform(galaxies) */ name
FROM galaxies
WHERE id<5
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 10 ROWS ONLY WITH TARGET ACCURACY 90;


SELECT /*+ vector_index_transform(galaxies galaxies_hnsw_idx
in_filter_with_join_back) */ name
FROM galaxies
WHERE id<5
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]'), COSINE )
FETCH FIRST 10 ROWS ONLY WITH TARGET ACCURACY 90;
```

> **Note**
>
> - The auxiliary tables for vector indexes are regular heap tables and require up-to-date optimizer statistics for accurate costing. The automatic optimizer statistics collection task gathers statistics for the auxiliary tables only when collecting statistics on their base tables.
> - Using the `DBMS_STATS` package, you can perform manual statistics gathering operations on the auxiliary tables. However, `GATHER_INDEX_STATS` on vector indexes is not permitted. The in-memory HNSW index automatically maintains its statistics in-memory.

> **See Also**
>
> - *Oracle AI Database SQL Tuning Guide* for information about configuring automatic optimizer statistics collection

The SQL examples included previously are based on the following:

```
DROP TABLE galaxies PURGE;
CREATE TABLE galaxies (id NUMBER, name VARCHAR2(50), doc VARCHAR2(500),
EMBEDDING VECTOR);
-- or CREATE TABLE galaxies (id NUMBER, name VARCHAR2(50), doc VARCHAR2(500),
EMBEDDING VECTOR(5,INT8));

INSERT INTO galaxies VALUES (1, 'M31', 'Messier 31 is a barred spiral galaxy
in the Andromeda constellation which has a lot of barred spiral galaxies.',
'[0,2,2,0,0]');
INSERT INTO galaxies VALUES (2, 'M33', 'Messier 33 is a spiral galaxy in the
Triangulum constellation.', '[0,0,1,0,0]');
INSERT INTO galaxies VALUES (3, 'M58', 'Messier 58 is an intermediate barred
spiral galaxy in the Virgo constellation.', '[1,1,1,0,0]');
INSERT INTO galaxies VALUES (4, 'M63', 'Messier 63 is a spiral galaxy in the
Canes Venatici constellation.', '[0,0,1,0,0]');
INSERT INTO galaxies VALUES (5, 'M77', 'Messier 77 is a barred spiral galaxy
in the Cetus constellation.', '[0,1,1,0,0]');
INSERT INTO galaxies VALUES (6, 'M91', 'Messier 91 is a barred spiral galaxy
in the Coma Berenices constellation.', '[0,1,1,0,0]');
INSERT INTO galaxies VALUES (7, 'M49', 'Messier 49 is a giant elliptical
galaxy in the Virgo constellation.', '[0,0,0,1,1]');
INSERT INTO galaxies VALUES (8, 'M60', 'Messier 60 is an elliptical galaxy in
the Virgo constellation.', '[0,0,0,0,1]');
INSERT INTO galaxies VALUES (9, 'NGC1073', 'NGC 1073 is a barred spiral
galaxy in Cetus constellation.', '[0,1,1,0,0]');
```

For HNSW indexes:

```
CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies (embedding)
ORGANIZATION INMEMORY NEIGHBOR GRAPH
DISTANCE COSINE
WITH TARGET ACCURACY 95;
```

For IVF indexes:

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies(embedding)
ORGANIZATION NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 95;
```

Approximate Similarity Search Examples

Review these examples to see how you can perform an approximate similarity search using vector indexes.

- Approximate Search Using HNSW
  This example shows how you can create the Hierarchical Navigable Small World (HNSW) index and run an approximate search using that index.
- Approximate Search Using IVF
  This example shows how you can create the Inverted File Flat (IVF) index and run an approximate search using that index.

Approximate Search Using HNSW

This example shows how you can create the Hierarchical Navigable Small World (HNSW) index and run an approximate search using that index.

```
create table galaxies (id number, name varchar2(50), doc varchar2(500),
embedding vector(5,INT8));

insert into galaxies values (1, 'M31', 'Messier 31 is a barred spiral galaxy
in the Andromeda constellation which has a lot of barred spiral galaxies.',
'[0,2,2,0,0]');
insert into galaxies values (2, 'M33', 'Messier 33 is a spiral galaxy in the
Triangulum constellation.', '[0,0,1,0,0]');
insert into galaxies values (3, 'M58', 'Messier 58 is an intermediate barred
spiral galaxy in the Virgo constellation.', '[1,1,1,0,0]');
insert into galaxies values (4, 'M63', 'Messier 63 is a spiral galaxy in the
Canes Venatici constellation.', '[0,0,1,0,0]');
insert into galaxies values (5, 'M77', 'Messier 77 is a barred spiral galaxy
in the Cetus constellation.', '[0,1,1,0,0]');
insert into galaxies values (6, 'M91', 'Messier 91 is a barred spiral galaxy
in the Coma Berenices constellation.', '[0,1,1,0,0]');
insert into galaxies values (7, 'M49', 'Messier 49 is a giant elliptical
galaxy in the Virgo constellation.', '[0,0,0,1,1]');
insert into galaxies values (8, 'M60', 'Messier 60 is an elliptical galaxy in
the Virgo constellation.', '[0,0,0,0,1]');
insert into galaxies values (9, 'NGC1073', 'NGC 1073 is a barred spiral
galaxy in Cetus constellation.', '[0,1,1,0,0]');

...

commit;
```

In the galaxies example, this is how you can create the HNSW index and how you would run an approximate search using that index:

```
CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies (embedding) ORGANIZATION
INMEMORY NEIGHBOR GRAPH
DISTANCE COSINE
WITH TARGET ACCURACY 95;

SELECT name
FROM galaxies
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8),
COSINE )
FETCH FIRST 3 ROWS ONLY;
```

This approximate search example inherits a target accuracy of 95 as it is set when the index is defined. You can override the `TARGET ACCURACY` of your search by running the following query examples:

```
SELECT name
FROM galaxies
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8),
COSINE  )
FETCH FIRST 3 ROWS ONLY WITH TARGET ACCURACY 90;


SELECT name
FROM galaxies
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8),
COSINE )
FETCH APPROXIMATE FIRST 3 ROWS ONLY WITH TARGET ACCURACY PARAMETERS (efsearch
500);
```

> **Note**
>
> If the vector format is not specified or is defined as `FLOAT32` (the default vector format) during table creation, the inclusion of the constructor, `TO_VECTOR`, is not required in subsequent queries

> **Note**
>
> The `APPROX` and `APPROXIMATE` keywords are optional. If omitted while connected to an ADB-S instance, an approximate search using a vector index is attempted if one exists.

You can also create the index using the following syntax:

```
CREATE VECTOR INDEX galaxies_hnsw_idx ON galaxies (embedding) ORGANIZATION
INMEMORY NEIGHBOR GRAPH
```

```
DISTANCE COSINE
WITH TARGET ACCURACY 90 PARAMETERS (type HNSW, neighbors 40, efconstruction
500);
```

> **See Also**
>
> *Oracle AI Database SQL Language Reference* for the full syntax of the `ROW_LIMITING_CLAUSE`

Approximate Search Using IVF

This example shows how you can create the Inverted File Flat (IVF) index and run an approximate search using that index.

```
create table galaxies (id number, name varchar2(50), doc varchar2(500),
embedding vector(5,INT8));

insert into galaxies values (1, 'M31', 'Messier 31 is a barred spiral galaxy
in the Andromeda constellation which has a lot of barred spiral galaxies.',
'[0,2,2,0,0]');
insert into galaxies values (2, 'M33', 'Messier 33 is a spiral galaxy in the
Triangulum constellation.', '[0,0,1,0,0]');
insert into galaxies values (3, 'M58', 'Messier 58 is an intermediate barred
spiral galaxy in the Virgo constellation.', '[1,1,1,0,0]');
insert into galaxies values (4, 'M63', 'Messier 63 is a spiral galaxy in the
Canes Venatici constellation.', '[0,0,1,0,0]');
insert into galaxies values (5, 'M77', 'Messier 77 is a barred spiral galaxy
in the Cetus constellation.', '[0,1,1,0,0]');
insert into galaxies values (6, 'M91', 'Messier 91 is a barred spiral galaxy
in the Coma Berenices constellation.', '[0,1,1,0,0]');
insert into galaxies values (7, 'M49', 'Messier 49 is a giant elliptical
galaxy in the Virgo constellation.', '[0,0,0,1,1]');
insert into galaxies values (8, 'M60', 'Messier 60 is an elliptical galaxy in
the Virgo constellation.', '[0,0,0,0,1]');
insert into galaxies values (9, 'NGC1073', 'NGC 1073 is a barred spiral
galaxy in Cetus constellation.', '[0,1,1,0,0]');

...

commit;
```

In the galaxies example, this is how you can create the IVF index and how you can run an approximate search using that index:

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding) ORGANIZATION
NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 95;

SELECT name
FROM galaxies
```

```
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8),
COSINE )
FETCH FIRST 3 ROWS ONLY;
```

If the index is used by the optimizer, then this approximate search example inherits a target accuracy of 95 as it is set when the index is defined. You can override the TARGET ACCURACY of your search by running the following query examples:

```
SELECT name
FROM galaxies
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8),
COSINE )
FETCH APPROXIMATE FIRST 3 ROWS ONLY WITH TARGET ACCURACY 90;


SELECT name
FROM galaxies
WHERE name <> 'NGC1073'
ORDER BY VECTOR_DISTANCE( embedding, to_vector('[0,1,1,0,0]', 5, INT8) )
FETCH FIRST 3 ROWS ONLY WITH TARGET ACCURACY PARAMETERS ( NEIGHBOR PARTITION
PROBES 10 );
```

Note

If the vector format is not specified or is defined as FLOAT32 (the default vector format) during table creation, the inclusion of the constructor, TO_VECTOR, is not required in subsequent queries

Note

The APPROX and APPROXIMATE keywords are optional. If omitted while connected to an ADB-S instance, an approximate search using a vector index is attempted if one exists.

You can also create the index using the following syntax:

```
CREATE VECTOR INDEX galaxies_ivf_idx ON galaxies (embedding) ORGANIZATION
NEIGHBOR PARTITIONS
DISTANCE COSINE
WITH TARGET ACCURACY 90 PARAMETERS (type IVF, neighbor partitions 100);
```

See Also

Oracle AI Database SQL Language Reference for the full syntax of the ROW_LIMITING_CLAUSE

Perform Multi-Vector Similarity Search

Another major use-case of vector search is multi-vector search. Multi-vector search is typically associated with a multi-document search, where documents are split into chunks that are individually embedded into vectors.

A multi-vector search consists of retrieving top-K vector matches using grouping criteria known as partitions based on the documents' characteristics. This ability to score documents based on the similarity of their chunks to a query vector being searched is facilitated in SQL using the partitioned row limiting clause.

With multi-vector search, it is easier to write SQL statements to answer the following type of question:

- If they exist, what are the four best matching sentences found in the three best matching paragraphs of the two best matching books?

For example, imagine if each book in your database is organized into paragraphs containing sentences which have vector embedding representations, then you can answer the previous question using a single SQL statement such as:

```
SELECT bookId, paragraphId, sentence
FROM books
ORDER BY vector_distance(sentence_embedding, :sentence_query_vector)
FETCH EXACT FIRST 2 PARTITIONS BY bookId, 3 PARTITIONS BY paragraphId, 4 ROWS
ONLY;
```

You can also use an approximate similarity search instead of an exact similarity search as shown in the following example:

```
SELECT bookId, paragraphId, sentence
FROM books
ORDER BY vector_distance(sentence_embedding, :sentence_query_vector)
FETCH FIRST 2 PARTITIONS BY bookId, 3 PARTITIONS BY paragraphId, 4 ROWS ONLY
WITH TARGET ACCURACY 90;
```

> **Note**
>
> All the rows returned are ordered by `VECTOR_DISTANCE()` and not grouped by the partition clause.

> **Note**
>
> The `APPROX` and `APPROXIMATE` keywords are optional. If omitted while connected to an ADB-S instance, an approximate search using a vector index is attempted if one exists.

Semantically, the previous SQL statement is interpreted as:

- Sort all records in the books table in descending order of the vector distance between the sentences and the query vector.

- For each record in this order, check its `bookId` and `paragraphId`. This record is produced if the following three conditions are met:
  1. Its `bookId` is one of the first two distinct `bookId` in the sorted order.
  2. Its `paragraphId` is one of the first three distinct `paragraphId` in the sorted order within the same `bookId`.
  3. Its record is one of the first four records within the same `bookId` and `paragraphId` combination.
- Otherwise, this record is filtered out.

Multi-vector similarity search is not just for documents and can be used to answer the following questions too:

- Return the top K closest matching photos but ensure that they are photos of different people.
- Find the top K songs with two or more audio segments that best match this sound snippet.

> **Note**
>
> - This partition row-limiting clause extension is a generic extension of the SQL language. It does not have to apply just to vector searches.
> - Multi-vector search is currently supported with the IVF index.

- Multi-Vector Search Using IVF Indexes
  In many real-world use cases, multi-vector similarity searches must be performed on large datasets, requiring efficient indexing to improve performance. The Oracle AI Database offers support for accelerated similarity searches using IVF vector indexes, which can significantly reduce latency and resource utilization compared to scanning entire tables.
- Set Up Schema and IVF Index for Multi-Vector Search
  This section describes how to prepare your data model for multi-vector queries by defining a base table that stores vector embeddings alongside the business identifiers used for multi-vector grouping and optional filter columns. It also explains how to create an IVF vector index.

> **See Also**
>
> *Oracle AI Database SQL Language Reference* for the full syntax of the `ROW_LIMITING_CLAUSE`

Multi-Vector Search Using IVF Indexes

In many real-world use cases, multi-vector similarity searches must be performed on large datasets, requiring efficient indexing to improve performance. The Oracle AI Database offers support for accelerated similarity searches using IVF vector indexes, which can significantly reduce latency and resource utilization compared to scanning entire tables.

Why Use IVF Indexes for Multi-Vector Search?

IVF indexes partition the vector space into a configurable number of centroids, allowing each query to focus only on the most relevant portions of the data. By searching within these relevant partitions, IVF indexes can achieve fast response times for multi-vector search workloads that would otherwise take significantly longer with a full scan.

> **See Also**
>
> Understand Inverted File Flat Vector Indexes

The following are the advantages of using IVF indexes for multi-vector search:

- **Fast approximate similarity search**: IVF indexing improves vector search efficiency by organizing vectors into clusters and storing them in centroid-aligned partitions. At query time, the database first identifies the nearest centroids to the query vector, and then searches only the corresponding centroid partitions, rather than scanning the entire base table. This approach significantly reduces the number of rows read, the number of vector distance computations, and the amount of sorting work required to return the top results.
- **Integration with SQL**: The IVF index works seamlessly with SQL syntax, supporting approximate searches, including multi-level partitioned row limiting.
- **Supported features**: IVF multi-vector search supports included columns, multiple partition columns, and expression-based indexing to match real-world schemas and query patterns. It is also compatible with Exadata offload for improved performance.

Set Up Schema and IVF Index for Multi-Vector Search

This section describes how to prepare your data model for multi-vector queries by defining a base table that stores vector embeddings alongside the business identifiers used for multi-vector grouping and optional filter columns. It also explains how to create an IVF vector index.

Define the Table Schema

Create a table to store your vector-embedded data along with the attributes that you will use for partitioning (for example, `book_id` and `paragraph_id`) and query filtering (for example, `create_date`). In this example, each book contains multiple paragraphs, and each row stores an individually embedded sentence vector.

```
CREATE TABLE books (
  record_id     NUMBER,
  book_id       NUMBER,
  paragraph_id  NUMBER,
  sentence      VECTOR,
  create_date   DATE
);
```

Create IVF Indexes

An IVF vector index speeds up vector similarity search by clustering vectors into centroids and searching only the most relevant centroid partitions at query time. For multi-vector queries, you can further improve performance by defining included columns in the index. Included columns are non-vector attributes that are stored with the IVF index structures, enabling the optimizer to satisfy eligible queries without reading the base table.

Use included columns when your multi-vector queries commonly:

- project non-vector attributes (for example, `book_id`, `paragraph_id`)
- filter on those attributes (for example, `create_date`)
- use them in `PARTITIONS BY` clauses

The following example creates an IVF index on the `sentence` column to speed up the multi-vector similarity search:

```
CREATE VECTOR INDEX ivf_idx ON books (sentence)
  ORGANIZATION NEIGHBOR PARTITIONS
  WITH TARGET ACCURACY 95
  DISTANCE COSINE
  PARAMETERS (type IVF, neighbor partitions 100);
```

The IVF index is created on the `sentence` column of the `books` table, clustering vectors into 100 partitions for efficient similarity search with 95% accuracy using squared Euclidean distance.

> **Note**
>
> `COSINE` is used here as an example distance function. Multi-vector similarity search works with any supported distance function.

Example: IVF Index With Included Columns

The following example creates an IVF vector index on the `sentence` vector column and includes `book_id`, `paragraph_id`, and `create_date` to support common multi-vector grouping and filtering patterns:

```
CREATE VECTOR INDEX ivf_idx ON books (sentence)
  INCLUDE (book_id, paragraph_id, create_date)
  ORGANIZATION NEIGHBOR PARTITIONS
  WITH TARGET ACCURACY 95
  DISTANCE COSINE
  PARAMETERS (type IVF, neighbor partitions 100);
```

Example: No-Filter Multi-Vector Query

With your schema and IVF index in place, you can perform efficient top-K queries with multi-level partitioning.

```
SELECT book_id, paragraph_id
FROM books
ORDER BY vector_distance(sentence, :query_vector, COSINE)
FETCH FIRST 2 PARTITIONS BY book_id,
                   3 PARTITIONS BY paragraph_id,
                   4 ROWS ONLY;
```

The query retrieves the two most relevant books (based on vector distance), then the three best paragraphs within each selected book, and finally the four closest sentences per (book, paragraph). The database can use the IVF index to identify the nearest centroids and search only the most relevant centroid partitions, improving response time by avoiding a full table scan and large global sort.

Example: Pre-Filter Multi-Vector Query (Filter on an Included Column)

With your schema and IVF index (including `create_date`) in place, you can add a filter on an included column while still using multi-level partitioning.

```
SELECT book_id, paragraph_id
FROM books
WHERE EXTRACT(YEAR FROM create_date) = 2020
ORDER BY VECTOR_DISTANCE(sentence, :query_vector, COSINE)
FETCH FIRST 2 PARTITIONS BY book_id,
                   3 PARTITIONS BY paragraph_id,
                   4 ROWS ONLY;
```

This query restricts the search to rows from year 2020 first, then applies the same multi-vector logic that return results from the two most relevant books, then the three best paragraphs per selected book, and finally the four closest sentences per (book, paragraph). Because `create_date` is an included column in the IVF index, the database can apply the filter while leveraging the IVF index to focus on the most relevant centroid partitions, reducing the need for a full table scan and large global sort.

Example: Multi-Vector Query with an Expression on a Partitioned Column

You can also use an expression in the `PARTITIONS BY` clause to group results at a derived level (for example, by year), while still ranking by vector distance.

```
SELECT book_id,
       paragraph_id,
       TRUNC(create_date, 'YYYY') AS create_year
FROM books
ORDER BY VECTOR_DISTANCE(sentence, :query_vector, COSINE)
FETCH FIRST 2 PARTITIONS BY book_id,
                   3 PARTITIONS BY TRUNC(create_date, 'YYYY'),
                   4 ROWS ONLY;
```

Here, the IVF index is used to prune the search space early by restricting evaluation to the most relevant centroid partitions, and then the multi-vector logic applies expression-based partitioning to enforce grouping in the final output.

Example: Multi-Vector Similarity Search as a Subquery

The following example demonstrates how a full multi-vector partitioned similarity search, leveraging the IVF vector index for efficient top-K retrieval at multiple grouping levels, can be wrapped as a subquery:

```
SELECT *
 FROM (
        SELECT /*+ vector_index_transform(CHUNKDOC_TAB) */
               pub_id, doc_id, chunk_id, sell_region, rating,
               vector_distance(embedding, :query_vec_1, COSINE) AS distance
        FROM   chunkdoc_tab
        ORDER BY distance
        FETCH FIRST 2 PARTITIONS BY pub_id,
```

```
                        3 PARTITIONS BY doc_id,
                        4 PARTITIONS BY chunk_id,
                        2 ROWS ONLY
    )
 WHERE distance > 0.5
 ORDER BY pub_id, doc_id;
```

The outer query then applies additional filtering (such as distance > 0.5) and sorting. This pattern makes it easy to filter or organize your results after performing a fast, flexible vector search across groups in your data.

Example: Multi-Vector Join Query

You can also perform multi-vector similarity search in combination with a join to enrich your results with additional attributes from another table. For example, to join the results of the vector search on the `books` table with publisher region information from a `publisher_info` table:

```
SELECT /*+ VECTOR_INDEX_TRANSFORM(b) */
       b.book_id,
       b.paragraph_id,
       b.sentence,
       p.publisher_name,
       p.region,
       VECTOR_DISTANCE(b.sentence, :query_vector, COSINE) AS similarity
FROM   books b
JOIN   publisher_info p ON b.book_id = p.book_id
ORDER BY similarity
FETCH FIRST 3 PARTITIONS BY (p.region),
            2 PARTITIONS BY (b.book_id),
            2 ROWS ONLY;
```

This query finds, for each region, the two most relevant books, and for each book, the two closest sentences, all ranked by vector similarity to the query. By joining with `publisher_info`, you can include publisher metadata in your search results.

Perform Hybrid Search

Hybrid search is an advanced information retrieval technique that lets you search documents by *keywords* and *vectors*, to achieve more relevant search results.

- Understand Hybrid Search
  With hybrid search, you can search through your documents by performing a combination of full-text queries and vector-based similarity queries, using out-of-the-box or custom scoring techniques.
- Query Hybrid Vector Indexes End-to-End Example
  In this example, you can see an end-to-end hybrid search workflow. First, you run the `CREATE HYBRID VECTOR INDEX` SQL statement that prepares, chunks, embeds, stores, and indexes your input data. You then perform vector search alongside keyword search using the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL query API.

Understand Hybrid Search

With hybrid search, you can search through your documents by performing a combination of full-text queries and vector-based similarity queries, using out-of-the-box or custom scoring techniques.

Here are the key points to note when using hybrid search:

- Hybrid searches are run against hybrid vector indexes (as explained in Manage Hybrid Vector Indexes). You use the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL function to query a hybrid vector index.
- After you create a hybrid vector index, you have five possibilities to use the index by querying it in multiple **search modes** summarized here:
  - Pure Semantic in Document Mode
  - Pure Semantic in Chunk Mode
  - Pure Keyword in Document Mode
  - Keyword and Semantic in Document Mode
  - Keyword and Semantic in Chunk Mode
- When performing hybrid search by both keywords and vectors, the results are combined (or fused) into a single result set.

  Similarity search uses the notion of `VECTOR_DISTANCE` values to decide on the ranking of chunks, whereas the traditional Oracle Text search uses the notion of **keyword score** also known as `CONTAINS` score. These two metrics are very different and one cannot be used directly to compare to the other. Hence, similarity search distances are converted or normalized into a `CONTAINS`-score equivalent that is called **semantic score** so its value will range between 100 (best) to 0 (worse). That way, keyword score and semantic score are comparable when running a hybrid search.

  After the results are combined (or fused) into a single result set, hybrid search employs any one of the following search_scorer methods to evaluate the search scores from both keyword and semantic search results. The results of a hybrid search are based on this scoring mechanism that is important to understand to be able to interpret the results.

  - RSF (Relative Score Fusion) : RSF computes a weighted sum of the normalized scores from the keyword and semantic search results. Use this method when you want a simple, weighted blend of scores directly proportional to their quality and relevance.
    Formula

$$RSF(r) = \frac{\sum_{s \in S} \text{weight}_s \times \text{score}_s(r)}{\sum_{s \in S} \text{weight}_s}$$

    When Oracle Database calculates the RSF search scores for hybrid search results, it uses the values specified in the `text.score_weight` and `vector.score_weight` fields provided in the JSON search request as the weights for the text and vector scores during the computation. Here is an example of how RSF is calculated for each document in a hybrid search result set.

Table 8-3 RSF Example

Document	Vector Score	Vector Weight	Vector RS	Text Score	Text Weight	Text RS	Sum RS	RSF
1	90	5	450.0000	4	1	4.0000	454.0000	75.67
2	50	5	250.0000	95	1	95.0000	345.0000	57.50
3	33	5	165.0000	80	1	80.0000	245.0000	40.83
MAX RFS	100	5	500.0000	100	1	100.0000	600.0000	100.00

– RRF (Reciprocal Rank Fusion): RRF uses the reciprocal of the rank positions from both keyword and semantic search methods, adjusted by a penalty (rank constant), to blend relevance. Use this method when you want to emphasize highly ranked results from both search types, rather than raw score magnitude.
Formula

$$RRF(r) = \frac{\sum_{s \in S} \frac{1}{penalty_s + rank_s(r)}}{\sum_{s \in S} \frac{1}{penalty_s + 1}}$$

When Oracle Database calculates the RRF search scores for hybrid search results, it uses the values specified in the `text.rank_penalty` and `vector.rank_penalty` fields provided in the JSON search request as the penalty values for the text and vector scores during the computation. The following table illustrates the calculation of the RRF for various documents with given rankings in the vector and text results.

Table 8-4 RRF Example

Document	Vector Rank	Vector Penalty	Vector RR	Text Rank	Text Penalty	Text RR	Sum RR	RRF
1	1	1	0.5000	4	5	0.1111	0.6111	91.67
2	5	1	0.1667	100	5	0.0095	0.1762	26.43
MAX RRF	1	1	0.5000	1	5	0.1667	0.6667	100.00

– WRRF (Weighted Reciprocal Rank Fusion): WRRF is an extension of RRF, introducing weights to balance the relative influence of keyword and semantic results. Use this method when more granular control of the influence of each modality (keyword vs. semantic) is needed.
Formula

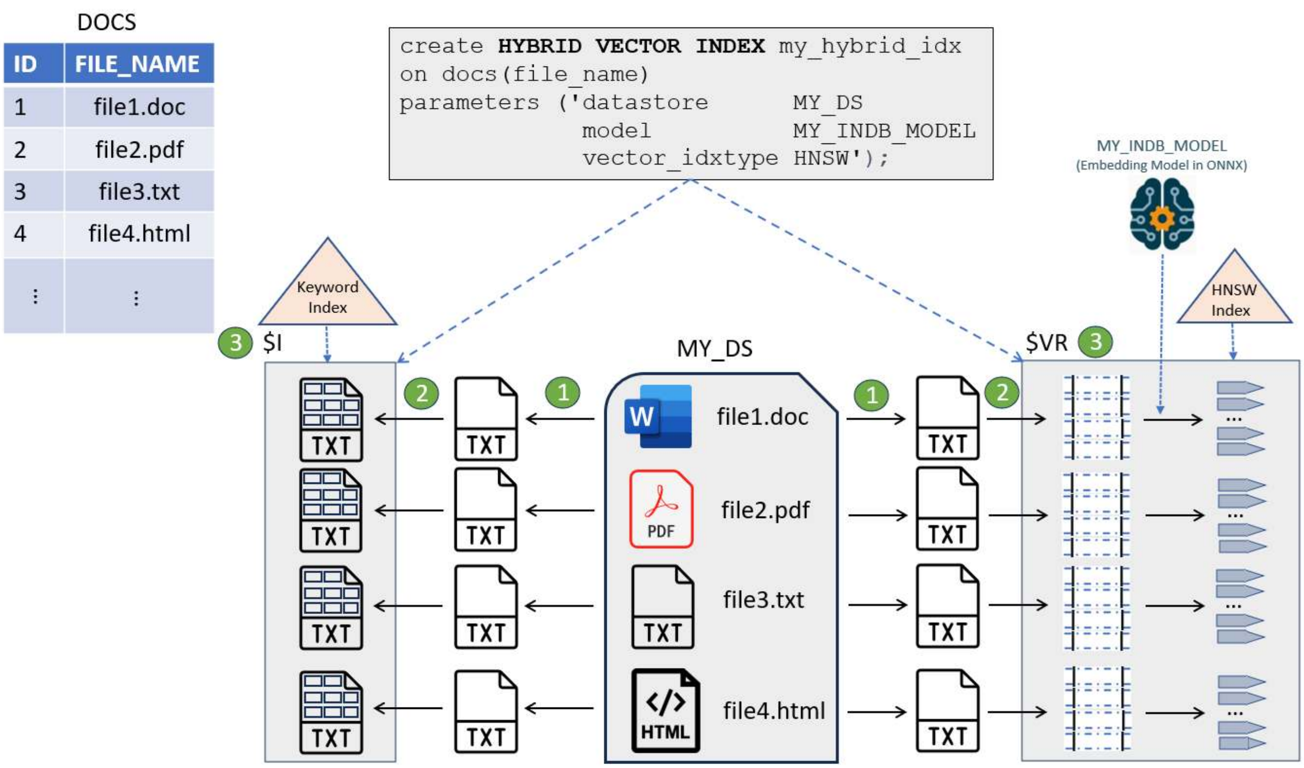
$$WRRF(r) = \frac{\displaystyle\sum_{s \in S} weight_s \times \frac{1}{penalty_s + rank_s(r)}}{\displaystyle\sum_{s \in S} weight_s \times \frac{1}{penalty_s + 1}}$$

When Oracle Database calculates the WRRF search scores for hybrid search results, it uses the values specified in the `text.score_weight` and `vector.score_weight` fields provided in the JSON search request as the weights values for the text and vector scores during the computation. Similarly, it uses the values specified in the `text.rank_penalty` and `vector.rank_penalty` fields provided in the JSON search request as the penalty values for the text and vector scores during the computation. The following table illustrates the calculation of the WRRF for various documents with given rankings in the vector and text results.

Table 8-5 WRRF Example

Document	Vector Rank	Vector Penalty	Vector Weight	Vector WRR	Text Rank	Text Penalty	Text Weight	Text WRR	Sum WRR	WRRF
1	1	60	5	0.0820	4	60	1	0.0156	0.0976	99.22
2	5	60	5	0.0769	100	60	1	0.0063	0.0832	84.56
MAX WRRF	1	60	5	0.0820	1	60	1	0.0164	0.0984	100.00

For the sake of the following explanations, we are using the same use case as seen in Understand Hybrid Vector Indexes:



Pure Semantic in Document Mode

The pure semantic in document mode performs vector-only search to fetch document-level results.

The following SQL statement queries a hybrid vector index in this mode:

```
select json_Serialize(
  DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "galaxies formation and massive black holes",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "AVG"
          },
         "return":
          {
             "values"        : [ "rowid", "score", "vector_score" ],
             "topN"          : 10
          }
      }'
    )
  ) RETURNING CLOB pretty);
```

The result of this query may look like the following, where you see the `ROWID`s of the `DOCS` table rows corresponding to documents as well as its vector score (`vector_score`). Here, the final score (`score`) is the same as the vector score because there is no keyword score in this case.

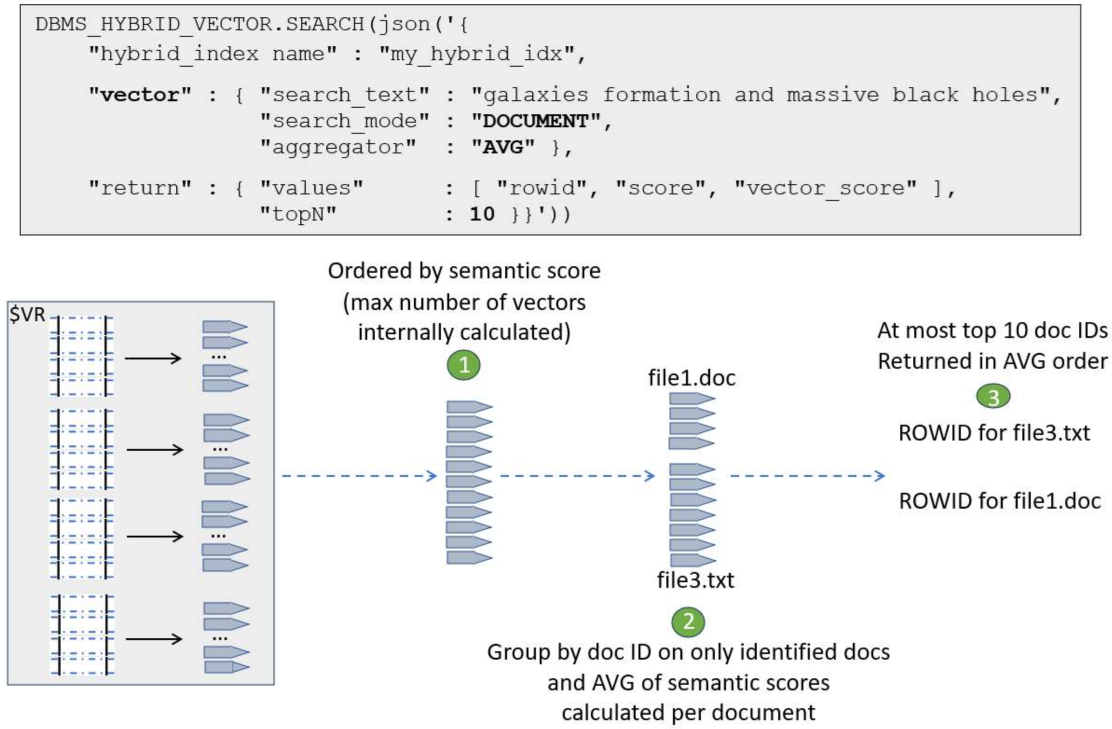
Here is an excerpt from the results:

```
[
{
    "rowid"        : "AAASBEAAEAAAUpqAAB",
    "score"        : 71.04,
    "vector_score" : 71.04
  },
  {
    "rowid"        : "AAASBEAAEAAAWBKAAE",
    "score"        : 67.82,
    "vector_score" : 67.82
  },
]
```

Here is how to interpret this statement:

1. The system runs a similarity search on all the vectors and extracts the top k ones at most. The value k is internally calculated. Each is given a vector score.
2. These k vectors (at most) are then grouped by document IDs and for each identified document, the semantic score of each associated vector found for that document is used to compute the average (in this example) of these scores for that particular document.
3. The top 10 documents (at most) with the highest averages are returned.

This is illustrated here:



Pure Semantic in Chunk Mode

The pure semantic in chunk mode is the SQL semantic search equivalent. This mode performs a vector-only search to fetch chunk results.

The following SQL statement queries a hybrid vector index in this mode:

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "galaxies formation and massive black holes",
             "search_mode"   : "CHUNK"
          },
         "return":
          {
             "values"        : [ "score", "chunk_text", "chunk_id" ],
             "topN"          : 3
          }
      }'
    )
  ) RETURNING CLOB pretty);
```

Here is how to interpret this statement:

1. The system runs a similarity search on all the vectors and extracts the top k ones at most. The value k is internally calculated. Each is given a vector score.

2. The top 3 chunks (at most) with the highest scores are returned.

The result of this query may look like the following, where you can see the chunks corresponding to their semantic scores:

```
[
  {
    "score"      : 61,
    "chunk_text" : "Galaxies form through a complex process that begins with
small fluctuations in the density of matter in the early
universe. Massive black holes, typically found at the centers of galaxies,
are believed to play a crucial role in their formation and evolution.",
    "chunk_id"   : "1"
  },
  {
    "score"      : 56.64,
    "chunk_text" : "The presence of massive black holes in galaxies is
closely linked to their morphological characteristics and star formation
rates.
Observations suggest that as galaxies evolve, their central black holes grow
in tandem with their host galaxy's mass.",
    "chunk_id"   : "3"
  },
  {
    "score"      : 55.75,
    "chunk_text" : "Black holes grow by accreting gas and merging with other
black holes. Their gravitational influence can regulate star
formation and drive powerful jets of energy, which can impact the surrounding
galaxy.",
    "chunk_id"   : "2"
  }
]
```

Pure Keyword in Document Mode

The pure keyword search in document mode is equivalent to the traditional `CONTAINS` query using Oracle Text. This mode performs a text-only search to fetch document-level results.

The following SQL statement queries a hybrid vector index in this mode:

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "galaxies, black holes"
          },
         "return":
          {
             "values"        : [ "rowid", "score" ],
             "topN"          : 3
          }
      }'
    )
  ) RETURNING CLOB pretty);
```

Here is how to interpret this statement:

1. The system runs a `CONTAINS` query that returns a maximum number of documents. This maximum number is internally calculated. Each document is given a keyword score.
2. The top 3 documents (at most) with the highest scores are returned.

The result of this query may look like the following, where you can see the `ROWID`s of the `DOCS` table rows corresponding to documents as well as their keyword scores:

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 68
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAA",
    "score" : 35
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 2
  }
]
```

Keyword and Semantic in Document Mode

Let us examine a non-pure case of hybrid search where keyword scores and semantic scores are combined.

The following SQL statement performs a keyword and semantic search to fetch document-level results:

```
select json_Serialize(
  DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{
         "hybrid_index_name" : "my_hybrid_idx",
         "search_scorer"     : "rsf",
         "search_fusion"     : "UNION",
         "vector":
          {
             "search_text"   : "How can I search with hybrid vector indexes?",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 1,
             "rank_penalty"  : 5
          },
         "text":
          {
             "contains"      : "hybrid AND vector AND index"
             "score_weight"  : 10,
             "rank_penalty"  : 1
          },
         "return":
          {
             "values"        : [ "rowid", "score", "vector_score",
```

```
"text_score" ],
             "topN"          : 10
          }
       }'
     )
  ) RETURNING CLOB pretty);
```

Here is how to interpret this statement:

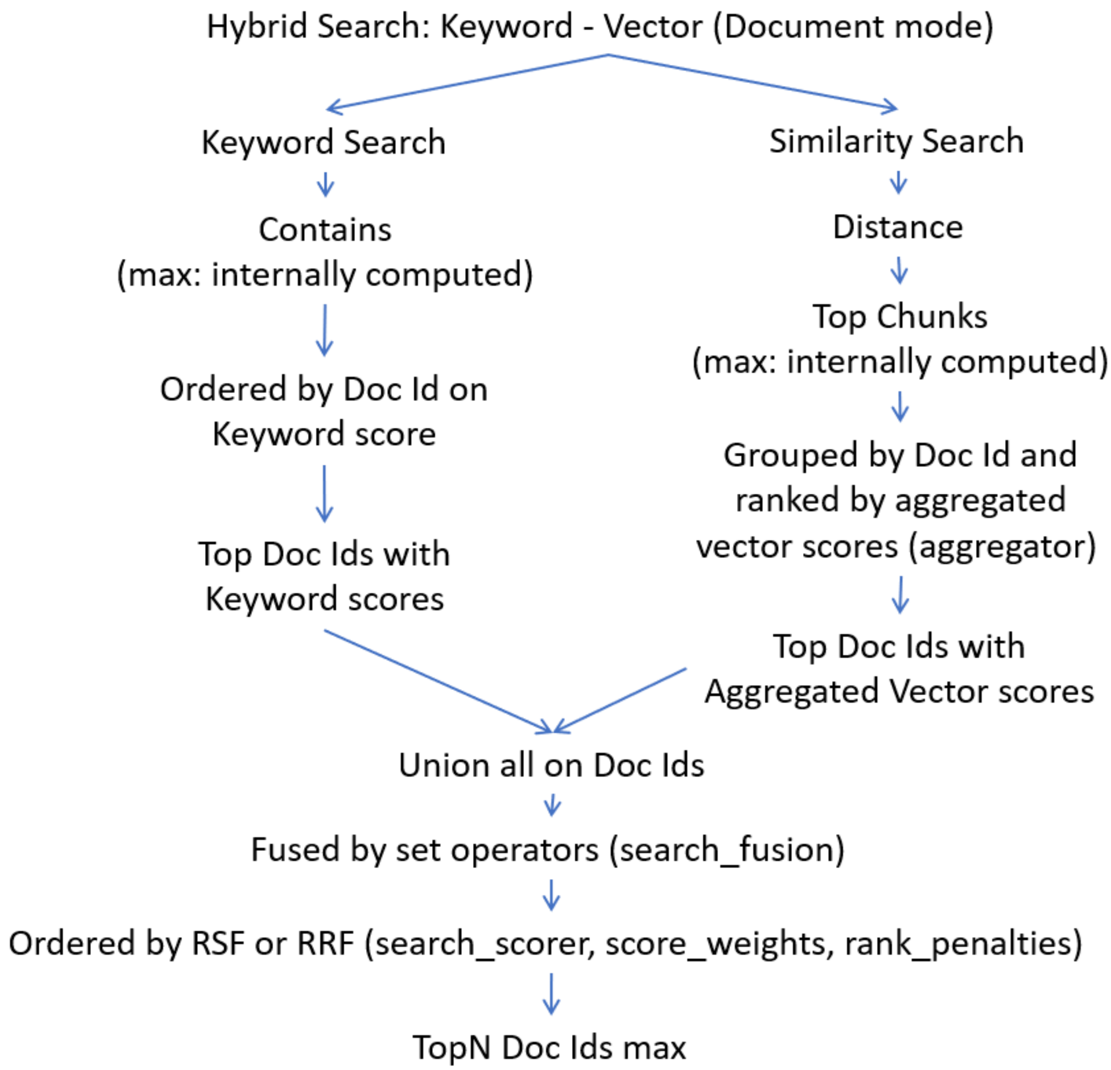
In document mode, the result of your search is a list of ROWIDs from your base table corresponding to the list of best files identified.

To get to this list, two searches are conducted:

- **Keyword search**: During this search, the system uses the CONTAINS string representing the Oracle Text CONTAINS search expression for the searched keywords. The result of this operation is a list of document identifiers satisfying your CONTAINS expression. The numbers of document identifiers to retrieve, at max, is internally calculated.
- **Similarity search**: During this search, the system performs a similarity search with the query vector (created from the SEARCH_TEXT string) against the vector index of all the chunks of all your documents. The maximum number of chunks to retrieve is also internally calculated. It then assigns a vector score to each chunk retrieved. Because you want to run this search in DOCUMENT SEARCH_MODE, the result of this similarity search is first grouped by document identifier. The process now aggregates the vector scores for each document, identified using the AGGREGATOR function. The result of this similarity search is a list of document identifiers satisfying your SEARCH_TEXT similarity query string.

After the searches complete, the system needs to merge the results and score them as illustrated here:

Figure 8-2 Scoring for Keyword and Semantic Search in Document Mode

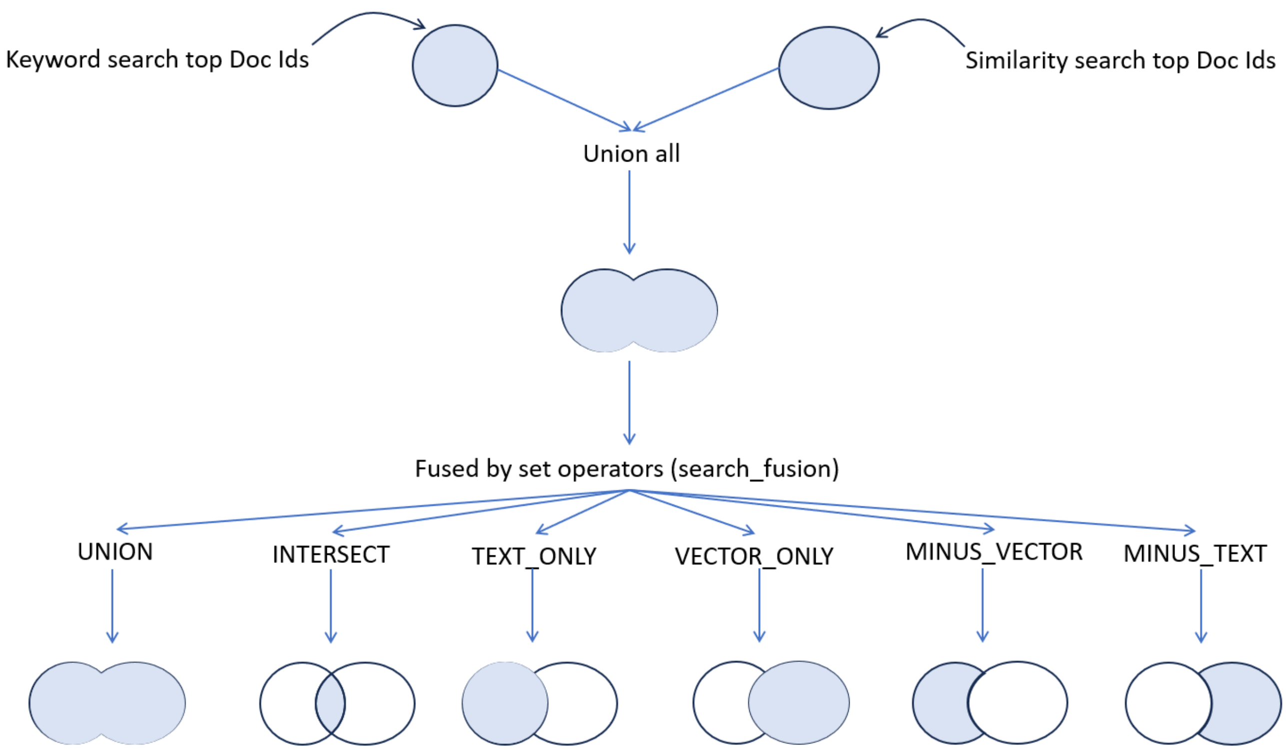


As outlined in the preceding diagram,

- First, both search results are added using a `UNION ALL` operation.
- Before final scoring, you have the possibility to define what you want to retain from this intermediate result set by specifying the `SEARCH_FUSION` operation.
- Then comes the time where final scoring is computed using the defined `SEARCH_SCORER` algorithm such as Reciprocal Rank Fusion (RRF) or Relative Score Fusion (RSF). The final scoring can use the specified `SCORE_WEIGHT` and `RANK_PENALTY` values for each retrieved document identifier, depending from which sort operation they are coming from.
- Finally, the defined `topN` document identifiers are returned at most.

The possible fuse operators are illustrated here:

Figure 8-3 Fusion Operation in Document Search Mode



Keyword and Semantic in Chunk Mode

Let us examine another non-pure case of hybrid search where keyword scores and semantic scores are combined to fetch chunk results.

The following SQL statement performs a keyword and semantic search in chunk mode:

```
select json_Serialize(
  DBMS_HYBRID_VECTOR.SEARCH(
    json(
         '{
            "hybrid_index_name" : "my_hybrid_vector_idx",
            "search_scorer"     : "rsf",
            "search_fusion"     : "UNION",
            "vector":
                      {
                        "search_text"   : "How can I search with hybrid
vector indexes?",
                        "search_mode"   : "CHUNK",
                        "score_weight"  : 1
                      },
            "text":
                      {
                       "contains"       : "hybrid AND vector AND index",
                       "score_weight"   : 1
                      },
            "return":
                      {
```

```
                        "values"        : [ "chunk_id", "score",
"vector_score", "text_score" ],
                        "topN"          : 10
                      }
          }'
    )
  ) RETURNING CLOB pretty);
```

Here is how to interpret this statement:

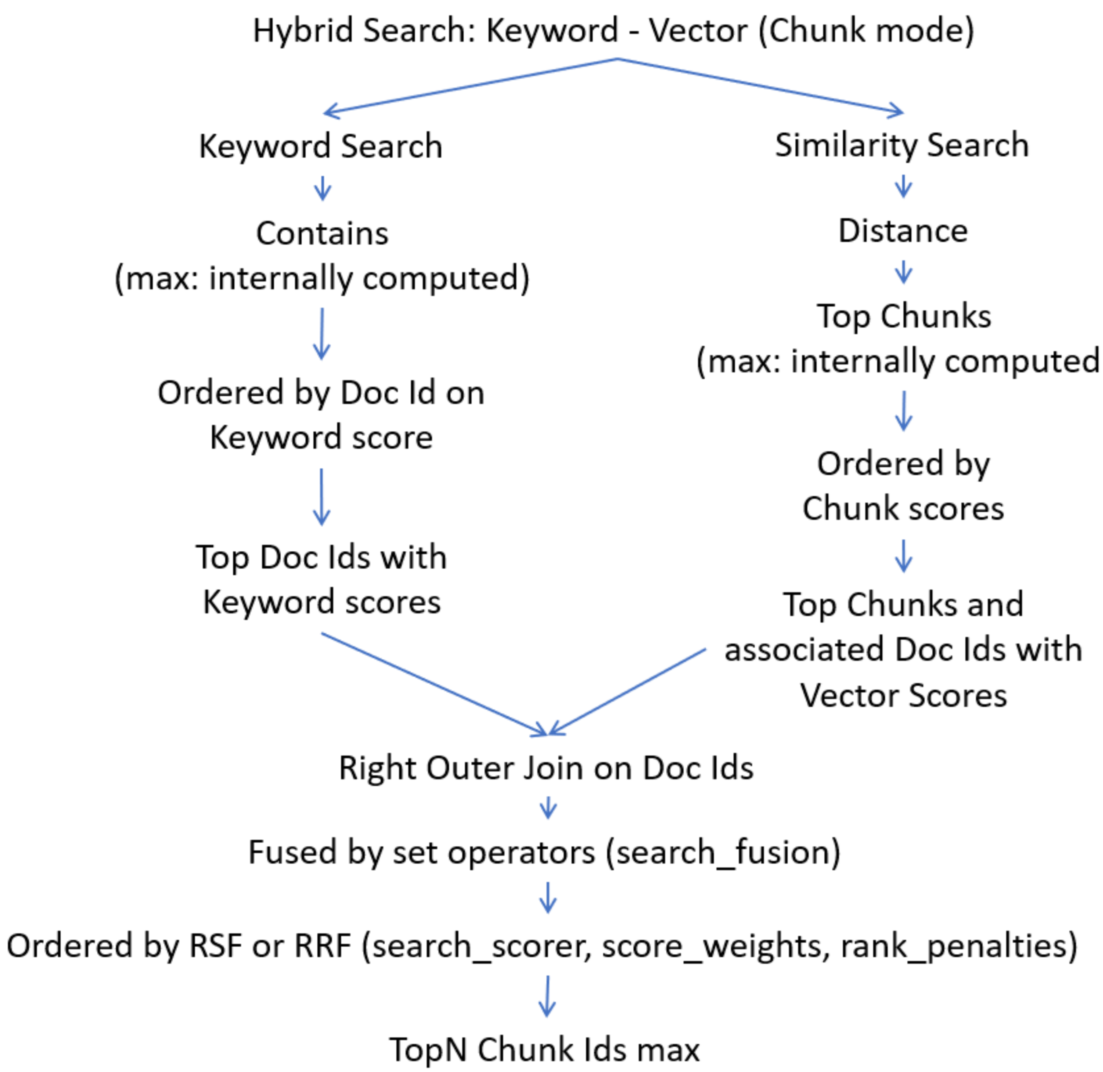
In chunk mode, the result of your search is a list of best chunk identifiers from the files stored using your base table.

To get to this list, two searches are conducted:

- **Keyword search**: During this search, the system uses the `CONTAINS` string representing the Oracle Text `CONTAINS` search expression for the searched keywords. The result of this operation is a list of document identifiers satisfying your `CONTAINS` expression. The numbers of document identifiers to retrieve, at max, is internally calculated.
- **Similarity search**: During this search, the system performs a similarity search with the query vector (created from the `SEARCH_TEXT` string) against the vector index of all the chunks of all your documents. The maximum number of chunks to retrieve is also internally calculated. It then assigns a vector score to each chunk retrieved. Because you want to do this search in `CHUNK SEARCH_MODE`, the result of this similarity search is ordered by chunk vector scores. The result of this similarity search is a list of chunk identifiers and associated document identifiers satisfying your `SEARCH_TEXT` similarity query string.

After the searches complete, the system needs to merge the results and score them as illustrated here:

Figure 8-4 Scoring for Keyword and Semantic Search in Chunk Mode

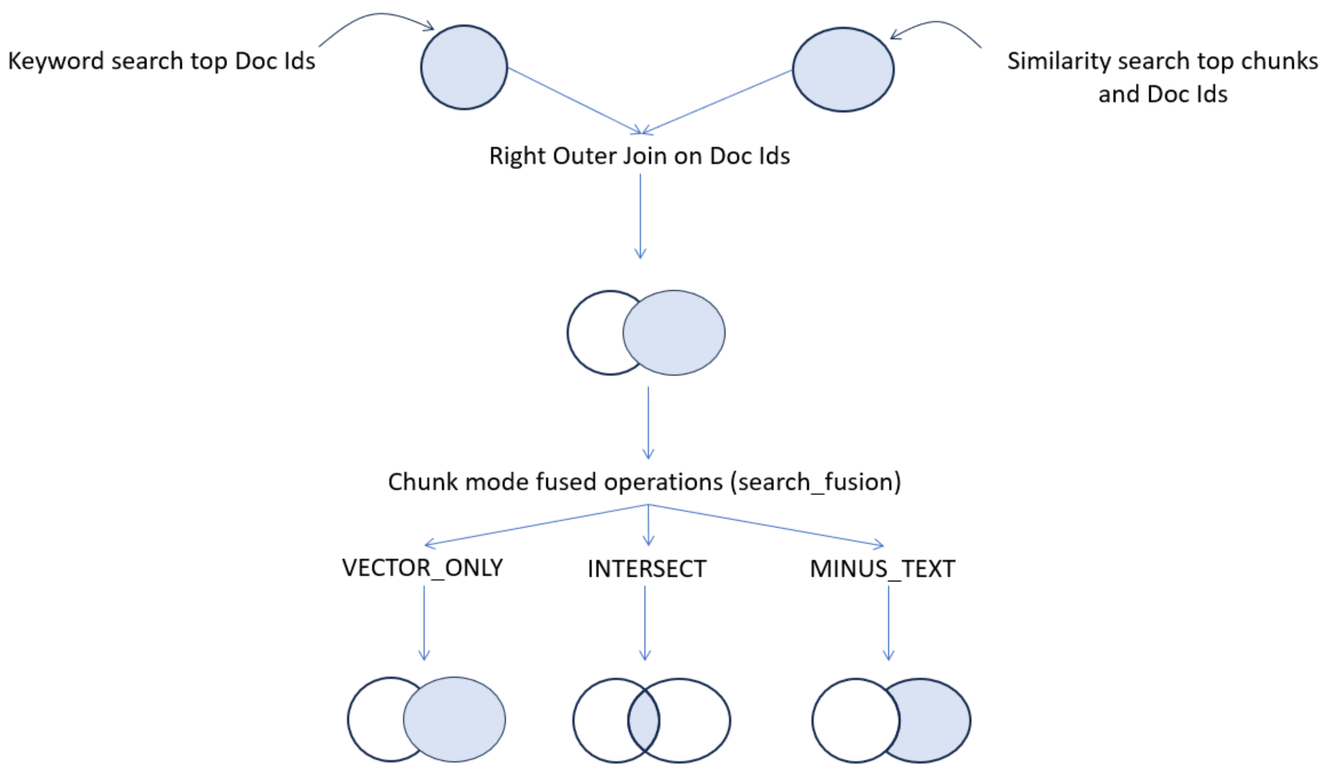


As outlined in the preceding diagram,

- First, both search results are submitted to a `RIGHT OUTER JOIN` operation on the document identifiers.
- Before final scoring, you have the possibility to define what you want to retain from this intermediate result set by specifying the `SEARCH_FUSION` operation.
- Then comes the time where final scoring is computed using the defined `SEARCH_SCORER` algorithm such as Reciprocal Rank Fusion (RRF) or Relative Score Fusion (RSF). The final scoring can use the specified `SCORE_WEIGHT` and `RANK_PENALTY` values for each retrieved document identifier, depending from which sort operation they are coming from.
- Finally, the defined `topN` chunk identifiers are returned at most.

The possible fuse operators are illustrated here:

Figure 8-5   Fusion Operation in Chunk Search Mode



`MINUS_VECTOR`, `UNION`, and `TEXT_ONLY` are ignored. `MINUS_VECTOR` would eliminate all results. `TEXT_ONLY` and `UNION` are not possible because the right outer join excludes the non-overlapping text results.

Related Topics

- Understand Hybrid Vector Indexes
  A hybrid vector index inherits all the information retrieval capabilities of Oracle Text search indexes and leverages the semantic search capabilities of Oracle AI Vector Search vector indexes.
- SEARCH
  Use the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL function to run textual queries, vector similarity queries, or hybrid queries against hybrid vector indexes.

Query Hybrid Vector Indexes End-to-End Example

In this example, you can see an end-to-end hybrid search workflow. First, you run the `CREATE HYBRID VECTOR INDEX` SQL statement that prepares, chunks, embeds, stores, and indexes your input data. You then perform vector search alongside keyword search using the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL query API.

Here, you can see how to use hybrid search in an HR Recruitment scenario, where you want to hire employees with specific technical skills (keyword search for "C, Python, and Database") but who also display certain personality and leadership traits (semantic search for "prioritize teamwork and leadership experience"). You can also see how to perform different kinds of pure and non-pure hybrid search queries in multiple search modes.

1. Connect to Oracle AI Database as a local user.

a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (`docuser`) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Create a local directory on your server (`DEMO_DIR`) to store your input data and model files. Grant necessary privileges:

```
create or replace directory DEMO_DIR as '/my_local_dir/';


grant read, write on directory DEMO_DIR to docuser;

commit;
```

d. Connect as the local user (`docuser`):

```
CONN docuser/password
```

2. Load an ONNX format embedding model into Oracle AI Database by calling the `load_onnx_model` procedure.

This embedding model is internally used to generate vector embeddings from your input data.

```
EXECUTE dbms_vector.drop_onnx_model(model_name => 'doc_model', force =>
true);


EXECUTE dbms_vector.load_onnx_model(
  'DEMO_DIR', 'my_embedding_model.onnx', 'doc_model',
   JSON('{"function" : "embedding", "embeddingOutput" : "embedding" ,
"input": {"input": ["DATA"]}}'));
```

In this example, the procedure loads an ONNX model file, named `my_embedding_model.onnx` from the `DEMO_DIR` directory, into the database as `doc_model`. You must replace `my_embedding_model.onnx` with an ONNX export of your embedding model and `doc_model` with the name under which the imported model is stored in the database.

> **Note**
>
> If you do not have an embedding model in ONNX format, then perform the steps listed in ONNX Pipeline Models: Text Embedding.

3. Create a relational table (for example, `doc_tab`) and insert all your documents in a textual column (for example, `text`).

```
DROP TABLE doc_tab purge;


CREATE TABLE doc_tab (id number, text varchar2(500));

insert into doc_tab values(1, 'Candidate-1: C Master. Optimizes low-level
system (i.e. Database) performance with C. Strong leadership skills in
guiding teams to deliver complex projects.');
insert into doc_tab values(2, 'Candidate-2: Full-Stack Developer. Skilled
in Database, C, HTML, JavaScript, and Python with experience in building
responsive web applications. Thrives in collaborative team environments.');
insert into doc_tab values(3, 'Candidate-3: DevOps Engineer. Manages CI/CD
pipelines (Jenkins, Gitlab) with expertise in cloud infrastructure (OCI,
AWS, GCP). Proven track record of streamlining deployments and ensuring
high availability.');
insert into doc_tab values(4, 'Candidate-4: Database Administrator (DBA).
Maintains and secures enterprise database (Oracle, MySql, SQL Server).
Passionate about data integrity and optimization. Strong mentor for junior
DBA(s).');
insert into doc_tab values(5, 'Candidate-5: C, Java, Python, and Database
(DBA) Guru. Develops scalable applications. Strong leadership, teamwork
and collaborative.');

commit;
```

4. Create a hybrid vector index (`my_hybrid_idx`) on the `text` column of the `doc_tab` table.

```
CREATE HYBRID VECTOR INDEX my_hybrid_idx on
  doc_tab(text)
  parameters('model doc_model');
```

   Here, `model` specifies the embedding model in ONNX format that you have imported into the database for generating embeddings. This is the only required indexing parameter to create a hybrid vector index.

   An index named `my_hybrid_idx` is created on the `text` column of the `doc_tab` table. The default vector index type is set to Inverted File Flat (IVF).

5. Query the hybrid vector index.

   In the next steps, we will explore all possible ways in which you can query the hybrid vector index using multiple search modes described in Understand Hybrid Search. You can then compare the results to examine the differences in scores and ranking for a comprehensive understanding.

   a. **Simple default query**:

      By default, hybrid search offers a simplified query with predefined fields. The minimum input parameters required are `hybrid_index_name` and `search_text`.

      The same search text (`C, Python, Database`) is used to query the vectorized chunk index and the text document index. This search text is transformed into a `CONTAINS` query for keyword search, and is vectorized for a `VECTOR_DISTANCE` query for semantic search.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
          '{
            "hybrid_index_name" : "my_hybrid_idx",
            "search_text"       : "C, Python, Database"
           }'
        )
  ) pretty)
from dual;
```

      This query returns the top 3 rows, ordered by score relevance. The highest final score (`69.54`) of Candidate-2 indicates the best match. All the return attributes are shown by default in this query.

```
[
  {
    "rowid"        : "AAAR9jAABAAAQeaAAB",
    "score"        : 69.54,
    "vector_score" : 69.69,
    "text_score"   : 68,
    "vector_rank"  : 1,
    "text_rank"    : 1,
    "chunk_text"   : "Candidate-2: Full-Stack Developer. Skilled in
Database, C, HTM
L, JavaScript, and Python with experience in building responsive web
application
```

```
s. Thrives in collaborative team environments.",
    "chunk_id"     : "1"
  },
  {
    "rowid"        : "AAAR9jAABAAAQeaAAA",
    "score"        : 62.77,
    "vector_score" : 65.55,
    "text_score"   : 35,
    "vector_rank"  : 4,
    "text_rank"    : 2,
    "chunk_text"   : "Candidate-1: C Master. Optimizes low-level system
(i.e. Da
tabase) performance with C. Strong leadership skills in guiding teams
to deliver
 complex projects.",
    "chunk_id"     : "1"
  },
  {
    "rowid"        : "AAAR9jAABAAAQeaAAD",
    "score" : 62.15,
    "vector_score" : 68.17,
    "text_score"   : 2,
    "vector_rank"  : 2,
    "text_rank"    : 3,
    "chunk_text"   : "Candidate-4: Database Administrator (DBA).
Maintains and
secures enterprise database (Oracle, MySql, SQL Server). Passionate
about data i
ntegrity and optimization. Strong mentor for junior DBA(s).",
    "chunk_id"     : "1"
  }
]
```

b. **Pure semantic in document mode (text as query input)**:

This is a pure vector-based similarity query to fetch document-level search results. Here, the `search_text` string is vectorized into a query vector for a `VECTOR_DISTANCE` semantic query.

A vector search operates at the chunk level, where the query first extracts the top candidates of vectorized chunks, aggregates them by document using the `aggregator` (`MAX`) function, produces a combined score (according to the `aggregator`), and finally returns the top n documents. To know more about how this works, see Pure Semantic in Document Mode.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
            {
             "search_text"   : "prioritize teamwork and leadership
experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX"
            },
         "return":
```

```
            {
             "values"        : [ "rowid", "score" ],
             "topN": 3
            }
      }'
    )
  ) pretty)
from dual;
```

The top 3 documents are returned with the corresponding ROWIDs of the doc_tab table rows as well as their vector scores.

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAA",
    "score" : 61
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 56.64
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 55.75
  }
]
```

c. **Pure semantic in document mode (embedding as query input)**:

As compared to the previous pure semantic query in document mode (where you used the search_text parameter to specify the query text for vector search), here you use the search_vector parameter to directly pass a vector embedding (query vector) as input for your vector search.

In the same manner, this query retrieves the top vectorized chunk results, aggregates them by document using the aggregator (MAX) function, and finally returns the top n documents.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
           "search_vector"   : vector_serialize(
                                          vector_embedding(doc_model
                                                      using
                                                      "C, Python,
Database"
                                                      as data)
                                              RETURNING CLOB),
           "search_mode"     : "DOCUMENT",
           "aggregator"      : "MAX"
          },
         "return":
          {
```

```
            "values"     : [ "rowid", "score" ],
            "topN": 3
           }
       }'
     )
   ) pretty)
from dual;
```

The top 3 documents are returned with the corresponding `ROWID`s of the `doc_tab` table rows as well as their vector scores.

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 69.69
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 68.17
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAC",
    "score" : 67.48
  }
]
```

d. **Pure semantic in chunk mode**:

This is a pure vector-based similarity query to retrieve chunk results. Note that this is the Oracle AI Vector Search semantic search equivalent. Here, vector search retrieves the best vectorized chunks from the same or different set of documents and then internally computes their vector scores. The top n chunks with the highest vector scores are returned. To know more about how this works, see Pure Semantic in Chunk Mode.

As compared to the previous pure semantic search in document mode, aggregation of chunk results is not performed in chunk mode, so the `aggregator` function is not needed.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "prioritize teamwork and leadership
experience",
             "search_mode"   : "CHUNK"
          },
         "return":
          {
             "values"        : [ "rowid", "score", "chunk_text",
"chunk_id" ],
             "topN"          : 3
          }
      }'
```

```
    )
  ) pretty)
from dual;
```

The top 3 chunks with the highest scores are returned along with chunk IDs and chunk texts. You can also see the corresponding `ROWID`s of the `doc_tab` table rows. The highest score (`61`) of Candidate-1 with chunk ID 1 indicates the top match:


```
[
  {
    "rowid"      : "AAAR9jAABAAAQeaAAA",
    "score"      : 61,
    "chunk_text" : "Candidate-1: C Master. Optimizes low-level system
(i.e. Da
tabase) performance with C. Strong leadership skills in guiding teams
to deliver
 complex projects.",
    "chunk_id"   : "1"
  },
  {
    "rowid"      : "AAAR9jAABAAAQeaAAD",
    "score"      : 56.64,
    "chunk_text" : "Candidate-4: Database Administrator (DBA).
Maintains and
secures enterprise database (Oracle, MySql, SQL Server). Passionate
about data i
ntegrity and optimization. Strong mentor for junior DBA(s).",
    "chunk_id"   : "1"
  },
  {
    "rowid"      : "AAAR9jAABAAAQeaAAB",
    "score"      : 55.75,
    "chunk_text" : "Candidate-2: Full-Stack Developer. Skilled in
Database, C, HTM
L, JavaScript, and Python with experience in building responsive web
application
s. Thrives in collaborative team environments.",
    "chunk_id"   : "1"
  }
]
```


e. **Pure keyword in document mode**:

This is a pure text-based keyword query to fetch document-level results. Note that this is equivalent to the traditional `CONTAINS` query using Oracle Text. Here, full-text search retrieves the best documents and then internally computes their keyword scores. The top n documents with the highest keyword scores are returned. To know more about how this works, see Pure Keyword in Document Mode.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
           "contains"        : "C, Python, Database"
```

```
          },
         "return":
          {
           "values"          : [ "rowid", "score" ],
           "topN": 3
          }
      }'
    )
  ) pretty)
from dual;
```

The top 3 documents with the highest scores are returned, where you can see the corresponding ROWIDs of the doc_tab table as well as their keyword scores:

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 68
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAA",
    "score" : 35
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 2
  }
]
```

f. **Keyword and semantic in document mode (common query string)**:

This is a hybrid query that conducts both keyword search and vector search on the data, and then combines the keyword scores and semantic scores to fetch document-level results.

Here, the same text string (C, Python, Database) is used for both:

- a keyword query on the document text index by converting the SEARCH_TEXT string into a CONTAINS ACCUM operator syntax
- and a semantic query on the vectorized chunk index by vectorizing the SEARCH_TEXT string for a VECTOR_DISTANCE query

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_text"       : "C, Python, Database",
         "search_fusion"     : "INTERSECT",
         "search_scorer"     : "rsf",
         "vector":
            {
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX"
            },
         "return":
            {
```

```
             "values"        : [ "rowid", "score" ],
             "topN"          : 3
            }
      }'
    )
  ) pretty)
from dual;
```

For vector search, this query searches the query vector (created from the `SEARCH_TEXT` string) against the vector index. A vector search operates at the chunk level, where the query first extracts the top candidates of vectorized chunks, aggregates them by document using the `aggregator` (`MAX`) function, produces a combined vector score (according to the `aggregator`), and finally returns the top n documents.

For scoring, first both search results are added using a `UNION ALL` operation. Then the results are fused using the `SEARCH_FUSION` operator, `INTERSECT` that combines all distinct rows selected by both the searches. The final scoring is computed using the defined `SEARCH_SCORER` algorithm, RSF. Finally, the defined `topN` doc IDs are returned at most.

To know more about how the scores are calculated, see Keyword and Semantic in Document Mode.

The top 3 documents with the highest scores are returned, where you can see a list of `ROWID`s from your base table (`doc_tab`) corresponding to the list of best files identified.

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 69.54
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAA",
    "score" : 62.77
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 62.15
  }
]
```

g. **Keyword and semantic in document mode (separate query strings)**:

As compared to the previous keyword and semantic search in document mode (where you used the same `search_text` string for both text search and vector search), here you specify two separate search texts for both the search types.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_fusion"     : "INTERSECT",
         "search_scorer"     : "rsf",
         "vector":
          {
             "search_text"   : "prioritize teamwork and leadership
experience",
```

```
                "search_mode"   : "DOCUMENT",
                "aggregator"    : "MAX"
             },
            "text":
             {
                "contains"      : "C, Python, Database"
             },
            "return":
             {
                "values"        : [ "rowid", "score" ],
                "topN"          : 3
             }
        }'
      )
    ) pretty)
from dual;
```

In the same manner, the keyword search retrieves a list of doc IDs satisfying your `CONTAINS` text query string (`C, Python, Database`). The vector search retrieves a list of doc IDs satisfying your `SEARCH_TEXT` similarity query string (`prioritize teamwork and leadership experience`).

For scoring, first both search results are added using a `UNION ALL` operation. Then the results are fused using the `SEARCH_FUSION` operator, `INTERSECT`. The final scoring is computed using the defined `SEARCH_SCORER` algorithm, RSF. Finally, the defined `topN` doc IDs are returned at most.

To know more about how the scores are calculated, see [Keyword and Semantic in Document Mode](#).

The top 3 documents with the highest scores are returned, where you can see a list of `ROWID`s from your base table (`doc_tab`) corresponding to the list of best files identified.

```
[
  {
    "rowid" : "AAAR9jAABAAAQeaAAA",
    "score" : 58.64
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAB",
    "score" : 56.86
  },
  {
    "rowid" : "AAAR9jAABAAAQeaAAD",
    "score" : 51.67
  }
]
```

h. **Keyword and semantic in chunk mode**:

This is a hybrid query that conducts both keyword search and vector search on the data, and then combines the keyword scores and semantic scores to fetch chunk-level results.

```
select json_Serialize(
  dbms_hybrid_vector.search(
    json(
```

```
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "prioritize teamwork and leadership
experience",
             "search_mode"   : "CHUNK"
          },
         "text":
          {
             "contains"      : "C, Python, Database"
          },
         "return":
          {
             "values"        : [ "rowid", "score", "chunk_text",
"chunk_id" ],
             "topN": 3
          }
      }'
    )
  ) pretty)
from dual;
```

The keyword search retrieves a list of doc IDs satisfying your `CONTAINS` text query string (`C, Python, Database`).

The vector search performs a similarity query with the query vector (created from the `SEARCH_TEXT` string: `prioritize teamwork and leadership experience`) against the vector index of all the chunks of all your documents. It retrieves a list of chunk IDs and associated doc IDs satisfying your `SEARCH_TEXT` similarity query string.

Aggregation of chunk results is not performed in chunk mode, so the `AGGREGATOR` function is not applicable.

For scoring, first the text document results are added into the chunk results using the `RIGHT OUTER JOIN` operation on doc IDs. Any text document results that are not in the vector candidate chunks are not returned. Then the results are fused using the `SEARCH_FUSION` operator, `INTERSECT` that combines all distinct rows selected by both the searches. The final scoring is computed using the defined `SEARCH_SCORER` algorithm, RSF. Finally, the defined `topN` chunk IDs are returned at most.

To know how the scores are calculated, see Keyword and Semantic in Chunk Mode.

A list of top 3 chunk IDs are returned from the files stored in your base table (`doc_tab`), along with chunk texts and the corresponding `ROWIDs`. The highest final score (`58.64`) of Candidate-1 with chunk ID 1 indicates the top match:

```
[
  {
    "rowid"      : "AAAR9jAABAAAQeaAAA",
    "score"      : 58.64,
    "chunk_text" : "Candidate-1: C Master. Optimizes low-level system
(i.e. Da
tabase) performance with C. Strong leadership skills in guiding teams
to deliver
 complex projects.",
    "chunk_id"   : "1"
  },
  {
```

```
    "rowid"      : "AAAR9jAABAAAQeaAAB",
    "score"      : 56.86,
    "chunk_text" : "Candidate-2: Full-Stack Developer. Skilled in
Database, C, HTM
L, JavaScript, and Python with experience in building responsive web
application
s. Thrives in collaborative team environments.",
    "chunk_id"   : "1"
  },
  {
    "rowid"      : "AAAR9jAABAAAQeaAAD",
    "score"      : 51.67,
    "chunk_text" : "Candidate-4: Database Administrator (DBA).
Maintains and
secures enterprise database (Oracle, MySql, SQL Server). Passionate
about data i
ntegrity and optimization. Strong mentor for junior DBA(s).",
    "chunk_id"   : "1"
  }
]
```

Related Topics

- Manage Hybrid Vector Indexes
  Learn how to manage a hybrid vector index, which is a single index for searching by *similarity* and *keywords*, to enhance the accuracy of your search results.
- SEARCH
  Use the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL function to run textual queries, vector similarity queries, or hybrid queries against hybrid vector indexes.

9

Work with LLM-Powered APIs and Retrieval Augmented Generation

You can use Vector Utility PL/SQL APIs for prompting Large Language Models (LLMs) with textual prompts and images, using LLM-powered interfaces. You can also communicate with LLMs through the implementation of Retrieval Augmented Generation (RAG), which helps to generate more accurate and informative responses.

- Use LLM-Powered APIs to Generate Summary and Text
  Run these end-to-end examples to see how you can summarize or describe textual inputs and images.
- Use Retrieval Augmented Generation to Complement LLMs
  RAG lets you mitigate the inaccuracies and hallucinations faced when using LLMs. Oracle AI Vector Search enables RAG within Oracle AI Database using the `DBMS_VECTOR_CHAIN` PL/SQL package or through popular frameworks (such as LangChain).
- Supported Third-Party Provider Operations and Endpoints
  Review a list of third-party REST providers and REST endpoints that are supported for various vector generation, summarization, text generation, and reranking operations.

Use LLM-Powered APIs to Generate Summary and Text

Run these end-to-end examples to see how you can summarize or describe textual inputs and images.

- Generate Summary
  In these examples, you can see how to summarize a given textual extract.
- Generate Text Response
  In these examples, you can see how to generate a textual answer, description, or summary based on the specified task in a given prompt.
- Describe Image Content
  In these examples, you can see how to generate a textual analysis or description of the contents of a given image.

Generate Summary

In these examples, you can see how to summarize a given textual extract.

A summary is a short and concise extract with key features of a document that best represents what the document is about as a whole. A summary can be free-form paragraphs or bullet points based on the format that you specify.

> **Note**
>
> You can also generate a summary using Oracle AI Database as the service provider. In this case, Oracle Text is internally used to generate a summary. However, that is outside the scope of these examples. See UTL_TO_SUMMARY.

- Generate Summary Using Public REST Providers
  Perform a text-to-summary transformation, using publicly hosted third-party text summarization models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI.
- Generate Summary Using the Local REST Provider Ollama
  Perform a text-to-summary transformation by accessing open LLMs, using the local host REST endpoint provider Ollama.

Generate Summary Using Public REST Providers

Perform a text-to-summary transformation, using publicly hosted third-party text summarization models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI.

Here, you call the chainable utility function `UTL_TO_SUMMARY`.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To summarize a textual extract, using an external LLM:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Set proxy if one exists.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to docuser for allowing connection to the host, using the DBMS_NETWORK_ACL_ADMIN procedure.

   This example uses * to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Set up your credentials for the REST provider that you want to access and then call UTL_TO_SUMMARY.

   - **Using Generative AI**:

> **Note**
>
> Currently, UTL_TO_SUMMARY does not work for Generative AI because the model and summary endpoint supported for Generative AI have been retired. It will be available in a subsequent release.

a. Run DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL to create and store an OCI credential (OCI_CRED).

   Generative AI requires the following authentication parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
```

```
"fingerprint"     : "<fingerprint>"
}
```

You will later refer to this credential name when declaring JSON parameters for the `UTL_TO_SUMMARY` call.

> **Note**
>
> The generated private key may appear as:
>
> ```
> -----BEGIN RSA PRIVATE KEY-----
> <private key string>
> -----END RSA PRIVATE KEY-----
> ```
>
> You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

```
exec dbms_vector_chain.drop_credential('OCI_CRED');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('user_ocid','<user ocid>');
  jo.put('tenancy_ocid','<tenancy ocid>');
  jo.put('compartment_ocid','<compartment ocid>');
  jo.put('private_key','<private key>');
  jo.put('fingerprint','<fingerprint>');
  dbms_vector_chain.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

Replace all the authentication parameter values.

For example:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();

jo.put('user_ocid','ocid1.user.oc1..aabbalbbaa1112233aabbaabb1111222
aa1111bb');

jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaaaalbbbb1112233aaaabbaa1
111222aaa111a');

jo.put('compartment_ocid','ocid1.compartment.oc1..ababalabab1112233a
bababab1111222aba11ab');
```

```
  jo.put('private_key','AAAaaaBBB11112222333...AAA111AAABBB222aaa1a/
+');

jo.put('fingerprint','01:1a:a1:aa:12:a1:12:1a:ab:12:01:ab:a1:12:ab:1
a');
  dbms_vector_chain.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

b. Run `DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY`:

Here, the cohere.command-r-16k model is used. You can replace the `model` value, as required.

> **Note**
>
> For a list of all REST endpoint URLs and models that are supported to use with Generative AI, see Supported Third-Party Provider Operations and Endpoints.

```
-- select example

var params clob;
exec :params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k"
}';

select dbms_vector_chain.utl_to_summary(
  'A transaction is a logical, atomic unit of work that contains
one or more SQL
    statements.
    An RDBMS must be able to group SQL statements so that they are
either all
    committed, which means they are applied to the database, or all
rolled back, which
    means they are undone.
    An illustration of the need for transactions is a funds
transfer from a savings account to
    a checking account. The transfer consists of the following
separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed
or fail as a unit. For
    example, if a hardware failure prevents a statement in the
transaction from executing,
```

```
    then the other statements must be rolled back.
    Transactions set Oracle AI Database apart from a file system.
If you
    perform an atomic operation that updates several files, and if
the system fails halfway
    through, then the files will not be consistent. In contrast, a
transaction moves an
    Oracle AI Database from one consistent state to another. The
basic principle of a
    transaction is all or nothing: an atomic operation succeeds or
fails as a whole.',
  json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'A transaction is a logical, atomic unit of work that
contains one or more SQL
    statements.
    An RDBMS must be able to group SQL statements so that they are
either all
    committed, which means they are applied to the database, or all
rolled back, which
    means they are undone.
    An illustration of the need for transactions is a funds
transfer from a savings account to
    a checking account. The transfer consists of the following
separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed
or fail as a unit. For
    example, if a hardware failure prevents a statement in the
transaction from executing,
    then the other statements must be rolled back.
    Transactions set Oracle AI Database apart from a file system.
If you
    perform an atomic operation that updates several files, and if
the system fails halfway
    through, then the files will not be consistent. In contrast, a
transaction moves an
    Oracle AI Database from one consistent state to another. The
basic principle of a
    transaction is all or nothing: an atomic operation succeeds or
fails as a whole.';

  params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
```

```
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k"
}';

  output := dbms_vector_chain.utl_to_summary(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

Optionally, you can specify additional REST provider-specific parameters. For example:

```
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k",
  "length"         : "MEDIUM",
  "format"         : "PARAGRAPH",
  "temperature"    : 1.0
}
```

- **Using Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI**:
  a. Run `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` to create and store a credential.

     Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

     `{ "access_token": "<access token>" }`

     You will later refer to this credential name when declaring JSON parameters for the `UTL_TO_SUMMARY` call.

```
exec dbms_vector_chain.drop_credential('<credential name>');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', '<access token>');
  dbms_vector_chain.create_credential(
    credential_name   => '<credential name>',
    params            => json(jo.to_string));
end;
/
```

Replace the `access_token` and `credential_name` values. For example:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', 'AbabA1B123aBc123AbabAb123a1a2ab');
  dbms_vector_chain.create_credential(
    credential_name   => 'HF_CRED',
    params            => json(jo.to_string));
end;
/
```

b. Run `DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY`:

```
-- select example

var params clob;
exec :params := '
{
  "provider": "<REST provider>",
  "credential_name": "<credential name>",
  "url": "<REST endpoint URL for text summarization service>",
  "model": "<REST provider text summarization model name>"
}';

select dbms_vector_chain.utl_to_summary(
  'A transaction is a logical, atomic unit of work that contains 
one or more SQL
    statements.
    An RDBMS must be able to group SQL statements so that they are 
either all
    committed, which means they are applied to the database, or all 
rolled back, which
    means they are undone.
    An illustration of the need for transactions is a funds 
transfer from a savings account to
    a checking account. The transfer consists of the following 
separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed 
or fail as a unit. For
    example, if a hardware failure prevents a statement in the 
transaction from executing,
    then the other statements must be rolled back.
    Transactions set Oracle AI Database apart from a file system. 
If you
    perform an atomic operation that updates several files, and if 
the system fails halfway
    through, then the files will not be consistent. In contrast, a 
transaction moves an
    Oracle AI Database from one consistent state to another. The 
basic principle of a
    transaction is "all or nothing": an atomic operation succeeds
```

```
or fails as a whole.',
  json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'A transaction is a logical, atomic unit of work that
contains one or more SQL
    statements.
    An RDBMS must be able to group SQL statements so that they are
either all
    committed, which means they are applied to the database, or all
rolled back, which
    means they are undone.
    An illustration of the need for transactions is a funds
transfer from a savings account to
    a checking account. The transfer consists of the following
separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed
or fail as a unit. For
    example, if a hardware failure prevents a statement in the
transaction from executing,
    then the other statements must be rolled back.
    Transactions set Oracle AI Database apart from a file system.
If you
    perform an atomic operation that updates several files, and if
the system fails halfway
    through, then the files will not be consistent. In contrast, a
transaction moves an
    Oracle AI Database from one consistent state to another. The
basic principle of a
    transaction is "all or nothing": an atomic operation succeeds
or fails as a whole.';

  params := '
{
  "provider": "<REST provider>",
  "credential_name": "<credential name>",
  "url": "<REST endpoint URL for text summarization service>",
  "model": "<REST provider text summarization model name>"
}';

  output := dbms_vector_chain.utl_to_summary(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
```

```
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

Note

For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Replace `provider`, `credential_name`, `url`, and `model` with your own values. Optionally, you can specify additional REST provider parameters. This is shown in the following examples:

Cohere example:

```
{
  "provider"        : "cohere",
  "credential_name" : "COHERE_CRED",
  "url"             : "https://api.cohere.ai/v1/chat",
  "model"           : "command",
  "length"          : "medium",
  "format"          : "paragraph",
  "temperature"     : 1.0
}
```

Google AI example:

```
{
  "provider"         : "googleai",
  "credential_name"  : "GOOGLEAI_CRED",
  "url"              : "https://generativelanguage.googleapis.com/
v1beta/models/",
  "model"            : "gemini-pro:generateContent",
  "generation_config": {
                        "temperature"    : 0.9,
                        "topP"           : 1,
                        "candidateCount" : 1,
                        "maxOutputTokens": 256
                       }
}
```

Hugging Face example:

```
{
  "provider"         : "huggingface",
  "credential_name"  : "HF_CRED",
  "url"              : "https://api-inference.huggingface.co/
models/",
  "model"            : "facebook/bart-large-cnn"
}
```

OpenAI example:

```
{
  "provider"        : "openai",
  "credential_name" : "OPENAI_CRED",
  "url"             : "https://api.openai.com/v1/chat/completions",
  "model"           : "gpt-4o-mini",
  "max_tokens"      : 256,
  "temperature"     : 1.0
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://LOCATION-
aiplatform.googleapis.com/v1/projects/PROJECT/locations/LOCATION/
publishers/google/models/",
  "model"            : "gemini-1.0-pro:generateContent",
  "generation_config": {
                        "temperature"    : 0.9,
                        "topP"           : 1,
                        "candidateCount" : 1,
                        "maxOutputTokens": 256
                       }
}
```

A generated summary may appear as:

```
A transaction is a logical unit of work that groups one or more SQL
statements
that must be executed as a unit, with all statements succeeding, or all
statements being rolled back. Transactions are a fundamental concept in
relational database management systems (RDBMS), and Oracle AI Database is
specifically designed to manage transactions, ensuring database
consistency and
integrity. Transactions differ from file systems in that they maintain
atomicity, ensuring that all related operations succeed or fail as a whole,
maintaining database consistency regardless of intermittent failures.
Transactions move a database from one consistent state to another, and the
fundamental principle is that a transaction is committed or rolled back as
a
whole, upholding the "all or nothing" principle.

PL/SQL procedure successfully completed.
```

This example uses the default settings for each provider. For detailed information on additional parameters, refer to your third-party provider's documentation.

Related Topics

- About Chainable Utility Functions and Common Use Cases
  These are intended to be a set of chainable and flexible "stages" through which you pass your input data to transform into a different representation, including vectors.

- UTL_TO_SUMMARY
  Use the DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY chainable utility function to generate a summary for textual documents.

Generate Summary Using the Local REST Provider Ollama

Perform a text-to-summary transformation by accessing open LLMs, using the local host REST endpoint provider Ollama.

Ollama is a free and open-source command-line interface tool that allows you to run open LLMs (such as Llama 3, Phi 3, Mistral, or Gemma 2) locally and privately on your Linux, Windows, or macOS systems. You can access Ollama as a service using SQL and PL/SQL commands.

Here, you call the chainable utility function UTL_TO_SUMMARY.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To generate a concise and informative summary of a textual extract, by calling a local LLM using Ollama:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the SYS user, connecting as SYSDBA:

```
conn sys/password as sysdba

CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;

SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;

CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;

GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Install Ollama and run a model locally.

a. Download and run the Ollama application from https://ollama.com/download.

You can either install Ollama as a service that runs in the background or as a standalone binary with a manual install. For detailed installation-specific steps, see Quick Start in the Ollama Documentation.

Note the following:

- The Ollama server needs to be able to connect to the internet so that it can download the models. If you require a proxy server to access the internet, remember to set the appropriate environment variables before running the Ollama server. For example, to set for Linux:

```
-- set a proxy if you require one

export https_proxy=<proxy-hostname>:<proxy-port>
export http_proxy=<proxy-hostname>:<proxy-port>
export no_proxy=localhost,127.0.0.1,.example.com
export ftp_proxy=<proxy-hostname>:<proxy-port>
```

- If you are running Ollama and the database on different machines, then on the database machine, you must change the URL to refer to the host name or IP address that is running Ollama instead of the local host.
- You may need to change your firewall settings on the machine that is running Ollama to allow the port through.

b. If running Ollama as a standalone binary from a manual install, then start the server:

```
ollama serve
```

c. Run a model using the `ollama run <model_name>` command.

For example, to call the Llama 3 model:

```
ollama run llama3
```

For detailed information on this step, see Ollama Readme.

d. Verify that Ollama is running locally by using a cURL command.

For example:

```
-- get summary

curl http://localhost:11434/api/generate -d '{
  "model" : "llama3",
  "prompt": "What is Oracle AI Vector Search?"
  "stream": false}'
```

3. Set proxy if one exists.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

4. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

   This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

5. Call `UTL_TO_SUMMARY`.

   The Ollama service has a REST API endpoint for summarizing text. Specify the URL and other configuration parameters in a JSON object.

```
var gent_ollama_params clob;
exec :gent_ollama_params := '{
  "provider": "ollama",
  "host"    : "local",
  "url"     : "http://localhost:11434/api/generate",
  "model"   : "llama3"
}';

select dbms_vector_chain.utl_to_summary(
  'A transaction is a logical, atomic unit of work that contains one or
more SQL
    statements.
    An RDBMS must be able to group SQL statements so that they are either
all
    committed, which means they are applied to the database, or all rolled
back, which
    means they are undone.
    An illustration of the need for transactions is a funds transfer from
a savings account to
    a checking account. The transfer consists of the following separate
operations:
    1. Decrease the savings account.
    2. Increase the checking account.
```

```
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed or
fail as a unit. For
    example, if a hardware failure prevents a statement in the transaction
from executing,
    then the other statements must be rolled back.
    Transactions set Oracle AI Database apart from a file system. If you
    perform an atomic operation that updates several files, and if the
system fails halfway
    through, then the files will not be consistent. In contrast, a
transaction moves an
    Oracle AI Database from one consistent state to another. The basic
principle of a
    transaction is "all or nothing": an atomic operation succeeds or fails
as a whole.',
  json(:gent_ollama_params)) from dual;
```

You can replace the `url` and `model` with your own values, as required.

> **Note**
>
> For a complete list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

A generated summary may appear as:

```
A transaction in an RDBMS (Relational Database Management System) is a
self-contained unit of work that
consists of one or more SQL statements. This ensures that all changes made
by the transaction are either
committed (applied to the database) or rolled back (undone). Oracle AI
Database is specifically designed to
manage transactions, ensuring database consistency and integrity.
Transactions differ from file systems in that they maintain atomicity,
ensuring that all related operations
succeed or fail as a whole, maintaining database consistency regardless of
intermittent failures.
Transactions move a database from one consistent state to another, and the
fundamental principle is that a
transaction is committed or rolled back as a whole, upholding the "all or
nothing" principle.
```

Related Topics

- About Chainable Utility Functions and Common Use Cases
  These are intended to be a set of chainable and flexible "stages" through which you pass your input data to transform into a different representation, including vectors.
- UTL_TO_SUMMARY
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY` chainable utility function to generate a summary for textual documents.

Generate Text Response

In these examples, you can see how to generate a textual answer, description, or summary based on the specified task in a given prompt.

A prompt can be an input text string, such as a question that you ask an LLM. For example, "`What is Oracle Text?`". A prompt can also be a set of instructions or a command, such as "`Summarize the following ...`", "`Draft an email asking for ...`", or "`Rewrite the following ...`", and can include results from a search.

- Generate Text Using Public REST Providers
  Perform a text-to-text transformation, using publicly hosted third-party text generation models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI. The input is a textual prompt, and the generated output is a textual answer or description based on the specified task in that prompt.
- Generate Text Using the Local REST Provider Ollama
  Perform a text-to-text transformation by accessing open LLMs, using the local host REST endpoint provider Ollama. The input is a textual prompt, and the generated output is a textual answer or description based on the specified task in that prompt.

Generate Text Using Public REST Providers

Perform a text-to-text transformation, using publicly hosted third-party text generation models by Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI. The input is a textual prompt, and the generated output is a textual answer or description based on the specified task in that prompt.

A prompt can be a text string (such as a question that you ask an LLM or a command), and can include results from a search.

Here, you can use the `UTL_TO_GENERATE_TEXT` function from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To generate a text response for the prompt "`What is Oracle Text?`", using an external LLM:

1. Connect to Oracle AI Database as a local user.

a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (`docuser`) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (`docuser`):

```
CONN docuser/password
```

2. Set proxy if one exists.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

   This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Set up your credentials for the REST provider that you want to access and then call `UTL_TO_GENERATE_TEXT`:

   - **Using Generative AI**:

     a. Call `CREATE_CREDENTIAL` to create and store an OCI credential (`OCI_CRED`).

        Generative AI requires the following authentication parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
"fingerprint"     : "<fingerprint>"
}
```

        You will later refer to this credential name when declaring JSON parameters for the `UTL_TO_GENERATE_TEXT` call.

> **Note**
>
> The generated private key may appear as:
>
> ```
> -----BEGIN RSA PRIVATE KEY-----
> <private key string>
> -----END RSA PRIVATE KEY-----
> ```
>
> You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

```
exec dbms_vector_chain.drop_credential('OCI_CRED');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('user_ocid','<user ocid>');
  jo.put('tenancy_ocid','<tenancy ocid>');
  jo.put('compartment_ocid','<compartment ocid>');
  jo.put('private_key','<private key>');
  jo.put('fingerprint','<fingerprint>');
  dbms_vector_chain.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

        Replace all the authentication parameter values. For example:

```
declare
  jo json_object_t;
```

```
begin

  -- create an OCI credential
  jo := json_object_t();

jo.put('user_ocid','ocid1.user.oc1..aabbalbbaa1112233aabbaabb1111222
aa1111bb');

jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaaaalbbbb1112233aaaabbaa1
111222aaa111a');

jo.put('compartment_ocid','ocid1.compartment.oc1..ababalabab1112233a
bababab1111222aba11ab');
  jo.put('private_key','AAAaaaBBB11112222333...AAA111AAABBB222aaa1a/
+');

jo.put('fingerprint','01:1a:a1:aa:12:a1:12:1a:ab:12:01:ab:a1:12:ab:1
a');
  dbms_vector_chain.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

b. Call `UTL_TO_GENERATE_TEXT`:

Here, the cohere.command-r-16k model is used. You can replace `model` with your own value, as required.

> **Note**
>
> For a list of all REST endpoint URLs and models that are supported to use with Generative AI, see Supported Third-Party Provider Operations and Endpoints.

```
-- select example

var params clob;
exec :params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

select dbms_vector_chain.utl_to_generate_text(
 'What is Oracle Text?',json(:params)) from dual;

-- PL/SQL example
```

```
declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';

  params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

  output := dbms_vector_chain.utl_to_generate_text(input,
json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

Optionally, you can specify additional REST provider-specific parameters.

> **Note**
>
> If you want to pass any additional REST provider-specific parameters, then you must enclose those in `chatRequest`.

- **Using Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI**:

  a. Call `CREATE_CREDENTIAL` to create and store a credential.

     Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

     `{ "access_token": "<access token>" }`

You will later refer to this credential name when declaring JSON parameters for the `UTL_TO_GENERATE_TEXT` call.

```
exec dbms_vector_chain.drop_credential('<credential name>');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', '<access token>');
  dbms_vector_chain.create_credential(
    credential_name   => '<credential name>',
    params            => json(jo.to_string));
end;
/
```

Replace the `access_token` and `credential_name` values. For example:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', 'AbabA1B123aBc123AbabAb123a1a2ab');
  dbms_vector_chain.create_credential(
    credential_name   => 'HF_CRED',
    params            => json(jo.to_string));
end;
/
```

b. Call `UTL_TO_GENERATE_TEXT`:

```
-- select example

var params clob;
exec :params := '
{
  "provider"       : "<REST provider>",
  "credential_name": "<credential name>",
  "url"            : "<REST endpoint URL for text generation
service>",
  "model"          : "<REST provider text generation model name>"
}';

select dbms_vector_chain.utl_to_generate_text(
 'What is Oracle Text?',json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';
```

```
  params := '
{
  "provider"       : "<REST provider>",
  "credential_name": "<credential name>",
  "url"            : "<REST endpoint URL for text generation
service>",
  "model"          : "<REST provider text generation model name>"
}';

  output := dbms_vector_chain.utl_to_generate_text(input,
json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

> **Note**
>
> For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Replace `provider`, `credential_name`, `url`, and `model` with your own values. Optionally, you can specify additional REST provider parameters. This is shown in the following examples:

Cohere example:

```
{
  "provider"       : "Cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.ai/v1/chat",
  "model"          : "command"
}
```

Google AI example:

```
{
  "provider"       : "googleai",
  "credential_name": "GOOGLEAI_CRED",
  "url"            : "https://generativelanguage.googleapis.com/
v1beta/models/",
  "model"          : "gemini-pro:generateContent"
}
```

Hugging Face example:

```
{
  "provider"       : "huggingface",
  "credential_name": "HF_CRED",
  "url"            : "https://api-inference.huggingface.co/models/",
  "model"          : "gpt2"
}
```

OpenAI example:

```
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60,
  "temperature"    : 1.0
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://LOCATION-
aiplatform.googleapis.com/v1/projects/PROJECT/locations/LOCATION/
publishers/google/models/",
  "model"            : "gemini-1.0-pro:generateContent",
  "generation_config": {
                        "temperature"    : 0.9,
                        "topP"           : 1,
                        "candidateCount" : 1,
                        "maxOutputTokens": 256
                       }
}
```

A response to your prompt may appear as:

```
BMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT(:INPUT,JSON(:PARAMS))
---------------------------------------------------------------------------
-----
Oracle Text is a powerful tool that enhances Oracle AI Database with
integrated
text mining and text analytics capabilities.

It enables users to extract valuable insights and make informed decisions
by
analyzing unstructured text data stored within the database.

Here are some enhanced capabilities offered by Oracle Text:

1. Full-Text Search: Enables powerful and rapid full-text searches across
large
```

```
collections of documents. This helps users find relevant information
quickly
and effectively, even within massive datasets.

2. Natural Language Processing: Implements advanced language processing
techn
iques to analyze text and extract meaningful information. This includes
capabilities
like tokenization, stemming, lemmatization, and part-of-speech tagging,
which collectively facilitate efficient text processing and understanding.

3. Sentiment Analysis: Provides a deeper understanding of sentiment
expressed
 in text. It enables businesses to automatically analyze customer
opinions, feed
back, and reviews, helping them gain valuable insights into customer
sentiment,
satisfaction levels, and potential trends.

4. Entity Recognition: Automatically identifies and categorizes entities
with
in text, such as names of people, organizations, locations, or any other
specific
terms of interest. This is useful in applications like customer
relationship
management, where linking relevant information to individuals or
organizations is
crucial.

5. Contextual Analysis: Delivers insights into the context and
relationships
between entities and concepts in textual data. It helps organizations
better und
erstand the broader implications and associations between entities,
facilitating
 a deeper understanding of their data.

These features collectively empower various applications, enhancing the
function
ality of the Oracle AI Database platform to allow businesses and
organizations to
derive maximum value from their unstructured text data.

Let me know if you'd like to dive deeper into any of these specific
capabilities
, or if there are other aspects of Oracle Text you'd like to explore
further.
```

This example uses the default settings for each provider. For detailed information on additional parameters, refer to your third-party provider's documentation.

Related Topics

- UTL_TO_GENERATE_TEXT
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.

- DBMS_VECTOR
  The DBMS_VECTOR package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The DBMS_VECTOR_CHAIN package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Generate Text Using the Local REST Provider Ollama

Perform a text-to-text transformation by accessing open LLMs, using the local host REST endpoint provider Ollama. The input is a textual prompt, and the generated output is a textual answer or description based on the specified task in that prompt.

A prompt can be a text string (such as a question that you ask an LLM or a command), and can include results from a search.

Ollama is a free and open-source command-line interface tool that allows you to run open LLMs (such as Llama 3, Phi 3, Mistral, or Gemma 2) locally and privately on your Linux, Windows, or macOS systems. You can access Ollama as a service using SQL and PL/SQL commands.

Here, you can use the UTL_TO_GENERATE_TEXT function from either the DBMS_VECTOR or the DBMS_VECTOR_CHAIN package, depending on your use case.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To generate a descriptive answer for the prompt "What is Oracle Text?", by calling a local LLM using Ollama:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the SYS user, connecting as SYSDBA:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
```

```
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Install Ollama and run a model locally.

   a. Download and run the Ollama application from https://ollama.com/download.

   You can either install Ollama as a service that runs in the background or as a standalone binary with a manual install. For detailed installation-specific steps, see Quick Start in the Ollama Documentation.

   Note the following:

   - The Ollama server needs to be able to connect to the internet so that it can download the models. If you require a proxy server to access the internet, remember to set the appropriate environment variables before running the Ollama server. For example, to set for Linux:

     ```
     -- set a proxy if you require one

     export https_proxy=<proxy-hostname>:<proxy-port>
     export http_proxy=<proxy-hostname>:<proxy-port>
     export no_proxy=localhost,127.0.0.1,.example.com
     export ftp_proxy=<proxy-hostname>:<proxy-port>
     ```

   - If you are running Ollama and the database on different machines, then on the database machine, you must change the URL to refer to the host name or IP address that is running Ollama instead of the local host.
   - You may need to change your firewall settings on the machine that is running Ollama to allow the port through.

b. If running Ollama as a standalone binary from a manual install, then start the server:

```
ollama serve
```

c. Run a model using the `ollama run <model_name>` command.

For example, to call the Llama 3 model:

```
ollama run llama3
```

For detailed information on this step, see Ollama Readme.

d. Verify that Ollama is running locally by using a cURL command.

For example:

```
-- generate text

curl -X POST http://localhost:11434/api/generate -d '{
  "model" : "llama3",
  "prompt": "Why is the sky blue?",
  "stream": false }'
```

3. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

4. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

5. Call `UTL_TO_GENERATE_TEXT`.

The Ollama service has a REST API endpoint for generating text. Specify the URL and other configuration parameters in a JSON object.

```
var gent_ollama_params clob;
exec :gent_ollama_params := '{
   "provider": "ollama",
   "host"    : "local",
   "url"     : "http://localhost:11434/api/generate",
   "model"   : "llama3"
}';
```

```
select dbms_vector.utl_to_generate_text('What is Oracle Text?',
json(:gent_ollama_params)) from dual;
```

You can replace the `url` and `model` with your own values, as required.

> **Note**
>
> For a complete list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

The response to your prompt may appear as:

```
Oracle Text (formerly known as Oracle InterMedia) is a suite of text
search and retrieval tools within Oracle AI Database. It allows you to
index and query
unstructured text data, such as documents, emails, and other text-based
content.

With Oracle Text, you can:

1. Index text: Create indexes on text columns or external files, making it
possible to efficiently search and retrieve relevant text data.
2. Query text: Use SQL syntax to query the indexed text data, allowing you
to find specific words, phrases, or patterns within large volumes of
text.
3. Full-text search: Perform full-text searches on unstructured text data,
returning relevant results based on keyword matches, proximity, and
relevance.

Oracle Text supports various indexing schemes, including:

1. Basic Indexing: A simple, fast index for searching exact keywords.
2. Phrase Indexing: An index that allows you to search for phrases (e.g.,
"John Smith").
3. Thesaurus Indexing: An index that enables searches based on synonyms
and related words.

Oracle Text also includes various text analysis and processing features,
such as:

1. Tokenization: Breaking down text into individual words or tokens.
2. Stemming: Reducing words to their base form (e.g., "running" becomes
"run").
3. Stopword removal: Eliminating common words like "the," "and," and "a"
that don't add much value to the search.

Oracle Text is particularly useful in scenarios where you need to search,
analyze, or retrieve unstructured text data, such as:

1. Content management: Searching and retrieving documents, articles, or
other content.
2. Email archiving: Indexing and searching email messages.
3. Search engines: Building custom search solutions for specific domains
or industries.
```

```
In summary, Oracle Text is a powerful tool within the Oracle AI Database
that enables you to index, query, and retrieve unstructured text data with
ease.
```

Related Topics

- UTL_TO_GENERATE_TEXT
  Use the `DBMS_VECTOR.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.
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- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Describe Image Content

In these examples, you can see how to generate a textual analysis or description of the contents of a given image.

Here, you supply an image along with a text question as the prompt (for example, "`What is this image about?`" or "`How many birds are there in this image?`"). The LLM responds with a textual answer or description based on the specified task in the prompt, which can then be used for image classification, object detection, or similarity search.

- Describe Images Using Public REST Providers
  Perform an image-to-text transformation by supplying an image along with a text question as the prompt, using publicly hosted third-party LLMs by Google AI, Hugging Face, OpenAI, or Vertex AI.
- Describe Images Using the Local REST Provider Ollama
  Perform an image-to-text transformation by supplying an image along with a text question as the prompt by accessing open LLMs, using the local host REST endpoint provider Ollama.

Describe Images Using Public REST Providers

Perform an image-to-text transformation by supplying an image along with a text question as the prompt, using publicly hosted third-party LLMs by Google AI, Hugging Face, OpenAI, or Vertex AI.

Here, you can use the `UTL_TO_GENERATE_TEXT` function from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To generate a text response by prompting with the following image and the text question "`Describe this image?`", using an external LLM:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba
```

```
CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;
```

```
SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (docuser) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (docuser):

```
CONN docuser/password
```

2. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

3. Grant connect privilege to docuser for allowing connection to the host, using the DBMS_NETWORK_ACL_ADMIN procedure.

   This example uses * to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

4. Create a local directory (DEMO_DIR) to store your image file:

```
create or replace directory DEMO_DIR as '/my_local_dir/';

create or replace function load_blob_from_file(directoryname varchar2,
filename varchar2)
  return blob
is
  filecontent blob := null;
  src_file bfile := bfilename(directoryname, filename);
  offset number := 1;
begin
  dbms_lob.createtemporary(filecontent, true, dbms_lob.session);
  dbms_lob.fileopen(src_file, dbms_lob.file_readonly);
  dbms_lob.loadblobfromfile(filecontent, src_file,
    dbms_lob.getlength(src_file), offset, offset);
  dbms_lob.fileclose(src_file);
  return filecontent;
end;
/
```

Upload the image file (for example, `bird.jpg`) to the directory.

5. Set up your credentials for the REST provider that you want to access and then call `UTL_TO_GENERATE_TEXT`:

   a. Call `CREATE_CREDENTIAL` to create and store a credential.

      Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

      `{ "access_token": "<access token>" }`

      You will later refer to this credential name when declaring JSON parameters for the `UTL_TO_GENERATE_TEXT` call.

```
exec dbms_vector_chain.drop_credential('<credential name>');


declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', '<access token>');
  dbms_vector_chain.create_credential(
    credential_name   => '<credential name>',
    params            => json(jo.to_string));
end;
/
```

      Replace `access_token` and `credential_name` with your own values. For example:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', 'AbabA1B123aBc123AbabAb123a1a2ab');
  dbms_vector_chain.create_credential(
    credential_name   => 'HF_CRED',
    params            => json(jo.to_string));
end;
/
```

   b. Call `UTL_TO_GENERATE_TEXT`:

```
-- select example

var input clob;
var media_data blob;
var media_type clob;
var params clob;

begin
  :input := 'Describe this image';
  :media_data := load_blob_from_file('DEMO_DIR', 'bird.jpg');
  :media_type := 'image/jpeg';
  :params := '
{
```

```
  "provider"       : "<REST provider>",
  "credential_name": "<credential name>",
  "url"            : "<REST endpoint URL for text generation service>",
  "model"          : "<REST provider text generation model name>",
  "max_tokens"     : <maximum number of tokens in the output text>
}';
end;
/

select
dbms_vector_chain.utl_to_generate_text(:input, :media_data, :media_type,
 json(:params));

-- PL/SQL example

declare
  input clob;
  media_data blob;
  media_type varchar2(32);
  params clob;
  output clob;
begin
  input := 'Describe this image';
  media_data := load_blob_from_file('DEMO_DIR', 'bird.jpg');
  media_type := 'image/jpeg';
  params := '
{
  "provider"       : "<REST provider>",
  "credential_name": "<credential name>",
  "url"            : "<REST endpoint URL for text generation service>",
  "model"          : "<REST provider text generation model name>",
  "max_tokens"     : <maximum number of tokens in the output text>
}';

  output := dbms_vector_chain.utl_to_generate_text(
    input, media_data, media_type, json(params));
  dbms_output.put_line(output);

  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
  if media_data is not null then
    dbms_lob.freetemporary(media_data);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

> **Note**
>
> For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Replace `provider`, `credential_name`, `url`, and `model` with your own values. Optionally, you can specify additional REST provider parameters. This is shown in the following examples:

Google AI example:

```
{
  "provider"       : "googleai",
  "credential_name": "GOOGLEAI_CRED",
  "url"            : "https://generativelanguage.googleapis.com/v1beta/
models/",
  "model"          : "gemini-pro:generateContent"
}
```

Hugging Face example:

```
{
  "provider"       : "huggingface",
  "credential_name": "HF_CRED",
  "url"            : "https://api-inference.huggingface.co/models/",
  "model"          : "gpt2"
}
```

> **Note**
>
> Hugging Face uses an image captioning model, which does not require a prompt. If you input a prompt along with an image, then the prompt will be ignored.

OpenAI example:

```
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://LOCATION-aiplatform.googleapis.com/v1/
projects/PROJECT/locations/LOCATION/publishers/google/models/",
```

```
      "model"            : "gemini-1.0-pro:generateContent",
      "generation_config": {
                             "temperature"    : 0.9,
                             "topP"           : 1,
                             "candidateCount" : 1,
                             "maxOutputTokens": 256
                            }
    }
```

A generated text response to your question may appear as:

```
This image showcases a stylized, artistic depiction of a hummingbird in
mid-flight, set against a vibrant red background. The bird is illustrated
with a
mix of striking colors and details - its head and belly are shown in
white, with
a black patterned detailing that resembles stripes or scales. Its long,
slender
beak is depicted in a darker color, extending forwards. The hummingbird's
wings
and tail are rendered in eye-catching shades of orange, purple, and red,
with
texture that suggests a rough, perhaps brush-like stroke.

The background features abstract shapes resembling clouds or wind currents
in a white line
pattern, which adds a sense of motion or air dynamics around the bird. The
overall use of vivid colors and dynamic patterns gives the image an
energetic
and modern feel.
```

This example uses the default settings for each provider. For detailed information on additional parameters, refer to your third-party provider's documentation.

Related Topics

- UTL_TO_GENERATE_TEXT
  Use the DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.
- DBMS_VECTOR
  The DBMS_VECTOR package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The DBMS_VECTOR_CHAIN package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Describe Images Using the Local REST Provider Ollama

Perform an image-to-text transformation by supplying an image along with a text question as the prompt by accessing open LLMs, using the local host REST endpoint provider Ollama.

Ollama is a free and open-source command-line interface tool that allows you to run open LLMs (such as Llama 3, Llava, Phi 3, Mistral, or Gemma 2) locally and privately on your Linux, Windows, or macOS systems. You can access Ollama as a service using SQL and PL/SQL commands.

Here, you can use the `UTL_TO_GENERATE_TEXT` function from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To describe the contents of an image, by prompting with the following image of a bird and the text question "`Describe this image?`", by calling a local LLM using Ollama:

1. Connect to Oracle AI Database as a local user.
   a. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

```
conn sys/password as sysdba


CREATE TABLESPACE tbs1
DATAFILE 'tbs5.dbf' SIZE 20G AUTOEXTEND ON
```

```
EXTENT MANAGEMENT LOCAL
SEGMENT SPACE MANAGEMENT AUTO;


SET ECHO ON
SET FEEDBACK 1
SET NUMWIDTH 10
SET LINESIZE 80
SET TRIMSPOOL ON
SET TAB OFF
SET PAGESIZE 10000
SET LONG 10000
```

b. Create a local user (`docuser`) and grant necessary privileges:

```
DROP USER docuser cascade;


CREATE USER docuser identified by docuser DEFAULT TABLESPACE tbs1 quota
unlimited on tbs1;


GRANT DB_DEVELOPER_ROLE, create credential to docuser;
```

c. Connect as the local user (`docuser`):

```
CONN docuser/password
```

2. Install Ollama and run a model locally.

   a. Download and run the Ollama application from https://ollama.com/download.

   You can either install Ollama as a service that runs in the background or as a standalone binary with a manual install. For detailed installation-specific steps, see Quick Start in the Ollama Documentation.

   Note the following:

   - The Ollama server needs to be able to connect to the internet so that it can download the models. If you require a proxy server to access the internet, remember to set the appropriate environment variables before running the Ollama server. For example, to set for Linux:

     ```
     -- set a proxy if you require one

     export https_proxy=<proxy-hostname>:<proxy-port>
     export http_proxy=<proxy-hostname>:<proxy-port>
     export no_proxy=localhost,127.0.0.1,.example.com
     export ftp_proxy=<proxy-hostname>:<proxy-port>
     ```

   - If you are running Ollama and the database on different machines, then on the database machine, you must change the `URL` to refer to the host name or IP address that is running Ollama instead of the local host.
   - You may need to change your firewall settings on the machine that is running Ollama to allow the port through.

b. If running Ollama as a standalone binary from a manual install, then start the server:

```
ollama serve
```

c. Run a model using the `ollama run <model_name>` command.

   For example, to call the llava model:

```
ollama run llava
```

   For detailed information on this step, see Ollama Readme.

d. Verify that Ollama is running locally by using a cURL command.

   For example:

```
-- generate text

curl -X POST http://localhost:11434/api/generate -d '{
  "model" : "llava",
  "prompt": "Why is the sky blue?",
  "stream": false }'
```

3. Set the HTTP proxy server, if configured.

```
EXEC UTL_HTTP.SET_PROXY('<proxy-hostname>:<proxy-port>');
```

4. Grant connect privilege to `docuser` for allowing connection to the host, using the `DBMS_NETWORK_ACL_ADMIN` procedure.

   This example uses `*` to allow any host. However, you can explicitly specify the host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'docuser',
                       principal_type => xs_acl.ptype_db));
END;
/
```

5. Create a local directory (`DEMO_DIR`) to store your image file:

```
create or replace directory DEMO_DIR as '/my_local_dir/';

create or replace function load_blob_from_file(directoryname varchar2,
filename varchar2)
  return blob
is
  filecontent blob := null;
  src_file bfile := bfilename(directoryname, filename);
  offset number := 1;
begin
  dbms_lob.createtemporary(filecontent, true, dbms_lob.session);
  dbms_lob.fileopen(src_file, dbms_lob.file_readonly);
```

```
  dbms_lob.loadblobfromfile(filecontent, src_file,
    dbms_lob.getlength(src_file), offset, offset);
  dbms_lob.fileclose(src_file);
  return filecontent;
end;
/
```

Upload the image file (for example, `bird.jpg`) to the directory.

6. Call `UTL_TO_GENERATE_TEXT`.

   The Ollama service has a REST API endpoint for generating text. Specify the URL and other configuration parameters in a JSON object.

```
var input clob;
var media_data blob;
var media_type clob;
var gent_ollama_params clob;

  :input := 'Describe this image';
  :media_data := load_blob_from_file('DEMO_DIR', 'bird.jpg');
  :media_type := 'image/jpeg';
  :gent_ollama_params := '{
     "provider": "ollama",
     "host"    : "local",
     "url"     : "http://localhost:11434/api/generate",
     "model"   : "llava"
  }';

select
dbms_vector_chain.utl_to_generate_text(:input, :media_data, :media_type,
json(:gent_ollama_params)) from dual;
```

   You can replace `url` and `model` with your own values, as required.

   > **Note**
   >
   > For a complete list of all supported REST endpoint URLs, see [Supported Third-Party Provider Operations and Endpoints]().

   A generated text response to your question may appear as:

```
This is an image of a stylized, artistic depiction of a hummingbird in
mid-flight, set against a vibrant red background. The bird is illustrated
with a
mix of striking colors and details - its head and belly are shown in
white, with
a black patterned detailing that resembles stripes or scales. Its long,
slender
beak is depicted in a darker color, extending forwards. The hummingbird's
wings
and tail are rendered in eye-catching shades of orange, purple, and red,
with
texture that suggests a rough, perhaps brush-like stroke.
```

```
The background features abstract shapes resembling clouds or wind currents
in a white line
pattern, which adds a sense of motion or air dynamics around the bird. The
overall use of vivid colors and dynamic patterns gives the image an
energetic
and modern feel.
```

Related Topics

- UTL_TO_GENERATE_TEXT
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.
- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Use Retrieval Augmented Generation to Complement LLMs

RAG lets you mitigate the inaccuracies and hallucinations faced when using LLMs. Oracle AI Vector Search enables RAG within Oracle AI Database using the `DBMS_VECTOR_CHAIN` PL/SQL package or through popular frameworks (such as LangChain).

- About Retrieval Augmented Generation
  Oracle AI Vector Search supports Enterprise Retrieval Augmented Generation (RAG) to enable sophisticated queries that can combine vectors with relational data, graph data, spatial data, and JSON collections.
- SQL RAG Example
  This scenario allows you to run a similarity search for specific documentation content based on a user query. Once documentation chunks are retrieved, they are concatenated and a prompt is generated to ask an LLM to answer the user question using retrieved chunks.
- Oracle AI Vector Search Integration with LangChain
  LangChain is a powerful and flexible open source orchestration framework that helps developers build applications that leverage the advanced capabilities of large language models (LLMs).
- Oracle AI Vector Search Integration with LlamaIndex
  LlamaIndex is an open-source data framework designed to simplify the process of building applications that leverage large language models (LLMs) with custom data. Basically, LlamaIndex acts as a bridge between custom data sources and LLMs such as Cohere Command models or OpenAI GPTs models.
- Use Reranking for Better RAG Results
  Reranking models are primarily used to reassess and reorder an initial set of search results. This helps to improve the relevance and quality of search results in both similarity search and Retrieval Augmented Generation (RAG) scenarios.

About Retrieval Augmented Generation

Oracle AI Vector Search supports Enterprise Retrieval Augmented Generation (RAG) to enable sophisticated queries that can combine vectors with relational data, graph data, spatial data, and JSON collections.

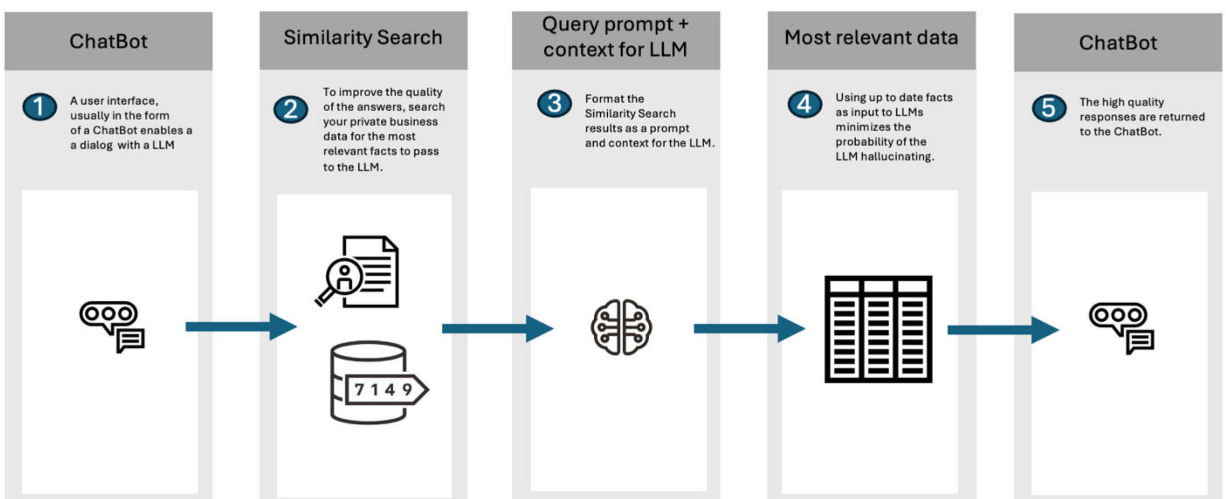
Retrieval Augmented Generation is an approach developed to address the limitations of LLMs. RAG combines the strengths of pretrained language models with the ability to retrieve recent and accurate information from a dataset or database in real-time during the generation of responses.

By communicating with LLMs through the implementation of RAG, the knowledge of LLMs is increased with business data found through AI Vector Search.

The primary problem with LLMs, such as GPT (Generative Pretrained Transformer), is that they generate responses based solely on the patterns and data they were trained on up to the point of their last update. This means that they inherently lack the ability to access or incorporate new, real-time information after their training is cut off, potentially limiting their responses to outdated or incomplete information. LLMs do not know about your private company data. Consequently, LLMs can make up answers (hallucinate) when they do not have enough relevant and up-to-date facts.

By providing your LLM with up-to-date facts from your company, you can minimize the probability that an LLM will hallucinate.

Figure 9-1   Example RAG Workflow



Here is how RAG improves upon the issues with traditional LLMs:

- **Access to External and Private Information:** RAG can pull in data from external and private sources during its response generation process. This allows it to provide answers that are up-to-date and grounded in the latest available information, which is crucial for queries requiring current knowledge or specific details not included in its original training data.
- **Factually More Accurate and Detailed Responses:** While traditional LLMs are trained on older data, RAG incorporates real-time retrieved information, meaning that generated responses are not only contextually rich but also factually more up-to-date and accurate as

time goes on. This is particularly beneficial for queries that require precision and detail, such as scientific facts, historical data, or specific statistics.

- **Reduced Hallucination:** LLMs can sometimes "hallucinate" information, as in generate plausible but false or unverified content. RAG mitigates this by grounding responses in retrieved documents, thereby enhancing the reliability of the information provided.

Oracle AI Vector Search enables RAG within Oracle AI Database using the `DBMS_VECTOR_CHAIN` PL/SQL package. Alternatively, you can implement RAG externally by using popular frameworks such as LangChain.

LangChain is a popular open source framework that encapsulates popular LLMs, vector databases, document stores, and embedding models. `DBMS_VECTOR_CHAIN` is a PL/SQL package that provides the ability to create RAG solutions, all within the database. With `DBMS_VECTOR_CHAIN`, your data never needs to leave the security of Oracle AI Database.

> **See Also**
>
> - *Oracle AI Database PL/SQL Packages and Types Reference* for details about the `DBMS_VECTOR_CHAIN` package

SQL RAG Example

This scenario allows you to run a similarity search for specific documentation content based on a user query. Once documentation chunks are retrieved, they are concatenated and a prompt is generated to ask an LLM to answer the user question using retrieved chunks.

1. Start SQL*Plus and connect to Oracle AI Database as a local test user.

   a. Log in to SQL*Plus as the `sys` user, connecting as `sysdba`:

   ```
   conn sys/password AS sysdba


   SET SERVEROUTPUT ON;
   SET ECHO ON;
   SET LONG 100000;
   ```

   b. Create a local test user (`vector`) and grant necessary privileges:

   ```
   DROP USER vector cascade;


   CREATE USER vector identified by <my vector password>


   GRANT DB_DEVELOPER_ROLE, CREATE CREDENTIAL TO vector;
   ```

   c. Set the proxy if one exists:

   ```
   EXEC UTL_HTTP.SET_PROXY('<my proxy full name>:<my proxy port>');
   ```

d. Grant connect privilege for a host using the `DBMS_NETWORK_ACL_ADMIN` procedure. This example uses `*` to allow any host. However, you can explicitly specify each host that you want to connect to.

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'VECTOR',
                       principal_type => xs_acl.ptype_db));
END;
/
```

e. Connect to Oracle AI Database as the test user.

```
conn docuser/password;
```

2. Create a credential for Oracle Cloud Infrastructure Generative AI:

a. Run `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` to create and store an OCI credential (`OCI_CRED`).

OCIGenAI requires the following parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
"fingerprint"     : "<fingerprint>"
}
```

> **Note**
>
> The generated private key may appear as:
>
> ```
> -----BEGIN RSA PRIVATE KEY-----
> <private key string>
> -----END RSA PRIVATE KEY-----
> ```
>
> You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

```
BEGIN
  DBMS_VECTOR_CHAIN.DROP_CREDENTIAL(credential_name => 'OCI_CRED');
EXCEPTION
  WHEN OTHERS THEN NULL;
END;
/


DECLARE
  jo json_object_t;
```

```
BEGIN
  jo := json_object_t();
  jo.put('user_ocid', '<user ocid>');
  jo.put('tenancy_ocid', '<tenancy ocid>');
  jo.put('compartment_ocid', '<compartment ocid>');
  jo.put('private_key', '<private key>');
  jo.put('fingerprint', '<fingerprint>');
  DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL(
    credential_name => 'OCID_CRED',
    params          => json(jo.to_string));
END;
/
```

b. Check credential creation:

```
col owner format a15
col credential_name format a20
col username format a20


SELECT owner, credential_name, username
FROM all_credentials
ORDER BY owner, credential_name, username;
```

3. Generate a prompt using similarity search results:

> **Note**
>
> For information about loading the ONNX format model into the database as `doc_model`, see Import ONNX Models into Oracle AI Database End-to-End Example.

```
SET SERVEROUTPUT ON;

VAR prompt CLOB;
VAR user_question CLOB;
VAR context CLOB;

BEGIN
  -- initialize the concatenated string
  :context := '';

  -- read this question from the user
  :user_question := 'what are vector indexes?';

  -- cursor to fetch chunks relevant to the user's query
  FOR rec IN (SELECT EMBED_DATA
              FROM doc_chunks
              WHERE DOC_ID = 'Vector User Guide'
              ORDER BY vector_distance(embed_vector, vector_embedding(
                  doc_model using :user_question as data), COSINE)
              FETCH EXACT FIRST 10 ROWS ONLY)
  LOOP
    -- concatenate each value to the string
```

```
    :context := :context || rec.embed_data;
  END LOOP;

  -- concatenate strings and format it as an enhanced prompt to the LLM
  :prompt := 'Answer the following question using the supplied context
              assuming you are a subject matter expert. Question: '
              || :user_question || ' Context: ' || :context;

  DBMS_OUTPUT.PUT_LINE('Generated prompt: ' || :prompt);
END;
/
```

4. Issue the GenAI call:

> **Note**
>
> For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

```
DECLARE
  input CLOB;
  params CLOB;
  output CLOB;
BEGIN
  input := :prompt;
  params := '{
    "provider" : "ocigenai",
    "credential_name" : "OCI_CRED",
    "url" : "https://inference.generativeai.us-chicago-1.oci.
            oraclecloud.com/20231130/actions/generateText",
    "model" : "cohere.command"
  }';

  output := DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT(input, json(params));
  DBMS_OUTPUT.PUT_LINE(output);
  IF output IS NOT NULL THEN
    DBMS_LOB.FREETEMPORARY(output);
  END IF;
EXCEPTION
  WHEN OTHERS THEN
    DBMS_OUTPUT.PUT_LINE(SQLERRM);
    DBMS_OUTPUT.PUT_LINE(SQLCODE);
END;
/
```

Oracle AI Vector Search Integration with LangChain

LangChain is a powerful and flexible open source orchestration framework that helps developers build applications that leverage the advanced capabilities of large language models (LLMs).

LangChain provides essential tools for managing workflows, maintaining context, and integrating with external systems. For example, the LangChain framework allows you to create

Chatbots and agent applications. LangChain primarily supports Python but also has support for JavaScript and TypeScript.

Oracle AI Vector Search integrates with LangChain at various levels:

- Document Loaders
- Text Splitter
- Embeddings
- Summary
- Vector Store

For more information about each of these components, see LangChain Oracle AI Vector Search documentation.

> **See Also**
>
> - LangChain documentation for an introduction to the LangChain framework
> - LangChain component documentation for a list of LangChain components
> - LangChain installation documentation to learn how to install LangChain packages in Python

Oracle AI Vector Search Integration with LlamaIndex

LlamaIndex is an open-source data framework designed to simplify the process of building applications that leverage large language models (LLMs) with custom data. Basically, LlamaIndex acts as a bridge between custom data sources and LLMs such as Cohere Command models or OpenAI GPTs models.

Oracle AI Vector Search is integrated with LlamaIndex in several ways to enable powerful semantic search and retrieval capabilities.

Here are the key aspects of this integration:

- **Embedding Generation**

  Oracle AI Vector Search provides embedding capabilities that can be used with LlamaIndex:

  - The `OracleEmbeddings` class from LlamaIndex can be used to generate embeddings using Oracle's embedding models.
  - Multiple embedding methods are supported, including locally-hosted ONNX models and third-party APIs such as Generative AI and Hugging Face.
  - Embeddings can be generated for documents and queries to enable semantic similarity search.

  For more information on how to use this integration for generating embeddings, see Oracle AI Vector Search: Use Embedding Generation Capabilities.

- **Vector Storage**

  LlamaIndex can leverage Oracle AI Database as a vector store:

  - Vector embeddings can be stored alongside business data in Oracle AI Database tables using the `VECTOR` data type.

- This allows combining semantic search on unstructured data with relational queries on structured data in a single system.

  For more information on how to use this integration for vector storage, see Oracle AI Vector Search: Use Vector Storage Capabilities.

- **Indexing and Search**

  You can utilize Oracle's vector indexing and search capabilities:

  - Vector indexes can be created on the embeddings to enable fast similarity search.
  - LlamaIndex can use Oracle's native SQL operations for similarity search to retrieve relevant data.
  - Various distance metrics, such as dot product, cosine similarity, Euclidean distance, and more are supported.

  For more information about how to use this integration for indexing and search, see Oracle AI Vector Search: Use Document Processing Capabilities and Oracle AI Vector Search: End-to-End Pipeline with Document Processing.

- **RAG Pipeline Integration**

  The integration enables building end-to-end Retrieval Augmented Generation (RAG) pipelines. You can embed and store unstructured data in Oracle AI Database. LlamaIndex can query the vector store to retrieve relevant context. The retrieved information can be used to generate prompts for LLMs.

  LlamaIndex provides several libraries and classes to integrate Oracle AI Vector Search capabilities. Here are the key components available:

  - `OracleEmbeddings`: Supports multiple embedding methods, including locally hosted ONNX models and third-party APIs such as Generative AI and Hugging Face.
  - `OracleReader`: Used for loading documents from various sources, including Oracle AI Database.
  - `OracleSummary`: Provides functionality for summarizing documents within or outside the database.
  - `OracleTextSplitter`: Offers advanced Oracle capabilities for chunking documents according to different requirements.
  - `OraLlamaVS`: Used for storing, indexing, and querying vector embeddings.

  For more information about how to use this integration for building end-to-end RAG pipelines, see Oracle AI Vector Search: End-to-End Pipeline with Document Processing.

Benefits

The Oracle AI Vector Search integration with LlamaIndex provides a powerful foundation for developing sophisticated AI applications that can leverage both structured and unstructured data within the Oracle ecosystem.

By integrating Oracle AI Vector Search with LlamaIndex, developers can:

- Leverage Oracle AI Database's enterprise features such as scalability, security, and transactions.
- Combine semantic search with relational queries in one single system.
- Utilize Oracle's optimized vector operations for efficient similarity search.
- Build AI-powered applications using familiar SQL and PL/SQL interfaces.

Use Reranking for Better RAG Results

Reranking models are primarily used to reassess and reorder an initial set of search results. This helps to improve the relevance and quality of search results in both similarity search and Retrieval Augmented Generation (RAG) scenarios.

In a RAG scenario, reranking plays a crucial role in improving the quality of information ingested into an LLM by ensuring that the most relevant documents or chunks are prioritized. This can reduce hallucinations and improve the accuracy of generated outputs. The reranking step is typically performed after an initial search that uses a faster but less precise embedding model. The reranker helps to identify the most pertinent information for a given query, but is more expensive in terms of resources because it often employs sophisticated matching methods that go beyond simple vector comparisons.

Here, you can use the `RERANK` function from either the `DBMS_VECTOR` or the `DBMS_VECTOR_CHAIN` package, depending on your use case.

Oracle AI Vector Search supports reranking models provided by Cohere and Vertex AI.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

To rerank results for the query "`What are some interesting characteristics of the Jovian satellites?`", using a third-party reranking model:

1. Log in to SQL*Plus as the `SYS` user, connecting as `SYSDBA`:

   ```
   conn sys/password as sysdba
   ```

2. Create a local user (`vector`) and grant necessary privileges:

   ```
   DROP USER vector cascade;

   CREATE USER vector identified by <my vector password>

   GRANT DB_DEVELOPER_ROLE, CREATE CREDENTIAL TO vector;
   ```

3. Set proxy if one exists.

   ```
   EXEC UTL_HTTP.SET_PROXY('<my proxy full name>:<my proxy port>');
   ```

4. Grant connect privilege for a host using the `DBMS_NETWORK_ACL_ADMIN` procedure. This example uses `*` to allow any host. However, you can explicitly specify each host that you want to connect to:

```
BEGIN
  DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
    host => '*',
    ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                       principal_name => 'VECTOR',
                       principal_type => xs_acl.ptype_db));
END;
/
```

5. Connect to Oracle AI Database as the test user:

```
conn vector/password;
```

6. Simulate the initial retrieval step that returns, in the specified order, potentially relevant documents for the following query: "`What are some interesting characteristics of the Jovian satellites?`":

```
set echo on

set serveroutput on

var query clob;

var initial_retrieval_docs clob;

exec :query := 'What are some interesting characteristics of the Jovian satellites?';

begin
  :initial_retrieval_docs := '
{
   "documents": [
                 "Jupiter boasts an impressive system of 95 known moons, 
including the four largest Galilean satellites.",
                 "Jupiter's immense mass, 318 times that of Earth, 
significantly influences the orbits of other bodies in the Solar System.",
                 "Io, one of Jupiter's Galilean moons, is the most 
volcanically active body in our solar system.",
                 "The gas giant completes one orbit around the Sun in just 
under 12 years, traveling at an average speed of 13 kilometers per 
second.",
                 "Jupiter's composition is similar to that of the Sun, and 
it could have become a brown dwarf if its mass had been 80 times greater."
                ]
}';
end;
/
```

7. Set up your credentials for the REST provider that you want to access, that is, Cohere or Vertex AI. Here, replace <*access token*> with your own values:

- Cohere example:

```
EXEC DBMS_VECTOR.DROP_CREDENTIAL('COHERE_CRED');

DECLARE
   jo json_object_t;
BEGIN
   jo := json_object_t();
   jo.put('access_token', '<access token>');
   DBMS_VECTOR.CREATE_CREDENTIAL(
          credential_name => 'COHERE_CRED',
          params => json(jo.to_string));
END;
/
```

- Vertex AI example:

```
begin
   dbms_vector_chain.drop_credential(credential_name  =>
'VERTEXAI_CRED');
exception
   when others then null;
end;
/

declare
   jo json_object_t;
begin
   jo := json_object_t();
   jo.put('access_token', '<access token>');
   dbms_vector_chain.create_credential(
          credential_name   =>  'VERTEXAI_CRED',
          params            => json(jo.to_string));
end;
/
```

8. Rerank the initial retrieval:

> **Note**
>
> For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

- Using the Cohere `rerank-english-v3.0` model:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "cohere",
  "credential_name": "COHERE_CRED",
  "url": "https://api.cohere.com/v1/rerank",
```

```
  "model": "rerank-english-v3.0",
  "return_documents": true,
  "top_n": 3
}';

  reranked_output := dbms_vector_chain.utl_to_rerank(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

- Using the Vertex AI `semantic-ranker-512` model:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url": "https://discoveryengine.googleapis.com/v1/projects/
1085581009881/locations/global/rankingConfigs/
default_ranking_config:rank",
  "model": "semantic-ranker-512@latest",
  "ignoreRecordDetailsInResponse": false,
  "topN": 3
}';

  reranked_output := dbms_vector_chain.utl_to_rerank(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

Using the above Cohere model, the reranked results appear as follows:

```
[
   {
     "index" : "0",
     "score" : "0.059319142",
     "content" : "Jupiter boasts an impressive system of 95 known moons,
including the four largest Galilean satellites."
   },
   {
     "index" : "2",
     "score" : "0.04352814",
     "content" : "Io, one of Jupiter's Galilean moons, is the most
volcanically active body in our solar system."
   },
   {
     "index" : "4",
     "score" : "0.04138472",
     "content" : "Jupiter's composition is similar to that of the Sun, and it
could have become a brown dwarf if its mass had been 80 times greater."
```

```
    }
]
```

Related Topics

- UTL_TO_RERANK
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_RERANK` function to reassess and reorder an initial set of results to retrieve more relevant search output.
- DBMS_VECTOR
  The `DBMS_VECTOR` package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The `DBMS_VECTOR_CHAIN` package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

Supported Third-Party Provider Operations and Endpoints

Review a list of third-party REST providers and REST endpoints that are supported for various vector generation, summarization, text generation, and reranking operations.

- Public or Remote REST Endpoint Providers
- Local REST Endpoint Provider
- REST Operations
- REST Endpoints
- Models Supported for Generative AI

Public or Remote REST Endpoint Providers

The supported publicly-hosted, third-party REST endpoint providers are:

- Cohere
- Google AI
- Hugging Face
- Oracle Cloud Infrastructure (OCI) Generative AI
- OpenAI
- Vertex AI
- Mistral AI
- Anthropic

Local REST Endpoint Provider

In addition to using a publicly-hosted, third-party REST endpoint provider, you can also use a provider that is running locally. The supported third-party REST endpoint providers are as follows:

- **Private AI**

The Private AI Services Container enables secure and private local AI infrastructure. The container can run in your data center and does not require access to a public cloud or public internet. Using the Private AI Services Container allows you to offload expensive AI computation, such as vector embedding generation, outside of the database. For more information, see *Oracle Private AI Services Container User's Guide*.

- **Ollama**

  You can use Ollama as a third-party REST endpoint provider, locally and privately on your Linux, Windows, and macOS systems.

  Ollama is a free and open-source command-line interface tool that allows you to run open LLMs (such as Llama 3, Phi 3, Mistral, and Gemma 2) and embedding models (such as mxbai-embed-large, nomic-embed-text, or all-minilm). You can access Ollama using SQL and PL/SQL commands.

  You can download and run the Ollama application from https://ollama.com/download. You can either install Ollama as a service that runs in the background or as a standalone binary with a manual install. For detailed installation-specific steps, see Quick Start in the Ollama Documentation.

- **Other Open-AI Compatible Provides (for example, Llamafile, vLLM)**

REST Operations

These are the supported third-party REST operations and APIs along with their corresponding REST providers:

Operation	Provider	API
Generate embedding: Convert textual documents and images to one or more vector embeddings	• For text input: All supported public and local providers • For image input: Vertex AI	• DBMS_VECTOR.UTL_TO_EMBEDDING and DBMS_VECTOR.UTL_TO_EMBEDDINGS • DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING and DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS
Generate summary: Extract a brief and comprehensive summary from textual documents	All supported public and local providers	• DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY
Generate text: Retrieve a descriptive response for textual prompts and images through conversations with LLMs	• For text input: All supported public and local providers • For image input: Google AI, Hugging Face, OpenAI, Ollama, and Vertex AI	• DBMS_VECTOR.UTL_TO_GENERATE_TEXT • DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT
Rerank results: Reassess and reorder search results to retrieve a more relevant output	Cohere and Vertex AI	• DBMS_VECTOR.UTL_TO_RERANK • DBMS_VECTOR_CHAIN.UTL_TO_RERANK

REST Endpoints

These are the supported REST endpoints for all third-party REST providers:

API	Provider	Endpoint
UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS	Cohere	https://api.cohere.ai/v1/embed
	Generative AI	https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/20231130/actions/embedText
	Google AI	https://generativelanguage.googleapis.com/v1beta/models/
	Hugging Face	https://api-inference.huggingface.co/pipeline/feature-extraction/
	Ollama	http://localhost:11434/api/embeddings
	OpenAI	https://api.openai.com/v1/embeddings
	OpenAI-Compatible Providers	**Note:** This is not an exhaustive list of supported third-party REST endpoints, only an example selection. You must set provider to openai to use these third-party endpoints. • Llamafile: http://localhost:8080/v1/embeddings • vLLM: http://localhost:8000/v1/embeddings • Ollama: http://localhost:11434/v1/embeddings
	Vertex AI	https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/locations/LOCATION/publishers/google/models/
	Mistral	https://api.mistral.ai/v1/embeddings
	Private AI	https://host:port/v1/embeddings
UTL_TO_SUMMARY	Cohere	https://api.cohere.ai/v1/chat https://api.cohere.ai/v1/summarize
	Generative AI	https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/20231130/actions/chat
	Google AI	https://generativelanguage.googleapis.com/v1beta/models/
	Hugging Face	https://api-inference.huggingface.co/models/
	Ollama	http://localhost:11434/api/generate
	OpenAI	https://api.openai.com/v1/chat/completions https://api.openai.com/v1/completions
	OpenAI-Compatible Providers	**Note:** This is not an exhaustive list of supported third-party REST endpoints, only an example selection. You must set provider to openai to use these third-party endpoints. • Llamafile: http://localhost:8080/v1/chat/completions • vLLM: http://localhost:8000/v1/chat/completions • Ollama: http://localhost:11434/v1/chat/completions
	Vertex AI	https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/locations/LOCATION/publishers/google/models/
	Mistral AI	https://api.mistral.ai/v1/chat/completions

API	Provider	Endpoint
	Anthropic	`https://api.anthropic.com/v1/messages`
`UTL_TO_GENERATE_TEXT`	Cohere	`https://api.cohere.ai/v1/chat` `https://api.cohere.ai/v1/generate`
	Generative AI	`https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/20231130/actions/chat`
	Google AI	`https://generativelanguage.googleapis.com/v1beta/models/`
	Hugging Face	`https://api-inference.huggingface.co/models/`
	Ollama	`http://localhost:11434/api/generate`
	OpenAI	`https://api.openai.com/v1/chat/completions` `https://api.openai.com/v1/completions`
	OpenAI-Compatible Providers	**Note:** This is not an exhaustive list of supported third-party REST endpoints, only an example selection. You must set `provider` to `openai` to use these third-party endpoints. • Llamafile: `http://localhost:8080/v1/chat/completions` • vLLM: `http://localhost:8000/v1/chat/completions` • Ollama: `http://localhost:11434/v1/chat/completions`
	Vertex AI	`https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/locations/LOCATION/publishers/google/models/`
	Mistral AI	`https://api.mistral.ai/v1/chat/completions`
	Anthropic	`https://api.anthropic.com/v1/messages`
`UTL_TO_RERANK`	Cohere	`https://api.cohere.com/v1/rerank`
	Vertex AI	`https://discoveryengine.googleapis.com/v1/projects/PROJECT/locations/global/rankingConfigs/default_ranking_config:rank`

Models Supported for OCI Generative AI

These are the third-party models that are supported to use with OCI Generative AI corresponding to each REST API. Keep in mind that these models are updated periodically. For currently available models, check Pretrained Foundational Models in Generative AI.

API	Model

`UTL_TO_EMBEDDING`
and
`UTL_TO_EMBEDDINGS`
	cohere.embed-english-v3.0
	cohere.embed-multilingual-v3.0
	cohere.embed-english-light-v3.0
	cohere.embed-multilingual-light-v3.0

`UTL_TO_SUMMARY`	cohere.command-r-08-2024 cohere.command-r-plus-08-2024 meta.llama-3.1-70b-instruct meta.llama-3.1-405b-instruct

API	Model
UTL_TO_GENERATE_TEXT	cohere.command-r-08-2024 cohere.command-r-plus-08-2024 meta.llama-3.1-70b-instruct meta.llama-3.1-405b-instruct

10 Supported Clients and Languages

For more information about Oracle AI Vector Search support using some of Oracle's available clients and languages, see the included reference material.

Clients and Languages	Reference Material
PL/SQL	*Oracle AI Database PL/SQL Language Reference*
MLE JavaScript	*Oracle AI Database JavaScript Developer's Guide*
JDBC	*Oracle AI Database JDBC Developer's Guide*
Node.js	node-oracledb documentation
Python	python-oracledb documentation
Oracle Call Interface	*Oracle Call Interface Developer's Guide*
ODP.NET	*Oracle Data Provider for .NET Developer's Guide*
SQL*Plus	*SQL*Plus User's Guide and Reference*

Oracle AI Database 26ai supports binding with native VECTOR types for all Oracle clients. Applications that use earlier Oracle Client 26ai libraries can connect to Oracle AI Database 26ai in the following ways:

- Using the `TO_VECTOR()` SQL function to insert vector data, as shown in the following example:

  ```
  INSERT INTO vecTab VALUES(TO_VECTOR('[1.1, 2.9, 3.14]'));
  ```

- Using the `FROM_VECTOR()` SQL function to fetch vector data, as shown in the following example:

  ```
  SELECT FROM_VECTOR(dataVec) FROM vecTab;
  ```

11 Vector Diagnostics

AI Vector Search includes several views, statistics, and parameters that can be used to help understand how vector search is performing for your workload.

- Oracle AI Vector Search Views
  These are a set of data dictionary views related to Oracle AI Vector Search.
- Oracle AI Vector Search Statistics
  These are a set of statistics related to Oracle AI Vector Search.
- Oracle AI Vector Search Parameters
  This is a set of parameters related to Oracle AI Vector Search.

Oracle AI Vector Search Views

These are a set of data dictionary views related to Oracle AI Vector Search.

- Text Processing Views
  These views display language-specific data (abbreviation token details) and vocabulary data related to the Oracle AI Vector Search SQL and PL/SQL utilities.
- Vector Memory Pool Views
  Review the various vector memory pool views.
- Vector Index and Hybrid Vector Index Views
  These views allow you to query tables related to vector indexes and hybrid vector indexes.

Text Processing Views

These views display language-specific data (abbreviation token details) and vocabulary data related to the Oracle AI Vector Search SQL and PL/SQL utilities.

- ALL_VECTOR_ABBREV_TOKENS
  The `ALL_VECTOR_ABBREV_TOKENS` view displays a list of abbreviation tokens from all supported languages.
- ALL_VECTOR_LANG
  The `ALL_VECTOR_LANG` view displays a list of all supported languages, distributed by default.
- DBA_VECTOR_HITCOUNTS
  The `DBA_VECTOR_HITCOUNTS` view tracks calls to third parties for observability.
- USER_VECTOR_ABBREV_TOKENS
  The `USER_VECTOR_ABBREV_TOKENS` view displays a list of abbreviation tokens from all languages loaded by the current user.
- USER_VECTOR_HITCOUNTS
  The `USER_VECTOR_HITCOUNTS` view tracks calls to third parties for observability for the current user.
- USER_VECTOR_LANG
  The `USER_VECTOR_LANG` view displays all languages loaded by the current user.

- USER_VECTOR_VOCAB
  The USER_VECTOR_VOCAB view displays all custom token vocabularies created by the current user.
- USER_VECTOR_VOCAB_TOKENS
  The USER_VECTOR_VOCAB_TOKENS view displays tokens from all custom token vocabularies created by the current user.
- ALL_VECTOR_VOCAB
  The ALL_VECTOR_VOCAB view displays all custom token vocabularies.
- ALL_VECTOR_VOCAB_TOKENS
  The ALL_VECTOR_VOCAB_TOKENS view displays tokens from all custom token vocabularies.

Related Topics

- Configure Chunking Parameters
  Oracle AI Vector Search provides many parameters for chunking text data, such as SPLIT [BY], OVERLAP, or NORMALIZE. In these examples, you can see how to configure these parameters to define your own chunking specifications and strategies, so that you can create meaningful chunks.

ALL_VECTOR_ABBREV_TOKENS

The ALL_VECTOR_ABBREV_TOKENS view displays a list of abbreviation tokens from all supported languages.

Column Name	Data Type	Description
ABBREV_OWNER	VARCHAR2(128)	Owner of the abbreviation token (for example, PUBLIC)
ABBREV_LANGUAGE	NUMBER	Language ID for the language (for example, 1 for American)
ABBREV_TOKEN	NVARCHAR2(255)	List of all abbreviation tokens corresponding to each language

ALL_VECTOR_LANG

The ALL_VECTOR_LANG view displays a list of all supported languages, distributed by default.

Column Name	Data Type	Description
LANG_OWNER	VARCHAR2(128)	Owner of the language (for example, PUBLIC)
LANG_LANG	NUMBER	Language ID for the language (for example, 1 for American)
LANG_NAME	VARCHAR2(128)	Name of the language (for example, AMERICAN)

DBA_VECTOR_HITCOUNTS

The DBA_VECTOR_HITCOUNTS view tracks calls to third parties for observability.

Column Name	Data Type	Description
USER_ID	NUMBER	ID of the current user

Column Name	Data Type	Description
CREDENTIAL_ID	NUMBER	Credential ID for the third party call used by the user
PROVIDER_NAME	VARCHAR2(128)	Name of the third-party provider
METHOD_NAME	VARCHAR2(128)	Name of the method calling the third-party API(s) (such as UTL_TO_SUMMARY, UTL_TO_EMBEDDING, and so on)
TOTAL_COUNT	NUMBER	Total number of calls made to the third-party provider
LAST_HIT	TIMESTAMP(6)	Time of the last call to the third-party provider

USER_VECTOR_ABBREV_TOKENS

The USER_VECTOR_ABBREV_TOKENS view displays a list of abbreviation tokens from all languages loaded by the current user.

Column Name	Data Type	Description
ABBREV_LANGUAGE	NUMBER	Language ID for the language (for example, 1 for American)
ABBREV_TOKEN	NVARCHAR2(255)	List of all abbreviation tokens corresponding to each language

USER_VECTOR_HITCOUNTS

The USER_VECTOR_HITCOUNTS view tracks calls to third parties for observability for the current user.

Column Name	Data Type	Description
USER_ID	NUMBER	ID of the current user
CREDENTIAL_ID	NUMBER	Credential ID for the third-party call used by the user
CREDENTIAL_NAME	VARCHAR2(128)	Credential name for the third-party call used by the user
PROVIDER_NAME	VARCHAR2(128)	Name of the third-party provider
METHOD_NAME	VARCHAR(128)	Name of the method calling the third-party API(s) (such as UTL_TO_SUMMARY, UTL_TO_EMBEDDING, and so on)
TOTAL_COUNT	NUMBER	Total number of calls made to the third-party provider.
LAST_HIT	TIMESTAMP(6)	Time of the last call to the third-party provider

USER_VECTOR_LANG

The USER_VECTOR_LANG view displays all languages loaded by the current user.

Column Name	Data Type	Description
LANG_LANG	NUMBER	Language ID for the language (for example, 1 for American)
LANG_NAME	VARCHAR2(128)	Name of the language (for example, AMERICAN)

USER_VECTOR_VOCAB

The USER_VECTOR_VOCAB view displays all custom token vocabularies created by the current user.

Column Name	Data Type	Description
VOCAB_NAME	VARCHAR2(128)	User-specified name of the vocabulary (for example, DOC_VOCAB)
FORMAT	VARCHAR2(4)	Format of the vocabulary, such as XLM, BERT, or GPT2
CASED	VARCHAR2(7)	Character-casing of the vocabulary, that is, vocabulary to be treated as cased or uncased

USER_VECTOR_VOCAB_TOKENS

The USER_VECTOR_VOCAB_TOKENS view displays tokens from all custom token vocabularies created by the current user.

Column Name	Data Type	Description
VOCAB_NAME	VARCHAR2(128)	User-specified name of the vocabulary (for example, DOC_VOCAB)
VOCAB_TOKEN	VARCHAR2(255)	Tokens contained in the vocabulary

ALL_VECTOR_VOCAB

The ALL_VECTOR_VOCAB view displays all custom token vocabularies.

Column Name	Data Type	Description
VOCAB_OWNER	VARCHAR2(128)	Owner of the vocabulary (for example, SYS)
VOCAB_NAME	VARCHAR2(128)	User-specified name of the vocabulary (for example, DOC_VOCAB)
FORMAT	VARCHAR2(4)	Format of the vocabulary, such as XLM, BERT, or GPT2

Column Name	Data Type	Description
CASED	VARCHAR2(7)	Character-casing of the vocabulary, that is, vocabulary to be treated as cased or uncased

ALL_VECTOR_VOCAB_TOKENS

The ALL_VECTOR_VOCAB_TOKENS view displays tokens from all custom token vocabularies.

Column Name	Data Type	Description
VOCAB_OWNER	VARCHAR2(128)	Owner of the vocabulary (for example, SYS)
VOCAB_NAME	VARCHAR2(128)	User-specified name of the vocabulary (for example, DOC_VOCAB)
VOCAB_TOKEN	VARCHAR2(255)	Tokens contained in the vocabulary

Vector Memory Pool Views

Review the various vector memory pool views.

- V$VECTOR_MEMORY_POOL
  This view contains information about the space allocation for Vector Memory.

V$VECTOR_MEMORY_POOL

This view contains information about the space allocation for Vector Memory.

The Vector Memory Pool area is used primarily to maintain in-memory vector indexes or metadata useful for vector-related operations. The Vector Memory Pool is subdivided into two pools: a 1MB pool used to store In-Memory Neighbor Graph Index allocations; and a 64K pool used to store metadata. The amount of available memory in each pool is visible in the V$VECTOR_MEMORY_POOL view. The relative size of the two pools is determined by internal heuristics. The size of the Vector Memory Pool is controlled by the `vector_memory_size` parameter. Area in the Vector Memory Pool is also allocated to accelerate Neighbor Partition Index access by storing centroid vectors.

Column Name	Data Type	Description
POOL	VARCHAR2(26)	Name of the pools in the Vector Memory Pool (64K or 1MB)
ALLOC_BYTES	NUMBER	Total amount of memory allocated to this pool
USED_BYTES	NUMBER	Amount of memory currently used in this pool
POPULATE_STATUS	VARCHAR2(26)	Status of the vector memory store —whether it is being populated, is done populating etc.

Column Name	Data Type	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values are: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root. • n: Where n is the applicable container ID for the rows containing data.

Example

```
select CON_ID, POOL, ALLOC_BYTES/1024/1024 as ALLOC_BYTES_MB,
USED_BYTES/1024/1024 as USED_BYTES_MB
from V$VECTOR_MEMORY_POOL order by 1,2;


    CON_ID POOL             ALLOC_BYTES_MB USED_BYTES_MB
---------- ---------------- -------------- -------------
         1 1MB POOL                    319             0
         1 64KB POOL                   144             0
         1 IM POOL METADATA             32            32
         2 1MB POOL                    320             0
         2 64KB POOL                   144             0
         2 IM POOL METADATA             16            16
         3 1MB POOL                    320             0
         3 64KB POOL                   144             0
         3 IM POOL METADATA             16            16
         4 1MB POOL                    320             0
         4 64KB POOL                   144             0
         4 IM POOL METADATA             16            16

12 rows selected.

SQL>
```

Vector Index and Hybrid Vector Index Views

These views allow you to query tables related to vector indexes and hybrid vector indexes.

- V$VECTOR_INDEX
  This fixed view provides diagnostic information about vector indexes and is available with Autonomous AI Database Serverless (ADB-S).
- V$VECTOR_INDEX_INST_MAP
  This fixed view provides cluster mapping and distribution metadata for distributed HNSW vector indexes.

- V$VECTOR_GRAPH_INDEX
  This fixed view provides diagnostic information about In-Memory Neighbor Graph vector indexes and is available with Autonomous AI Database Serverless (ADB-S).
- V$VECTOR_GRAPH_INDEX_CHKPT
  This fixed view provides diagnostic information about In-Memory Neighbor Graph vector index disk checkpoints and is available with Autonomous AI Database Serverless (ADB-S).
- V$VECTOR_GRAPH_INDEX_SNAPSHOT
  This fixed view provides diagnostic information about In-Memory Neighbor Graph vector index snapshots and is available with Autonomous AI Database Serverless (ADB-S).
- V$VECTOR_PARTITIONS_INDEX
  This fixed view provides diagnostic information about Inverted File Flat indexes and is available with Autonomous AI Database Serverless (ADB-S).
- <index name>$VECTORS
  This dictionary table provides information about the vector index part of a hybrid vector index, showing the contents of the `$VR` table with row ids, chunks, and embeddings.
- VECSYS.VECTOR$INDEX
  This catalog table contains detailed information about vector indexes. This table is not available when using Autonomous AI Database Serverless (ADB-S) or Dedicated (ADB-D). Oracle recommends using the fixed view, `V$VECTOR_INDEX`, instead.
- VECSYS.VECTOR$INDEX$CHECKPOINTS
  This catalog table provides detailed information about Hierarchical Navigable Small World (HNSW) full checkpoints at the database level. This table is not available when using Autonomous AI Database Serverless (ADB-S) or Dedicated (ADB-D). Oracle recommends using the fixed view, `V$VECTOR_GRAPH_INDEX_CHKPT`, instead.

V$VECTOR_INDEX

This fixed view provides diagnostic information about vector indexes and is available with Autonomous AI Database Serverless (ADB-S).

Column Name	Data Type	Description
OWNER	VARCHAR(129)	User name of the vector index owner
INDEX_NAME	VARCHAR(129)	Name of the vector index
PARTITION_NAME	VARCHAR(129)	Object partition name (set to NULL for non-partitioned objects)
INDEX_ORGANIZATION	VARCHAR(129)	The vector index organization. The value can be one of the following: • NEIGHBOR PARTITIONS • INMEMORY NEIGHBOR GRAPH • DISTRIBUTED INMEMORY NEIGHBOR GRAPH
INDEX_OBJN	NUMBER	Index object number
ALLOCATED_BYTES	NUMBER	Total amount of memory allocated to this vector index, in bytes. When the vector index is an IVF index, the value is 0.

Column Name	Data Type	Description
USED_BYTES	NUMBER	Total amount of memory currently used by the vector index, in bytes. When the vector index is an IVF index, the value is 0.
NUM_VECTORS	NUMBER	The number of vectors currently indexed in the main-memory storage of the index as of the latest snapshot
NUM_REPOP	NUMBER	Number of times this index has been repopulated
INDEX_USED_COUNT	NUMBER	Number of times this vector index has been used by queries
DISTANCE_TYPE	VARCHAR2(129)	Possible distance values: • EUCLIDEAN • EUCLIDEAN_SQUARED • COSINE • DOT • MANHATTAN • HAMMING
INDEX_DIMENSIONS	NUMBER	Number of dimensions of the indexed vector
INDEX_DIM_TYPE	VARCHAR2(129)	Type of the dimensions of the index vector. Possible values: • FLOAT16 • FLOAT32 • FLOAT64 • INT8
DEFAULT_ACCURACY	NUMBER	The accuracy to achieve when performing approximate search on this vector index if a target query accuracy is not provided
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include the following: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root. • n: Where n is the applicable container ID for the rows containing data.

V$VECTOR_INDEX_INST_MAP

This fixed view provides cluster mapping and distribution metadata for distributed HNSW vector indexes.

Column Name	Data Type	Description
OWNER	VARCHAR2(129)	User name of the distributed HNSW vector index owner
INDEX_NAME	VARCHAR2(129)	Name of the distributed HNSW vector index
INDEX_OBJN	NUMBER	Distributed HNSW vector index object number
VIRTUAL_INST_LIST	VARCHAR2(129)	Virtual instance list. It's a comma-separated list of virtual instance IDs associated with the distributed HNSW vector index.
INSTANCE_LIST	VARCHAR2(129)	Physical instance list. It's a comma-separated list of physical cluster instance IDs associated with the distributed HNSW vector index.
CLUSTER_INCARNATION	NUMBER	Cluster incarnation number
CON_ID	NUMBER	Container ID

V$VECTOR_GRAPH_INDEX

This fixed view provides diagnostic information about In-Memory Neighbor Graph vector indexes and is available with Autonomous AI Database Serverless (ADB-S).

Column Name	Data Type	Description
OWNER	VARCHAR2(129)	User name of the vector graph index owner
INDEX_NAME	VARCHAR2(129)	Name of the vector graph index
PARTITION_NAME	VARCHAR2(129)	Object partition name (set to NULL for non-partitioned objects)
INDEX_OBJN	NUMBER	Index object number
ANCHOR_ADDRESS	RAW(8)	The address of the anchor structure of the in-memory neighbor graph index in the vector pool
INDEX_GRAPH_TYPE	VARCHAR2(129)	Type of the constructed graph of this index. Possible values: • HNSW: Hierarchical Navigable Small Worlds. Currently only this graph is supported. • DISTRIBUTED_HNSW [AUTO/ROWID_RANGE/ PARTITION/ SUBPARTITION]

Column Name	Data Type	Description
NUM_LAYERS	NUMBER	Number of layers in the constructed graph
NUM_VECTORS	NUMBER	Number of indexed vectors in the vector graph index
SPARSE_LAYER_VECTORS	NUMBER	The total number of vectors in the sparse layers. Note that a sparse layer is defined as any layer above the bottom most layer.
NUM_NEIGHBORS	NUMBER	Maximum number of neighbors a vector can have in the sparse layers
EF_CONSTRUCTION	NUMBER	Maximum number of closest vector candidates considered at each step of the search during insertion
TOTAL_EDGES	NUMBER	Number of total edges in the constructed graph as of the latest snapshot
REF_COUNT	NUMBER	Number used to track readers of the inmemory neighbor graph index. The inmemory neighbor graph index can only be dropped once the `refcount` is 0.
QUERY_DIST_COUNT	NUMBER	Number of distance computations done as part of queries that used the inmemory neighbor graph index
CREATION_DIST_COUNT	NUMBER	Number of distance computations done as part of the inmemory neighbor graph index creation
PRUNED_NEIGHBORS	NUMBER	Number of neighbors pruned out during the neighbor selection phase of the inmemory neighbor graph index creation
NUM_SNAPSHOTS	NUMBER	Number of snapshots currently tracked for the inmemory neighbor graph index
MAX_SNAPSHOT	NUMBER	The total number of snapshots created so far for the inmemory graph index
ALLOCATED_BYTES	NUMBER	Total amount of memory allocated to this vector graph index, in bytes
USED_BYTES	NUMBER	Total amount of memory currently used by the vector graph index, in bytes
COVERING_COLS[1]	VARCHAR2(3999)	Comma-separated list of included columns for the index.

Column Name	Data Type	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include the following: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • n: Where n is the applicable container ID for the rows containing data

1 This column is available starting with Oracle AI Database 26ai, Release Update 23.26.1.

V$VECTOR_GRAPH_INDEX_CHKPT

This fixed view provides diagnostic information about In-Memory Neighbor Graph vector index disk checkpoints and is available with Autonomous AI Database Serverless (ADB-S).

Column Name	Data Type	Description
INDEX_OBJN	NUMBER	Index object number
CHECKPOINT_ID	NUMBER	Unique identifier for the checkpoint
CHECKPOINT_SCN	NUMBER	The SCN at which the checkpoint was created
CHECKPOINT_TYPE	NUMBER	Indicates whether the checkpoint is a full checkpoint or incremental checkpoint. The value of this column is always 1, which indicates a full checkpoint.
VERSION_NUMBER	NUMBER	Version number for the checkpoint
TABLESPACE_NUMBER	NUMBER	TSN for the tablespace storing the checkpoint
NUM_VECTORS	NUMBER	Number of vectors currently present in the checkpoint
NUM_DELETES	NUMBER	Number of vertices marked as deleted in the checkpoint graph
MAX_VERTEX_ID	NUMBER	Maximum vertex ID for the checkpoint
NUM_LAYERS	NUMBER	Number of graph layers in the checkpoint

Column Name	Data Type	Description
NUM_COMPONENTS	NUMBER	Number of subcomponents in the multi-layered graph checkpoint. This value can be used to track the multiple BLOBs into which the graph checkpoint is divided.
CHECKPOINT_SIZE	NUMBER	Total size of the checkpoint, including serialized BLOBs for each graph layer (in bytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root. • *n*: Where *n* is the applicable container ID for the rows containing data

V$VECTOR_GRAPH_INDEX_SNAPSHOT

This fixed view provides diagnostic information about In-Memory Neighbor Graph vector index snapshots and is available with Autonomous AI Database Serverless (ADB-S).

Column Name	Data Type	Description
OWNER	VARCHAR2(129)	User name of the vector graph index owner
INDEX_NAME	VARCHAR(129)	Index name
INDEX_OBJN	NUMBER	Index object number
SNAPSHOT_ID	NUMBER	Unique identifier for the snapshot
FLAGS	NUMBER	State of the snapshot, used for debugging
BUILD_SCN	NUMBER	The SCN at which the snapshot was built
VALID_SCN	NUMBER	The SCN at which the snapshot is open for queries
REFCOUNT	NUMBER	Number of active readers using the snapshot
NUM_LAYERS	NUMBER	Number of graph layers for this snapshot
NUM_VECTORS	NUMBER	Number of vectors currently present in the graph for the snapshot

Column Name	Data Type	Description
NUM_DELETES	NUMBER	Number of vertices marked as deleted in the graph for the snapshot
START_VERTEX_ID	NUMBER	Starting vertex ID for the snapshot
MAX_VERTEX_ID	NUMBER	Maximum vertex ID for the snapshot
AVG_QUERY_DIST_COUNT	NUMBER	Average number of distance computations performed as part of queries that used the snapshot
CREATION_DIST_COUNT	NUMBER	Number of distance computations performed as part of the In-Memory Neighbor Graph index creation
SNAPSHOT_SIZE	NUMBER	The additional number of bytes that were used in the vector pool to host this snapshot.
ALLOC_BYTES	NUMBER	The additional amount of memory, spanning all previous snapshots, that has been allocated from the vector pool, in bytes.
USED_BYTES	NUMBER	Total amount of memory, spanning all previous snapshots, that is currently being used from the vector pool, in bytes
MIN_DELETE_SCN	NUMBER	Minimum SCN at which a delete happened for the snapshot
MIN_INSERT_SCN	NUMBER	Minimum SCN at which an insert happened the snapshot
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root. • *n*: Where *n* is the applicable container ID for the rows containing data

V$VECTOR_PARTITIONS_INDEX

This fixed view provides diagnostic information about Inverted File Flat indexes and is available with Autonomous AI Database Serverless (ADB-S).

Column Name	Data Type	Description
OWNER	VARCHAR2(129)	User name of the vector partition index owner
INDEX_NAME	VARCHAR2(129)	Name of the vector partition index
PARTITION_NAME	VARCHAR2(129)	Object partition name (set to NULL for non-partitioned objects)
INDEX_OBJN	NUMBER	Index object number
CENTROIDS_ANCHOR_ADDRESS	RAW(8)	The address of the anchor structure hosting the centroids for the in-memory neighbor partitions index in the vector pool. CENTROIDS_ANCHOR_ADDRESS is only enabled if in-memory centroids are enabled on the index. Set to 0 if in-memory centroids are unavailable for the index.
INDEX_PARTITIONS_TYPE	VARCHAR2(129)	The partition type of this neighbor partitions vector index. Possible values include: • IVF: Inverted File Index
NUM_NEIGHBOR_PARTITIONS	NUMBER	Number of neighbor partitions created for the neighbor partitions vector index
NUM_VECTORS	NUMBER	Number of indexed vectors in this vector index
MIN_PARTITION_VECTORS_COUNT	NUMBER	Number of vectors in the smallest neighbor partition
MAX_PARTITION_VECTORS_COUNT	NUMBER	Number of vectors in the largest neighbor partition
INDEX_USED_COUNT	NUMBER	Number of times this vector partition index has been used by queries
AVG_PARTITIONS_SCANNED	NUMBER	Average number of neighbor partitions scanned in order to answer queries. This is an average over the NEIGHBOR PROBES query parameter supported for an Inverted File neighbor partitions index

Column Name	Data Type	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include the following: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • n: Where n is the applicable container ID for the rows containing data

<index name>$VECTORS

This dictionary table provides information about the vector index part of a hybrid vector index, showing the contents of the $VR table with row ids, chunks, and embeddings.

> **Note**
>
> The *<index name>*$VECTORS view resides in a user's schema. Therefore, if a hybrid vector index is named IDX, then the view name is IDX$VECTORS.
> *<index name>*$VECTORS view is not supported for Local Hybrid Vector Indexes.

Column Name	Data Type	Description
DOC_ROWID	ROWID	Document table's row ID, excluding lazy deletes
DOC_CHUNK_ID	NUMBER	ID for each chunk text
DOC_CHUNK_COUNT	NUMBER	Number of chunks into which each document is split
DOC_CHUNK_OFFSET	NUMBER	Original position of each chunk in the source document, relative to the start of document (which has a position of 1)
DOC_CHUNK_LENGTH	NUMBER	Character length of each chunk text
DOC_CHUNK_TEXT	VARCHAR2(4000)	Human-readable content in each chunk
DOC_CHUNK_EMBEDDING	VECTOR(*, *)	Generated vector embedding for each chunk

Related Topics

- Manage Hybrid Vector Indexes
  Learn how to manage a hybrid vector index, which is a single index for searching by *similarity* and *keywords*, to enhance the accuracy of your search results.

VECSYS.VECTOR$INDEX

This catalog table contains detailed information about vector indexes. This table is not available when using Autonomous AI Database Serverless (ADB-S) or Dedicated (ADB-D). Oracle recommends using the fixed view, `V$VECTOR_INDEX`, instead.

Column Name	Data Type	Description
`IDX_OBJN`	`NUMBER`	Object number of the vector index
`IDX_OBJD`	`NUMBER`	ID of the vector index object. This ID can be used to rebuild the vector index.
`IDX_OWNER#`	`NUMBER`	Owner ID of the vector index. Refer user$ entry
`IDX_NAME`	`VARCHAR2(128)`	Name of the vector index.
`IDX_BASE_TABLE_OBJN`	`NUMBER`	Base table object number
`IDX_PARAMS`	`JSON`	Vector index creation parameters such as vector column indexed, index distance, vector dimension datatype, number of dimensions, efConstruction, and number of neighbors for in-memory neighbor graph HNSW index or the number of centroids for Inverted File Flat vector indexes.
`IDX_AUXILIARY_TABLES`	`JSON`	Names and object IDs of auxiliary tables used to support rowid-to-vertexid conversion information or names of a centroid table and its associated partitions table for Inverted File Flat vector indexes.

Example

```
SQL> SELECT JSON_SERIALIZE(IDX_PARAMS returning varchar2 PRETTY)
  2* FROM VECSYS.VECTOR$INDEX where IDX_NAME = 'DOCS_HNSW_IDX';

JSON_SERIALIZE(IDX_PARAMSRETURNINGVARCHAR2PRETTY)
{
  "type" : "HNSW",
  "num_neighbors" : 32,
  "efConstruction" : 300,
  "upcast_dtype" : 1,
  "distance" : "COSINE",
  "accuracy" : 95,
  "vector_type" : "FLOAT32",
  "vector_dimension" : 384,
  "degree_of_parallelism" : 1,
  "indexed_col" : "EMBED_VECTOR"
}
SQL>
SQL> select * from vecsys.vector$index;

IDX_OBJN IDX_OBJD  IDX_OWNER# IDX_NAME       IDX_BASE_TABLE_OBJN
```

```
IDX_PARAMS
IDX_AUXILIARY_TABLES
                          IDX_SPARE1 IDX_SPARE2
-------- --------  ---------- ------------- -------------------
--------------------------
--------------------
                          ----------  ---------
   74051     143              DOCS_HNSW_IDX             73497
{"type":"HNSW",                        {"rowid_vid_map_objn":74052,

"num_neighbors":32,
"shared_journal_transaction_commits_objn":74054,

"efConstruction":300,
"shared_journal_change_log_objn":74057,

"upcast_dtype":1,
"rowid_vid_map_name":"VECTOR$DOCS_HNSW_IDX$HNSW_ROWID_VID_MAP",

"distance":"COSINE",
"shared_journal_transaction_commits_name":"VECTOR$DOCS_HNSW_IDX$HNSW_SHARED_JO
URNAL_TRANSACTION_COMMITS",

"accuracy":95,
"shared_journal_change_log_name":"VECTOR$DOCS_HNSW_IDX$HNSW_SHARED_JOURNAL_CHA
NGE_LOG"}

"vector_type":"FLOAT32",

"vector_dimension":384,

"degree_of_parallelism":1,

"indexed_col":"EMBED_VECTOR"}

   74072      143            GALAXIES_HNSW_IDX          74069
{"type":"HNSW",                         {"rowid_vid_map_objn":74073,

"num_neighbors":32,
"shared_journal_transaction_commits_objn":74075,

"efConstruction":300,
"shared_journal_change_log_objn":74078,

"upcast_dtype":0,
"rowid_vid_map_name":"VECTOR$GALAXIES_HNSW_IDX$HNSW_ROWID_VID_MAP",

"distance":"COSINE",
"shared_journal_transaction_commits_name":"VECTOR$GALAXIES_HNSW_IDX$HNSW_SHARE
D_JOURNAL_TRANSACTION_COMMITS",

"accuracy":95,
"shared_journal_change_log_name":"VECTOR$GALAXIES_HNSW_IDX$HNSW_SHARED_JOURNAL
_CHANGE_LOG"}

"vector_type":"INT8",
```

```
"vector_dimension":5,

"degree_of_parallelism":1,

"indexed_col":"EMBEDDING"}


SQL>
```

VECSYS.VECTOR$INDEX$CHECKPOINTS

This catalog table provides detailed information about Hierarchical Navigable Small World (HNSW) full checkpoints at the database level. This table is not available when using Autonomous AI Database Serverless (ADB-S) or Dedicated (ADB-D). Oracle recommends using the fixed view, `V$VECTOR_GRAPH_INDEX_CHKPT`, instead.

Column Name	Data Type	Description
`INDEX_OBJN`	`NUMBER`	The index object number to uniquely identify the HNSW index.
`INDEX_OWNER_ID`	`NUMBER`	The owner id for the vector index.
`CHECKPOINT_ID`	`NUMBER`	A monotonically increasing ID tracking various checkpoints for this HNSW index.
`CHECKPOINT_SCN`	`NUMBER`	The SCN as of which the HNSW checkpoint is taken.
`CHECKPOINT_TYPE`	`NUMBER`	Full Checkpoint is the only checkpoint type supported.
`VERSION_NUMBER`	`NUMBER`	The checkpoint format may change in between releases. Thus, you can use a version number to decide whether to reload the HNSW index using a particular checkpoint.
`TABLESPACE_NUMBER`	`NUMBER`	The tablespace number. You can view a tablespace number to decide whether to store full checkpoints of your HNSW graphs in the same tablespace or in a different one.

Related Topics

- HNSW RAC Duplication
  Learn how Hierarchical Navigable Small World (HNSW) indexes are populated during index creation, index repopulation, or instance startup in an Oracle Real Application Clusters (Oracle RAC) or a non-RAC environment.

Oracle AI Vector Search Statistics

These are a set of statistics related to Oracle AI Vector Search.

- Oracle AI Vector Search Dictionary Statistics
  A set of dictionary statistics related to Oracle AI Vector Search.
- Oracle Machine Learning Static Dictionary Views
  Lists data dictionary views related to Oracle Machine Learning models.

Oracle AI Vector Search Dictionary Statistics

A set of dictionary statistics related to Oracle AI Vector Search.

- **Vector simd dist single calls**: The number of vector distance function calls invoked by the user, where both the inputs are a single vector.
- **Vector simd dist point calls**: The number of vector distance function calls invoked by the user, where one input is a single vector, and the other input is an array of vectors.
- **Vector simd dist array calls**: The number of vector distance function calls invoked by the user, where both the inputs are an array of vectors.
- **Vector simd dist flex calls**: The number of vector distance function calls invoked by the user, where at least one input is with FLEX vector data type (no dimension or storage data type specified).
- **Vector simd dist total rows:** The number of rows processed by the vector distance function.
- **Vector simd dist flex rows**: The number of rows processed by the vector distance function with FLEX vector data type.
- **Vector simd topK single calls**: The number of `topK` distance function calls invoked by the user, where both the inputs are a single vector.
- **Vector simd topK point calls**: The number of `topK` distance function calls invoked by the user, where one input is a single vector and the other input is an array of vectors.
- **Vector simd topK array calls**: The number of `topK` distance function calls invoked by the user, where both the inputs are an array of vectors.
- **Vector simd topK flex calls**: The number of `topK` distance function calls invoked by the user, where at least one input is with FLEX vector data type (no dimension or storage data type specified).
- **Vector simd topK total rows**: The number of rows processed by the `topK` distance function.
- **Vector simd topK flex rows**: The number of rows processed by the `topK` distance function with FLEX vector data type.
- **Vector simd topK selected total rows**: The number of distance results returned by the `topK` vector distance function.

> **Note**
>
> This is typically `sum(K)` for all `topK` queries.

- **Vector simd construction num of total calls**: The number of vector construction function calls invoked by the user.
- **Vector simd construction total result rows**: The number of vector rows returned by the vector construction function.

- **Vector simd construction num of ASCII calls**: The number of vector construction function calls invoked by the user, where the input encoding is ASCII.
- **Vector simd construction num of flex calls**: The number of vector construction function calls invoked by the user, where the output vector is of FLEX vector data type.
- **Vector simd construction result rows for flex**: The number of vector rows returned by the vector construction function, where the output vector is of FLEX vector data type.
- **Vector simd vector conversion num of total calls**: The number of vector conversion function calls invoked by the user.
- **VECTOR NEIGHBOR GRAPH HNSW build HT Element Allocated:** The number of hash table elements allocated for vector indexes having organization `Inmemory Neighbor Graph` and type `HNSW` that were created by a user.
- **VECTOR NEIGHBOR GRAPH HNSW build HT Element Freed:** The number of hash table elements freed for vector indexes having organization `Inmemory Neighbor Graph` and type `HNSW` that were created by a user.
- **VECTOR NEIGHBOR GRAPH HNSW reload attempted:** The number of reload operations attempted in the background for vector indexes having organization `Inmemory Neighbor Graph` and type `HNSW`.
- **VECTOR NEIGHBOR GRAPH HNSW reload successful:** The number of reload operations completed in the background for vector indexes having organization `Inmemory Neighbor Graph` and type `HNSW`.
- **VECTOR NEIGHBOR GRAPH HNSW build computed layers:** The total number of layers created in an HNSW index.
- **VECTOR NEIGHBOR GRAPH HNSW build indexed vectors:** The total number of vector indexes in the HNSW index.
- **VECTOR NEIGHBOR GRAPH HNSW build computed distances:** The total number of distance computations executed during the HNSW index build phase.
- **VECTOR NEIGHBOR GRAPH HNSW build sparse layers computed distances:** The total number of distance computations executed during the HNSW index build phase, in all the layers, except the bottom one.
- **VECTOR NEIGHBOR GRAPH HNSW build dense layer computed distances:** The total number of distance computations executed during the HNSW index phase in the bottom layer.
- **VECTOR NEIGHBOR GRAPH HNSW build prune operation computed distances:** The total number of distance computations that were executed during the pruning operations required to find the closest neighbors for a vector in the HNSW index build phase.
- **VECTOR NEIGHBOR GRAPH HNSW build pruned neighbor lists:** The total number of neighbors pruned during the HNSW index build phase.
- **VECTOR NEIGHBOR GRAPH HNSW build pruned neighbors:** The total number of neighbors pruned during the HNSW index build phase.
- **VECTOR NEIGHBOR GRAPH HNSW build created edges:** The total number of edges created in an HNSW index.
- **VECTOR NEIGHBOR GRAPH HNSW search executed approximate:** The total number of query searches executed using approximate search in an HNSW index.
- **VECTOR NEIGHBOR GRAPH HNSW search executed exhaustive:** The total number of query searches executed using exact search in an HNSW index.

- **VECTOR NEIGHBOR GRAPH HNSW search computed distances dense layer:** The total number of distance computations executed in the bottom layer of an HNSW index during query searches.
- **VECTOR NEIGHBOR GRAPH HNSW search computed distances sparse layer:** The total number of distance computations executed in all the layers except the bottom layer of an HNSW index during query searches.
- **VECTOR NEIGHBOR PARTITIONS IVF build HT Element Allocated:** The number of hash table elements allocated for vector indexes having organization `Neighbor Partitions` and type `IVF` that were created by a user.
- **VECTOR NEIGHBOR PARTITIONS IVF build HT Element Freed:** The number of hash table elements freed for vector indexes having organization `Neighbor Partitions` and type `IVF` that were created by a user.
- **VECTOR NEIGHBOR PARTITIONS IVF background Population Started:** The number of in-memory centroids background population operations started for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF background Population Succeeded:** The number of in-memory centroids background population operations completed for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF background Cleanup Started:** The number of in-memory centroids background cleanup operations started for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF XIC Population Started:** The number of in-memory centroids cross-instance population operations started for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF XIC Population Succeeded:** The number of in-memory centroids cross-instance population operations completed for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF XIC Cleanup Started:** The number of in-memory centroids cross-instance cleanup operations started for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF XIC Cleanup Succeeded:** The number of in-memory centroids cross-instance cleanup operations completed for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF im centroids in scan:** The number of times that in-memory centroids are used in scan for vector indexes having organization `Neighbor Partitions` and type `IVF`.
- **VECTOR NEIGHBOR PARTITIONS IVF im centroids in dmls:** The number of times that in-memory centroids are used in the DML that happens on the base table with vector indexes having organization `Neighbor Partitions` and type `IVF`.

Oracle Machine Learning Static Dictionary Views

Lists data dictionary views related to Oracle Machine Learning models.

Query the following views to learn more about the machine learning models:

- ALL_MINING_MODEL_ATTRIBUTES
- ALL_MINING_MODELS

Oracle AI Vector Search Parameters

This is a set of parameters related to Oracle AI Vector Search.

- **VECTOR_MEMORY_SIZE**

  Syntax: `VECTOR_MEMORY_SIZE = integer[K | M | G]`

  The initialization parameter `VECTOR_MEMORY_SIZE` specifies either the current size of the Vector Pool (at CDB level) or the maximum Vector Pool usage allowed by a PDB (at PDB level).

  For more information about this parameter, see Size the Vector Pool.

- **VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD**

  Syntax: `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD = [RESTART | OFF]` (default `RESTART`)

  The initialization parameter `VECTOR_INDEX_NEIGHBOR_GRAPH_RELOAD` is used to manage automatic recreation of HNSW indexes. You can enable or disable both the HNSW duplication and reload mechanisms in Oracle RAC and non-RAC environments.

  For more information about this parameter and the duplication and reload mechanisms, see HNSW RAC Duplication.

- **VECTOR_QUERY_CAPTURE**

  Syntax: `VECTOR_QUERY_CAPTURE = [ON | OFF]` (default `ON`)

  The initialization parameter `VECTOR_QUERY_CAPTURE` is used to enable or disable capture of query vectors.

  For more information about this parameter and about capturing query vectors, see Index Accuracy Report.

- **COMPATIBLE**

  Syntax: `COMPATIBLE = release_number`

  The initialization parameter enables you to use a new release of Oracle while ensuring the ability to downgrade the database to an earlier release. To use the `VECTOR` data type and its related features, `COMPATIBLE` must be set to 23.4.0 or higher.

  Certain vector features require the `COMPATIBLE` parameter to be manually updated to a higher value in order to be available for use. To find requirements for a particular feature, see the applicable Release Update page at Oracle AI Database 26ai Release Updates.

  For more information about this parameter, see *Oracle AI Database Reference*.

12

Vector Search PL/SQL Packages

The DBMS_VECTOR, DBMS_VECTOR_CHAIN, and DBMS_HYBRID_VECTOR PL/SQL APIs are available to support Oracle AI Vector Search capabilities.

- DBMS_VECTOR
  The DBMS_VECTOR package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.
- DBMS_VECTOR_CHAIN
  The DBMS_VECTOR_CHAIN package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.
- DBMS_HYBRID_VECTOR
  The DBMS_HYBRID_VECTOR package contains a JSON-based query API SEARCH, which lets you query against hybrid vector indexes.
- DBMS_VECTOR_ADMIN
  The DBMS_VECTOR_ADMIN package provides administrative interfaces for managing database configuration settings, such as modifying system parameters and performing privileged operations.

DBMS_VECTOR

The DBMS_VECTOR package simplifies common operations with Oracle AI Vector Search, such as extracting chunks or embeddings from user data, generating text for a given prompt or an image, creating a vector index, or reporting on index accuracy.

This table lists the DBMS_VECTOR subprograms and briefly describes them.

Table 12-1 DBMS_VECTOR Package Subprograms

Subprogram	Description
ONNX Model Related Procedures:	
These procedures enable you to load an ONNX model into Oracle AI Database and drop the ONNX model.	
LOAD_ONNX_MODEL	Loads an ONNX model into the database
LOAD_ONNX_MODEL_CLOUD	Loads an ONNX model from object storage into the database
DROP_ONNX_MODEL Procedure	Drops the ONNX model
INMEMORY_ONNX_MODEL	Controls the in-memory sharing of initializers for an ONNX model with external initializers within the database
Chainable Utility (UTL) Functions:	
These functions are a set of modular and flexible functions within vector utility PL/SQL packages. You can chain these together to automate end-to-end data transformation and similarity search operations.	
UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS	Converts text or an image to one or more vector embeddings

Table 12-1 (Cont.) DBMS_VECTOR Package Subprograms

Subprogram	Description
UTL_TO_GENERATE_TEXT	Generates text for a prompt (input string) or an image
Credential Helper Procedures: These procedures enable you to securely manage authentication credentials in the database. You require these credentials to enable access to third-party service providers for making REST calls.	
CREATE_CREDENTIAL	Creates a credential name
DROP_CREDENTIAL	Drops an existing credential name
Data Access Functions: These functions enable you to retrieve data, create index, and perform simple similarity search operations.	
CREATE_INDEX	Creates a vector index
REBUILD_INDEX	Rebuilds a vector index
GET_INDEX_STATUS	Describes the status of a vector index creation
ENABLE_CHECKPOINT	Enables the Checkpoint feature for a vector index user and index name
DISABLE_CHECKPOINT	Disables the Checkpoint feature for a vector index user and index name
INDEX_VECTOR_MEMORY_ADVISOR	Determines the vector memory size that is needed for a vector index
QUERY	Performs a similarity search query
UTL_TO_RERANK	Reorders search results for a more relevant output
Accuracy Reporting Function: These functions enable you to determine the accuracy of existing search indexes and to capture accuracy values achieved by approximate searches performed by past workloads.	
INDEX_ACCURACY_QUERY	Verifies the accuracy of a vector index
INDEX_ACCURACY_REPORT	Captures accuracy values achieved by approximate searches

Note

`DBMS_VECTOR` is a lightweight package that does not support text processing or summarization operations. Therefore, the `UTL_TO_TEXT` and `UTL_TO_SUMMARY` chainable utility functions and all the chunker helper procedures are available only in the advanced `DBMS_VECTOR_CHAIN` package.

- CREATE_CREDENTIAL
  Use the `DBMS_VECTOR.CREATE_CREDENTIAL` credential helper procedure to create a credential name for storing user authentication details in Oracle AI Database.
- CREATE_INDEX
  Use the `DBMS_VECTOR.CREATE_INDEX` procedure to create a vector index.
- DISABLE_CHECKPOINT
  Use the `DISABLE_CHECKPOINT` procedure to disable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name. This operation purges all older checkpoints for the HNSW index. It also disables the creation of future checkpoints as part of the HNSW graph refresh.

- DROP_CREDENTIAL
  Use the `DBMS_VECTOR.DROP_CREDENTIAL` credential helper procedure to drop an existing credential name from the data dictionary.
- DROP_ONNX_MODEL Procedure
  This procedure deletes the specified ONNX model.
- ENABLE_CHECKPOINT
  Use the `ENABLE_CHECKPOINT` procedure to enable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name.
- GET_INDEX_STATUS
  Use the `GET_INDEX_STATUS` procedure to query the status of a vector index creation.
- INDEX_ACCURACY_QUERY
  Use the `DBMS_VECTOR.INDEX_ACCURACY_QUERY` function to verify the accuracy of a vector index for a given query vector, top-K, and target accuracy.
- INDEX_ACCURACY_REPORT
  Use the `DBMS_VECTOR.INDEX_ACCURACY_REPORT` function to capture from your past workloads, accuracy values achieved by approximate searches using a particular vector index for a certain period of time.
- INDEX_VECTOR_MEMORY_ADVISOR
  Use the `INDEX_VECTOR_MEMORY_ADVISOR` procedure to determine the vector memory size needed for a particular vector index. This helps you evaluate the number of indexes that can fit for each simulated vector memory size.
- INMEMORY_ONNX_MODEL
  This procedure enables you to control the in-memory sharing of initializers for an ONNX model with external initializers within the database. When you enable in-memory sharing, the database loads the model's initializers once into shared memory, so that all sessions using the model benefit from reduced aggregate memory consumption. You must load or import the model before calling this procedure. Attempting to enable in-memory sharing on a model that has not been imported raises an error.
- LOAD_ONNX_MODEL
  This procedure enables you to load an ONNX-format embedding model or a rerank model into your database.
- LOAD_ONNX_MODEL_CLOUD
  This procedure enables you to load an ONNX-format embedding model from Oracle Object Storage into your database.
- QUERY
  Use the `DBMS_VECTOR.QUERY` function to perform a similarity search operation which returns the top-k results as a JSON array.
- REBUILD_INDEX
  Use the `DBMS_VECTOR.REBUILD_INDEX` procedure to rebuild a vector index.
- UTL_TO_RERANK
  Use the `DBMS_VECTOR.UTL_TO_RERANK` function to reassess and reorder an initial set of results to retrieve more relevant search output.
- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the `DBMS_VECTOR.UTL_TO_EMBEDDING` and `DBMS_VECTOR.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.
- UTL_TO_GENERATE_TEXT
  Use the `DBMS_VECTOR.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.

CREATE_CREDENTIAL

Use the `DBMS_VECTOR.CREATE_CREDENTIAL` credential helper procedure to create a credential name for storing user authentication details in Oracle AI Database.

Purpose

To securely manage authentication credentials in the database. You require these credentials to enable access during REST API calls to your chosen third-party service provider, such as Cohere, Google AI, Hugging Face, Oracle Cloud Infrastructure (OCI) Generative AI, OpenAI, or Vertex AI.

A credential name holds authentication parameters, such as user name, password, access token, private key, or fingerprint.

Note that if you are using Oracle AI Database as the service provider, then you do not need to create a credential.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

```
DBMS_VECTOR.CREATE_CREDENTIAL (
    CREDENTIAL_NAME     IN VARCHAR2,
    PARAMS              IN JSON DEFAULT NULL
);
```

CREDENTIAL_NAME

Specify a name of the credential that you want to create for holding authentication parameters.

PARAMS

Specify authentication parameters in JSON format, based on your chosen service provider.

Generative AI requires the following authentication parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
```

```
"fingerprint"     : "<fingerprint>"
}
```

Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

```
{ "access_token": "<access token>" }
```

Table 12-2 Parameter Details

Parameter	Description
`user_ocid`	Oracle Cloud Identifier (OCID) of the user, as listed on the User Details page in the OCI console.
`tenancy_ocid`	OCID of your tenancy, as listed on the Tenancy Details page in the OCI console.
`compartment_ocid`	OCID of your compartment, as listed on the Compartments information page in the OCI console.

`private_key`	OCI private key.
	Note: The generated private key may appear as:
	`-----BEGIN RSA PRIVATE KEY-----`
	`<private key string>`
	`-----END RSA PRIVATE KEY-----`
	You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

`fingerprint`	Fingerprint of the OCI profile key, as listed on the User Details page under API Keys in the OCI console.
`access_token`	Access token obtained from your third-party service provider.

Required Privilege

You need the `CREATE CREDENTIAL` privilege to call this API.

Examples

- For Generative AI:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();

jo.put('user_ocid','ocid1.user.oc1..aabbalbbaa1112233aabbaabb1111222aa1111b
b');

jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaaaalbbbb1112233aaaabbaa1111222a
aa111a');

jo.put('compartment_ocid','ocid1.compartment.oc1..ababalabab1112233abababab
1111222aba11ab');
  jo.put('private_key','AAAaaaBBB11112222333...AAA111AAABBB222aaa1a/+');
  jo.put('fingerprint','01:1a:a1:aa:12:a1:12:1a:ab:12:01:ab:a1:12:ab:1a');
```

```
  dbms_vector.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

- For Cohere:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', 'A1Aa0abA1AB1a1Abc123ab1A123ab123AbcA12a');
  dbms_vector.create_credential(
    credential_name   => 'COHERE_CRED',
    params            => json(jo.to_string));
end;
/
```

End-to-end examples:

To run end-to-end example scenarios using this procedure, see Use LLM-Powered APIs to Generate Summary and Text.

CREATE_INDEX

Use the `DBMS_VECTOR.CREATE_INDEX` procedure to create a vector index.

Purpose

To create a vector index such as Hierarchical Navigable Small World (HNSW) vector index or Inverted File Flat (IVF) vector index.

Syntax

```
DBMS_VECTOR.CREATE_INDEX (
    idx_name                   IN VARCHAR2,
    table_name                 IN VARCHAR2,
    idx_vector_col             IN VARCHAR2,
    idx_include_cols           IN VARCHAR2 DEFAULT NULL,
    idx_partitioning_scheme    IN VARCHAR2 DEFAULT 'LOCAL',
    idx_organization           IN VARCHAR2,
    idx_distance_metric        IN VARCHAR2 DEFAULT 'COSINE',
    idx_accuracy               IN NUMBER DEFAULT 90,
    idx_parameters             IN CLOB,
    idx_parallel_creation      IN NUMBER DEFAULT 1,
    idx_quantization_type      IN VARCHAR2 DEFAULT 'NONE',
    idx_compression_ratio      IN NUMBER DEFAULT NULL,
    idx_distribute_parameters  IN CLOB DEFAULT NULL,
    idx_duplicate_parameters   IN CLOB DEFAULT NULL,
    idx_online_build           IN BOOLEAN DEFAULT FALSE
);
```

Parameters

Parameter	Description
`idx_name`	Name of the index to create.
`table_name`	Table on which to create the index.
`idx_vector_col`	Vector column on which to create the index.
`idx_include_cols`	A comma-separated list of column names to be covered by the index.
`idx_partitioning_scheme`	Partitioning scheme for IVF indexes: • `GLOBAL` • `LOCAL` All IVF indexes are partitioned by centroid. Additionally, IVF indexes support both global and local indexes on partitioned tables. You can choose to create a local IVF index, which provides a one-to-one relationship between the base table partitions or subpartitions and the index partitions. If the value passed for this parameter is `NULL`, then `GLOBAL` is used as the partitioning scheme. For detailed information on these partitioning schemes, see Partition Maintenance Operations and Vector indexes.
`idx_organization`	Index organization: • `NEIGHBOR PARTITIONS` • `INMEMORY NEIGHBOR GRAPH` For detailed information on these organization types, see Manage the Different Categories of Vector Indexes.
`idx_distance_metric`	Distance metric or mathematical function used to compute the distance between vectors: • `COSINE` (default) • `MANHATTAN` • `HAMMING` • `JACCARD` • `DOT` • `EUCLIDEAN` • `L2_SQUARED` • `EUCLIDEAN_SQUARED` For detailed information on each of these metrics, see Vector Distance Functions and Operators.

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>idx_accuracy</code></td><td>Target accuracy at which the approximate search should be performed when running an approximate search query.

As explained in Understand Approximate Similarity Search Using Vector Indexes, you can specify non-default target accuracy values either by specifying a percentage value or by specifying internal parameters values, depending on the index type you are using.

• For an HNSW approximate search:
In the case of an HNSW approximate search, you can specify a target accuracy percentage value to influence the number of candidates considered to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose a similar result as an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the <code>EFSEARCH</code> parameter to impose a certain maximum number of candidates to be considered while probing the index. The higher that number, the higher the accuracy.
For detailed information, see Understand Hierarchical Navigable Small World Indexes.

• For an IVF approximate search:
In the case of an IVF approximate search, you can specify a target accuracy percentage value to influence the number of partitions used to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the <code>NEIGHBOR PARTITION PROBES</code> parameter to impose a certain maximum number of partitions to be probed by the search. The higher that number, the higher the accuracy.
For detailed information, see Understand Inverted File Flat Vector Indexes.</td></tr>
</table>

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>idx_parameters</code></td><td>
Type of vector index and associated parameters.
Specify the indexing parameters in JSON format:

For HNSW indexes:

<code>type</code>: Type of vector index to create, that is, <code>HNSW</code>
<code>neighbors</code>: Maximum number of connections permitted per vector in the HNSW graph
<code>efConstruction</code>: Maximum number of closest vector candidates considered at each step of the search during insertion
<code>offload_credential_name</code>: The identifier of the credential that should be used when authenticating with the offload service of the Private AI Services Container.
<code>offload_url</code>: The Private AI Services Container service endpoint for index creation.
<code>rescore factor</code>: The default rescore factor to be used for quantized indexes.
<code>algorithm</code>: The quantization algorithm to be used for quantized indexes.

For example:
<pre>
{
    "type"   : "HNSW",
    "neighbors"   : 3,
    "efConstruction"   : 4,
    "offload_credential_name" : "privateai",
    "offload_url"   : "https://
<myprivateaiservicehostname>/v1/index",
    "rescore factor"   : 2,
    "algorithm"   :
"uniform_quantization"
}
</pre>
For detailed information on these parameters, see Hierarchical Navigable Small World Index Syntax and Parameters.
For information about the Private AI Services Container vector index service, see Oracle Private AI Services Container User's Guide.

For IVF indexes:

<code>type</code>: Type of vector index to create, that is, <code>IVF</code>
<code>partitions</code>: Neighbor partition or cluster in which you want to divide your vector space
<code>neighbor partition grouping</code>: Enables, in addition to partitioning, further data grouping in the storage layer for centroids in the IVF index. This can result in query performance benefits in serial, parallel, and multi-user (concurrent) queries due to additional pruning during the scan. This storage optimization is only supported for global IVF indexes. The default value is <code>OFF</code>.

For example:
<pre>
{
    "type"   : "IVF",
    "partitions"   : 5,
</pre>

</td></tr>
</table>

Parameter	Description

	`"neighbor partition grouping" : "ON"`
	`}`
	
	For detailed information on these parameters, see Inverted File Flat Index Syntax and Parameters.

`idx_parallel_creation`	Parallel degree used for index construction.
`idx_quantization_type`	Optional type of quantization to apply to the indexed vectors. The supported values are `'NONE'` and `'SCALAR'`.
`idx_compression_ratio`	Optional compression ratio to use with vector quantization. This is provided either as a numeric value or `NULL` and is only valid when quantization is enabled.
`idx_distribute_parameters`	Optional JSON object that is used to control distribution for distributed HNSW vector indexes. The value is provided as `NULL` or as a JSON `CLOB` that includes `distribute_method`. The following are supported as the value for `distribute_method`: • `'ROWID RANGE'` • `'PARTITION'` • `'SUBPARTITION'` • `'DISTRIBUTE'` • `'AUTO'` This parameter is only supported for HNSW indexes and can only be used if `idx_duplicate_parameters` is passed as `NULL`.
`idx_duplicate_parameters`	Optional JSON object that controls duplication for distributed HNSW vector indexes. The value is expected to be either `NULL` or a JSON `CLOB` that includes `duplicate_method`. The value provided for `duplicate_method` must be `'ALL'`. This parameter is only supported for HNSW indexes and can only be used if `idx_distribute_parameters` is passed as `NULL`.
`idx_online_build`	Specifies whether the vector index should be built online. The values accepted are `TRUE` and `FALSE`. The default value is `FALSE`.

Examples

- Specify neighbors and efConstruction for HNSW indexes:

```
BEGIN
  DBMS_VECTOR.CREATE_INDEX(
    idx_name                => 'v_hnsw_01',
    table_name              => 'vpt01',
    idx_vector_col          => 'EMBEDDING',
    idx_include_cols        => NULL,
    idx_partitioning_scheme => NULL,
    idx_organization        => 'INMEMORY NEIGHBOR GRAPH',
    idx_distance_metric     => 'EUCLIDEAN',
    idx_accuracy            => 95,
    idx_parameters          => '{"type" : "HNSW",
                                 "neighbors" : 3,
                                 "efConstruction" : 4}'
```

```
  );
END;
/
```

- Specify the number of partitions for IVF indexes:

```
BEGIN
  DBMS_VECTOR.CREATE_INDEX(
    idx_name                => 'V_IVF_01',
    table_name              => 'vpt01',
    idx_vector_col          => 'EMBEDDING',
    idx_include_cols        => NULL,
    idx_partitioning_scheme => 'GLOBAL',
    idx_organization        => 'NEIGHBOR PARTITIONS',
    idx_distance_metric     => 'EUCLIDEAN',
    idx_accuracy            => 95,
    idx_parameters          => '{"type" : "IVF",
                                  "partitions" : 5,
                                  "neighbor partition grouping" : "ON"}',
    idx_online_build        => TRUE
  );
END;
/
```

- Create a scalar quantized HNSW index:

```
BEGIN
  DBMS_VECTOR.CREATE_INDEX(
    idx_name                 => 'V_HNSW_SQ_01',
    table_name               => 'VPT01',
    idx_vector_col           => 'EMBEDDING',
    idx_include_cols         => NULL,
    idx_partitioning_scheme  => NULL,
    idx_organization         => 'INMEMORY NEIGHBOR GRAPH',
    idx_distance_metric      => 'EUCLIDEAN',
    idx_accuracy             => 90,
    idx_quantization_type    => 'SCALAR',
    idx_compression_ratio    => 4,
    idx_parameters           => '{
                                   "type" : "HNSW",
                                   "neighbors" : 32,
                                   "efConstruction" : 200,
                                   "rescore factor" : 2,
                                   "algorithm" : "uniform_quantization"
                                 }'
  );
END;
/
```

DISABLE_CHECKPOINT

Use the `DISABLE_CHECKPOINT` procedure to disable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name. This operation purges all older checkpoints for the HNSW index. It also disables the creation of future checkpoints as part of the HNSW graph refresh.

Syntax

```
DBMS_VECTOR.DISABLE_CHECKPOINT('INDEX_USER',['INDEX_NAME']);
```

INDEX_USER

Specify the user name of the HNSW vector index owner.

INDEX_NAME

Specify the name of the HNSW vector index for which you want to disable the Checkpoint feature.

The `INDEX_NAME` clause is optional. If you do not specify the index name, then this procedure disables the Checkpoint feature for all HNSW vector indexes under the given user.

Examples

- Using both the index name and index user:

  ```
  DBMS_VECTOR.DISABLE_CHECKPOINT('VECTOR_USER','VIDX1');
  ```

- Using only the index user:

  ```
  DBMS_VECTOR.DISABLE_CHECKPOINT('VECTOR_USER');
  ```

Related Topics

- *Oracle AI Database AI Vector Search User's Guide*
- ENABLE_CHECKPOINT
  Use the `ENABLE_CHECKPOINT` procedure to enable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name.

DROP_CREDENTIAL

Use the `DBMS_VECTOR.DROP_CREDENTIAL` credential helper procedure to drop an existing credential name from the data dictionary.

Syntax

```
DBMS_VECTOR.DROP_CREDENTIAL (
    CREDENTIAL_NAME      IN VARCHAR2
);
```

CREDENTIAL_NAME

Specify the credential name that you want to drop.

Examples

- For Generative AI:

```
exec dbms_vector.drop_credential('OCI_CRED');
```

- For Cohere:

```
exec dbms_vector.drop_credential('COHERE_CRED');
```

DROP_ONNX_MODEL Procedure

This procedure deletes the specified ONNX model.

Syntax

```
DBMS_VECTOR.DROP_ONNX_MODEL (model_name IN VARCHAR2,
                                  force      IN BOOLEAN DEFAULT FALSE);
```

Parameters

Table 12-3 DROP_ONNX_MODEL Procedure Parameters

Parameter	Description
`model_name`	Name of the machine learning ONNX model in the form [*schema_name*.]*model_name*. If you do not specify a schema, then your own schema is used.
`force`	Forces the machine learning ONNX model to be dropped even if it is invalid. An ONNX model may be invalid if a serious system error interrupted the model build process.

Usage Note

To drop an ONNX model, you must be the owner or you must have the `DB_DEVELOPER_ROLE`.

Example

You can use the following command to delete a valid ONNX model named `doc_model` that exists in your schema.

```
BEGIN
  DBMS_VECTOR.DROP_ONNX_MODEL(model_name => 'doc_model');
END;
/
```

ENABLE_CHECKPOINT

Use the `ENABLE_CHECKPOINT` procedure to enable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name.

> **Note**
>
> - This procedure only allows the index to create checkpoints. The checkpoint is created as part of the next HNSW graph refresh.
> - By default, HNSW checkpointing is enabled. If required, you can disable it using the `DBMS_VECTOR.DISABLE_CHECKPOINT` procedure.

Syntax

```
DBMS_VECTOR.ENABLE_CHECKPOINT('INDEX_USER',['INDEX_NAME']);
```

INDEX_USER

Specify the user name of the HNSW vector index owner.

INDEX_NAME

Specify the name of the HNSW vector index for which you want to enable the Checkpoint feature.

The `INDEX_NAME` clause is optional. If you do not specify the index name, then this procedure enables the Checkpoint feature for all HNSW vector indexes under the given user.

Examples

- Using both the index name and index user:

  ```
  DBMS_VECTOR.ENABLE_CHECKPOINT('VECTOR_USER','VIDX1');
  ```

- Using only the index user:

  ```
  DBMS_VECTOR.ENABLE_CHECKPOINT('VECTOR_USER');
  ```

Related Topics

- *Oracle AI Database AI Vector Search User's Guide*
- DISABLE_CHECKPOINT
  Use the `DISABLE_CHECKPOINT` procedure to disable the Checkpoint feature for a given Hierarchical Navigable Small World (HNSW) index user and HNSW index name. This operation purges all older checkpoints for the HNSW index. It also disables the creation of future checkpoints as part of the HNSW graph refresh.

GET_INDEX_STATUS

Use the `GET_INDEX_STATUS` procedure to query the status of a vector index creation.

Syntax

```
DBMS_VECTOR.GET_INDEX_STATUS ('USER_NAME','INDEX_NAME');
```

USER_NAME

Specify the user name of the vector index owner.

INDEX_NAME

Specify the name of the vector index. You can query the index creation status for both Hierarchical Navigable Small World (HNSW) indexes and Inverted File Flat (IVF) indexes.

Usage Notes

- You can use the `GET_INDEX_STATUS` procedure only during a vector index creation.
- The `Percentage` value is shown in the output only for Hierarchical Navigable Small World (HNSW) indexes (and not for Inverted File Flat (IVF) indexes).
- The possible values of `Stage` for HNSW vector indexes are:

Value	Description
`HNSW Index Initialization`	Initialization phase for the HNSW vector index creation
`HNSW Index Auxiliary Tables Creation`	Creation of the internal auxiliary tables for the HNSW Neighbor Graph vector index
`HNSW Index Graph Allocation`	Allocation of memory from the vector memory pool for the HNSW graph
`HNSW Index Loading Vectors`	Loading of the base table vectors into the vector pool memory
`HNSW Index Graph Construction`	Creation of the multi-layered HNSW graph with the previously loaded vectors
`HNSW Index Load From Checkpoints`	HNSW vector index is loading from checkpoints
`HNSW Index Graph Complete`	HNSW vector index graph is complete
`HNSW Index Creation Completed`	HNSW vector index creation finished

- The possible values of `Stage` for IVF vector indexes are:

Value	Description
`IVF Index Initialization`	Initialization phase for the IVF vector index creation
`IVF Index Centroids Creation`	The K-means clustering phase that computes the cluster centroids on a sample of base table vectors
`IVF Index Centroid Partitions Creation`	Centroids assignment phase for the base table vectors
`IVF Index Creation Completed`	IVF vector index creation completed

Examples

```
exec DBMS_VECTOR.GET_INDEX_STATUS('VECTOR_USER','VIDX_HNSW');
```

Result:

```
"VECTOR_USER"."IDX1"  inst# 2 HNSW Index Initialization
"VECTOR_USER"."IDX1"  inst# 1 HNSW Index Graph Construction (76.25%)
```

```
exec DBMS_VECTOR.GET_INDEX_STATUS('VECTOR_USER','VIDX_HNSW');
```

Result:

```
"VECTOR_USER"."IDX1"  inst# 2 HNSW Index Load Vectors (22%)
"VECTOR_USER"."IDX1"  inst# 1 HNSW Index Graph Construction (100%)
```

```
exec DBMS_VECTOR.GET_INDEX_STATUS('VECTOR_USER','VIDX_HNSW');
```

Result:

```
"VECTOR_USER"."IDX1" HNSW Index Creation Completed
```

INDEX_ACCURACY_QUERY

Use the `DBMS_VECTOR.INDEX_ACCURACY_QUERY` function to verify the accuracy of a vector index for a given query vector, top-K, and target accuracy.

Syntax

```
DBMS_VECTOR.INDEX_ACCURACY_QUERY (
    OWNER_NAME          IN VARCHAR2,
    INDEX_NAME          IN VARCHAR2,
    QV                  IN VECTOR,
    TOP_K               IN NUMBER,
    TARGET_ACCURACY     IN NUMBER
) return VARCHAR2;

DBMS_VECTOR.INDEX_ACCURACY_QUERY (
    OWNER_NAME          IN VARCHAR2,
    INDEX_NAME          IN VARCHAR2,
    QV                  IN VECTOR,
    TOP_K               IN NUMBER,
    QUERY_PARAM         IN JSON
) return VARCHAR2;
```

Parameters

Table 12-4 INDEX_ACCURACY_QUERY (IN) Parameters of DBMS_VECTOR

Parameter	Description
`owner_name`	The name of the vector index owner.
`index_name`	The name of the vector index.
`qv`	Specifies the query vector.
`top_k`	The `top_k` value for accuracy computation.
`target_accuracy`	The target accuracy value for the vector index.
`query_param`	A JSON input that can be used to specify query parameters to be used during computation of vector index accuracy. The supported values are as follows: • `accuracy`: The target accuracy. • `neighbor partition probes`: The neighbor partition probes in case of an IVF index. • `efsearch`: The `efsearch` in case of an HNSW index. • `rescore factor`: Used with quantized vector indexes. Currently, `rescore factor` is only supported for quantized HNSW indexes.

For information about determining the accuracy of your vector indexes, see Index Accuracy Report in *Oracle AI Database AI Vector Search User's Guide*.

Usage Notes

When you specify the `rescore factor` field in the `query_param` parameter, the achieved accuracy is calculated for the given index using the this value. This allows you to manually choose the appropriate rescore factor for your scalar-quantized HNSW index based on your accuracy requirements.

For more information about quantization, see Scalar Quantized HNSW Indexes in *Oracle AI Database AI Vector Search User's Guide*.

Examples

Examples with IVF indexes:

```
DECLARE
  q_v    VECTOR;
  report VARCHAR2(128);
BEGIN
  q_v := TO_VECTOR('[1, 1, 1]', 3, FLOAT64);
  report := DBMS_VECTOR.INDEX_ACCURACY_QUERY(owner_name => 'VEC',
                                 index_name => 'IVF_IDX',
                                 qv => q_v,
                                 top_K => 10,
                                 query_param => json('{"accuracy": 80 }'));
  DBMS_OUTPUT.PUT_LINE(report);
END;
/
```

Result:

```
Accuracy achieved (100%) is 20% higher than the Target Accuracy requested
(80%)
```

```
DECLARE
  q_v    VECTOR;
  report VARCHAR2(128);
BEGIN
  q_v := TO_VECTOR('[1, 1, 1]', 3, FLOAT64);
  report := DBMS_VECTOR.INDEX_ACCURACY_QUERY(owner_name => 'VEC',
                                 index_name => 'IVF_IDX',
                                 qv => q_v,
                                 top_K => 10,
                                 query_param => json('{"neighbor partition
probes": 2}'));
  DBMS_OUTPUT.PUT_LINE(report);
END;
/
```

Result:

```
Accuracy achieved (100%) for NEIGHBOR PARTITION PROBES (2)
```

Examples with HNSW indexes:

```
DECLARE
  q_v    VECTOR;
  report VARCHAR2(128);
BEGIN
  q_v := TO_VECTOR('[1, 1, 1]', 3, FLOAT64);
  report := DBMS_VECTOR.INDEX_ACCURACY_QUERY(owner_name => 'VEC',
                                 index_name => 'HNSW_IDX',
                                 qv => q_v,
                                 top_K => 10,
                                 query_param => json('{"efsearch": 32}'));
  DBMS_OUTPUT.PUT_LINE(report);
END;
/
```

Result:

```
Accuracy achieved (100%) for EFSEARCH (32)
```

Examples with quantized HNSW indexes:

```
DECLARE
  q_v    VECTOR;
  report VARCHAR2(128);
BEGIN
  q_v := TO_VECTOR('[1, 1, 1]', 3, FLOAT64);
  report := DBMS_VECTOR.INDEX_ACCURACY_QUERY(owner_name => 'VEC',
```

```
                                index_name => 'QUANTIZED_HNSW_IDX',
                                qv => q_v,
                                top_K => 10,
                                query_param => json('{"efsearch": 32,
"rescore factor " : 2}'));
  DBMS_OUTPUT.PUT_LINE(report);
END;
/
```

Result:

```
Accuracy achieved (100%) for EFSEARCH (32), RESCORE FACTOR (2)
```

INDEX_ACCURACY_REPORT

Use the `DBMS_VECTOR.INDEX_ACCURACY_REPORT` function to capture from your past workloads, accuracy values achieved by approximate searches using a particular vector index for a certain period of time.

Syntax

```
DBMS_VECTOR.INDEX_ACCURACY_REPORT (
    OWNER_NAME          IN VARCHAR2,
    INDEX_NAME          IN VARCHAR2,
    START_TIME          IN TIMESTAMP WITH TIME ZONE,
    END_TIME            IN TIMESTAMP WITH TIME ZONE
) return NUMBER;
```

Parameters

Table 12-5 INDEX_ACCURACY_REPORT (IN) Parameters of DBMS_VECTOR

Parameter	Description
`owner_name`	The name of the vector index owner.
`index_name`	The name of the vector index.
`start_time`	Specifies from what time to capture query vectors to consider for the accuracy computation. A `NULL` `start_time` uses query vectors captured in the last 24 hours.
`end_time`	Specifies an end point up until which query vectors are considered for accuracy computation. A `NULL` `end_time` uses query vectors captured from `start_time` until the current time.

For information about determining the accuracy of your vector indexes, see Index Accuracy Report in *Oracle AI Database AI Vector Search User's Guide*.

INDEX_VECTOR_MEMORY_ADVISOR

Use the `INDEX_VECTOR_MEMORY_ADVISOR` procedure to determine the vector memory size needed for a particular vector index. This helps you evaluate the number of indexes that can fit for each simulated vector memory size.

Syntax

- Using the number and type of vector dimensions that you want to store in your vector index.

```
DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
    INDEX_TYPE         IN     VARCHAR2,
    NUM_VECTORS        IN     NUMBER,
    DIM_COUNT          IN     NUMBER,
    DIM_TYPE           IN     VARCHAR2,
    PARAMETER_JSON     IN     CLOB,
    RESPONSE_JSON      OUT    CLOB);
```

- Using the table and vector column on which you want to create your vector index:

```
DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
    TABLE_OWNER         IN     VARCHAR2,
    TABLE_NAME          IN     VARCHAR2,
    COLUMN_NAME         IN     VARCHAR2,
    INDEX_TYPE          IN     VARCHAR2,
    PARAMETER_JSON      IN     CLOB,
    RESPONSE_JSON       OUT    CLOB);
```

Table 12-6 Syntax Details: INDEX_VECTOR_MEMORY_ADVISOR

Parameter	Description
`INDEX_TYPE`	Type of vector index: • `IVF` for Inverted File Flat (IVF) vector indexes • `HNSW` for Hierarchical Navigable Small World (HNSW) vector indexes
`NUM_VECTORS`	Number of vectors that you plan to create the vector index with.
`DIM_COUNT`	Number of dimensions of a vector as a `NUMBER`.
`DIM_TYPE`	Type of dimensions of a vector. Possible values are: • `FLOAT16` • `FLOAT32` • `FLOAT64` • `INT8`
`TABLE_OWNER`	Owner name of the table on which to create the vector index.
`TABLE_NAME`	Table name on which to create the vector index.
`COLUMN_NAME`	Name of the vector column on which to create the vector index.

Table 12-6 (Cont.) Syntax Details: INDEX_VECTOR_MEMORY_ADVISOR

Parameter	Description
PARAMETER_JSON	Input parameter in JSON format. You can specify only one of the following form: • PARAMETER_JSON=>{"accuracy":value} • INDEX_TYPE=>IVF, parameter_json=>{"neighbor_partitions":value} • INDEX_TYPE=>HNSW, parameter_json=>{"neighbors":value} **Note**: You cannot specify values for accuracy along with neighbor_partitions or neighbors.
RESPONSE_JSON	JSON-formatted response string.

Examples

- Using neighbors in the parameters list:

```
SET SERVEROUTPUT ON;
DECLARE
    response_json CLOB;
BEGIN
    DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
        INDEX_TYPE=>'HNSW',
        NUM_VECTORS=>10000,
        DIM_COUNT=>768,
        DIM_TYPE=>'FLOAT32',
        PARAMETER_JSON=>'{"neighbors":32}',
        RESPONSE_JSON=>response_json);
END;
/
```

  Result:

```
Suggested vector memory pool size: 59918628 Bytes
```

- Using accuracy in the parameters list:

```
SET SERVEROUTPUT ON;
DECLARE
    response_json CLOB;
BEGIN
    DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
        INDEX_TYPE=>'HNSW',
        NUM_VECTORS=>10000,
        DIM_COUNT=>768,
        DIM_TYPE=>'FLOAT32',
        PARAMETER_JSON=>'{"accuracy":90}',
        RESPONSE_JSON=>response_json);
END;
/
```

Result:

```
Suggested vector memory pool size: 53926765 Bytes
```

- Using the table and vector column on which you want to create the vector index:

```
SET SERVEROUTPUT ON;
DECLARE
    response_json CLOB;
BEGIN
    DBMS_VECTOR.INDEX_VECTOR_MEMORY_ADVISOR(
        'VECTOR_USER',
        'VECTAB',
        'DATA_VECTOR',
        'HNSW',
        RESPONSE_JSON=>response_json);
END;
/
```

Example result:

```
Using default accuracy: 90%
Suggested vector memory pool size: 76396251 Bytes
```

Related Topics

- *Oracle AI Database AI Vector Search User's Guide*

INMEMORY_ONNX_MODEL

This procedure enables you to control the in-memory sharing of initializers for an ONNX model with external initializers within the database. When you enable in-memory sharing, the database loads the model's initializers once into shared memory, so that all sessions using the model benefit from reduced aggregate memory consumption. You must load or import the model before calling this procedure. Attempting to enable in-memory sharing on a model that has not been imported raises an error.

Note that only models with external data can be shared using `INMEMORY_ONNX_MODEL`. Models without external initializers cannot be enabled for in-memory sharing; attempting to do so will raise an error.

Syntax

```
DBMS_VECTOR.INMEMORY_ONNX_MODEL(
   model_name IN VARCHAR2,
   enable     IN BOOLEAN DEFAULT TRUE);
```

Parameters

Table 12-7 INMEMORY_ONNX_MODEL Procedure Parameters

Parameter	Description
`model_name`	Name of the model in the form `[schema_name.]model_name`. If you do not specify a schema, then your own schema is used. See DBMS_DATA_MINING.IMPORT_ONNX_MODEL Procedure for restrictions on the name of the model.
`enable`	Specify whether to enable or disable in-memory sharing for the ONNX model. Possible options: • `TRUE`: When you set this parameter to `TRUE` (the default), the database loads and shares the model's initializers in shared memory, allowing all processes to use the same copy during scoring. • `FALSE`: When you set it to `FALSE`, the database disables in-memory sharing and reclaims the shared memory after all processes finish using the model.

Example

The following example demonstrates how you can enable in-memory sharing for an imported ONNX embedding model named `MINILML6`. After enabling, the model's initializers are loaded into shared memory when you call the procedure and can be used concurrently by multiple processes with minimal memory overhead.

```
BEGIN
  DBMS_VECTOR.INMEMORY_ONNX_MODEL(
    model_name => 'MINILML6');
END;
/
```

This code snippet sets the model `MINILML6` to use in-memory sharing. After calling the procedure, you can confirm the model's in-memory status and track its usage by querying the `ALL_MINING_MODELS` dictionary view. If `INMEMORY` returns `YES`, the model's initializers are now shared and accessible from shared memory for any scoring operations.

Usage Notes

- The name of the model follows the same restrictions as those used for other machine learning models. For example, if a schema is not specified, your schema is used.
- The procedure manages in-memory sharing of the initializers of a specified ONNX model that was imported with external initializers (`ALL_MINING_MODELS.EXTERNAL_DATA = YES`). Models without external initializers cannot be enabled for in-memory sharing; attempting to do so will raise an error.
- You must have the `ALTER MINING MODEL` privilege on the specified model.
- When `enable` is set to `TRUE` (the default), the procedure immediately populates the model's initializers into shared memory. The call blocks until the full shared-memory population succeeds or fails; there is no post-call delay. Upon successful enablement, the model's shared-initializer status is displayed in the `ALL_MINING_MODELS` view as `INMEMORY = YES`.
- When `enable` is set to `FALSE`, the model is disabled for in-memory sharing. The call blocks until cleanup of the shared registry and in-memory buffers completes, so the model is fully disabled by the time control returns to the caller. Memory allocated for the shared

initializers is reclaimed after all sessions finish using the model. The model's status changes to `INMEMORY = NO` in dictionary views.

- In non-RAC environments, if there is not enough memory to populate the model in-memory, the procedure raises an error. The database does not queue the request. You must call the procedure again after freeing up memory.
- In Oracle RAC environments, enabling a model for in-memory sharing triggers the operation on the instance where you call the procedure. The database updates the dictionary views and immediately populates the model into memory. On other instances in the cluster, the database updates the dictionary views, but it delays population until the first inference that uses the model on each instance. Memory usage and population status (including possible instance-specific errors) are tracked separately per instance.
  – If there isn't enough memory, the population (and therefore the inference) will fail. An error is raised to indicate the insufficient memory issue and the model will be in the `MEM_EXHAUSTED` state. Memory usage and population status, including possible instance-specific errors, are tracked separately per instance.
- Models that are in-memory enabled can be exported.
- Upon import, models are always set to `INMEMORY = NO` and must be explicitly re-enabled.
- Use the dictionary and fixed views (`ALL_MINING_MODELS`, `V$IM_ONNX_MODEL`, and `V$IM_ONNX_SEGMENT`) to audit in-memory status, population details, and memory consumption for ONNX models. See ALL_MINING_MODELS, V$IM_ONNX_MODEL, and V$IM_ONNX_SEGMENT in *Oracle AI Database Reference* for more information.
- You may consider disabling an in-memory ONNX model (by setting `ENABLE` to `FALSE`) in the following scenarios:
  – When session isolation is required. For example, for multi-tenant or security-sensitive deployments where sharing model weights in memory between sessions may pose risks.
  – When you need a guaranteed fresh load of the model for each session, for example, during testing or troubleshooting.
  – In environments with very limited memory, to avoid retaining shared model data across sessions longer than necessary.
  – If models are reloaded or updated often during runtime, disabling sharing ensures each session gets the current version and reduces cache management complexity.

See Also

Oracle Machine Learning for SQL User's Guide for more information about external initializers

LOAD_ONNX_MODEL

This procedure enables you to load an ONNX-format embedding model or a rerank model into your database.

Syntax

```
DBMS_VECTOR.LOAD_ONNX_MODEL (
     directory                IN  VARCHAR2,
     file_name                IN  VARCHAR2,
```

```
     model_name               IN  VARCHAR2,
     metadata                 IN  JSON DEFAULT JSON('{"function" : "embedding", '||
                                      '"embeddingOutput" : "embedding", "input":
{"input":["DATA"]}}'),
     external_data_file_name  IN  VARCHAR2  DEFAULT NULL);

DBMS_VECTOR.LOAD_ONNX_MODEL(
    model_name         IN  VARCHAR2,
    model_data         IN  BLOB,
    metadata           IN  JSON DEFAULT JSON('{"function" : "embedding", '||
                                '"embeddingOutput" : "embedding", "input":
{"input":["DATA"]}}'));
```

Parameters

Table 12-8 LOAD_ONNX_MODEL Procedure Parameters

Parameter	Description
`directory`	The directory name of the data dump. For example, `DM_DUMP`.
`file_name`	A `VARCHAR2` type parameter that specifies the file name of the ONNX model.
`model_name`	The user-defined name of the model in the form `[schema_name.]model_name`. This is the name the model will have in OML as a first-class database object. If you do not specify a schema, then your own schema is used.
`model_data`	It is a `BLOB` holding the ONNX representation of the model. The `BLOB` contains the identical byte sequence as the one stored in an ONNX file.
`metadata`	A JSON description of the metadata describing the model. The metadata at minimum must describe the machine learning function supported by the model. The model's metadata parameters are described in JSON Metadata Parameters for ONNX Models.
`external_data_file_name`	A `VARCHAR2` type parameter used to provide the file name of the external data file metadata if the file name does not match the expected format, which is `<file_name>_external_data.json`. The default value is `NULL`.

Examples

The following examples illustrates a code snippet that uses the `DBMS_VECTOR.LOAD_ONNX_MODEL` procedure. The complete step-by-step example is illustrated in Import ONNX Models and Generate Embeddings.

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
    directory  => 'DM_DUMP',
    file_name  => 'my_embedding_model.onnx',
    model_name => 'doc_model',
    metadata   => JSON('{"function" : "embedding",
               "embeddingOutput" : "embedding",
               "input": {"input": ["DATA"]}}'));
```

```
DBMS_VECTOR.LOAD_ONNX_MODEL(
    model_name => 'my_embedding_model.onnx',
    model_data => :blob_bind_variable,
    metadata   => JSON('{"function" : "embedding",
```

```
                        "embeddingOutput" : "embedding",
                        "input":{"input": ["DATA"]}}'));
```

For a complete example to illustrate how you can define a `BLOB` variable and use it in the `LOAD_ONNX_MODEL` procedure, you can have the following:

```
CREATE OR REPLACE MY_LOAD_EMBEDDING_MODEL(embedding_model_name VARCHAR2,
onnx_blob BLOB) IS
BEGIN
DBMS_VECTOR.LOAD_ONNX_MODEL(embedding_model_name,
                            onnx_blob,
                            JSON('{"function" : "embedding",
                                   "embeddingOutput" : "embedding" ,
                                   "input":{"input": ["DATA"]}}'));
END;
/
```

The following example illustrates a code snippet that uses the `DBMS_VECTOR.LOAD_ONNX_MODEL` procedure to load a rerank model:

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL(
    directory  => 'DM_DUMP',
    file_name  => 'my_rerank_model.onnx',
    model_name => 'rerank_model',
    metadata   => JSON('{"function" : "regression"}'));
```

Usage Notes

- The name of the model follows the same restrictions as those used for other machine learning models, namely:
  - The schema name, if provided, is limited to 128 characters.
  - The model name is limited to 123 characters and must follow the rules of unquoted identifiers: they contain only alphanumeric characters, the underscore (_), dollar sign ($), and pound sign (#). The initial character must be alphabetic.
- The model size is limited to 2 GB.
- There are default input and output names for input and output attributes for models that are prepared by the Python utility. You can load those models without the JSON parameters. For example:

  ```
  EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL('DM_DUMP', 'my_embedding_model.onnx',
  'doc_model'));
  ```

- To be loaded as an ONNX model with external initializers, there must exist in the specified directory a file with the format `<file_name>_external_data.json`. If a file with that name does not exist, and the `external_data_file_name` parameter has not been used to specify an alternate file name, the ONNX model will load without external initializers. If the file does exist, it is verified and the model is imported as an ONNX model with external initializers. The cumulative size of the `.onnx` file and each of the external data files can be more than 2 GB. However, loading may fail if there is not enough PGA aggregate memory. For information about the parameter used to control the PGA limit, see PGA_AGGREGATE_LIMIT in *Oracle AI Database Reference*.

- When the `LOAD_ONNX_MODEL` syntax using a `BLOB` format for `model_data` is used, the `model_data` argument can be up to 2 GB in size. Loading may fail if there is not enough PGA aggregate memory to load the model. This version of the procedure syntax can only load a model *without* external initializers.
- JSON Metadata Parameters for ONNX Models
  When importing models using the `IMPORT_ONNX_MODEL` (`DBMS_DATA_MINING`), `LOAD_ONNX_MODEL` (`DBMS_VECTOR`), or `LOAD_ONNX_MODEL_CLOUD` (`DBMS_VECTOR`) procedures, you supply metadata as JSON parameters.

> **See Also**
>
> - *Oracle Machine Learning for SQL User's Guide* for examples of using ONNX models for machine learning tasks
> - *Oracle Machine Learning for SQL User's Guide* for information about external initializers

JSON Metadata Parameters for ONNX Models

When importing models using the `IMPORT_ONNX_MODEL` (`DBMS_DATA_MINING`), `LOAD_ONNX_MODEL` (`DBMS_VECTOR`), or `LOAD_ONNX_MODEL_CLOUD` (`DBMS_VECTOR`) procedures, you supply metadata as JSON parameters.

Parameters

Field	Value Type	Description
`function`	String	Specify regression, classification, clustering, or embedding. This is a mandatory setting. **NOTE**: The only JSON parameter required when importing the model is the machine learning function.
`input`	NA	Describes the model input mapping. See "Input" in Usage Notes.
`regressionOutput`	String	The name of the regression model output that stores the regression results. The output is expected to be a tensor of supported shape of any supported regression output type. See "Output" in Usage Notes.
`classificationProbOutput`	String	The name of the classification model output storing probabilities. The output is expected to be a tensor value of type float (width 32/64) of supported shape. See "Automatic normalization of output probabilities" in Usage Notes.
`clusteringDistanceOutput`	String	The name of the clustering model output storing distances. The output is of type float (width 16/32/64) of supported shape.
`clusteringProbOutput`	String	The name of the clustering model output storing probabilities. The output is of type float (width 16/32/64) of supported shape.

Field	Value Type	Description
`classificationLabelOutput`	String	The name of the model output holding label information. You have the following metadata parameters to specify the labels for classification: • `labels`: specify the labels directly in the JSON metadata • `classificationLabelOutput`: specify the model output that provides labels If you do not specify any value for this parameter or the function of the model is not classification, you will receive an error. The user can specify to use labels from the model directly by setting `classificationLabelOutput` to the model output holding the label information. The tensor output holding the label information must be the same size as the number of classes and must be of integer or string type. If the tensor that holds the labels is of string type, the returned type of the `PREDICTION` operator is `VARCHAR2`. If the tensor that holds the labels is of integer type, the returned type of the `PREDICTION` operator is `NUMBER`.
`normalizeProb`	String	Describes automatic normalization of output probabilities. See "Automatic normalization of output probabilities" in Usage Notes.
`labels`	NA	The labels used for classification. If you want to use custom labels, specify the labels using the labels field in the JSON metadata. The field can be set to an array of length equal to the number of classes. The labels for the class i must be stored at index i of the label array. If an array of strings is used, the returned type of the `PREDICTION` operator is `VARCHAR2`. The size of the string labels specified by the user cannot exceed *4000* bytes. If an array of numbers is used, the returned type of the `PREDICTION` operator is `NUMBER`. If you do not specify labels or `classificationLabelOutput`, classes are identified by integers in the range 1 to N where N is the number of classes. In this case, the returned type of the `PREDICTION` operator is `NUMBER`.
`embeddingOutput`	String	The model output that holds the generated embeddings.
`suitableDistanceMetrics`	String	An array of names of suitable distance metrics for the model. The names must be the names of the distance metrics used for the Oracle `VECTOR_DISTANCE` operator. To know the supported distance metrics, see Vector Distance Metrics. This parameter is for informational purposes only.

Field	Value Type	Description
`normalization`	Boolean	A boolean value indicates if normalization is applied to the output vector. The value 1 means normalization is applied. Normalization is process of converting an embedding vector so that it's norm or length equals 1. A normalized vector maintains its direction but its length becomes 1. The resulting vector is often called a unit vector.
`maxSequenceLength`	Number	The maximum length of the token (input) sequence that is meaningful for the model. This parameter sets a limit on the number of tokens, words, or elements in each input sequence that the model will process. This ensures uniform input size for the model. For example, the value could be 128, or 512 to 4096 depending on the task for which the parameter is used. A machine translation model might have a `maxSequenceLength` of 512, accommodating sentences or paragraphs up to 512 tokens for translation tasks. This parameter is for informational purposes only.
`pooling`	String	Indicates the pooling function performed on the output vector. This parameter is for informational purposes only.
`modelDescription`	Object	A JSON object that allows users to add additional descriptions to the models complementing the existing ONNX metadata for model description. This parameter is for informational purposes only.
`languages`	String	A comma-separated list of language name or abbreviation, as described in "A.1 Languages" of *Oracle AI Database Globalization Support Guide*. If you import multi-lingual embedding model, specify the language or the language abbreviation as the metadata. This parameter is for informational purposes only.
`tokenizer`	String	Tokenizers help in transforming text into words. There are several tokenizers available, including: bert, gpt2, bpe, wordpiece, sentencepiece, and clip. This parameter is for informational purposes only.
`embeddingLayer`	String	An identifier for the embedding layer. An embedding layer, serving as a hidden layer in neural networks, transforms input data from high to lower dimensions, enhancing the network's understanding of input relationships and data processing efficiency. Embedding layer helps in processing and analyzing categorical or discrete data. It achieves this by transforming categories into continuous embeddings, capturing the essential semantic relationships and similarities between them. For example the last hidden state in some transformer, or a layer in a resnet network. This parameter is for informational purposes only.

Field	Value Type	Description
`defaultOnNull`	NA	Specify the replacement of missing values in the JSON using the `defaultOnNull` field. If `defaultOnNull` is not specified, the replacement of missing values is not performed. The `defaultOnNull` sets the missing values to NULL by default. You can override the default value of NULL by providing meaningful default values to substitute for NULL. The field must be a JSON object literal, whose fields are the input attribute names and whose values are the default values for the input. Note that the default value is of type string and must be a valid Oracle PL/SQL NVL value for the given datatype.

Note: The parameters are case-sensitive. A number of default conventions for output parameter names and default values allows to minimize the information that you may have to provide. The parameters such as `suitableDistanceMetrics` are informational only and you are not expected to provide this information while importing the model. The JSON descriptor may specify only one input attribute. If more are specified, you will receive an error. You will receive an error if the `normalizeProb` field is specified as the JSON metadata parameter.

Usage Notes

The name of the model follows the same restrictions as those used for other machine learning models, namely:

- **Input**

  When importing a model from an ONNX representation, you must specify the name of the attribute used for scoring and how it maps to actual ONNX inputs. A scoring operator uses these attribute names to identify the columns to be used. (For example, `PREDICTION`). Follow these conventions to specify the attribute names using the input field:

  not specified: When the field input is not specified, attribute names are mapped directly to model inputs by name. That is, if the attribute name is not specified in the JSON metadata, then the name of the input tensor is used as an attribute name. Each model input must have dimension `[batch_size, value]`. If you do not specify `input` in the JSON metatdata, the value must be 1. You don't have to specify extra metadata if the input of the model already conforms to the format. For an embedding model, a single input is provided that may be used in batches. Here, if the `input` parameter is not specified in the JSON metadata, the valid model will have `[batch_size, 1]`.

  You must ensure that all attribute names, whether implied by the model or explicitly set by you through the input field, are valid Oracle AI Database identifiers for column names. Each attribute name within a model must be unique, ensuring no duplicates exist.

  You can explicitly specify attribute name for model that use input tensors that have a dimension larger than 1 (for example, (batch_size, 2)). In this case, you must specify a name for each of these values for them to be interpreted as independent attribute name. This can be done for regression, classification, clustering which are models whose scoring operation can take multiple input attributes.

- **Output**

  As models might have multiple outputs, you can specify which output is of interest for a specific machine learning technique. You have the following ways to specify model outputs:

- Specify the output name of interest in the JSON during model import. If the specified name is not a valid model output (see the table with valid outputs for a given machine learning function), you will receive an error.
- If the model produces an output that matches the expected output name for the given machine learning technique (for example, `classificationProbOutput`) and you didn't explicitly specify it, the output is automatically assumed.
- If you do not specify any output name and the model has a single output, the system assumes that the single output corresponds to a default specific to the machine learning technique. For an embedding machine learning function, the default value is `embeddingOutput`.
  The system reports an error if you do not specify model outputs or if you supply outputs that the specified machine learning function does not support. The following table displays supported outputs for a specific machine learning function:

Machine learning function	Output
regression	`regressionOutput`
classification	`classificationProbOutput`
clustering	`clusteringDistanceOutput`
embedding	`embeddingOutput`

  If none of the mentioned model outputs are specified, or if you supply outputs that are not supported by the specified machine learning function, you will receive an error.

- **Automatic Normalization of Output Probabilities**

  Many users widely employ the softmax function to normalize the output of multi-class classification models, as it enables to easily interpret the results of these models. The **softmax function** is a mathematical function that converts a vector of real numbers into a probability distribution. It is also known as the softargmax, or normalized exponential function. This function is available to you to specify at the model import-time that a softmax normalization must be applied to the tensor holding output probabilities such as `classificationProbOutput` and `clusteringProbOutput`. Specify `normalizeProb` to define the normalization that must be applied for softmax normalization. The default setting is `none`, indicating that no normalization is applied. You can choose `softmax` to apply a softmax function to the probability output. Specifying any other value for this field will result in an error during import. Additionally, specifying this field for models other than classification and clustering will also lead to an error.

Example: Specifying JSON Metadata Parameters for Embedding Models

The following example illustrates a simple case of how you can specify JSON metadata parameters while importing an ONNX embedding model into the Database using the `DBMS_VECTOR.IMPORT_ONNX_MODEL` procedure.

```
DBMS_VECTOR.IMPORT_ONNX_MODEL('my_embedding_model.onnx', 'doc_model',
                JSON('{"function" : "embedding",
                      "embeddingOutput" : "embedding" ,
                       "input":{"input": ["DATA"]}}'));
```

Example: Specifying Complete JSON Metadata Parameters for Embedding Models

The following example illustrates how to provide a complete JSON metadata parameters, with an exception of `embeddingLayer`, for importing embedding models.

```
DECLARE
  metadata JSON;
  mdtxt varchar2(4000);
BEGIN
  metadata := JSON(q'#
           {
             "function"                : "embedding",
             "embeddingOutput"         : "embedding",
             "input"                   : { "input" : ["txt"]},
             "maxSequenceLength"       : 512,
             "tokenizer"               : "bert",
             "suitableDistanceMetrics" : [ "DOT", "COSINE", "EUCLIDEAN"],
             "pooling"                 : "Mean Pooling",
             "normalization"           : true,
             "languages"               : ["US"],
             "modelDescription"        : {
                 "description" : "This model was tuned for semantic search:
Given a query/question, if can find relevant passages. It was trained on a
large and diverse set of (question, a
nswer) pairs.",
                 "url" : "https://example.co/sentence-transformers/
my_embedding_model"}
           }
           #');
  -- load the onnx model
    DBMS_VECTOR.IMPORT_ONNX_MODEL('my_embedding_model.onnx', 'doc_model',
metadata);
END;
/
```

> **See Also**
>
> *Oracle Machine Learning for SQL User's Guide* for examples of using ONNX models for machine learning tasks

LOAD_ONNX_MODEL_CLOUD

This procedure enables you to load an ONNX-format embedding model from Oracle Object Storage into your database.

Syntax

```
DBMS_VECTOR.LOAD_ONNX_MODEL_CLOUD (
     model_name        IN  VARCHAR2,
     credential        IN  VARCHAR2,
     uri               IN  VARCHAR2,
     metadata          IN  JSON DEFAULT JSON('{"function" : "embedding", '||
                              '"embeddingOutput" : "embedding", "input": {"input":
["DATA"]}}')
);
```

Parameters

Table 12-9   LOAD_ONNX_MODEL_CLOUD Procedure Parameters

Parameter	Description
`model_name`	The user-defined name of the model in the form `[schema_name.]model_name`. This is the name the model will have in OML as a first-class database object. If you do not specify a schema, then your own schema is used.
`credential`	The name of the credential to be used to access Oracle Object Storage.
`uri`	The URI of the ONNX model. When the value of `uri` is a pre-authenticated URI, the `credential` argument should be passed as NULL.
`metadata`	A JSON description of the metadata describing the model. The metadata at minimum must describe the machine learning function supported by the model. The model's metadata parameters are described in JSON Metadata Parameters for ONNX Models.

Examples

The following example includes a code snippet that uses the `DBMS_VECTOR.LOAD_ONNX_MODEL_CLOUD` procedure.

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL_CLOUD(
    model_name => 'database',
    credential => 'MYCRED',
    uri => 'https://objectstorage.us-phoenix-1.oraclecloud.com/n/namespace-string/b/
bucketname/o/all-MiniLM-L6-v2.onnx',
    metadata => JSON('{"function" : "embedding", "embeddingOutput" : "embedding" , "input":
{"input": ["DATA"]}}')
);
```

Usage Notes

- The name of the model follows the same restrictions as those used for other machine learning models, namely:
  - The schema name, if provided, is limited to 128 characters.

  - The model name is limited to 123 characters and must follow the rules of unquoted identifiers: they contain only alphanumeric characters, the underscore (_), dollar sign ($), and pound sign (#). The initial character must be alphabetic.
- The model size is limited to 2 GB.
- There are default input and output names for input and output attributes for models that are prepared by the Python utility. You can load those models without the JSON parameters. For example:

```
EXECUTE DBMS_VECTOR.LOAD_ONNX_MODEL_CLOUD(
    model_name => 'database',
    credential => 'MYCRED',
    uri => 'https://objectstorage.us-phoenix-1.oraclecloud.com/n/namespace-string/b/bucketname/o/all-MiniLM-L6-v2.onnx'
);
```

- To use ONNX models with external initializers, follow these steps:
  1. Create an Oracle Object Storage bucket.
  2. Create an ONNX model with OML4Py.
  3. Upload the model to the object storage bucket. If the model is a single-file model, there is only the one `.onnx` file to upload. If the model has external data, upload the `.onnx` file, `.json` metadata file, and `.data` files into the same bucket.
  4. Create a pre-authenticated request (PAR). In case of external data, the PAR must be valid either for the entire bucket or for the prefix that matches all of the uploaded files. You can skip this step and use credentials instead, in which case you must provide a valid `credential` along with the full URL to the `.onnx` file in the next step.
  5. Call the `LOAD_ONNX_MODEL_CLOUD` procedure with `credential` set to `NULL` and `uri` set to the full URL to the `.onnx` file (from the PAR).

See Also

- *Oracle Machine Learning for SQL User's Guide* for examples of using ONNX models for machine learning tasks
- *Oracle Machine Learning for SQL User's Guide* for information about external initializers

QUERY

Use the `DBMS_VECTOR.QUERY` function to perform a similarity search operation which returns the top-k results as a JSON array.

Syntax

Query is overloaded and supports a version with `query_vector` passed in as a `VECTOR` type in addition to `CLOB`.

```
DBMS_VECTOR.QUERY (
    TAB_NAME            IN VARCHAR2,
    VEC_COL_NAME        IN VARCHAR2,
    QUERY_VECTOR        IN CLOB,
```

```
    TOP_K                IN NUMBER,
    VEC_PROJ_COLS        IN JSON_ARRAY_T DEFAULT NULL,
    IDX_NAME             IN VARCHAR2 DEFAULT NULL,
    DISTANCE_METRIC      IN VARCHAR2 DEFAULT 'COSINE',
    USE_INDEX            IN BOOLEAN DEFAULT TRUE,
    ACCURACY             IN NUMBER DEFAULT 90,
    IDX_PARAMETERS       IN CLOB DEFAULT NULL
) return JSON_ARRAY_T;

DBMS_VECTOR.QUERY (
    TAB_NAME             IN VARCHAR2,
    VEC_COL_NAME         IN VARCHAR2,
    QUERY_VECTOR         IN VECTOR,
    TOP_K                IN NUMBER,
    VEC_PROJ_COLS        IN JSON_ARRAY_T DEFAULT NULL,
    IDX_NAME             IN VARCHAR2 DEFAULT NULL,
    DISTANCE_METRIC      IN VARCHAR2 DEFAULT 'COSINE',
    USE_INDEX            IN BOOLEAN DEFAULT TRUE,
    ACCURACY             IN NUMBER DEFAULT 90,
    IDX_PARAMETERS       IN CLOB DEFAULT NULL
) return JSON_ARRAY_T;
```

Parameters

Table 12-10 DBMS_VECTOR.QUERY Parameters

Parameter	Description
`tab_name`	Table name to query
`vec_col_name`	Vector column name
`query_vector`	Query vector passed in as `CLOB` or `VECTOR`.
`top_k`	Number of results to be returned.
`vec_proj_cols`	Columns to be projected as part of the result.
`idx_name`	Optional index hint.
`distance_metric`	Distance computation metric. Defaults to `COSINE`. Can also be `MANHATTAN`, `HAMMING`, `DOT`, `EUCLIDEAN`, `L2_SQUARED`, `EUCLIDEAN_SQUARED`, or `JACCARD`.
`use_index`	Specifies whether the search is an approximate search or exact search. Defaults to TRUE (that is, approximate).
`accuracy`	Specifies the minimum desired query accuracy.
`idx_parameters`	Specifies values of `efsearch`, `neighbor partition probes`, and `rescore factor` passed in, formatted as JSON.

REBUILD_INDEX

Use the `DBMS_VECTOR.REBUILD_INDEX` procedure to rebuild a vector index.

Purpose

To rebuild a vector index such as Hierarchical Navigable Small World (HNSW) vector index or Inverted File Flat (IVF) vector index. If `idx_rebuild_mode` is passed as non-`NULL`, the index is rebuilt using HNSW refresh support. If it is passed as NULL, the index is dropped and then recreated using the new index parameters.

Syntax

```
DBMS_VECTOR.REBUILD_INDEX (
    idx_name                    IN VARCHAR2,
    table_name                  IN VARCHAR2 DEFAULT NULL,
    idx_vector_col              IN VARCHAR2 DEFAULT NULL,
    idx_include_cols            IN VARCHAR2 DEFAULT NULL,
    idx_partitioning_scheme     IN VARCHAR2 DEFAULT NULL,
    idx_organization            IN VARCHAR2 DEFAULT NULL,
    idx_distance_metric         IN VARCHAR2 DEFAULT NULL,
    idx_accuracy                IN NUMBER   DEFAULT NULL,
    idx_parameters              IN CLOB     DEFAULT NULL,
    idx_parallel_creation       IN NUMBER   DEFAULT NULL,
    idx_rebuild_mode            IN VARCHAR2 DEFAULT NULL,
    idx_quantization_type       IN VARCHAR2 DEFAULT NULL,
    idx_compression_ratio       IN NUMBER   DEFAULT NULL,
    idx_distribute_parameters   IN CLOB     DEFAULT NULL,
    idx_duplicate_parameters    IN CLOB     DEFAULT NULL,
    idx_online_build            IN BOOLEAN  DEFAULT NULL,
    idx_owner                   IN VARCHAR2 DEFAULT NULL
);
```

Parameters

Parameter	Description
idx_name	Name of the index to rebuild.
table_name	Table on which to create the index.
idx_vector_col	Vector column on which to create the index.
idx_include_cols	A comma-separated list of column names to be covered by the index.
idx_partitioning_scheme	Partitioning scheme for IVF indexes: • GLOBAL • LOCAL IVF indexes support both global and local indexes on partitioned tables. By default, these indexes are globally partitioned by centroid. You can choose to create a local IVF index, which provides a one-to-one relationship between the base table partitions or subpartitions and the index partitions. For detailed information on these partitioning schemes, see Inverted File Flat Vector Indexes Partitioning Schemes.
idx_organization	Index organization: • NEIGHBOR PARTITIONS • INMEMORY NEIGHBOR GRAPH For detailed information on these organization types, see Manage the Different Categories of Vector Indexes.

Parameter	Description
`idx_distance_metric`	Distance metric or mathematical function used to compute the distance between vectors: • `COSINE` • `MANHATTAN` • `HAMMING` • `JACCARD` • `DOT` • `EUCLIDEAN` • `L2_SQUARED` • `EUCLIDEAN_SQUARED` For detailed information on each of these metrics, see Vector Distance Functions and Operators.
`idx_accuracy`	Target accuracy at which the approximate search should be performed when running an approximate search query. As explained in Understand Approximate Similarity Search Using Vector Indexes, you can specify non-default target accuracy values either by specifying a percentage value or by specifying internal parameters values, depending on the index type you are using. • For an HNSW approximate search: In the case of an HNSW approximate search, you can specify a target accuracy percentage value to influence the number of candidates considered to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose a similar result as an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the `EFSEARCH` parameter to impose a certain maximum number of candidates to be considered while probing the index. The higher that number, the higher the accuracy. For detailed information, see Understand Hierarchical Navigable Small World Indexes. • For an IVF approximate search: In the case of an IVF approximate search, you can specify a target accuracy percentage value to influence the number of partitions used to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the `NEIGHBOR PARTITION PROBES` parameter to impose a certain maximum number of partitions to be probed by the search. The higher that number, the higher the accuracy. For detailed information, see Understand Inverted File Flat Vector Indexes.

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>idx_parameters</code></td><td>
Type of vector index and associated parameters.
Specify the indexing parameters in JSON format:

For HNSW indexes:

<code>type</code>: Type of vector index to create, that is, <code>HNSW</code>
<code>neighbors</code>: Maximum number of connections permitted per vector in the HNSW graph
<code>efConstruction</code>: Maximum number of closest vector candidates considered at each step of the search during insertion
<code>offload_credential_name</code>: The identifier of the credential that should be used when authenticating with the offload service of the Private AI Services Container. Typically, this should match the <code>offload_credential_name</code> used when the vector index was created (prefixed by a schema if the credential schema is not the same as the currently active user schema).
<code>offload_url</code>: The Private AI Services Container service endpoint for index creation.

For example:
<pre>{
    "type"   : "HNSW",
    "neighbors"   : 3,
    "efConstruction"   : 4,
    "offload_credential_name" : "privateai",
    "offload_url"   : "https://
<myprivateaiservicehostname>/v1/index"
}</pre>
For detailed information on these parameters, see Hierarchical Navigable Small World Index Syntax and Parameters.
For information about the Private AI Services Container vector index service, see Oracle Private AI Services Container User's Guide.

For IVF indexes:

<code>type</code>: Type of vector index to create, that is, <code>IVF</code>
<code>partitions</code>: Neighbor partition or cluster in which you want to divide your vector space

For example:
<pre>{
    "type"   : "IVF",
    "partitions" : 5
}</pre>
For detailed information on these parameters, see Inverted File Flat Index Syntax and Parameters.

</td></tr>
<tr><td><code>idx_parallel_creation</code></td><td>Number of parallel threads used for index construction.</td></tr>
<tr><td><code>idx_rebuild_mode</code></td><td>
Supported values:

<code>FULL</code>: Perform a full rebuild.
<code>INCREMENTAL</code>: Perform an incremental build.

By default, this parameter is set to <code>NULL</code>.
This parameter is only supported for HNSW vector indexes.
</td></tr>
</table>

Parameter	Description
idx_quantization_type	Optional replacement quantization type to apply to the rebuilt index. The supported values are NULL, 'NONE', and 'SCALAR'.
idx_compression_ratio	Optional replacement compression ratio for a quantized vector index. This is provided either as a numeric value or NULL and is only valid when quantization is enabled.
idx_distribute_parameters	Optional replacement distribution settings for a distributed HNSW vector index. The value is provided as NULL or as a JSON CLOB that includes distribute_method. The following are supported as the value for distribute_method: • 'ROWID RANGE' • 'PARTITION' • 'SUBPARTITION' • 'DISTRIBUTE' • 'AUTO' This parameter is only supported for HNSW indexes and can only be used if idx_duplicate_parameters is passed as NULL.
idx_duplicate_parameters	Optional replacement duplication settings for a distributed HNSW vector index. The value is expected to be either NULL or a JSON CLOB that includes duplicate_method. The value provided for duplicate_method must be 'ALL'. This parameter is only supported for HNSW indexes and can only be used if idx_distribute_parameters is passed as NULL.
idx_online_build	Optional replacement online build setting to use when recreating the vector index. The values accepted are TRUE, FALSE, and NULL. The default value is NULL.
idx_owner	Specify the owning schema of the index to be rebuilt. The index will be rebuilt in the same schema. Use the default value, NULL, if the index is present in the current schema.

Usage Notes

For all parameters besides idx_name, if the value is passed as NULL (the default), the procedure uses the same value that was used by the index before the rebuild.

Examples

- Specify neighbors and efConstruction for HNSW indexes:

```
BEGIN
  DBMS_VECTOR.REBUILD_INDEX(
    idx_name                 => 'v_hnsw_01',
    table_name               => 'vpt01',
    idx_vector_col           => 'EMBEDDING',
    idx_include_cols         => NULL,
    idx_partitioning_scheme  => NULL,
    idx_organization         => 'INMEMORY NEIGHBOR GRAPH',
    idx_distance_metric      => 'EUCLIDEAN',
    idx_accuracy             => 95,
```

```
    idx_parameters          => '{
                                  "type" : "HNSW",
                                  "neighbors" : 3,
                                  "efConstruction" : 4
                                }'
  );
END;
/
```

- Specify the number of partitions for IVF indexes:

```
BEGIN
  DBMS_VECTOR.REBUILD_INDEX(
    idx_name                => 'V_IVF_01',
    table_name              => 'vpt01',
    idx_vector_col          => 'EMBEDDING',
    idx_include_cols        => NULL,
    idx_partitioning_scheme => NULL,
    idx_organization        => 'NEIGHBOR PARTITIONS',
    idx_distance_metric     => 'EUCLIDEAN',
    idx_accuracy            => 95,
    idx_parameters          => '{
                                 "type" : "IVF",
                                 "partitions" : 5
                               }'
  );
END;
/
```

- To enable a full graph rebuild:

```
BEGIN
  DBMS_VECTOR.REBUILD_INDEX(
    idx_name                => 'v_hnsw_01',
    table_name              => 'vpt01',
    idx_vector_col          => 'EMBEDDING',
    idx_include_cols        => NULL,
    idx_partitioning_scheme => NULL,
    idx_organization        => 'INMEMORY NEIGHBOR GRAPH',
    idx_distance_metric     => 'EUCLIDEAN',
    idx_accuracy            => 95,
    idx_parameters          => '{
                                 "type" : "HNSW",
                                 "neighbors" : 3,
                                 "efConstruction" : 4
                               }',
    idx_rebuild_mode        => 'FULL'
  );
END;
/
```

- Specify a scalar quantized HNSW index rebuild:

```
BEGIN
  DBMS_VECTOR.REBUILD_INDEX(
    idx_name              => 'V_HNSW_01',
```

```
        idx_vector_col        => 'EMBEDDING',
        idx_quantization_type => 'SCALAR',
        idx_compression_ratio => 4,
        idx_parameters        => '{
                                   "type" : "HNSW",
                                   "neighbors" : 32,
                                   "efConstruction" : 200,
                                   "rescore factor" : 2,
                                   "algorithm" : "uniform_quantization"
                                 }'
      );
    END;
    /
```

UTL_TO_RERANK

Use the DBMS_VECTOR.UTL_TO_RERANK function to reassess and reorder an initial set of results to retrieve more relevant search output.

Purpose

To improve the relevance and quality of search results in both similarity search and Retrieval Augmented Generation (RAG) scenarios.

Reranking improves the quality of information ingested into an LLM by ensuring that the most relevant documents or chunks are prioritized. This helps to reduce hallucinations and improves the accuracy of generated outputs.

For this operation, Oracle AI Vector Search supports reranking models provided by Cohere, Vertex AI, ocigenai, and ONNX (Oracle AI Database as the service provider). However, ocigenai is available only in Dedicated mode.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

```
DBMS_VECTOR.UTL_TO_RERANK(
                QUERY      IN CLOB,
                DOCUMENTS  IN JSON,
                PARAMS     IN JSON default NULL
) return JSON;
```

This function accepts the input containing a query as `CLOB` and a list of documents in `JSON` format. It then processes this information to generate a `JSON` object containing a reranked list of documents, sorted by score.

For example, a reranked output includes:

```
{
    "index"   : "1",
    "score"   : "0.99",
    "content" : "Jupiter boasts an impressive system of 95 known moons."
}
```

Where,

- `index` specifies the position of the document in the list of input text.
- `score` specifies the relevance score.
- `content` specifies the input text corresponding to the index.

QUERY

Specify the search query (typically from an initial search) as `CLOB`.

DOCUMENTS

Specify a JSON array of strings (list of potentially relevant documents to rerank) in the following format:

```
{
  "documents": [
  "string1",
  "string2",
    ...
  ]
}
```

PARAMS

Specify the following list of parameters in JSON format. All these parameters are mandatory.

```
{
  "provider"          : "<service provider>",
  "credential_name"   : "<credential name>",
  "url"               : "<REST endpoint URL for reranking>",
  "model"             : "<reranking model name>",
  ...
}
```

Table 12-11   UTL_TO_RERANK Parameter Details

Parameter	Description
`provider`	Supported REST provider to access for reranking: • `cohere` • `vertexai`

Table 12-11 (Cont.) UTL_TO_RERANK Parameter Details

Parameter	Description
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the reranking model in the form: *`schema.model_name`* If the model name is not schema-qualified, then the schema of the procedure invoker is used.

Additional REST provider parameters:

Optionally, specify additional provider-specific parameters for reranking.

> **Important**
>
> - The following examples are for illustration purposes. For accurate and up-to-date information on additional parameters to use, refer to your third-party provider's documentation.
> - For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

Cohere example:

```
{
  "provider"        : "cohere",
  "credential_name" : "COHERE_CRED",
  "url"             : "https://api.cohere.example.com/rerank",
  "model"           : "rerank-english-v3.0",
  "return_documents": false,
  "top_n"           : 3
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://googleapis.example.com/
default_ranking_config:rank",
  "model"            : "semantic-ranker-512@latest",
```

```
  "ignoreRecordDetailsInResponse" : true,
  "topN"             : 3
  }
```

ONNX (Oracle AI Database as the service provider) example:

```
{
   "provider": "database",
   "model": "<model_name>",
   "return_docs": true,
   "top_n": 3
}
```

OCI Generative AI (Dedicated mode) example:

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/rerankText",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "topN": 3,
  "isEcho": true
}
```

> **Note**
>
> For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Table 12-12 Additional REST Provider Parameter Details

Parameter	Description
`return_documents`	Whether to return search results with original documents or input text (`content`): • `false` (default, also recommended) to not return any input text (return only index and score) • `true` to return input text along with index and score **Note**: With Cohere as the provider, Oracle recommends that you keep this option disabled for better performance. You may choose to enable it for debugging purposes when you need to view the original text.
`ignoreRecordDetailsInResponse`	Whether to return search results with original record details or input text (`content`): • `false` (default) to return input text along with index and score • `true` (recommended) to not return any input text (return only index and score) **Note**: With Vertex AI as the provider, Oracle recommends that you keep this option enabled for better performance. You may choose to disable it for debugging purposes when you need to view the original text.
`top_n` or `topN`	The number of most relevant documents to return.

Examples

- Using Cohere:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "cohere",
  "credential_name": "COHERE_CRED",
  "url": "https://api.cohere.com/v1/rerank",
  "model": "rerank-english-v3.0",
  "return_documents": true,
  "top_n": 3
}';

  reranked_output := dbms_vector.utl_to_rerank(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

- Using Vertex AI:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url": "https://discoveryengine.googleapis.com/v1/projects/1085581009881/
locations/global/rankingConfigs/default_ranking_config:rank",
  "model": "semantic-ranker-512@latest",
  "ignoreRecordDetailsInResponse": false,
  "topN": 3
}';

  reranked_output := dbms_vector.utl_to_rerank(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

End-to-end example:

To run an end-to-end example scenario using this function, see Use Reranking for Better RAG Results.

UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS

Use the `DBMS_VECTOR.UTL_TO_EMBEDDING` and `DBMS_VECTOR.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.

Purpose

To automatically generate one or more vector embeddings from textual documents and images.

- Text to Vector:

  You can perform a text-to-embedding transformation by accessing either Oracle AI Database or a third-party service provider:

  - Oracle AI Database as the service provider (default setting):

    This API calls an ONNX format embedding model that you load into the database.

  - Third-party embedding model:

    This API makes a REST API call to your chosen remote service provider (Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI) or local service provider (Private AI, Ollama).

- Image to Vector:

  You can also perform an image-to-embedding transformation. This API makes a REST call to your chosen image embedding model or multimodal embedding model by Vertex AI. Note that currently Vertex AI is the only supported service provider for this operation.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

- Text to Vector:

```
DBMS_VECTOR.UTL_TO_EMBEDDING (
    DATA           IN CLOB,
    PARAMS         IN JSON default NULL
) return VECTOR;


DBMS_VECTOR.UTL_TO_EMBEDDINGS (
    DATA           IN VECTOR_ARRAY_T,
```

```
    PARAMS         IN JSON default NULL
) return VECTOR_ARRAY_T;
```

- Image to Vector:

```
DBMS_VECTOR.UTL_TO_EMBEDDING (
    DATA           IN BLOB,
    MODALITY       IN VARCHAR2,
    PARAMS         IN JSON default NULL
) return VECTOR;
```

DATA

- Text to Vector:

  UTL_TO_EMBEDDING accepts the input as CLOB containing textual data (text strings or small documents). It then converts the text to a single embedding (VECTOR).

  UTL_TO_EMBEDDINGS converts an array of chunks (VECTOR_ARRAY_T) to an array of embeddings (VECTOR_ARRAY_T).

> **Note**
>
> Although data is a CLOB or a VECTOR_ARRAY_T of CLOB, the maximum input is 4000 characters. If you have input that is greater, you can use UTL_TO_CHUNKS to split the data into smaller chunks before passing in.

- Image to Vector:

  UTL_TO_EMBEDDING accepts the input as BLOB containing media data for media files such as images. It then converts the image input to a single embedding (VECTOR).

A generated embedding output includes:

```
{
    "embed_id"    :  NUMBER,
    "embed_data"  : "VARCHAR2(4000)",
    "embed_vector": "CLOB"
}
```

Where,

- embed_id displays the ID number of each embedding.
- embed_data displays the input text that is transformed into embeddings.
- embed_vector displays the generated vector representations.

MODALITY

For BLOB inputs, specify the type of content to vectorize. The only supported value is image.

> **Note**
>
> For image BLOB input, standalone baseline 8-bit lossy JPEG images are accepted. Unsupported or malformed image input does not produce a successful image decode. At SQL scoring time, this can result in a NULL embedding result. Perform any required image preprocessing or format conversion externally before embedding creation.
>
> For more information about image input restrictions, see ONNX Pipeline Models: Image Embedding.

PARAMS

Specify input parameters in JSON format, depending on the service provider that you want to use.

If using Oracle AI Database as the provider:

```
{
  "provider" : "database",
  "model"    : "<in-database ONNX embedding model name>"
}
```

Table 12-13 Database Provider Parameter Details

Parameter	Description
provider	Specify database (default setting) to use Oracle AI Database as the provider. With this setting, you must load an ONNX format embedding model into the database.
model	User-specified name under which the imported ONNX embedding model is stored in Oracle AI Database. If you do not have an embedding model in ONNX format, then perform the steps listed in Convert Pretrained Models to ONNX Format.

If using a third-party provider:

Set the following parameters along with additional embedding parameters specific to your provider:

- For UTL_TO_EMBEDDING:

  ```
  {
    "provider"        : "<AI service provider>",
    "credential_name" : "<credential name>",
    "url"             : "<REST endpoint URL for embedding service>",
    "model"           : "<REST provider embedding model name>",
    "transfer_timeout": <maximum wait time for the request to complete>,
    "max_count": "<maximum calls to the AI service provider>",
    "<additional REST provider parameter>": "<REST provider parameter
  value>"
  }
  ```

- For `UTL_TO_EMBEDDINGS`:

```
{
  "provider"         : "<AI service provider>",
  "credential_name" : "<credential name>",
  "url"              : "<REST endpoint URL for embedding service>",
  "model"            : "<REST provider embedding model name>",
  "transfer_timeout": <maximum wait time for the request to complete>,
  "batch_size"       : "<number of vectors to request at a time>",
  "max_count": "<maximum calls to the AI service provider>",
  "<additional REST provider parameter>": "<REST provider parameter
value>"
}
```

These are the details on each of the input parameters for `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING` and `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS` when you specify a third-party provider.

When using with Private AI, specify either a `credential_name` for remote service authentication or use `local` in the host key for local, unauthenticated access.

Table 12-14 Third-Party Provider Parameter Details

Parameter	Description
`provider`	Third-party service provider that you want to access for this operation. A REST call is made to the specified provider to access its embedding model. For image input, specify `vertexai`. For text input, specify one of the following values: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `ollama` • `privateai`
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`host`	When using a local service provider, `host` can be used to disable credential by setting the value to `local`. When the `host` parameter is specified, `credential_name` should be omitted. The `host` parameter is applicable **only if** `provider` is one of the following: • `openai` • `ollama` • `privateai`

Table 12-14 (Cont.) Third-Party Provider Parameter Details

Parameter	Description
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the third-party embedding model in the form: *`schema.model_name`* If you do not specify a schema, then the schema of the procedure invoker is used. **Note**: • For Generative AI, all the supported third-party models are listed in Supported Third-Party Provider Operations and Endpoints. • If `privateai` is specified as the `provider`, `model` is not a required parameter. If `model` is not provided, the default model for `privateai` is used. Omitting `model` with any other `provider` value results in an error. • For accurate results, ensure that the chosen text embedding model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. • To get image embeddings, you can use any image embedding model or multimodal embedding model supported by Vertex AI. Multimodal embedding is a technique that vectorizes data from different modalities such as text and images. When using a multimodal embedding model to generate embeddings, ensure that you use the same model to vectorize both types of content (text and images). By doing so, the resulting embeddings are compatible and situated in the same vector space, which allows for effective comparison between the two modalities during similarity searches.
`transfer_timeout`	Maximum time to wait for the request to complete. The default value is `60` seconds. You can increase this value for busy web servers.
`batch_size`	Maximum number of vectors to request at a time. For example, for a batch size of `50`, if `100` chunks are passed, then this API sends two requests with an array of `50` strings each. If `30` chunks are passed (which is lesser than the defined batch size), then the API sends those in a single request. For REST calls, it is more efficient to send a batch of inputs at a time rather than requesting a single input per call. Increasing the batch size can provide better performance, whereas reducing the batch size may reduce memory and data usage, especially if your provider has a rate limit. The default or maximum allowed value depends on the third-party provider settings.
`max_count`	Maximum number of times the API can be called for a given third-party provider. When set to an integer *n*, `max_count` stops the execution of the API for the given provider beyond *n* times. This prevents accidental calling of a third-party over some limit, for example to avoid surpassing the service amount that you have purchased.

Additional third-party provider parameters:

Optionally, specify additional provider-specific parameters.

Table 12-15 Additional REST Provider Parameter Details

Parameter	Description
`input_type`	Type of input to vectorize.

Let us look at some example configurations for all third-party providers:

Important

- The following examples are for illustration purposes. For accurate and up-to-date information on the parameters to use, refer to your third-party provider's documentation.
- For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.
- The generated embedding results may be different between requests for the same input and configuration, depending on your embedding model or floating point precision. However, this does not affect your queries (and provides semantically correct results) because the vector distance will be similar.

Cohere example:

```
{
  "provider"       : "cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.example.com/embed",
  "model"          : "embed-english-light-v2.0",
  "input_type"     : "search_query"
}
```

OCI Generative AI (On-Demand mode) example:

```
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://generativeai.oci.example.com/embedText",
  "model"          : "cohere.embed-english-v3.0",
  "batch_size"     : 10
}
```

OCI Generative AI (Dedicated mode) example:

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/embedText",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1..."
}
```

Note

For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Google AI example:

```
{
  "provider"       : "googleai",
  "credential_name": "GOOGLEAI_CRED",
  "url"            : "https://googleapis.example.com/models/",
  "model"          : "embedding-001",
  "max_count"      : 500
}
```

Hugging Face example:

```
{
  "provider"       : "huggingface",
  "credential_name": "HF_CRED",
  "url"            : "https://api.huggingface.example.com/",
  "model"          : "sentence-transformers/all-MiniLM-L6-v2"
}
```

Ollama example:

```
{
  "provider"       : "ollama",
  "host"           : "local",
  "url"            : "http://localhost:11434/api/embeddings",
  "model"          : "phi3:mini"
}
```

OpenAI example:

```
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.example.com/embeddings",
  "model"          : "text-embedding-3-small"
}
```

OpenAI-compatible provider example:

> **Note**
>
> This is just one example of how to set up the configuration for a number of OpenAI-compatible third-party providers, such as Llamafile and vLLM. The `provider` value must specify `openai` while the `url` value depends on your chosen third-party provider's REST endpoint.
>
> Set `host` to `local` to disable credential. Set `model` to the desired model if the third-party provider requires a model name (for example, Ollama, vLLM, and so on). Otherwise, you can set `model` to any string such as `any` (for example, if using Llamafile).

```
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/chat/completions",
  "host": "local",
  "model": "any"
}
```

Vertex AI example:

```
{
  "provider"       : "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url"            : "https://googleapis.example.com/models/",
  "model"          : "textembedding-gecko:predict"
}
```

Mistral AI example:

```
{
  "provider": "mistralai",
  "credential_name": "MISTRALAI_CRED",
  "url": "https://api.mistral.ai/v1/embeddings",
  "model": "mistral-embed"
}
```

Private AI example:

```
{
  "provider": "privateai",
  "url": "http://host:port/v1/embeddings",
  "host": "local",
  "model": "all-minilm-l12-v2"
}
```

Examples

You can use `UTL_TO_EMBEDDING` in a `SELECT` clause and `UTL_TO_EMBEDDINGS` in a `FROM` clause, as follows:

UTL_TO_EMBEDDING:

- **Text to vector using Generative AI**:

  The following examples use `UTL_TO_EMBEDDING` to generate an embedding with `Hello world` as the input.

  Here, the cohere.embed-english-v3.0 model is used by accessing Generative AI as the provider. You can replace the `model` value with any other supported model that you want to use with Generative AI, as listed in Supported Third-Party Provider Operations and Endpoints.

```
-- declare embedding parameters

var params clob;

begin
 :params := '
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/embedText",
  "model": "cohere.embed-english-v3.0",
  "batch_size": 10
}';
end;
/

-- get text embedding: PL/SQL example

declare
  input clob;
  v vector;
begin
  input := 'Hello world';

  v := dbms_vector.utl_to_embedding(input, json(:params));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/

-- get text embedding: select example

select dbms_vector.utl_to_embedding('Hello world', json(:params)) from
dual;
```

- **Image to vector using Vertex AI**:

  The following examples use `UTL_TO_EMBEDDING` to generate an embedding by accessing the Vertex AI's multimodal embedding model.

Here, the input is `parrots.jpg`, `VEC_DUMP` is a local directory that stores the `parrots.jpg` file, and the modality is specified as `image`.

```
-- declare embedding parameters

var params clob;

begin
  :params := '
{
  "provider": "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url": "https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/
locations/LOCATION/publishers/google/models/",
  "model": "multimodalembedding:predict"
}';
end;
/

-- get image embedding: PL/SQL example

declare
  v vector;
  output clob;
begin
  v := dbms_vector.utl_to_embedding(
    to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
  output := vector_serialize(v);
  dbms_output.put_line('vector data=' || dbms_lob.substr(output, 100) ||
'...');
end;
/

-- get image embedding: select example

select dbms_vector.utl_to_embedding(
  to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
```

- **Text to vector using in-database embedding model**:

  The following example uses `UTL_TO_EMBEDDING` to generate a vector embedding by calling an ONNX format embedding model (`doc_model`) loaded into Oracle AI Database.

  Here, the provider is `database`, and the input is `hello`.

```
var params clob;
exec :params := '{"provider":"database", "model":"doc_model"}';

select dbms_vector.utl_to_embedding('hello', json(:params)) from dual;
```

  For complete example, see Convert Text String to Embedding Within Oracle AI Database.

- **Text to vector using Private AI with HTTP and HTTPS**:

  The first part of this example demonstrates how to use Private AI if it is configured for HTTPS with a credential. It uses `UTL_TO_EMBEDDING` to generate an embedding with a string as input. The `provider` is specified as `privateai` to indicate that the Private AI Services Container should be used as the local REST endpoint provider.

For more information about the Private AI Services Container, see *Oracle Private AI Services Container User's Guide*.

```
define host=localhost
define port=9092
define wallet_path=<path to wallet>
define wallet_password=<wallet password>

exec utl_http.set_wallet('file:&wallet_path', '&wallet_password');

declare
  input clob;
  v vector;
begin
  input := 'The quick brown fox jumped over the fence.';
  v := dbms_vector.utl_to_embedding(
    input,
    json('{"provider": "privateai",
      "url": "https://&host:&port/v1/embeddings",
      "credential_name": "ORACLEAI_CRED",
      "model": "all-MiniLM-L12-v2" }'));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

The second part of this example implements a similar embedding generation but uses Private AI configured for HTTP without a credential. Note the use of `host=local` to skip passing a credential.

```
define host=localhost
define port=9091

declare
  input clob;
  v vector;
begin
  input := 'The quick brown fox jumped over the fence.';
  v := dbms_vector.utl_to_embedding(
    input,
    json('{"provider": "privateai",
      "url": "http://&host:&port/v1/embeddings",
      "host": "local",
      "model": "all-MiniLM-L12-v2" }'));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
```

```
end;
/
```

- **Text to vector using Private AI, end-to-end:**

  This example demonstrates how to use Private AI to generate vector embeddings with `UTL_TO_EMBEDDING`, including creating a wallet, adding a signed certificate, and connecting to a host before generating text embeddings.

  For more information about the Private AI Services Container, see *Oracle Private AI Services Container User's Guide*.

  1. Use the command-line tool `orapki` to create an Oracle wallet and add the signed certificate.

```
mkdir /path/to/wallet
orapki wallet create -wallet /path/to/wallet -pwd <walletpassword>
orapki wallet add -wallet /path/to/wallet -trusted_cert -cert cert.pem -
pwd <walletpassword>
```

  2. Add permissions to connect to a host and access the wallet.

```
sqlplus / as sysdba

BEGIN
    DBMS_NETWORK_ACL_ADMIN.APPEND_HOST_ACE(
        host => 'your-fully-qualified-hostname',
        lower_port => 8080,
        upper_port => 8080,
        ace => xs$ace_type(privilege_list => xs$name_list('connect'),
                           principal_name => 'vectordb',
                           principal_type => xs_acl.ptype_db));
END;
/


BEGIN
    DBMS_NETWORK_ACL_ADMIN.APPEND_WALLET_ACE(
        wallet_path => 'file:/path/to/wallet',
        ace => xs$ace_type(
           privilege_list => xs$name_list('use_client_certificates',
'use_passwords'),
           principal name => 'vectordb',
           principal_type => xs_acl.ptype_db));
END;
/
```

  3. Now you can get the text embeddings:

```
sqlplus vectordb/vectord@pdb

-- create credential

exec dbms_vector.drop_credential('ORACLEAI_CRED');

declare
  jo json_object_t;
```

```
begin
  jo := json_object_t();
  jo.put('access_token', '<api key>');
  dbms_vector.create_credential(
    credential_name   => 'ORACLEAI_CRED',
    params            => json(jo.to_string));
end;
/

set serveroutput on

-- set wallet
exec utl_http.set_wallet('file:/path/to/wallet', '<walletpassword>');
-- get text embeddings
declare
  input clob;
  v vector;
begin
  input := 'The quick brown fox jumped over the fence.';
  v := dbms_vector.utl_to_embedding(
    input,
    json('{"provider": "privateai",
      "url": "https://<host>:<port>/v1/embeddings",
      "credential_name": "ORACLEAI_CRED",
      "model": "all-MiniLM-L12-v2" }'));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

- **End-to-end examples**:

  To run various end-to-end example scenarios using `UTL_TO_EMBEDDING`, see Generate Embedding.

UTL_TO_EMBEDDINGS:

- **Text to vector using in-database embedding model**:

  The following example uses `UTL_TO_EMBEDDINGS` to generate an array of embeddings by calling an ONNX format embedding model (`doc_model`) loaded into Oracle AI Database.

  Here, the provider is `database`, and the input is a PDF document stored in the `documentation_tab` table. As you can see, you first use `UTL_TO_CHUNKS` to split the data into smaller chunks before passing in to `UTL_TO_EMBEDDINGS`.

```
CREATE TABLE doc_chunks as
(select dt.id doc_id, et.embed_id, et.embed_data,
to_vector(et.embed_vector) embed_vector
 from
   documentation_tab dt,
   dbms_vector.utl_to_embeddings(
       dbms_vector.utl_to_chunks(dbms_vector.utl_to_text(dt.data),
json('{"normalize":"all"}')),
       json('{"provider":"database", "model":"doc_model"}')) t,
```

```
   JSON_TABLE(t.column_value, '$[*]' COLUMNS (embed_id NUMBER PATH
'$.embed_id', embed_data VARCHAR2(4000) PATH '$.embed_data', embed_vector
CLOB PATH '$.embed_vector')) et
);
```

For complete example, see SQL Quick Start Using a Vector Embedding Model Uploaded into the Database.

- **End-to-end examples**:

  To run various end-to-end example scenarios using `UTL_TO_EMBEDDINGS`, see Perform Chunking With Embedding.

UTL_TO_GENERATE_TEXT

Use the `DBMS_VECTOR.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.

Purpose

To communicate with Large Language Models (LLMs) through natural language conversations. You can generate a textual answer, description, or summary for prompts and images, given as input to LLM-powered chat interfaces.

- Prompt to Text:

  A prompt can be an input text string, such as a question that you ask an LLM. For example, "`What is Oracle Text?`". A prompt can also be a command, such as "`Summarize the following ...`", "`Draft an email asking for ...`", or "`Rewrite the following ...`", and can include results from a search. The LLM responds with a textual answer or description based on the specified task in the prompt.

  For this operation, this API makes a REST call to your chosen remote third-party provider (Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI) or local third-party provider (Ollama).

- Image to Text:

  You can also prompt with a media file, such as an image, to extract text from pictures or photos. You supply a text question as the prompt (such as "`What is this image about?`" or "`How many birds are there in this painting?`") along with the image. The LLM responds with a textual analysis or description of the contents of the image.

  For this operation, this API makes a REST call to your chosen remote third-party provider (Google AI, Hugging Face, OpenAI, or Vertex AI) or local third-party provider (Ollama).

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

This function accepts the input as `CLOB` containing text data (for textual prompts) or as `BLOB` containing media data (for media files such as images). It then processes this information to generate a new `CLOB` containing the generated text.

- Prompt to Text:

```
DBMS_VECTOR.UTL_TO_GENERATE_TEXT (
        DATA          IN CLOB,
        PARAMS        IN JSON default NULL
) return CLOB;
```

- Image to Text:

```
DBMS_VECTOR.UTL_TO_GENERATE_TEXT(
          TEXT_DATA      IN CLOB,
          MEDIA_DATA     IN BLOB,
          MEDIA_TYPE     IN VARCHAR2 default 'image/jpeg',
          PARAMS         IN JSON default NULL
) return CLOB;
```

DATA and TEXT_DATA

Specify the textual prompt as `CLOB` for the `DATA` or `TEXT_DATA` clause.

> **Note**
>
> Hugging Face uses an image captioning model that does not require a prompt, when giving an image as input. If you input a prompt along with an image, then the prompt will be ignored.

MEDIA_DATA

Specify the `BLOB` file, such as an image or a visual PDF file.

MEDIA_TYPE

Specify the image format for the given image or visual PDF file (`BLOB` file) in one of the supported image data MIME types. For example:

- For PNG: `image/png`
- For JPEG: `image/jpeg`
- For PDF: `application/pdf`

> **Note**
>
> For a complete list of the supported image formats, refer to your third-party provider's documentation.

PARAMS

Specify the following input parameters in JSON format, depending on the service provider that you want to access for text generation:

```
{
  "provider"         : "<AI service provider>",
  "credential_name"  : "<credential name>",
  "url"              : "<REST endpoint URL for text generation service>",
  "model"            : "<text generation model name>",
  "transfer_timeout" : <maximum wait time for the request to complete>,
  "max_count": "<maximum calls to the AI service provider>",
  "<additional REST provider parameter>": "<REST provider parameter value>"
}
```

Table 12-16 UTL_TO_GENERATE_TEXT Parameter Details

Parameter	Description
`provider`	Supported REST provider that you want to access to generate text. Specify one of the following values: For `CLOB` input: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `anthropic` • `ollama` For `BLOB` input: • `googleai` • `huggingface` • `openai` • `vertexai` • `mistralai` • `anthropic` • `ollama`

Table 12-16 (Cont.) UTL_TO_GENERATE_TEXT Parameter Details

Parameter	Description
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`host`	When using a local service provider, `host` can be used to disable credential by setting the value to `local`. When the `host` parameter is specified, `credential_name` should be omitted. The `host` parameter is applicable **only if** `provider` is one of the following: • `openai` • `ollama`
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the third-party text generation model in the form: *`schema.model_name`* If the model name is not schema-qualified, then the schema of the procedure invoker is used. **Note**: For Generative AI, all the supported third-party models are listed in Supported Third-Party Provider Operations and Endpoints.
`transfer_timeout`	Maximum time to wait for the request to complete. The default value is `60` seconds. You can increase this value for busy web servers.
`max_count`	Maximum number of times the API can be called for a given third-party provider. When set to an integer *n*, `max_count` stops the execution of the API for the given provider beyond *n* times. This prevents accidental calling of a third-party over some limit, for example to avoid surpassing the service amount that you have purchased.

Additional third-party provider parameters:

Optionally, specify additional provider-specific parameters.

Table 12-17 Additional REST Provider Parameter Details

Parameter	Description
`max_tokens`	Maximum number of tokens in the output text.
`temperature`	Degree of randomness used when generating the output text, in the range of `0.0-5.0`. To generate the same output for a prompt, use `0`. To generate a random new text for that prompt, increase the temperature. **Note**: Start with the temperature set to `0`. If you do not require random results, a recommended temperature value is between `0` and `1`. A higher value is not recommended because a high temperature may produce creative text, which might also include hallucinations.

Table 12-17 (Cont.) Additional REST Provider Parameter Details

Parameter	Description
`topP`	Probability of tokens in the output, in the range of `0.0–1.0`. A lower value provides less random responses and a higher value provides more random responses.
`candidateCount`	Number of response variations to return, in the range of `1-4`.
`maxOutputTokens`	Maximum number of tokens to generate for each response.

Let us look at some example configurations for all third-party providers:

> **Important**
>
> - The following examples are for illustration purposes. For accurate and up-to-date information on additional parameters to use, refer to your third-party provider's documentation.
> - For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Cohere example:

```
{
  "provider"       : "cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.example.com/chat",
  "model"          : "command"
}
```

OCI Generative AI (On-Demand mode) example:

> **Note**
>
> For Generative AI, if you want to pass any additional REST provider-specific parameters, then you must enclose those in `chatRequest`.

```
{
  "provider": "ocigenai",
  "credential_name" : "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "meta.llama-3.3-70b-instruct",
  "chatRequest": {
    "maxTokens": 256
  }
}
```

OCI Generative AI (Dedicated mode) examples:

For Cohere models, use "apiFormat":"COHERE"

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "chatRequest": {
    "apiFormat":"COHERE"
  }
}
```

For other models, use "apiFormat":"GENERIC"

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "chatRequest": {
    "apiFormat":"GENERIC"
  }
}
```

Note

For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Google AI example:

```
{
  "provider"        : "googleai",
  "credential_name" : "GOOGLEAI_CRED",
  "url"             : "https://googleapis.example.com/models/",
  "model"           : "gemini-pro:generateContent"
}
```

Hugging Face example:

```
{
  "provider"        : "huggingface",
  "credential_name" : "HF_CRED",
  "url"             : "https://api.huggingface.example.com/models/",
  "model"           : "gpt2"
}
```

Ollama example:

```
{
  "provider"       : "ollama",
  "host"           : "local",
  "url"            : "http://localhost:11434/api/generate",
  "model"          : "phi3:mini"
}
```

OpenAI example:

```
{
  "provider"        : "openai",
  "credential_name" : "OPENAI_CRED",
  "url"             : "https://api.openai.example.com",
  "model"           : "gpt-4o-mini",
  "max_tokens"      : 60,
  "temperature"     : 1.0
}
```

OpenAI-compatible provider examples:

> **Note**
>
> This is just one example of how to set up the configuration for a number of OpenAI-compatible third-party providers, such as Llamafile, vLLM, and xAI. The `provider` value must specify `openai` while the `url` value depends on your chosen third-party provider's REST endpoint.
>
> Set `host` to `local` to disable credential. Set `model` to the desired model if the third-party provider requires a model name (for example, Ollama, vLLM, and so on). Otherwise, you can set `model` to any string such as `any` (for example, if using Llamafile).

```
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/chat/completions",
  "host": "local",
  "model": "any"
}
```

```
{
   "provider": "openai",
   "credential_name": "XAI_CRED",
   "url": "https://api.x.ai/v1/chat/completions",
   "model": "grok-4"
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://googleapis.example.com/models/",
  "model"            : "gemini-1.0-pro:generateContent",
  "generation_config": {
                        "temperature"    : 0.9,
                        "topP"           : 1,
                        "candidateCount" : 1,
                        "maxOutputTokens": 256
                       }
}
```

Mistral AI example:

```
{
   "provider": "mistralai",
   "credential_name": "MISTRALAI_CRED",
   "url": "https://api.mistral.ai/v1/chat/completions",
   "model": "mistral-small-latest"
}
```

Anthropic example:

```
{
   "provider": "anthropic",
   "credential_name": "ANTHROPIC_CRED",
   "url": "https://api.anthropic.com/v1/messages",
   "model": "claude-3-opus-20240229",
   "max_tokens": 256
}
```

Examples

- Prompt to Text:

  The following statements generate a text response by making a REST call to Generative AI. The prompt given here is "`What is Oracle Text?`".

  Here, the cohere.command-r-16k and meta.llama-3.1-70b-instruct models are used. You can replace the `model` value with any other supported model that you want to use with Generative AI, as listed in Supported Third-Party Provider Operations and Endpoints.

  Using the cohere.command-r-16k model:

```
-- select example

var params clob;
exec :params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
```

```
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

select dbms_vector.utl_to_generate_text(
 'What is Oracle Text?',
 json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';

  params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

  output := dbms_vector.utl_to_generate_text(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

Using the meta.llama-3.1-70b-instruct model:

```
-- select example

var params clob;
exec :params := '
{
   "provider"       : "ocigenai",
   "credential_name": "OCI_CRED",
   "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
   "model"          : "meta.llama-3.1-70b-instruct",
```

```
   "chatRequest"    : {
                       "topK" : 1
                      }
}';

select dbms_vector.utl_to_generate_text(
 'What is Oracle Text?',
 json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';

  params := '
{
   "provider"       : "ocigenai",
   "credential_name": "OCI_CRED",
   "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
   "model"          : "meta.llama-3.1-70b-instruct",
   "chatRequest"    : {
                       "topK" : 1
                      }
}';

  output := dbms_vector.utl_to_generate_text(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

End-to-end examples:

To run end-to-end example scenarios, see Generate Text Response.

- Image to Text:

  The following statements generate a text response by making a REST call to OpenAI. Here, the input is an image (`sample_image.jpeg`) along with the prompt "`Describe this image?`".

```
-- select example

var input clob;
var media_data blob;
var media_type clob;
```

```
var params clob;

begin
  :input := 'Describe this image';
  :media_data := load_blob_from_file('DEMO_DIR', 'sample_image.jpeg');
  :media_type := 'image/jpeg';
  :params := '
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60
}';
end;
/

select dbms_vector.utl_to_generate_text(:input, :media_data, :media_type,
json(:params));

-- PL/SQL example

declare
  input clob;
  media_data blob;
  media_type varchar2(32);
  params clob;
  output clob;

begin
  input := 'Describe this image';
  media_data := load_blob_from_file('DEMO_DIR', 'image_file');
  media_type := 'image/jpeg';
  params := '
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60
}';

  output := dbms_vector.utl_to_generate_text(
    input, media_data, media_type, json(params));
  dbms_output.put_line(output);

  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
  if media_data is not null then
    dbms_lob.freetemporary(media_data);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
```

```
end;
/
```

End-to-end examples:

To run end-to-end example scenarios, see Describe Image Content.

DBMS_VECTOR_CHAIN

The DBMS_VECTOR_CHAIN package enables advanced operations with Oracle AI Vector Search, such as chunking and embedding data along with text generation and summarization capabilities. It is more suitable for text processing with similarity search and hybrid search, using functionality that can be pipelined together for an end-to-end search.

This table lists the DBMS_VECTOR_CHAIN subprograms and briefly describes them.

Table 12-18 DBMS_VECTOR_CHAIN Package Subprograms

Subprogram	Description
Chainable Utility (UTL) Functions: These functions are a set of modular and flexible functions within vector utility PL/SQL packages. You can chain these together to automate end-to-end data transformation and similarity search operations.	
UTL_TO_TEXT	Extracts plain text data from documents
UTL_TO_CHUNKS	Splits data into smaller pieces or chunks
UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS	Converts text or an image to one or more vector embeddings
UTL_TO_SUMMARY	Extracts a summary from documents
UTL_TO_GENERATE_TEXT	Generates text for a prompt (input string) or an image
Credential Helper Procedures: These procedures enable you to securely manage authentication credentials in the database. You require these credentials to enable access to third-party service providers for making REST calls.	
CREATE_CREDENTIAL	Creates a credential name
DROP_CREDENTIAL	Drops an existing credential name
Preference Helper Procedures: These procedures enable you to manage vectorizer preferences, to be used with the CREATE_HYBRID_VECTOR_INDEX and ALTER_INDEX SQL statements when creating or managing hybrid vector indexes.	
CREATE_PREFERENCE	Creates a vectorizer preference
DROP_PREFERENCE	Drops an existing vectorizer preference
Chunker Helper Procedures: These procedures enable you to configure vocabulary and language data (abbreviations), to be used with the VECTOR_CHUNKS SQL function or UTL_TO_CHUNKS PL/SQL function.	
CREATE_VOCABULARY	Loads your token vocabulary file into the database
DROP_VOCABULARY	Removes existing vocabulary data
CREATE_LANG_DATA	Loads your language data file into the database
DROP_LANG_DATA	Removes existing abbreviation data
Data Access Function: This function enables you to enhance search operations.	

Table 12-18 (Cont.) DBMS_VECTOR_CHAIN Package Subprograms

Subprogram	Description
UTL_TO_RERANK	Reorders search results for a more relevant output

> **Note**
>
> The `DBMS_VECTOR_CHAIN` package requires you to install the `CONTEXT` component of Oracle Text, an Oracle AI Database technology that provides indexing, term extraction, text analysis, text summarization, word and theme searching, and other utilities.
>
> Due to underlying dependance on the text processing capabilities of Oracle Text, note that both the `UTL_TO_TEXT` and `UTL_TO_SUMMARY` chainable utility functions and all the chunker helper procedures are available only in this package through Oracle Text.

- CREATE_CREDENTIAL
  Use the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` credential helper procedure to create a credential name for storing user authentication details in Oracle AI Database.
- CREATE_LANG_DATA
  Use the `DBMS_VECTOR_CHAIN.CREATE_LANG_DATA` chunker helper procedure to load your own language data file into the database.
- CREATE_PREFERENCE
  Use the `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` helper procedure to create a vectorizer preference, to be used when creating or updating hybrid vector indexes.
- CREATE_VOCABULARY
  Use the `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY` chunker helper procedure to load your own token vocabulary file into the database.
- DROP_CREDENTIAL
  Use the `DBMS_VECTOR_CHAIN.DROP_CREDENTIAL` credential helper procedure to drop an existing credential name from the data dictionary.
- DROP_LANG_DATA
  Use the `DBMS_VECTOR_CHAIN.DROP_LANG_DATA` chunker helper procedure to remove abbreviation data from the data dictionary.
- DROP_PREFERENCE
  Use the `DBMS_VECTOR_CHAIN.DROP_PREFERENCE` preference helper procedure to remove an existing Vectorizer preference.
- DROP_VOCABULARY
  Use the `DBMS_VECTOR_CHAIN.DROP_VOCABULARY` chunker helper procedure to remove vocabulary data from the data dictionary.
- UTL_TO_RERANK
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_RERANK` function to reassess and reorder an initial set of results to retrieve more relevant search output.
- UTL_TO_CHUNKS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS` chainable utility function to split a large plain text document into smaller chunks of text.

- UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING` and `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.
- UTL_TO_GENERATE_TEXT
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.
- UTL_TO_SUMMARY
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY` chainable utility function to generate a summary for textual documents.
- UTL_TO_TEXT
  Use the `DBMS_VECTOR_CHAIN.UTL_TO_TEXT` chainable utility function to convert an input document (for example, PDF, DOC, JSON, XML, or HTML) to plain text.
- Supported Languages and Data File Locations
  These are the supported languages for which language data files are distributed by default in the specified directories.

CREATE_CREDENTIAL

Use the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` credential helper procedure to create a credential name for storing user authentication details in Oracle AI Database.

Purpose

To securely manage authentication credentials in the database. You require these credentials to enable access during REST API calls to your chosen third-party service provider, such as Cohere, Google AI, Hugging Face, Oracle Cloud Infrastructure (OCI) Generative AI, OpenAI, or Vertex AI.

A credential name holds authentication parameters, such as user name, password, access token, private key, or fingerprint.

Note that if you are using Oracle AI Database as the service provider, then you do not need to create a credential.

Warning

Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.

Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

```
DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL (
    CREDENTIAL_NAME     IN VARCHAR2,
    PARAMS              IN JSON DEFAULT NULL
);
```

CREDENTIAL_NAME

Specify a name of the credential that you want to create for holding authentication parameters.

PARAMS

Specify authentication parameters in JSON format, based on your chosen service provider.

Generative AI requires the following authentication parameters:

```
{
"user_ocid"       : "<user ocid>",
"tenancy_ocid"    : "<tenancy ocid>",
"compartment_ocid": "<compartment ocid>",
"private_key"     : "<private key>",
"fingerprint"     : "<fingerprint>"
}
```

Cohere, Google AI, Hugging Face, OpenAI, and Vertex AI require the following authentication parameter:

```
{ "access_token": "<access token>" }
```

Table 12-19 Parameter Details

Parameter	Description
`user_ocid`	Oracle Cloud Identifier (OCID) of the user, as listed on the User Details page in the OCI console.
`tenancy_ocid`	OCID of your tenancy, as listed on the Tenancy Details page in the OCI console.
`compartment_ocid`	OCID of your compartment, as listed on the Compartments information page in the OCI console.

`private_key`	OCI private key.
	Note: The generated private key may appear as:
	`-----BEGIN RSA PRIVATE KEY-----`
	`<private key string>`
	`-----END RSA PRIVATE KEY-----`
	You pass the `<private key string>` value (excluding the `BEGIN` and `END` lines), either as a single line or as multiple lines.

`fingerprint`	Fingerprint of the OCI profile key, as listed on the User Details page under API Keys in the OCI console.
`access_token`	Access token obtained from your third-party service provider.

Required Privilege

You need the `CREATE CREDENTIAL` privilege to call this API.

Examples

- For Generative AI:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();

jo.put('user_ocid','ocid1.user.oc1..aabbalbbaa1112233aabbaabb1111222aa1111b
b');

jo.put('tenancy_ocid','ocid1.tenancy.oc1..aaaaalbbbb1112233aaaabbaa1111222a
aa111a');

jo.put('compartment_ocid','ocid1.compartment.oc1..ababalabab1112233abababab
1111222aba11ab');
  jo.put('private_key','AAAaaaBBB11112222333...AAA111AAABBB222aaa1a/+');
  jo.put('fingerprint','01:1a:a1:aa:12:a1:12:1a:ab:12:01:ab:a1:12:ab:1a');
  dbms_vector_chain.create_credential(
    credential_name   => 'OCI_CRED',
    params            => json(jo.to_string));
end;
/
```

- For Cohere:

```
declare
  jo json_object_t;
begin
  jo := json_object_t();
  jo.put('access_token', 'A1Aa0abA1AB1a1Abc123ab1A123ab123AbcA12a');
  dbms_vector_chain.create_credential(
    credential_name   => 'COHERE_CRED',
    params            => json(jo.to_string));
end;
/
```

- For OpenAI:

```
begin
  dbms_vector_chain.create_credential (
    credential_name => 'OPENAI_CRED',
    params => json('{ "access_token": "
A1Aa0abA1AB1a1Abc123ab1A123ab123AbcA12a"}') );
end;
/
```

End-to-end examples:

To run end-to-end example scenarios using this procedure, see Use LLM-Powered APIs to Generate Summary and Text.

CREATE_LANG_DATA

Use the `DBMS_VECTOR_CHAIN.CREATE_LANG_DATA` chunker helper procedure to load your own language data file into the database.

Purpose

To create custom language data for your chosen language (specified using the `language` chunking parameter).

A language data file contains language-specific abbreviation tokens. You can supply this data to the chunker to help in accurately determining sentence boundaries of chunks, by using knowledge of the input language's end-of-sentence (EOS) punctuations, abbreviations, and contextual rules.

Usage Notes

- All supported languages are distributed with the default language-specific abbreviation dictionaries. You can create a language data based on the abbreviation tokens loaded in the `schema.table.column`, using a user-specified language data name (`PREFERENCE_NAME`).
- After loading your language data, you can use language-specific chunking by specifying the `language` chunking parameter with `VECTOR_CHUNKS` or `UTL_TO_CHUNKS`.
- You can query these data dictionary views to access existing language data:
  - `ALL_VECTOR_LANG` displays all available languages data.
  - `USER_VECTOR_LANG` displays languages data from the schema of the current user.
  - `ALL_VECTOR_ABBREV_TOKENS` displays abbreviation tokens from all available language data.
  - `USER_VECTOR_ABBREV_TOKENS` displays abbreviation tokens from the language data owned by the current user.

Syntax

```
DBMS_VECTOR_CHAIN.CREATE_LANG_DATA (
    PARAMS       IN JSON default NULL
);
```

PARAMS

Specify the input parameters in JSON format:

```
{
    table_name,
    column_name,
    language,
    preference_name
}
```

Table 12-20 Parameter Details

Parameter	Description	Required	Default Value
`table_name`	Name of the table (along with the optional table owner) in which you want to load the language data	Yes	No value
`column_name`	Column name in the language data table in which you want to load the language data	Yes	No value
`language`	Any supported language name, as listed in Supported Languages and Data File Locations	Yes	No value
`preference_name`	User-specified preference name for this language data	Yes	No value

Example

```
declare
    params CLOB := '{"table_name"      : "eos_data_1",
                     "column_name"     : "token",
                     "language"        : "indonesian",
                     "preference_name" : "my_lang_1"}';
begin
    DBMS_VECTOR_CHAIN.CREATE_LANG_DATA(
        JSON (params));
end;
/
```

End-to-end example:

To run an end-to-end example scenario using this procedure, see Create and Use Custom Language Data.

Related Topics

- VECTOR_CHUNKS
- UTL_TO_CHUNKS
- Text Processing Views

CREATE_PREFERENCE

Use the `DBMS_VECTOR_CHAIN.CREATE_PREFERENCE` helper procedure to create a vectorizer preference, to be used when creating or updating hybrid vector indexes.

Purpose

To create a vectorizer preference.

This allows you to customize vector search parameters of a hybrid vector indexing pipeline. The goal of a vectorizer preference is to provide you with a straightforward way to configure how to chunk or embed your documents, without requiring a deep understanding of various chunking or embedding strategies.

Usage Notes

A **vectorizer** preference is a JSON object that collectively holds user-specified values related to the following chunking, embedding, or vector index creation parameters:

- Chunking (`UTL_TO_CHUNKS` and `VECTOR_CHUNKS`)
- Embedding (`UTL_TO_EMBEDDING`, `UTL_TO_EMBEDDINGS`, and `VECTOR_EMBEDDING`)
- Vector index creation (`distance`, `accuracy`, and `vector_idxtype`)

All vector index preferences follow the same JSON syntax as defined for their corresponding `DBMS_VECTOR` and `DBMS_VECTOR_CHAIN` APIs.

After creating a vectorizer preference, you can use the `VECTORIZER` parameter to pass this preference name in the `paramstring` of the `PARAMETERS` clause for `CREATE_HYBRID_VECTOR_INDEX` and `ALTER_INDEX` SQL statements.

Creating a preference is optional. If you do not specify any optional preference, then the index is created with system defaults.

Syntax

```
DBMS_VECTOR_CHAIN.CREATE_PREFERENCE (
    PREF_NAME     IN VARCHAR2,
    PREF_TYPE     IN VARCHAR2,
    PARAMS        IN JSON default NULL
);
```

PREF_NAME

Specify the name of the vectorizer preference to create.

PREF_TYPE

Type of preference. The only supported preference type is:

`DBMS_VECTOR_CHAIN.VECTORIZER`

PARAMS

Specify vector search-specific parameters in JSON format:

- Embedding Parameter
- Chunking Parameters
- Vector Index Parameters
- Paths Parameter

Embedding Parameter:

```
{ "model" : <embedding_model_for_vector_generation> }
```

For example:

```
{ "model" : MY_INDB_MODEL }
```

`model` specifies the name under which your ONNX embedding model is stored in the database.

If you do not have an in-database embedding model in ONNX format, then perform the steps listed in *Oracle AI Database AI Vector Search User's Guide*.

If creating a vectorizer preference with externally hosted vector embedding models:

```
{
  "embedder_spec": {
    "provider"          : provider_name,
    "url"               : embedding_endpoint_url,
    "proxy"             : proxy_host_and_port,
    "no_proxy"          : no_proxy_list,
    "credential_name"   : db_credential_name,
    "model"             : embedding_model_name
  }
}
```

Set the following parameters. In addition to the embedding parameters documented below, you may include extra parameters that are specific to your embedding provider/model. These additional parameters are optional.

Table 12-21 Embedding parameters

Parameter	Description
`provider`	Third-party service provider that you want to access for this operation. A REST call is made to the specified provider to access its embedding model. For image input, specify `vertexai`. For text input, specify one of the following values: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `ollama` • `privateai`
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`proxy`	Optional. Proxy host:port used for outbound HTTP(S) calls to the embedding endpoint (common in enterprise networks).
`no_proxy`	Optional. Comma-separated list of hosts/domains/IP ranges that should bypass the proxy (for example, local addresses and internal domains).

Table 12-21 (Cont.) Embedding parameters

Parameter	Description
`credential_name`	Name of the credential. A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL. **Note** When using privateai, a credential is not required.
`model`	Name of the third-party embedding model. **Note**: • For Generative AI, all the supported third-party models are listed in Supported Third-Party Provider Operations and Endpoints. • For accurate results, ensure that the chosen text embedding model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. • To get image embeddings, you can use any image embedding model or multimodal embedding model supported by Vertex AI. Multimodal embedding is a technique that vectorizes data from different modalities such as text and images. When using a multimodal embedding model to generate embeddings, ensure that you use the same model to vectorize both types of content (text and images). By doing so, the resulting embeddings are compatible and situated in the same vector space, which allows for effective comparison between the two modalities during similarity searches.

Chunking Parameters:

```
{
    "by"           :  mode,
    "max"          :  max,
    "overlap"      :  overlap,
    "split"        :  split_condition,
    "vocabulary"   :  vocabulary_name,
    "language"     :  nls_language,
```

```
    "normalize"    :  normalize_mode,
    "extended"     :  boolean
}
```

For example:

```
JSON(
 '{ "by"           :    "vocabulary",
    "max"          :    "100",
    "overlap"      :    "0",
    "split"        :    "none",
    "vocabulary"   :    "myvocab",
    "language"     :    "american",
    "normalize"    :    "all"
  }')
```

If you specify `split` as `custom` and `normalize` as `options`, then you must additionally specify the `custom_list` and `norm_options` parameters, respectively:

```
JSON(
 '{ "by"           :    "vocabulary",
    "max"          :    "100",
    "overlap"      :    "0",
    "split"        :    "custom",
    "custom_list"  :    [ "<p>" , "<s>" ],
    "vocabulary"   :    "myvocab",
    "language"     :    "american",
    "normalize"    :    "options",
    "norm_options" :    [ "whitespace" ]
  }')
```

The following table describes all the chunking parameters:

Parameter	JSON Value Type	Description and Acceptable Values
`limit`	Number, or string containing a number	Specifies the maximum number of bytes for a chunk. Use this parameter to avoid exceeding the maximum size of the output `VARCHAR2` chunk text. If `limit` is omitted, the maximum size is `4000` bytes unless `extended` is enabled, in which case the maximum size is `32767` bytes. **Default value**: omitted

Parameter	JSON Value Type	Description and Acceptable Values
by	String	Specify a mode for splitting your data, that is, to split by counting the number of characters, words, or vocabulary tokens. **Valid values**: • `characters` (or `chars`): Splits by counting the number of characters. • `words`: Splits by counting the number of words. Words are defined as sequences of alphabetic characters, sequences of digits, individual punctuation marks, or symbols. For segmented languages without white space word boundaries (such as Chinese, Japanese, or Thai), each native character is considered a word (that is, unigram). • `vocabulary`: Splits by counting the number of vocabulary tokens. Vocabulary tokens are words or word pieces, recognized by the vocabulary of the tokenizer that your embedding model uses. You can load your vocabulary file using the chunker helper API `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY`. **Note**: For accurate results, ensure that the chosen model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. **Default value**: `words`
`vocabulary`	String	Specify vocabulary when `"by": "vocabulary"` is used.

Parameter	JSON Value Type	Description and Acceptable Values
`max`	Number, or string containing a number	Specify a limit on the maximum size of each chunk. This setting splits the input text at a fixed point where the maximum limit occurs in the larger text. The units of `max` correspond to the `by` mode, that is, to split data when it reaches the maximum size limit of a certain number of characters, words, numbers, punctuation marks, or vocabulary tokens. **Valid values**: • `by characters`: `50` to `4000` characters • `by words`: `10` to `1000` words • `by vocabulary`: `10` to `1000` tokens **Default value**: `100`

<table>
<tr><th>Parameter</th><th>JSON Value Type</th><th>Description and Acceptable Values</th></tr>
<tr><td><code>split [by]</code></td><td>String</td><td>
Specify where to split the input text when it reaches the maximum size limit. This helps to keep related data together by defining appropriate boundaries for chunks.
Valid values:

 <code>none</code>:
Splits at the <code>max</code> limit of characters, words, or vocabulary tokens.
 <code>newline</code>, <code>blankline</code>, and <code>space</code>:
These are single-split character conditions that split at the last split character before the <code>max</code> value.
Use <code>newline</code> to split at the end of a line of text. Use <code>blankline</code> to split at the end of a blank line (sequence of characters, such as two newlines). Use <code>space</code> to split at the end of a blank space.
 <code>recursively</code>:
This is a multiple-split character condition that breaks the input text using an ordered list of characters (or sequences).
<code>recursively</code> is predefined as <code>BLANKLINE</code>, <code>newline</code>, <code>space</code>, <code>none</code> in this order:
1. If the input text is more than the <code>max</code> value, then split by the first split character.
2. If that fails, then split by the second split character.
3. And so on.
4. If no split characters exist, then split by <code>max</code> wherever it appears in the text.
 <code>sentence</code>:
This is an end-of-sentence split condition that breaks the input text at a sentence boundary.
This condition automatically determines sentence boundaries by using knowledge of the input language's sentence punctuation and contextual rules. This language-specific condition relies mostly on end-of-sentence (EOS) punctuations and common abbreviations.
Contextual rules are based on word information, so this condition is only valid when splitting the text by words or vocabulary (not by characters).
Note: This condition obeys the <code>by word</code> and <code>max</code> settings, and thus may not determine accurate sentence boundaries in some cases. For example, when a sentence is larger than the <code>max</code> value, it splits the sentence at <code>max</code>. Similarly, it includes multiple sentences in the text only when they fit within the <code>max</code> limit.
 <code>custom</code>:
Splits based on a custom split characters list. You can provide custom sequences up to a limit of <code>16</code> split character strings, with a maximum length of <code>10</code> each.
Specify an array of valid text literals using the <code>custom_list</code> parameter.
<pre>
{
    "split"   : "custom",
    "custom_list" : ["split_chars1", ...]
}
</pre>

</td></tr>
</table>

Parameter	JSON Value Type	Description and Acceptable Values

		For example:
		{
		    "split" : "custom",
		    "custom_list" : ["<p>" , "<s>"]
		}
		Note: You can omit sequences only for tab (\t), newline (\n), and linefeed (\r).
		Default value: recursively

custom_list	Array of strings	Specifies the custom split character strings to use when "split": "custom" is specified.

Parameter	JSON Value Type	Description and Acceptable Values
`overlap`	Number, or string containing a number	Specify the amount (as a positive integer literal or zero) of the preceding text that the chunk should contain, if any. This helps in logically splitting up related text (such as a sentence) by including some amount of the preceding chunk text. The amount of overlap depends on how the maximum size of the chunk is measured (in characters, words, or vocabulary tokens). The overlap begins at the specified `split` condition (for example, at `newline`). **Valid value**: `5%` to `20%` of `max` **Default value**: `0`

<table>
<tr><th>Parameter</th><th>JSON Value Type</th><th>Description and Acceptable Values</th></tr>
<tr><td><code>language</code></td><td>String</td><td>
Specify the language of your input data.
This clause is important, especially when your text contains certain characters (for example, punctuations or abbreviations) that may be interpreted differently in another language.
Valid values:

NLS-supported language name or its abbreviation, as listed in Oracle AI Database Globalization Support Guide.
Custom language name or its abbreviation, as listed in Supported Languages and Data File Locations. You use the <code>DBMS_VECTOR_CHAIN.CREATE_LANG_DATA</code> chunker helper API to load language-specific data (abbreviation tokens) into the database, for your specified language.

Note: You must use escape characters with any language abbreviation that is also a SQL reserved word (for example, language abbreviations such as <code>IN</code>, <code>AS</code>, <code>OR</code>, <code>IS</code>).
For example:
<pre>
SELECT dbms_vector_chain.utl_to_chunks('this is an example',
   JSON('{ "language" : "\"in\"" }'))
from dual;

SELECT dbms_vector_chain.utl_to_chunks('this is an example',
   JSON_OBJECT('language' value '"in"' RETURNING JSON))
from dual;
</pre>
Default value: <code>NLS_LANGUAGE</code> from session
</td></tr>
</table>

<table>
<tr><th>Parameter</th><th>JSON Value Type</th><th>Description and Acceptable Values</th></tr>
<tr><td>normalize</td><td>String</td><td>Automatically pre-processes or post-processes issues (such as multiple consecutive spaces and smart quotes) that may arise when documents are converted into text. Oracle recommends you to use a normalization mode to extract high-quality chunks.

Valid values:

• none:
Applies no normalization.
• all:
Normalizes common multi-byte (unicode) punctuation to standard single-byte.
• options:
Specify an array of normalization options using the norm_options parameter.

{
    "normalize"   : "options",
    "norm_options" : ["normalize_option1", ...]
}

– punctuation:
Converts quotes, dashes, and other punctuation characters supported in the character set of the text to their common ASCII form. For example:
* U+2013 (En Dash) maps to U+002D (Hyphen-Minus)
* U+2018 (Left Single Quotation Mark) maps to U+0027 (Apostrophe)
* U+2019 (Right Single Quotation Mark) maps to U+0027 (Apostrophe)
* U+201B (Single High-Reversed-9 Quotation Mark) maps to U+0027 (Apostrophe)
– whitespace:
Minimizes whitespace by eliminating unnecessary characters.
For example, retain blank lines, but remove any extra newlines and interspersed spaces or tabs: " \n \n " => "\n\n"
– widechar:
Normalizes wide, multi-byte digits and (a-z) letters to single-byte.
These are multi-byte equivalents for 0-9 and a-z A-Z, which can show up in Chinese, Japanese, or Korean text.
For example:

{
    "normalize"   : "options",
    "norm_options" : ["whitespace"]
}

Default value: all</td></tr>
</table>

Parameter	JSON Value Type	Description and Acceptable Values
`norm_options`	Array of strings	Specify an array of normalization options when `normalize : options` is specified. Valid values: • `whitespace`: This option generally reduces sequences of whitespace characters into the minimum, such as 20 spaces reduces to 1. • `punctuation`: This option generally replaces "smart" or non-standard Unicode punctuation into the simple ASCII equivalents. • `widechar`: This option replaces the Unicode "wide" multi-byte characters, common in CJK, with their simple single-byte ASCII equivalents.
`join`	String	Specifies word-joining behavior for white space normalization. Use `join` with the `whitespace` normalization option to identify characters that should not have white space added around them. This can improve chunk-boundary selection and the normalized `chunk_data` text. **Valid values**: • `none`: Do not apply word-joining characters. • `default`: Use the default word-joining characters. • A string containing custom word-joining characters.
`extended`	Boolean, number, or string	Increases the output limit of a `VARCHAR2` string to `32767` bytes, without requiring you to set the `max_string_size` parameter to `extended`. This parameters accepts standard JSON Boolean values, and additionally supports several numeric and string equivalents for compatibility, such as `1`, `"1"`, `"ON"`, `"TRUE"`, `0`, `"0"`, `"OFF"`, and `"FALSE"`. **Default value**: `4000` or `32767` (when `max_string_size=extended`)

Parameter	JSON Value Type	Description and Acceptable Values
`aggregate`	Number, or string containing a number	Specifies the maximum size of an aggregated chunk. If `aggregate` is greater than `0`, adjacent chunks can be combined into larger aggregated chunks when they satisfy the aggregation threshold. The value of aggregate must be greater than the value of `max`. A value of `0` disables aggregation. **Default value**: `0`
`aggregate_mode`	String	Specifies the comparison mode used to determine whether adjacent chunks are similar enough to aggregate. **Valid values**: • `words`:Compares adjacent chunks using whole-word overlap. • `substr`: Compares adjacent chunks using word-substring overlap. This is similar to `words`, but substring matches are included, such as `MEET` and `MEETING`.

Parameter	JSON Value Type	Description and Acceptable Values
`aggregate_thresh`	Number, or string containing a number	Specifies the maximum distance percentage between two adjacent chunks for them to be aggregated. **Valid values**: `0` to `100` • `0`: Only identical chunks are aggregated. • `100`: Any adjacent chunks can be aggregated, subject to the aggregate size limit. Lower values require greater similarity between chunks. Higher values allow less-similar chunks to be aggregated.

Vector Index Parameters:

```
{
  "distance"       :  <vector_distance>,
  "accuracy"       :  <vector_accuracy>,
  "vector_idxtype" :  <vector_idxtype>
}
```

For example:

```
{
  "distance"        :  COSINE,
  "accuracy"        :  95,
  "vector_idxtype"  :  HNSW
}
```

Parameter	Description
`distance`	Distance metric or mathematical function used to compute the distance between vectors: • `COSINE` • `MANHATTAN` • `DOT` • `EUCLIDEAN` • `L2_SQUARED` • `EUCLIDEAN_SQUARED` **Note**: Currently, the `HAMMING` and `JACCARD` vector distance metrics are not supported with hybrid vector indexes. For detailed information on each of these metrics, see Vector Distance Functions and Operators. **Default value**: `COSINE`

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>accuracy</code></td><td>Target accuracy at which the approximate search should be performed when running an approximate search query using vector indexes.

As explained in Understand Approximate Similarity Search Using Vector Indexes, you can specify non-default target accuracy values either by specifying a percentage value or by specifying internal parameters values, depending on the index type you are using.
For a Hierarchical Navigable Small World (HNSW) approximate search:

In the case of an HNSW approximate search, you can specify a target accuracy percentage value to influence the number of candidates considered to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose a similar result as an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the <code>EFSEARCH</code> parameter to impose a certain maximum number of candidates to be considered while probing the index. The higher that number, the higher the accuracy.

For detailed information, see Understand Hierarchical Navigable Small World Indexes.
For an Inverted File Flat (IVF) approximate search:

In the case of an IVF approximate search, you can specify a target accuracy percentage value to influence the number of partitions used to probe the search. This is automatically calculated by the algorithm. A value of 100 will tend to impose an exact search, although the system may still use the index and will not perform an exact search. The optimizer may choose to still use an index as it may be faster to do so given the predicates in the query. Instead of specifying a target accuracy percentage value, you can specify the <code>NEIGHBOR PARTITION PROBES</code> parameter to impose a certain maximum number of partitions to be probed by the search. The higher that number, the higher the accuracy.

For detailed information, see Understand Inverted File Flat Vector Indexes.
Valid range for both HNSW and IVF vector indexes is:

<code>> 0 and <= 100</code>

Default value: None</td></tr>
<tr><td><code>vector_idxtype</code></td><td>Type of vector index to create:
<code>HNSW</code> for an HNSW vector index
<code>IVF</code> for an IVF vector index
For detailed information on each of these index types, see Manage the Different Categories of Vector Indexes.

Default value: <code>IVF</code></td></tr>
</table>

Paths Parameter:

This field allows specification of an array of path objects. There can be many path objects and each path object must specify a `type` and a `path_list`.

> **Note**
>
> If the user does not specify the `paths` field, the whole document would be considered.

```
"paths":[
        {"type"      : "<path_type>",
         "path_list" : ["<path_list_array>"]
        }
        ]
```

Let us consider a sample JSON document:

```
 {
      "person":
       {
         "bio": "James is a data scientist who specializes in natural
language .. ",
         "profile":
          {
              "text" : "James is a data scientist with expertise in Python
programming...",
              "embedding" :
[1.60541728E-001,5.76677322E-002,4.0473938E-003,1.2037459E-001,-5.98970801E-00
4, ..]
          },
         "avatar": "https://example.com/images/James.jpg"
       },
      "product":
      {
        "description": "A social media analytics tool.", "It helps brands
track...",
        "image": "https://example.com/images/data_tool.jpg",
        "embedding" :
[1.60541728E-001,5.76677322E-002,4.0473938E-003,1.2037459E-001,-5.98970801E-00
4, ..]
      }
 }
```

And a `path_list` corresponding to the above JSON is provided here:

```
"paths": [
         {"type"      : "VECTOR",
          "path_list" : ["$.person.profile.embedding", "$.product.embedding"]
         },
         {"type"      : "STRING",
          "path_list" : ["$.person.bio", "$.product.description"]
         }
         ]
```

The following table describes the details of `paths` parameter:

Parameter	Accepted Values
`type`	The possible values for this field are: • `STRING` - for fields that need to be converted to vectors. • `VECTOR` - for fields that are already vectors.
`path_list`	Accepts an array of paths with at least one path in valid JSON format - (`$.a.b.c.d`). **Note:** For the `VECTOR` option, Oracle currently accepts one vector array per path.

Example

```
begin
  DBMS_VECTOR_CHAIN.CREATE_PREFERENCE(
    'my_vec_spec',
     DBMS_VECTOR_CHAIN.VECTORIZER,
     json('{ "vector_idxtype" :  "hnsw",
             "model"          :  "my_doc_model",
             "by"             :  "words",
             "max"            :  100,
             "overlap"        :  10,
             "split"          :  "recursively,
             "language"       :  "english",
             "paths":          : [
                                 {
                                  "type"      : "VECTOR",
                                  "path_list" : ["$.person.profile.embedding"]
                                  }
                                 ]
              }'));
end;
/

CREATE HYBRID VECTOR INDEX my_hybrid_idx on
    doc_table(text_column)
    parameters('VECTORIZER my_vec_spec');
```

Related Topics

- CREATE HYBRID VECTOR INDEX

CREATE_VOCABULARY

Use the `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY` chunker helper procedure to load your own token vocabulary file into the database.

Purpose

To create custom token vocabulary that is recognized by the tokenizer used by your vector embedding model.

A vocabulary contains a set of tokens (words and word pieces) that are collected during a model's statistical training process. You can supply this data to the chunker to help in accurately selecting the text size that approximates the maximum input limit imposed by the tokenizer of your embedding model.

Usage Notes

- Usually, the supported vocabulary files (containing recognized tokens) are included as part of a model's distribution. Oracle recommends to use the vocabulary files associated with your model.

  If a vocabulary file is not available, then you may download one of the following files depending on the tokenizer type:

  – **WordPiece**:

    Vocabulary file (`vocab.txt`) for the "bert-base-uncased" (English) or "bert-base-multilingual-cased" model

  – **Byte-Pair Encoding (BPE)**:

    Vocabulary file (`vocab.json`) for the "GPT2" model

    Use the following python script to extract the file:

    ```
    import json
    import sys

    with open(sys.argv[1], encoding="utf-8") as f:
      d = json.load(f)
      for term in d:
        print(term)
    ```

  – **SentencePiece**:

    Vocabulary file (`tokenizer.json`) for the "xlm-roberta-base" model

    Use the following python script to extract the file:

    ```
    import json
    import sys

    with open(sys.argv[1], encoding="utf-8") as f:
      d = json.load(f)
      for entry in d["model"]["vocab"]:
        print(entry[0])
    ```

  Ensure to save your vocabulary files in `UTF-8` encoding.

- You can create a vocabulary based on the tokens loaded in the `schema.table.column`, using a user-specified vocabulary name (`vocabulary_name`).

  After loading your vocabulary data, you can use the `by vocabulary` chunking mode (with `VECTOR_CHUNKS` or `UTL_TO_CHUNKS`) to split input data by counting the number of tokens.

- You can query these data dictionary views to access existing vocabulary data:

  – `ALL_VECTOR_VOCAB` displays all available vocabularies.

  – `USER_VECTOR_VOCAB` displays vocabularies from the schema of the current user.

  – `ALL_VECTOR_VOCAB_TOKENS` displays a list of tokens from all available vocabularies.

– `USER_VECTOR_VOCAB_TOKENS` displays a list of tokens from the vocabularies owned by the current user.

Syntax

```
DBMS_VECTOR_CHAIN.CREATE_VOCABULARY(
    PARAMS      IN JSON default NULL
);
```

PARAMS

Specify the input parameters in JSON format:

```
{
    table_name,
    column_name,
    vocabulary_name,
    format,
    cased
}
```

Table 12-22 Parameter Details

Parameter	Description	Required	Default Value
`table_name`	Name of the table (along with the optional table owner) in which you want to load the vocabulary file	Yes	No value
`column_name`	Column name in the vocabulary table in which you want to load the vocabulary file	Yes	No value
`vocabulary_name`	User-specified name of the vocabulary, along with the optional owner name (if other than the current owner)	Yes	No value
`format`	• `xlm` for SentencePiece tokenization • `bert` for WordPiece tokenization • `gpt2` for BPE tokenization	Yes	No value
`cased`	Character-casing of the vocabulary, that is, vocabulary to be treated as cased or uncased	No	`false`

Example

```
DECLARE
  params clob := '{"table_name"       : "doc_vocabtab",
                   "column_name"      : "token",
                   "vocabulary_name"  : "doc_vocab",
                   "format"           : "bert",
                   "cased"            : false}';

BEGIN
  dbms_vector_chain.create_vocabulary(json(params));
END;
/
```

End-to-end example:

To run an end-to-end example scenario using this procedure, see Create and Use Custom Vocabulary.

Related Topics

- VECTOR_CHUNKS
- UTL_TO_CHUNKS
- Text Processing Views

DROP_CREDENTIAL

Use the `DBMS_VECTOR_CHAIN.DROP_CREDENTIAL` credential helper procedure to drop an existing credential name from the data dictionary.

Syntax

```
DBMS_VECTOR_CHAIN.DROP_CREDENTIAL (
    CREDENTIAL_NAME      IN VARCHAR2
);
```

CREDENTIAL_NAME

Specify the credential name that you want to drop.

Examples

- For Generative AI:

  ```
  exec dbms_vector_chain.drop_credential('OCI_CRED');
  ```

- For Cohere:

  ```
  exec dbms_vector_chain.drop_credential('COHERE_CRED');
  ```

DROP_LANG_DATA

Use the `DBMS_VECTOR_CHAIN.DROP_LANG_DATA` chunker helper procedure to remove abbreviation data from the data dictionary.

Syntax

```
DBMS_VECTOR_CHAIN.DROP_LANG_DATA(
    PREF_NAME     IN VARCHAR2
);
```

LANG

Specify the name of the language data that you want to drop for a given language.

Example

```
DBMS_VECTOR_CHAIN.DROP_LANG_DATA('indonesian');
```

DROP_PREFERENCE

Use the `DBMS_VECTOR_CHAIN.DROP_PREFERENCE` preference helper procedure to remove an existing Vectorizer preference.

Syntax

```
DBMS_VECTOR_CHAIN.DROP_PREFERENCE (PREF_NAME);
```

PREF_NAME

Name of the Vectorizer preference to drop.

Example

```
DBMS_VECTOR_CHAIN.DROP_PREFERENCE ('scott_vectorizer');
```

DROP_VOCABULARY

Use the `DBMS_VECTOR_CHAIN.DROP_VOCABULARY` chunker helper procedure to remove vocabulary data from the data dictionary.

Syntax

```
DBMS_VECTOR_CHAIN.DROP_VOCABULARY(
    VOCABULARY_NAME    IN VARCHAR2
);
```

VOCAB_NAME

Specify the name of the vocabulary that you want to drop, in the form:

`vocabulary_name`

or

`owner.vocabulary_name`

Example

```
DBMS_VECTOR_CHAIN.DROP_VOCABULARY('MY_VOCAB_1');
```

UTL_TO_RERANK

Use the `DBMS_VECTOR_CHAIN.UTL_TO_RERANK` function to reassess and reorder an initial set of results to retrieve more relevant search output.

Purpose

To improve the relevance and quality of search results in both similarity search and Retrieval Augmented Generation (RAG) scenarios.

Reranking improves the quality of information ingested into an LLM by ensuring that the most relevant documents or chunks are prioritized. This helps to reduce hallucinations and improves the accuracy of generated outputs.

For this operation, Oracle AI Vector Search supports reranking models provided by Cohere, Vertex AI, ocigenai, and ONNX (Oracle AI Database as the service provider). However, ocigenai is available only in Dedicated mode.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

```
DBMS_VECTOR_CHAIN.UTL_TO_RERANK(
                QUERY      IN CLOB,
                DOCUMENTS  IN JSON,
                PARAMS     IN JSON default NULL
) return JSON;
```

This function accepts the input containing a query as `CLOB` and a list of documents in `JSON` format. It then processes this information to generate a `JSON` object containing a reranked list of documents, sorted by score.

For example, a reranked output includes:

```
{
    "index"   : "1",
    "score"   : "0.99",
    "content" : "Jupiter boasts an impressive system of 95 known moons."
}
```

Where,

- `index` specifies the position of the document in the list of input text.
- `score` specifies the relevance score.
- `content` specifies the input text corresponding to the index.

QUERY

Specify the search query (typically from an initial search) as `CLOB`.

DOCUMENTS

Specify a JSON array of strings (list of potentially relevant documents to rerank) in the following format:

```
{
  "documents": [
  "string1",
  "string2",
    ...
  ]
}
```

PARAMS

Specify the following list of parameters in JSON format. All these parameters are mandatory.

```
{
  "provider"          : "<service provider>",
  "credential_name"   : "<credential name>",
  "url"               : "<REST endpoint URL for reranking>",
  "model"             : "<reranking model name>",
  ...
}
```

Table 12-23 UTL_TO_RERANK Parameter Details

Parameter	Description
`provider`	Supported REST provider to access for reranking: • `cohere` • `vertexai`
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the reranking model in the form: *`schema.model_name`* If the model name is not schema-qualified, then the schema of the procedure invoker is used.

Additional REST provider parameters:

Optionally, specify additional provider-specific parameters for reranking.

Important

- The following examples are for illustration purposes. For accurate and up-to-date information on additional parameters to use, refer to your third-party provider's documentation.
- For a list of all supported REST endpoints, see Supported Third-Party Provider Operations and Endpoints.

Cohere example:

```
{
  "provider"        : "cohere",
  "credential_name" : "COHERE_CRED",
  "url"             : "https://api.cohere.example.com/rerank",
  "model"           : "rerank-english-v3.0",
  "return_documents": false,
  "top_n"           : 3
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://googleapis.example.com/
default_ranking_config:rank",
  "model"            : "semantic-ranker-512@latest",
  "ignoreRecordDetailsInResponse" : true,
  "topN"             : 3
  }
```

ONNX (Oracle AI Database as the service provider) example:

```
{
   "provider": "database",
   "model": "<model_name>",
   "return_docs": true,
   "top_n": 3
}
```

OCI Generative AI (Dedicated mode) example:

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/rerankText",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "topN": 3,
  "isEcho": true
}
```

> **Note**
>
> For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Table 12-24 Additional REST Provider Parameter Details

Parameter	Description
`return_documents`	Whether to return search results with original documents or input text (`content`): • `false` (default, also recommended) to not return any input text (return only index and score) • `true` to return input text along with index and score **Note**: With Cohere as the provider, Oracle recommends that you keep this option disabled for better performance. You may choose to enable it for debugging purposes when you need to view the original text.
`ignoreRecordDetailsInResponse`	Whether to return search results with original record details or input text (`content`): • `false` (default) to return input text along with index and score • `true` (recommended) to not return any input text (return only index and score) **Note**: With Vertex AI as the provider, Oracle recommends that you keep this option enabled for better performance. You may choose to disable it for debugging purposes when you need to view the original text.
`top_n` or `topN`	The number of most relevant documents to return.

Examples

- Using Cohere:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "cohere",
  "credential_name": "COHERE_CRED",
  "url": "https://api.cohere.com/v1/rerank",
  "model": "rerank-english-v3.0",
  "return_documents": true,
  "top_n": 3
}';

  reranked_output := dbms_vector_chain.UTL_TO_RERANK(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

- Using Vertex AI:

```
declare
  params clob;
  reranked_output json;
begin
  params := '
{
  "provider": "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url": "https://discoveryengine.googleapis.com/v1/projects/1085581009881/
locations/global/rankingConfigs/default_ranking_config:rank",
  "model": "semantic-ranker-512@latest",
  "ignoreRecordDetailsInResponse": false,
  "topN": 3
}';

  reranked_output := dbms_vector_chain.UTL_TO_RERANK(:query,
json(:initial_retrieval_docs), json(params));
  dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

Using ONNX (Oracle AI Database as the service provider):

```
declare
   params clob;
   reranked_output json;
begin
   params := '
{
   "provider": "database",
   "model": "reranker_model",
   "return_docs": true,
   "top_n": 3
}';
   reranked_output := dbms_vector_chain.utl_to_rerank(:query,
json(:initial_retrieval_docs), json(params));
   dbms_output.put_line(json_serialize(reranked_output));
end;
/
```

End-to-end example:

To run an end-to-end example scenario using this function, see Use Reranking for Better RAG Results.

UTL_TO_CHUNKS

Use the `DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS` chainable utility function to split a large plain text document into smaller chunks of text.

Purpose

To perform a text-to-chunks transformation. This chainable utility function internally calls the `VECTOR_CHUNKS` SQL function for the operation.

To embed a large document, you may first need to split it into multiple appropriate-sized segments or chunks through a splitting process known as chunking (as explained in Understand the Stages of Data Transformations). A chunk can be words (to capture specific words or word pieces), sentences (to capture a specific meaning), or paragraphs (to capture broader themes). A single document may be split into multiple chunks, each transformed into a vector.

Syntax

```
DBMS_VECTOR_CHAIN.UTL_TO_CHUNKS (
    DATA         IN  CLOB | VARCHAR2,
    PARAMS       IN  JSON  default  NULL
) return VECTOR_ARRAY_T;
```

DATA

This function accepts the input data type as `CLOB` or `VARCHAR2`.

It returns an array of `CLOB`s, where each `CLOB` contains a chunk along with its metadata in JSON format, as follows:

```
{
    "chunk_id"     : NUMBER,
    "chunk_offset" : NUMBER,
    "chunk_length" : NUMBER,
    "chunk_data"   : "VARCHAR2(4000)"
}
```

For example:

```
{"chunk_id":1,"chunk_offset":1,"chunk_length":6,"chunk_data":"sample"}
```

Where,

- `chunk_id` specifies the chunk ID for each chunk.
- `chunk_offset` specifies the original position of each chunk in the source document, relative to the start of document which has a position of `1`.
- `chunk_length` specifies the character length of each chunk.
- `chunk_data` displays text pieces from each chunk.

PARAMS

Specify input parameters in JSON format.

```
{
    "limit"           :     limit,
    "by"              :     mode,
    "max"             :     max,
    "overlap"         :     overlap,
    "split"           :     split_condition,
    "custom_list"     :     [ split_chars1, ... ],
    "vocabulary"      :     vocabulary_name,
    "language"        :     nls_language,
    "normalize"       :     normalize_mode,
    "norm_options"    :     [ normalize_option1, ... ],
    "join"            :     join_mode_or_characters,
    "extended"        :     boolean,
    "aggregate"       :     aggregate_size,
    "aggregate_mode"  :     aggregation_mode,
    "aggregate_thresh":     aggregate_threshold
}
```

For example:

```
JSON('
  { "by"              :    "vocabulary",
    "vocabulary"      :    "myvocab",
    "max"             :    "100",
    "overlap"         :    "0",
    "split"           :    "custom",
    "custom_list"     :    [ "<p>" , "<s>" ],
    "language"        :    "american",
    "normalize"       :    "options",
    "norm_options"    :    [ "whitespace" ]
    "join"            :    "default",
    "aggregate"       :    "400",
    "aggregate_mode"  :    "words",
    "aggregate_thresh":    "20"
  }')
```

Here is a complete description of these parameters:

Parameter	JSON Value Type	Description and Acceptable Values
`limit`	Number, or string containing a number	Specifies the maximum number of bytes for a chunk. Use this parameter to avoid exceeding the maximum size of the output `VARCHAR2` chunk text. If `limit` is omitted, the maximum size is `4000` bytes unless `extended` is enabled, in which case the maximum size is `32767` bytes. **Default value**: omitted

Parameter	JSON Value Type	Description and Acceptable Values
`by`	String	Specify a mode for splitting your data, that is, to split by counting the number of characters, words, or vocabulary tokens. **Valid values**: • `characters` (or `chars`): Splits by counting the number of characters. • `words`: Splits by counting the number of words. Words are defined as sequences of alphabetic characters, sequences of digits, individual punctuation marks, or symbols. For segmented languages without white space word boundaries (such as Chinese, Japanese, or Thai), each native character is considered a word (that is, unigram). • `vocabulary`: Splits by counting the number of vocabulary tokens. Vocabulary tokens are words or word pieces, recognized by the vocabulary of the tokenizer that your embedding model uses. You can load your vocabulary file using the chunker helper API `DBMS_VECTOR_CHAIN.CREATE_VOCABULARY`. **Note**: For accurate results, ensure that the chosen model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. **Default value**: `words`
`vocabulary`	String	Specify vocabulary when `"by": "vocabulary"` is used.

Parameter	JSON Value Type	Description and Acceptable Values
`max`	Number, or string containing a number	Specify a limit on the maximum size of each chunk. This setting splits the input text at a fixed point where the maximum limit occurs in the larger text. The units of `max` correspond to the `by` mode, that is, to split data when it reaches the maximum size limit of a certain number of characters, words, numbers, punctuation marks, or vocabulary tokens. **Valid values**: • `by characters`: `50` to `4000` characters • `by words`: `10` to `1000` words • `by vocabulary`: `10` to `1000` tokens **Default value**: `100`

<table>
<tr><th>Parameter</th><th>JSON Value Type</th><th>Description and Acceptable Values</th></tr>
<tr><td><code>split [by]</code></td><td>String</td><td>
Specify where to split the input text when it reaches the maximum size limit. This helps to keep related data together by defining appropriate boundaries for chunks.
Valid values:

<code>none</code>:
Splits at the <code>max</code> limit of characters, words, or vocabulary tokens.
<code>newline</code>, <code>blankline</code>, and <code>space</code>:
These are single-split character conditions that split at the last split character before the <code>max</code> value.
Use <code>newline</code> to split at the end of a line of text. Use <code>blankline</code> to split at the end of a blank line (sequence of characters, such as two newlines). Use <code>space</code> to split at the end of a blank space.
<code>recursively</code>:
This is a multiple-split character condition that breaks the input text using an ordered list of characters (or sequences).
<code>recursively</code> is predefined as <code>BLANKLINE</code>, <code>newline</code>, <code>space</code>, <code>none</code> in this order:
1. If the input text is more than the <code>max</code> value, then split by the first split character.
2. If that fails, then split by the second split character.
3. And so on.
4. If no split characters exist, then split by <code>max</code> wherever it appears in the text.
<code>sentence</code>:
This is an end-of-sentence split condition that breaks the input text at a sentence boundary.
This condition automatically determines sentence boundaries by using knowledge of the input language's sentence punctuation and contextual rules. This language-specific condition relies mostly on end-of-sentence (EOS) punctuations and common abbreviations.
Contextual rules are based on word information, so this condition is only valid when splitting the text by words or vocabulary (not by characters).
Note: This condition obeys the <code>by word</code> and <code>max</code> settings, and thus may not determine accurate sentence boundaries in some cases. For example, when a sentence is larger than the <code>max</code> value, it splits the sentence at <code>max</code>. Similarly, it includes multiple sentences in the text only when they fit within the <code>max</code> limit.
<code>custom</code>:
Splits based on a custom split characters list. You can provide custom sequences up to a limit of <code>16</code> split character strings, with a maximum length of <code>10</code> each.
Specify an array of valid text literals using the <code>custom_list</code> parameter.
<pre>{
    "split"   : "custom",
    "custom_list"   : ["split_chars1", ...]
}</pre>

</td></tr>
</table>

Parameter	JSON Value Type	Description and Acceptable Values
		For example: `{ "split" : "custom", "custom_list" : [ "<p>" , "<s>" ] }` **Note**: You can omit sequences only for tab (`\t`), newline (`\n`), and linefeed (`\r`). **Default value**: `recursively`
`custom_list`	Array of strings	Specifies the custom split character strings to use when `"split": "custom"` is specified.

Parameter	JSON Value Type	Description and Acceptable Values
`overlap`	Number, or string containing a number	Specify the amount (as a positive integer literal or zero) of the preceding text that the chunk should contain, if any. This helps in logically splitting up related text (such as a sentence) by including some amount of the preceding chunk text. The amount of overlap depends on how the maximum size of the chunk is measured (in characters, words, or vocabulary tokens). The overlap begins at the specified `split` condition (for example, at `newline`). **Valid value**: `5%` to `20%` of `max` **Default value**: `0`

<table>
<tr><th>Parameter</th><th>JSON Value Type</th><th>Description and Acceptable Values</th></tr>
<tr><td><code>language</code></td><td>String</td><td>Specify the language of your input data.

This clause is important, especially when your text contains certain characters (for example, punctuations or abbreviations) that may be interpreted differently in another language.

Valid values:
NLS-supported language name or its abbreviation, as listed in Oracle AI Database Globalization Support Guide.
Custom language name or its abbreviation, as listed in Supported Languages and Data File Locations. You use the <code>DBMS_VECTOR_CHAIN.CREATE_LANG_DATA</code> chunker helper API to load language-specific data (abbreviation tokens) into the database, for your specified language.
Note: You must use escape characters with any language abbreviation that is also a SQL reserved word (for example, language abbreviations such as <code>IN</code>, <code>AS</code>, <code>OR</code>, <code>IS</code>).

For example:<pre>
SELECT dbms_vector_chain.utl_to_chunks('this is an example',
   JSON('{ "language" : "\"in\"" }'))
from dual;
</pre><pre>
SELECT dbms_vector_chain.utl_to_chunks('this is an example',
   JSON_OBJECT('language' value '"in"' RETURNING JSON))
from dual;
</pre>Default value: <code>NLS_LANGUAGE</code> from session</td></tr>
</table>

Parameter	JSON Value Type	Description and Acceptable Values

`normalize`	String	Automatically pre-processes or post-processes issues (such as multiple consecutive spaces and smart quotes) that may arise when documents are converted into text. Oracle recommends you to use a normalization mode to extract high-quality chunks.
		Valid values:
		• `none`:
		Applies no normalization.
		• `all`:
		Normalizes common multi-byte (unicode) punctuation to standard single-byte.
		• `options`:
		Specify an array of normalization options using the `norm_options` parameter.
		`{`
		`    "normalize"    :  "options",`
		`    "norm_options" :  [ "normalize_option1", ... ]`
		`}`
		– `punctuation`:
		Converts quotes, dashes, and other punctuation characters supported in the character set of the text to their common ASCII form. For example:
		* U+2013 (En Dash) maps to U+002D (Hyphen-Minus)
		* U+2018 (Left Single Quotation Mark) maps to U+0027 (Apostrophe)
		* U+2019 (Right Single Quotation Mark) maps to U+0027 (Apostrophe)
		* U+201B (Single High-Reversed-9 Quotation Mark) maps to U+0027 (Apostrophe)
		– `whitespace`:
		Minimizes whitespace by eliminating unnecessary characters.
		For example, retain blank lines, but remove any extra newlines and interspersed spaces or tabs: `" \n \n " => "\n\n"`
		– `widechar`:
		Normalizes wide, multi-byte digits and (a-z) letters to single-byte.
		These are multi-byte equivalents for `0-9` and `a-z A-Z`, which can show up in Chinese, Japanese, or Korean text.
		For example:
		`{`
		`    "normalize"    :  "options",`
		`    "norm_options" :  [ "whitespace" ]`
		`}`
		Default value: `all`

Parameter	JSON Value Type	Description and Acceptable Values
`norm_options`	Array of strings	Specify an array of normalization options when `normalize : options` is specified. Valid values: • `whitespace`: This option generally reduces sequences of whitespace characters into the minimum, such as 20 spaces reduces to 1. • `punctuation`: This option generally replaces "smart" or non-standard Unicode punctuation into the simple ASCII equivalents. • `widechar`: This option replaces the Unicode "wide" multi-byte characters, common in CJK, with their simple single-byte ASCII equivalents.
`join`	String	Specifies word-joining behavior for white space normalization. Use `join` with the `whitespace` normalization option to identify characters that should not have white space added around them. This can improve chunk-boundary selection and the normalized `chunk_data` text. **Valid values**: • `none`: Do not apply word-joining characters. • `default`: Use the default word-joining characters. • A string containing custom word-joining characters.
`extended`	Boolean, number, or string	Increases the output limit of a `VARCHAR2` string to `32767` bytes, without requiring you to set the `max_string_size` parameter to `extended`. This parameters accepts standard JSON Boolean values, and additionally supports several numeric and string equivalents for compatibility, such as `1`, `"1"`, `"ON"`, `"TRUE"`, `0`, `"0"`, `"OFF"`, and `"FALSE"`. **Default value**: `4000` or `32767` (when `max_string_size=extended`)

Parameter	JSON Value Type	Description and Acceptable Values
`aggregate`	Number, or string containing a number	Specifies the maximum size of an aggregated chunk. If `aggregate` is greater than `0`, adjacent chunks can be combined into larger aggregated chunks when they satisfy the aggregation threshold. The value of aggregate must be greater than the value of `max`. A value of `0` disables aggregation. **Default value**: `0`
`aggregate_mode`	String	Specifies the comparison mode used to determine whether adjacent chunks are similar enough to aggregate. **Valid values**: • `words`:Compares adjacent chunks using whole-word overlap. • `substr`: Compares adjacent chunks using word-substring overlap. This is similar to `words`, but substring matches are included, such as `MEET` and `MEETING`.

Parameter	JSON Value Type	Description and Acceptable Values
`aggregate_thresh`	Number, or string containing a number	Specifies the maximum distance percentage between two adjacent chunks for them to be aggregated. **Valid values**: `0` to `100` • `0`: Only identical chunks are aggregated. • `100`: Any adjacent chunks can be aggregated, subject to the aggregate size limit. Lower values require greater similarity between chunks. Higher values allow less-similar chunks to be aggregated.

Example

```
SELECT D.id doc,
    JSON_VALUE(C.column_value, '$.chunk_id' RETURNING NUMBER) AS id,
    JSON_VALUE(C.column_value, '$.chunk_offset' RETURNING NUMBER) AS pos,
    JSON_VALUE(C.column_value, '$.chunk_length' RETURNING NUMBER) AS siz,
    JSON_VALUE(C.column_value, '$.chunk_data') AS txt
FROM docs D,
   dbms_vector_chain.utl_to_chunks(D.text,
   JSON('{ "by"       : "words",
           "max"      : "100",
           "overlap"  : "0",
```

```
"split"     : "recursively",
"language" : "american",
"normalize": "all" }')) C;
```

End-to-end examples:

To run end-to-end example scenarios using this function, see Perform Chunking With Embedding and Configure Chunking Parameters.

Related Topics

- VECTOR_CHUNKS

UTL_TO_EMBEDDING and UTL_TO_EMBEDDINGS

Use the `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING` and `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS` chainable utility functions to generate one or more vector embeddings from textual documents and images.

Purpose

To automatically generate one or more vector embeddings from textual documents and images.

- Text to Vector:

  You can perform a text-to-embedding transformation by accessing either Oracle AI Database or a third-party service provider:

  - Oracle AI Database as the service provider (default setting):

    This API calls an ONNX format embedding model that you load into the database.

  - Third-party embedding model:

    This API makes a REST API call to your chosen remote service provider (Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI) or local service provider (Private AI, Ollama).

- Image to Vector:

  You can also perform an image-to-embedding transformation. This API makes a REST call to your chosen image embedding model or multimodal embedding model by Vertex AI. Note that currently Vertex AI is the only supported service provider for this operation.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

- Text to Vector:

```
DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING (
    DATA           IN CLOB,
    PARAMS         IN JSON default NULL
) return VECTOR;


DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS (
    DATA           IN VECTOR_ARRAY_T,
    PARAMS         IN JSON default NULL
) return VECTOR_ARRAY_T;
```

- Image to Vector:

```
DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING (
    DATA           IN BLOB,
    MODALITY       IN VARCHAR2,
    PARAMS         IN JSON default NULL
) return VECTOR;
```

DATA

- Text to Vector:

  `UTL_TO_EMBEDDING` accepts the input as `CLOB` containing textual data (text strings or small documents). It then converts the text to a single embedding (`VECTOR`).

  `UTL_TO_EMBEDDINGS` converts an array of chunks (`VECTOR_ARRAY_T`) to an array of embeddings (`VECTOR_ARRAY_T`).

> **Note**
>
> Although data is a `CLOB` or a `VECTOR_ARRAY_T` of `CLOB`, the maximum input is 4000 characters. If you have input that is greater, you can use `UTL_TO_CHUNKS` to split the data into smaller chunks before passing in.

- Image to Vector:

  `UTL_TO_EMBEDDING` accepts the input as `BLOB` containing media data for media files such as images. It then converts the image input to a single embedding (`VECTOR`).

A generated embedding output includes:

```
{
    "embed_id"    : NUMBER,
    "embed_data"  : "VARCHAR2(4000)",
    "embed_vector": "CLOB"
}
```

Where,

- `embed_id` displays the ID number of each embedding.

- `embed_data` displays the input text that is transformed into embeddings.
- `embed_vector` displays the generated vector representations.

MODALITY

For `BLOB` inputs, specify the type of content to vectorize. The only supported value is `image`.

> **Note**
>
> For image `BLOB` input, standalone baseline 8-bit lossy JPEG images are accepted. Unsupported or malformed image input does not produce a successful image decode. At SQL scoring time, this can result in a `NULL` embedding result. Perform any required image preprocessing or format conversion externally before embedding creation.
>
> For more information about image input restrictions, see ONNX Pipeline Models: Image Embedding.

PARAMS

Specify input parameters in JSON format, depending on the service provider that you want to use.

If using Oracle AI Database as the provider:

```
{
  "provider" : "database",
  "model"    : "<in-database ONNX embedding model name>"
}
```

Table 12-25 Database Provider Parameter Details

Parameter	Description
`provider`	Specify `database` (default setting) to use Oracle AI Database as the provider. With this setting, you must load an ONNX format embedding model into the database.
`model`	User-specified name under which the imported ONNX embedding model is stored in Oracle AI Database. If you do not have an embedding model in ONNX format, then perform the steps listed in Convert Pretrained Models to ONNX Format.

If using a third-party provider:

Set the following parameters along with additional embedding parameters specific to your provider:

- For `UTL_TO_EMBEDDING`:

```
{
  "provider"        : "<AI service provider>",
  "credential_name" : "<credential name>",
  "url"             : "<REST endpoint URL for embedding service>",
  "model"           : "<REST provider embedding model name>",
  "transfer_timeout": <maximum wait time for the request to complete>,
  "max_count": "<maximum calls to the AI service provider>",
```

```
  "<additional REST provider parameter>": "<REST provider parameter
value>"
}
```

- For `UTL_TO_EMBEDDINGS`:

```
{
  "provider"         : "<AI service provider>",
  "credential_name" : "<credential name>",
  "url"              : "<REST endpoint URL for embedding service>",
  "model"            : "<REST provider embedding model name>",
  "transfer_timeout": <maximum wait time for the request to complete>,
  "batch_size"       : "<number of vectors to request at a time>",
  "max_count": "<maximum calls to the AI service provider>",
  "<additional REST provider parameter>": "<REST provider parameter
value>"
}
```

These are the details on each of the input parameters for `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDING` and `DBMS_VECTOR_CHAIN.UTL_TO_EMBEDDINGS` when you specify a third-party provider.

When using with Private AI, specify either a `credential_name` for remote service authentication or use `local` in the host key for local, unauthenticated access.

Table 12-26 Third-Party Provider Parameter Details

Parameter	Description
`provider`	Third-party service provider that you want to access for this operation. A REST call is made to the specified provider to access its embedding model. For image input, specify `vertexai`. For text input, specify one of the following values: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `ollama` • `privateai`
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.

Table 12-26 (Cont.) Third-Party Provider Parameter Details

Parameter	Description
`host`	When using a local service provider, `host` can be used to disable credential by setting the value to `local`. When the `host` parameter is specified, `credential_name` should be omitted. The `host` parameter is applicable **only if** `provider` is one of the following: • `openai` • `ollama` • `privateai`
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the third-party embedding model in the form: *`schema.model_name`* If you do not specify a schema, then the schema of the procedure invoker is used. **Note**: • For Generative AI, all the supported third-party models are listed in Supported Third-Party Provider Operations and Endpoints. • For accurate results, ensure that the chosen text embedding model matches the vocabulary file used for chunking. If you are not using a vocabulary file, then ensure that the input length is defined within the token limits of your model. • To get image embeddings, you can use any image embedding model or multimodal embedding model supported by Vertex AI. Multimodal embedding is a technique that vectorizes data from different modalities such as text and images. When using a multimodal embedding model to generate embeddings, ensure that you use the same model to vectorize both types of content (text and images). By doing so, the resulting embeddings are compatible and situated in the same vector space, which allows for effective comparison between the two modalities during similarity searches.
`transfer_timeout`	Maximum time to wait for the request to complete. The default value is `60` seconds. You can increase this value for busy web servers.
`batch_size`	Maximum number of vectors to request at a time. For example, for a batch size of `50`, if `100` chunks are passed, then this API sends two requests with an array of `50` strings each. If `30` chunks are passed (which is lesser than the defined batch size), then the API sends those in a single request. For REST calls, it is more efficient to send a batch of inputs at a time rather than requesting a single input per call. Increasing the batch size can provide better performance, whereas reducing the batch size may reduce memory and data usage, especially if your provider has a rate limit. The default or maximum allowed value depends on the third-party provider settings.
`max_count`	Maximum number of times the API can be called for a given third-party provider. When set to an integer *n*, `max_count` stops the execution of the API for the given provider beyond *n* times. This prevents accidental calling of a third-party over some limit, for example to avoid surpassing the service amount that you have purchased.

Additional third-party provider parameters:

Optionally, specify additional provider-specific parameters.

Table 12-27 Additional REST Provider Parameter Details

Parameter	Description
`input_type`	Type of input to vectorize.

Let us look at some example configurations for all third-party providers:

> **Important**
>
> - The following examples are for illustration purposes. For accurate and up-to-date information on the parameters to use, refer to your third-party provider's documentation.
> - For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.
> - The generated embedding results may be different between requests for the same input and configuration, depending on your embedding model or floating point precision. However, this does not affect your queries (and provides semantically correct results) because the vector distance will be similar.

Cohere example:

```
{
  "provider"       : "cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.example.com/embed",
  "model"          : "embed-english-light-v2.0",
  "input_type"     : "search_query"
}
```

OCI Generative AI (On-Demand mode) example:

```
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://generativeai.oci.example.com/embedText",
  "model"          : "cohere.embed-english-v3.0",
  "batch_size"     : 10
}
```

OCI Generative AI (Dedicated mode) example:

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/embedText",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1..."
}
```

> **Note**
>
> For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Google AI example:

```
{
  "provider"       : "googleai",
  "credential_name": "GOOGLEAI_CRED",
  "url"            : "https://googleapis.example.com/models/",
  "model"          : "embedding-001"
}
```

Hugging Face example:

```
{
  "provider"       : "huggingface",
  "credential_name": "HF_CRED",
  "url"            : "https://api.huggingface.example.com/",
  "model"          : "sentence-transformers/all-MiniLM-L6-v2"
}
```

Ollama example:

```
{
  "provider"       : "ollama",
  "host"           : "local",
  "url"            : "http://localhost:11434/api/embeddings",
  "model"          : "phi3:mini"
}
```

OpenAI example:

```
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.example.com/v1/embeddings",
  "model"          : "text-embedding-3-small"
}
```

OpenAI-compatible provider example:

Note

This is just one example of how to set up the configuration for a number of OpenAI-compatible third-party providers, such as Llamafile and vLLM. The `provider` value must specify `openai` while the `url` value depends on your chosen third-party provider's REST endpoint.

Set `host` to `local` to disable credential. Set `model` to the desired model if the third-party provider requires a model name (for example, Ollama, vLLM, and so on). Otherwise, you can set `model` to any string such as `any` (for example, if using Llamafile).

```
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/embeddings",
  "host": "local",
  "model": "any"
}
```

Vertex AI example:

```
{
  "provider"       : "vertexai",
  "credential_name": "VERTEXAI_CRED",
  "url"            : "https://googleapis.example.com/models/",
  "model"          : "textembedding-gecko:predict"
}
```

Mistral AI example:

```
{
   "provider": "mistralai",
   "credential_name": "MISTRALAI_CRED",
   "url": "https://api.mistral.ai/v1/embeddings",
   "model": "mistral-embed"
}
```

Private AI example:

```
{
  "provider": "privateai",
  "url": "http://host:port/v1/embeddings",
  "host": "local",
  "model": "all-minilm-l12-v2"
}
```

Examples

You can use `UTL_TO_EMBEDDING` in a `SELECT` clause and `UTL_TO_EMBEDDINGS` in a `FROM` clause, as follows:

UTL_TO_EMBEDDING:

- **Text to vector using Generative AI**:

  The following examples use `UTL_TO_EMBEDDING` to generate an embedding with `Hello world` as the input.

  Here, the cohere.embed-english-v3.0 model is used by accessing Generative AI as the provider. You can replace the `model` value with any other supported model that you want to use with Generative AI, as listed in Supported Third-Party Provider Operations and Endpoints.

```
-- declare embedding parameters

var params clob;

begin
  :params := '
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/embedText",
  "model": "cohere.embed-english-v3.0"
}';
end;
/

-- get text embedding: PL/SQL example

declare
  input clob;
  v vector;
begin
  input := 'Hello world';

  v := dbms_vector_chain.utl_to_embedding(input, json(:params));
  dbms_output.put_line(vector_serialize(v));
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/

-- get text embedding: select example

select dbms_vector_chain.utl_to_embedding('Hello world', json(:params))
from dual;
```

- **Text to vector using Llamafile**:

  The following example uses `UTL_TO_EMBEDDING` to generate an embedding with `Hello world` as the input.

  Here, Llamafile is the OpenAI-compatible provider, which is specified in the preferences using the relevant third-party REST endpoint as the `url` and `openai` as the `provider`. This example also uses `DBMS_LOB.SUBSTR` to truncate large vectors, printing only the first 100 characters.

The value `openai` can be provided as the `provider` for a number of OpenAI-compatible third-party providers, such as Llamafile and vLLM. Examples of supported third-party endpoints can be found in Supported Third-Party Provider Operations and Endpoints.

```
-- declare embedding parameters

var params clob;

begin
  :params := '
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/embeddings",
  "host": "local",
  "model": "any"
}';
end;
/

-- get text embedding: PL/SQL example

declare
  input clob;
  v vector;
begin
  input := 'Hello world';

  v := dbms_vector_chain.utl_to_embedding(input, json(:params));
  dbms_output.put_line(dbms_lob.substr(to_clob(v), 100) || '...');
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/

-- get text embedding: select example

select dbms_vector_chain.utl_to_embedding('Hello world', json(:params))
from dual;
```

- **Image to vector using Vertex AI**:

  The following examples use `UTL_TO_EMBEDDING` to generate an embedding by accessing the Vertex AI's multimodal embedding model.

  Here, the input is `parrots.jpg`, `VEC_DUMP` is a local directory that stores the `parrots.jpg` file, and the modality is specified as `image`.

```
-- declare embedding parameters

var params clob;

begin
  :params := '
{
  "provider": "vertexai",
```

```
  "credential_name": "VERTEXAI_CRED",
  "url": "https://LOCATION-aiplatform.googleapis.com/v1/projects/PROJECT/
locations/LOCATION/publishers/google/models/",
  "model": "multimodalembedding:predict"
}';
end;
/

-- get image embedding: PL/SQL example

declare
  v vector;
  output clob;
begin
  v := dbms_vector_chain.utl_to_embedding(
    to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
  output := vector_serialize(v);
  dbms_output.put_line('vector data=' || dbms_lob.substr(output, 100) ||
'...');
end;
/

-- get image embedding: select example

select dbms_vector_chain.utl_to_embedding(
  to_blob(bfilename('VEC_DUMP', 'parrots.jpg')), 'image', json(:params));
```

- **Text to vector using in-database embedding model**:

  The following example uses `UTL_TO_EMBEDDING` to generate a vector embedding by calling an ONNX format embedding model (`doc_model`) loaded into Oracle AI Database.

  Here, the provider is `database`, and the input is `hello`.

  ```
  var params clob;
  exec :params := '{"provider":"database", "model":"doc_model"}';

  select dbms_vector_chain.utl_to_embedding('hello', json(:params)) from
  dual;
  ```

  For complete example, see Convert Text String to Embedding Within Oracle AI Database.

- **End-to-end examples**:

  To run various end-to-end example scenarios using `UTL_TO_EMBEDDING`, see Generate Embedding.

UTL_TO_EMBEDDINGS:

- **Text to vector using in-database embedding model**:

  The following example uses `UTL_TO_EMBEDDINGS` to generate an array of embeddings by calling an ONNX format embedding model (`doc_model`) loaded into Oracle AI Database.

Here, the provider is `database`, and the input is a PDF document stored in the `documentation_tab` table. As you can see, you first use `UTL_TO_CHUNKS` to split the data into smaller chunks before passing in to `UTL_TO_EMBEDDINGS`.

```
CREATE TABLE doc_chunks as
(select dt.id doc_id, et.embed_id, et.embed_data,
to_vector(et.embed_vector) embed_vector
 from
   documentation_tab dt,
   dbms_vector_chain.utl_to_embeddings(

dbms_vector_chain.utl_to_chunks(dbms_vector_chain.utl_to_text(dt.data),
json('{"normalize":"all"}')),
       json('{"provider":"database", "model":"doc_model"}')) t,
   JSON_TABLE(t.column_value, '$[*]' COLUMNS (embed_id NUMBER PATH
'$.embed_id', embed_data VARCHAR2(4000) PATH '$.embed_data', embed_vector
CLOB PATH '$.embed_vector')) et
);
```

For complete example, see SQL Quick Start Using a Vector Embedding Model Uploaded into the Database.

- **End-to-end examples**:

  To run various end-to-end example scenarios using `UTL_TO_EMBEDDINGS`, see Perform Chunking With Embedding.

UTL_TO_GENERATE_TEXT

Use the `DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT` chainable utility function to generate a text response for a given prompt or an image, by accessing third-party text generation models.

Purpose

To communicate with Large Language Models (LLMs) through natural language conversations. You can generate a textual answer, description, or summary for prompts and images, given as input to LLM-powered chat interfaces.

- Prompt to Text:

  A prompt can be an input text string, such as a question that you ask an LLM. For example, "`What is Oracle Text?`". A prompt can also be a command, such as "`Summarize the following ...`", "`Draft an email asking for ...`", or "`Rewrite the following ...`", and can include results from a search. The LLM responds with a textual answer or description based on the specified task in the prompt.

  For this operation, this API makes a REST call to your chosen remote third-party provider (Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI) or local third-party provider (Ollama).

- Image to Text:

  You can also prompt with a media file, such as an image, to extract text from pictures or photos. You supply a text question as the prompt (such as "`What is this image about?`" or "`How many birds are there in this painting?`") along with the image. The LLM responds with a textual analysis or description of the contents of the image.

  For this operation, this API makes a REST call to your chosen remote third-party provider (Google AI, Hugging Face, OpenAI, or Vertex AI) or local third-party provider (Ollama).

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

This function accepts the input as `CLOB` containing text data (for textual prompts) or as `BLOB` containing media data (for media files such as images). It then processes this information to generate a new `CLOB` containing the generated text.

- Prompt to Text:

```
DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT (
        DATA          IN CLOB,
        PARAMS        IN JSON default NULL
) return CLOB;
```

- Image to Text:

```
DBMS_VECTOR_CHAIN.UTL_TO_GENERATE_TEXT(
          TEXT_DATA      IN CLOB,
          MEDIA_DATA     IN BLOB,
          MEDIA_TYPE     IN VARCHAR2 default 'image/jpeg',
          PARAMS         IN JSON default NULL
) return CLOB;
```

DATA and TEXT_DATA

Specify the textual prompt as `CLOB` for the `DATA` or `TEXT_DATA` clause.

> **Note**
>
> Hugging Face uses an image captioning model that does not require a prompt, when giving an image as input. If you input a prompt along with an image, then the prompt will be ignored.

MEDIA_DATA

Specify the `BLOB` file, such as an image or a visual PDF file.

MEDIA_TYPE

Specify the image format for the given image or visual PDF file (`BLOB` file) in one of the supported image data MIME types. For example:

- For PNG: `image/png`
- For JPEG: `image/jpeg`
- For PDF: `application/pdf`

> **Note**
>
> For a complete list of the supported image formats, refer to your third-party provider's documentation.

PARAMS

Specify the following input parameters in JSON format, depending on the service provider that you want to access for text generation:

```
{
  "provider"        : "<AI service provider>",
  "credential_name" : "<credential name>",
  "url"             : "<REST endpoint URL for text generation service>",
  "model"           : "<text generation model name>",
  "transfer_timeout": <maximum wait time for the request to complete>,
  "max_count": "<maximum calls to the AI service provider>",
  "<additional REST provider parameter>": "<REST provider parameter value>"
}
```

Table 12-28 UTL_TO_GENERATE_TEXT Parameter Details

Parameter	Description
`provider`	Supported REST provider that you want to access to generate text. Specify one of the following values: For `CLOB` input: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `anthropic` • `ollama` For `BLOB` input: • `googleai` • `huggingface` • `openai` • `vertexai` • `mistralai` • `anthropic` • `ollama`

Table 12-28 (Cont.) UTL_TO_GENERATE_TEXT Parameter Details

Parameter	Description
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`host`	When using a local service provider, `host` can be used to disable credential by setting the value to `local`. When the `host` parameter is specified, `credential_name` should be omitted. The `host` parameter is applicable **only if** `provider` is one of the following: • `openai` • `ollama`
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the third-party text generation model in the form: *`schema.model_name`* If the model name is not schema-qualified, then the schema of the procedure invoker is used. **Note**: For Generative AI, all the supported third-party models are listed in Supported Third-Party Provider Operations and Endpoints.
`transfer_timeout`	Maximum time to wait for the request to complete. The default value is `60` seconds. You can increase this value for busy web servers.
`max_count`	Maximum number of times the API can be called for a given third-party provider. When set to an integer *n*, `max_count` stops the execution of the API for the given provider beyond *n* times. This prevents accidental calling of a third-party over some limit, for example to avoid surpassing the service amount that you have purchased.

Additional third-party provider parameters:

Optionally, specify additional provider-specific parameters.

Table 12-29 Additional REST Provider Parameter Details

Parameter	Description
`max_tokens`	Maximum number of tokens in the output text.
`temperature`	Degree of randomness used when generating the output text, in the range of `0.0-5.0`. To generate the same output for a prompt, use `0`. To generate a random new text for that prompt, increase the temperature. **Note**: Start with the temperature set to `0`. If you do not require random results, a recommended temperature value is between `0` and `1`. A higher value is not recommended because a high temperature may produce creative text, which might also include hallucinations.

Table 12-29 (Cont.) Additional REST Provider Parameter Details

Parameter	Description
`topP`	Probability of tokens in the output, in the range of `0.0–1.0`. A lower value provides less random responses and a higher value provides more random responses.
`candidateCount`	Number of response variations to return, in the range of `1-4`.
`maxOutputTokens`	Maximum number of tokens to generate for each response.

Let us look at some example configurations for all third-party providers:

> **Important**
>
> - The following examples are for illustration purposes. For accurate and up-to-date information on additional parameters to use, refer to your third-party provider's documentation.
> - For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Cohere example:

```
{
  "provider"       : "cohere",
  "credential_name": "COHERE_CRED",
  "url"            : "https://api.cohere.example.com/chat",
  "model"          : "command"
}
```

OCI Generative AI (On-Demand mode) example:

> **Note**
>
> For Generative AI, if you want to pass any additional REST provider-specific parameters, then you must enclose those in `chatRequest`.

```
{
  "provider": "ocigenai",
  "credential_name" : "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "meta.llama-3.3-70b-instruct",
  "chatRequest": {
    "maxTokens": 256
  }
}
```

OCI Generative AI (Dedicated mode) examples:

For Cohere models, use `"apiFormat":"COHERE"`

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "chatRequest": {
    "apiFormat":"COHERE"
  }
}
```

For other models, use `"apiFormat":"GENERIC"`

```
{
  "provider": "ocigenai",
  "credential_name": "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "ocid1.generativeaiendpoint.oc1.us-chicago-1...",
  "chatRequest": {
    "apiFormat":"GENERIC"
  }
}
```

> **Note**
>
> For Dedicated mode, specify the model OCID in the `model` parameter instead of the model name.

Google AI example:

```
{
  "provider"        : "googleai",
  "credential_name" : "GOOGLEAI_CRED",
  "url"             : "https://googleapis.example.com/models/",
  "model"           : "gemini-pro:generateContent"
}
```

Hugging Face example:

```
{
  "provider"        : "huggingface",
  "credential_name" : "HF_CRED",
  "url"             : "https://api.huggingface.example.com/models/",
  "model"           : "gpt2"
}
```

Ollama example:

```
{
  "provider"        : "ollama",
  "host"            : "local",
  "url"             : "http://localhost:11434/api/generate",
  "model"           : "phi3:mini"
}
```

OpenAI example:

```
{
  "provider"        : "openai",
  "credential_name" : "OPENAI_CRED",
  "url"             : "https://api.openai.example.com/v1/chat/completions",
  "model"           : "gpt-4o-mini",
  "max_tokens"      : 60,
  "temperature"     : 1.0
}
```

OpenAI-compatible provider examples:

> **Note**
>
> This is just one example of how to set up the configuration for a number of OpenAI-compatible third-party providers, such as Llamafile, vLLM, and xAI. The `provider` value must specify `openai` while the `url` value depends on your chosen third-party provider's REST endpoint.
>
> Set `host` to `local` to disable credential. Set `model` to the desired model if the third-party provider requires a model name (for example, Ollama, vLLM, and so on). Otherwise, you can set `model` to any string such as `any` (for example, if using Llamafile).

```
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/chat/completions",
  "host": "local",
  "model": "any"
}
```

```
{
   "provider": "openai",
   "credential_name": "XAI_CRED",
   "url": "https://api.x.ai/v1/chat/completions",
   "model": "grok-4"
}
```

Vertex AI example:

```
{
  "provider"         : "vertexai",
  "credential_name"  : "VERTEXAI_CRED",
  "url"              : "https://googleapis.example.com/models/",
  "model"            : "gemini-1.0-pro:generateContent",
  "generation_config": {
                        "temperature"    : 0.9,
                        "topP"           : 1,
                        "candidateCount" : 1,
                        "maxOutputTokens": 256
                       }
}
```

Mistral AI example:

```
{
   "provider": "mistralai",
   "credential_name": "MISTRALAI_CRED",
   "url": "https://api.mistral.ai/v1/chat/completions",
   "model": "mistral-small-latest"
}
```

Anthropic example:

```
{
   "provider": "anthropic",
   "credential_name": "ANTHROPIC_CRED",
   "url": "https://api.anthropic.com/v1/messages",
   "model": "claude-3-opus-20240229",
   "max_tokens": 256
}
```

Examples

- Prompt to Text:

  The following statements generate a text response by making a REST call to Generative AI. The prompt given here is "`What is Oracle Text?`".

  Here, the cohere.command-r-16k and meta.llama-3.1-70b-instruct models are used. You can replace the `model` value with any other supported model that you want to use with Generative AI, as listed in Supported Third-Party Provider Operations and Endpoints.

  Using the cohere.command-r-16k model:

  ```
  -- select example

  var params clob;
  exec :params := '
  {
    "provider"       : "ocigenai",
    "credential_name": "OCI_CRED",
    "url"            : "https://inference.generativeai.us-
  chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  ```

```
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

select dbms_vector_chain.utl_to_generate_text(
 'What is Oracle Text?',
 json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';

  params := '
{
  "provider"       : "ocigenai",
  "credential_name": "OCI_CRED",
  "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
  "model"          : "cohere.command-r-16k",
  "chatRequest"    : {
                      "maxTokens": 256
                     }
}';

  output := dbms_vector_chain.utl_to_generate_text(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

Using the meta.llama-3.1-70b-instruct model:

```
-- select example

var params clob;
exec :params := '
{
   "provider"       : "ocigenai",
   "credential_name": "OCI_CRED",
   "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
   "model"          : "meta.llama-3.1-70b-instruct",
```

```
   "chatRequest"    : {
                        "topK" : 1
                       }
}';

select dbms_vector_chain.utl_to_generate_text(
 'What is Oracle Text?',
 json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'What is Oracle Text?';

  params := '
{
   "provider"       : "ocigenai",
   "credential_name": "OCI_CRED",
   "url"            : "https://inference.generativeai.us-
chicago-1.oci.oraclecloud.com/20231130/actions/chat",
   "model"          : "meta.llama-3.1-70b-instruct",
   "chatRequest"    : {
                        "topK" : 1
                       }
}';

  output := dbms_vector_chain.utl_to_generate_text(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

As a third option, you can set the `provider` value to `openai` and then use the `url` to specify the third-party REST endpoint of a number of compatible providers, for example Llamafile and vLLM (in this case Llamafile):

```
set serveroutput on

declare
  preferences clob;
  input clob;
  output clob;
begin
  preferences := '
{
```

```
  "provider": "openai",
  "url": "http://localhost:8080/v1/chat/completions",
  "host": "local",
  "model": "any"
}';
  input := 'What is Oracle Text?';

  output := dbms_vector_chain.utl_to_generate_text(input,
json(preferences));
  dbms_output.put_line(output);

  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

End-to-end examples:

To run end-to-end example scenarios, see Generate Text Response.

- Image to Text:

  The following statements generate a text response by making a REST call to OpenAI. Here, the input is an image (`sample_image.jpeg`) along with the prompt "`Describe this image?`".

```
-- select example

var input clob;
var media_data blob;
var media_type clob;
var params clob;

begin
  :input := 'Describe this image';
  :media_data := load_blob_from_file('DEMO_DIR', 'sample_image.jpeg');
  :media_type := 'image/jpeg';
  :params := '
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60
}';
end;
/

select
dbms_vector_chain.utl_to_generate_text(:input, :media_data, :media_type,
json(:params));
```

```
-- PL/SQL example

declare
  input clob;
  media_data blob;
  media_type varchar2(32);
  params clob;
  output clob;

begin
  input := 'Describe this image';
  media_data := load_blob_from_file('DEMO_DIR', 'image_file');
  media_type := 'image/jpeg';
  params := '
{
  "provider"       : "openai",
  "credential_name": "OPENAI_CRED",
  "url"            : "https://api.openai.com/v1/chat/completions",
  "model"          : "gpt-4o-mini",
  "max_tokens"     : 60
}';

  output := dbms_vector_chain.utl_to_generate_text(
    input, media_data, media_type, json(params));
  dbms_output.put_line(output);

  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
  if media_data is not null then
    dbms_lob.freetemporary(media_data);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

End-to-end examples:

To run end-to-end example scenarios, see Describe Image Content.

UTL_TO_SUMMARY

Use the `DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY` chainable utility function to generate a summary for textual documents.

A summary is a short and concise extract with key features of a document that best represents what the document is about as a whole. A summary can be free-form paragraphs or bullet points based on the format that you specify.

Purpose

To perform a text-to-summary transformation by accessing either Oracle AI Database or a third-party service provider:

- Oracle AI Database as the service provider (default setting):

  Uses the in-house implementation with Oracle AI Database, where Oracle Text is internally used to extract a summary (gist) from your document using the Oracle Text PL/SQL procedure `CTX_DOC.GIST`.

- Third-party summarization model:

  Makes a REST API call to your chosen remote service provider (Cohere, Generative AI, Google AI, Hugging Face, OpenAI, or Vertex AI) or local service provider (Ollama).

> **Note**
>
> Currently, `UTL_TO_SUMMARY` does not work for Generative AI because the model and summary endpoint supported for Generative AI have been retired. It will be available in a subsequent release.

> **Warning**
>
> Certain features of the database may allow you to access services offered separately by third-parties, for example, through the use of JSON specifications that facilitate your access to REST APIs.
>
> Your use of these features is solely at your own risk, and you are solely responsible for complying with any terms and conditions related to use of any such third-party services. Notwithstanding any other terms and conditions related to the third-party services, your use of such database features constitutes your acceptance of that risk and express exclusion of Oracle's responsibility or liability for any damages resulting from such access.

Syntax

```
DBMS_VECTOR_CHAIN.UTL_TO_SUMMARY (
    DATA          IN CLOB,
    PARAMS        IN JSON default NULL
) return CLOB;
```

DATA

This function accepts the input data type in plain text as `CLOB`.

It returns a summary of the input document also as `CLOB`.

PARAMS

Specify summary parameters in JSON format, depending on the service provider that you want to use for document summarization.

If using Oracle AI Database as the provider:

```
{
  "provider"     : "database",
  "glevel"       : "<summary format>",
  "numParagraphs": <number in the range 1-16>,
```

```
  "maxPercent"   : <number in the range 1-100>,
  "num_themes"   : <number in the range 1-50>,
  "language"     : "<name of the language>"
}
```

Table 12-30 Database Provider Parameter Details

<table>
<tr><th>Parameter</th><th>Description</th></tr>
<tr><td><code>provider</code></td><td>Specify <code>database</code> (default setting) to access Oracle AI Database as the provider. Leverages the document <code>gist</code> or summary generated by Oracle Text.</td></tr>
<tr><td><code>glevel</code></td><td>Format to display the summary:
• <code>SENTENCE</code> | <code>S</code>: As a list of sentences
• <code>PARAGRAPH</code> | <code>P</code>: In a free-form paragraph</td></tr>
<tr><td><code>numParagraphs</code></td><td>Maximum number of document paragraphs (or sentences) selected for the summary. The default value is <code>16</code>.
The <code>numParagraphs</code> parameter is used only when this parameter yields a smaller summary size than the summary size yielded by the <code>maxPercent</code> parameter, because the function always returns the smallest size summary.</td></tr>
<tr><td><code>maxPercent</code></td><td>Maximum number of document paragraphs (or sentences) selected for the summary, as a percentage of the total paragraphs (or sentences) in the document. The default value is <code>10</code>.
The <code>maxPercent</code> parameter is used only when this parameter yields a smaller summary size than the summary size yielded by the <code>numParagraphs</code> parameter, because the function always returns the smallest size summary.</td></tr>
<tr><td><code>num_themes</code></td><td>Number of theme summaries to produce. For example, if you specify <code>10</code>, then this function returns the top 10 theme summaries. If you specify <code>0</code> or <code>NULL</code>, then this function returns all themes in a document.
The default value is <code>50</code>. If the document contains more than 50 themes, only the top 50 themes show conceptual hierarchy.</td></tr>
<tr><td><code>language</code></td><td>Language name of your summary text, as listed in Supported Languages and Data File Locations.</td></tr>
</table>

For example:

```
{
  "provider"          : "database",
  "glevel"            : "sentence",
  "numParagraphs"     : 1
}
```

If using a third-party provider:

Set the following parameters along with additional summarization parameters specific to your provider:

```
{
  "provider"          : "<AI service provider>",
  "credential_name"   : "<credential name>",
  "url"               : "<REST endpoint URL for summarization service>",
  "model"             : "<REST provider summarization model name>",
  "transfer_timeout"  : <maximum wait time for the request to complete>,
  "max_count": "<maximum calls to the AI service provider>",
```

```
  "<additional REST provider parameter>": "<REST provider parameter value>"
}
```

Table 12-31 Third-Party Provider Parameter Details

Parameter	Description
`provider`	Third-party service provider that you want to access to get the summary. A REST call is made to the specified provider to access its text summarization model. Specify one of the following values: • `cohere` • `googleai` • `huggingface` • `ocigenai` • `openai` • `vertexai` • `mistralai` • `anthropic`
`credential_name`	Name of the credential in the form: *`schema.credential_name`* A credential name holds authentication credentials to enable access to your provider for making REST API calls. You need to first set up your credential by calling the `DBMS_VECTOR_CHAIN.CREATE_CREDENTIAL` helper function to create and store a credential, and then refer to the credential name here. See CREATE_CREDENTIAL.
`url`	URL of the third-party provider endpoint for each REST call, as listed in Supported Third-Party Provider Operations and Endpoints.
`model`	Name of the third-party text summarization model in the form: *`schema.model_name`* If the model name is not schema-qualified, then the schema of the procedure invoker is used. **Note**: For Generative AI, you must specify *`schema.model_name`*. All the third-party models supported for Generative AI are listed in Supported Third-Party Provider Operations and Endpoints.
`transfer_timeout`	Maximum time to wait for the request to complete. The default value is `60` seconds. You can increase this value for busy web servers.
`max_count`	Maximum number of times the API can be called for a given third-party provider. When set to an integer *n*, `max_count` stops the execution of the API for the given provider beyond *n* times. This prevents accidental calling of a third-party over some limit, for example to avoid surpassing the service amount that you have purchased.

Additional third-party provider parameters:

Optionally, specify additional provider-specific parameters.

Table 12-32 Additional REST Provider Parameter Details

Parameter	Description
`length`	Approximate length of the summary text: • `SHORT`: Roughly up to 2 sentences • `MEDIUM`: Between 3 and 5 sentences • `LONG`: 6 or more sentences • `AUTO`: The model chooses a length based on the input size **Note**: For Generative AI, you must enter this value in uppercase.
`format`	Format to display the summary: • `PARAGRAPH`: In a free-form paragraph • `BULLETS`: In bullet points **Note**: For Generative AI, you must enter this value in uppercase.
`temperature`	Degree of randomness used when generating output text, in the range of `0.0-5.0`. To generate the same output for a prompt, use `0`. To generate a random new text for that prompt, increase the temperature. Default temperature is `1` and the maximum temperature is `5`. **Note**: To summarize a text, start with the temperature set to `0`. If you do not require random results, a recommended temperature value is `0.2` for Generative AI and between `0` and `1` for Cohere. Use a higher value if for example you plan to perform a selection of the various summaries afterward. Do not use a high temperature for summarization because a high temperature encourages the model to produce creative text, which might also include hallucinations.
`extractiveness`	How much to reuse the input in the summary: • `LOW`: Summaries with low extractiveness tend to paraphrase. • `HIGH`: Summaries with high extractiveness lean toward reusing sentences verbatim. **Note**: For Generative AI, you must enter this value in uppercase.
`max_tokens`	Maximum number of tokens in the output text.
`topP`	Probability of tokens in the output, in the range of `0.0–1.0`. A lower value provides less random responses and a higher value provides more random responses.
`candidateCount`	Number of response variations to return, in the range of `1-4`.
`maxOutputTokens`	Maximum number of tokens to generate for each response.

> **Note**
>
> When specifying the `length`, `format`, and `extractiveness` parameters for Generative AI, ensure to enter the values in uppercase letters.

Let us look at some example configurations for all third-party providers:

> **Important**
>
> - The following examples are for illustration purposes. For accurate and up-to-date information on additional parameters to use, refer to your third-party provider's documentation.
> - For a list of all supported REST endpoint URLs, see Supported Third-Party Provider Operations and Endpoints.

Cohere example:

```
{
  "provider"          : "cohere",
  "credential_name"   : "COHERE_CRED",
  "url"               : "https://api.cohere.example.com/summarize",
  "model"             : "command",
  "length"            : "medium",
  "format"            : "paragraph",
  "temperature"       : 1.0
}
```

Generative AI example:

```
{
 "provider"           : "ocigenai",
 "credential_name"    : "OCI_CRED",
 "url"                : "https://generativeai.oci.example.com/summarizeText",
 "model"              : "cohere.command-r-16k",
 "length"             : "MEDIUM",
 "format"             : "PARAGRAPH"
}
```

Google AI example:

```
{
  "provider"          : "googleai",
  "credential_name"   : "GOOGLEAI_CRED",
  "url"               : "https://googleapis.example.com/models/",
  "model"             : "gemini-pro:generateContent",
  "generation_config" : {
    "temperature"     : 0.9,
    "topP"            : 1,
    "candidateCount"  : 1,
    "maxOutputTokens" : 256
  }
}
```

Hugging Face example:

```
{
  "provider"          : "huggingface",
  "credential_name"   : "HF_CRED",
```

```
  "url"               : "https://api.huggingface.example.co/models/",
  "model"             : "facebook/bart-large-cnn"
}
```

OCI Generative AI (ocigenai) example

```
{
  "provider": "ocigenai",
  "credential_name" : "OCI_CRED",
  "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
  "model": "meta.llama-3.3-70b-instruct",
  "temperature": "0.0",
  "length": "SHORT",
  "format": "PARAGRAPH",
  "chatRequest": {
    "maxTokens": 256
  }
}
```

Ollama example:

```
{
  "provider"          : "ollama",
  "host"              : "local",
  "url"               : "http://localhost:11434/api/generate",
  "model"             : "phi3:mini"
}
```

OpenAI example:

```
{
  "provider"          : "openai",
  "credential_name"   : "OPENAI_CRED",
  "url"               : "https://api.openai.example.com/v1/chat/completions",
  "model"             : "gpt-4o-mini",
  "max_tokens"        : 256,
  "temperature"       : 1.0
}
```

OpenAI-compatible provider examples:

> **Note**
>
> This is just one example of how to set up the configuration for a number of OpenAI-compatible third-party providers, such as Llamafile, vLLM, and xAI. The `provider` value must specify `openai` while the `url` value depends on your chosen third-party provider's REST endpoint.
>
> Set `host` to `local` to disable credential. Set `model` to the desired model if the third-party provider requires a model name (for example, Ollama, vLLM, and so on). Otherwise, you can set `model` to any string such as `any` (for example, if using Llamafile).

```
{
  "provider": "openai",
  "url": "http://localhost:8080/v1/chat/completions",
  "host": "local",
  "model": "any"
}
```

```
{
   "provider": "openai",
   "credential_name": "XAI_CRED",
   "url": "https://api.x.ai/v1/chat/completions",
   "model": "grok-4"
}
```

Vertex AI example:

```
{
  "provider"          : "vertexai",
  "credential_name"   : "VERTEXAI_CRED",
  "url"               : "https://googleapis.example.com/models/",
  "model"             : "gemini-1.0-pro:generateContent",
  "generation_config" : {
    "temperature"     : 0.9,
    "topP"            : 1,
    "candidateCount"  : 1,
    "maxOutputTokens" : 256
  }
}
```

Mistral AI example:

```
{
   "provider": "mistralai",
   "credential_name": "MISTRALAI_CRED",
   "url": "https://api.mistral.ai/v1/chat/completions",
   "model": "mistral-small-latest"
}
```

Anthropic example:

```
{
   "provider": "anthropic",
   "credential_name": "ANTHROPIC_CRED",
   "url": "https://api.anthropic.com/v1/messages",
   "model": "claude-3-opus-20240229",
   "max_tokens": 256
}
```

Examples

- **Generate summary using Oracle AI Database**:

  This statement specifies `database` as the provider. Here, the Oracle Text PL/SQL procedure `CTX_DOC.GIST` is internally called to generate a summary of an extract on "Transactions".

```
-- select example

set serveroutput on

var params clob;
begin
  :params := '
{
  "provider": "database",
  "glevel": "sentence",
  "numParagraphs": 1
}';
end;
/

select dbms_vector_chain.utl_to_summary(
 'A transaction is a logical, atomic unit of work that contains one or
more SQL statements.  An RDBMS must be able to group SQL statements so
that they are either all committed, which means they are applied to the
database, or all rolled back, which means they are undone.  An
illustration of the need for transactions is a funds transfer from a
savings account to a checking account. The transfer consists of the
following separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed or
fail as a unit. For example, if a hardware failure prevents a statement in
the transaction from executing, then the other statements must be rolled
back.
    Transactions set Oracle AI Database apart from a file system. If you
perform an atomic operation that updates several files, and if the system
fails halfway through, then the files will not be consistent. In contrast,
a transaction moves an Oracle AI database from one consistent state to
another. The basic principle of a transaction is "all or nothing": an
atomic operation succeeds or fails as a whole.',
 json(:params)) from dual;
```

```
-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'A transaction is a logical, atomic unit of work that contains
one or more SQL statements.  An RDBMS must be able to group SQL statements
so that they are either all committed, which means they are applied to the
database, or all rolled back, which means they are undone.  An
illustration of the need for transactions is a funds transfer from a
savings account to a checking account. The transfer consists of the
following separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed or
fail as a unit. For example, if a hardware failure prevents a statement in
the transaction from executing, then the other statements must be rolled
back.
    Transactions set Oracle AI Database apart from a file system. If you
perform an atomic operation that updates several files, and if the system
fails halfway through, then the files will not be consistent. In contrast,
a transaction moves an Oracle AI database from one consistent state to
another. The basic principle of a transaction is "all or nothing": an
atomic operation succeeds or fails as a whole.';

  params := '
  {
   "provider": "database",
   "glevel": "sentence",
   "numParagraphs": 1
  }';

  output := dbms_vector_chain.utl_to_summary(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

- **Generate summary using Generative AI**:

  These statements generate a summary of an extract on "Transactions" by accessing Generative AI as the provider.

Here, the cohere.command-r-16k model is used for the summarization operation. You can replace the `model` value with any other supported model that you want to use with Generative AI, as listed in Supported Third-Party Provider Operations and Endpoints.

```
-- select example

var params clob;
begin
  :params := '
{
 "provider": "ocigenai",
 "credential_name": "OCI_CRED",
 "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
 "model": "cohere.command-r-16k",
 "temperature": "0.0",
 "extractiveness": "LOW"
}';
end;
/

select dbms_vector_chain.utl_to_summary(
 'A transaction is a logical, atomic unit of work that contains one or
more SQL statements.  An RDBMS must be able to group SQL statements so
that they are either all committed, which means they are applied to the
database, or all rolled back, which means they are undone.  An
illustration of the need for transactions is a funds transfer from a
savings account to a checking account. The transfer consists of the
following separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed or
fail as a unit. For example, if a hardware failure prevents a statement in
the transaction from executing, then the other statements must be rolled
back.
    Transactions set Oracle AI Database apart from a file system. If you
perform an atomic operation that updates several files, and if the system
fails halfway through, then the files will not be consistent. In contrast,
a transaction moves an Oracle AI database from one consistent state to
another. The basic principle of a transaction is "all or nothing": an
atomic operation succeeds or fails as a whole.',
 json(:params)) from dual;

-- PL/SQL example

declare
  input clob;
  params clob;
  output clob;
begin
  input := 'A transaction is a logical, atomic unit of work that contains
one or more SQL statements.  An RDBMS must be able to group SQL statements
so that they are either all committed, which means they are applied to the
database, or all rolled back, which means they are undone.  An
illustration of the need for transactions is a funds transfer from a
```

```
savings account to a checking account. The transfer consists of the
following separate operations:
    1. Decrease the savings account.
    2. Increase the checking account.
    3. Record the transaction in the transaction journal.
    Oracle AI Database guarantees that all three operations succeed or
fail as a unit. For example, if a hardware failure prevents a statement in
the transaction from executing, then the other statements must be rolled
back.
    Transactions set Oracle AI Database apart from a file system. If you
perform an atomic operation that updates several files, and if the system
fails halfway through, then the files will not be consistent. In contrast,
a transaction moves an Oracle AI database from one consistent state to
another. The basic principle of a transaction is "all or nothing": an
atomic operation succeeds or fails as a whole.';

  params := '
  {
   "provider": "ocigenai",
   "credential_name": "OCI_CRED",
   "url": "https://inference.generativeai.us-chicago-1.oci.oraclecloud.com/
20231130/actions/chat",
   "model": "cohere.command-r-16k",
   "length": "MEDIUM",
   "format": "PARAGRAPH",
   "temperature": 1.0
  }';

  output := dbms_vector_chain.utl_to_summary(input, json(params));
  dbms_output.put_line(output);
  if output is not null then
    dbms_lob.freetemporary(output);
  end if;
exception
  when OTHERS THEN
    DBMS_OUTPUT.PUT_LINE (SQLERRM);
    DBMS_OUTPUT.PUT_LINE (SQLCODE);
end;
/
```

- **End-to-end examples**:

  To run end-to-end example scenarios using this function, see Generate Summary.

UTL_TO_TEXT

Use the `DBMS_VECTOR_CHAIN.UTL_TO_TEXT` chainable utility function to convert an input document (for example, PDF, DOC, JSON, XML, or HTML) to plain text.

Purpose

To perform a file-to-text transformation by using the Oracle Text component (`CONTEXT`) of Oracle AI Database.

Syntax

```
DBMS_VECTOR_CHAIN.UTL_TO_TEXT (
    DATA          IN CLOB | BLOB,
    PARAMS        IN JSON default NULL
) return CLOB;
```

DATA

This function accepts the input data type as `CLOB` or `BLOB`. It can read documents from a remote location or from files stored locally in the database tables.

It returns a plain text version of the document as `CLOB`.

Oracle Text supports around 150 file types. For a complete list of all the supported document formats, see *Oracle Text Reference*.

PARAMS

Specify the following input parameter in JSON format:

```
{
    "plaintext": "true or false",
    "charset": "UTF8 | EUCJP | <other_valid_charset>",
    "format": "BINARY | TEXT | IGNORE"
}
```

Table 12-33 Parameter Details

Parameter	Description
`plaintext`	Plain text output. The default value for this parameter is `true`, that is, by default the output format is plain text. If you do not want to return the document as plain text, then set this parameter to `false`. When you set it to `false`, then the document is returned as HTML markup instead of plain text.
`charset`	Character set encoding. The default value for this parameter is the current database character set. That is, by default, the input is assumed to use the same character set as the database. If your input uses a different character set, specify that character set using this parameter.
`format`	Format type of the content to be processed. Valid values are: • `BINARY`: For rich content like PDF/Word. The default value of the `format` parameter. • `TEXT`: For plain text • `IGNORE`: No conversion

Example

```
select DBMS_VECTOR_CHAIN.UTL_TO_TEXT (
    t.blobdata,
    json('{
            "plaintext": "true",
            "charset"  : "UTF8",
```

```
              "format"    : "TEXT"
            }')
) from tab t;
```

End-to-end example:

To run an end-to-end example scenario using this function, see Convert File to Text to Chunks to Embeddings Within Oracle AI Database.

Supported Languages and Data File Locations

These are the supported languages for which language data files are distributed by default in the specified directories.

Language Name	Abbreviation	Data File
AFRIKAANS	af	ctx/data/eos/dreosaf.txt
AMERICAN	us	ctx/data/eos/dreosus.txt
ARABIC	ar	ctx/data/eos/dreosar.txt
BASQUE	eu	ctx/data/eos/dreoseu.txt
BELARUSIAN	be	ctx/data/eos/dreosbe.txt
BRAZILIAN PORTUGUESE	ptb	ctx/data/eos/dreosptb.txt
BULGARIAN	bg	ctx/data/eos/dreosbg.txt
CANADIAN FRENCH	frc	ctx/data/eos/dreosfrc.txt
CATALAN	ca	ctx/data/eos/dreosca.txt
CROATIAN	hr	ctx/data/eos/dreoshr.txt
CYRILLIC SERBIAN	csr	ctx/data/eos/dreoscsr.txt
CZECH	cs	ctx/data/eos/dreoscs.txt
DANISH	dk	ctx/data/eos/dreosdk.txt
DARI	prs	ctx/data/eos/dreosprs.txt
DUTCH	nl	ctx/data/eos/dreosnl.txt
EGYPTIAN	eg	ctx/data/eos/dreoseg.txt
ENGLISH	gb	ctx/data/eos/dreosgb.txt
ESTONIAN	et	ctx/data/eos/dreoset.txt
FINNISH	sf	ctx/data/eos/dreossf.txt
FRENCH	f	ctx/data/eos/dreosf.txt
GALICIAN	ga	ctx/data/eos/dreosga.txt
GERMAN	d	ctx/data/eos/dreosd.txt
GERMAN DIN	din	ctx/data/eos/dreosdin.txt
GREEK	el	ctx/data/eos/dreosel.txt
HEBREW	iw	ctx/data/eos/dreosiw.txt
HINDI	hi	ctx/data/eos/dreoshi.txt
HUNGARIAN	hu	ctx/data/eos/dreoshu.txt
ICELANDIC	is	ctx/data/eos/dreosis.txt

Language Name	Abbreviation	Data File
INDONESIAN	in	`ctx/data/eos/dreosin.txt`
ITALIAN	i	`ctx/data/eos/dreosi.txt`
JAPANESE	ja	`ctx/data/eos/dreosja.txt`
KOREAN	ko	`ctx/data/eos/dreosko.txt`
LATIN AMERICAN SPANISH	esa	`ctx/data/eos/dreosesa.txt`
LATIN BOSNIAN	lbs	`ctx/data/eos/dreoslbs.txt`
LATIN SERBIAN	lsr	`ctx/data/eos/dreoslsr.txt`
LATVIAN	lv	`ctx/data/eos/dreoslv.txt`
LITHUANIAN	lt	`ctx/data/eos/dreoslt.txt`
MACEDONIAN	mk	`ctx/data/eos/dreosmk.txt`
MALAY	ms	`ctx/data/eos/dreosms.txt`
MEXICAN SPANISH	esm	`ctx/data/eos/dreosesm.txt`
NORWEGIAN	n	`ctx/data/eos/dreosn.txt`
NYNORSK	nn	`ctx/data/eos/dreosnn.txt`
PERSIAN	fa	`ctx/data/eos/dreosfa.txt`
POLISH	pl	`ctx/data/eos/dreospl.txt`
PORTUGUESE	pt	`ctx/data/eos/dreospt.txt`
ROMANIAN	ro	`ctx/data/eos/dreosro.txt`
RUSSIAN	ru	`ctx/data/eos/dreosru.txt`
SIMPLIFIED CHINESE	zhs	`ctx/data/eos/dreoszhs.txt`
SLOVAK	sk	`ctx/data/eos/dreossk.txt`
SLOVENIAN	sl	`ctx/data/eos/dreossl.txt`
SPANISH	e	`ctx/data/eos/dreose.txt`
SWEDISH	s	`ctx/data/eos/dreoss.txt`
THAI	th	`ctx/data/eos/dreosth.txt`
TRADITIONAL CHINESE	zht	`ctx/data/eos/dreoszht.txt`
TURKISH	tr	`ctx/data/eos/dreostr.txt`
UKRAINIAN	uk	`ctx/data/eos/dreosuk.txt`
URDU	ur	`ctx/data/eos/dreosur.txt`

Related Topics

- CREATE_LANG_DATA
  Use the `DBMS_VECTOR_CHAIN.CREATE_LANG_DATA` chunker helper procedure to load your own language data file into the database.

DBMS_HYBRID_VECTOR

The `DBMS_HYBRID_VECTOR` package contains a JSON-based query API `SEARCH`, which lets you query against hybrid vector indexes.

- SEARCH
  Use the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL function to run textual queries, vector similarity queries, or hybrid queries against hybrid vector indexes.
- GET_SQL
  Use the `DBMS_HYBRID_VECTOR.GET_SQL` PL/SQL function to return the internal SQL query that is generated from the parameters.
- SEARCHPIPELINE
  Use the standard table function `DBMS_HYBRID_VECTOR.SEARCHPIPELINE` to return a pipeline of row records.

SEARCH

Use the `DBMS_HYBRID_VECTOR.SEARCH` PL/SQL function to run textual queries, vector similarity queries, or hybrid queries against hybrid vector indexes.

Purpose

To search by vectors and keywords. This function lets you perform the following tasks:

- Facilitate a combined (hybrid) query of textual documents and vectorized chunks:

  You can query a hybrid vector index in multiple vector and keyword search combinations called search modes, as described in Understand Hybrid Search. This API accepts a JSON specification for all query parameters.
- Fuse and reorder the search results:

  The search results of a hybrid query are fused into a unified result set as `CLOB` using the specified fusion set operator, and reordered by a combined score using the specified scoring algorithm.
- Run a default query for a simplified search experience:

  The minimum input parameters required are the `hybrid_index_name` and `search_text`. The same text string is used to query against a vectorized chunk index and a text document index.

Syntax

```
DBMS_HYBRID_VECTOR.SEARCH(
   json(
     '{  "hybrid_index_name"     :  "<hybrid_vector_index_name>",
         "partition_name"        :  "<index partition name>",
         "search_text"           :  "<query string for keyword-and-semantic
search>",
         "search_fusion"         :  one of these values : "INTERSECT | UNION
| TEXT_ONLY | VECTOR_ONLY | MINUS_TEXT |
                                       MINUS_VECTOR | RERANK",
         "search_scorer"         :  one of these values : "RRF | RSF | WRRF",
         "score_calc"            :  "<a mathematical expression for custom
scoring calculation>",
         "vector":
           {
            "search_text"            :  "<query string for semantic
search>",
            "search_vector"          :  "<vector_embedding>",
            "search_mode"            :  one of thse values : "DOCUMENT |
```

```
CHUNK",
            "aggregator"              :  one of these values : "COUNT | SUM |
MIN | MAX | AVG | MEDIAN | BONUSMAX | WINAVG |
                                         ADJBOOST | MAXAVGMED",

            "result_max"              :  <maximum number of vector results>,
            "score_weight"            :  <weight of vector score for RSF>,
            "rank_penalty"            :  <penalty of vector ranking for RRF>,
            "inpath"                  :  <an array of valid JSON paths>,
            "accuracy"                :  <target accuracy for semantic
search>,
            "index_probes"            :  <neighbor partitions for semantic
search>,
            "index_efsearch"          :  <efsearch for semantic search>,
            "filter_type"             :  one of these values : "IN_WO | IN_W
| PRE_WO | PRE_W | POST_WO | DEFAULT"
           },
         "text":
           {
            "contains"                :  "<query string for keyword
search>",
            "search_text"             :  "<alternative text to use to
construct a contains query automatically>",
            "json_textcontains"       :  <an array of valid JSON path and a
query string>,
            "score_weight"            :  <weight of text score for RSF>,
            "rank_penalty"            :  <penalty of text ranking for RRF>,
            "result_max"              :  <maximum number of document results>,
            "inpath"                  :  <array of valid JSON paths>,
            "snippet"                 :  <length of the snippet>
           },
         "filter_by":
           {
            "op"                      :  one of these values: "< | > | <= |
>= | = | != | ^= | <> | LIKE | LIKEC | LIKE2 | LIKE4 |
                                         REGEXP_LIKE | BETWEEN | EXISTS |
INSTR | INSTRC | INSTR2 | INSTR4 | STSTR | STSTR2 | STSTR4 |
                                         STSTRB | STSTRC | <ANY | >ANY |
<=ANY | >=ANY | =ANY | !=ANY | <SOME | >SOME |
                                         <=SOME | >=SOME | =SOME | !=SOME |
<ALL | >ALL | <=ALL | >=ALL | =ALL | !=ALL | IN |
                                         AND | OR | NOT | NOTOR |
NOTAND",
            "type"                    :  one of these values : "number |
string | date | timestamp",
            "col"                     :  "<base table column name>",
            "path"                    :  "<JSON path dot notation within a
base table JSON column>",
            "func"                    :  one of these values : "ABS | FLOOR |
LENGTH | CEILING | UPPER | LOWER | TO_BOOLEAN |
                                         TO_DATE | TO_DOUBLE |
TO_BINARYDOUBLE | TO_NUMBER | TO_CHAR | TO_TIMESTAMP",
            "args"                    :  <an array of arguments to the
operator>,
            "passing"                 :  <an array of variable bindings for
the EXISTS operator path expression>
```

```
          }
         "return":
          {
           "topN"                    :
<topN_value>,
           "values"                  :  one or more of these values : "rowid
| score | vector_score | text_score | vector_rank |
                                        text_rank | chunk_text | chunk_id |
paths",
           "format"                  :  one of these values : "JSON | XML"

          }
     }'
   )
)
```

> **Note**
>
> This API supports two constructs of search. One where you specify a single `search_text` field for both semantic search and keyword search (default setting). Another where you specify separate `search_text` and `contains` query fields using `vector` and `text` sub-elements for semantic search and keyword search, respectively. You cannot use both of these search constructs in one query.

hybrid_index_name

Specify the name of the hybrid vector index to use.

For information on how to create a hybrid vector index if not already created, see Manage Hybrid Vector Indexes.

partition_name

Specify the index partition name so that the results are returned only from the specified partition. Please refer to Manage Hybrid Vector Indexes to understand creation of Local Hybrid Vector Indexes which allows you to create local partitions.

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "partition_name"    : "SYS_IDX_P1",
            "search_text"       : "leadership experience"
          }'))
FROM DUAL;
```

> **Note**
>
> `SYS_IDX_P1` is a sample partition name. Specify the custom partition name or system generated partition name which you used when creating a local hybrid vector index.

search_text

Specify a search text string (your query input) for both semantic search and keyword search.

The same text string is used for a keyword query on document text index (by converting the `search_text` into a `CONTAINS ACCUM` operator syntax) and a semantic query on vectorized chunk index (by vectorizing or embedding the `search_text` for a `VECTOR_DISTANCE` search).

For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "search_text"       : "C, Python"
          }'))
FROM DUAL;
```

> **Note**
>
> The `search_text` parameter uses the same embedding model specified in the vectorizer during index creation. This parameter converts the textual input in `search_text` into a vector at query time. However, do not use the `search_text` parameter at query time for semantic searches in cases where an embedding model is not specified in the vectorizer because the input JSON data already contains vector fields. In such cases, you should explicitly specify the query vector by using the search_vector parameter.

search_fusion

Specify a fusion sort operator to define what you want to retain from the combined set of keyword-and-semantic search results.

> **Note**
>
> This search fusion operation is applicable only to non-pure hybrid search cases. Vector-only and text-only searches do not fuse any results.

Parameter	Description
`INTERSECT`	Returns only the rows that are common to both text search results and vector search results. Score condition: `text_score > 0 AND vector_score > 0`
`UNION` (default)	Combines all distinct rows from both text search results and vector search results. Score condition: `text_score > 0 OR vector_score > 0`
`TEXT_ONLY`	Returns all distinct rows from text search results plus the ones that are common to both text search results and vector search results. Thus, the fused results contain the text search results that appear in text search, including those that appear in both. Score condition: `text_score > 0`
`VECTOR_ONLY`	Returns all distinct rows from vector search results plus the ones that are common to both text search results and vector search results. Thus, the fused results contain the vector search results that appear in vector search, including those that appear in both. Score condition: `vector_score > 0`
`MINUS_TEXT`	Returns all distinct rows from vector search results minus the ones that are common to both text search results and vector search results. Thus, the fused results contain the vector search results that appear in vector search, excluding those that appear in both. Score condition: `text_score = 0`

Parameter	Description
MINUS_VECTOR	Returns all distinct rows from text search results minus the ones that are common to both text search results and vector search results. Thus, the fused results contain the text search results that appear in text search, excluding those that appear in both. Score condition: `vector_score = 0`
RERANK	Returns all distinct rows from text search ordered by the aggregated vector score of their respective vectors. There is no score condition for this field since the text search is followed by the use of the aggregated document vector scores.

See Also

Usage Notes

For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name"     : "my_hybrid_idx",
            "search_fusion"         : "UNION",
            "vector":
                    { "search_text" : "leadership experience" },
             "text":
                    { "contains"    : "C and Python" }
          }'))
FROM DUAL;
```

Usage Notes

When using `DBMS_HYBRID_VECTOR.SEARCH` with `search_fusion` set to `RERANK` and `search_scorer` set to `RRF` (Reciprocal Rank Fusion) or `WRRF` (Weighted Reciprocal Rank Fusion), the results may not reflect the intended semantic ordering. Instead, the final ranking may prioritize rows based on their text rank rather than the semantic rerank score derived from vector distances. This behavior can lead to misleading results where rows with lower vector scores are ranked higher due to their higher text ranks.

`RERANK` mode is not a fusion search mode, despite being specified via the `search_fusion` parameter. Instead, `RERANK` performs a text keyword search and then uses vector scores to rerank the results. When `RRF` or `WRRF` is used with `RERANK` mode these scorers do not properly aggregate vector distances or semantic scores. Instead, it relies solely on the reciprocal text ranks and does not incorporate the semantic scores, even though they are computed. This behavior can lead to misleading results where rows with lower semantic scores (vector distances) are ranked higher due to their higher text ranks. Additionally, `RERANK` only considers documents retrieved by the initial text search (limited by `text.result_max`), meaning better semantically scoring documents may exist but are not evaluated. As a result, the final output may prioritize rows with higher text ranks, even if their vector scores are lower.

Applications relying on `DBMS_HYBRID_VECTOR.SEARCH` with `RERANK` and `RRF` or `WRRF` may receive results that appear valid but do not reflect the intended semantic ordering. This can lead to:

- Inaccurate rankings: Rows with higher semantic relevance (higher vector scores) may be ranked lower than less relevant rows with higher text ranks.

- Suboptimal search performance: Users may experience misleading or incorrect results, impacting data correctness and application reliability.

To achieve proper semantic reranking, avoid using `RRF` or `WRRF` scorers with `search_fusion=RERANK`. It is recommended to use `search_scorer=RSF` (Relative Score Fusion) for reranking, as it correctly prioritizes rows based on their vector scores.

search_scorer

Specify a method to evaluate the combined "fusion" search scores from both keyword and semantic search results.

- `RSF` (default) to use the Relative Score Fusion (RSF) algorithm
- `RRF` to use the Reciprocal Rank Fusion (RRF) algorithm
- `WRRF` to use the Weighted Reciprocal Rank Fusion (WRRF) algorithm

For a deeper understanding of how these algorithms work in hybrid search modes, see Understand Hybrid Search.

For example:

With a single search text string for hybrid search:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_text"       : "C, Python",
         "search_scorer"     : "rsf"
      }'))
FROM DUAL;
```

With separate vector and text search strings:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_scorer"     : "rsf",
         "vector":
          { "search_text"    : "leadership experience" },
         "text":
          { "contains"       : "C and Python" }
      }'))
FROM DUAL;
```

score_calc

While you can use the standard algorithms such as RSF, RRF and WRRF to evaluate the combined "fusion" search scores from both keyword and semantic search results, you can also define and use your own custom score calculations. The `score_calc` field allows you to define the score computation.

> **Note**
>
> You can only use one of the two fields to obtain the search score. Either use the field `search_scorer` which uses the standard RSF, RRF or WRRF algorithms. Or, use the `score_calc` to define a custom scoring algorithm.

The score_calc field accepts either a math expression or a case expression.

The expression consists of two parts: an operator (op) and one or more operands (opnds), depending on the chosen operator. The format of both math and case expressions are shown below:

A **mathematical expression** contains a math operator specified as `op` and corresponding operands specified as `opnds`. The number of operands depends on the choice of the operator. The table below lists the possible operators and operands.

```
MATH_EXPR := { “op” : “MATH_OPERATOR”, “opnds” : [ “OPERAND1”, ... ] }
OPERAND := COLUMN_NAME | PARAMETER_NAME | NUMBER | EXPR
```

The following table specifies the possible values for operators (`op`) and operands (`OPERAND`) specified in the `score_calc` expression.

Table 12-34 Possible Values for Operators and Operands

	Math Expression	Case Expression
operator - `op`	One of the following mathematical operators: • `MUL, ADD, SUB, DIV` - requires two or more operands to be specified. • `ABS, CEIL, EXP, FLOOR, LN, SQRT` - requires only one operands to be specified. • `LOG, MOD, POWER, REMAINDER, ROUND` - requires only two operands to be specified. • `LEAST, GREATEST` - requires two or more operands to be specified.	`case`

Table 12-34 (Cont.) Possible Values for Operators and Operands

	Math Expression	Case Expression
operands - `OPERAND`	The `OPERAND` can one of the following : `COLUMN_NAME`, `PARAMETER_NAME`, `NUMBER` or a `EXPR` The number of required operands depends on the operator. Possible column names (`COLUMN_NAME`) : • `text_score` representing the text contains score • `vector_score` representing the vector distance score (aggregated if `DOCUMENT` mode) • `text_rank` representing the rank of the text result • `vector_rank` representing the rank of the vector result parameter names : • `text_weight` refers to `text.score_weight` parameter • `text_penalty` refers to `text.rank_penalty` parameter • `vector_weight` refers to `vector.score_weight` parameter • `vector_penalty` refers to `vector.rank_penalty` parameter `NUMBER` can contain only numbers `EXPR` is a sub-expression	

A **case expression** has a `CASE` operator and must have pairs of conditional expressions (`COND_EXPR`) and `OPERAND`s as shown below :

```
CASE_EXPR := { “op” : “case”, “opnds” : [ "COND_EXPR", “OPERAND1”, ..., "else",
OPERAND-ELSE ] }
```

The possible `OPERAND` values are specified in the table above. The `COND_EXPR` is also a JSON element form containing operators (`op`) and operands (`opnds`). The possible formats of the `COND_EXPR` is specified below:

```
COND_EXPR  :=  { “op” : “COMPARATIVE_OPERATOR”, “opnds” :
[“ARGUMENT1”, ... ] }
COND_EXPR  :=  { “op” : “LOGICAL_OPERATOR”, “opnds” : [“COND_EXPR”, ... ] }
```

Possible values of `COMPARATIVE_OPERATOR`:

- `EQ` - equal (=)
- `LT` - less than (<)
- `LTE` - less than equal (<=)
- `GT` - greater than (>)
- `GTE` - greater than equal (>=)
- `NEQ` - not equal (!=)

Possible values of `LOGICAL_OPERATOR`:

- AND
- OR
- NOT
- NOTAND - short cut for NOT (x AND ... z)

- NOTOR - short cut for NOT (x OR ... y)

> **Note**
>
> The comparative operator ***ARGUMENT*** can be any of the ***OPERAND*** types specified in the table above, except for a sub-expression (EXPR).

For example, this code defines a custom score depending on one of the following cases:

- If the search is a vector only search (in this case, text_score =0), then score is 50+25% of the vector score.
- If the search is a text only search (in this case, vector_score =0), then score is 25+25% of the text score.
- Else, it is a hybrid search (in this case both text_score > 0 and vector_score > 0), then the score is 75+12.5% of each score.

```
SELECT dbms_hybrid_vector.search(
       JSON('{ "hybrid_index_name" : "my_hybrid_idx",
               "vector" : { "search_text" : "database leadership" },
               "text" : { "contains" : "strong AND database" },
               "search_fusion" : "UNION",
               "score_calc" :
                  { "op" : "case",
                    "opnds" : [ { "op" : "eq", "opnds" : [ "text_score", 0 ]},
                                { "op" : "add", "opnds" : [
                                    50,
                                    { "op" : "mul",
                                      "opnds" : [ "vector_score",
0.25 ] } ] },
                                { "op" : "eq", "opnds" : [ "vector_score",
0]},
                                { "op" : "add", "opnds" : [
                                    25,
                                    { "op" : "mul",
                                      "opnds" : [ "text_score", 0.25 ] } ] },
                                "ELSE",
                                { "op" : "add", "opnds" : [
                                    75,
                                    { "op" : "mul",
                                      "opnds" : [ "text_score", 0.125] },
                                    { "op" : "mul",
                                      "opnds" : [ "vector_score", 0.125] } ] }
                               ] }
             }'))
FROM DUAL;
```

The score_calc defined in the above example translates to :

```
       (CASE WHEN tscr > 0 AND vscr > 0 THEN
                  75 + (tscr * 0.125) + (vscr * 0.125)
             WHEN tscr = 0 THEN
                  50 + (vscr * 0.25)
             WHEN vscr = 0 THEN
```

```
        25 + (tscr * 0.25)
    ELSE 0.0 END)
```

vector

Specify query parameters for semantic search against the vector index part of your hybrid vector index:

- `search_text`: Search text string (query text).

  This string is converted into a query vector (embedding), and is used in a `VECTOR_DISTANCE` query to search against the vectorized chunk index.

  For example:

  ```
  SELECT DBMS_HYBRID_VECTOR.SEARCH(
      json('{ "hybrid_index_name" : "my_hybrid_idx",
              "vector":
                      { "search_text" : "C, Python" }
            }'))
  FROM DUAL;
  ```

- `search_vector`: Vector embedding (query vector).

  This embedding is directly used in a `VECTOR_DISTANCE` query to search against the vectorized chunk index.

  > **Note**
  >
  > `search_vector` is an alternative to the above mentioned `search_text` when the semantic query is already available as a vector. The vector embedding that you pass here must be generated using the same embedding model used for semantic search by the specified hybrid vector index.

  For example:

  ```
  SELECT JSON_SERIALIZE(
           DBMS_HYBRID_VECTOR.SEARCH(
              json_object( 'hybrid_index_name' value 'my_hybrid_idx',
                   'vector' value json_object( 'search_vector' value
  vector_serialize(
                                              vector_embedding(doc_model
                                                          using
                                                          'C, Python,
  Database'
                                                          as data)
                                                      RETURNING CLOB)
                                               RETURNING JSON)
                    RETURNING JSON))
             RETURNING CLOB PRETTY)
  FROM dual;
  ```

- `search_mode`: Document or chunk search mode in which you want to query the hybrid vector index:

Parameter	Description
DOCUMENT (default)	Returns document-level results. In document mode, the result of your search is a list of document IDs from the base table corresponding to the list of best documents identified.
CHUNK	Returns chunk-level results. In chunk mode, the result of your search is a list of chunk identifiers and associated document IDs from the base table corresponding to the list of best chunks identified, regardless of whether the chunks come from the same document or different documents. The content from these chunk texts can be used as input for LLMs to formulate responses.

For example, semantic search in chunk mode:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "CHUNK"
          }
      }'))
FROM DUAL;
```

- `aggregator`: Aggregate function to apply for ranking the vector scores for each document in `DOCUMENT SEARCH_MODE`.

Parameter	Description
MAX (default)	Standard database aggregate function that selects the top chunk score as the result score.
AVG	Standard database aggregate function that sums the chunk scores and divides by the count.
MEDIAN	Standard database aggregate function that computes the middle value or an interpolated value of the sorted scores.
BONUSMAX	This function combines the maximum chunk score with the remainder multiplied by the average score of the other top scores.
WINAVG	This function computes the maximum average of the rolling window (of size `windowSize`) of chunk scores.
ADJBOOST	This function computes the average "boosted" chunk score. The chunk scores are boosted with the `BOOSTFACTOR` multiplied by average score of the surrounding chunk's scores (if they exist).
MAXAVGMED	This function computes a weighted sum of the `MAX`, `AVGN`, and `MEDN` values.

For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "AVG"
          }
```

```
      }'))
FROM DUAL;
```

- `result_max`: The maximum number of vector results in distance order to fetch (approx) from the vector index.

  Value : Any positive integer greater then `0` (zero)

  Default: If the field is not specified, by default, the maximum is computed based on topN.

  For Example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 5,
             "result_max"    : 100
          }
      }'))
FROM DUAL;
```

- `score_weight`: Relative weight (degree of importance or preference) to assign to the semantic `VECTOR_DISTANCE` query. This value is used when combining the results of RSF ranking.

  Value: Any positive integer greater than `0` (zero)

  Default: `10` (implies 10 times more importance to vector query than text query)

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 5
          }
      }'))
FROM DUAL;
```

- `rank_penalty`: Penalty (denominator in RRF, represented as `1/(rank+penalty)`) to assign to vector query. This can help in balancing the relevance score by reducing the importance of unnecessary or repetitive words in a document. This value is used when combining the results of RRF ranking.

  Value: `0` (zero) or any positive integer

  Default: `1`

For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_scorer"     : "rrf",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 5,
             "rank_penalty"  : 2
          }
      }'))
FROM DUAL;
```

- `inpath`: Valid JSON paths

  `vector.inpath` uses the vectorizer paths as you have in the document. Providing this parameter will restrict the search to the paths specified in this field. Accepts an array of paths in valid JSON format - (`$.a.b.c.d`).

  The list of paths match against VECTORIZER index path lists to form a query constraint on the vector index search. Simple wild cards on the paths are supported such as $.main.*.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_scorer"     : "rrf",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 5,
             "rank_penalty"  : 2,
             "inpath"   : ["$.person.*", "$.product.*"]
          }
      }'))
FROM DUAL;
```

- `accuracy`: Target accuracy to assign to the semantic `VECTOR_DISTANCE` query.

  Value: Any positive integer between `0` (zero) and `100`.

  Default: `0`(zero). The value `0` indicates that the internal default for vector_distance query will be assigned to the field.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
```

```
                "search_text"   : "leadership experience",
                "search_mode"   : "DOCUMENT",
                "aggregator"    : "MAX",
                "score_weight"  : 5,
                "rank_penalty"  : 2,
                "inpath"        : ["$.person.*", "$.product.*"],
                "accuracy"      : 95
             }
       }'))
FROM DUAL;
```

- `index_probes`: Number of probes to assign to the semantic `VECTOR_DISTANCE` query.

  Value: Any positive integer greater than `0` (zero).

  Default: `0`(zero). The value `0` indicates that the internal default number of probes will be assigned to the field.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "vector":
          {
             "search_text"   : "leadership experience",
             "search_mode"   : "DOCUMENT",
             "aggregator"    : "MAX",
             "score_weight"  : 5,
             "rank_penalty"  : 2,
             "inpath"        : ["$.person.*", "$.product.*"],
             "accuracy"      : 95,
             "index_probes"  : 3
          }
      }'))
FROM DUAL;
```

- `index_efsearch`: `efs` to assign to the semantic `VECTOR_DISTANCE` query.

  Value: Any positive integer greater than `0` (zero). The value `0` indicates that the internal default for vector_distance query will be assigned to the field.

  Default: `0`(zero)

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name"   : "my_hybrid_idx",
         "vector":
          {
             "search_text"     : "leadership experience",
             "search_mode"     : "DOCUMENT",
             "aggregator"      : "MAX",
             "score_weight"    : 5,
             "rank_penalty"    : 2,
             "inpath"          : ["$.person.*", "$.product.*"],
             "accuracy"        : 95,
             "index_probes"    : 3,
```

```
              "index_efsearch"  : 500,
           }
       }'))
FROM DUAL;
```

- `filter_type`: Vector index hint filter type. For more information on optimizer plans, hints and filter types for vector indexes, please refer to Optimizer Plans for Vector Indexes and Vector Index Hints.

  Value: The `filter_type` field could take one of the following values:

  – `PRE_W` - Pre-filter with join back. This applies only to HNSW indexes.

  – `PRE_WO` - Pre-filter without join back. This applies to both HNSW and IVF indexes.

  – `IN_W` - In-filter with join back. This applies only to HNSW indexes.

  – `IN_WO` - In-filter without join back. This applies only to HNSW indexes.

  – `POST_WO` - Post-filter without join back. This applies only to IVF indexes.

  Default: No filter type hint.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name"   : "my_hybrid_idx",
         "vector":
          {
             "search_text"     : "leadership experience",
             "search_mode"     : "DOCUMENT",
             "aggregator"      : "MAX",
             "score_weight"    : 5,
             "rank_penalty"    : 2,
             "inpath"          : ["$.person.*", "$.product.*"],
             "accuracy"        : 95,
             "index_probes"    : 3,
             "index_efsearch"  : 500,
             "filter_type"     : "IN_WO"
          }
      }'))
FROM DUAL;
```

text

Specify query parameters for keyword search against the Oracle Text index part of your hybrid vector index:

- `contains`: Search text string (query text).

  This string is converted into an Oracle Text `CONTAINS` query operator syntax for keyword search.

  You can use `CONTAINS` query operators to specify query expressions for full-text search, such as OR (|), AND (&), STEM ($), MINUS (-), and so on. For a complete list of all such operators to use, see *Oracle Text Reference*.

  For example:

With a text contains string for pure keyword search:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "text":
                   { "contains" : "C and Python" }
          }'))
FROM DUAL;
```

With separate search texts using `vector` and `text` sub-elements for hybrid search. One search text or a vector embedding to run a `VECTOR_DISTANCE` query for semantic search. A second search text to run a `CONTAINS` query for keyword search. This query conducts two separate keyword and semantic queries, where keyword scores and semantic scores are combined:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "vector":
                    { "search_text" : "leadership experience" },
             "text":
                    { "contains" : "C and Python" }
          }'))
FROM DUAL;
```

- `search_text`: The alternative search text to use to construct a contains query automatically.

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "text":
                   { "contains"    : "C and Python",
                     "search_text" : "data science skills"
                   }
          }'))
FROM DUAL;
```

- `json_textcontains`: An alternate JSON expression to use instead of `contains` AND `search_text`.

> **Note**
>
> It is an error to specify `json_textcontains` **WITH** either `text.contains` or `text.search_text`.

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "text":
                   { "json_textcontains"    : ["$.person", "$C
and $Python"]
                   }
          }'))
FROM DUAL;
```

- score_weight: Relative weight (degree of importance or preference) to assign to the text CONTAINS query. This value is used when combining the results of RSF ranking.

  Value: Any positive integer greater than 0 (zero)

  Default: 1 (implies neutral weight)

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "C and Python",
             "score_weight"  : 1
          }
      }'))
FROM DUAL;
```

- rank_penalty: Penalty (denominator in RRF, represented as 1/(rank+penalty) to assign to keyword query.

  This can help in balancing the relevance score by reducing the importance of unnecessary or repetitive words in a document. This value is used when combining the results of RRF ranking.

  Value: 0 (zero) or any positive integer

  Default: 5

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "C and Python",
             "rank_penalty"  : 5
          }
      }'))
FROM DUAL;
```

- inpath: Valid JSON paths

  Providing this parameter will restrict the search to the paths specified in this field. Accepts an array of paths in valid JSON format - ($.a.b.c.d).

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "C and Python",
             "rank_penalty"  : 5,
             "inpath"    : ["$.person.*","$.product.*"]
          }
```

```
      }'))
FROM DUAL;
```

- `result_max`: The maximum number of document results (ordered by score) to retrieve from the document index. If not provided, the maximum is computed based on the `topN`.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "C and Python",
             "rank_penalty"  : 5,
             "inpath"        : ["$.person.*","$.product.*"],
             "result_max"    : 100
          }
      }'))
FROM DUAL;
```

  If the `result_max` value is not provided, it is calculated based on the topN. The following table summarizes the computed `result_max` values for the various search modes and search fusion combinations.

  Table 12-35 `result_max` Computation

Search Type	Search Mode	Search Fusion	`result_max` Computation (Text)	`result_max` Computation (Vector)
Pure keyword (text)	N/A	N/A	`return.topN`	N/A
Pure semantic (vector)	`CHUNK`	N/A	N/A	`return.topN`
Pure semantic (vector)	`DOCUMENT`	N/A	N/A	`return.topN × 10`
Hybrid	`CHUNK`	N/A	`return.topN × 100`	`return.topN`
Hybrid	`DOCUMENT`	`UNION, VECTOR_ONLY, MINUS_TEXT`	`return.topN × 20`	`return.topN × 4`
Hybrid	`DOCUMENT`	`INTERSECT, TEXT_ONLY, MINUS_VECTOR`	`return.topN × 100`	`return.topN × 30`
Keyword re-ranking	`DOCUMENT`	`RERANK`	`return.topN`	N/A

  `result_max` calculation guidelines:

  It is recommended to use `result_max` Computation table as a guidance and perform custom calculations to determine an appropriate `result_max` value to address the following business use cases:

  – Aggregating vector chunks to documents: Vector search often returns vector chunks (smaller segments of data) that are later aggregated to document results. When aggregating vector chunks into document results, `vector result_max` needs to be set to a value larger than `topN`. This is because each document may consist of multiple chunks, and you need to ensure that enough chunks are retrieved to cover the `top N` documents.

  – Combining vector and text search results: When combining vector search results with text search results, `result_max` needs to be set higher to ensure the intersection yields enough results. For certain `search_fusion` operators that require both the text and vector search results to match (for example, `INTERSECT`), using a small result set for each could give you little or no overlap—meaning you do not get enough (or any) results after intersection. Higher `result_max` values increase your chances of getting the number of results you want.

- `snippet`: Enables a text snippet for text-only (non-hybrid) search results OR text-only hybrid search results in `UNION`, `TEXT_ONLY`, `MINUS_VECTOR`, and `RERANK` mode. This parameter takes the desired maximum length of the snippet as an input. A value of 0 would disable this feature. The snippet is generated using the CTX_DOC.SNIPPET procedure which returns one or more most relevant fragments for a document that contains the query term. The Query term in this case is specified by either explicit `text.contains` value or derived from `text.search_text` or common `search_text`, plus `inpath` modifications. The resulting snippet is returned in `chunk_text`.

  For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "text":
          {
             "contains"      : "C and Python",
             "rank_penalty"  : 5,
             "inpath"        : ["$.person.*","$.product.*"],
             "result_max"    : 100,
             "snippet"       : 250
          }
      }'))
FROM DUAL;
```

whether to enable the `snippet` depends on both the search_mode and the search_fusion settings that you specify.

`search_mode` determines how to query the hybrid vector index. In the `CHUNK` mode, results are fused with vector information. In the `DOCUMENT` mode, the results are not fused with vector information. The `CHUNK` mode does not require snippet as the results are fused with vector information. The `DOCUMENT` mode may require snippet depending upon the `search_fusion` value that you specify to combine the text and vector search results.

`search_fusion` determines how the results from text and vector searches are combined. Different fusion modes affect whether the combined (fused) result set contains text search results or vector search results. Text search results require a text snippet to provide meaningful summaries to users. The following table illustrates the seven `search_fusion` values, how the results are returned for each value, and whether a text snippet is required:

Table 12-36 Fusion Modes and Snippet Generation

`search_fusion`	Description	Only Text Search Results	Only Vector Search Results	Both Text and Vector Search Results	Vector `chunk_text`	Text Snippet Requirement
`INTERSECT`	The fused results contain both text and vector search results.	No	No	Yes	Yes	No
`UNION`	The fused results contain either text or vector search results, or both.	Yes	Yes	Yes	Maybe	Maybe
`TEXT_ONLY`	The fused results contain the text search results that appear in text search, including those that appear in both text and vector search results.	Yes	No	Yes	Maybe	Maybe
`VECTOR_ONLY`	The fused results contain the vector search results that appear in vector search, including those that appear in both vector and text search results.	No	Yes	Yes	Yes	No
`MINUS_TEXT`	The fused results contain the vector search results that appear in vector search, excluding those that appear in both text and vector search results.	No	Yes	No	Yes	No

Table 12-36 (Cont.) Fusion Modes and Snippet Generation

search_fusion	Description	Only Text Search Results	Only Vector Search Results	Both Text and Vector Search Results	Vector chunk_text	Text Snippet Requirement
MINUS_VECTOR	The fused results contain the text search results that appear in text search, excluding those that appear in both text and vector search results.	Yes	No	No	No	Yes
RERANK	Text search results, re-ranked based on their vector score.	Yes	No	No	No	Yes

> **Note**
>
> Alternatively snippets can be generated outside of the API using a syntax like below:
>
> ```
> SELECT NVL(chunk_text, CTX_DOC.SNIPPET(...)) chunk_text FROM
> JSON_TABLE(dbms_hybrid_vector.search(params), COLUMNS ...)
> ```
>
> Using this syntax is beneficial in the following cases:
>
> - Require snippets for both text and vector search results: You want to create snippets for both text and vector search results to provide a uniform experience or to ignore semantic content in previews. When the snippet parameter is enabled, Oracle generates text snippets for text search results that are not fused with vector information. Therefore, you can use this API to generate snippets for both text and vector search results.
> - Performance optimization: Generating snippets during the search can add overhead. You may prefer to skip snippet generation initially and produce snippets only on demand.

filter_by

To constrain the search results via standard relational logical constraints :

Parameter	Value
op	Logical comparison operator. Accepted values - One of these operators :

- **Simple comparison operators** : `'<'`, `'>'`, `'<='`, `'>='`, `'='`, `'!='`, `'^='`, `'<>'`, `'LIKE'`, `'LIKEC'`, `'LIKE2'`, `'LIKE4'`, `'INSTR'`, `'INSTR2'`, `'INSTR4'`,`'INSTRB'`, `'INSTRC'`, `'STSTR'`, `'STSTR2'`, `'STSTR4'`,`'STSTRB'`, `'STSTRC'`, `'REGEXP_LIKE'`, `'BETWEEN'`, `'EXISTS'`

> **Note**
>
> 1. `STSTR` is the only non-standard operator in the list. It stands for "**START STRING**" and is analogous to `INSTR`, but the result must be equal to position 1.
> 2. The EXISTS operator translates into a `JSON_EXISTS(col, arg1)` condition. The first string argument is the path-expression. The full SQL `JSON_EXISTS` supports other parameters, including the `PASSING` clause which provides bind variables to reference in the path expression. To support the passing clause, the `filter_by` element has an optional `passing` parameter, which is explained here.

- **Group comparison operators** :
  - 18 combinations of `'<'`, `'>'`, `'<='`, `'>='`, `'='`, `'!='` with `ANY`, `SOME`, `ALL`
  - `IN`
- **Logical operators** : `'AND'`, `'OR'`, `'NOT'`, `'NOTAND'`, `'NOTOR'`

	Note `"NOTAND" and "NOTOR"` are short-hand for the following expressions, useful in reducing the JSON expression tree. `NOTOR` is NOT (arg1 OR arg2 ...). `NOTAND` is NOT (arg1 AND arg2 ...)
`col`	Base table column name. **Note** • No `column` is required for logical operators. • Only one of `col` or `path` could be specified in the same element.
`path`	The JSON path dot notation within a base table JSON column.

	Note • No `path` is required for logical operators. • Only one of `col` or `path` could be specified in the same element. • If the base table has a JSON column called `data`, then the syntax would be "`data.path`" where the path is case-sensitive matching the JSON data schema. For more details, see [JSON dot notation]().
`type`	The data type of the column. Accepted types include : `number, date, timestamp` and `string`.
`func`	For the comparison operators, an optional function can be applied to the column value before the comparison. These functions are the standard SQL functions. The one exception is "`TO_DOUBLE`" is provided as an alias to the full name "`TO_BINARY_DOUBLE`" Accepted values include : `ABS, FLOOR, LENGTH, CEILING, UPPER, LOWER, TO_BOOLEAN, TO_DATE, TO_DOUBLE, TO_BINARY_DOUBLE, TO_NUMBER, TO_CHAR, TO_TIMESTAMP`.
`args`	An array of arguments to the operator: • For simple comparison operators, the `args` contains a single literal value. It is an error to provide 0 or more than 1 arguments.

	• For group comparison operators, the `args` contains 1 or more literal values. It is an error to provide 0 arguments. • For logical operators, the `args` contains sub-elements of the same structure, forming an expression tree.

`passing`	An array of variable bindings for the `EXISTS` operator path expression (ignored if not `EXISTS`). Each array element has three required attributes: `var`, `type`, `val`. For example, the following `filter_by` parameters translates to `JSON_EXISTS(COLUMN_NAME, ARG1, PASSING TO_TYPE('VALUE') AS "VARIABLE", ....)`
	
	`{ "op" : "EXISTS",`
	`"col" : COLUMN_NAME,`
	`"type" : "STRING",`
	`"args" : [ ARG1 ],`
	`"passing" : [ { "var" : VARIABLE,`
	`"type" : TYPE, "val" : VALUE }, ... ]}`
	
	More details on JSON_EXISTS condition can be found here.

For example: Using simple comparison operators

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "filter_by":
                    { "op"   : "<",
                      "col"  : "price",
                      "type" : "number",
                      "func" : "ABS"
                      "args" : ["10"] }
          }'))
FROM DUAL;
```

For example: Using group comparison operators

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "filter_by":
                    { "op"   :  "IN",
                      "path" : "DATA.brand",
                      "type" " "string",
                      "args" : ["nike", "adidas"] }
          }'))
FROM DUAL;
```

For example: Using logical operators

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "filter_by":
                    { "op"   :  "AND",
                      "args" : [
                      {"op" : "IN", "col" : "brand", "type" : "string",
"args" : ["nike", "adidas"]},
                      {"op" : "<", "col" : "price", "type" : "number",
"args" : ["10"]}]
                    }
          }'))
FROM DUAL;
```

For example: Passing a JSON array as a variable binding

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json('{ "hybrid_index_name" : "my_hybrid_idx",
            "filter_by":
                    { "op"      :  "EXISTS",
                      "col"     :  "data",
                      "type"    :  "string",
                      "args"    :  [ "$?(@.dateline == $v1[*])" ],
                      "passing" :  [ {"var" : "v1", "type" : "JSON", "val" :
"[ \"HOUSTON (AP)\" ]" }]
                    }
          }'))
FROM DUAL;
```

return

Specify which fields to appear in the result set:

Parameter	Description
`topN`	Maximum number of best-matched results to be returned Value: Any integer greater than `0` (zero) Default: `20`
`values`	Return attributes for the search results. Values for scores range between 100 (best) to 0 (worse). • `rowid`: Row ID associated with the source document. • `score`: Final score computed from keyword-and-semantic search scores. • `vector_score`: Semantic score from vector search results. • `text_score`: Keyword score from text search results. • `vector_rank`: Ranking of chunks retrieved from semantic or `VECTOR_DISTANCE` search. • `text_rank`: Ranking of documents retrieved from keyword or `CONTAINS` search. • `chunk_text`: Human-readable content from each chunk. • `chunk_id`: ID of each chunk text. • `paths`: Paths from which the result occurred. Default: All the above return attributes EXCEPT `paths` are shown by default. As there are no paths for non-JSON, you need to explicitly specify the `paths` field.

Parameter	Description
`format`	Format of the results as: • `JSON` (default) • `XML`

For example:

```
SELECT DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_text"       : "C, Python",
         "return":
          {
            "values"         : [ "rowid", "score", "paths" ],
            "topN"           : 10,
            "format"         : "JSON"

          }
      }'))
FROM DUAL;
```

Complete Example With All Query Parameters

The following example shows a hybrid search query that performs separate text and vector searches against `my_hybrid_idx`. This query specifies the `search_text` for vector search using the `vector_distance` function as `prioritize teamwork and leadership experience` and the keyword for text search using the `contains` operator as `C and Python`. The search mode is `DOCUMENT` to return the search results as topN documents.

```
SELECT JSON_SERIALIZE(
  DBMS_HYBRID_VECTOR.SEARCH(
    json(
      '{ "hybrid_index_name" : "my_hybrid_idx",
         "search_fusion"     : "INTERSECT",
         "search_scorer"     : "rsf",
         "vector":
          {
            "search_text"       : "prioritize teamwork and leadership
experience",
            "search_mode"       : "DOCUMENT",
            "score_weight"      : 10,
            "rank_penalty"      : 1,
            "aggregator"        : "SCORE_AGGR",
            "aggregator_params" : ["AVGN", 5, 50],
            "inpath"            : ["$.main.body", "$.main.summary"],
            "accuracy"          : 95
          },
         "text":
          {
            "contains"     : "C and Python",
            "score_weight" : 1,
            "rank_penalty" : 5,
            "inpath"       : ["$.main.body"]
```

```
          },
          "return":
          {
             "format"        : "JSON",
             "topN"          : 3,
             "values"        : [ "rowid", "score", "vector_score",
                                 "text_score", "vector_rank",
                                 "text_rank", "chunk_text", "chunk_id",
"paths" ]
          }
      }'
    )
  ) pretty)
FROM DUAL;
```

The top 3 rows are ordered by relevance, with higher scores indicating a better match. All the return attributes are shown by default:

```
[
  {
    "rowid"         : "AAAR9jAABAAAQeaAAA",
    "score"         : 58.64,
    "vector_score"  : 61,
    "text_score"    : 35,
    "vector_rank"   : 1,
    "text_rank"     : 2,
    "chunk_text"    : "Candidate 1: C Master. Optimizes low-level system
(i.e. Database)
                       performance with C. Strong leadership skills in
guiding teams to
                       deliver complex projects.",
    "chunk_id"      : "1",
    "paths"         : ["$.main.body","$.main.summary"]
  },
  {
    "rowid"         : "AAAR9jAABAAAQeaAAB",
    "score"         : 56.86,
    "vector_score"  : 55.75,
    "text_score"    : 68,
    "vector_rank"   : 3,
    "text_rank"     : 1,
    "chunk_text"    : "Candidate 3: Full-Stack Developer. Skilled in
Database, C, HTML,
                       JavaScript, and Python with experience in building
responsive web
                       applications. Thrives in collaborative team
environments.",
    "chunk_id"      : "1",
    "paths"         : ["$.main.body", "$.main.summary"]
  },
  {
    "rowid"         : "AAAR9jAABAAAQeaAAD",
    "score"         : 51.67,
    "vector_score"  : 56.64,
    "text_score"    : 2,
```

```
    "vector_rank"   : 2,
    "text_rank"     : 3,
    "chunk_text"    : "Candidate 2: Database Administrator (DBA). Maintains 
and secures
                       enterprise database (Oracle, MySql, SQL Server). 
Passionate about
                       data integrity and optimization. Strong mentor for 
junior DBA(s).",
    "chunk_id"      : "1",
    "paths"         : ["$.main.body", "$.main.summary"]
  }
]
```

End-to-end example:

To see how to create a hybrid vector index and explore all types of queries against the index, see Query Hybrid Vector Indexes End-to-End Example.

Related Topics

- Perform Hybrid Search

GET_SQL

Use the `DBMS_HYBRID_VECTOR.GET_SQL` PL/SQL function to return the internal SQL query that is generated from the parameters.

When calling the `DBMS_HYBRID_VECTOR` Search function, the API is called using the `JSON` Document format. Using the `GET_SQL` procedure shows the SQL that the `DBMS_HYBRID_VECTOR.SEARCH` API has generated. The resulting SQL can be used to view the query execution plan to view the index chosen for the hybrid search operation. An example is shown below:

```
SET LINESIZE 200;
SET PAGESIZE 1000;
SET TAB OFF;
SET TRIMSPOOL ON;
DECLARE
    res CLOB;
BEGIN
    res := dbms_hybrid_vector.get_sql(JSON('{"hybrid_index_name" : 
"trecvol2j_idx",
                                            "search_text" : "offers",
                                            "return" : { "values" : 
[ "score" ] } }'));
    execute immediate 'EXPLAIN PLAN FOR '||res;
END;
/


SELECT PLAN_TABLE_OUTPUT FROM TABLE(DBMS_XPLAN.DISPLAY(NULL,NULL,'ADVANCED'));
```

SEARCHPIPELINE

Use the standard table function `DBMS_HYBRID_VECTOR.SEARCHPIPELINE` to return a pipeline of row records.

This pipeline function accepts valid JSON query input and returns a pipeline of row records. The syntax is as shown below:

```
FUNCTION SEARCHPIPELINE(qparams JSON)
RETURN results PIPELINED;
```

The `results` is of type `RECORD`. The `results` contains the following fields:

Field	Type
`doc_rowid`	varchar2(18)
`score`	number
`vector_score`	number
`text_score`	number
`vector_rank`	number
`text_rank`	number
`chunk_text`	varchar2(32767)
`chunk_id`	varchar2(4000)
`paths`	varchar2(4000)

The record members are the column names in the `SELECT` statement. These names are the same as the JSON field names that are returned in `DBMS_HYBRID_VECTOR_SEARCH()`, except that `paths` is a list of field IDs in the record, where as the JSON result maps the ids to their actual paths (in an array). Also note that the result record could not have a member named `rowid` nor a member with a `rowid` type.

Example 12-1

```
SELECT
   chartorowid(doc_rowid) as doc_rowid,
   score,
   vector_score,
   text_score,
   vector_rank,
   text_rank,
   chunk_text,
   chunk_id,
```

```
   paths
FROM dbms_hybrid_vector.searchpipeline(JSON('{"hybrid_index_name" : "idx",
                                              "search_text" : "teamwork" }'));
```

If you do not wish to use the table function `DBMS_HYBRID_VECTOR.SEARCHPIPELINE()`, the original SEARCH API can be wrapped in a `JSON_TABLE` specification. This is shown in the example below:

```
SELECT jt.*
FROM
   JSON_TABLE(
        dbms_hybrid_vector.search(
                      json_object('hybrid_index_name' value 'idx',
                      'search_text' value 'teamwork'
                      RETURNING JSON)
                   ),
                   '$[*]' COLUMNS idx for ORDINALITY,
                                  doc_rowid PATH '$.rowid',
                                  score NUMBER PATH '$.score',
                                  vector_score NUMBER PATH '$.vector_score',
                                  text_score NUMBER PATH '$.text_score',
                                  vector_rank NUMBER PATH '$.vector_rank',
                                  text_rank NUMBER PATH '$.text_rank',
                                  chunk_text PATH '$.chunk_text',
                                  chunk_id PATH '$.chunk_id',
                                  paths PATH '$.paths'
                    ) jt
ORDER by idx ASC
```

Usage Notes

When using `DBMS_HYBRID_VECTOR.SEARCHPIPELINE` with `search_fusion` set to `RERANK` and `search_scorer` set to `RRF` (Reciprocal Rank Fusion) or `WRRF` (Weighted Reciprocal Rank Fusion), the results may not reflect the intended semantic ordering. Instead, the final ranking may prioritize rows based on their text rank rather than the semantic rerank score derived from vector distances. This behavior can lead to misleading results where rows with lower vector scores are ranked higher due to their higher text ranks.

`RERANK` mode is not a fusion search mode, despite being specified via the `search_fusion` parameter. Instead, `RERANK` performs a text keyword search and then uses vector scores to rerank the results. When `RRF` or `WRRF` is used with `RERANK` mode these scorers do not properly aggregate vector distances or semantic scores. Instead, it relies solely on the reciprocal text ranks and does not incorporate the semantic scores, even though they are computed. This behavior can lead to misleading results where rows with lower semantic scores (vector distances) are ranked higher due to their higher text ranks. Additionally, `RERANK` only considers documents retrieved by the initial text search (limited by `text.result_max`), meaning better semantically scoring documents may exist but are not evaluated. As a result, the final output may prioritize rows with higher text ranks, even if their vector scores are lower.

Applications relying on `DBMS_HYBRID_VECTOR.SEARCHPIPELINE` with `RERANK` and `RRF` or `WRRF` may receive results that appear valid but do not reflect the intended semantic ordering. This can lead to:

- Inaccurate rankings: Rows with higher semantic relevance (higher vector scores) may be ranked lower than less relevant rows with higher text ranks.

- Suboptimal search performance: Users may experience misleading or incorrect results, impacting data correctness and application reliability.

To achieve proper semantic reranking, avoid using `RRF` or `WRRF` scorers with `search_fusion=RERANK`. It is recommended to use `search_scorer=RSF` (Relative Score Fusion) for reranking, as it correctly prioritizes rows based on their vector scores.

DBMS_VECTOR_ADMIN

The `DBMS_VECTOR_ADMIN` package provides administrative interfaces for managing database configuration settings, such as modifying system parameters and performing privileged operations.

> **Note**
>
> In order to run the `DBMS_VECTOR_ADMIN` package, the user must be granted the `PDB_DBA` role.

Table 12-37 DBMS_VECTOR_ADMIN PACKAGE

Subprogram	Description
SET_PARAMETER	Sets a value for the specified input parameter.
GET_PARAMETER	Gets the current value for the specified input parameter.

- SET_PARAMETER
  Use the `DBMS_VECTOR_ADMIN.SET_PARAMETER` to set the value of the configuration parameters to facilitate the automatic IVF index rebuilds.
- GET_PARAMETER
  Use the `DBMS_VECTOR_ADMIN.GET_PARAMETER` procedure to get the current value for the specified input parameter.

SET_PARAMETER

Use the `DBMS_VECTOR_ADMIN.SET_PARAMETER` to set the value of the configuration parameters to facilitate the automatic IVF index rebuilds.

Syntax

```
DBMS_VECTOR_ADMIN.SET_PARAMETER (
    parameter       IN NUMBER,
    value           IN NUMBER
);
```

Parameters

Table 12-38 SET_PARAMETER Procedure Parameters

Parameter	Description
`parameter`	The name of the parameter for which you want to set a value.

Table 12-38 (Cont.) SET_PARAMETER Procedure Parameters

Parameter	Description
`value`	The value to assign to the specified parameter.

Example

The following is an example of setting the background optimization interval for vector indexes to 1800 seconds:

```
DECLARE
BEGIN
   DBMS_VECTOR_ADMIN.SET_PARAMETER(parameter =>
DBMS_VECTOR_ADMIN.VECTOR_INDEX_OPTIMIZATION_BACKGROUND_INTERVAL,
   value => 1800);
END;
/
```

GET_PARAMETER

Use the `DBMS_VECTOR_ADMIN.GET_PARAMETER` procedure to get the current value for the specified input parameter.

Syntax

```
DBMS_VECTOR_ADMIN.GET_PARAMETER (
    parameter      IN  NUMBER,
    value          OUT NUMBER
);
```

Parameters

Table 12-39 GET_PARAMETER Procedure Parameters

Parameter	Description
`parameter`	The name of the parameter for which you want to get the current value.
`value`	The current value of the specified parameter. This value is returned by the procedure.

Examples

The following is an example of retrieving and displaying the current background optimization interval setting for vector indexes:

```
DECLARE
  v_param_val NUMBER;
BEGIN
  DBMS_VECTOR_ADMIN.GET_PARAMETER(parameter =>
DBMS_VECTOR_ADMIN.VECTOR_INDEX_OPTIMIZATION_BACKGROUND_INTERVAL,
  value => v_param_val);
```

```
  DBMS_OUTPUT.PUT_LINE('parameter value = '|| v_param_val);
END;
/
```

```
parameter value = 1800
```

The following is an example of retrieving and displaying the current status of background optimization for vector indexes:

```
DECLARE
  v_param_val VARCHAR2;
BEGIN
  DBMS_VECTOR_ADMIN.GET_PARAMETER(parameter =>
DBMS_VECTOR_ADMIN.VECTOR_INDEX_OPTIMIZATION_BACKGROUND, value =>
  v_param_val);
  DBMS_OUTPUT.PUT_LINE('parameter value = '|| v_param_val);
END;
/
```

```
parameter value = 2
```

where, the output value 2 indicates that background optimization for vector indexes is enabled globally.

A

Python Classes to Convert Pretrained Models to ONNX Models (Deprecated)

This topic provides the functions, attributes and example usage of the deprecated python classes `EmbeddingModelConfig` and `EmbeddingModel`.

> **Note**
>
> This API is updated for 23.7. The version used for 23.6 and below used python packages EmbeddingModel and EmbeddingModelConfig. These packages are replaced with ONNXPipeline and ONNXPipelineConfig respectively. Oracle recommends that you use the latest version. If a you choose to use a deprecated class, a warning message will be shown indicating that the classes will be removed in the future and advising the user to switch to the new class. You can find the details of the updated python classes in Import Pretrained Models in ONNX Format

Examples for converting pretrained text models to ONNX format using EmbeddingModel, EmbeddingModelConfig

1. To load the python classes on the OML4Py client :

   ```
   from oml.utils import EmbeddingModel, EmbeddingModelConfig
   ```

2. You can get a list of all preconfigured models by running the following:

   ```
   EmbeddingModelConfig.show_preconfigured()
   ```

3. To get a list of available templates:

   ```
   EmbeddingModelConfig.show_templates()
   ```

4. Generate an ONNX file from the preconfigured model "sentence-transformers/all-MiniLM-L6-v2":

   ```
   em = EmbeddingModel(model_name="sentence-transformers/all-MiniLM-L6-v2")
   em.export2file("your_preconfig_file_name",output_dir=".")
   ```

5. Generate an ONNX model from the preconfigured model "sentence-transformers/all-MiniLM-L6-v2" in the database:

   ```
   em = EmbeddingModel(model_name="sentence-transformers/all-MiniLM-L6-v2")
   em.export2db("your_preconfig_model_name")
   ```

6. Generate an ONNX file using the provided text template:

```
config = EmbeddingModelConfig.from_template("text",max_seq_length=512)
em = EmbeddingModel(model_name="intfloat/e5-small-v2",config=config)
em.export2file("your_template_file_name",output_dir=".")
```

Functions and Attributes of EmbeddingModelConfig

The `EmbeddingModelConfig` class contains the properties required for the package to perform downloading, exporting, augmenting, validation, and storing of an ONNX model. The class provides access to configuration properties using the dot operator. As a convenience, well-known configurations are provided as templates.

Parameters

This table describes the functions and properties of the `EmbeddingModelConfig` class.

Functions	Parameter Type	Returns	Description
`from_template(name, **kwargs)`	• `name` (`String`): The name of the template • `**kwargs`: template properties to override or add	Instance of `EmbeddingModelConfig`	A static function that creates an `EmbeddingModelConfig` object based on a predefined template given by the name parameter. You can use named arguments to override the template properties.
`show_templates()`	NA	List of existing templates	A static function that returns a list of existing templates by name.

Functions	Parameter Type	Returns	Description
`show_preconfigured()`	• `include_properties` (`bool,optional`): A flag indicating whether properties should be included in the results. Defaults to `False` so only names will be included by default. • `model_name` (`str,optional`): A model name to filter by when including properties. This argument will be ignored if `include_properties` is `False`. Otherwise only the properties of this model will be included in the results.	A list of preconfigured model names or properties.	Shows a list of preconfigured model names, or properties. By default, this function returns a list of names only. If the properties are required, pass the `include_properties` parameter as `True`. The returned list will contain a single dict where each key of the dict is the name of a preconfigured model and the value is the property set for that model. Finally, if only a single set of properties for a specific model is required, pass the name of the model in the `model_name` parameter (the `include_properties` parameter should also be `True`). This will return a list of a single dict with the properties for the specified model.

Template Properties

The text template has configuration properties shown below:

```
"do_lower_case": true,
"post_processors":[{"name":"Pooling","type":"mean"},{"name":"Normalize"}]
```

> **Note**
>
> All other properties in the Properties table will take the default values. Any property without a default value must be provided when creating the `EmbeddingModelConfig` instance.

Properties

This table shows all properties that can be configured. preconfigured models already have these properties set to specific values. Templates will use the default values unless a user overrides it when using the `from_template` function on `EmbeddingModelConfig`.

Property	Description
`post_processors`	An array of post_processors that will be loaded after the model is loaded or initialized. The list of known and supported post_processors is provided later in this section. Templates may define a list of post_processors for the types of models they support. Otherwise, an empty array is the default.
`max_seq_length`	This property is applicable for text-based models only. The maximum length of input to the model as number of tokens. There is no default value. Specify this value for models that are not preconfigured.
`do_lower_case`	Specifies whether or not to lowercase the input when tokenizing. The default value is `True`.
`quantize_model`	Perform quantization on the model. This could greatly reduce the size of the model as well as speed up the process. It may however result in different results for the embedding vector (against the original model) and possibly small reduction in accuracy. The default value is `False`.
`distance_metrics`	An array of names of suitable distance metrics for the model. The names must be name of distance metrics used for Oracle vector distance operator. Only used when exporting the model to the database. Supported list is ["`EUCLIDEAN`","`COSINE`","`MANHATTAN`","`HAMMING`","`DOT`","`EUCLIDEAN_SQUARED`", "`JACCARD`"]. The default value is an empty array.
`languages`	An array of language (Abbreviation) supported in the Database. Only used when exporting the model to the database. For a supported list of languages, see Languages. The default value is an empty array.
`use_float16`	Specifies whether or not to convert the exported onnx model to float16. The default value is `False`.

Properties of post_processors

This table describes the built-in post_processors and their configuration parameters.

post_processor	Parameters	Description
`Pooling`	• `name`: `Pooling`. • `type`: Valid values should be `mean`(Default), `max`, `cls`	The Pooling post_processor summarizes the output of the transformer model into a fixed-length vector.
`Normalize`	• `name`: Specify `Normalize`	The Normalize post_processor bounds the vector values to a range using L2 normalization.

post_processor	Parameters	Description
`Dense`	• `name`: Dense • `in_features`: Input feature size • `out_features`: Output feature size • `bias`: Whether to learn an additive bias. The default value is `True`. • `activation_function`: Activation function of the dense layer. Currently only supports `Tanh` as the activation function.	Applies transformation to the incoming data.

Example: Configure post_processors

In this example, you override the post_processors in the sentence-transformers template with a `Max` Pooling post_processor followed by Normalization.

```
config = EmbeddingModelConfig.from_template("text")
config.post_processors = [{"name":"Pooling","type":"max"},
{"name":"Normalize"}]
```

Functions and Attributes of EmbeddingModel

Use the `EmbeddingModel` class to convert transformer models to the ONNX format with post_processing steps embedded into the final model.

Parameters

This table describes the signature and properties of the `EmbeddingModel` class.

Functions	Parameters	Description
`EmbeddingModel(model_name, configuration=None,settings={})`	• `model_name`: The name of the model to be used. For example, `medicalai/ClinicalBERT` • `configuration`: An initialized `EmbeddingModelConfig` object. This parameter must be specified when using a template. If not specified, the model will be assumed to be a preconfigured model. • `settings`: A dictionary of various settings that are global and control various operations such as logging levels and locations for files.	Creates a new instance of the `EmbeddingModel` class.

Settings

The settings object is a dictionary passed to the `EmbeddingModel` class. It provides global properties for the `EmbeddingModel` class that are used for non-model-specific operations, such as logging.

Property	Default Value	Description
`cache_dir`	`$HOME/.cache/OML`	The base directory used for downloads. Model files will be downloaded from the repository to directories relative to the `cache_dir`. If the `cache_dir` does not exist at time of execution, it will be created.
`logging_level`	`ERROR`	The level for logging. Valid values are ['DEBUG', 'INFO', 'WARNING', 'ERROR', 'CRITICAL']. **Note:** This log level is also applied globally to all python packages and is also mapped to the ONNXRuntime libraries.
`force_download`	`False`	Forces download of model files instead of reloading from cache.
`ignore_checksum_error`	`False`	Ignores any errors caused by mismatch in checksums when using preconfigured models.

Functions

This table describes the function and properties of the `EmbeddingModel` class.

Function	Parameters	Description
`export2file(export_name,output_dir=None)`	• `export_name(string)`: The name of the file. The file will be saved with the file extension `.onnx` • `output_dir(string)`: An optional output directory. If not specified the file will be saved to the current directory	Exports the model to a file.
`export2db(export_name)`	• `export_name(string)`: The name that will be used for the mining model object. This name must be compliant with existing rules for object names in the database.	Exports the model to the database.

Example: Preconfigured Model

This example illustrates the preconfigured embedding model that comes with the Python package. You can use this model without any additional configurations.

```
"sentence-transformers/distiluse-base-multilingual-cased-v2": {
        "max_seq_length": 128,
```

```
        "do_lower_case": false,
        "post_processors":[{"name":"Pooling","type":"mean"},
{"name":"Dense","in_features":768, "out_features":512, "bias":true,
"activation_function":"Tanh"}],
        "quantize_model":true,
        "distance_metrics": ["COSINE"],
        "languages": ["ar", "bg", "ca", "cs", "dk", "d", "us", "el", "et",
"fa", "sf", "f", "frc", "gu", "iw", "hi", "hr", "hu", "hy", "in", "i", "ja",
"ko", "lt",
                      "lv", "mk", "mr", "ms", "n", "nl", "pl", "pt", "ptb",
"ro", "ru", "sk", "sl", "sq", "lsr", "s", "th", "tr", "uk", "ur", "vn",
"zhs", "zht"]
    }
```

Glossary

dimension

A dimension refers to an array element of a vector.

dimension format

The dimensions of a vector can be represented using numbers of varying types and precisions, called the dimension format. The dimensions supported by vector embeddings in Oracle AI Database are `INT8` (1-byte signed integer), `FLOAT32` (4-byte single-precision floating point number), and `FLOAT64` (8-byte double-precision floating point number). All dimensions of a vector must have the same dimension format.

distance metric

Distance metric refers to the mathematical function used to compute distance between vectors. Popular distance metrics supported by Oracle AI Vector search include Euclidean distance, Cosine distance, and Manhattan distance, among others.

embedding model

Embedding models are Machine Learning algorithms that are trained to capture semantic information of unstructured data and represent it as vectors in multidimensional space. Different embedding models exist for different types of unstructured data, for example, BERT for Text data, ResNet-50 for Image data, and so on.

HNSW slice graph

An HNSW graph built over a subset of vector data, referred to as a "slice". The subset of vector data (slice) can represent any one of the following:

- Entire table
- Individual partition or sub partition of a table
- One or more vector distribution units (corresponding to vector data assigned to a single RAC instance in case of HNSW distribution on Oracle RAC)

hybrid search

Hybrid search is an advanced information retrieval technique that lets you search documents by both *keywords* and *vectors*. Hybrid searches are run against hybrid vector indexes by querying it in various search modes. By integrating traditional keyword-based text search with vector-based similarity search, you can improve the overall search experience and provide users with more relevant information.

hybrid vector index

A hybrid vector index is a class of specialized Domain Index that combines the existing Oracle Text search indexes and Oracle AI Vector Search vector indexes into one unified structure. A single index contains both textual and vector fields for a document, enabling you to perform a combination of keyword-based text search and vector-based similarity search simultaneously.

large language model

Large language models (LLMs) are advanced Machine Learning models designed to understand, process, and generate natural language for rich human interaction. They are typically built using deep learning algorithms and are pretrained on vast amounts of data. Popular examples include Open AI's GPT-4, Cohere's Command R+, and Meta's LLaMa 3.

multi-vector

Multi-vector refers to a scenario where multiple vectors correspond to a single entity. For example, a large text document can be chunked into paragraphs and every paragraph can be embedded into a separate vector. A similarity search query could retrieve matching documents based on the most similar paragraph (closest vectors) per document to a given query vector. Oracle AI Vector Search has the option of Partitioned Row Limiting Fetch syntax to enable efficient multi-vector searches.

neighbor graph

A neighbor graph is a graph-based data structure used in vector indexes. For example, a Hierarchical Navigable Small World (HNSW) vector index leverages a multilayer neighbor graph index. In a neighbor graph, each vertex of the graph represents a vector in the data set, and edges are created between vertices representing similar vectors.

query accuracy

Query accuracy is an intuitive indicator of the quality of an approximate query result obtained from a vector index search. Consider a query vector, for which an exact search, that searches through all vectors in the data set, returns Top 5 matches as `{ID1, ID3, ID5, ID7, ID9}`, and an approximate vector index search returns Top 5 matches as `{ID1, ID3, ID5, ID9, ID10}`. Since the approximate result has 4 out of 5 correct matches, the query accuracy is 80%.

query vector

Query vector refers to the vector embedding representing the item for which the user wants to find similar items using similarity search. For example, while searching for movies similar to a user's favorite movie, the vector embedding representing the user's favorite movie is the query vector.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is a popular technique for enhancing the accuracy of responses generated by [Large Language Models] by augmenting the user-provided prompt with relevant, up-to-date, enterprise-specific content retrieved using AI Vector Search. Applications, such as Chat Assistants, built using RAG are often more accurate, reliable, and cost-effective.

similarity search

Similarity Search is a common operation in information retrieval to find items in a data set that are similar to a user-provided query item. For example, finding movies similar to a user's favorite movie is an example of similarity search. Vectors can enable efficient similarity searches by leveraging the property that the mathematical distance between vectors is a proxy for similarity, as in, the more similar two items are, the smaller the distance between the vectors.

vector

A vector is a mathematical entity that has a magnitude and a direction. It is typically represented as an array of numbers, which are coordinates that define its position in a multidimensional space.

vector distance

Vector distance refers to the mathematical distance between two vectors in a multidimensional space. The vector distance between similar items is smaller than the vector distance between dissimilar items. Vector distance is meaningful only if the vectors being compared are generated by the same embedding model.

Vector Distribution Unit

A subset of vector dataset that is partitioned for distribution across the cluster based on the distribution strategy specified in the `CREATE VECTOR INDEX` DDL.

vector embedding

A vector embedding is a numerical representation of text, image, audio, or video data that encodes the semantic content of the data, and not the underlying words or pixels. The terms *vector* and *vector embedding* are often used interchangeably in AI Vector Search.

vector index

Vector indexes are a class of specialized indexing data structures that are designed to accelerate similarity searches using high-dimensional vectors. They use techniques such as clustering, partitioning, and neighbor graphs to group vectors representing similar items, which drastically reduces the search space, thereby making the search process extremely efficient. Unlike traditional databases indexes, vector indexes enable approximate similarity searches, which allow users to trade off query accuracy for query performance to better suit application requirements.

Vector Memory Pool

Vector Memory Pool is a region of the System Global Area (SGA) that is dedicated to storing In-Memory Neighbor Graph Vector Indexes (HNSW index), as well as metadata for Neighbor Partition Vector Indexes. It can be specified by using the `VECTOR_MEMORY_SIZE` parameter.